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(54) **PROTEASE-ACTIVATING CD45-GATE CAR**

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CPC **C07K 14/7051** (2013.01); **A61K 40/11** (2025.01); **A61K 40/15** (2025.01); **A61K 40/31** (2025.01); **A61K 40/4224** (2025.01); **A61K 40/4252** (2025.01); **A61K 40/4257** (2025.01); **C07K 14/70521** (2013.01); **C07K 14/70596** (2013.01); **C07K 16/28** (2013.01); **C07K 16/289** (2013.01); **C07K 19/00** (2013.01); **C12N 5/0634** (2013.01); **C12N 5/0636** (2013.01); **C12N 5/10** (2013.01); **C12N 15/79** (2013.01); **A61K 2239/31** (2023.05); **A61K 2239/38** (2023.05); **A61K 2239/49** (2023.05); **A61K 2239/55** (2023.05); **A61K 2239/59**

(2023.05); **C07K 2317/22** (2013.01); **C07K 2317/24** (2013.01); **C07K 2317/31** (2013.01); **C07K 2317/569** (2013.01); **C07K 2319/01** (2013.01); **C07K 2319/31** (2013.01); **C07K 2319/50** (2013.01)

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See application file for complete search history.

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(57)

ABSTRACT

A reversibly gated effector polypeptide e.g. a chimeric antigen receptor (protease-activating CD45-gate CAR) comprising an extracellular CD45 recruiting domain, a protease-cleavable linker, and a polypeptide comprising an extracellular ligand binding domain, a transmembrane domain, and an intracellular domain. Nucleic acids including vectors and expression vectors that encode the protease-activating CD45-gate CAR and cells including immune cells such as T cells that comprise and express the nucleic acids. Methods of treatment of various conditions including various forms of cancer comprising administering the cells including CAR T cell therapy. In some embodiments, the CD45 gate at least partially inhibits activation of the protease-activating CD45-gate CAR when the protease-activating CD45-gate CAR binds antigen. The inhibition is at least partially diminished, relieved and/or eliminated when the protease-activating CD45-gate CAR is exposed to a protease that can cleave the linker.

35 Claims, 27 Drawing Sheets

Specification includes a Sequence Listing.

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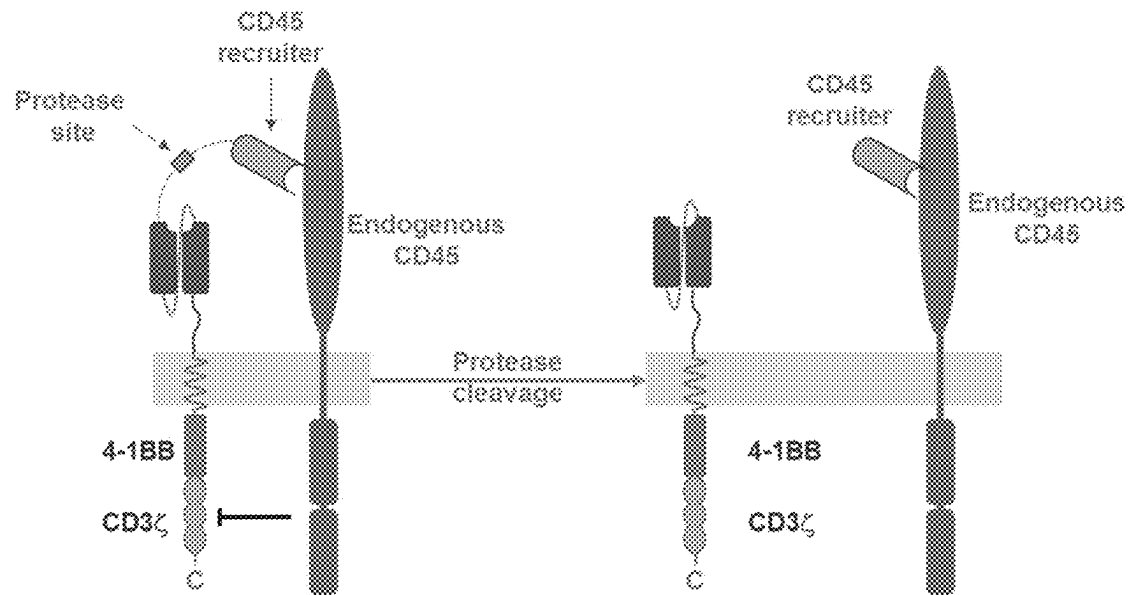


Fig. 1A

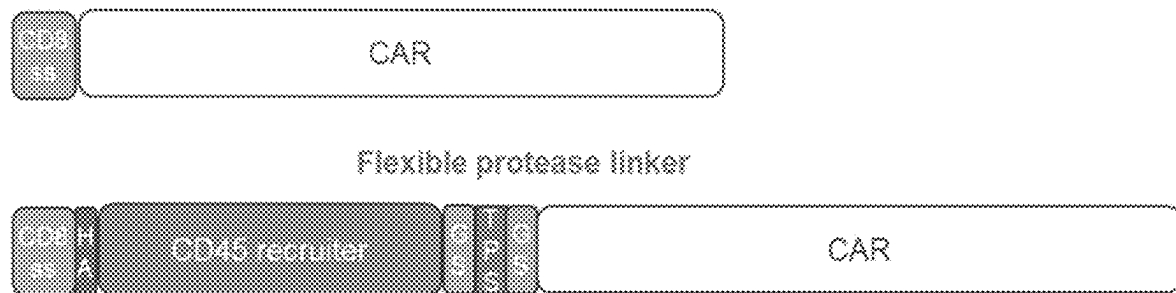


Fig. 1B

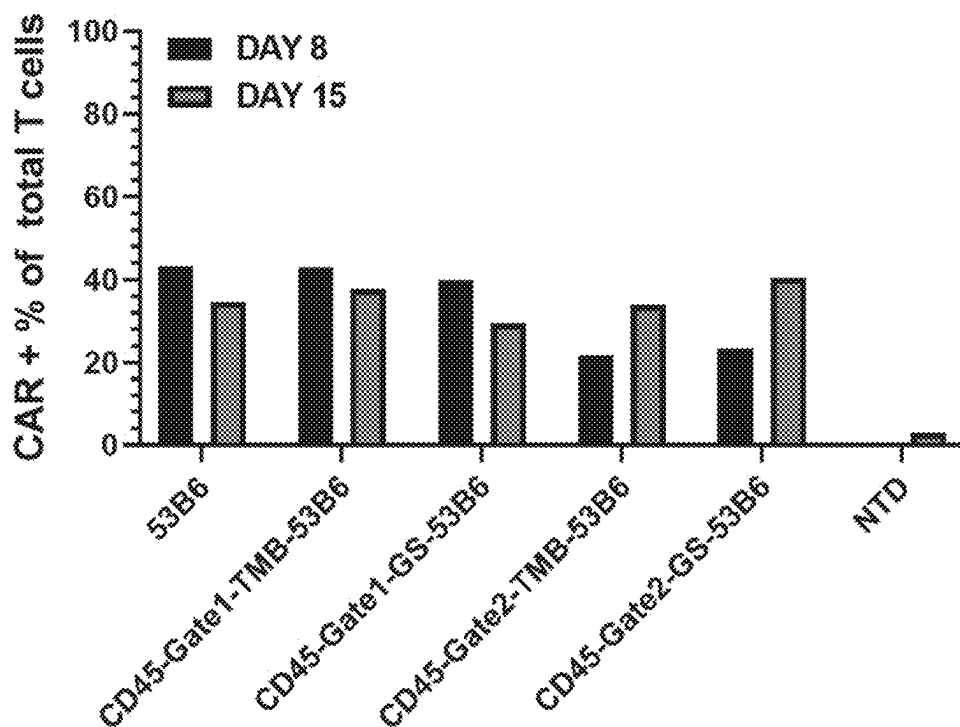


Fig. 2A

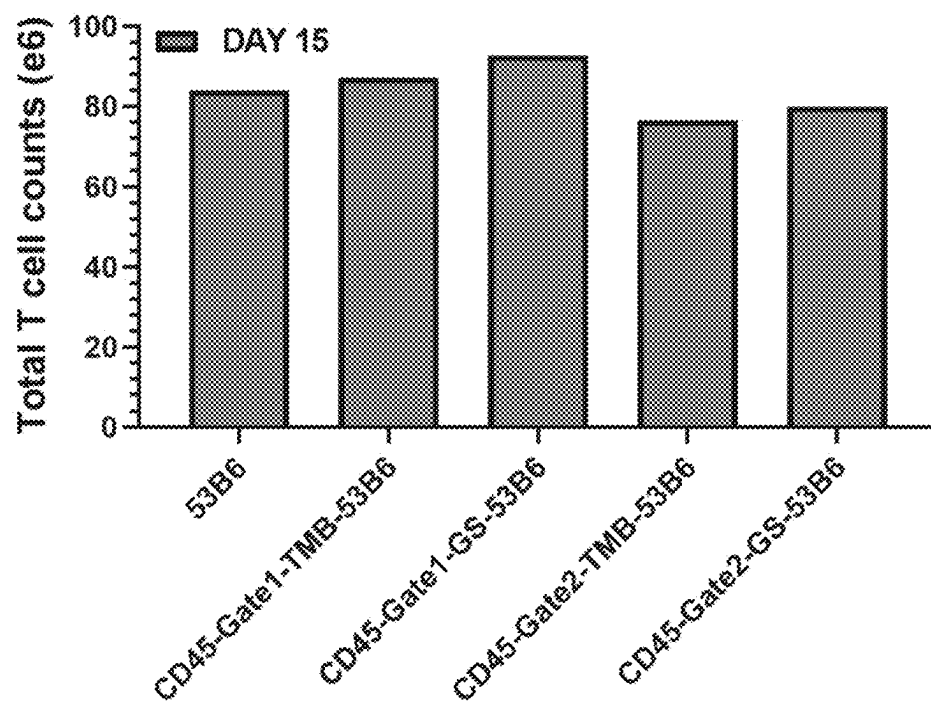


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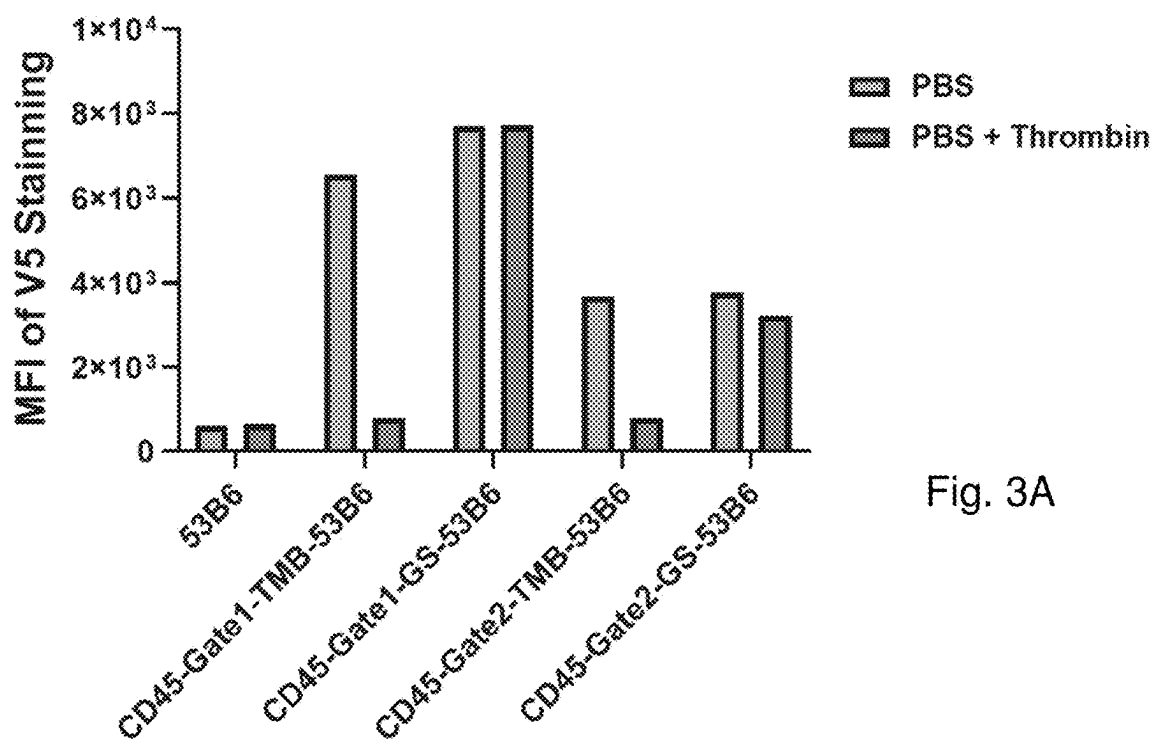


Fig. 3A

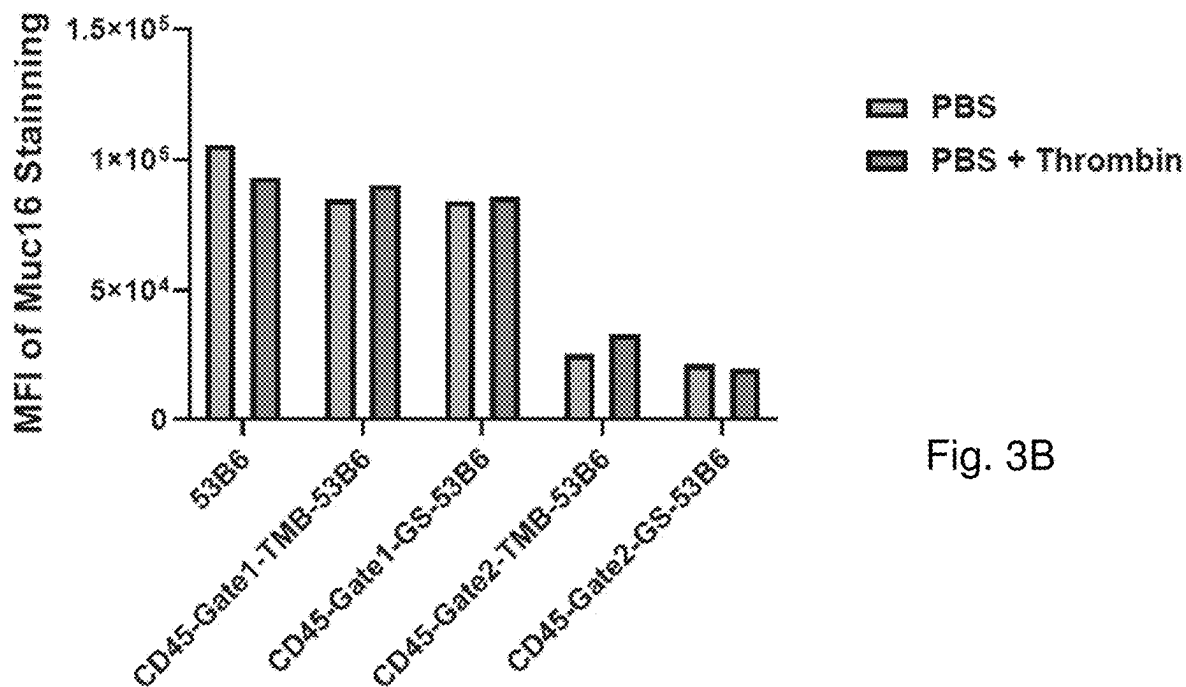


Fig. 3B

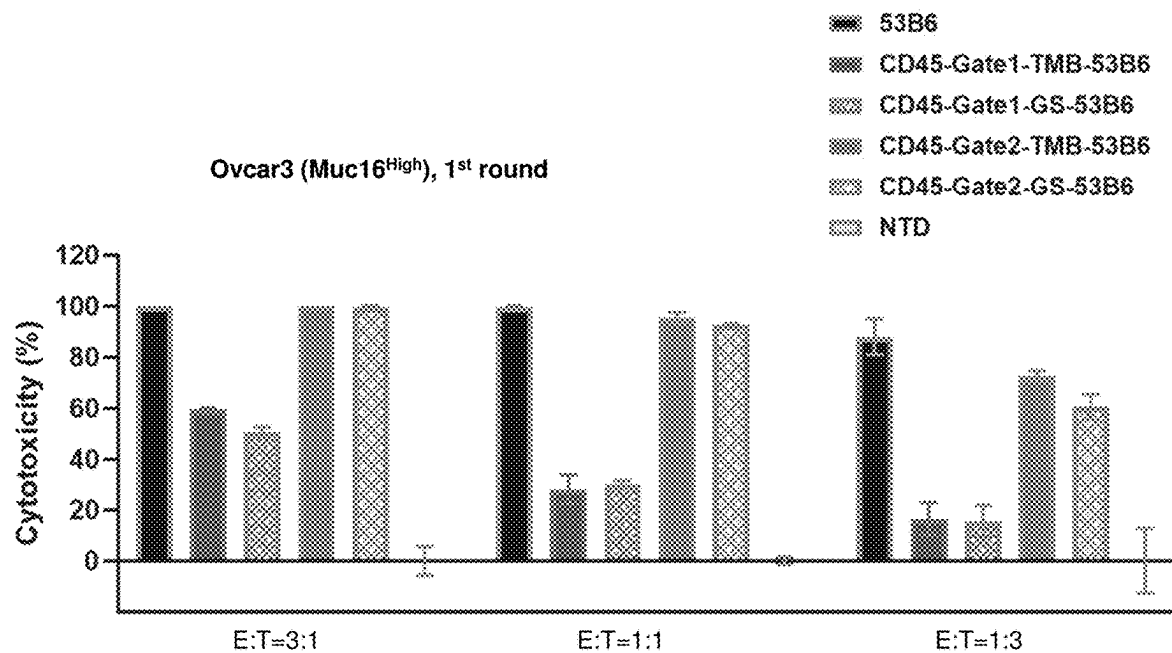


Fig. 4A

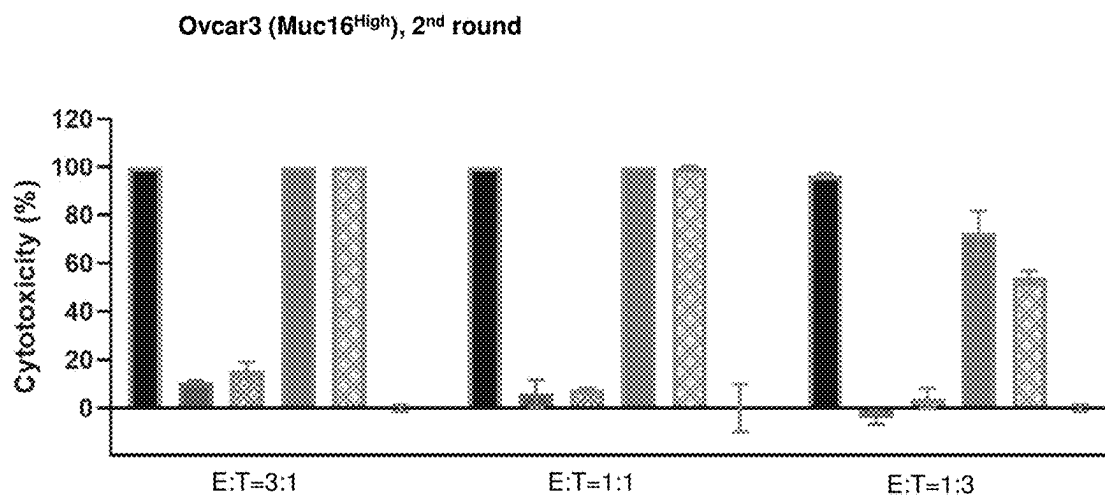


Fig. 4B

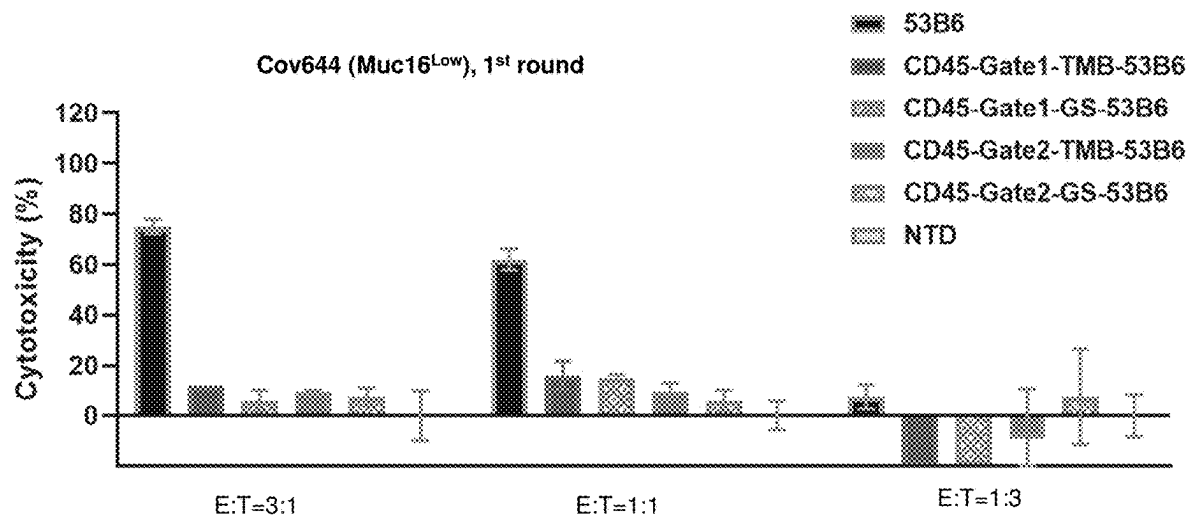


Fig. 5A

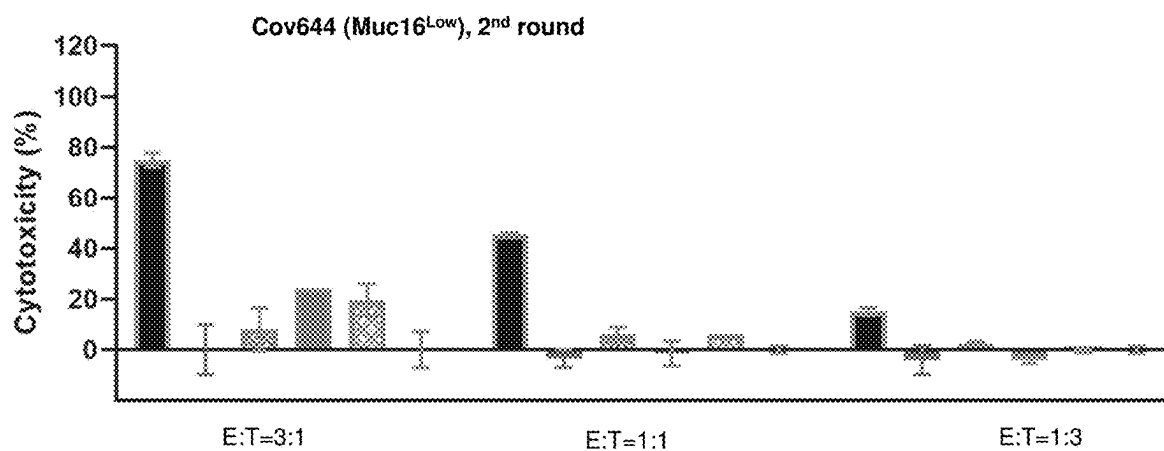


Fig. 5B

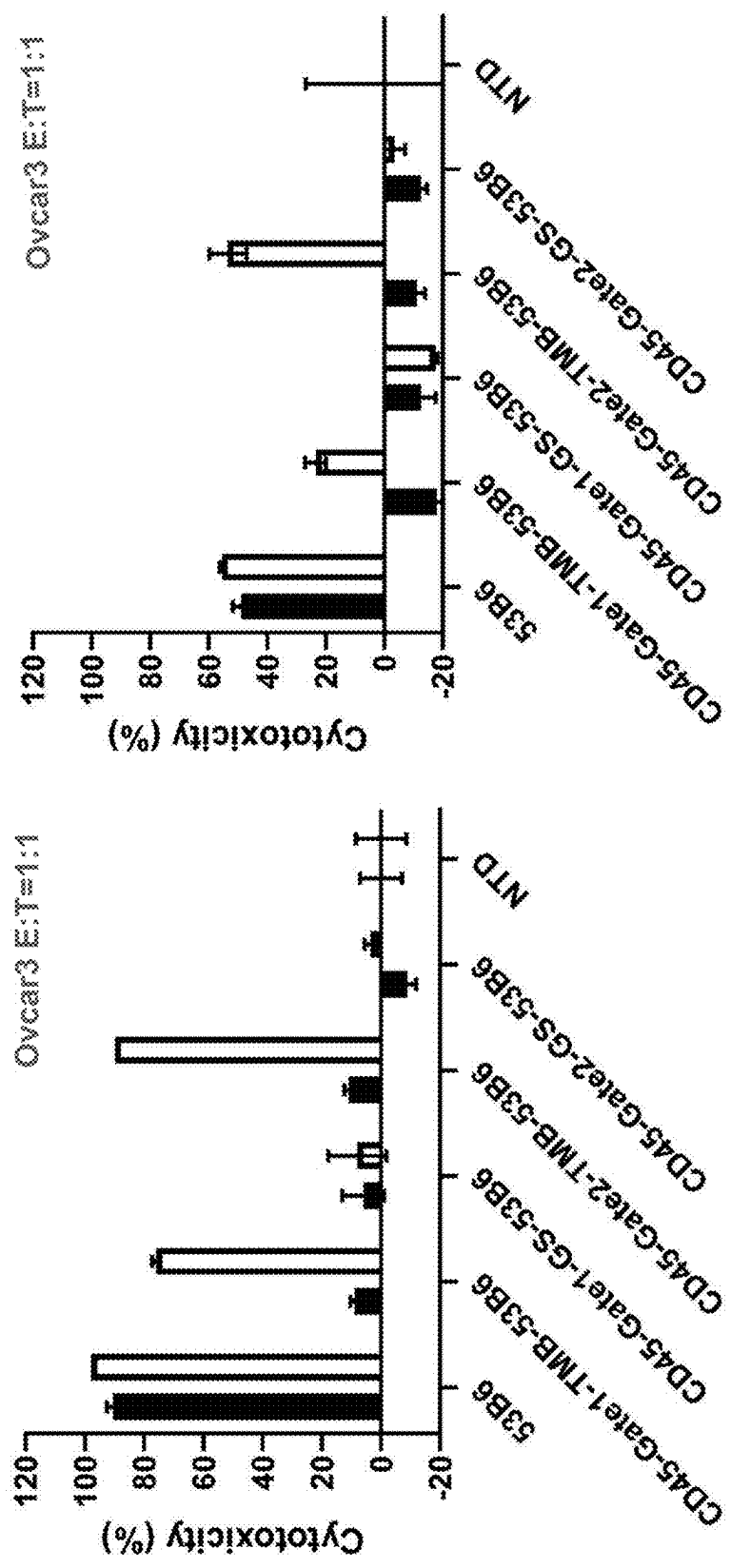
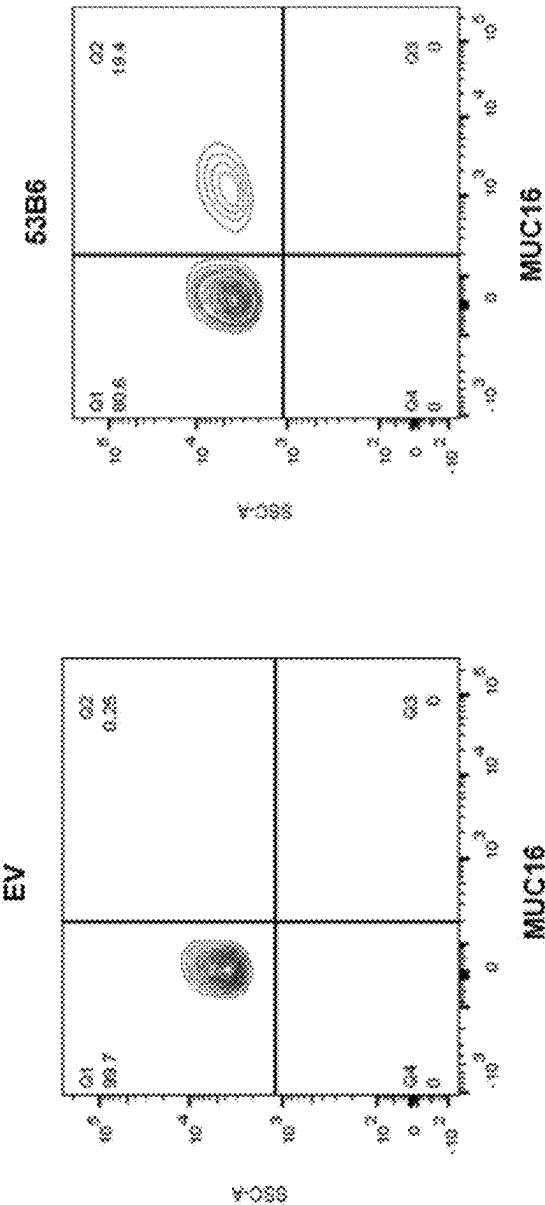
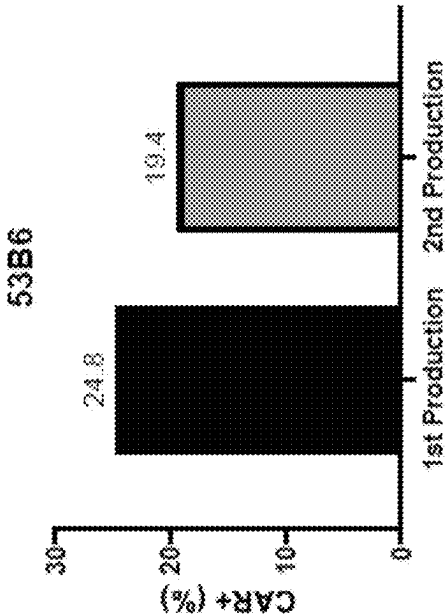


Fig. 6B

Fig. 6A

Fig. 7A



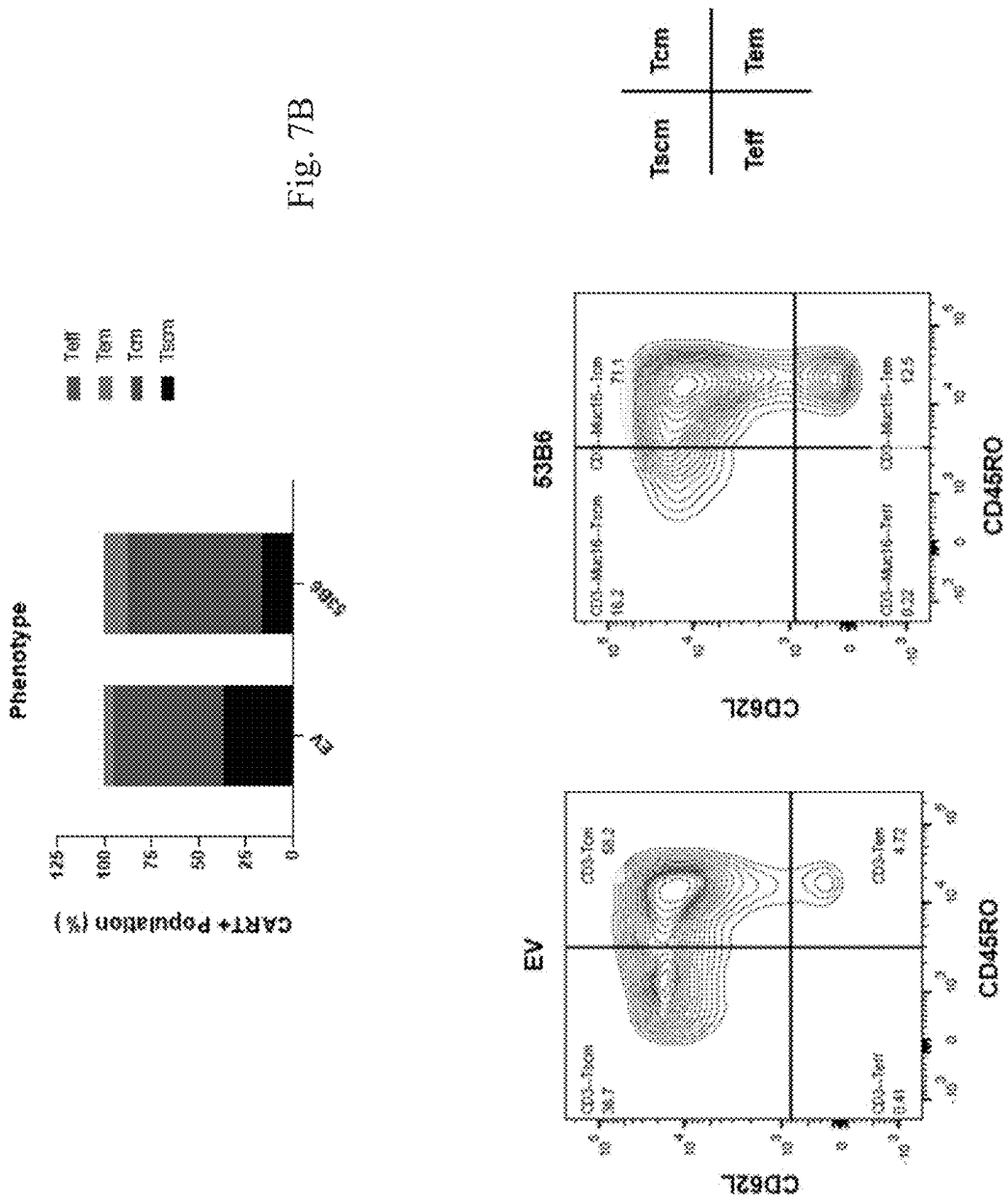


Fig. 7B

Fig. 8B

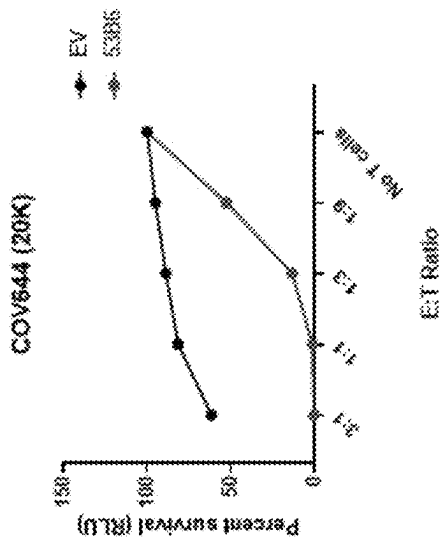


Fig. 8A

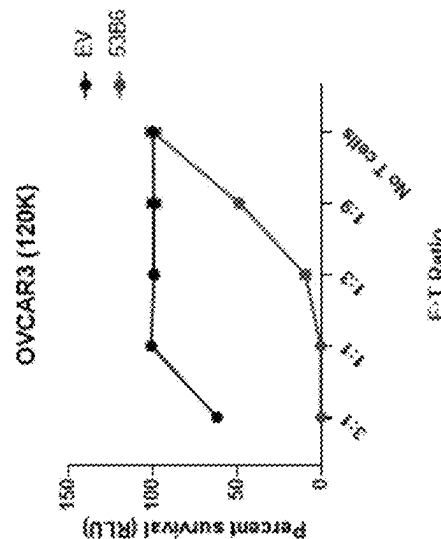
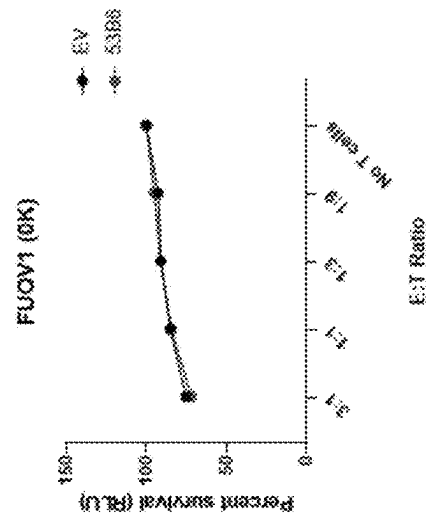


Fig. 8C



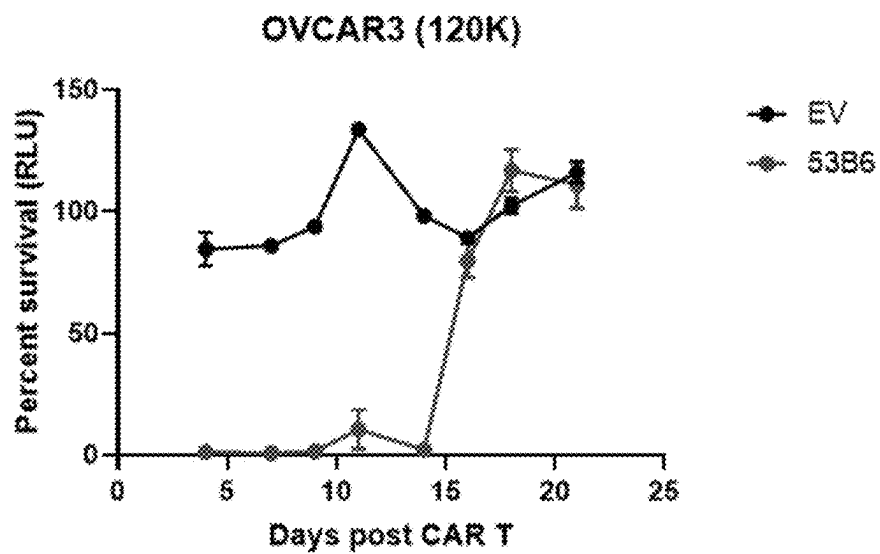


Fig. 9A

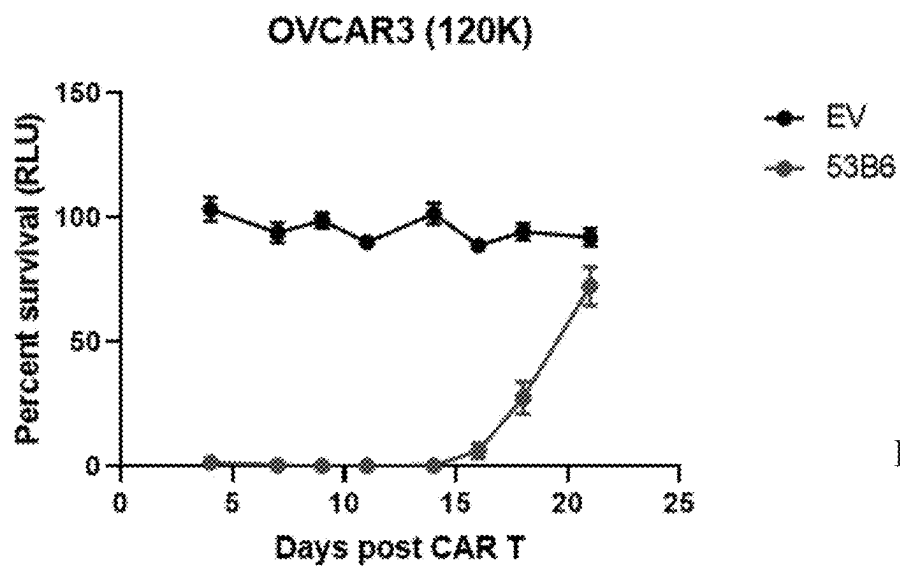


Fig. 9B

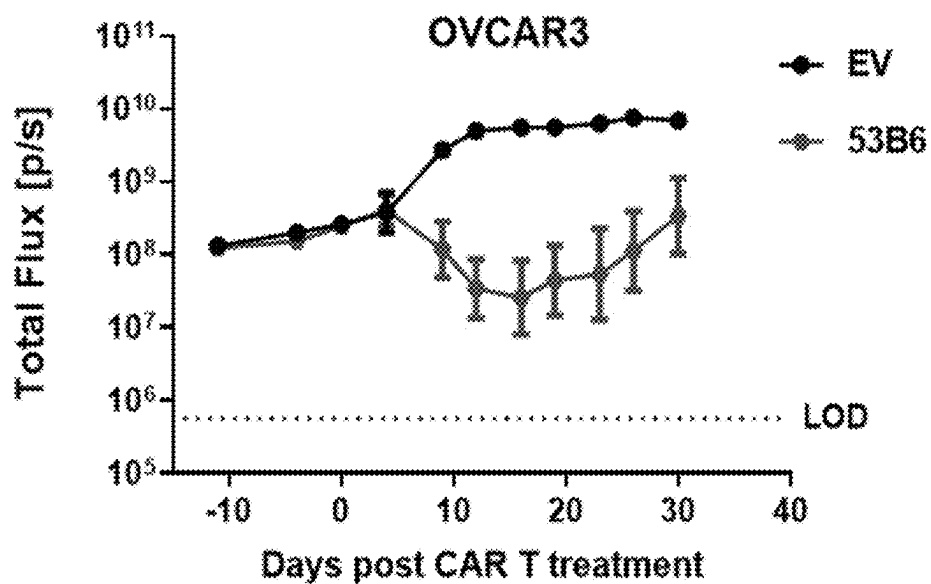


Fig. 10A

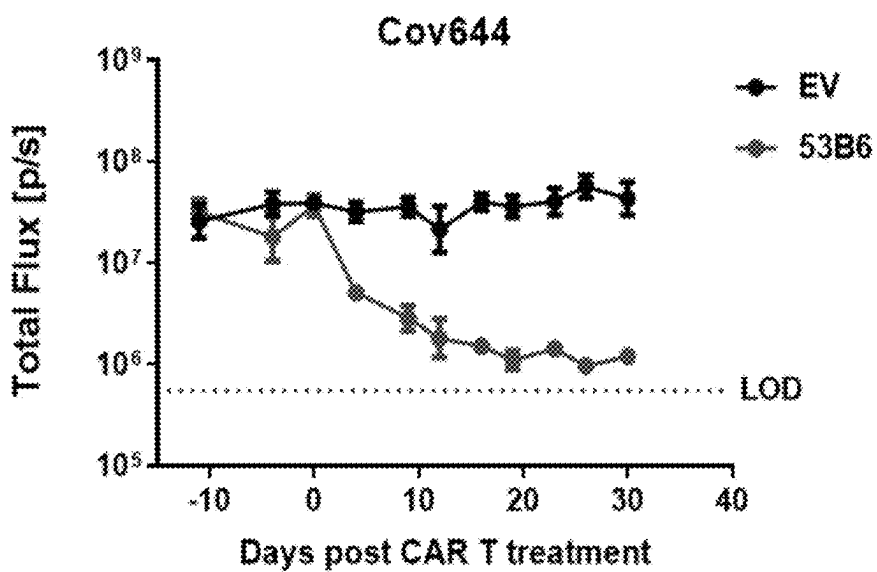


Fig. 10B

FIG. 11

Description	Matriptase cleavage site	MMP cleavage site	uPA cleavage site	Linker length	Schematic
53B6 naked CAR				None	
Non-cleavable GS				30aa	
Non-cleavable V5				45aa	
2xTPS3	✓	✓	✓	45aa	
TPS4	✓	✓		45aa	
TPS6	✓			45aa	
2xTPS6	✓			45aa	
2xTPS6	✓			40aa	
2xTPS6	✓			35aa	
2xTPS6	✓			30aa	
2xTPS6	✓			25aa	

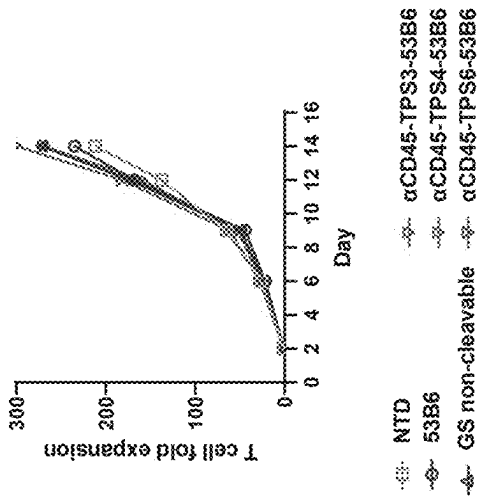


FIG. 12B

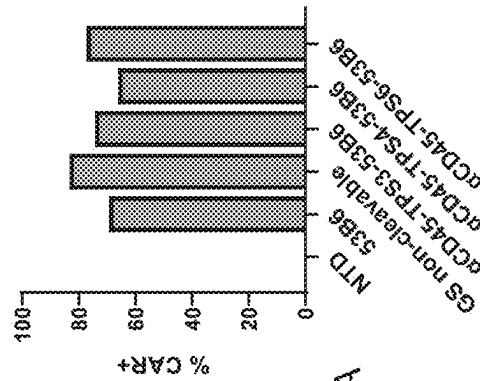


FIG. 12A

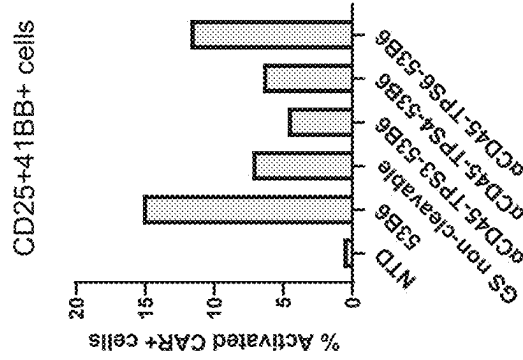


FIG. 12D

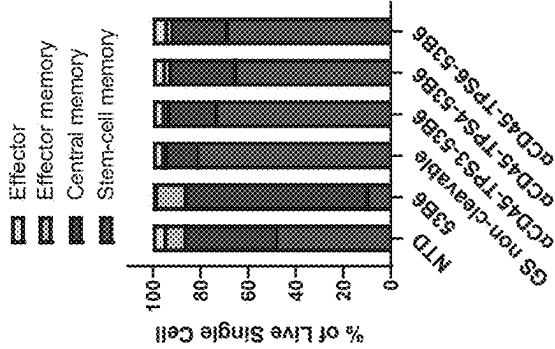


FIG. 12C

FIG. 13A

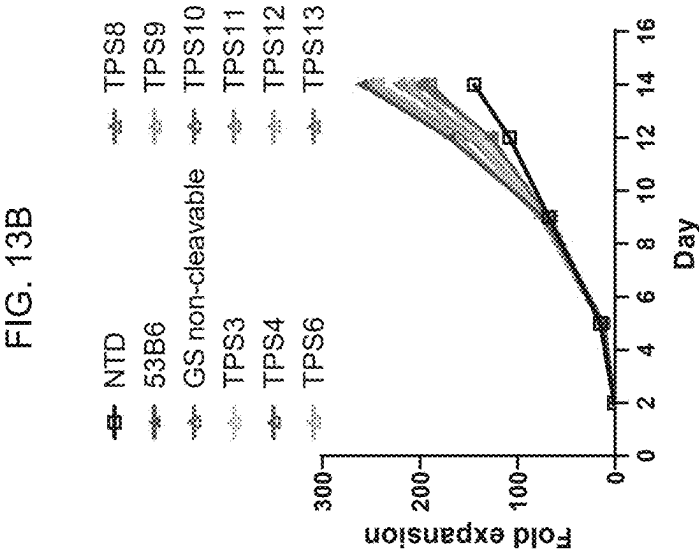
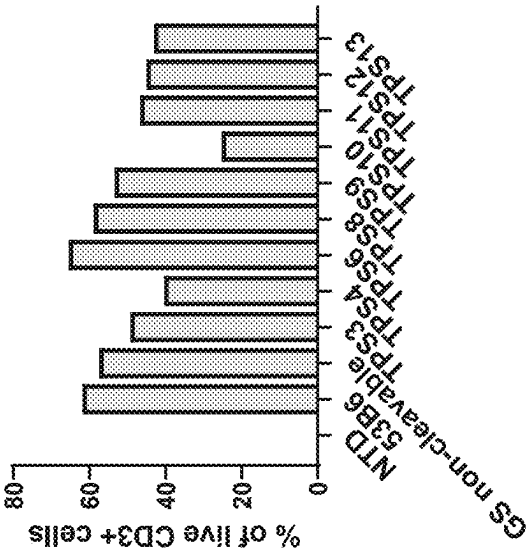


FIG. 13D

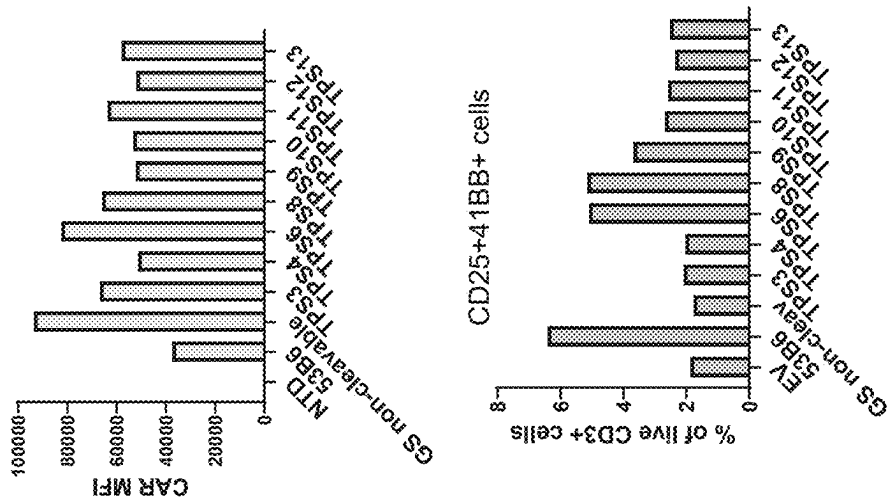
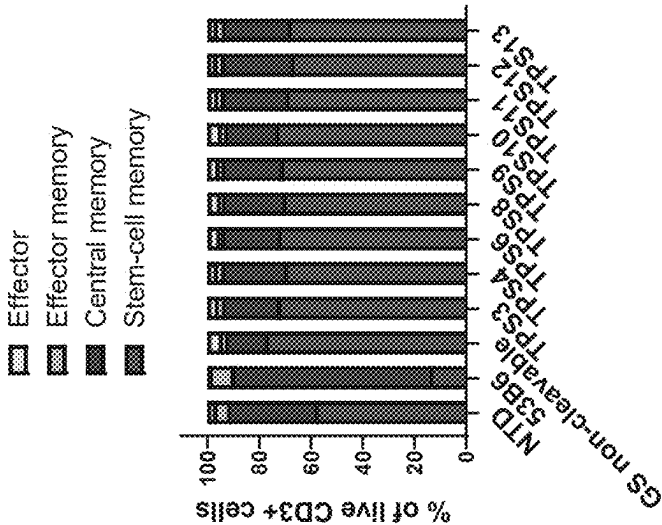


FIG. 13C



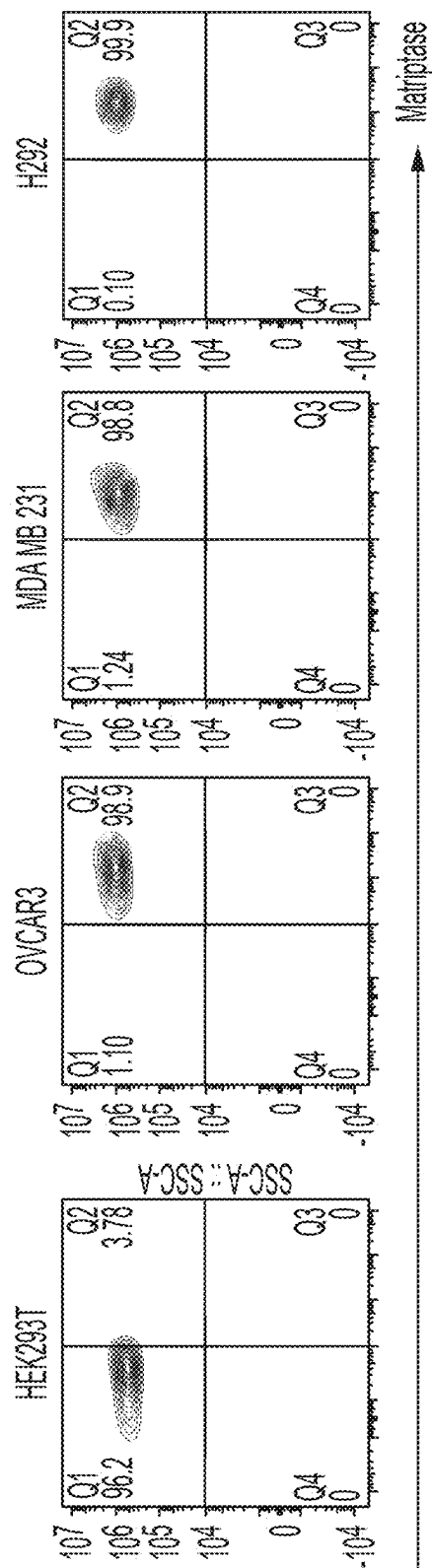


FIG. 14A

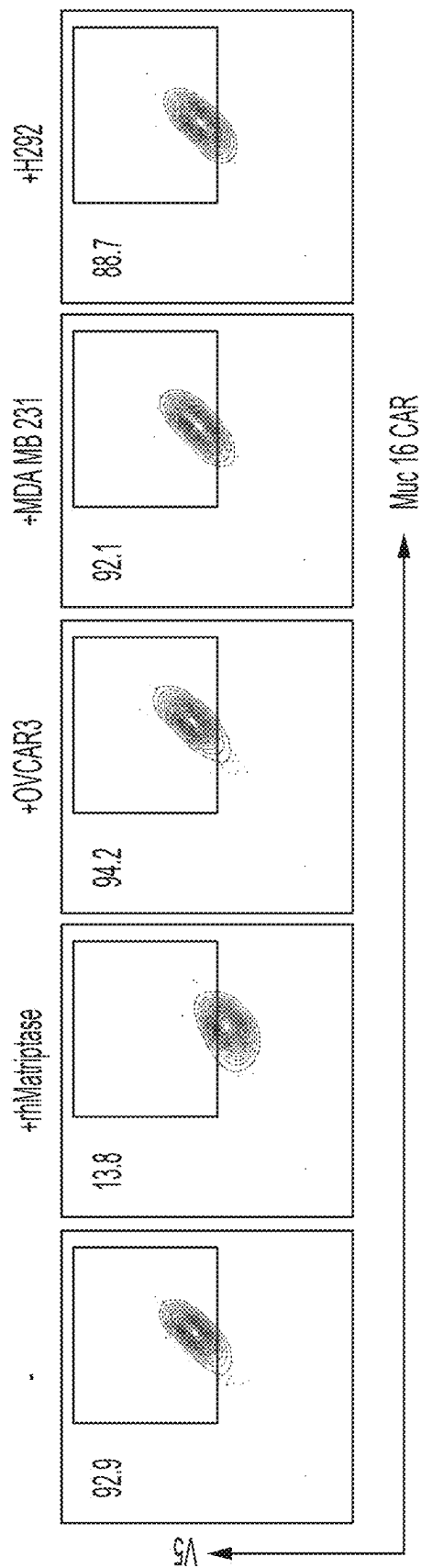


FIG. 14B

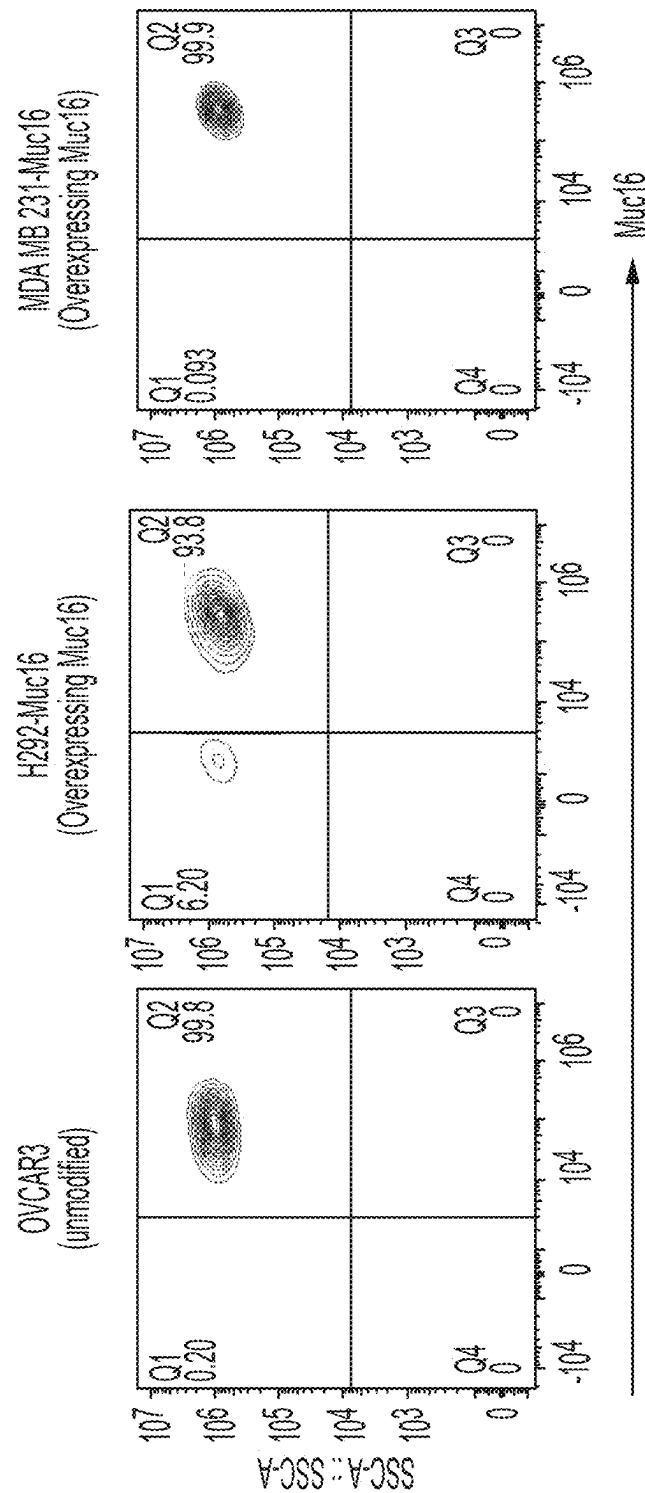


FIG. 14C

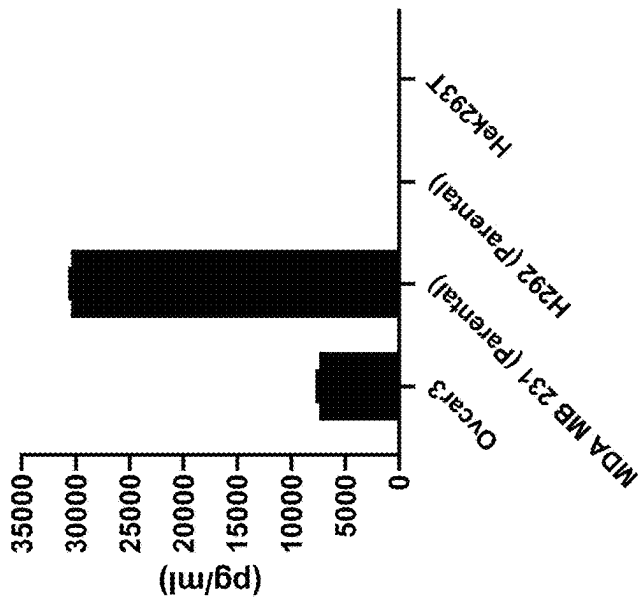


FIG. 14D

	Target expression	Protease detection in vitro			
	MUC16	Matrilptase (membrane-bound)	uPA	MMP-9	
OVCAR3	Endogenous (122K)	+ / ++	+	-	
MDA-MB-231- Muc16	Overexpressed	++	+++	-	
H292-Muc16	Overexpressed	++	-	-	

FIG. 14E

FIG. 15A

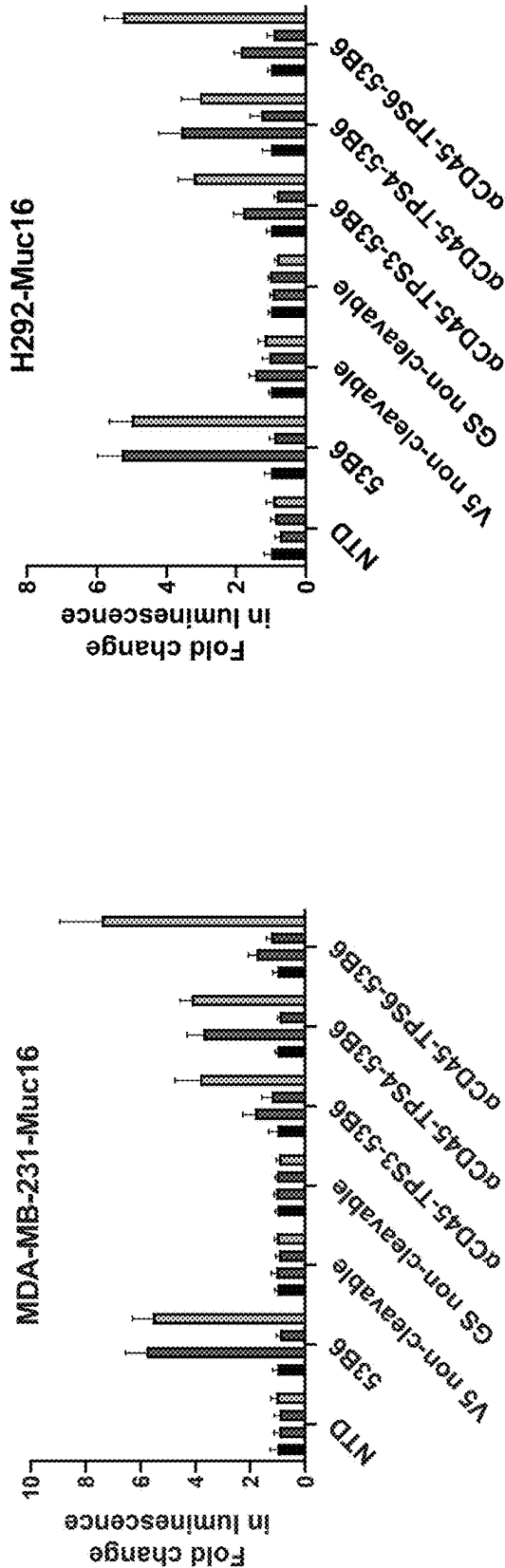
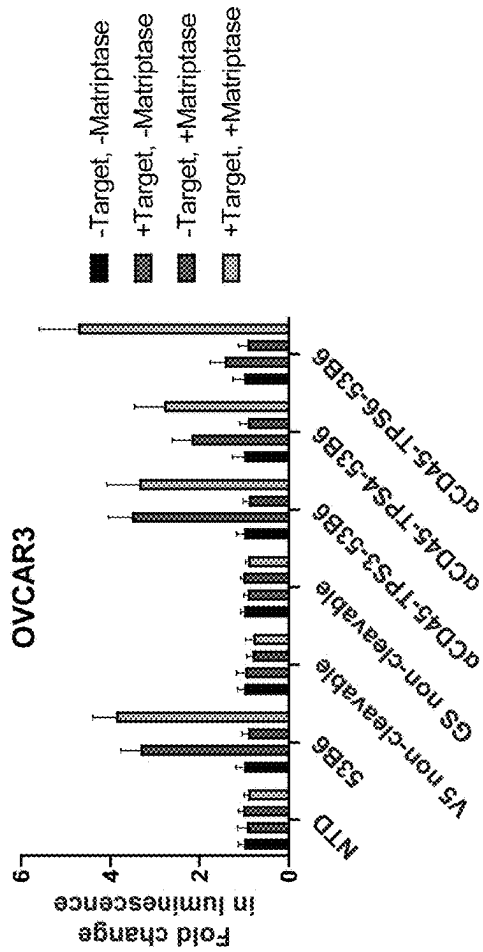


FIG. 15B

FIG. 15C

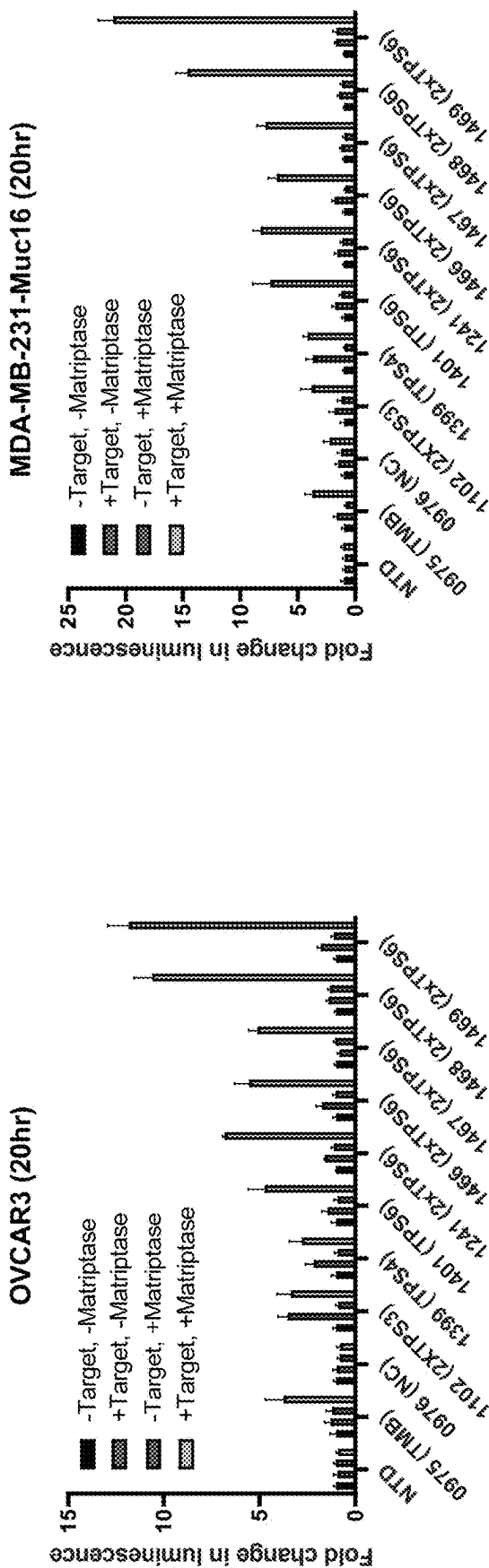


FIG. 15E

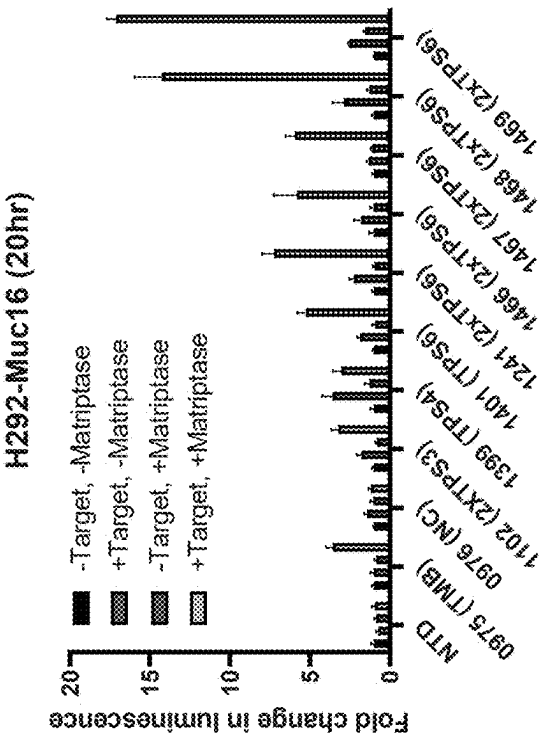


FIG. 15F

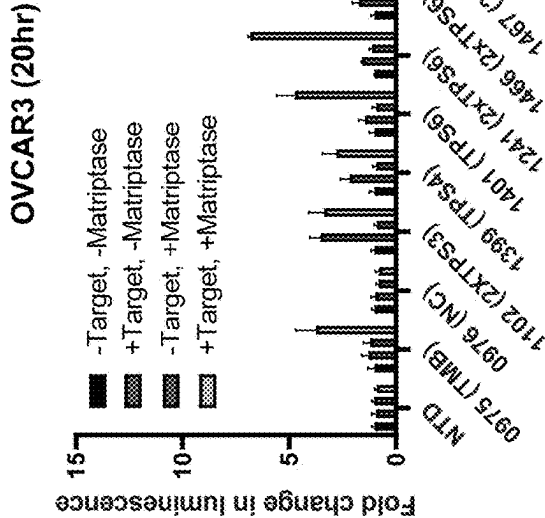


FIG. 15D

FIG. 16B

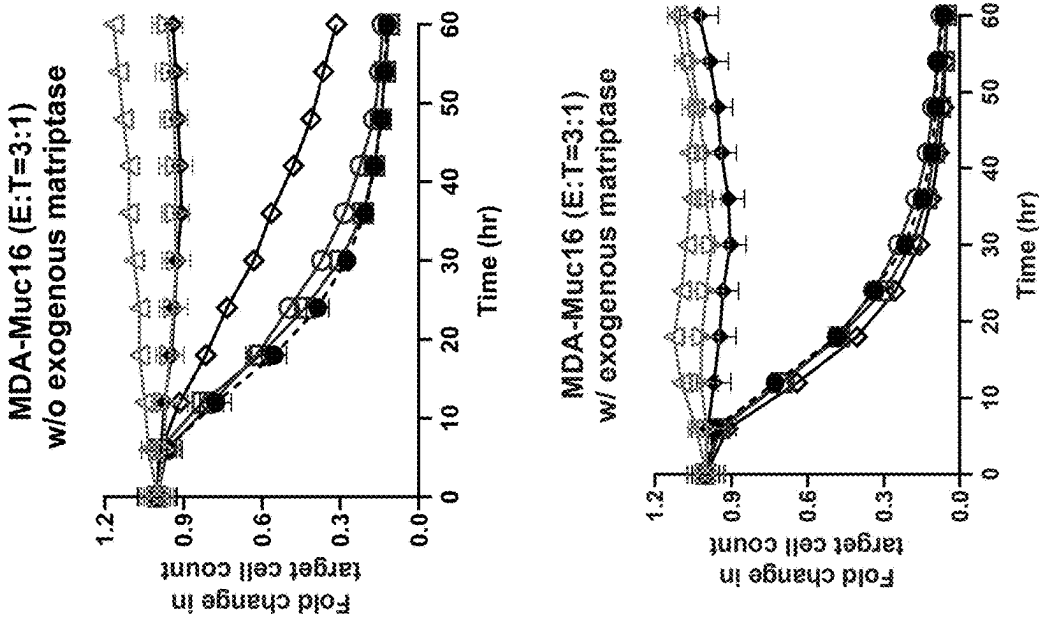


FIG. 16A

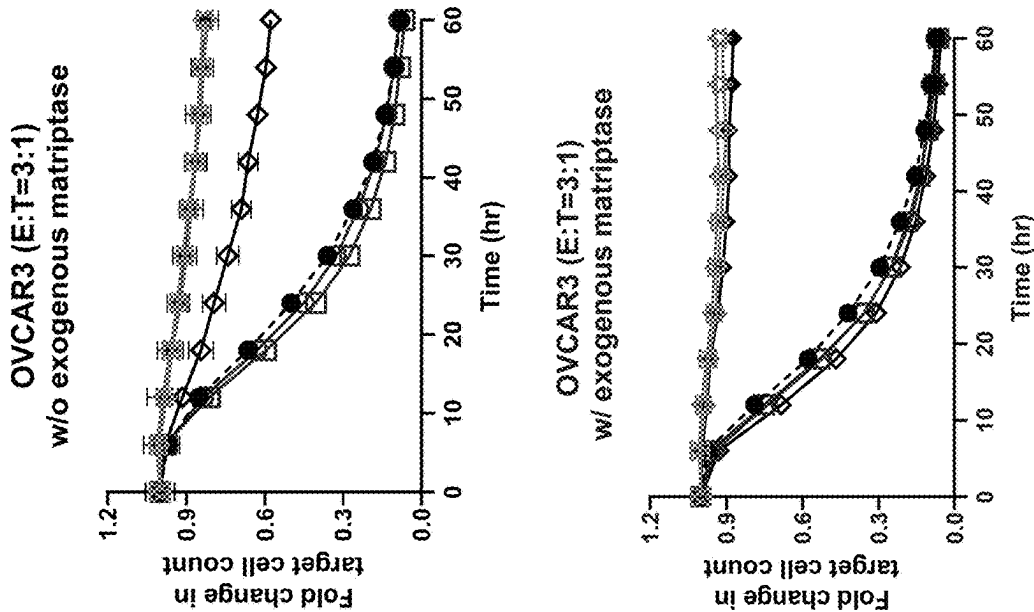
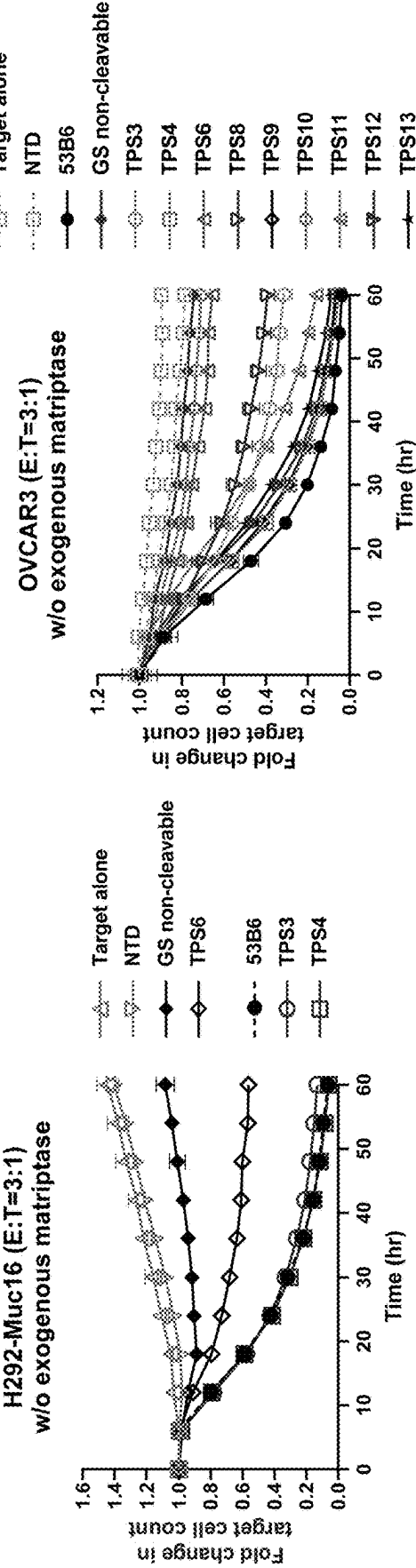


FIG. 16C



H292-Muc16 (E:T=3:1)
w/ exogenous matriptase

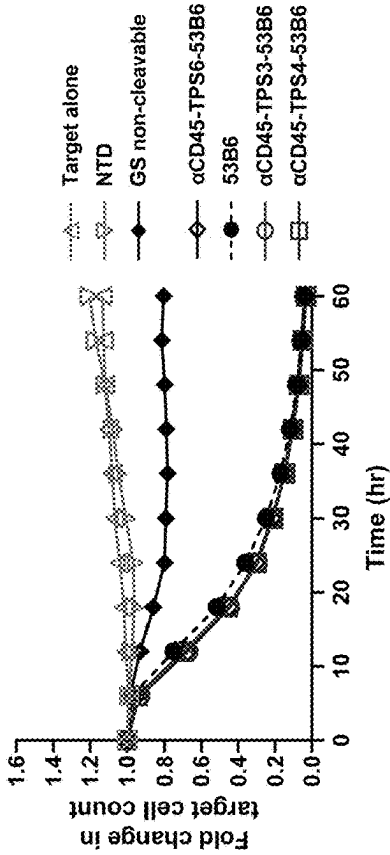
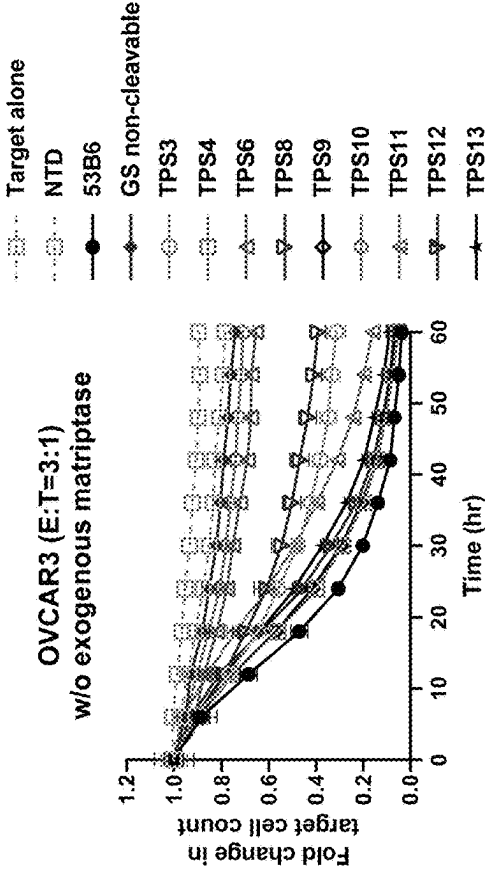


FIG. 16D



OVCAR3 (E:T=3:1)
w/ exogenous matriptase

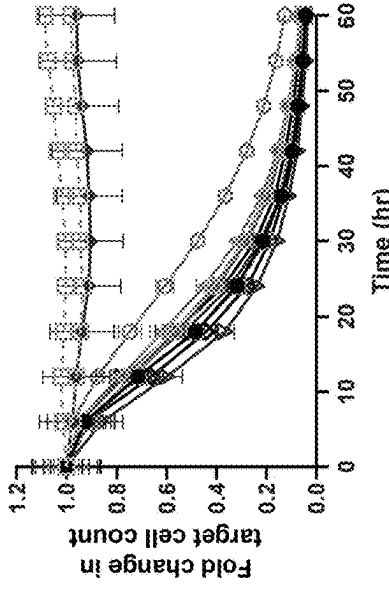


FIG. 16E

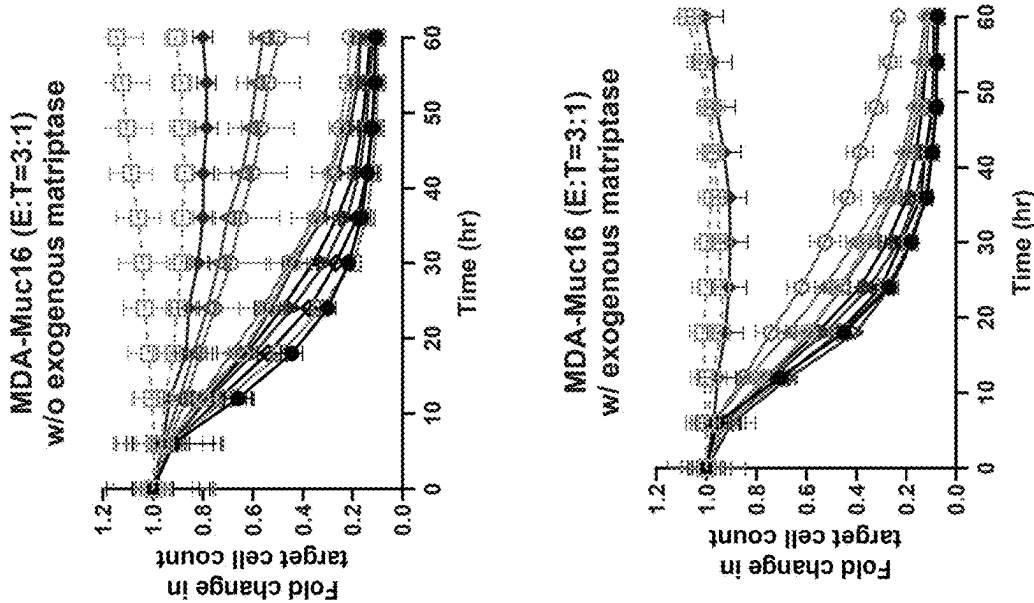


FIG. 16F

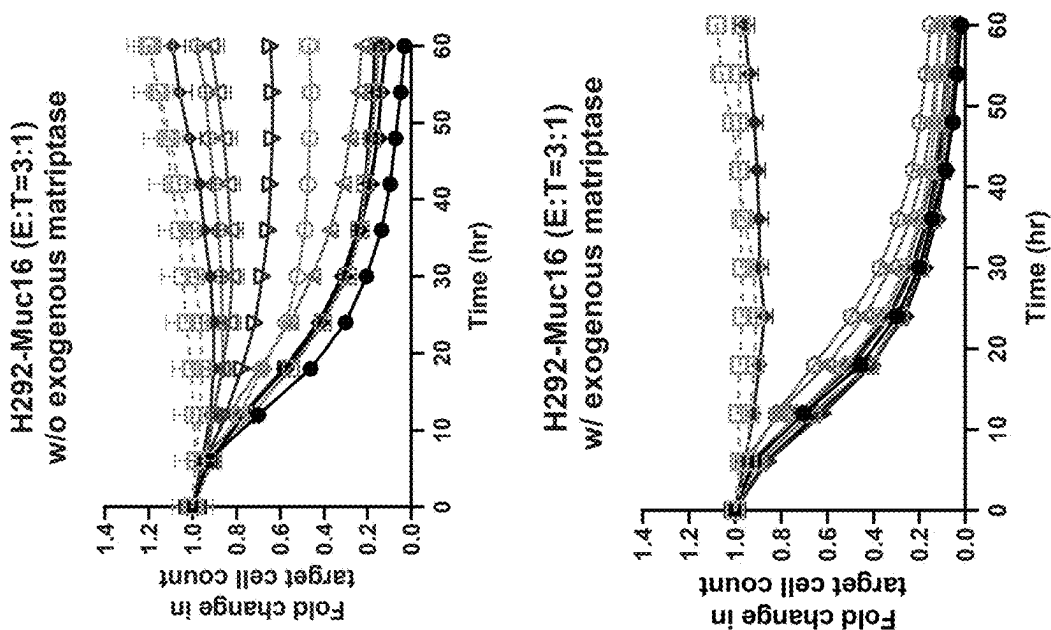


FIG. 17A

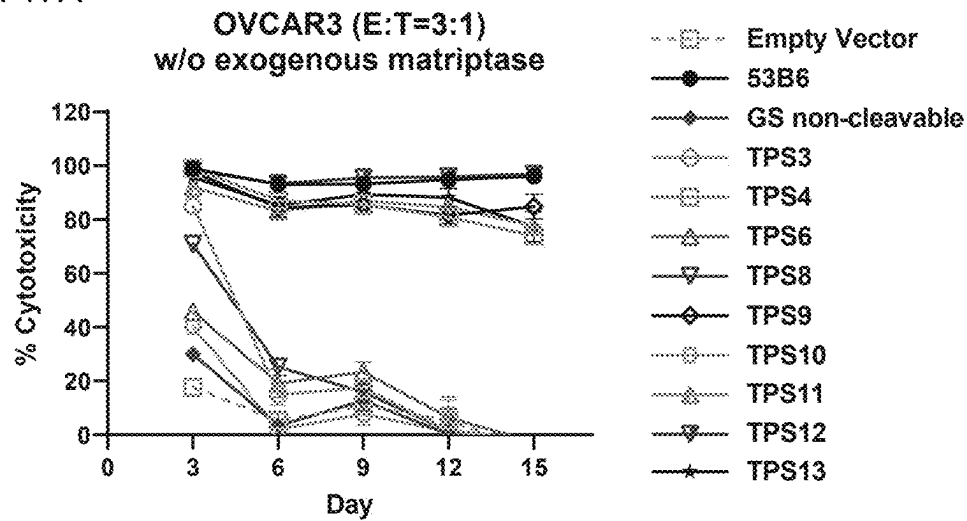
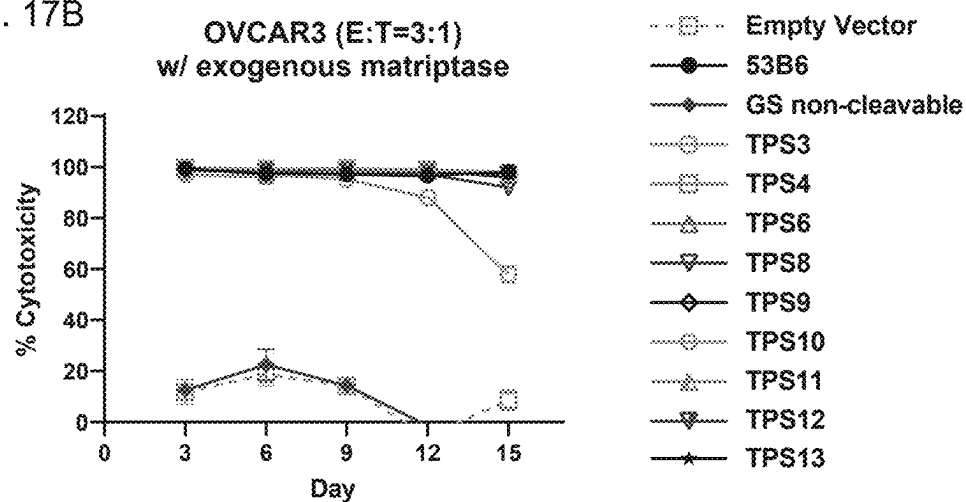


FIG. 17B



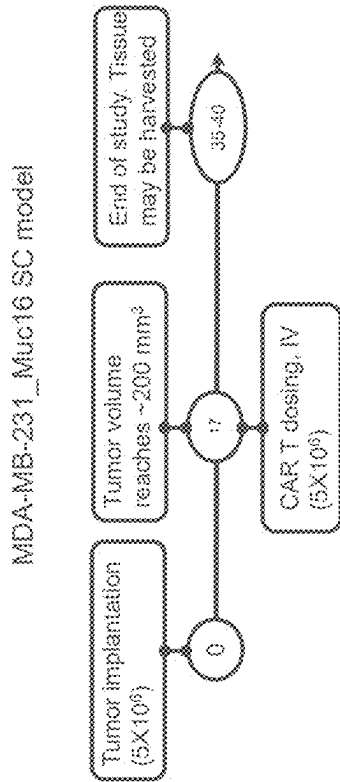


FIG. 18A

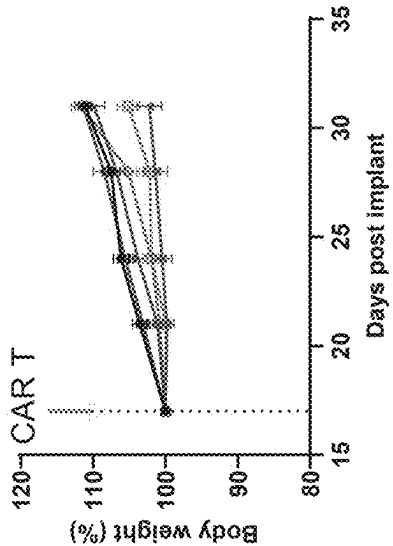


FIG. 18C

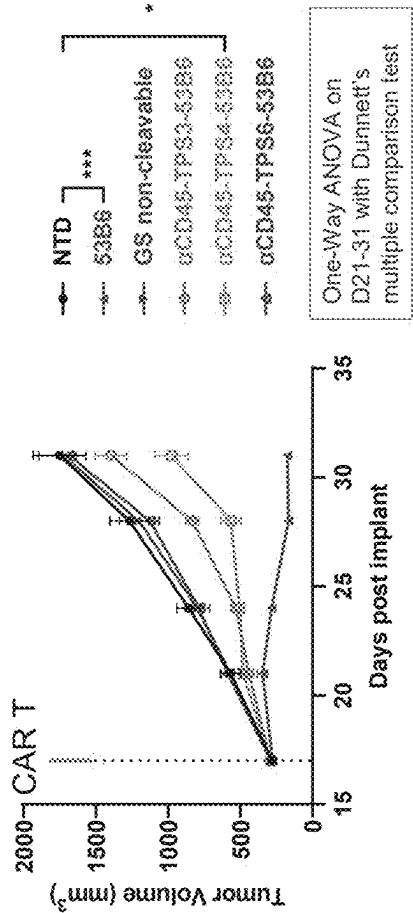


FIG. 18B

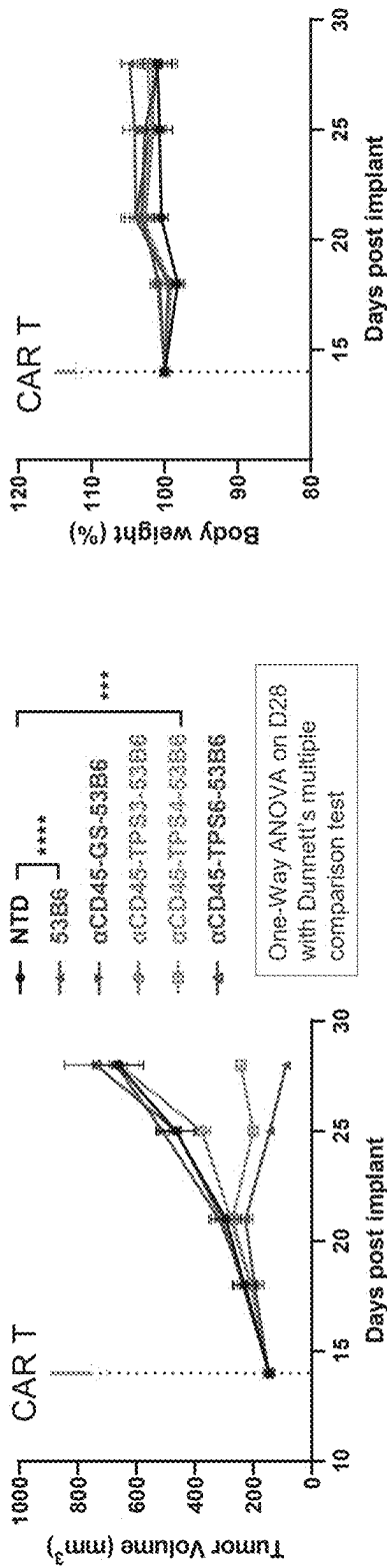
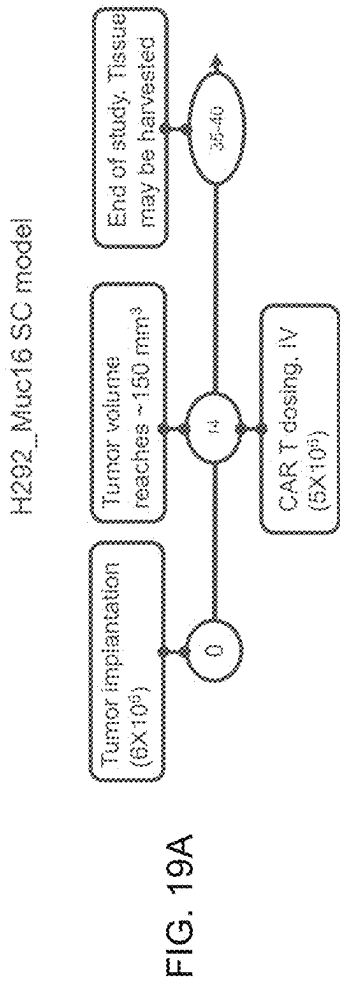


FIG. 19B

FIG. 19C

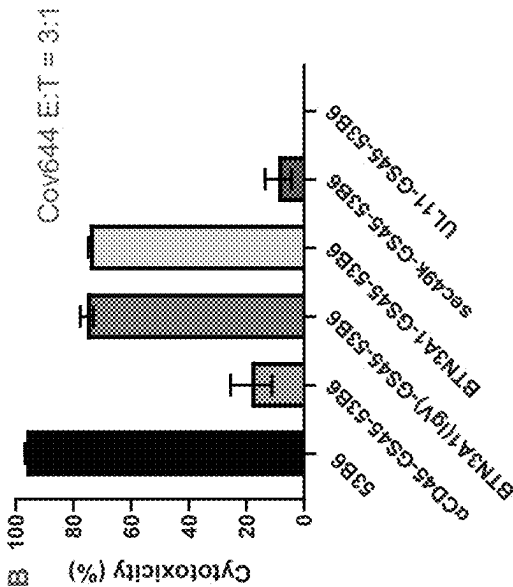


FIG. 20B

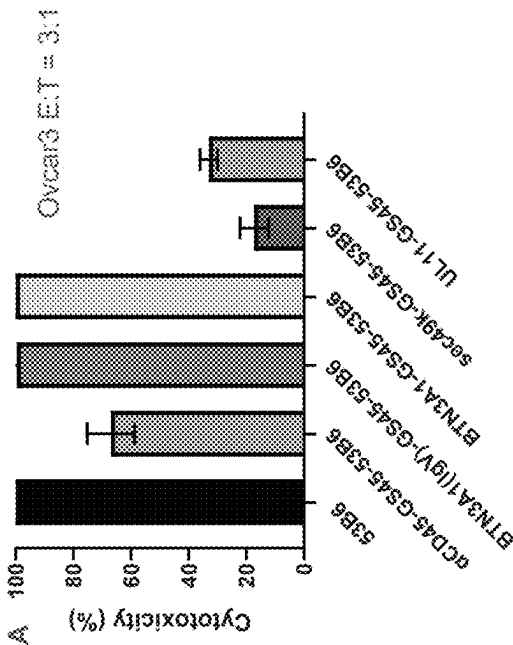


FIG. 20A

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PROTEASE-ACTIVATING CD45-GATE CAR**CROSS REFERENCE TO RELATED APPLICATIONS**

The present application claims the benefit of priority to U.S. Provisional Application No. 63/128,667, filed on Dec. 21, 2020; and U.S. Provisional Application No. 63/289,984, filed on Dec. 15, 2021, the contents of both of which are hereby incorporated by reference in their entireties.

SEQUENCE LISTING

This application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Dec. 19, 2021, is named AT-043_03US_SL.txt and is 556,722 bytes in size.

TECHNICAL FIELD

The present disclosure relates generally to the use of immune cells (e.g., T cells) engineered to express a chimeric antigen receptor (CAR) to treat a disease.

BACKGROUND

Tumor specificity is one of the major hurdles for the development of cancer therapy, such as CAR T cell therapy, in solid tumor indications. This differentiates solid tumor targets from many hematologic targets (e.g. CD19, CD20, BCMA), in that expression of these hematologic targets is heavily restricted to the hematologic lineage. Although their expression can be found on healthy hematologic cells and this can lead to the destruction of both healthy and cancerous cells, repopulation of normal hematologic cells from their progenitor stem cells makes these targets acceptable. However, many solid tumor targets exhibit an appreciable level of expression in some healthy organs or tissues, and the collateral destruction of these healthy tissues by CAR T cells may lead to severe adverse side effects and even death in patients ("on-target off-tumor toxicity"). Hence, it is useful to develop effective strategies and improve the anti-tumor specificity of CAR T cells to address potential on-target off-tumor toxicities. There exists a need for a cell-based cancer therapy, such as CAR T cells, with reduced on-target off-tumor toxicity.

SUMMARY

The present disclosure provides, among other things, chimeric antigen receptors (CARs) modified to improve target specificity in CAR T cell therapy, and related nucleic acids, expression vectors, cells, compositions and methods of treatment of cancer and other conditions amenable to CAR T cell therapy. The present disclosure provides a modification to CARs, the CD45 gate, that at least partially suppresses the modified CAR's activation in the presence of the CAR's antigen, thereby reducing the CAR's activity relative to the CAR not so modified or relative to the CAR not modified. This suppression and reduction in activity is partially or fully relieved when the modified CAR is in the presence of certain proteases, as occurs at the site of a tumor that is the CAR T cell therapy target. Disclosed herein are at least the following aspects and embodiments of the disclosure. Also provided is an anti-MUC16 CAR, including an

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anti-MUC16 CAR for use as a component of the protease-activating CD45-gate CAR disclosed herein.

CD45 is a phosphatase that plays a dynamic role in regulating T cell activation (1). It was found to directly activate TCR signaling by dephosphorylating other kinases, e.g. LCK, which activates these kinases and the downstream TCR signaling (2). CD45 is also able to directly dephosphorylate CD3zeta, which leads to the deactivation of TCR signaling (3).

Accordingly, in one aspect, provided herein is a protease-activating CD45-gate effector polypeptide comprising a CD45 recruiting domain connected to an effector entity optionally by a linker. In some embodiments, the effector entity can be reversibly modulated by CD45. In some embodiments, the effector entity is activated when the CD45 recruiting domain is released from the protease-activating CD45-gate effector by protease cleavage. In some embodiments, the CD45 recruiter is released by protease cleavage at the linker. In some embodiments, the effector entity comprises a chimeric antigen receptor (CAR). In some embodiments, the effector entity comprises a T cell receptor (TCR). In some embodiments, the CD45 recruiting domain (e.g. an anti-CD45 scFv) is prepared, using no more than ordinary skill in the art, from the CD45 antibodies disclosed in, for example, WO2005/026210, WO20131048804, WO2017009473, WO2020092654, EP1664122B1, and U.S. Pat. No. 6,106,834.

In one aspect, provided herein is a protease-activating CD45-gate chimeric antigen receptor (CD45-gate CAR) comprising an extracellular domain, a transmembrane domain, and an intracellular domain, wherein the extracellular domain comprises: a CD45 recruiting domain, a linker comprising one or more protease cleavage sites that can be cleaved by at least one protease, and an antigen binding domain. In some embodiments of the protease-activating CD45-gate CAR, the intracellular domain comprises at least one signaling domain that can be reversibly inactivated by CD45. In an embodiment, the linker is between the CD45 recruiting domain and the antigen binding domain. In an embodiment, the CD45 recruiting domain comprises one or more linkers. In an embodiment, the CD45 recruiting domain comprises an anti-CD45 antibody comprising a heavy chain variable domain (V_H) and a light chain variable domain (V_L). In some embodiments, the linker is between the V_H and V_L of the anti-CD45 antibody. In some embodiments, the protease-activating CD45-gate CAR comprises additional flexible regions e.g. placed between domains.

In a further aspect, provided herein is a protease-activating CD45-gate chimeric antigen receptor (CD45-gate CAR) comprising an extracellular domain, a transmembrane domain, and an intracellular domain, wherein the extracellular domain comprises:

a CD45 recruiting domain,
an antigen binding domain, and
a linker connecting the carboxy terminus of the CD45 recruiting domain to the amino terminus of the antigen binding domain, wherein the linker comprises at least one protease cleavage site that is recognized by a protease,
and further wherein the intracellular domain comprises at least one (e.g. one or more than one) signaling domain that can be reversibly inactivated by CD45.

In some embodiments, the CD45 recruiting domain comprises one or more of an anti-CD45 antibody antigen binding domain, an anti-CD45 NANOBODY®, an anti-CD45 scFv, a viral protein binder of CD45, a truncated viral protein binder of CD45, an anti-CD45 Fab, an anti-CD45 camelid

VHH, a CD45-binding protein, an endogenous CD45 binder, or a truncated endogenous CD45 binder. In some embodiments, the CD45 recruiting domain comprises one or more of a peptide or protein that interacts with CD45, for example, full-length or truncated UL11, full-length or truncated sec49K, and full-length or truncated BTN3A1. In some embodiments, the CD45 recruiting domain comprises a truncated BTN3A1, e.g. a truncated BTN3A1 having the amino acid sequence of SEQ ID NO: 14 or of SEQ ID NO: 15.

In some embodiments, the CD45 recruiting domain comprises an anti-CD45 scFv. In some embodiments, the CD45 recruiting domain comprises an anti-CD45 antibody antigen binding domain. In some embodiments, the CD45 recruiting domain comprises one or more of truncated UL11, truncated sec49K, and truncated BTN3A1. In some embodiments, the CD45 recruiting domain comprises the amino acid sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14 or SEQ ID NO: 15.

In some embodiments, the antigen binding domain specifically binds BCMA, MUC16 (also known as CA125), EGFR, EGFRvIII, MUC1, Flt-3, WT-1, CD20, CD23, CD30, CD38, CD70, CD33, CD133, MHC-WT1, TSPAN10, MHC-PRAME, MHC-NY-ESO1, HER2 (ERBB2), CAIX (Carbonic anhydrase IX), LIV1, ADAM10, CHRNA2, LeY, NKG2D, CS1, CD44v6, ROR1, CD19, Claudin-18.2 (Claudin-18A2, or Claudin18 isoform 2), PSCA, DLL3 (Delta-like protein 3, *Drosophila* Delta homolog 3, Delta3), Mud 7 (Mucin17, Muc3, Muc3), FAP alpha (Fibroblast Activation Protein alpha), Ly6G6D (Lymphocyte antigen 6 complex locus protein G6d, c6orf23, G6D, MEGT1, NG25), PSMA, MSLN, or RNF43 (E3 ubiquitin-protein ligase RNF43, RING finger protein 43).

In some embodiments, the antigen binding domain specifically binds MUC16. In some embodiments, the antigen binding domain specifically binds MUC16 and the protease-activating CD45-gate CAR comprises the amino acid sequence of any of SEQ ID NOs: 11, 56, 75-79, and 121-170, each of which can have or not have the signal sequence of SEQ ID NO:1. In some embodiments, the antigen binding domain specifically binds MUC16 and the protease-activating CD45-gate CAR comprises the amino acid sequence of SEQ ID NO: 75, with or without the signal sequence of SEQ ID NO:1. In some embodiments, the antigen binding domain specifically binds MUC16 and the protease-activating CD45-gate CAR comprises the amino acid sequence of SEQ ID NO: 76, with or without the signal sequence of SEQ ID NO:1. In some embodiments, the extracellular domain comprises the amino acid sequence of any one of SEQ ID NOs: 3, 4, 5, 6, 12, 13, 14, 15, 56, and 56 without the signal sequence of SEQ ID NO:1.

In some embodiments, the antigen binding domain specifically binds to an antigen expressed by a certain type of tumor and the protease cleavage sites can be recognized and cleaved by a protease that the same type of tumor expresses. In some embodiments, the protease-activating CD45-gate CAR both (1) can specifically recognize and bind to an antigen expressed by a tumor and (2) comprises a CD45 recruiting domain-CAR linker that can be cleaved by a protease that the tumor secretes and/or that is present and/or active in the tumor microenvironment.

In some embodiments, the antigen binding domain specifically binds to a breast cancer tumor antigen or a colorectal cancer tumor antigen and one or more of the at least one protease cleavage sites can be recognized and cleaved by uPA. In some embodiments, the protease-activating CD45-

gate CAR's antigen binding domain specifically recognizes and binds to an antigen characteristic of any of cervical, breast, ovarian and colorectal cancers and at least one of its protease cleavage sites can be recognized and cleaved by MMP-2 and/or MMP-9. In some embodiments, the protease-activating CD45-gate CAR's antigen binding domain specifically recognizes and binds to an antigen characteristic of any of breast and ovarian cancers and at least one of its protease cleavage sites can be recognized and cleaved by matriptase.

In some embodiments, the linker comprises a GS sequence on either or both ends of each of the one or more protease cleavage sites. In some embodiments, the GS sequence is a polypeptide that consist of only glycine and serine residues. In some embodiments, each GS sequence comprises a series of linked GS peptides, a GS peptide being, for example, a peptide having an amino acid sequence comprising only glycine and serine residues, e.g. a peptide having an amino acid sequence comprising one or more glycine residues followed by one or more serine residues, for example, a peptide having an amino acid sequence of GS, GGS, GGGs (SEQ ID NO: 184), or GGGGS (SEQ ID NO:178), wherein the linked GS peptides are linked in any order and in any combination. In some embodiments, the GS sequence comprises any combination of 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 GS peptides or comprises between 1 and 15 GS peptides in any order and combination.

In some embodiments, the linker of the protease-activating CD45-gate CAR disclosed herein has a length of 15-100 amino acids, 30-100 amino acids, 15-75 amino acids, 20-50 amino acids, 20 amino acids, 25 amino acids, 30 amino acids, 35 amino acids, 40 amino acids, 45 amino acids, or 50 amino acids, or about 20 amino acids, about 25 amino acids, about 30 amino acids, about 35 amino acids, about 40 amino acids, about 45 amino acids, or about 50 amino acids.

In some embodiments of the protease-activating CD45-gate CAR disclosed herein, the linker comprises one or more protease cleavage sites that comprise the amino acid sequence of SEQ ID NO: 32, 91-98, 103-105 or SEQ ID NO: 106. In some embodiments, the linker comprises one or more amino acid sequences of SEQ ID NO: 53, 89, or 90. In some embodiments, the linker comprises one or more amino acid sequences of SEQ ID NO: 8-10, 99-102, 107-120, 172-176 or SEQ ID NO: 177. In some embodiments, the linker comprises two or more protease cleavage sites and each cleavage site is the same as or different from any of the other cleavage sites. In some embodiments, the number of protease cleavage sites is 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or is between 1 and 15. In some embodiments, when the linker comprises more than one protease cleavage site, each protease cleavage site is independently the same as or different from one or more of the other protease cleavage sites.

In some embodiments, the linker comprises one protease cleavage site (e.g. any of the protease cleavage sites disclosed herein, e.g. a cleavage site having the amino acid sequence of, for example, any of SEQ ID NOs: 32, 89-98 and 103-106) (SEQ ID NOs: 89 and 90 each have two protease cleavage sites, a MMP2/9 site and a MTSP site). In some embodiments, the protease cleavage site comprises the amino acid sequence of SEQ ID NO: 32 [LVPRGS]. In some embodiments, the protease cleavage site comprises the amino acid sequence of SEQ ID NO: 91. In some embodiments, the protease cleavage site comprises the amino acid sequence of SEQ ID NO: 92. In some embodiments, the protease cleavage site comprises the amino acid sequence of SEQ ID NO:93. In some embodiments, the protease cleavage site comprises the amino acid sequence of SEQ ID

NO:94. In some embodiments, the protease cleavage site comprises the amino acid sequence of SEQ ID NO:95. In some embodiments, the protease cleavage site comprises the amino acid sequence of SEQ ID NO:96. In some embodiments, the protease cleavage site comprises the amino acid sequence of SEQ ID NO:97. In some embodiments, the protease cleavage site comprises the amino acid sequence of SEQ ID NO:98. In some embodiments, the protease cleavage site comprises the amino acid sequence of SEQ ID NO:103. In some embodiments, the protease cleavage site comprises the amino acid sequence of SEQ ID NO:104. In some embodiments, the protease cleavage site comprises the amino acid of any one of SEQ ID NOs: 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 103, 104, 105, and 106. In some embodiments, the linker comprises multiple protease cleavage sites (for example, any of the protease cleavage sites known and/or disclosed herein, e.g. a cleavage site having the amino acid sequence of any of SEQ ID NOs: 32, 89-98 and 103-104), such as two, three, four, five, six or more protease cleavage sites, e.g. up to ten or up to fifteen protease cleavage sites, and each cleavage site is the same as or different from any of the other cleavage sites. In some embodiments, one or more than one of the multiple protease cleavage sites comprises the amino acid sequence of SEQ ID NO: 32 (amino acid sequence LVPRGS). In some embodiments, the CD45 recruiting domain comprises one or more than one protease cleavage sites. In some embodiments, the linker amino acid sequence is or comprises the amino acid sequence of any one or more of (e.g. 1, 2, 3, 4, 5, 6 or more) of SEQ ID NOs: 99-102, 172-177, and 108-120. In some embodiments, the linker comprises one copy or more than one copy (e.g. 1, 2, 3, 4, 5, 6 or more) of the amino acid sequence of any of SEQ ID NOs: 89-98 and 103-106.

In some embodiments, the protease cleavage site, or one or more of the multiple protease cleavage sites, is recognized by one or more (e.g. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or 15, or any range between any two of these values) of thrombin, trypsin, plasmin, prostate-specific antigen (PSA), urokinase plasminogen activator (uPA), urokinase plasminogen activator receptor (uPAR), matrix metalloproteinase (MMP), matriptase (MT-SP1), legumain, a disintegrin and metalloproteinase (ADAM), and transmembrane Serine Protease (TMPRSS). Alternatively stated, in some embodiments, the protease that recognizes the protease cleavage site, or that recognizes one or more of the multiple protease cleavage sites, is one or more of thrombin, trypsin, plasmin, prostate-specific antigen (PSA), urokinase plasminogen activator (uPA), urokinase plasminogen activator receptor (uPAR), matrix metalloproteinase (MMP), matriptase (MT-SP1), legumain, a disintegrin and metalloproteinase (ADAM), transmembrane Serine Protease (TMPRSS), Granzyme B, activated protein C, Caspase, Cathepsin, Chymase, Elastase, Guanidinobenzoate, HtrA1, Human Neutrophil Elastase, Lactoferrin, Marapsin, NS3/4A, PACE4, tissue plasminogen activator (tPA), DESC1, DPP-4, FAP, Hepsin, Matriptase-2, secretase, kallikrein-related peptidase (KLK), and trypsin, or a serine protease, a cysteine-type lysosomal protease, a metalloproteinase, a coagulation factor protease, or an aspartyl-type lysosomal protease. In some embodiments, the protease cleavage site comprises an amino acid sequence that is cleaved by a protease or type of protease listed herein. In some embodiments, the protease cleavage site or one or more of the multiple protease cleavage sites is recognized by an endogenous protease or by a protease which is present in a tumor microenvironment. In some embodiments, the protease that recognizes the protease cleavage site, or that

recognizes one or more of the multiple protease cleavage sites, is a protease which is present in a tumor microenvironment.

In some embodiments, the linker comprises, in any order, including e.g. from the amino to the carboxy terminus and from the carboxy to the amino terminus, any of the following combinations of cleavage sites: for example, uPA and MMP, uPA and matriptase, MMP and matriptase, matriptase and MMP and uPA. In some embodiments, the linker further comprises one or more spacer sequences at either or both ends of one or more of each protease cleavage site to provide flexibility. In some embodiments, the linker further comprises one or more of a GS sequence, e.g. a GS sequence as disclosed herein (e.g. any one or more of SEQ ID NOs: 52, 54, 57), at either or both ends of one or more of each protease cleavage site. In some embodiments, the CD45-gate CAR comprises one or more of the linkers disclosed herein (e.g. SEQ ID NOs: 7-10, 99-102, 172-177, 108-120), e.g. 1, 2, 3, 4, 5, 15, or any intervening integer value or any range between any two such values. The linkers can be placed, e.g., between the CD45 recruiting domain, e.g., an anti-CD45 scFv, and an antigen binding domain, e.g., an antigen binding domain of a CAR. In some embodiments, the CD45 recruiting domain comprises an anti-CD45 antibody such as scFv, and the linkers can be placed between V_L and V_H or between V_H and V_L of the CD45 binding domain. In some embodiments, the antigen binding domain comprises 53B6 as disclosed herein. In some embodiments, the GS sequence provides a spacer sequence that creates flexibility between functional domains. In some embodiments, either region or both regions flanking the protease cleavage site, or the spacer sequence providing flexibility is not a GS sequence, as many flexible sequences that may function equally well are known in the art. See for example, Whitlow M. et al., An improved linker for single-chain Fv with reduced aggregation and enhanced proteolytic stability. Protein Eng. 1993 November; 6(8):989-95. doi: 10.1093/protein/6.8.989. PMID: 8309948. In some embodiments, the spacer sequence creates flexibility and does not interfere with the activities of each functional domain. In some embodiments, the linker can be any suitable length, e.g. from 15-100 amino acids, 30-100 amino acids, 20-80 amino acids, 20-50 amino acids, or 25, 30, 35, 40, 45, 50, or 55 amino acids in length, or about 25, about 30, about 35, about 40, about 45, about 50, or about 55 amino acids in length.

In some embodiments, the protease cleavage site or one or more of the multiple protease cleavage sites is recognized by one or more of a serine protease, a cysteine-type lysosomal protease, an aspartyl-type lysosomal protease and a metalloproteinase. In some embodiments, a protease that recognizes the protease cleavage site, or that recognizes one or more of the multiple protease cleavage sites, is one or more of a serine protease, a cysteine-type lysosomal protease, an aspartyl-type lysosomal protease and a metalloproteinase.

In some embodiments, the linker comprises between 1 and 15 or between 1 and 10 or 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 cleavage sites, each one of which may be the same as or different from one or more of the other cleavage sites, and/or the number of proteases that recognize the protease cleavage site or sites is between 1 and 15 or between 1 and 10 or is 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10.

In some embodiments, the linker comprises an amino GS peptide, a peptide comprising a protease cleavage site, and a carboxy GS peptide. In some embodiments, the linker comprises the amino acid sequence of SEQ ID NO: 9, which comprises an amino GS peptide (SEQ ID NO: 52), a cleavage peptide (e.g. SEQ ID NO: 53) comprising a pro-

tease cleavage site, and a carboxy GS peptide (SEQ ID NO: 54). In some embodiments, the linker comprises the same amino and carboxy GS peptides and a different cleavage peptide comprising the same or a different protease cleavage site.

In some embodiments, the linker of the protease-activating CD45-gate CAR disclosed herein comprises the amino acid sequence of any one or more of SEQ ID NOs: 9, 10, 102, 172-177, and 108-120. In some embodiments, the linker comprises one or more protease cleavage sites having an amino acid sequence selected from the group consisting of SEQ ID NOs: 32, 89-98 and 103-104, optionally wherein the number of protease cleavage sites is 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or between 1 and 15, and, when the linker comprises more than one protease cleavage site, each protease cleavage site is independently the same as or different from one or more of the other protease cleavage sites.

In some embodiments, the linker of the protease-activating CD45-gate CAR disclosed herein comprises, on either or both ends of, and contiguous with, one or more of the protease cleavage sites, one or more sets of contiguous GS sequences, wherein each GS sequence has an amino acid sequence of GS, GGS, GGGS (SEQ ID NO: 184), or GGGGS (SEQ ID NO: 178), and the sets of contiguous GS sequences comprise GS sequences in any combination, optionally wherein a set of contiguous GS sequences comprises any combination of 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 GS sequences or comprises between 1 and 15 GS sequences in any combination. Exemplary embodiments of linkers comprising such sets of contiguous GS sequences include SEQ ID NOs: 9, 10, 102, 172-177, and 107-120.

In some embodiments, the protease-activating CD45-gate CAR comprises more than one linker comprising a protease cleavage site, wherein each linker has an amino acid sequence independently selected from SEQ ID NOs: 9, 10, 102, 172-177, and 107-120, and optionally wherein the number of linkers is 2, 3, 4, 5, or an integer value between 2 and 10. Exemplary embodiments of protease-activating CD45-gate CARs comprising more than one linker include SEQ ID NOs: 125-128 and 142-145.

In some embodiments, the extracellular domain comprises a stalk domain between the antigen binding domain and the transmembrane domain.

In some embodiments, the intracellular signaling domain comprises an activating domain. In some embodiments, the intracellular signaling domain comprises a costimulatory domain. In some embodiments, the intracellular signaling domain comprises an activating domain and a costimulatory domain. In some embodiments, the intracellular signaling domain comprises an activating domain such as an ITAM-containing domain. In some embodiments, the intracellular signaling domain comprises the CD3 zeta intracellular domain. In some embodiments, the intracellular signaling domain comprises a CD3 zeta domain that comprises the amino acid sequence of SEQ ID NO: 34 or a fragment thereof.

In some embodiments, an intracellular signaling domain for use in a protease-activating CD45-gate CAR can be the cytoplasmic sequences of, for example without limitation, the T cell receptor and co-receptors that act in concert to initiate signal transduction following antigen receptor engagement, as well as any derivative or variant of these sequences and any synthetic sequence that has the same functional capability.

In some embodiments, the intracellular domain comprises or further comprises at least one costimulatory domain. In some embodiments, the at least one costimulatory domain is

a cytoplasmic signaling region or domain of CD28, OX-40, 4-1BB/CD137, CD2, CD7, CD27, CD30, CD40, programmed death-1 (PD-1), inducible T cell costimulator (ICOS), lymphocyte function-associated antigen-1 (LFA-1 (CD1 la/CD 18), CD3 gamma, CD3 delta, CD3 epsilon, CD247, CD276 (B7-H3), LIGHT, (TNFSF14), NKG2C, Ig alpha (CD79a), DAP-10, Fc gamma receptor, MHC class I molecule, TNF receptor proteins, an Immunoglobulin protein, cytokine receptor, integrins, Signaling Lymphocytic Activation Molecules (SLAM proteins), activating NK cell receptors, BTLA, a Toll ligand receptor, ICAM-1, B7-H3, CDS, ICAM-1, GITR, BAFR, LIGHT, HVEM (LIGHTR), KIRDS2, SLAMF7, NKp80 (KLRF1), NKp44, NKp30, NKp46, CD19, CD4, CD8alpha, CD8beta, IL-2R beta, IL-2R gamma, IL-7R alpha, ITGA4, VLA1, CD49a, ITGA4, CD49D, ITGA6, VLA-6, CD49f, ITGAD, CD1 Id, ITGAE, CD103, ITGAL, CD1 la, LFA-1, ITGAM, CD1 lb, ITGAX, CD1 lc, ITGB1, CD29, ITGB2, CD 18, LFA-1, ITGB7, NKG2D, TNFR2, TRANCE/RANKL, DNAMI (CD226), SLAMF4 (CD244, 2B4), CD84, CD96 (Tactile), CEACAM1, CRT AM, Ly9 (CD229), CD 160 (BY55), PSGLI1, CD100 (SEMA4D), CD69, SLAMF6 (NTB-A, Lyl08), SLAM (SLAMF1, CD150, IPO-3), BLAME (SLAMF8), SELPLG (CD162), LTBR, LAT, GADS, SLP-76, PAG/Cbp, CD 19a, a ligand that specifically binds with CD83, or any combination thereof.

In some embodiments, the protease-activating CD45-gate CAR comprises a signal sequence. In some embodiments, the signal sequence is the CD8 signal sequence. In some embodiments, the signal sequence comprises the amino acid sequence of SEQ ID NOs: 1 or 44.

In some embodiments, the amino acid sequence of the protease-activating CD45-gate CAR is or comprises the amino acid sequence of any one of SEQ ID NOs: 16-31 and 121-170, or is or comprises the amino acid sequence of any one of SEQ ID NOs: 16-31 and 121-170 but which lacks any one or more of: the signal sequence, hemagglutinin (HA) tag, and/or V5 tag.

In an alternative aspect of the protease-activating CD45-gate CAR disclosed herein, the CD45 recruiting domain itself comprises one or more than one protease cleavage sites (e.g. protease cleavage sites as disclosed herein), and the linker that joins the CD45 recruiting domain to the antigen binding domain either comprises no protease cleavage site or comprises one or more than one protease cleavage sites (e.g. protease cleavage sites as disclosed herein). In an embodiment, protease cleavage produces a CAR attached to a fragment of the CD45 protein, and the protease cleavage partially or wholly relieves the CD45-induced inhibition of the CAR's activity.

Also provided herein is a CAR that specifically recognizes and binds to MUC16. In an embodiment, the anti-MUC16 CAR comprises the amino acid sequence of any of SEQ ID NO: 11, 75, 76, 77, 78, and 79.

The present disclosure further provides a protease-activating CD45-gate T cell receptor (TCR). In an embodiment, the protease-activating CD45-gate TCR comprises a modified TCR a, 3, 7, or 6 polypeptide, wherein the modification comprises a CD45 recruiting domain joined by a protease-cleavable linker peptide to the amino terminus of the TCR a, 3, 7, or 6 polypeptide. In an embodiment, the protease-activating CD45-gate TCR comprises a modified single-chain TCR, wherein the modification comprises a CD45 recruiting domain joined by a protease-cleavable linker peptide to the amino terminus of a single-chain TCR. In an embodiment, the polypeptide is a CD45-TCR a, J, 7, or 6 polypeptide. In an embodiment, the polypeptide is a CD45-

single chain TCR. In another aspect, provided herein is a nucleic acid that encodes the CD45-gate TCR disclosed herein.

In another aspect, provided herein is a nucleic acid encoding the extracellular domain of the protease-activating CD45-gate CAR disclosed herein. In some embodiments, the encoded extracellular domain comprises a CD45 recruiting domain such as but not limited to a CD45 recruiting domain disclosed herein, the extracellular domain of a CAR such as but not limited to a CAR disclosed herein, and a linker connecting the carboxy terminus of the CD45 recruiting domain to the amino terminus of the CAR extracellular domain, such as but not limited to a linker disclosed herein.

In another aspect, provided herein is a nucleic acid that encodes the CD45-gate CAR disclosed herein.

Also provided herein is a nucleic acid that encodes a polypeptide that comprises the amino acid sequence of any of SEQ ID NOs: 11, 56, 75, 76, 77, 78, 79 and 121-170. In some embodiments, the polypeptide is an anti-MUC16 CAR as disclosed herein. In some embodiments, the polypeptide is a protease-activating CD45-gate CAR as disclosed herein.

In another aspect, provided herein is a vector comprising a nucleic acid as disclosed herein. In an embodiment, the vector is an expression vector. In an embodiment, the vector is a viral vector, a retroviral vector, a DNA vector, a plasmid, a RNA vector, an adenoviral vector, an adenovirus associated vector, a lentiviral vector, or any combination thereof.

In a further aspect, provided herein is an engineered cell e.g. an engineered immune cell comprising a protease-activating CD45-gate CAR as disclosed herein.

In a further aspect, provided herein is an engineered immune cell comprising a nucleic acid as disclosed herein.

In another aspect, provided herein is an engineered immune cell comprising a vector as disclosed herein.

In various embodiments, any of the engineered cells e.g. any of the engineered immune cells disclosed herein functionally express a protease-activating CD45-gate CAR as disclosed herein. In various embodiments, any of the engineered cells e.g. any of the engineered immune cells disclosed herein functionally express a protease-activating CD45-gate CAR that comprises the amino acid sequence of any of SEQ ID NOs: 11, 56, 75, 76, 77, 78, 79, and 121-170. Also provided herein are engineered cells e.g. engineered immune cells that functionally express a MUC16-specific CAR that comprises the amino acid sequence of any of SEQ ID NOs: 11, 56, 75, 76, 77, 78, 79 and 121-170. In various embodiments, any of the engineered cells e.g. any of the engineered immune cells disclosed herein is an isolated cell.

In various embodiments, the engineered cell e.g. engineered immune cell disclosed herein functionally expresses the protease-activating CD45-gate CAR disclosed herein from a nucleic acid encoding the protease-activating CD45-gate CAR. In various embodiments, the engineered cell e.g. engineered immune cell disclosed herein functionally expresses CD45, e.g. functionally expresses the cell's endogenous CD45 gene or functionally expresses CD45, or a fragment or variant of CD45, from an exogenous nucleic acid that encodes CD45 and that was introduced into the cell. In various embodiments, the engineered cell e.g. engineered immune cell disclosed herein functionally expresses the anti-MUC16 CAR disclosed herein from a nucleic acid encoding the anti-MUC16 CAR. In some embodiments, the anti-MUC16 CAR is a protease-activating CD45-gate MUC16CAR.

In various embodiments, the engineered immune cell disclosed herein is an engineered B cell, mast cell, myeloid-derived phagocyte, T cell e.g. an alpha/beta and/or gamma/

delta T cell, tumor infiltrating lymphocyte (TIL), NK cell, TCR-expressing cell, dendritic cell, or NK-T cell. In some embodiments, the cell is and/or is derived from an autologous T cell. In some embodiments, the cell is and/or is derived from an allogeneic T cell or an iPSC (induced pluripotent stem cell). In some embodiments, the cell is and/or is derived from an iPSC-derived T cell.

In an embodiment, an engineered immune cell as disclosed herein comprises or further comprises one or more genomic modifications of one or more of the endogenous CD52 gene and the endogenous TCR genes, e.g. the TCR α gene. In an embodiment, an engineered immune cell as disclosed herein is a T cell.

In another aspect, provided herein is a population of immune cells comprising one or more of the engineered immune cells disclosed herein. In an embodiment, the population comprises about or at least about 1×10^4 , about or at least about 1×10^5 , about or at least about 1×10^6 , about or at least about 1×10^7 , or about or at least about 1×10^8 engineered cells, e.g. engineered immune cells, as disclosed herein, optionally wherein the population does not comprise more than about 1×10^{10} or more than about 1×10^9 or more than about 5×10^9 engineered cells, e.g. engineered immune cells, as disclosed herein. In an embodiment, a population of immune cells as disclosed herein is enriched for the engineered immune cell as disclosed herein. In various embodiments, the population of immune cells is at least 50%, e.g. 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or more than 95% engineered cells that are T cells, e.g., alpha/beta T cells and gamma/delta T cells, B cells, natural killer (NK) cells, natural killer T (NKT) cells, mast cells, and/or myeloid-derived phagocytes. In various embodiments, the population of immune cells is at least 50%, e.g. 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or more than 95% engineered T cells.

In another aspect, provided herein is a composition comprising an engineered cell e.g. an engineered immune cell as disclosed herein or a population of cells as disclosed herein, and a pharmaceutically acceptable carrier.

Also provided herein is the use of any of the engineered immune cells e.g. engineered T cells provided herein in the manufacture of a medicament for the treatment of cancer or for inhibiting tumor growth or progression in a subject in need thereof. Also provided herein is any of the engineered immune cells e.g. engineered T cells provided herein for use as a medicament and for use in treating any disease or condition amenable to treatment by CAR T cell therapy. In some embodiments, the disease or condition is a cancer, autoimmune disease, or infection. In some embodiments, the disease or condition is a form of cancer.

In another aspect, provided herein is a method of treating a condition in a patient comprising administering to the patient an engineered immune cell as disclosed herein. In an embodiment of the method, the engineered immune cell is an allogeneic engineered immune cell derived from a donor other than the patient.

In another aspect, provided herein is a method of treating a condition in a patient comprising administering to the patient a population of engineered immune cells as disclosed herein. In an embodiment of the method, the engineered immune cells of the population are derived from one or more allogeneic immune cells from a donor other than the patient.

In another aspect, provided herein is a method of treating a condition in a patient comprising administering to the patient a pharmaceutical composition as disclosed herein. In an embodiment of the method, the composition comprises

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one or more engineered allogeneic immune cells derived from a donor other than the patient.

In embodiments of the methods of treating a condition in a patient disclosed herein, the condition may be any disease or condition amenable to treatment by CAR T cell therapy. In some embodiments, the disease or condition is a cancer, autoimmune disease, or infection. In some embodiments, the disease or condition is a form of cancer. In some embodiments, the cancer is a hematological malignancy or non-solid tumor. In some embodiments, the cancer is a solid cancer or solid tumor. In some embodiments, the cancer is a hematological malignancy that is acute lymphoblastic leukemia (ALL), acute myeloid leukemia (AML), chronic myelogenous leukemia (CML), chronic eosinophilic leukemia (CEL), myelodysplasia syndrome (MDS), non-Hodg-
 5 kin's lymphoma (NHL), or multiple myeloma (MM). In some embodiments, the cancer is a solid cancer that is biliary cancer, bladder cancer, bone or soft tissue carcinoma, brain tumor, breast cancer, cervical cancer, colon cancer, colorectal adenocarcinoma, colorectal cancer, desmoid tumor, embryonal cancer, endometrial cancer, esophageal cancer, gastric cancer, gastric adenocarcinoma, glioblastoma multiforme, gynecological tumor, head and neck squamous cell carcinoma, hepatic cancer, lung cancer, malignant mela-
 20 noma, osteosarcoma, ovarian cancer, pancreatic cancer, pancreatic ductal adenocarcinoma, primary astrocytic tumor, primary thyroid cancer, prostate cancer, renal cancer, renal cell carcinoma, rhabdomyosarcoma, skin cancer, soft tissue sarcoma, testicular germ-cell tumor, urothelial cancer, uterine sarcoma, or uterine cancer.

In another aspect, provided herein is a method of reducing on-target off-tumor toxicity of CAR T cells comprising administering to a patient a cell as disclosed herein, a population of cells as disclosed herein, or a composition as disclosed herein, wherein the on-target, off-tumor toxicity is lower than the on-target, off-tumor toxicity of control cells, a population of control cells or a composition of control cells, respectively, administered under the same conditions, wherein the control cells, population of control cells, or control cells of the composition comprise a CAR that lacks at least one of a CD45 recruiting domain and a linker that comprises a protease cleavage site.

In another aspect, provided herein is a method of increasing the efficiency of CAR T cells or CAR T cell therapy comprising administering to a patient a cell as disclosed herein, a population of cells as disclosed herein, or a composition as disclosed herein, wherein the efficiency is greater than the efficiency of control cells, a population of control cells or a composition of control cells, respectively, administered under the same conditions, wherein the control cells, population of control cells, or control cells of the composition comprise a CAR that lacks at least one of a CD45 recruiting domain and a linker that comprises a protease cleavage site.

In another aspect, provided herein is a method of increasing the efficacy of CAR T cell therapy comprising administering to a patient a cell as disclosed herein, a population of cells as disclosed herein, or a composition as disclosed herein, wherein the efficacy is greater than the efficacy of control cells, a population of control cells or a composition of control cells, respectively, administered under the same conditions, wherein the control cells, population of control cells, or control cells of the composition comprise a CAR that lacks at least one of a CD45 recruiting domain and a linker that comprises a protease cleavage site.

In another aspect, provided herein is a method of increasing the efficacy of CAR T cell therapy against a solid tumor

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comprising administering to a patient having a solid tumor a cell as disclosed herein, a population of cells as disclosed herein, or a composition as disclosed herein, wherein the efficacy against the solid tumor is greater than the efficacy against the solid tumor of control cells, a population of control cells or a composition of control cells, respectively, administered under the same conditions, wherein the control cells, population of control cells, or control cells of the composition comprise a CAR that lacks at least one of a CD45 recruiting domain and a linker that comprises a protease cleavage site.

In another aspect, provided herein is a method of reducing the incidence of side effects in CAR T cell therapy comprising administering to a patient a cell as disclosed herein, a population of cells as disclosed herein, or a composition as disclosed herein, wherein the incidence of side effects is lower than the incidence of side effects when control cells, a population of control cells or a composition of control cells, respectively, are administered under the same conditions, wherein the control cells, population of control cells, or control cells of the composition comprise a CAR that lacks at least one of a CD45 recruiting domain and a linker that comprises a protease cleavage site.

In another aspect, provided herein is a method of effecting reversible colocalization of CD45 and a CD45-gate CAR in a CAR T cell, the reversible colocalization comprising a CD45-gate CAR recruiting domain binding to a CD45 protein on a CAR T cell surface, resulting in colocalization, which binding and colocalization can be disrupted by a protease cleaving a protease cleavage site in the CD45-gate CAR's linker, thereby reversing the colocalization, the method comprising the steps of:

- (a) providing a T cell that expresses CD45,
- (b) introducing a nucleic acid encoding a protease-activating CD45-gate CAR as disclosed herein into the T cell,
- (c) maintaining the T cell under conditions in which both CD45 and the CAR are functionally expressed at the cell surface, resulting in a CD45-gate CAR recruiting domain binding to a CD45 protein on the CAR T cell surface and colocalization of the CAR and the CD45 protein, which colocalization can be reversed by a protease cleaving a protease cleavage site in the CD45-gate CAR's linker, wherein the cleaving produces a functional CAR no longer connected by the linker to the CD45-gate CAR recruiting domain.

In another aspect, provided herein is a method of effecting reversible reduction of CAR activity by a CD45 protein in a CAR T cell, the reversible reduction of CAR activity comprising a CD45-gate CAR recruiting domain binding to a CD45 protein on a CAR T cell surface, resulting in inactivation of the chimeric antigen receptor by the CD45, which inactivation can be disrupted by a protease cleaving a protease cleavage site in the CD45-gate CAR's linker, resulting in the disassociation of the chimeric antigen receptor and the CD45 protein and ending the inactivation of the chimeric antigen receptor by the CD45, the method comprising the steps of:

- (a) providing a T cell that expresses CD45,
- (b) introducing a nucleic acid encoding a CD45-gate CAR into the T cell, (c) maintaining the T cell under conditions in which both CD45 and the CAR are functionally expressed at the cell surface,

resulting in a CD45-gate CAR recruiting domain binding to a CD45 protein on the CAR T cell surface, association of the CAR and the CD45 protein, and reduction of the CAR's activity by the CD45 protein, which

activity reduction can be reversed by a protease cleaving a protease cleavage site in the CD45-gate CAR's linker, thereby ending the association of the CAR and the CD45 protein, and ending the CD45 protein's reduction of or inhibition of CAR activity, wherein the cleaving produces a functional CAR no longer connected by the linker to the CD45-gate CAR recruiting domain.

In another aspect, provided herein is a method of treating with CAR T cell therapy a patient who has a tumor characterized by a protease-rich tumor microenvironment, comprising administering to the patient a cell as disclosed herein, a population of cells as disclosed herein, or a composition as disclosed herein, wherein CAR T cell activity is lower outside the protease-rich tumor microenvironment than in the protease-rich tumor microenvironment.

In another aspect, provided herein is a method of regulating cytotoxic activity of a CAR T cell that comprises a CAR as disclosed herein, the method comprising inhibiting the cytotoxic activity of the CAR T cell by associating the CAR with CD45 of the CAR T cell, and activating the cytotoxic activity of the CAR T cell by subjecting and/or exposing the CAR T cell to a protease that recognizes and cleaves the protease cleavage site.

In embodiments of the methods of treating a condition in a patient disclosed herein, the cell, population of cells or composition can be administered to the subject on one occasion or can be administered to the subject on two or more occasions spaced at least about 1, 2, 3, 4, 5, 6, 7, or more days apart. In some embodiments, the disorder can be a viral disease, a bacterial disease, a cancer, an inflammatory disease, an immune disease, or an aging-associated disease.

Also provided herein are chimeric antigen receptors (CARs) that bind to MUC16 ("MUC16-specific CARs" or "anti-MUC16 CARs" or "MUC16 CARs"). It is demonstrated that the expression of MUC16 specific CARs in T cells is effective to activate the T cells upon contact with MUC16. The MUC16 specific CARs provided herein bind human MUC16 and exhibit cytotoxic activity upon contact with MUC16-expressing cells.

Accordingly, in another aspect, provided herein is a MUC16 specific chimeric antigen receptor (CAR) comprising an extracellular ligand-binding domain, a first transmembrane domain, and an intracellular signaling domain, wherein the extracellular domain comprises a single chain variable fragment (scFv) binding to the extracellular domain of MUC16.

The extracellular domain of MUC16 specific CARs comprises a scFv, wherein the scFv comprises a heavy chain variable (V_H) region and a light chain variable (V_L) region, the V_H and V_L regions each comprising three complementarity determining regions (CDRs) specific for MUC16.

In some embodiments, the V_H region comprises (i) a V_H complementarity determining region one (CDR1) having the amino acid sequence shown in SEQ ID NO: 60 or 63; (ii) a V_H complementarity determining region two (CDR2) having the amino acid sequence shown in SEQ ID NO: 61 or 64; and (iii) a V_H complementarity determining region three (CDR3) having the amino acid sequence shown in SEQ ID NO: 62 or 65; and/or the V_L region comprises (i) a V_L complementarity determining region one (CDR1) having the amino acid sequence shown in SEQ ID NO: 66 or 69; (ii) a V_L complementarity determining region two (CDR2) having the amino acid sequence shown in SEQ ID NO: 67 or 70; and (iii) a V_L complementarity determining region three (CDR3) having the amino acid sequence shown in SEQ ID NO: 68 or 71.

In some embodiments, provided are MUC16 specific CARs comprising a scFv, wherein the scFv comprises a V_H region having the sequence shown in SEQ ID NO: 58.

In some embodiments, provided are MUC16 specific CARs comprising a scFv, wherein the scFv comprises a V_L region having the sequence shown in SEQ ID NO: 59.

In some embodiments, provided are MUC16 specific CARs comprising a scFv comprising a V_H region and a V_L region, wherein the V_H region has the sequence shown in SEQ ID NO: 58, and the V_L region has the sequence shown in SEQ ID NO: 59.

In some embodiments, the V_H region comprises the sequence shown in SEQ ID NO: 58, or a variant thereof with one or several conservative amino acid substitutions in residues that are not within a CDR and/or the V_L region comprises the amino acid sequence shown in SEQ ID NO: 59, or a variant thereof with one or several amino acid substitutions in amino acids that are not within a CDR.

In some embodiments, the V_H region comprises the amino acid sequence shown in SEQ ID NO: 58 and the V_L region comprises the amino acid sequence shown in SEQ ID NO: 59.

In some embodiments, each CDR is defined in accordance with the Kabat definition, the Chothia definition, the combination of the Kabat definition and the Chothia definition, the AbM definition, or the contact definition of CDR.

In some embodiments, the MUC16 specific CAR intracellular signaling domain comprises a CD3 ζ signalling domain. In some embodiments, the intracellular signaling domain comprises a 4-1BB domain. In some embodiments, the MUC16 specific CAR comprises a second intracellular signaling domain. In some embodiments, the second intracellular signaling domain comprises a 4-1BB domain.

In some embodiments, the MUC16 specific CARs disclosed herein may comprise a stalk domain between the extracellular ligand-binding domain and the first transmembrane domain. In some embodiments, the stalk domain is selected from the group consisting of: a CD8 α hinge, an IgG1 hinge, and an Fc γ RIII α hinge. In some embodiments, the stalk domain is a human CD8 α hinge, a human IgG1 hinge, or a human Fc γ RIII α hinge.

In some embodiments, the MUC16 specific CARs disclosed herein may comprise one or more epitopes or mimotopes specific for one or more monoclonal antibodies. In some embodiments, the one or more epitopes or mimotopes specific for a monoclonal antibody includes a CD20 epitope or mimotope. In some embodiments, the CD20 epitope or mimotope comprises the amino acid sequence shown in SEQ ID NO: 50 or SEQ ID NO: 51.

In some embodiments, the MUC16 specific CAR comprises the amino acid sequence shown in any of SEQ ID NO: 11, 75, 76, 77, 78, and 79, with or without a signal sequence. In some embodiments, the MUC16 specific CAR comprises the amino acid sequence shown in SEQ ID NO: 75, with or without a signal sequence. In some embodiments, the MUC16 specific CAR comprises the amino acid sequence shown in SEQ ID NO: 76, with or without a signal sequence.

In some embodiments, the first transmembrane domain comprises a CD8 α chain transmembrane domain.

In some embodiments, the MUC16 specific CAR can comprise another extracellular ligand-binding domain which is not specific for MUC16.

In some embodiments, the extracellular ligand-binding domain(s), the first transmembrane domain, and intracellular signaling domain(s) are on a single polypeptide.

In some embodiments, the CAR can comprise a second transmembrane domain, wherein the first transmembrane

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domain and the extracellular ligand-binding domain(s) are on a first polypeptide, and wherein the second transmembrane domain and the intracellular signaling domain(s) are on a second polypeptide. In an exemplary embodiment, the first transmembrane domain comprises a transmembrane domain from the α chain of the high-affinity IgE receptor (Fc ϵ RI) and the second transmembrane domain comprises a transmembrane domain from the γ or β chain of Fc ϵ RI.

In some embodiments, the CAR can comprise a third polypeptide comprising a third transmembrane domain fused to an intracellular signaling domain from a co-stimulatory molecule, wherein the third transmembrane domain comprises a transmembrane domain from the γ or β chain of Fc ϵ RI.

In another aspect, provided herein is an isolated polynucleotide comprising a nucleic acid sequence encoding the MUC16 specific CAR described herein.

In another aspect, provided herein is an expression vector comprising the polynucleotide encoding the MUC16 specific CAR described herein.

In another aspect, provided herein is an engineered immune cell expressing at its cell surface membrane a MUC16 specific CAR described herein. In some embodiments, the engineered immune cell can comprise another CAR which is not specific for MUC16. In some embodiments, the engineered immune cell can comprise a polynucleotide encoding a suicide polypeptide. In some embodiments, the suicide polypeptide is RQR8.

In some embodiments, the engineered immune cell is derived from an inflammatory T-lymphocyte, a cytotoxic T-lymphocyte, a regulatory T-lymphocyte, or a helper T-lymphocyte. In some embodiments, the engineered immune cell is an inflammatory T-lymphocyte, a cytotoxic T-lymphocyte, a regulatory T-lymphocyte, or a helper T-lymphocyte.

In some embodiments, the engineered immune cell can comprise a disruption of one or more endogenous genes, wherein the endogenous gene encodes TCR α , TCR β , CD52, glucocorticoid receptor (GR), deoxycytidine kinase (dCK), or an immune checkpoint protein such as for example programmed death-1 (PD-1).

In some embodiments, the engineered immune cell is obtained from a healthy donor. In some embodiments, the engineered immune cell is obtained from an individual afflicted with a disease or disorder.

In another aspect, provided herein is an engineered immune cell expressing at its cell surface membrane a MUC16 specific CAR as described herein for use as a medicament. In another aspect, provided herein is a MUC16 specific antibody for use as a medicament. In some embodiments, the medicament comprising the MUC16 specific CAR expressing immune cells or MUC16 specific antibodies is for use in treatment of a MUC16 associated disease or disorder.

In one aspect, provided herein is a population of cells comprising engineered immune cells expressing MUC16 specific CARs described herein, wherein (ii) said population of cells comprises a percentage of stem cell memory and central memory cells greater than 20%, 30% or 40%, and/or (iii) said population of cells achieves a percentage lysis of MUC16 expressing cells greater than 10%, 20%, 30% or 40%.

In another aspect, provided herein is a method of engineering an immune cell expressing any one of the MUC16 specific CARs described herein, the method comprising: providing an immune cell; and introducing into the cell at

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least one polynucleotide encoding said MUC16 specific CAR; whereby said immune cell expresses said MUC16 specific CAR.

In some embodiments, the method comprises providing an immune cell;

introducing into the cell at least one polynucleotide encoding said MUC16 specific CAR; and introducing at least one polynucleotide encoding a CAR which is not specific for MUC16.

In another aspect, provided herein is a method of treating a subject suffering from a MUC16 associated disease or disorder, the method comprising: providing an immune cell expressing at the surface a MUC16 specific CAR as described herein; and administering said immune cells to said subject. The disclosure also provides methods of treating subjects suffering from a MUC16 associated disease or disorder, the method comprising providing MUC16 specific antibodies described herein and administering said antibodies to said subject.

In some embodiments, provided herein is a pharmaceutical composition comprising an engineered immune cell expressing MUC16 specific CARs as described herein. In other embodiments, the disclosure provides pharmaceutical compositions comprising any of the MUC16 specific antibodies described herein.

In another aspect, provided herein is a method of treating a condition associated with malignant cells expressing MUC16 in a subject comprising administering to a subject in need thereof an effective amount of the pharmaceutical composition comprising an engineered immune cell as described herein. In some embodiments, the condition is a cancer.

In another aspect, provided herein is a method of inhibiting tumor growth or progression in a subject who has malignant cells expressing MUC16, comprising administering to the subject in need thereof an effective amount of the pharmaceutical composition comprising an engineered immune cell as described herein or a MUC16 specific antibody as described herein to the subject.

In another aspect, provided herein is a method of inhibiting metastasis of malignant cells expressing MUC16 in a subject, comprising administering to the subject in need thereof an effective amount of the pharmaceutical composition comprising an engineered immune cell as described herein or a MUC16 specific antibody as described herein to the subject.

In another aspect, provided herein is a method of inducing tumor regression in a subject who has malignant cells expressing MUC16, comprising administering to the subject in need thereof an effective amount of the pharmaceutical composition comprising an engineered immune cell as described herein or a MUC16 specific antibody as described herein to the subject.

In some embodiments, any of the above methods further comprises administering one or more additional therapies, such as for example, a monoclonal antibody and/or a chemotherapeutic. In some embodiments, the monoclonal antibody can be, for example, an antibody that binds to a checkpoint inhibitor such as, for example, an anti-PD-1 antibody or an anti-PD-L1 antibody. In some embodiments, any of the above methods further comprises administering a Receptor Tyrosine Kinase inhibitor, an mTOR inhibitor, an epigenetic modulator, a proteasome inhibitor, an immunomodulatory agent such as lenalidomide, a Hedgehog inhibitor or an Isocitrate Dehydrogenase (IDH) inhibitors, to the subject.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A-1B. FIG. 1A (left side) illustrates a potential mechanism of action for an exemplary protease-activating CD45-gate CAR. In the cell membrane of a T cell, for example, a linker peptide tethers (i.e. joins or connects) a CD45 recruiter domain to a CAR. The linker peptide comprises at least one protease site. The recruiter recognizes and binds to a CD45 protein, thus holding the CAR and CD45 protein in relatively close proximity to each other. This close proximity permits the CD45 to inactivate the CD3zeta domain. Thus, the CAR activity is reduced even when it binds to its target antigen, so long as the linker peptide is intact. When a protease cleaves the linker peptide (FIG. 1A, right side) (e.g. when the CAR T cell is in the environment of a tumor that expresses the protease that recognizes the protease site in the linker), the CAR is "released" from the CD45 and ceases to be inactivated by the CD45. The CAR T cell thus selectively becomes active primarily in the vicinity of its target tumor, thereby reducing on-target off-tumor toxicity. FIG. 1B presents schematic representations of the domains of exemplary CARs/CAR constructs. Upper: An exemplary CAR preprotein comprises a CD8 signal sequence and a CAR polypeptide. Lower: An exemplary protease-activating CD45-gate CAR preprotein comprises a CD8 signal sequence, optional hemagglutinin (HA) tag, CD45 recruiter domain, linker comprising glycine-serine (GS) rich sequences flanking a thrombin protease cleavage site, and a CAR.

FIGS. 2A-2B. CAR T cells comprising vectors encoding protease-activating CD45-gate CARs grow and express the transgenic protein. T cells were transduced with a vector that encodes a CAR that is not a protease-activating CD45-gate CAR (53B6, an anti-MUC16 CAR), with a vector that encodes a protease-activating CD45-gate CAR (CD45-Gate1-TMB-53B6, CD45-Gate2-TMB-53B6), or with a vector that encodes analogs of the protease-activating CD45-gate CARs that comprise a non-cleavable linker (CD45-Gate1-GS-53B6, CD45-Gate2-GS-53B6). The linker of the cleavable protein comprised: GS-region-thrombin cleavage site-V5 peptide-thrombin cleavage site-GS-region. The linker of the non-cleavable protein comprised: GS-region-V5 peptide-GS-region. FIG. 2A. Percentage of CAR+ cells at days 8 and 15 of production. FIG. 2B. Cell number at day 15 of production. NTD, not transduced.

FIG. 3A. Exposing the CAR T cells of FIGS. 2A-2B to thrombin resulted in reduced detection of the V5 tag in the cleavable CARs but not in the non-cleavable CARs. FIG. 3B. The ability of the CAR T cells of FIGS. 2A-2B to bind to soluble MUC16 was not significantly affected by the presence or absence of the CD45 gate.

FIGS. 4A-4B. Results of cytotoxicity assays used to assess inhibitory activity of CD45 gate. Assays were performed in R10 medium without thrombin or other exogenous protein present. FIG. 4A. Ovar3 (MUC16^{High}), 1st round—48H cytotoxicity. FIG. 4B. Ovar3 (MUC16^{High}), 2nd round—2×48H cytotoxicity. NTD: not transduced; Ovar3 (MUC16^{High}): cell line that expresses a high density of the target antigen MUC16.

FIGS. 5A-5B. Results of cytotoxicity assays used to assess inhibitory activity of CD45 gate. Assays were performed in R10 medium without thrombin or other exogenous protein present. FIG. 5A. Cov644 (MUC16^{Low}), 1st round—48H cytotoxicity. FIG. 5B. Cov644 (MUC16^{Low}), 2nd round—2×48H cytotoxicity. NTD: not transduced; Cov644 (MUC16^{Low}): cell line that expresses a low density of the target antigen MUC16.

FIGS. 6A-6B. Results of cytotoxicity assays used to assess reversibility of CD45 gate inhibitory activity. Assays were performed in serum-free medium. Black bars: no protease present. White bars: protease (thrombin) present. E:T: ratio of number of effector cells to number of target cells; NTD: not transduced.

FIG. 7A. 53B6 CAR T cells were successfully generated from two different T cell donors using lentiviral transfer vector containing the 53B6 scFv that targets the MUC16 protein. CAR+ T cell population was detected with recombinant MUC16 protein (SEA1-4) using flow cytometry. FIG. 7B. Phenotypes were assigned according to CD62L and CD45RO expression within the CAR+ T cell population as follows: stem cell memory (Tscm; CD45RO-/CD62L+), central memory (Tcm; CD45RO+/CD62L+), effector memory (Tem; CD45RO+/CD62L-), effector cells (Teff; CD45RO-/CD62L-). SEA=sperm protein, enterokinase, agrin; EV: empty vector.

FIGS. 8A-8C. 1×10⁷ luciferase-expressing ovarian target cells (OVCAR3, COV644, FUOV₁) with different MUC16 surface protein expression levels (indicated in parentheses) were co-cultured with 53B6 CAR+ T cells for 72 hours at various Effector:Target (E:T) ratio. Empty vector (EV) control T cells were also included in the study. Percent target cell survival after being exposed to T cells was determined by comparing to target cells alone.

FIGS. 9A-9B. 1×10⁷ luciferase-expressing OVCAR3 target cells were co-cultured with 53B6 CAR+ T cells at Effector:Target ratio of 1:1 in a serial killing assay, where the T cells are exposed to fresh target cells every 2-3 days. Empty vector (EV) control T cells were also included in the study. Percent target cell survival after being exposed to T cells was determined by comparing to target cells alone. Data presented are the activity of 53B6 CAR T cells from two different donors.

FIGS. 10A-10B. 53B6 anti-MUC16 CAR T cells display anti-tumor activity against two different orthotopic ovarian tumor models. NSG mice (n=5) intra-peritoneally (IP) injected with 3×10⁶ luciferase-labeled OVCAR3 (FIG. 10A) or COV644 (FIG. 10B) cells received IP injection of 1×10⁶ or 3×10⁶ 53B6 CAR+ T cells, respectively, 14 days later. Tumor burden was monitored by bioluminescence, and the results demonstrated anti-tumor activity in both models. Empty-vector (EV) control T cells were also included in the study. LOD=Limit of Detection, based on bioluminescence imaging of non-tumor bearing mice.

FIG. 11. Schematic diagram of exemplary CD45-gate CAR constructs. TPS3, TPS4 etc. refer to linkers of various lengths (e.g. TPS6 20aa, TPS6 25aa, etc.) as shown in Table 2; MMP=matrix metalloproteinase; uPA=urokinase plasminogen activator (uPA); ✓ indicates presence of the indicated site; white rectangle between 53B6vL and 41BB ICD indicates the transmembrane and hinge region.

FIGS. 12A-12D. Production of new CD45-gate CART cells. CD45-gate CAR T cells containing a selective panel of TPS linkers were successfully generated and characterized using flow cytometry according to methods described in Example 2. Autoactivation and memory phenotypes of the CAR T cells were also determined using flow cytometry by analyzing the expression of CD45/41BB and CD45RO/CD62L on Day 9 and Day 14, respectively. FIG. 12A. The percentages of CAR transduction are comparable across all CD45-gate CAR T cells. FIG. 12B. The overall expansion of T cells remains similar among all the clones and is comparable to non-transduced control (NTD), indicating no obvious fratricide. FIG. 12C. Overall, CD45-gate CAR T cells show higher percentage of stem-cell memory subset (CD45RO-/

CD62L+) than naked CAR T cells. FIG. 12D. CD45-gate CAR T cells exhibit less tonic signaling (e.g., CD25+/41BB+) than naked CAR T cells, suggesting a potential benefit of CD45-gate CARs in controlling T cell exhaustion. All cleavable linkers in FIGS. 12A-12D have a length of 45 amino acids and their sequences (e.g. TPS3 (45 aa), TPS4 (45aa), TPS6 (45 aa)) are set forth in Table 2. TPS3 (45aa) has the same amino acid sequence as SEQ ID NO:9, referred to in Table 2 as "GSTPS1 (45 aa)" (also known as "TPS1").

FIGS. 13A-13D. CD45-gate CAR T cells containing different TPS linkers were generated. FIG. 13A. The percentages of CAR transduction are comparable for all TPS linkers except TPS10. FIG. 13B. CD45-gate does not cause obvious fratricide. FIG. 13C. CD45-gate CARs show more stem-cell memory phenotype than non-transduced T cells and T cells transduced with 53B6 (anti-MUC) CAR (non-CD45-gate). FIG. 13D. CD45-gate CARs show higher mean fluorescence intensity (MFI, top panel), but show less tonic signaling than T cells transduced with 53B6 (bottom panel). FIG. 13A shows the results of flow cytometry. All cleavable linkers in FIGS. 13A-13D have a length of 45 amino acids.

FIGS. 14A-14E. Generation and analysis of reagent cell lines. Target cells were developed in-house based on three cell lines that are positive for tumor-associated proteases, including an ovarian cancer cell line OVCAR3, a breast cancer cell line MDA-MB-231, and a lung carcinoma cell line NCI-H292. FIG. 14A. Flow cytometry results confirming the surface expression of matriptase (anti-matriptase antibody used at 3 µg/mL: R&D Systems Cat. No. BAF3946) for each cell line. Vertical axis: site scatter area (SSC-A). FIG. 14B. Surface-expressed matriptase activity was assessed through the detection of a V5 tag that is attached to the N-terminus of Muc16 CAR 53B6 via a matriptase cleavable linker TPS6 45aa, as described in Example 1. Minimal protease activity was detected in vitro despite matriptase expression, as evidenced by the fact that the V5 signal coincides with Muc16 CAR signal in OVCAR3, MDA MB 231, and H292. This indicates that V5 is not cleaved off by matriptase. Recombinant human matriptase cleaved the V5 signal. FIG. 14C. Validation of the expression of the target protein MUC16 by flow cytometry for two additional cell lines, H292-Muc16 and MDA MB 231-Muc16, for use as CAR target cells. FIG. 14D. ELISA kit (R7D, DUPA00) results, detecting soluble urokinase (uPA) secretion in conditioned media for each cell line. FIG. 14E. Summary table of target cells and protease expression.

FIGS. 15A-15F. TPS3 and TPS4 show response to endogenous proteases (-Matriptase) in a Jurkat NFAT reporter assay while TPS6 mainly responds to exogenous matriptase (+Matriptase). A Jurkat-NFAT-luciferase reporter cell line transduced with lentivirus encoding CD45-gate CARs was used to evaluate the activity of CD45-gate CARs in the presence of endogenous matriptase only (-Matriptase) or in the presence of exogenous matriptase (+Matriptase). Exposure to MUC16 positive target cells led to the activation of 53B6 CAR, while no activation was detected with a CD45-gate CAR with a non-cleavable linker. However, the activation can be restored via the cleavable linkers upon exposure to endogenous or exogenous proteases. TPS3 and TPS4 show response to endogenous proteases in a Jurkat NFAT reporter assay while TPS6 mainly responds to exogenous matriptase. All cleavable linkers in FIGS. 15A-15C have a length of 45 amino acids. The following construct names in FIGS. 15D-15F have the amino acid sequence of the following SEQ ID NOs, respectively: 1102 (2×TPS3); SEQ ID NO:9; 1399 (TPS4); SEQ ID NO:137; 1401 (TPS6); SEQ ID NO:138; 1241 (2×TPS6); SEQ ID NO:144; 1466

(2×TPS6); SEQ ID NO:153; 1467 (2×TPS6); SEQ ID NO:154; 1468 (2×TPS6); SEQ ID NO:155; 1469 (2×TPS6); SEQ ID NO:156. This shows that the CD45-gate functions with linkers of different lengths.

FIGS. 16A-16F. The short-term cytotoxicity of CD45-gate CAR T cells against target cell lines MDA-MB-231-Muc16, H292-Muc16, and OVCAR3 was evaluated using a nuclear GFP assay. Average fold change in target cell count and standard deviation are shown in each of FIGS. 16A-16F (E:T=3:1 for each cell line). Top panels: CD45-gate CARs with various TPS linkers showed a spectrum of activity in the presence of endogenous proteases, with some of the linkers (e.g., TPS4) exhibiting activity comparable to conventional 53B6 CAR. Bottom panels: all CD45-gate CARs with TPS linkers showed cytotoxic activities in the presence of exogenous matriptase. Incorporation of a non-cleavable GS linker in a CD45-gate CAR effectively inhibited CAR function. All cleavable linkers in FIGS. 16A-16F have a length of 45 amino acids.

FIGS. 17A-17B. CARs with TPS linkers maintained long-term cytotoxicity in the presence of endogenous/exogenous proteases. FIG. 17A. Activity measured in the absence of exogenous matriptase. FIG. 17B. Activity measured in the presence of exogenous matriptase. Activity ranking: TPS12>TPS4, TPS9, TPS11, TPS13>TPS3, TPS8>TPS6, TPS10. All cleavable linkers in FIGS. 17A-17B have a length of 45 amino acids.

FIGS. 18A-18C. Evaluation of CD45-gate CART cells (MDA-MB-231-Muc16) in a breast cancer xenograft model in 7-8 weeks old NSG mice. FIG. 18A. Schematic summary of the experimental scheme for in vivo analysis. FIG. 18B. CD45-Gate CARs with cleavable TPS3 or TPS4 linker showed anti-tumor activity while the CD45-Gate CAR with TPS6 linker did not show activity in this assay. CD45-gate CAR with non-cleavable GS linker had no CAR activity. FIG. 18C. Treatment with CD45-gate CAR T cells had no effect on the body weight of the mice and no toxicities were observed with any of the cells used. All cleavable linkers in FIGS. 18B-18C have a length of 45 amino acids.

FIGS. 19A-19C. Evaluation of CD45-gate CART cells (NCI-H292-MUC16) in a non-small cell lung cancer xenograft model in NSH mice. FIG. 19A. Schematic summary of the experimental scheme for in vivo analysis. FIG. 19B. CD45-Gate CAR with cleavable TPS4 linker showed anti-tumor activity. CD45-gate CAR with non-cleavable GS linker (aCD45-GS-53B6; this is the same construct as "GS non-cleavable" in FIGS. 12A-15C and 16A-18C; the non-cleavable linker has a length of 30 amino acids) had no CAR activity. FIG. 19C. Treatment with CD45-gate CAR T cells had no effect on the body weight of the mice and no toxicities were observed with any of the cells used. All cleavable linkers in FIGS. 19B-19C have a length of 45 amino acids.

FIGS. 20A-20B. Evaluation of other CD45 ligands in protease-activating CD45-gate CAR. T cells were transduced with a vector that encodes a CAR that is not a protease-activating CD45-Gate CAR (53B6, an anti-Muc16 CAR), or with a vector that encodes the antibody-based CD45-Gate CAR (CD45-GS45-53B6), and other alternative CD45-Gate CARs that comprise a non-cleavable linker (BTN3A1(IgV)-GS45-53B6, BTN3A1-GS45-53B6, sec49k-GS45-53B6, UL11-GS45-53B6). The linker of the non-cleavable protein comprised a 45-mer G4S linker. Results of cytotoxicity assays used to assess inhibitory activity of CD45 gate. Assays were performed in R10 medium without thrombin or other exogenous protein present. FIG. 20A, OVCAR3 (Muc16^{High}) and FIG. 20B, Cov644

(Muc16^{Low}) 1st round—48H cytotoxicity. The constructs comprised the following domains as indicated by each construct's name: 53B6, which has the amino acid sequence of SEQ ID NO: 11; CD45, which has the amino acid sequence of SEQ ID NO: 3; GS45, which has the amino acid sequence of SEQ ID NO: 182; BTN3A1(IgV), which has the amino acid sequence of SEQ ID NO: 14; BTN3A1, which has the amino acid sequence of SEQ ID NO: 183; sec49k, which has the amino acid sequence of SEQ ID NO: 12; UL11, which has the amino acid sequence of SEQ ID NO: 13. The constructs have the following amino acid sequences: BTN3A1(IgV)-GS45-53B6: SEQ ID NO: 166; BTN3A1-GS45-53B6: SEQ ID NO: 168; sec49k-GS45-53B6: SEQ ID NO: 169; UL11-GS45-53B6: SEQ ID NO: 170.

DETAILED DESCRIPTION

The present disclosure provides a protease-activating CD45-gate CAR e.g. a chimeric antigen receptor modified to comprise a removable CD45-recruiting domain, which specifically binds to CD45. The disclosure further provides related nucleic acids, engineered cells, compositions, and methods. The CD45 binding domain is linked, joined to or tethered to the antigen binding domain of a CAR by a peptide linker that comprises at least one protease cleavage site. The CAR's antigen binding domain can specifically bind to an antigen expressed by a target cell such as a tumor cell. Whilst not wishing to be bound by theory, it is hypothesized that when the protease-activating CD45-gate CAR is functionally expressed in a cell that also functionally expresses CD45, the protease-activating CD45-gate CAR will bind to the CD45 protein. This binding is thought to hold CD45 in close proximity to the protease-activating CD45-gate CAR. This is thought to permit the CD45, a phosphatase, to dephosphorylate the CAR's intracellular signalling domain, thereby at least partially suppressing the CAR's activity. When the CAR binds to its antigen under this condition, it is not activated to the same extent that it would have been in the absence of the CD45 gate. Thus the CAR is in a "suppressed" or "reduced activity" state. The protease-activating CD45-gate CAR's extracellular linker protease cleavage site can be recognized and cleaved by a protease present in the microenvironment of one or more tumor cells that the CAR targets. In this circumstance, the protease disconnects the CAR from the CD45 recruiting domain. This permits the CAR to disassociate from the CD45, thus ending the CD45-mediated suppression of the CAR's activation. Under this condition, when the CAR binds to its antigen, it can become activated up to the same extent that it would have been if it had never comprised a CD45 gate. Thus, the partial suppression of the CAR's activity is at least partially relieved, in other words the protease-activating CD45-gate CAR is at least partially activated, when the protease-activating CD45-gate CAR-expressing cell is in the presence of its target cell. The protease-activating CD45-gate CAR thus is expected to provide more selective CAR activation than a CAR that lacks the gate.

General Techniques

The practice of the disclosure will employ, unless otherwise indicated, conventional techniques of molecular biology (including recombinant techniques), microbiology, cell biology, biochemistry and immunology, which are within the skill of the art. Such techniques are explained fully in the literature, such as, *Molecular Cloning: A Laboratory Manual*, second edition (Sambrook et al., 1989) Cold Spring Harbor Press; *Oligonucleotide Synthesis* (M. J. Gait, ed.,

1984); *Methods in Molecular Biology*, Humana Press; *Cell Biology: A Laboratory Notebook* (J. E. Cellis, ed., 1998) Academic Press; *Animal Cell Culture* (R. I. Freshney, ed., 1987); *Introduction to Cell and Tissue Culture* (J. P. Mather and P. E. Roberts, 1998) Plenum Press; *Cell and Tissue Culture: Laboratory Procedures* (A. Doyle, J. B. Griffiths, and D. G. Newell, eds., 1993-1998) J. Wiley and Sons; *Methods in Enzymology* (Academic Press, Inc.); *Handbook of Experimental Immunology* (D. M. Weir and C. C. Blackwell, eds.), Gene Transfer Vectors for Mammalian Cells (J. M. Miller and M. P. Calos, eds., 1987); *Current Protocols in Molecular Biology* (F. M. Ausubel et al., eds., 1987); *PCR: The Polymerase Chain Reaction*, (Mullis et al., eds., 1994); *Current Protocols in Immunology* (J. E. Coligan et al., eds., 1991); *Short Protocols in Molecular Biology* (Wiley and Sons, 1999); *Immunobiology* (C. A. Janeway and P. Travers, 1997); *Antibodies* (P. Finch, 1997); *Antibodies: a practical approach* (D. Catty, ed., IRL Press, 1988-1989); *Monoclonal antibodies: a practical approach* (P. Shepherd and C. Dean, eds., Oxford University Press, 2000); *Using antibodies: a laboratory manual* (E. Harlow and D. Lane (Cold Spring Harbor Laboratory Press, 1999); *The Antibodies* (M. Zanetti and J. D. Capra, eds., Harwood Academic Publishers, 1995). Gene editing techniques using TALENs, CRISPR/Cas9, and megaTAL nucleases, for example, are within the skill of the art and explained fully in the literature, such as T. Gaj et al., *Genome-Editing Technologies: Principles and Applications*, *Cold Spring Harb Perspect Biol* 2016; 8:a023754 and citations therein, C. Sommer et al, *Mol. Ther.* 27: 1126-38 (2019) (PMID: 31005597), C. Sommer et al., *Mol. Ther.* 28: 2237-51 (PMID: 32592688).

Definitions

As used herein "autologous" means that cells, a cell line, or population of cells used for treating subjects are originating from said subject.

As used herein "allogeneic" means that cells or population of cells used for treating subjects are not originating from said subject but from a donor.

As used herein, the term "endogenous" refers to any material from or produced inside an organism, cell, tissue or system.

As used herein, the term "exogenous" refers to any material introduced from or produced outside an organism, cell, tissue or system.

As used herein, "immune cell" refers to a cell of hematopoietic origin functionally involved in the initiation and/or execution of innate and/or adaptive immune response. Examples of immune cells include T cells, e.g., alpha/beta T cells and gamma/delta T cells, B cells, natural killer (NK) cells, natural killer T (NKT) cells, mast cells, and myeloid-derived phagocytes.

As used herein, the term "expression" refers to the transcription and/or translation of a particular nucleotide sequence driven by a promoter.

As used herein, "expression vector" refers to a vector comprising a recombinant polynucleotide comprising expression control sequences operatively linked to a nucleotide sequence to be expressed. Expression vectors include all those known in the art, including cosmids, plasmids (e.g., naked or contained in liposomes) and viruses (e.g., lentiviruses, retroviruses, adenoviruses, and adeno-associated viruses) that incorporate the recombinant polynucleotide.

As used herein, "operably linked" refers to the association of nucleic acid sequences on a single nucleic acid fragment so that the function of one is affected by the other. For

example, a promoter is operably linked with a coding sequence when it is capable of affecting the expression of that coding sequence (i.e., that the coding sequence is under the transcriptional control of the promoter).

As used herein, "expression control sequence" means a nucleic acid sequence that directs transcription of a nucleic acid. An expression control sequence can be a promoter, such as a constitutive or an inducible promoter, or an enhancer. The expression control sequence is operably linked to the nucleic acid sequence to be transcribed.

"Promoter" and "promoter sequence" are used interchangeably and refer to a DNA sequence capable of controlling the expression of a coding sequence or functional RNA. In general, a coding sequence is located 3' to a promoter sequence. It is understood by those skilled in the art that different promoters may direct the expression of a gene in different tissues or cell types, or at different stages of development, or in response to different environmental or physiological conditions.

In any of the vectors of the present disclosure, the vector optionally comprises a promoter disclosed herein.

A "host cell" includes an individual cell or cell culture that can be or has been a recipient for vector(s) for incorporation of polynucleotide inserts. Host cells include progeny of a single host cell, and the progeny may not necessarily be completely identical (in morphology or in genomic DNA complement) to the original parent cell due to natural, accidental, or deliberate mutation. A host cell includes cells transfected in vivo with a polynucleotide(s) of this disclosure.

The term "extracellular ligand-binding domain" as used herein refers to an oligo- or polypeptide that is capable of binding a ligand. Preferably, the domain will be capable of interacting with a cell surface molecule. For example, the extracellular ligand-binding domain may be chosen to recognize a ligand that acts as a cell surface marker on target cells associated with a particular disease state. The term "stalk domain" is used herein to refer to any oligo- or polypeptide that functions to link the transmembrane domain to the extracellular ligand-binding domain. In particular, stalk domains are used to provide more flexibility and accessibility for the extracellular ligand-binding domain.

The term "intracellular signaling domain" refers to the portion of a protein which transduces the effector signal function signal and directs the cell to perform a specialized function.

A "co-stimulatory molecule" as used herein refers to the cognate binding partner on a T cell that specifically binds with a co-stimulatory ligand, thereby mediating a co-stimulatory response by the cell, such as, but not limited to proliferation. Co-stimulatory molecules include, but are not limited to an MHC class I molecule, BTLA and Toll ligand receptor. Examples of costimulatory molecules include CD27, CD28, CD8, 4-1 BB (CD137), OX40, CD30, CD40, PD-1, ICOS, lymphocyte function-associated antigen-1 (LFA-1), CD2, CD7, LIGHT, NKG2C, B7-H3 and a ligand that specifically binds with CD83 and the like.

A "co-stimulatory ligand" refers to a molecule on an antigen presenting cell that specifically binds a cognate co-stimulatory signal molecule on a T cell, thereby providing a signal which, in addition to the primary signal provided by, for instance, binding of a TCR/CD3 complex with an MHC molecule loaded with peptide, mediates a T cell response, including, but not limited to, proliferation activation, differentiation and the like. A co-stimulatory ligand can include but is not limited to CD7, B7-1 (CD80), B7-2

(CD86), PD-L1, PD-L2, 4-1 BBL, OX40L, inducible costimulatory ligand (ICOS-L), intercellular adhesion molecule (ICAM, CD30L, CD40, CD70, CD83, HLA-G, MICA, M1 CB, HVEM, lymphotoxin R receptor, 3/TR6, ILT3, ILT4, an agonist or antibody that binds Toll ligand receptor and a ligand that specifically binds with B7-H3. A co-stimulatory ligand also encompasses, inter alia, an antibody that specifically binds with a co-stimulatory molecule present on a T cell, such as but not limited to, CD27, CD28, 4-1 BB, OX40, CD30, CD40, PD-1, ICOS, lymphocyte function-associated antigen-1 (LFA-1), CD2, CD7, LTGHT, NKG2C, B7-H3, a ligand that specifically binds with CD83.

An "antibody" is an immunoglobulin molecule capable of specific binding to a target, such as a carbohydrate, polynucleotide, lipid, polypeptide, etc., through at least one antigen recognition site, located in the variable region of the immunoglobulin molecule. As used herein, the term encompasses not only intact polyclonal or monoclonal antibodies, but also antigen-binding fragments thereof (such as Fab, Fab', F(ab')₂, and Fv), and any other modified configuration of the immunoglobulin molecule that comprises an antigen recognition site including, for example without limitation, single chain (scFv) and domain antibodies (including, for example, shark and camelid antibodies), and fusion proteins comprising an antibody. An antibody includes an antibody of any class, such as IgG, IgA, or IgM (or sub-class thereof), and the antibody need not be of any particular class. Depending on the antibody amino acid sequence of the constant region of its heavy chains, immunoglobulins can be assigned to different classes. There are five major classes of immunoglobulins: IgA, IgD, IgE, IgG, and IgM, and several of these may be further divided into subclasses (isotypes), e.g., IgG1, IgG2, IgG3, IgG4, IgA1 and IgA2. The heavy-chain constant regions that correspond to the different classes of immunoglobulins are called alpha, delta, epsilon, gamma, and mu, respectively. The subunit structures and three-dimensional configurations of different classes of immunoglobulins are well known.

The term "antigen-binding fragment" or "antigen binding portion" of an antibody, as used herein, refers to one or more fragments of an intact antibody that retain the ability to specifically bind to a given antigen. Antigen binding functions of an antibody can be performed by fragments of an intact antibody. Examples of binding fragments encompassed within the term "antigen binding fragment" of an antibody include Fab; Fab'; F(ab')₂; an Fd fragment consisting of the V_H and CH1 domains; an Fv fragment consisting of the V_L and V_H domains of a single arm of an antibody; a single domain antibody (dAb) fragment (Ward et al., Nature 341:544-546, 1989), and an isolated complementarity determining region (CDR).

An antibody, an antibody conjugate, or a polypeptide that "specifically binds" to a target is a term well understood in the art, and methods to determine such specific binding are also well known in the art. A molecule is said to exhibit "specific binding" if it reacts or associates more frequently, more rapidly, with greater duration and/or with greater affinity with a particular cell or substance than it does with alternative cells or substances. An antibody "specifically binds" to a target if it binds with greater affinity, avidity, more readily, and/or with greater duration than it binds to other substances. It is also understood that by reading this definition, for example, an antibody (or moiety or epitope or mimotope) that specifically binds to a first target may or may not specifically bind to a second target. As such, "specific binding" does not necessarily require (although it can include) exclusive binding.

A “variable region” of an antibody refers to the variable region of the antibody light chain or the variable region of the antibody heavy chain, either alone or in combination. As known in the art, the variable regions of the heavy and light chain each consist of four framework regions (FR) connected by three complementarity determining regions (CDRs) also known as hypervariable regions. The CDRs in each chain are held together in close proximity by the FRs and, with the CDRs from the other chain, contribute to the formation of the antigen binding site of antibodies. There are at least two techniques for determining CDRs: (1) an approach based on cross-species sequence variability (i.e., Kabat et al. Sequences of Proteins of Immunological Interest, (5th ed., 1991, National Institutes of Health, Bethesda MD)); and (2) an approach based on crystallographic studies of antigen-antibody complexes (Al-lazikani et al., 1997, J. Molec. Biol. 273:927-948). As used herein, a CDR may refer to CDRs defined by either approach or by a combination of both approaches.

A “CDR” of a variable domain are amino acid residues within the variable region that are identified in accordance with the definitions of the Kabat, Chothia, the accumulation of both Kabat and Chothia, AbM, contact, and/or conformational definitions or any method of CDR determination well known in the art. Antibody CDRs may be identified as the hypervariable regions originally defined by Kabat et al. See, e.g., Kabat et al., 1992, Sequences of Proteins of Immunological Interest, 5th ed., Public Health Service, NIH, Washington D.C. The positions of the CDRs may also be identified as the structural loop structures originally described by Chothia and others. See, e.g., Chothia et al., Nature 342:877-883, 1989. Other approaches to CDR identification include the “AbM definition,” which is a compromise between Kabat and Chothia and is derived using Oxford Molecular’s AbM antibody modeling software (now Accelrys®), or the “contact definition” of CDRs based on observed antigen contacts, set forth in MacCallum et al., J. Mol. Biol., 262:732-745, 1996. In another approach, referred to herein as the “conformational definition” of CDRs, the positions of the CDRs may be identified as the residues that make enthalpic contributions to antigen binding. See, e.g., Makabe et al., Journal of Biological Chemistry, 283:1 156-1 166, 2008. Still other CDR boundary definitions may not strictly follow one of the above approaches, but will nonetheless overlap with at least a portion of the Kabat CDRs, although they may be shortened or lengthened in light of prediction or experimental findings that particular residues or groups of residues or even entire CDRs do not significantly impact antigen binding. As used herein, a CDR may refer to CDRs defined by any approach known in the art, including combinations of approaches. The methods used herein may utilize CDRs defined according to any of these approaches. For any given embodiment containing more than one CDR, the CDRs may be defined in accordance with any of Kabat, Chothia, extended, AbM, contact, and/or conformational definitions.

Antibodies of the disclosure can be produced using techniques well known in the art, e.g., recombinant technologies, phage display technologies, synthetic technologies or combinations of such technologies or other technologies readily known in the art (see, for example, Jayasena, S. D., Clin. Chem., 45: 1628-50, 1999 and Fellouse, F. A., et al, J. Mol. Biol., 373(4):924-40, 2007).

As known in the art, “polynucleotide,” or “nucleic acid,” as used interchangeably herein, refer to chains of nucleotides of any length, and include DNA and RNA. The nucleotides can be deoxyribonucleotides, ribonucleotides, modified

nucleotides or bases, and/or their analogs, or any substrate that can be incorporated into a chain by DNA or RNA polymerase. A polynucleotide may comprise modified nucleotides, such as methylated nucleotides and their analogs. If present, modification to the nucleotide structure may be imparted before or after assembly of the chain. The sequence of nucleotides may be interrupted by non-nucleotide components. A polynucleotide may be further modified after polymerization, such as by conjugation with a labeling component. Other types of modifications include, for example, “caps”, substitution of one or more of the naturally occurring nucleotides with an analog, internucleotide modifications such as, for example, those with uncharged linkages (e.g., methyl phosphonates, phosphotriesters, phosphoramidates, carbamates, etc.) and with charged linkages (e.g., phosphorothioates, phosphorodithioates, etc.), those containing pendant moieties, such as, for example, proteins (e.g., nucleases, toxins, antibodies, signal peptides, poly-L-lysine, etc.), those with intercalators (e.g., acridine, psoralen, etc.), those containing chelators (e.g., metals, radioactive metals, boron, oxidative metals, etc.), those containing alkylators, those with modified linkages (e.g., alpha anomeric nucleic acids, etc.), as well as unmodified forms of the polynucleotide(s). Further, any of the hydroxyl groups ordinarily present in the sugars may be replaced, for example, by phosphonate groups, phosphate groups, protected by standard protecting groups, or activated to prepare additional linkages to additional nucleotides, or may be conjugated to solid supports. The 5' and 3' terminal OH can be phosphorylated or substituted with amines or organic capping group moieties of from 1 to 20 carbon atoms. Other hydroxyls may also be derivatized to standard protecting groups. Polynucleotides can also contain analogous forms of ribose or deoxyribose sugars that are generally known in the art, including, for example, 2'-O-methyl-, 2'-O-allyl-, 2'-fluoro- or 2'-azido-ribose, carbocyclic sugar analogs, alpha- or beta-anomeric sugars, epimeric sugars such as arabinose, xyloses or lyxoses, pyranose sugars, furanose sugars, sedoheptuloses, acyclic analogs and abasic nucleoside analogs such as methyl riboside. One or more phosphodiester linkages may be replaced by alternative linking groups. These alternative linking groups include, but are not limited to, embodiments wherein phosphate is replaced by P(O)S(“thioate”), P(S)S(“dithioate”), (O)NR2(“amidate”), P(O)R, P(O)OR', CO or CH2(“formacetal”), in which each R or R' is independently H or substituted or unsubstituted alkyl (1-20 C) optionally containing an ether (—O—) linkage, aryl, alkenyl, cycloalkyl, cycloalkenyl or araldyl. Not all linkages in a polynucleotide need be identical. The preceding description applies to all polynucleotides referred to herein, including RNA and DNA.

As used herein, “transfection” refers to the uptake of exogenous or heterologous RNA or DNA by a cell. A cell has been “transfected” by exogenous or heterologous RNA or DNA when such RNA or DNA has been introduced inside the cell. A cell has been “transformed” by exogenous or heterologous RNA or DNA when the transfected RNA or DNA effects a phenotypic change. The transforming RNA or DNA can be integrated (covalently linked) into chromosomal DNA making up the genome of the cell.

As used herein, “transformation” refers to the transfer of a nucleic acid fragment into the genome of a host organism, resulting in genetically stable inheritance. Host organisms containing the transformed nucleic acid fragments are referred to as “transgenic” or “recombinant” or “transformed” organisms.

As used herein, "substantially pure" refers to material which is at least 50% pure (i.e., free from contaminants), more preferably, at least 90% pure, more preferably, at least 95% pure, yet more preferably, at least 98% pure, and most preferably, at least 99% pure. The term "compete", as used herein with regard to an antibody, means that a first antibody, or an antigen binding fragment (or portion) thereof, binds to an epitope or mimotope in a manner sufficiently similar to the binding of a second antibody, or an antigen binding portion thereof, such that the result of binding of the first antibody with its cognate epitope or mimotope is detectably decreased in the presence of the second antibody compared to the binding of the first antibody in the absence of the second antibody. The alternative, where the binding of the second antibody to its epitope or mimotope is also detectably decreased in the presence of the first antibody, can, but need not be the case. That is, a first antibody can inhibit the binding of a second antibody to its epitope or mimotope without that second antibody inhibiting the binding of the first antibody to its respective epitope or mimotope. However, where each antibody detectably inhibits the binding of the other antibody with its cognate epitope or mimotope or ligand, whether to the same, greater, or lesser extent, the antibodies are said to "cross-compete" with each other for binding of their respective epitope(s) or mimotope(s). Both competing and cross-competing antibodies are encompassed by the disclosure. Regardless of the mechanism by which such competition or cross-competition occurs (e.g., steric hindrance, conformational change, or binding to a common epitope or mimotope, or portion thereof), the skilled artisan would appreciate, based upon the teachings provided herein, that such competing and/or cross-competing antibodies are encompassed and can be useful for the methods disclosed herein.

As used herein, "treatment" is an approach for obtaining beneficial or desired clinical results. For purposes of this disclosure, beneficial or desired clinical results include, but are not limited to, one or more of the following: reducing the proliferation of (or destroying) neoplastic or cancerous cells, inhibiting metastasis of neoplastic cells, shrinking or decreasing the size of tumor, remission of a disease (e.g., cancer), decreasing symptoms resulting from a disease (e.g., cancer), increasing the quality of life of those suffering from a disease (e.g., cancer), decreasing the dose of other medications required to treat a disease (e.g., cancer), delaying the progression of a disease (e.g., cancer), curing a disease (e.g., cancer), and/or prolong survival of subjects having a disease (e.g., cancer).

"Ameliorating" means a lessening or improvement of one or more symptoms as compared to not administering a treatment. "Ameliorating" also includes shortening or reduction in duration of a symptom. As used herein, an "effective dosage" or "effective amount" of drug, compound, or pharmaceutical composition is an amount sufficient to effect any one or more beneficial or desired results. For prophylactic use, beneficial or desired results include eliminating or reducing the risk, lessening the severity, or delaying the onset of the disease, including biochemical, histological and/or behavioral symptoms of the disease, its complications and intermediate pathological phenotypes presenting during development of the disease. For therapeutic use, beneficial or desired results include clinical results such as reducing incidence or amelioration of one or more symptoms of various diseases or conditions (such as for example cancer), decreasing the dose of other medications required to treat the disease, enhancing the effect of another medication, and/or delaying the progression of the disease. An effective

dosage can be administered in one or more administrations. For purposes of this disclosure, an effective dosage of drug, compound, or pharmaceutical composition is an amount sufficient to accomplish prophylactic or therapeutic treatment either directly or indirectly. As is understood in the clinical context, an effective dosage of a drug, compound, or pharmaceutical composition may or may not be achieved in conjunction with another drug, compound, or pharmaceutical composition. Thus, an "effective dosage" may be considered in the context of administering one or more therapeutic agents, and a single agent may be considered to be given in an effective amount if, in conjunction with one or more other agents, a desirable result may be or is achieved.

As used herein, a "subject" is any mammal, e.g. a human, or a monkey. Mammals include, but are not limited to, farm animals, sport animals, pets, primates, horses, dogs, cats, mice and rats. In an exemplary embodiment, the subject is a human. In an exemplary embodiment, the subject is a monkey, e.g. a cynomolgus monkey.

As used herein, "vector" means a construct, which is capable of delivering, and, preferably, expressing, one or more gene(s) or sequence(s) of interest in a host cell. Examples of vectors include, but are not limited to, viral vectors, naked DNA or RNA expression vectors, plasmid, cosmid or phage vectors, DNA or RNA expression vectors associated with cationic condensing agents, DNA or RNA expression vectors encapsulated in liposomes, and certain eukaryotic cells, such as producer cells.

As used herein, "pharmaceutically acceptable carrier" or "pharmaceutically acceptable excipient" includes any material which, when combined with an active ingredient, allows the ingredient to retain biological activity and is non-reactive with the subject's immune system. Examples include, but are not limited to, any of the standard pharmaceutical carriers such as a phosphate buffered saline solution, water, emulsions such as oil/water emulsion, and various types of wetting agents. Preferred diluents for aerosol or parenteral administration are phosphate buffered saline (PBS) or normal (0.9%) saline. Compositions of the disclosure comprising such carriers are formulated by well-known conventional methods (see, for example, Remington's Pharmaceutical Sciences, 18th edition, A. Gennaro, ed., Mack Publishing Co., Easton, PA, 1990; and Remington, The Science and Practice of Pharmacy 21 st Ed. Mack Publishing, 2005).

As used herein, "alloreactivity" refers to the ability of T cells to recognize MHC complexes that were not encountered during thymic development. Alloreactivity manifests itself clinically as host versus graft rejection and graft versus host disease.

Reference to "about" a value or parameter herein includes (and describes) embodiments that are directed to that value or parameter per se. For example, description referring to "about X" includes description of "X." Numeric ranges are inclusive of the numbers defining the range.

It is understood that wherever embodiments are described herein with the language "comprising," otherwise analogous embodiments described in terms of "consisting of" and/or "consisting essentially of" are also provided.

Where aspects or embodiments of the disclosure are described in terms of a Markush group or other grouping of alternatives, the disclosure encompasses not only the entire group listed as a whole, but each member of the group individually and all possible subgroups of the main group, but also the main group absent one or more of the group

members. The disclosure also envisages the explicit exclusion of one or more of any of the group members in the claimed disclosure.

Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. In case of conflict, the present specification, including definitions, will control. Throughout this specification and claims, the word “comprise,” or variations such as “comprises” or “comprising” will be understood to imply the inclusion of a stated integer or group of integers but not the exclusion of any other integer or group of integers. Unless otherwise required by context, singular terms shall include pluralities and plural terms shall include the singular.

Exemplary methods and materials are described herein, although methods and materials similar or equivalent to those described herein can also be used in the practice or testing of the disclosure. The materials, methods, and examples are illustrative only and not intended to be limiting.

An “antigen binding protein” comprises one or more antigen binding domains. An “antigen binding domain” as used herein means any polypeptide that binds a specified target antigen. In some embodiments, the antigen binding domain binds to an antigen on a tumor cell. In some embodiments, the antigen binding domain binds to an antigen on a cell involved in a hyperproliferative disease or to a viral or bacterial antigen.

Antigen binding domains include, but are not limited to, antibody binding regions that are immunologically functional fragments. The term “immunologically functional fragment” (or “fragment”) of an antigen binding domain is a species of antigen binding domain comprising a portion (regardless of how that portion is obtained or synthesized) of an antibody that lacks at least some of the amino acids present in a full-length chain, but which is still capable of specifically binding to a target antigen. Such fragments are biologically active in that they bind to the target antigen and can compete with other antigen binding domains, including intact antibodies, for binding to a given epitope or mimotope.

Immunologically functional immunoglobulin fragments include, but are not limited to, scFv fragments, Fab fragments (Fab', F(ab')₂, and the like), one or more complementarity determining regions (“CDRs”), a diabody (heavy chain variable domain on the same polypeptide as a light chain variable domain, connected via a short peptide linker that is too short to permit pairing between the two domains on the same chain), domain antibodies, bivalent antigen binding domains (comprises two antigen binding sites), multispecific antigen binding domains, and single-chain antibodies. These fragments can be derived from any mammalian source, including but not limited to human, mouse, rat, camelid or rabbit. As will be appreciated by one of skill in the art, an antigen binding domain can include non-protein components.

The variable regions typically exhibit the same general structure of relatively conserved framework regions (FR) joined by the 3 hypervariable regions (CDRs). The CDRs from the two chains of each pair typically are aligned by the framework regions, which can enable binding to a specific epitope or mimotope. From N-terminal to C-terminal, both light and heavy chain variable regions typically comprise the domains FR1, CDR1, FR2, CDR2, FR3, CDR3 and FR4. By convention, CDR regions in the heavy chain are typically

referred to as HC CDR1, CDR2, and CDR3. The CDR regions in the light chain are typically referred to as LC CDR1, CDR2, and CDR3.

In some embodiments, antigen binding domains comprise one or more complementarity binding regions (CDRs) present in the full-length light or heavy chain of an antibody, and in some embodiments comprise a single heavy chain and/or light chain or portion thereof. These fragments can be produced by recombinant DNA techniques or can be produced by enzymatic or chemical cleavage of antigen binding domains, including intact antibodies.

In some embodiments, the antigen binding domain is an antibody or fragment thereof, including one or more of the complementarity determining regions (CDRs) thereof. In some embodiments, the antigen binding domain is a single chain variable fragment (scFv), comprising light chain CDRs CDR1, CDR2 and CDR3, and heavy chain CDRs CDR1, CDR2 and CDR3.

The assignment of amino acids to each of the framework, CDR, and variable domains is typically in accordance with numbering schemes of Kabat numbering (see, e.g., Kabat et al. in *Sequences of Proteins of Immunological Interest*, 5th Ed., NIH Publication 91-3242, Bethesda Md. 1991), Chothia numbering (see, e.g., Chothia & Lesk, (1987), *J Mol Biol* 196: 901-917; Al-Lazikani et al., (1997) *J Mol Biol* 273: 927-948; Chothia et al., (1992) *J Mol Biol* 227: 799-817; Tramontano et al., (1990) *J Mol Biol* 215(1): 175-82; and U.S. Pat. No. 7,709,226), contact numbering, or the AbM scheme (Antibody Modeling program, Oxford Molecular).

In some embodiments, the antigen binding domain is a recombinant antigen receptor. The term “recombinant antigen receptor” as used herein refers broadly to a non-naturally occurring surface receptor that comprises an extracellular antigen-binding domain or an extracellular ligand-binding domain, a transmembrane domain and an intracellular domain. In some embodiments, the recombinant antigen receptor is a chimeric antigen receptor (CAR). Chimeric antigen receptors (CARs) are well-known in the art. A CAR is a fusion protein that comprises an extracellular domain comprising an antigen recognition moiety (also referred to herein as an antigen binding domain), a transmembrane domain and an intracellular domain comprising one or more T cell activation domains (see, e.g., Eshhar et al., *Proc. Natl. Acad. Sci. USA*, 90(2): 720-724 (1993), and Sadelain et al., *Curr. Opin. Immunol.*, 21(2): 215-223 (2009)). A hinge region or domain typically is situated between the CAR's antigen recognition domain and the transmembrane domain.

In some embodiments, the intracellular domain of a recombinant antigen receptor comprises a co-stimulatory domain and/or an ITAM-containing domain. In some embodiments, the intracellular domain of a recombinant antigen receptor comprises an intracellular protein or a functional variant thereof (e.g., truncation(s), insertion(s), deletion(s) or substitution(s)).

The term “extracellular ligand-binding domain” or “extracellular antigen-binding domain” as used herein refers to a polypeptide that is capable of binding a ligand or an antigen or capable of interacting with a cell surface molecule, such as a ligand or a surface antigen. For example, the extracellular ligand-binding or antigen-binding domain may be chosen to recognize a ligand that acts as a cell surface marker on target cells associated with a particular disease state, e.g., a tumor-specific antigen. In some embodiments, the antigen-binding domain comprises an antibody, or an antigen binding fragment or an antigen binding portion of an antibody. In some embodiments, the antigen binding domain

comprises an Fv or scFv, an Fab or scFab, an F(ab')₂ or a scF(ab')₂, an Fd, a monobody, an affibody, a camelid antibody, a VHH antibody, a single domain antibody, or a darpin. In some embodiments, the ligand-binding domain comprises a partner of a binding pair, such as a ligand that binds to a surface receptor, or an ectodomain of a surface receptor that binds to a ligand.

The terms "stalk domain" and "hinge domain" are used interchangeably herein to refer to any polypeptide that functions to link the transmembrane domain to the extracellular ligand-binding domain. In particular, stalk domains are used to provide more flexibility and accessibility for the extracellular ligand-binding domain.

The term "intracellular signaling domain" refers to the portion of a protein which transduces the effector signal function signal and directs the cell to perform a specialized function.

Vectors

Expression vectors and administration of polynucleotide compositions are further described herein.

In another aspect, the disclosure provides a method of making any of the polynucleotides described herein.

Polynucleotides complementary to any such sequences are also encompassed by the disclosure. Polynucleotides may be single-stranded (coding or antisense) or double-stranded, and may be DNA (genomic, cDNA or synthetic) or RNA molecules. RNA molecules include HnRNA molecules, which contain introns and correspond to a DNA molecule in a one-to-one manner, and mRNA molecules, which do not contain introns. Additional coding or non-coding sequences may, but need not, be present within a polynucleotide of the disclosure, and a polynucleotide may, but need not, be linked to other molecules and/or support materials.

Polynucleotides may comprise a native sequence (i.e., an endogenous sequence that encodes an antibody or a portion thereof) or may comprise a variant of such a sequence. Polynucleotide variants contain one or more substitutions, additions, deletions and/or insertions such that the immunoreactivity of the encoded polypeptide is not diminished, relative to a native immunoreactive molecule. The effect on the immunoreactivity of the encoded polypeptide may generally be assessed as described herein. Variants preferably exhibit at least about 70% identity, more preferably, at least about 80% identity, yet more preferably, at least about 90% identity, and most preferably, at least about 95% identity to a polynucleotide sequence that encodes a native antibody or a portion thereof. Two polynucleotide or polypeptide sequences are said to be "identical" if the sequence of nucleotides or amino acids in the two sequences is the same when aligned for maximum correspondence as described below. Comparisons between two sequences are typically performed by comparing the sequences over a comparison window to identify and compare local regions of sequence similarity. A "comparison window" as used herein, refers to a segment of at least about 20 contiguous positions, usually 30 to about 75, or 40 to about 50, in which a sequence may be compared to a reference sequence of the same number of contiguous positions after the two sequences are optimally aligned.

Optimal alignment of sequences for comparison may be conducted using the Megalign program in the Lasergene suite of bioinformatics software (DNASTAR, Inc., Madison, WI), using default parameters. This program embodies several alignment schemes described in the following references: Dayhoff, M. O., 1978, A model of evolutionary change in proteins—Matrices for detecting distant relation-

ships. In Dayhoff, M. O. (ed.) Atlas of Protein Sequence and Structure, National Biomedical Research Foundation, Washington DC Vol. 5, Suppl. 3, pp. 345-358; Hein J., 1990, Unified Approach to Alignment and Phylogenies pp. 626-645; Methods in Enzymology vol. 183, Academic Press, Inc., San Diego, CA; Higgins, D. G. and Sharp, P. M., 1989, CABIOS 5:151-153; Myers, E. W. and Muller W., 1988, CABIOS 4:11-17; Robinson, E. D., 1971, Comb. Theor. 1 1:105; Santou, N., Nes, M., 1987, Mol. Biol. Evol. 4:406-425; Sneath, P. H. A. and Sokal, R. R., 1973, Numerical Taxonomy the Principles and Practice of Numerical Taxonomy, Freeman Press, San Francisco, CA; Wilbur, W. J. and Lipman, D. J., 1983, Proc. Natl. Acad. Sci. USA 80:726-730.

Preferably, the "percentage of sequence identity" is determined by comparing two optimally aligned sequences over a window of comparison of at least 20 positions, wherein the portion of the polynucleotide or polypeptide sequence in the comparison window may comprise additions or deletions (i.e., gaps) of 20 percent or less, usually 5 to 15 percent, or 10 to 12 percent, as compared to the reference sequences (which does not comprise additions or deletions) for optimal alignment of the two sequences. The percentage is calculated by determining the number of positions at which the identical nucleic acid bases or amino acid residue occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the reference sequence (i.e. the window size) and multiplying the results by 100 to yield the percentage of sequence identity.

Variants may also, or alternatively, be substantially homologous to a native gene, or a portion or complement thereof. Such polynucleotide variants are capable of hybridizing under moderately stringent conditions to a naturally occurring DNA sequence encoding a native antibody (or a complementary sequence).

Suitable "moderately stringent conditions" include pre-washing in a solution of 5×SSC, 0.5% SDS, 1.0 mM EDTA (pH 8.0); hybridizing at 50° C.-65° C., 5×SSC, overnight; followed by washing twice at 65° C. for 20 minutes with each of 2×, 0.5× and 0.2×SSC containing 0.1% SDS.

As used herein, "highly stringent conditions" or "high stringency conditions" are those that: (1) employ low ionic strength and high temperature for washing, for example 0.015 M sodium chloride/0.0015 M sodium citrate/0.1% sodium dodecyl sulfate at 50° C.; (2) employ during hybridization a denaturing agent, such as formamide, for example, 50% (v/v) formamide with 0.1% bovine serum albumin/0.1% Ficoll/0.1% polyvinylpyrrolidone/50 mM sodium phosphate buffer at pH 6.5 with 750 mM sodium chloride, 75 mM sodium citrate at 42° C.; or (3) employ 50% formamide, 5×SSC (0.75 M NaCl, 0.075 M sodium citrate), 50 mM sodium phosphate (pH 6.8), 0.1% sodium pyrophosphate, 5×Denhardt's solution, sonicated salmon sperm DNA (50 g/m), 0.1% SDS, and 10% dextran sulfate at 42° C., with washes at 42° C. in 0.2×SSC (sodium chloride/sodium citrate) and 50% formamide at 55° C., followed by a high-stringency wash consisting of 0.1×SSC containing EDTA at 55° C. The skilled artisan will recognize how to adjust the temperature, ionic strength, etc. as necessary to accommodate factors such as probe length and the like.

It will be appreciated by those of ordinary skill in the art that, as a result of the degeneracy of the genetic code, there are many nucleotide sequences that encode a polypeptide as described herein. Some of these polynucleotides bear minimal homology to the nucleotide sequence of any native gene. Nonetheless, polynucleotides that vary due to differences in codon usage are specifically contemplated by the

disclosure. Further, alleles of the genes comprising the polynucleotide sequences provided herein are within the scope of the disclosure. Alleles are endogenous genes that are altered as a result of one or more mutations, such as deletions, additions and/or substitutions of nucleotides. The resulting mRNA and protein may, but need not, have an altered structure or function. Alleles may be identified using standard techniques (such as hybridization, amplification and/or database sequence comparison).

The polynucleotides of this disclosure can be obtained using chemical synthesis, recombinant methods, or PCR. Methods of chemical polynucleotide synthesis are well known in the art and need not be described in detail herein. One of skill in the art can use the sequences provided herein and a commercial DNA synthesizer to produce a desired DNA sequence.

For preparing polynucleotides using recombinant methods, a polynucleotide comprising a desired sequence can be inserted into a suitable vector, and the vector in turn can be introduced into a suitable host cell for replication and amplification, as further discussed herein. Polynucleotides may be inserted into host cells by any means known in the art. Cells are transformed by introducing an exogenous polynucleotide by direct uptake, endocytosis, transfection, F-mating or electroporation. Once introduced, the exogenous polynucleotide can be maintained within the cell as a non-integrated vector (such as a plasmid) or integrated into the host cell genome. The polynucleotide so amplified can be isolated from the host cell by methods well known within the art. See, e.g., Sambrook et al., 1989.

Alternatively, PCR allows reproduction of DNA sequences. PCR technology is well known in the art and is described in U.S. Pat. Nos. 4,683,195, 4,800,159, 4,754,065 and 4,683,202, as well as PCR: The Polymerase Chain Reaction, Mullis et al. eds., Birkauwer Press, Boston, 1994.

RNA can be obtained by using the isolated DNA in an appropriate vector and inserting it into a suitable host cell. When the cell replicates and the DNA is transcribed into RNA, the RNA can then be isolated using methods well known to those of skill in the art, as set forth in Sambrook et al., 1989, *supra*, for example.

Suitable cloning vectors may be constructed according to standard techniques, or may be selected from a large number of cloning vectors available in the art. While the cloning vector selected may vary according to the host cell intended to be used, useful cloning vectors will generally have the ability to self-replicate, may possess a single target for a particular restriction endonuclease, and/or may carry genes for a marker that can be used in selecting clones containing the vector. Suitable examples include plasmids and bacterial viruses, e.g., pUC18, pUC19, Bluescript (e.g., pBS SK+) and its derivatives, mp18, mp19, pBR322, pMB9, ColE1, pCR1, RP4, phage DNAs, and shuttle vectors such as pSA3 and pAT28. These and many other cloning vectors are available from commercial vendors such as BioRad, Stratagene, and Invitrogen.

Expression vectors generally are replicable polynucleotide constructs that contain a polynucleotide according to the disclosure. It is implied that an expression vector must be replicable in the host cells either as episomes or as an integral part of the chromosomal DNA. Suitable expression vectors include but are not limited to plasmids, viral vectors, including adenoviruses, adeno-associated viruses, retroviruses, cosmids, and expression vector(s) disclosed in PCT Publication No. WO 87/04462. Vector components may generally include, but are not limited to, one or more of the following: a signal sequence; an origin of replication; one or

more marker genes; suitable transcriptional controlling elements (such as promoters, enhancers and terminator). For expression (i.e., translation), one or more translational controlling elements are also usually required, such as ribosome binding sites, translation initiation sites, and stop codons.

The vectors containing the polynucleotides of interest can be introduced into the host cell by any of a number of appropriate means, including electroporation, transfection employing calcium chloride, rubidium chloride, calcium phosphate, DEAE-dextran, or other substances; microprojectile bombardment; lipofection; and infection (e.g., where the vector is an infectious agent such as vaccinia virus). The choice of introducing vectors or polynucleotides will often depend on features of the host cell.

A polynucleotide encoding a protease-activating CD45-gate CAR disclosed herein, its extracellular domain, or another fragment of the protease-activating CD45-gate CAR may exist in an expression cassette or expression vector (e.g., a plasmid for introduction into a bacterial host cell, or a viral vector such as a baculovirus vector for transfection of an insect host cell, or a plasmid or viral vector such as a lentivirus for transfection of a mammalian host cell). In some embodiments, a polynucleotide or vector can include a nucleic acid sequence encoding ribosomal skip sequences such as, for example without limitation, a sequence encoding a 2A peptide. 2A peptides, which were identified in the Aphthovirus subgroup of picornaviruses, cause a ribosomal "skip" from one codon to the next without the formation of a peptide bond between the two amino acids encoded by the codons (see (Donnelly and Elliott 2001; Atkins, Wills et al. 2007; Doronina, Wu et al. 2008)). By "codon" is meant three nucleotides on an mRNA (or on the sense strand of a DNA molecule) that are translated by a ribosome into one amino acid residue. Thus, two polypeptides can be synthesized from a single, contiguous open reading frame within an mRNA when the polypeptides are separated by a 2A oligopeptide sequence that is in frame. Such ribosomal skip mechanisms are well known in the art and are known to be used by several vectors for the expression of several proteins encoded by a single messenger RNA.

To direct transmembrane polypeptides into the secretory pathway of a host cell, in some embodiments, a secretory signal sequence (also known as a leader sequence, prepro sequence or pre sequence) is provided in a polynucleotide sequence or vector sequence. The secretory signal sequence is operably linked to the transmembrane nucleic acid sequence, i.e., the two sequences are joined in the correct reading frame and positioned to direct the newly synthesized polypeptide into the secretory pathway of the host cell. Secretory signal sequences are commonly positioned 5' to the nucleic acid sequence encoding the polypeptide of interest, although certain secretory signal sequences may be positioned elsewhere in the nucleic acid sequence of interest (see, e.g., Welch et al., U.S. Pat. No. 5,037,743; Holland et al., U.S. Pat. No. 5,143,830). Those skilled in the art will recognize that, in view of the degeneracy of the genetic code, considerable sequence variation is possible among these polynucleotide molecules. In some embodiments, nucleic acid sequences of the disclosure are codon-optimized for expression in mammalian cells, preferably for expression in human cells. Codon-optimization refers to the exchange in a sequence of interest of codons that are generally rare in highly expressed genes of a given species for codons that are generally frequent in highly expressed genes of such species, such codons encoding the same amino acids as the codons that are being exchanged.

Methods of preparing immune cells for use in immunotherapy are provided herein. In some embodiments, the methods comprise introducing a protease-activating CD45-gate CAR into immune cells, and expanding the cells. In some embodiments, the disclosure relates to a method of engineering an immune cell comprising: providing an immune cell and expressing at the surface of the cell at least one protease-activating CD45-gate CAR. In some embodiments, the method comprises: transfecting the cell with at least one polynucleotide encoding a protease-activating CD45-gate CAR, and expressing the at least one polynucleotide in the cell.

In some embodiments, the polynucleotide encoding the protease-activating CD45-gate CAR is present in an expression vector for stable expression in the cells. In some embodiments, the polynucleotide is present in a viral vector for stable expression in the cells. In some embodiments, the viral vector may be, for example, a lentiviral vector or adenoviral vector.

In some embodiments, polynucleotides encoding polypeptides according to the present disclosure can be mRNA which is introduced directly into the cells, for example by electroporation. In some embodiments, cytoPulse technology can be used to transiently permeabilize living cells for delivery of material into the cells. Parameters can be modified in order to determine conditions for high transfection efficiency with minimal mortality.

Also provided herein are methods of transfecting an immune cell e.g. a T cell. In some embodiments, the method comprises: contacting a T cell with RNA and applying to the T cell an agile pulse sequence consisting of: (a) an electrical pulse with a voltage range from about 2250 to 3000 V per centimeter; (b) a pulse width of 0.1 ms; (c) a pulse interval of about 0.2 to 10 ms between the electrical pulses of step (a) and (b); (d) an electrical pulse with a voltage range from about 2250 to 3000 V per centimeter with a pulse width of about 100 ms and a pulse interval of about 100 ms between the electrical pulse of step (b) and the first electrical pulse of step (c); and (e) four electrical pulses with a voltage of about 325 V with a pulse width of about 0.2 ms and a pulse interval of 2 ms between each of 4 electrical pulses. In some embodiments, a method of transfecting a T cell comprises contacting said T cell with RNA and applying to the T cell an agile pulse sequence comprising: (a) an electrical pulse with a voltage of about 2250, 2300, 2350, 2400, 2450, 2500, 2550, 2400, 2450, 2500, 2600, 2700, 2800, 2900 or 3000V per centimeter; (b) a pulse width of 0.1 ms; (c) a pulse interval of about 0.2, 0.5, 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 ms between the electrical pulses of step (a) and (b); (d) one electrical pulse with a voltage range from about 2250 to 3000 V per centimeter, e.g. of 2250, 2300, 2350, 2400, 2450, 2500, 2550, 2400, 2450, 2500, 2600, 2700, 2800, 2900 or 3000V per centimeter with a pulse width of 100 ms and a pulse interval of 100 ms between the electrical pulse of step (b) and the first electrical pulse of step (c); and (e) 4 electrical pulses with a voltage of about 325 V with a pulse width of about 0.2 ms and a pulse interval of about 2 ms between each of 4 electrical pulses. Any values included in the value range described above are disclosed in the present application. Electroporation medium can be any suitable medium known in the art. In some embodiments, the electroporation medium has conductivity in a range spanning about 0.01 to about 1.0 milliSiemens.

In some embodiments, the method can further comprise a step of genetically modifying a cell by inactivating or reducing the expression level of at least one gene expressing, for example without limitation, a component of the TCR, a

target for an immunosuppressive agent, an HLA gene, and/or an immune checkpoint protein such as, for example, PDCD1 or CTLA-4. By inactivating a gene it is intended that the gene of interest is not expressed in a functional protein form. In some embodiments, the gene to be inactivated is selected from the group consisting of, for example without limitation, TCR α , TCR β , β 2-microglobulin ("02m"), CD52, GR, deoxycytidine kinase (DCK), PD-1, and CTLA-4. In some embodiments the method comprises inactivating or reducing the expression level of one or more genes by introducing into the cells a rare-cutting endonuclease able to selectively inactivate a gene by selective DNA cleavage. In some embodiments the rare-cutting endonuclease can be, for example, a transcription activator-like effector nuclease (TALE-nuclease or TALEN), a megaTAL nuclease or a Cas9 endonuclease.

In another aspect, a step of genetically modifying immune cells e.g. T cells can comprise: modifying immune cells e.g. T cells by inactivating at least one gene expressing a target for an immunosuppressive agent, and; expanding the cells, optionally in the presence of the immunosuppressive agent. An immunosuppressive agent is an agent that suppresses immune function by one of several mechanisms of action. An immunosuppressive agent can diminish the extent and/or voracity of an immune response. Non-limiting examples of immunosuppressive agents include calcineurin inhibitors, targets of rapamycin, interleukin-2 α -chain blockers, inhibitors of inosine monophosphate dehydrogenase, inhibitors of dihydrofolic acid reductase, corticosteroids, and immunosuppressive antimetabolites. Some cytotoxic immunosuppressants act by inhibiting DNA synthesis. Others may act through activation of T cells or by inhibiting the activation of helper cells. The methods according to the disclosure allow conferring immunosuppressive resistance to e.g. T cells for immunotherapy by inactivating the target of the immunosuppressive agent in the T cells. As non-limiting examples, targets for an immunosuppressive agent can be a receptor for an immunosuppressive agent such as for example without limitation CD52, glucocorticoid receptor (GR), FKBP family gene members, and cyclophilin family gene members.

Compositions and methods for expressing a protease-activating CD45-gate CAR are provided herein. Also provided are uses of such compositions and methods for improving the functional activities of immune cells e.g. T cells, such as CAR-T cells. The methods and compositions provided herein are useful for improving activation specificity and therapeutic efficacy of immune cells e.g. T cells such as CAR-T cells.

Immune cells e.g. T cells provided herein express a protease-activating CD45-gate CAR as disclosed herein. Advantageously, the immune cells provided herein exhibit improved in vivo activation specificity relative to cells that express a non-gated CAR e.g. that express the same or comparable CAR except it lacks a functional protease cleavage site.

Protease-Activating CD45-Gate CAR

In one aspect, provided herein is a protease-activating CD45-gate chimeric antigen receptor (CD45-gate CAR) comprising an extracellular domain, a transmembrane domain, and an intracellular domain, wherein the extracellular domain comprises:

- a CD45 recruiting domain,
- an antigen binding domain, and
- a linker connecting the carboxy terminus of the CD45 recruiting domain to the amino terminus of the antigen

binding domain, wherein the linker comprises at least one protease cleavage site that is recognized by a protease,

and further wherein the intracellular domain comprises at least one signaling domain that can be reversibly inactivated by CD45.

In some embodiments, the extracellular domain comprises a stalk domain that joins the extracellular domain to the transmembrane domain.

In some embodiments, the intracellular signaling domain comprises an activating domain. In some embodiments, the intracellular signaling domain comprises a costimulatory domain. In some embodiments, the intracellular signaling domain comprises an activating domain and a costimulatory domain. In some embodiments, the intracellular signaling domain comprises an activating domain such as an ITAM-containing domain. In some embodiments, the intracellular signaling domain comprises the CD3 zeta intracellular domain. In some embodiments, the intracellular signaling domain comprises a CD3 zeta domain that comprises the amino acid sequence of SEQ ID NO: 34 or a fragment thereof.

In some embodiments the intracellular signaling domain comprises one, two or three ITAM domains selected from the group consisting of wildtype or variants of a CD37 ITAM, a CD36 ITAM, a CD3E ITAM, a CD3 ζ 1 ITAM, a CD3 ζ 2 ITAM, and a CD3 ζ 3 ITAM, provided that, if three ITAM domains are selected that are wildtype ITAM domains, the three ITAM domains are not CD3 ζ 1, CD3 ζ 2 and CD3 ζ 3; see U.S. Provisional Patent Appl. No. 63/054,701, incorporated herein by reference in its entirety.

In some embodiments, the intracellular domain comprises or further comprises at least one costimulatory domain. In some embodiments, the at least one costimulatory domain is a signaling region of CD28, OX-40, 4-1BB/CD137, CD2, CD7, CD27, CD30, CD40, programmed death-1 (PD-1), inducible T cell costimulator (ICOS), lymphocyte function-associated antigen-1 (LFA-1 (CD11a/CD18), CD3 gamma, CD3 delta, CD3 epsilon, CD247, CD276 (B7-H3), LIGHT, (TNFSF14), NKG2C, Ig alpha (CD79a), DAP-10, Fc gamma receptor, MHC class I molecule, TNF receptor proteins, an Immunoglobulin protein, cytokine receptor, integrins, Signaling Lymphocytic Activation Molecules (SLAM proteins), activating NK cell receptors, BTLA, a Toll ligand receptor, ICAM-1, B7-H3, CDS, ICAM-1, GITR, BAFTR, LIGHT, HVEM (LIGHTR), KIRDS2, SLAMF7, NKp80 (KLRF1), NKp44, NKp30, NKp46, CD19, CD4, CD8alpha, CD8beta, IL-2R beta, IL-2R gamma, IL-7R alpha, ITGA4, VLA1, CD49a, ITGA4, CD49D, ITGA6, VLA-6, CD49f, ITGAD, CD11d, ITGAE, CD103, ITGAL, CD11a, LFA-1, ITGAM, CD11b, ITGAX, CD11c, ITGB1, CD29, ITGB2, CD 18, LFA-1, ITGB7, NKG2D, TNFR2, TRANCE/RANKL, DNAMI (CD226), SLAMF4 (CD244, 2B4), CD84, CD96 (Tactile), CEACAM1, CRT AM, Ly9 (CD229), CD 160 (BY55), PSGL1, CD100 (SEMA4D), CD69, SLAMF6 (NTB-A, Ly108), SLAM (SLAMFI, CD150, IPO-3), BLAME (SLAMF8), SELPLG (CD162), LTBR, LAT, GADS, SLP-76, PAG/Cbp, CD 19a, a ligand that specifically binds with CD83, or any combination thereof.

In some embodiments, the CD45 recruiting domain comprises one or more of an anti-CD45 antibody antigen binding domain, an anti-CD45 NANOBODY®, an anti-CD45 scFv, a truncated viral protein binder of CD45, an anti-CD45 Fab, an anti-CD45 camelid VHH, a CD45-binding protein, or a truncated endogenous CD45 binder. In some embodiments,

the CD45 recruiting domain comprises one or more of truncated UL11, truncated sec49K, and truncated BTN3A1.

In some embodiments, the linker comprises one protease cleavage site. In some embodiments, the protease cleavage site comprises any of the amino acid sequences of SEQ ID NO: 32 and SEQ ID NOs: 89-102 (Table 2), which comprise (e.g. SEQ ID NOs: 99-102), or can be modified to comprise (e.g. any of SEQ ID NOs: 89-98), a GS linker at the amino and/or carboxy terminus when incorporated into the linker.

In some embodiments, the linker comprises multiple protease cleavage sites, such as two, three, four, five, six or more protease cleavage sites, e.g. up to ten or up to fifteen protease cleavage sites, and each cleavage site is the same as or different from any of the other cleavage sites. In some embodiments, the linker is or comprises the amino acid sequence of any one or more of (e.g. 2, 3, 4, 5 copies of) of SEQ ID NOs: 99-102, 172-177, and 108-120. In some embodiments, the linker comprises one copy or more than one copy (e.g. 2, 3, 4, 5, 6, 10 or 15) of the amino acid sequence of any of SEQ ID NOs: 89-98, 103 and 104.

In some embodiments, the protease cleavage site or one or more of the multiple protease cleavage sites is recognized by one or more of thrombin, trypsin, plasmin, prostate-specific antigen (PSA), urokinase plasminogen activator (uPA), urokinase plasminogen activator receptor (uPAR), matrix metalloproteinase (MMP), matrilysin (MT-SP1), legumain, a disintegrin and metalloproteinase (ADAM), and transmembrane Serine Protease (TMPRSS). Alternatively stated, in some embodiments, the protease that recognizes the protease cleavage site, or that recognizes one or more of the multiple protease cleavage sites, is one or more of thrombin, trypsin, plasmin, prostate-specific antigen (PSA), urokinase plasminogen activator (uPA), urokinase plasminogen activator receptor (uPAR), matrix metalloproteinase (MMP), matrilysin (MT-SP1), legumain, a disintegrin and metalloproteinase (ADAM), transmembrane Serine Protease (TMPRSS), Granzyme B, activated protein C, Caspase, Cathepsin, Chymase, Elastase, Guanidinobenzoate, HtrA1, Human Neutrophil Elastase, Lactoferrin, Marapsin, NS3/4A, PACE4, tissue plasminogen activator (tPA), DESC1, DPP-4, FAP, Hepsin, Matrilysin-2, secretase, kallikrein-related peptidase (KLK), and tryptase, or a serine protease, a cysteine-type lysosomal protease, a metalloproteinase, a coagulation factor protease, or an aspartyl-type lysosomal protease. In some embodiments, the protease cleavage site comprises an amino acid sequence that is cleaved by a protease or type of protease listed herein. In some embodiments, the protease cleavage site or one or more of the multiple protease cleavage sites is recognized by an endogenous protease. In some embodiments, the protease that recognizes the protease cleavage site, or that recognizes one or more of the multiple protease cleavage sites, is an endogenous protease.

In some embodiments, the protease cleavage site or one or more of the multiple protease cleavage sites is recognized by one or more of a serine protease, a cysteine-type lysosomal protease, an aspartyl-type lysosomal protease and a metalloproteinase. In some embodiments, the protease that recognizes the protease cleavage site, or that recognizes one or more of the multiple protease cleavage sites, is one or more of a serine protease, a cysteine-type lysosomal protease, an aspartyl-type lysosomal protease and a metalloproteinase.

In some embodiments, the linker comprises an amino GS peptide, a peptide comprising a protease cleavage site, and a carboxy GS peptide. In some embodiments, the linker comprises the amino acid sequence of SEQ ID NO: 9, which comprises an amino GS peptide (SEQ ID NO: 52), a TPS

cleavage peptide (SEQ ID NO: 53) comprising a protease cleavage site, and a carboxy GS peptide (SEQ ID NO: 54). In some embodiments, the linker comprises the same amino and carboxy GS peptides and a different cleavage peptide comprising the same or a (LHF) different protease cleavage site. Information on additional exemplary linkers is set forth in the tables below, which show the linker name, SEQ ID NO: protease cleavage sites that each linker comprises and their corresponding SEQ ID NOs: and the length (in amino acids ("a.a.)) of each linker. For example, the linker TPS4 45aa (SEQ ID NO: 107) comprises an MMP cleavage site having an amino acid sequence of SEQ ID NO: 93 and a matriptase cleavage site having an amino acid sequence of SEQ ID NO: 91. As shown in Table 2, TPS4 45aa (SEQ ID NO: 107) further comprises a GS peptide at the amino and carboxy termini. Additional linker information can be found at, for example, M. Geiger et al., Protease-activation using anti-idiotypic masks enables tumor specificity of afolate receptor 1-T cell bispecific antibody, Nature Commun 11: 3196 (2020); E. J. Kwon et al., Ultrasensitive tumour-penetrating nanosensors of protease activity, Nature Biomed Eng 1:0054 (2017); WO 2016/118629 A1, US 20170204139A1; WO 2013/130683 A2, US 2013 0165389A1.

TABLE 1A

Linker	Matriptase site	MMP site	Length
TPS1 45aa (SEQ ID NO: 9)	LSGRSDNH* (SEQ ID NO: 97)	SPLGLAGS (SEQ ID NO: 103)	45 a.a.
TPS3 45aa (SEQ ID NO: 9)	LSGRSDNH* (SEQ ID NO: 97)	SPLGLAGS (SEQ ID NO: 103)	45 a.a.
TPS6 45aa (SEQ ID NO: 113)	PMARK (SEQ ID NO: 95)		45 a.a.

*The sequence "LSGRSDNH" (SEQ ID NO 97) can be cleaved by both matriptase and uPA.

TABLE 1B

Linker	MMP site	Matriptase site	Length
TPS4 45aa (SEQ ID NO: 107)	VHMPGLGFLGP (SEQ ID NO: 93)	RQARVVNG (SEQ ID NO: 91)	45 a.a.
TPS8 45aa (SEQ ID NO: 115)	SPLGLAGS (SEQ ID NO: 103)	PMARK (SEQ ID NO: 95)	45 a.a.
TPS9 45aa (SEQ ID NO: 116)	VHMPGLGFLGP (SEQ ID NO: 93)	PMARK (SEQ ID NO: 95)	45 a.a.
TPS10 45aa (SEQ ID NO: 117)	PLGVRGK (SEQ ID NO: 104)	PMARK (SEQ ID NO: 95)	45 a.a.
TPS11 45aa (SEQ ID NO: 118)	SPLGLAGS (SEQ ID NO: 103)	RQARVVNG (SEQ ID NO: 91)	45 a.a.
TPS12 45aa (SEQ ID NO: 119)	VHMPGLGFLGP (SEQ ID NO: 93)	RQARVVNG (SEQ ID NO: 91)	45 a.a.

TABLE 1B-continued

Linker	MMP site	Matriptase site	Length
5 TPS13 45aa (SEQ ID NO: 120)	PLGVRGK (SEQ ID NO: 104)	RQARVVNG (SEQ ID NO: 91)	45 a.a.

In some embodiments, the antigen binding domain specifically binds BCMA, MUC16 (also known as CA125), EGFR, EGFRvIII, MUC1, Flt-3, WT-1, CD20, CD23, CD30, CD38, CD70, CD33, CD133, MHC-WT1, TSPAN10, MHC-PRAME, MHC-NY-ESO1, HER2 (ERBB2), CAIX (Carbonic anhydrase IX), LIV1, ADAM10, CHRNA2, LeY, NKG2D, CS1, CD44v6, ROR1, CD19, Claudin-18.2 (Claudin-18A2, or Claudin18 isoform 2), PSCA, DLL3 (Delta-like protein 3, Drosophila Delta homolog 3, Delta3), Mud 7 (Mucin17, Muc3, Muc3), FAP alpha (Fibroblast Activation Protein alpha), Ly6G6D (Lymphocyte antigen 6 complex locus protein G6d, c6orf23, G6D, MEGT1, NG25), PSMA, MSLN, or RNF43 (E3 ubiquitin-protein ligase RNF43, RING finger protein 43). CARs and/or antibodies that target the antigens are disclosed, for example, in the following: BCMA: WO201616630, WO2020150339, WO2019196713, WO2016014565, WO2017025038; MUC16: U.S. Pat. No. 9,169,328, WO2016149368, WO2020023888; EGFRvIII: WO2017125830, WO2016016341; Flt3: WO2018222935, WO2020010284, WO2017173410; CD20: WO2018145649, WO2020010235, WO2020123691; CD38: WO2017025323; CD70: WO2019152742, WO2018152181; CD33: WO2016014576; CD133: WO2018072025; CS1: WO2019030240; ROR1: WO2016115559; CD19: WO2020077029, U.S. Pat. No. 11,077,144; Claudin: WO2018006882, WO2021008463; DLL3: WO2020180591; WT1: US20160152725A1, U.S. Pat. No. 7,622,119B2; CD23: U.S. Pat. No. 6,011,138A, CN1568198A; CD30: U.S. Ser. No. 10/815,301B2, U.S. Ser. No. 10/808,035B2; PRAME: US20180148503A1, WO2020186204A1; LIV1: US20200231699A1; NKG2D: WO2021179353A1, US20210269501A1; FAP Alpha: US20200246383A1, US20210115102A1; PSMA: US20210277141A1, WO2020108646A1; MSLN: CN109680002A, CN109628492A.

In some embodiments, the antigen binding domain specifically binds MUC16. In some embodiments, the antigen binding domain specifically binds MUC16 and the protease-activating CD45-gate CAR comprises the amino acid sequence of any of SEQ ID NOs: 11, 56, 75, 76, 77, 78, 79, and 121-170, with or without a signal sequence. In some embodiments, the antigen binding domain specifically binds MUC16 and comprises the amino acid sequence of SEQ ID NO: 75 or 76, with or without a signal sequence. In some embodiments, the antigen binding domain specifically binds to an antigen expressed by a certain type of tumor and the protease cleavage sites can be recognized and cleaved by a protease that the same type of tumor expresses. In some embodiments, the protease-activating CD45-gate CAR both (1) can specifically recognize and bind to an antigen expressed by a tumor and (2) comprises a CD45-CAR linker that can be cleaved by a protease that the tumor secretes and/or that is present and/or active in the tumor microenvironment.

In some embodiments, the antigen binding domain specifically binds to a breast cancer tumor antigen or a colorectal cancer tumor antigen and one or more of the at least one protease cleavage sites can be recognized and cleaved by

uPA. In some embodiments, the protease-activating CD45-gate CAR's antigen binding domain specifically recognizes and binds to an antigen characteristic of any of cervical, breast, ovarian and colorectal cancers and at least one of its protease cleavage sites can be recognized and cleaved by MMP-2 and/or MMP-9. In some embodiments, the protease-activating CD45-gate CAR's antigen binding domain specifically recognizes and binds to an antigen characteristic of any of breast and ovarian cancers and at least one of its protease cleavage sites can be recognized and cleaved by

matriptase. In some embodiments, the protease-activating CD45-gate CAR comprises a signal sequence. In some embodiments, the signal sequence is the CD8 signal sequence. In some embodiments, the signal sequence comprises the amino acid sequence of SEQ ID NOs: 1 or 44.

Also provided herein is a CAR that specifically recognizes and binds to MUC16 (also known as CA125). In an embodiment, the anti-MUC16 CAR comprises the amino acid sequence of any of SEQ ID NOs: 11, 56, 75, 76, 77, 78, 79, and 121-170, or a variant thereof that does not comprise a signal sequence, or a conservative variant thereof.

MUC16 has been found to be an overexpressed antigen in several cancers, including ovarian, breast, pancreatic, non-small-cell lung cancer, intrahepatic cholangiocarcinoma-mass forming type, adenocarcinoma of the uterine cervix, and adenocarcinoma of the gastric tract. See, e.g., Haridas et al., *FASEB J.* 28: 4183-99 (2014). The present disclosure therefore provides, in some embodiments, a method of treating any of ovarian cancer, breast cancer, pancreatic cancer, non-small-cell lung cancer, intrahepatic cholangiocarcinoma-mass forming type, adenocarcinoma of the uterine cervix, and adenocarcinoma of the gastric tract comprising administering to a patient who has that condition one or more engineered immune cells that functionally express a MUC16-specific CAR disclosed herein e.g. one comprising the amino acid sequence of any of SEQ ID NOs: 11, 56, 75, 76, 77, 78, 79, and 121-170, and variants thereof described herein or that functionally express a protease-activating CD45-gate MUC16-specific CAR disclosed herein (e.g. one comprising the amino acid sequence of SEQ ID NOs: 11, 16, 18, 20-31, 56, 75, 76, 77, 78, 79, or 121-170, or a variant thereof described herein).

In various embodiments, the protease-activating CD45-gate CAR protein disclosed herein comprises the amino acid sequence of, e.g., any one or more of SEQ ID NOs: 1-178 and conservative variants thereof. In certain embodiments, the protease-activating CD45-gate CAR comprises an amino acid sequence which comprises at least 70%, for example at least 80%, or at least 90%, 95%, 97%, or 99% sequence identity with any one or more of SEQ ID NOs: 1-178.

In some embodiments, the protease-activating CD45-gate CAR disclosed herein comprises any one of the amino acid sequences of SEQ ID NOs: 11, 16, 17 (control sequence-contains no protease cleavage site), 18, 19 (control sequence-contains no protease cleavage site), 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 56, 75, 76, 77, 78, 79, and 121-170. Some of these sequences contain one or both of the HA (tag) (SEQ ID NO: 2) and V5 peptide motif (SEQ ID NO: 55). In some embodiments of the protease-activating CD45-gate CAR of the disclosure, either or both of the HA (tag) and V5 peptide motif, and/or the signal peptide, are excluded from the protease-activating CD45-gate CAR. The present disclosure therefore provides a protease-activating CD45-gate CAR that comprises a variant of the amino acid sequence of any one of SEQ ID NOs: 11, 16-31, 56, 75-79 and 121-170 wherein the variant excludes (does not com-

prise) the amino acid sequence of the HA tag of SEQ ID NO: 2 and/or the amino acid sequence of the V5 peptide motif of SEQ ID NO: 55 and/or the amino acid sequence of CD8ss SEQ ID NO: 1, or an amino acid sequence which comprises at least 70%, for example at least 80%, or at least 90%, 95%, 97%, 99% or 100% sequence identity therewith. Some embodiments of the protease-activating CD45-gate CAR disclosed herein contain one, two, or all three of the HA (tag) (SEQ ID NO: 2), V5 peptide motif (SEQ ID NO: 55) and CD8 signal sequence (SEQ ID NO: 1).

In some embodiments, the protease-activating CD45-gate CAR comprises a CD45 recruiting domain, a CAR, and a linker connecting the carboxy terminus of the CD45 recruiting domain to the amino terminus of the antigen binding domain of the CAR. In some embodiments, the protease-activating CD45-gate CAR comprises a CD45 recruiting domain, a CAR, and a linker connecting the carboxy terminus of the CD45 recruiting domain to the amino terminus of the antigen binding domain of the CAR wherein the CAR comprises the amino acid sequence of SEQ ID NO: 11.

In some embodiments, the protease-activating CD45-gate CAR extracellular CD45 recruiting domain comprises the amino acid sequence of SEQ ID NOs: 3, 4, 12, 13, 14, or 15.

In some embodiments, the protease-activating CD45-gate CAR extracellular CD45 recruiting domain together with the protease-cleavable linker comprises the amino acid sequence of SEQ ID NOs: 5 or 6.

In some embodiments, the protease-activating CD45-gate CAR extracellular protease-cleavable linker comprises the amino acid sequence of SEQ ID NOs: 8, 9, or 10 and/or the protease cleavage site having the amino acid sequence of SEQ ID NOs: 32 (thrombin cleavage site) or 53 (TPS cleavage site) and/or either or both of SEQ ID NOs: 52 or 54 (sequences flanking the cleavage site). In some embodiments, the protease-activating CD45-gate CAR comprises a non-cleavable linker; in some embodiments, the non-cleavable linker comprises the amino acid sequence of SEQ ID NO: 7.

In some embodiments, the protease-activating CD45-gate CAR extracellular antigen binding domain comprises the amino acid sequence of SEQ ID NO: 56.

In some embodiments, the protease-activating CD45-gate CAR transmembrane domain comprises the amino acid sequence of SEQ ID NO: 39. In some embodiments, the CAR's transmembrane and intracellular domain comprises the amino acid sequence of SEQ ID NO: 41.

In some embodiments, the protease-activating CD45-gate CAR intracellular domain comprises one or more of the amino acid sequences of SEQ ID NOs: 33, 40 and 43.

In some embodiments, the protease-activating CD45-gate CAR hinge domain comprises the amino acid sequence of SEQ ID NOs: 36, 37 or 38.

In some embodiments, the protease-activating CD45-gate CAR disclosed herein comprises an N-terminal signal sequence. In some embodiments, the signal sequence has the amino acid sequence of SEQ ID NOs: 1 or 44. In some embodiments, a nucleic acid that encodes the protease-activating CD45-gate CAR encodes the signal sequence, and, during processing within the cell, the signal sequence is removed, leaving the remainder of the encoded amino acid sequence in the mature form of the protein. This mature form of protease-activating CD45-gate CAR is within the scope of the disclosure. In some embodiments, the protease-activating CD45-gate CAR disclosed herein does not comprise an N-terminal signal sequence.

In another aspect, provided herein is a nucleic acid encoding the extracellular domain of the protease-activating

CD45-gate CAR disclosed herein. In another aspect, provided herein is a nucleic acid that encodes the protease-activating CD45-gate CAR disclosed herein. In certain embodiments, a nucleic acid of the disclosure encodes a protease-activating CD45-gate CAR that comprises an amino acid sequence which comprises at least 70%, for example at least 80%, or at least 90%, 95%, 97%, or 99% sequence identity with one or more of the amino acid sequences of SEQ ID NOs: 1-56. In certain embodiments, a nucleic acid of the disclosure encodes a protease-activating CD45-gate CAR that comprises an amino acid sequence which comprises at least 70%, for example at least 80%, or at least 90%, 95%, 97%, or 99% sequence identity with the amino acid sequence of SEQ ID NOs: 11, 16, 17 (control sequence-contains no protease cleavage site), 18, 19 (control sequence-contains no protease cleavage site), 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 56, 75, 76, 77, 78, 79, and 121-170, or a variant thereof that does not comprise an HA tag and/or a V5 sequence.

Also provided herein is a nucleic acid that encodes the amino acid sequence of any of SEQ ID NOs: 11, 56, 75, 76, 77, 78, 79, and 121-170. Also provided herein is a nucleic acid that encodes the amino acid sequence of SEQ ID NO: 56. In certain embodiments, a nucleic acid of the disclosure encodes an anti-MUC16 CAR that comprises an amino acid sequence which comprises at least 70%, for example at least 80%, or at least 90%, 95%, 97%, or 99% sequence identity with the amino acid sequence of SEQ ID NOs: 11, 56, 75, 76, 77, 78, 79, and 121-170.

In another aspect, provided herein is a vector comprising the nucleic acid disclosed herein. In an embodiment, the vector is an expression vector. In an embodiment, the vector is a viral vector, a retroviral vector, a DNA vector, a plasmid, a RNA vector, an adenoviral vector, an adenovirus associated vector, a lentiviral vector, or any combination thereof.

In a further aspect, provided herein is an engineered cell e.g. an engineered immune cell comprising the protease-

activating CD45-gate CAR disclosed herein. In a further aspect, provided herein is an engineered immune cell comprising the nucleic acid disclosed herein. In another aspect, provided herein is an engineered immune cell comprising the vector disclosed herein.

In another aspect, an immune cell e.g. T cell of the disclosure comprises e.g. expresses a polypeptide that consists of or comprises one or more amino acid sequences listed in Table 2. In some embodiments, an immune cell e.g. T cell of the disclosure comprises e.g. expresses a nucleic acid that encodes one or more amino acid sequences listed in Table 2.

In various embodiments, any of the engineered cells e.g. any of the engineered immune cells disclosed herein functionally express the protease-activating CD45-gate CAR disclosed herein. In various embodiments, any of the engineered cells e.g. any of the engineered immune cells disclosed herein functionally express a protease-activating CD45-gate CAR that comprises the amino acid sequence of SEQ ID NO: 11, 56, 75, 76, 77, 78, 79, and 121-170 or a variant thereof that lacks an N-terminal signal sequence. Also provided herein are engineered cells e.g. engineered immune cells that functionally express a CAR that comprises the amino acid sequence of SEQ ID NO: 11, 56, 75, 76, 77, 78, 79, and 121-170 or a variant thereof that lacks an N-terminal signal sequence. In various embodiments, any of the engineered cells e.g. any of the engineered immune cells disclosed herein is an isolated cell.

In various embodiments, the engineered cell e.g. engineered immune cell disclosed herein functionally expresses the protease-activating CD45-gate CAR disclosed herein from a nucleic acid encoding the protease-activating CD45-gate CAR. In various embodiments, the engineered cell e.g. engineered immune cell disclosed herein functionally expresses CD45, e.g. functionally expresses the cell's endogenous CD45 gene or functionally expresses CD45 from an exogenous nucleic acid that encodes CD45 and was introduced into cell. Table 2: Exemplary Protein Sequences

TABLE 2

Exemplary Protein Sequences	
SEQ ID NO: Name	Amino Acid Sequence
1 CD8ss	MALPVTALLLPLALLLHAARP
2 HA (tag)	GYPDVDPDYA
3 CD45-Gate1 (4131)	DIVMTQTPASVSEPVGGSVTIKQASQSFYNLLAWYQKPGQPPKLLIYDASDL ASGVPSRPFKSGSGTDFTLTISDLECADAAAYYCQADGSSYAFGGGTEVVVKG GGGSGGGSGGGSGGGSGGGSGGLEESGGGLVKPEGSLTLTCTASGVSFSSSYW IYVWRQAPGKLEWIACTYTGSSGSTYYASWAKGRFTVSETSTTVTLQMTSLT AADTATYFCARASAWTYGMDLWGPGLTVTVSS
4 CD45-Gate2 (4122)	DIVMTQTPASVSEPVGGTVMTCQASQSFYNLLAWYQKPGQPPKLLIYQASKL ASGVPSRPFKSGSGTEYTLTISDLECADAAAYYCQSYDGSNNVFFAFGGGTKV VVEGGGGSGGGSGGGSGGGSGGGSLLEESGGDLVKPGASLTTLTCTASGFSFSA GYNICWVRQAPGKLEWIACTYAGRSGSTYYANWVNGRFTIPTKSTTVTLQMT SLSGADTASYFCARGNAGVAVGALWGPGLTVTVSS
5 CD45-Gate1 withTPS	DIVMTQTPASVSEPVGGSVTIKQASQSFYNLLAWYQKPGQPPKLLIYDASDL ASGVPSRPFKSGSGTDFTLTISDLECADAAAYYCQADGSSYAFGGGTEVVVKG GGGSGGGSGGGSLGRSDNHSPLGLAGSGGGSGGGSGGLEESGGGLVKPEGSL TLCTASGVSFSSSYW IYVWRQAPGKLEWIACTYTGSSGSTYYASWAKGRFTV SETSTTVTLQMTSLTAADTATYFCARASAWTYGMDLWGPGLTVTVSS
6 CD45-Gate2 withTPS	DIVMTQTPASVSEPVGGTVMTCQASQSFYNLLAWYQKPGQPPKLLIYQASKL ASGVPSRPFKSGSGTEYTLTISDLECADAAAYYCQSYDGSNNVFFAFGGGTKV VVEGGGGSGGGSLGRSDNHSPLGLAGSGGGSGGGSLLEESGGDLVKPG ASLTTLCTASGFSFSA GYNICWVRQAPGKLEWIACTYAGRSGSTYYANWVNGR FTIPTKSTTVTLQMTSLSGADTASYFCARGNAGVAVGALWGPGLTVTVSS

TABLE 2-continued

Exemplary Protein Sequences	
SEQ ID NO: Name	Amino Acid Sequence
171 CD45-Gate3 (RB/RO)	DILLTQSPATLSLSPGERATLSCRASQNIQTSIQWYQQKPGQAPRLLIRSSSES ISGIPSRFSGSGSGTDFTLTISLLEPEDFAVYYCQQSNTWPFQGGTKLEIKG GGGSGGGSGGGSGGGSEVQLVESGAEVKKPGASVKVSCASGYTFTNYITH WVKQEPGQGLEWIGYFNPYNHGTQYNEKFKGRATLTADKSI STAYMELSSLRSE DTAVYYCARSGPYAWFDTWGQGTITVTVSS
11 MUC16 CAR1 (53B6)	QVQLQESGPGLVKPKSETLSLTCTVSGGSISSYSWTWVRQPAQGLEWIGRIHIS GVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAFDIW QGGTMVTVSSGGGSGGGSGGGSGGGSGGGSEIVLTQSPGTLTSLSPGERSTLSR ASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLTISR LEPEDFAVYYCQQYGSFPWTFGQGTKEIKTTTPAPRPPTPAPTIASQPLSLRP EACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLSLVITLYCKRGRKKLLY IFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCELRVKFSRSADAPAYQQGQNQLY NELNLGRREEYDVLDRRGRDPEMGKPRRKNPQEGLYNELQDKMAEAYSEIG MKGERRRGKGDGLYQGLSTATKDTYDALHMQALPPR
12 Sec49k	GFHTINATWWANITLVGPPDTPV TWYDTQGLWFCNGSRVKNPQIRHTCNDQNL LIHVNKTYERTYMGYNRQGTCKEDYKVVVIPPPATVKPQPEPEYVEVYMGENK TLEGPPGTPTVTFNQDGKKFCEGEKVLHPEFNHTCDQNLI LLFVNFTHDGAYL GYNQGTQORTHYEVTVLDLPDSDGQMKIENHSEETE QKND EHNWQKQGQKQG GQKTNQTKVNDRRKTAQKRPSKLKPATIEAMLVTVTAGSNLTVLGPKAEGKVTW FDGDLKRPCEPNYRLRHECNNQNLTLLINVT KDYEGTYGTNDKDEGRYRVKVN TTNSQSVKIQPYTRQTTDPQEHKFELQFETNGNYDSKIP
13 UL11	HDACIPVVGKIGTNTVTLNAVD FHPGDHVRWSYGP GGAGYMLCVYTGSWTEYKKP DII FKCLSNNSLLLINVTVNYTNTYRTLSLNNVHNQH HHHKFP GWNLDTCYSL TVNENGTFPTTTTKKPTTTTTRTTTTTKKTTTTRTTTAAKTTISTTHHKHSS PKKSTPNSHVEHHVGFETA AETPLQPSQPHQHVATH
14 BTN3A1-IgV	QFSVLGSPGPILAMVGEDADLPCHLEPTMSAETMELKQVSSSLRQVVNVYADGK EVEDRQSAFYRGRTSILRDGITAGKAALRIHNVTASDSGKYL CYFGDGFYEKA LVELKVA
15 BTN3A1-IgC	ALGSDLHVDVKGYKDGGIHLECRSTGWYPQPIQWSNNKGENIPTVEAPVVADG VGLYAVAASVIMRGSSGEGVSC TIRSSLLGLEKTASIS IADPFPSAQRWIAAL AGT
183 BTN3A1	QFSVLGSPGPILAMVGEDADLPCHLFPTMSAETMELKQVSSSLRQVVNVYADGK EVEDRQSAFYRGRTSILRDGITAGKAALRIHNVTASDSGKYL CYFGDGFYEKA LVELKVAALGSDLHVDVKGYKDGGIHLECRSTGWYPQPIQWSNNKGENIPTVE APVVADGVGLYAVAASVIMRGSSGEGVSC TIRSSLLGLEKTASIS IADPFPSA QRWIAALAGT
16 0975-N-HA-4131scFv-V5-2G4STMB-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGSVTIKC QASQSFYNLLAWYQQKPGQPPKLLIYDASDLASGVPSRFKSGSGSDFTLTISD LECDAAAYYCQSDAGSSYAFGGGTEVVVKGSGGSGGGSGGGSGGGSGGQEQ LEESGGGLVKPEGSLTLTCTASGVSFSSSYWIYVVRQAPGKGLEWIACIYTGSS GSTYYASWAKGRFTVSETSTTVTLQMTSLTAADTATYFCARASAWTYGMDLWG PGTLVTVSSGGGSGGGSGGGSGGGSGGSKIPNPLGLDSTLVPRGSGGGSGGGSGG QVQLQESGPGLVKPKSETLSLTCTVSGGSISSYSWTWVRQPAQGLEWIGRIHIS GVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAFDIW QGGTMVTVSSGGGSGGGSGGGSGGGSGGGSEIVLTQSPGTLTSLSPGERSTLSR ASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLTISR LEPEDFAVYYCQQYGSFPWTFGQGTKEIKTTTPAPRPPTPAPTIASQPLSLRP EACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLSLVITLYCKRGRKKLLY IFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCELRVKFSRSADAPAYQQGQNQLY NELNLGRREEYDVLDRRGRDPEMGKPRRKNPQEGLYNELQDKMAEAYSEIG MKGERRRGKGDGLYQGLSTATKDTYDALHMQALPPR
17 0976-N-HA-4131scFv-V5-3G4S-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGSVTIKC QASQSFYNLLAWYQQKPGQPPKLLIYDASDLASGVPSRFKSGSGSDFTLTISD LECDAAAYYCQSDAGSSYAFGGGTEVVVKGSGGSGGGSGGGSGGGSGGQEQ LEESGGGLVKPEGSLTLTCTASGVSFSSSYWIYVVRQAPGKGLEWIACIYTGSS GSTYYASWAKGRFTVSETSTTVTLQMTSLTAADTATYFCARASAWTYGMDLWG PGTLVTVSSGGGSGGGSGGGSGGGSGGSKIPNPLGLDSTGGSGSGGGSGGGSGG QVQLQESGPGLVKPKSETLSLTCTVSGGSISSYSWTWVRQPAQGLEWIGRIHIS GVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAFDIW QGGTMVTVSSGGGSGGGSGGGSGGGSGGGSEIVLTQSPGTLTSLSPGERSTLSR ASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLTISR LEPEDFAVYYCQQYGSFPWTFGQGTKEIKTTTPAPRPPTPAPTIASQPLSLRP EACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLSLVITLYCKRGRKKLLY IFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCELRVKFSRSADAPAYQQGQNQLY NELNLGRREEYDVLDRRGRDPEMGKPRRKNPQEGLYNELQDKMAEAYSEIG MKGERRRGKGDGLYQGLSTATKDTYDALHMQALPPR

TABLE 2-continued

Exemplary Protein Sequences	
SEQ ID NO: Name	Amino Acid Sequence
18 0977-N-HA-4122scFv-V5-2G4STMB-53B6	MALPVTALLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGTVTMC QASQSI SNWLAWYQQKPGQPPKLLIYQASKLASGVPSRFKSGSGTEYTLTISD LECADAATYYCQSYDSSGNVFFAFGGGKTVVVEGGGGSGGGSGGGSGGGG SLSLEESGGDLVKPGASLTLTCTASGFSFSAGYWCWVRQAPGKGLEWIACTYA GRSGSTYYANWVNGRFTIPKTSSTTVTLQMTSLSGADTASYFCARGNAGVAVGA LWGPGLTVTVSSGGGGSGGGSGGGSGGGSGGGSEIVLTQSPGTLSPGERSTL GGSQVQLQESGPGLVKPSSETLSLTCTVSGGSI SYSWTWVRQAPAGQGLEWIGRI HISGVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAF DIWQGTMTVTVSSGGGGSGGGSGGGSGGGSGGGSEIVLTQSPGTLSPGERSTL SCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLT ISRLEPEDFAVYYCQYGS SPWTFGQGTKEIKTTTPAPRPPTPAPTIASQPLS LRPEACRPAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKK LLYIFKQPFMRPVQTTQEEDGCSRFP EEEEGGCELRVKFSRSADAPAYQQGQN QLYNELNLGRREEYDVLDRRGRDPEMGGKPRRKNPQEGLYNELQDKMAEAYS EIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
19 0978-N-HA-4122scFv-V5-3G4S-53B6	MALPVTALLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGTVTMC QASQSI SNWLAWYQQKPGQPPKLLIYQASKLASGVPSRFKSGSGTEYTLTISD LECADAATYYCQSYDSSGNVFFAFGGGKTVVVEGGGGSGGGSGGGSGGGG SLSLEESGGDLVKPGASLTLTCTASGFSFSAGYWCWVRQAPGKGLEWIACTYA GRSGSTYYANWVNGRFTIPKTSSTTVTLQMTSLSGADTASYFCARGNAGVAVGA LWGPGLTVTVSSGGGGSGGGSGGGSGGGSGGGSEIVLTQSPGTLSPGERSTL GGSQVQLQESGPGLVKPSSETLSLTCTVSGGSI SYSWTWVRQAPAGQGLEWIGRI HISGVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAF DIWQGTMTVTVSSGGGGSGGGSGGGSGGGSGGGSEIVLTQSPGTLSPGERSTL SCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLT ISRLEPEDFAVYYCQYGS SPWTFGQGTKEIKTTTPAPRPPTPAPTIASQPLS LRPEACRPAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKK LLYIFKQPFMRPVQTTQEEDGCSRFP EEEEGGCELRVKFSRSADAPAYQQGQN QLYNELNLGRREEYDVLDRRGRDPEMGGKPRRKNPQEGLYNELQDKMAEAYS EIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
20 01082-N-HA-4131scFv-TPS1-53B6	MALPVTALLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGSVTIKC QASQSFYNLLAWYQQKPGQPPKLLIYDASDLASGVPSRFKSGSGTDFTLTISD LECADAAYYCQSDGSSYAFGGGTEVVVKGGGGSGGGSGGGSGGGSGGGQEQ LEESGGGLVKPEGLTLTCTASGVSFSSSYWIYVVRQAPGKGLEWIACTIYTGSS GSTYYASWAKGRFTVSETSTTVTLQMTSLTAADTATYFCARASAWTYGMDLWG PGTLVTVTVSSGGGGSGGGSGGGSGGLSGRSDNHSPLGLAGSGGGSGGGSGGGG QVQLQESGPGLVKPSSETLSLTCTVSGGSI SYSWTWVRQAPAGQGLEWIGRIHIS GVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAFDIW QGQTMVTVSSGGGGSGGGSGGGSGGGSGGGSEIVLTQSPGTLSPGERSTLSR ASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLTISR LEPEDFAVYYCQYGS SPWTFGQGTKEIKTTTPAPRPPTPAPTIASQPLSLRP EACRPAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKKLLY IFKQPFMRPVQTTQEEDGCSRFP EEEEGGCELRVKFSRSADAPAYQQGQNQLY NELNLGRREEYDVLDRRGRDPEMGGKPRRKNPQEGLYNELQDKMAEAYSEIG MKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
21 01092-N-HA-4122scFv-TPS1-53B6	MALPVTALLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGTVTMC QASQSI SNWLAWYQQKPGQPPKLLIYQASKLASGVPSRFKSGSGTEYTLTISD LECADAATYYCQSYDSSGNVFFAFGGGKTVVVEGGGGSGGGSGGGSGGGG SLSLEESGGDLVKPGASLTLTCTASGFSFSAGYWCWVRQAPGKGLEWIACTYA GRSGSTYYANWVNGRFTIPKTSSTTVTLQMTSLSGADTASYFCARGNAGVAVGA LWGPGLTVTVSSGGGGSGGGSGGGSGGLSGRSDNHSPLGLAGSGGGSGGGSGG GGSQVQLQESGPGLVKPSSETLSLTCTVSGGSI SYSWTWVRQAPAGQGLEWIGRI HISGVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAF DIWQGTMTVTVSSGGGGSGGGSGGGSGGGSGGGSEIVLTQSPGTLSPGERSTL SCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLT ISRLEPEDFAVYYCQYGS SPWTFGQGTKEIKTTTPAPRPPTPAPTIASQPLS LRPEACRPAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKK LLYIFKQPFMRPVQTTQEEDGCSRFP EEEEGGCELRVKFSRSADAPAYQQGQN QLYNELNLGRREEYDVLDRRGRDPEMGGKPRRKNPQEGLYNELQDKMAEAYS EIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
22 01098-N-HA-4131scFv-2XTPS1-53B6	MALPVTALLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGSVTIKC QASQSFYNLLAWYQQKPGQPPKLLIYDASDLASGVPSRFKSGSGTDFTLTISD LECADAAYYCQSDGSSYAFGGGTEVVVKGGGGSGGGSGGGSGGGSGGGQEQ LEESGGGLVKPEGLTLTCTASGVSFSSSYWIYVVRQAPGKGLEWIACTIYTGSS GSTYYASWAKGRFTVSETSTTVTLQMTSLTAADTATYFCARASAWTYGMDLWG PGTLVTVTVSSGGGSLSGRSDNHSPLGLAGSKPIPNPLLGLDSTLSGRSDNHSPL GLAGSGGGGQVQLQESGPGLVKPSSETLSLTCTVSGGSI SYSWTWVRQAPAGQ LEWIGRIHISGVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCAR SGGTYSAFDIWQGTMTVTVSSGGGGSGGGSGGGSGGGSGGGSEIVLTQSPGTLSP

TABLE 2-continued

Exemplary Protein Sequences	
SEQ ID NO: Name	Amino Acid Sequence
	PGERSTLSCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGS GTDFTLTIISRLPEDFAVYYCQQYGSPPWTFGQGTKVEIKTTTPAPRPPTPAPT IASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVI TLY CKRGRKKLLYIFKQPFMRPVQTTQEEDGCSRFPPEEEEGGCELRVKFSRSADAP AYQQGQNQLYNELNLGRREEYDVLDKRRGRDPGEMGGKPRRKNPQEGLYNELQKD KMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
23 01099-N-HA-4122scFv-V5-2XTPS1-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGTITMC QASQSFYNLLAWYQQKPGQPPKLLIYQASKLASGVPSRFKSGSGSTEYTLTISD LECADAATYYCQSYDSSGNVFFAFGGGTQVVEGGGGSGGGSGGGSGGGSGGGG SLSLSEESGGDLVKPGASLTLTCTASGFSFSAAGYWCWVRQAPGKLEWIACTYA GRSGSTYYANWVNGRFTIPKTSSTTVTLQMTSLSGADTASYFCARGNAGVAVGA LWGPGLTVTVSSGGGSLSGRSDNHSPLGLAGSKIPNPLLGLDSTLSGRSDNH SPLGLAGSGGGSGVQLQESGPGLVKPSSETLSLTCTVSGGSIYYSWTWVRQPA QGQLEWIGRIHISGVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFC ARSGGTYSAPDIWGQGTMTVTVSSGGGSGGGSGGGSGGGSGGGSGGGSEIVLTQSPGTL SLSPGERSTLSCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSG SGSGTDFTLTIISRLPEDFAVYYCQQYGSPPWTFGQGTKVEIKTTTPAPRPPTP APTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVI TLYCKRGRKKLLYIFKQPFMRPVQTTQEEDGCSRFPPEEEEGGCELRVKFSRSDA PAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPGEMGGKPRRKNPQEGLYNEL QDKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
24 01102-N-HA-4131-VL-RS-VH-TPS1-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGSVTIKC QASQSFYNLLAWYQQKPGQPPKLLIYDASDLASGVPSRFKSGSGSDFTLTIISD LECADAATYYCQSDGSSYAFGGGTQEVVVGKGGGGSGGGGSLSGRSDNHSPLGL AGSGGGSGGGSGGQLEESGGGLVKPEGSLTLTCTASGVFSFSSSYWYIYVWRQA PGKLEWIACTIYTGSSGSTYYASWAKGRFTVSETSTTVTLQMTSLTAADTATY FCARASAWTYGMDLWGPGLTVTVSSGGGSGGGSGGGSGGGSLSGRSDNHSPLGLA GSGGGSGGGSGGGSGVQLQESGPGLVKPSSETLSLTCTVSGGSIYYSWTWVR QPAGQLEWIGRIHISGVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAV YFCARSGGTYSAPDIWGQGTMTVTVSSGGGSGGGSGGGSGGGSGGGSEIVLTQSP GTLSPGERSTLSCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDR FSGSGSDFTLTIISRLPEDFAVYYCQQYGSPPWTFGQGTKVEIKTTTPAPRP PTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLS LVITLYCKRGRKKLLYIFKQPFMRPVQTTQEEDGCSRFPPEEEEGGCELRVKFS RSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPGEMGGKPRRKNPQEGLY NELQDKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
25 01103-N-HA-4122-VL-RS-VH-TPS1-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGTITMC QASQSFYNLLAWYQQKPGQPPKLLIYQASKLASGVPSRFKSGSGSTEYTLTISD LECADAATYYCQSYDSSGNVFFAFGGGTQVVEGGGGSGGGGSLVPRSGGGSGGG PLGLAGSGGGSGGGGSLSEESGGDLVKPGASLTLTCTASGFSFSAAGYWCWV RQAPGKLEWIACTYAGRSGSTYYANWVNGRFTIPKTSSTTVTLQMTSLSGADT ASYFCARGNAGVAVGALWGPGLTVTVSSGGGSGGGSGGGSGGGSLSGRSDNHSPL GLAGSGGGSGGGSGGGSGVQLQESGPGLVKPSSETLSLTCTVSGGSIYYSWT WVRQAPGQLEWIGRIHISGVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAAD TAVYFCARSGGTYSAPDIWGQGTMTVTVSSGGGSGGGSGGGSGGGSGGGSEIVLT QSPGTLSPGERSTLSCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGI PDRFSGSGSDFTLTIISRLPEDFAVYYCQQYGSPPWTFGQGTKVEIKTTTPA PRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGV LLSLVITLYCKRGRKKLLYIFKQPFMRPVQTTQEEDGCSRFPPEEEEGGCELRV KFSRSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPGEMGGKPRRKNPQ GLYNELQDKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALP PR
26 01112-N-HA-4131-VL-TMB-VH-V5-2G4STMB-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGSVTIKC QASQSFYNLLAWYQQKPGQPPKLLIYDASDLASGVPSRFKSGSGSDFTLTIISD LECADAATYYCQSDGSSYAFGGGTQEVVVGKGGGGSGGGGSLVPRSGGGSGGG GGQLEESGGGLVKPEGSLTLTCTASGVFSFSSSYWYIYVWRQAPGKLEWIACTI YTGSSGSTYYASWAKGRFTVSETSTTVTLQMTSLTAADTATYFCARASAWTY GMDLWGPGLTVTVSSGGGSGGGGSLVPRGSKIPNPLLGLDSTLVPRSGGGGG SGGGSGVQLQESGPGLVKPSSETLSLTCTVSGGSIYYSWTWVRQAPGQLEWIG RIHISGVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYS APDIWGQGTMTVTVSSGGGSGGGSGGGSGGGSGGGSEIVLTQSPGTLSPGER STLSCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSDFT LTIISRLPEDFAVYYCQQYGSPPWTFGQGTKVEIKTTTPAPRPPTPAPTIASQ PLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVI TLYCKRG RKLLYIFKQPFMRPVQTTQEEDGCSRFPPEEEEGGCELRVKFSRSADAPAYQQ GQNQLYNELNLGRREEYDVLDKRRGRDPGEMGGKPRRKNPQEGLYNELQDKMAE AYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR

TABLE 2-continued

Exemplary Protein Sequences		
SEQ ID NO: Name	Amino Acid Sequence	
27 01113-N-HA-4122-VL-TMB-VH-V5-2G4STMB-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGTITMCQASQSI SNWLAWYQKPGQPPKLLIYQASKLASGVP SRFKSGSGTEYTLTISDLECADAATYYCQSYDSSGNVFFAFGGGTQVVVEGGGGSGGGSLVPRGSGGGSGGGGGSLLEESGGDLVKPGASLTLTCTASGFSFSAGYIWCVRQAPGKGLEWIACTYAGRSGSTYYANWVNGRFTIPKTSSTTVTLQMTSLSGADTASYFCARGNAVAVGALWGPGTLVTVSSGGGGSGGGSLVPRGSKPIPNPLGLDSTLVPRGSGGGSGGGGQVQLQESGPGLVKPSSETLSLTCTVSGGSI SYSWTWVRQAPAGQGLEWIGRIHISGVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAFDIWGQGTMTVTVSSGGGGSGGGSGGGSGGGSEIVLTQSPGTLTSLSPGERSTLSCRASQSFSTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGGTDFTLTISRLEPEDFAVYYCQYGS SPWTFGQGTKEIKTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKKLLYIFKQPFMRPVQTTQEEDGCS CRFPEEEEGGCEL RVKFSRSADAPAYQQQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR	
28 01121-N-HA-sec49k-V5-2G4STMB-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYAGFHTINATWVANITLVGPPDTPV TWYDTQGLWFCNGSRVKNPQIRHTCNDQNLTLIHVNKTYERTYMGYNRQGTKE DYKVVVIPPATVTKPQPEPEYVFVYMGENKTEGPGTPTVTFNQDGKFCG EKV LHPFNHTCDKQNLILFVNFTHDGAYLGYNHQTQRTHYEVTVLDLFPDS GQMKIENHSEETE QKND EHNWQKQGQKQGGQKTNQTKVNDRRKTAQKRPSKL KPATIEAMLVTVTAGSNLTLVGPKAEGKVTWEDGDLKRPCENYRLRHECENNQN LTLINVT KDYEGTYGTNDKDEGRYRVKVNNTNSQSVKI QPYTRQTPDQEHK FELQFETNGYDSKIPGGGGSGGGSLVPRGSKPIPNPLGLDSTLVPRGSGGGSGGGGQVQLQESGPGLVKPSSETLSLTCTVSGGSI SYSWTWVRQAPAGQGLEWIGRIHISGVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAFDIWGQGTMTVTVSSGGGGSGGGSGGGSGGGSEIVLTQSPGTLTSLSPGERSTLSCRASQSFSTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGGTDFTLTISRLEPEDFAVYYCQYGS SPWTFGQGTKEIKTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKKLLYIFKQPFMRPVQTTQEEDGCS CRFPEEEEGGCEL RVKFSRSADAPAYQQQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR	
29 01122-N-HA-UL11-V5-2G4STMB-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYAHDACIPVVGKIGTNVTLNAVDFH PGDHVRWSYGGGAGYMLCVYTGSWTEYKKPDII FKCLSNNSLLLINVTVNYTN TYRTL SLNWNVHNQH HHKFPGWNLDTCYSLTVNENGTEPTTTTKPTTTTRTT TTTTTKTTTTTRTTTAAKTTISTTHKHSSPKSSTPNSHVEHHVGFEEATAE TPLQPSQHQHVATHGGGGSGGGSLVPRGSKPIPNPLGLDSTLVPRGSGGGSGGGGQVQLQESGPGLVKPSSETLSLTCTVSGGSI SYSWTWVRQAPAGQGLEWIGRIHISGVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAFDIWGQGTMTVTVSSGGGGSGGGSGGGSGGGSEIVLTQSPGTLTSLSPGERSTLSCRASQSFSTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGGTDFTLTISRLEPEDFAVYYCQYGS SPWTFGQGTKEIKTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKKLLYIFKQPFMRPVQTTQEEDGCS CRFPEEEEGGCEL RVKFSRSADAPAYQQQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR	
30 01207-N-HA-IgV-V5-2G4STMB-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYAQFSLVLPSPGPII AMVGEDADLPC HLEPTMSAETMELKWVSSSLRQVNVNADGKEVEDRQSAPYRGRTSILRDGITA GKAALRIHNVTASDSGKLYCYFQDGDYFEKALVELKVAGGGSGGGSLVPRGSKPIPNPLGLDSTLVPRGSGGGSGGGGQVQLQESGPGLVKPSSETLSLTCTVSGGSI SYSWTWVRQAPAGQGLEWIGRIHISGVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAFDIWGQGTMTVTVSSGGGGSGGGSGGGSEIVLTQSPGTLTSLSPGERSTLSCRASQSFSTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGGTDFTLTISRLEPEDFAVYYCQYGS SPWTFGQGTKEIKTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKKLLYIFKQPFMRPVQTTQEEDGCS CRFPEEEEGGCEL RVKFSRSADAPAYQQQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR	
31 01208-N-HA-IgC-V5-2G4STMB-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYAALGSDLHVDVKYKDGGIHLECR STGWYPQPIQWSNNKGENIPTVEAPVVDGVLGYAVAA SVIMRGSSGEGVSC TIRSSLLGLEKTASIS IADPFRRSAQRWIAALAGTGGGGSGGGSLVPRGSKPIPNPLGLDSTLVPRGSGGGSGGGGQVQLQESGPGLVKPSSETLSLTCTVSGGSI SYSWTWVRQAPAGQGLEWIGRIHISGVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAFDIWGQGTMTVTVSSGGGGSGGGSGGGSGGGSEIVLTQSPGTLTSLSPGERSTLSCRASQSFSTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGGTDFTLTISRLEPEDFAVYYCQYGS SPWTFGQGTKEIKTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKKLLYIFKQPFMRPVQTTQEEDGCS CRFPEEEEGGCEL RVKFSRSADAPAYQQQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPR	

TABLE 2-continued

Exemplary Protein Sequences	
SEQ ID NO: Name	Amino Acid Sequence
	RKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDAL HMQALPPR
33 4-1BB intra-cellular signaling domain (ISD)	KRGRKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCEL
34 CD3 ζ intra-cellular signaling domain (ISD)	RVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNP QEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQA LPPR
35 CD28-IC (CD28 RSKRSRGHSDYMNMTPRRPGPTRKHYQPYAPPRDFAAYRS co-stimulatory domain)	
36 Fc γ R1II α hinge	GLWSTISSFEPPGYQ
37 CD8 α hinge	TTTPAPRPPTPTIASQPLSLRPEACRPAAGGAVRTRGLDFACD
38 IgG1 hinge	EPKSPDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMTkRTPEVTCVVVDVSHED PEVKFNWYVDGVEVHNAETKPREEQYNSTYRVVSVLTVLHQDWLNGKEYNCKV SKALPAPIEKTIISKAKGQPREPQVITLPPSRDELTKNOVSLTCLVKGFIPSDIA VEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSQCSVMHEAL HNHYTOKSLSLSPGK
39 CD8 α transmembrane (TM) domain	IYIWAPLAGTCGVLLLSLVITLYG
40 CD3 ζ intra-cellular signaling domain (ISD)	RVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVIDKRRGRDPEMGGKPRRKNP QEGLYNELQKDKMAEAYSEIGMKGERRRGKGEDGLYQGLSTATKDTYDALHMQA LPPP
41 Fc ϵ R1 α -TM-IC (Fc ϵ R1 α chain transmembrane and intra-cellular domain)	FFIPLLVILFAVDTGLEFISTQQQVTFLLKIKRTRKGERLLNPHPKPNTKNN
42 Fc ϵ R1 β -A1TAM (Fc ϵ R1 β chain without ITAM)	MDTESNRRANLALPQEPSSVPAFEVLEISPQEVSSGRELKSASSPPLHTWLTVL KKEQEFGLGVTQILTAMICLCFGTVVCSVLDSHIEGDIFFSSFKAGYPFWGAIFF SISGMLSIISERRNATYLVRLGANTASSIAGGTGITILIINLKSLAYIH SCQKFFETKCFMSFSTEIVVMMLFLTILGLGSAVSLTICGAGEELKGNKVPE
43 CD23-IC (CD28 RSKRSRGHSDYMNMTPRRPGPTPKHYQPIPPPDFAAYRS co-stimulatory domain)	
44 Fc γ R1 γ -SP (signal peptide)	MIPAVVILLLLVEQAAA
45 Fc ϵ R1 γ -A1TAM (Fc ϵ R1 γ chain without ITAM)	LGEPQLCYILDAILFLIGIVLTLLICRLKIQVRKAAITSYEKS
46 GSG-P2A (GSG-GSGATNFSLLKQAGDVEENPGP P2A ribosomal skip poly-peptide)	
47 GSG-T2A (GSG-GSGERGSLLTCGDVEENPGP T2A ribosomal skip poly-peptide)	

TABLE 2-continued

Exemplary Protein Sequences	
SEQ ID NO: Name	Amino Acid Sequence
48 safety switch	CPYSNPSCSGGGSELPTQGTFSNVSTNVSPAKPTTTACPYSNPSCSGGGSPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCNHRNRRRVCKCPRPVV
49 safety switch-amino term.	MGTSLLCWMALCLLGADHADACPYSNPSCSGGGSELPTQGTFSNVSTNVSPAKPTTTACPYSNPSCSGGGSPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCNHRNRRRVCKCPRPVV
50 CD20 mimotope	CPYSNPSC
51 2xCD20 mimotope	GSGGGGSCPYSNPSCSGGGGSCPYSNPSCSGGGG
52 aminoGS	GGGSGGGSGGGSG
54 carboxyGS	GGGSGGGSGGGG
55 V5	KPIPNPLLGLDST
56 Leader-53B6 scEv	MALPVTALLPLALLHAARPQVQLQESGPGLVKPSSETLSLTCTVSGGSIYSYWTWVRQPAQGQLEWIGRIHISGVTHNPSTLKSRSMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAFDIWGQGTMTVSSGGGGSGGGSGGGSGGGSGEIVLTQSPGTLSTLSPGERSTLSCRASQSFSTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLTISRLEPEDFAVYYCQYGGSPWTFGQGTKVEIK
57 GS linker	GGGSGGGSGGGG
178 GS sequence (1)	GGGG
181 GS45 (GS sequence 2)	GGGSGGGSGGGSGSGGGSGGGSGGGSGGGSGGGSGGGSGGGG
182 GS45 (3)	GGGSGGGSGGGSGGGSGSGGGSGGGSGGGSGGGSGGGSGGGG
PROTEASE CLEAVAGE SITES:	
32 TMB cleavage site	LVPRGS
53 TPS cleavage	LSGRSDNHSPLGLAGS
89 MMP2/9/MTSP site (1)	VHMPGLGLGPRQARVVNG
90 MMP2/9/MTSP site (2)	VHMPGLGLGPGSARVVNG
91 MTSP (matriptase) site (1)	RQARVVNG
92 MTSP site (2)	RQARVGSG
93 MMP2/9 (1)	VHMPGLGLGP
94 MMP2/9 (2)	VHMPGLGLGP
95 MTSP (3)	PMAKK
96 MTSP (4)	PMAKGS
97 MTSP (5)	LSGRSDNH
98 MTSP (6)	LSGRSDSH
103 MMP (site 3)	SPLGLAGS
104 MMP (site 4)	PLGVRGK
105 MTSP (7)	PMAKG
106 MTSP (8)	GSARVVNG

TABLE 2-continued

Exemplary Protein Sequences	
SEQ ID NO:Name	Amino Acid Sequence
LINKER SEQUENCES WITH PROTEASE CLEAVAGE SITES	
7GS (45aa)	GGGSGGGSGGGSGSKIPNPLLGLDSTGGSGSGGGSGGGGS
8GSTMB (45aa)	GGGSGGGGSLVPRGSKIPNPLLGLDSTLVPRSGGGSGGGGS
9GSTPS1 (45aa) (also referred to as TPS1 (45aa); same sequence as TPS3 (45aa) and TPS2 (45aa))	GGGSGGGSGGGSGLSGRSDNHSPLGLAGSGGGSGGGSGGGGS
10GSTPS2 (55aa) (also referred to as "TPS2 (55aa)")	GGGSLSGRSDNHSPLGLAGSKIPNPLLGLDSTLSGRSDNHSPLGLAGSGGGGS
99TPS4 (25aa)	GGGGSVHMP LGFLGPRQARVVNGGS
100TPS5 (25aa)	GGGGSVHMP LGFLGPGSARVVNGGS
101TPS6 (25aa)	GGGSGGGGSPMAKKGGGSGGGGS
102TPS7 (25aa)	GGGSGGGGSPMAKSGGGSGGGGS
107TPS4 (45aa)	GGGSGGGSGGGGSVHMP LGFLGPRQARVVNGSGGGSGGGGS
172TPS8 (25aa)	GSSPLGLAGSGGGSGGGGSPMAKK
173TPS9 (25aa)	GGSVHMP LGFLGPGSGGGSPMAKK
174TPS10 (25aa)	GGSP LGVRGKGGGSGGGGSPMAKK
175TPS11 (25aa)	GGSSPLGLAGSGSGGSRQARVVNG
176TPS12 (25aa)	GGSVHMP LGFLGPGGSRQARVVNG
177TPS13 (25aa)	GGSP LGVRGKGGSGGSRQARVVNG
108TPS5 45aa	GGGSGGGSGGGGSVHMP LGFLGPGSARVVNGSGGGSGGGGS
109TPS6 20aa	GGGGSMAKKGGGSGGGGS
110TPS6 25aa	GGGSGGGGSPMAKKGGGSGGGGS
111TPS6 30aa	GGGSGGGGSPMAKKGGGSGGGSGGGGS
112TPS6 35aa	GGGSGGGSGGGGSPMAKKGGGSGGGSGGGGS
113TPS6 45aa	GGGSGGGSGGGSGGGGSPMAKKGGGSGGGSGGGSGGGGS
114TPS7 45aa	GGGSGGGSGGGSGGGGSPMAKSGGGSGGGSGGGSGGGGS
115TPS8 45aa	GGGSGGGSGGSSPLGLAGSGGGSGGGSPMAKKGGGSGGGGS
116TPS9 45aa	GGGSGGGSGGGSVHMP LGFLGPGSGGGSPMAKKGGGSGGGGS
117TPS10 45aa	GGGSGGGSGGSP LGVRGKGGSGGGSPMAKKGGGSGGGGS
118TPS11 45aa	GGGSGGGSGGSSPLGLAGSGSGGSRQARVVNGGGGSGGGGS
119TPS12 45aa	GGGSGGGSGGGSVHMP LGFLGPGGSRQARVVNGGGGSGGGGS
120TPS13 45aa	GGGSGGGSGGSP LGVRGKGGSGGSRQARVVNGGGGSGGGGS

TABLE 2-continued

Exemplary Protein Sequences	
SEQ ID NO: Name	Amino Acid Sequence
Exemplary protease-activating CD45-gate MUC16-specific CARs	
121N-HA-4131scFv-TPS4-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGSVTIKC QASQSFYNLLAWYQQKPGQPPKLLIYDASDLASGVPSRFKSGSGTDFTLTISD LECADAAAYYCQSDGSSYAFGGGTEVTVKGGGGSGGGSGGGSGGGSGGGSGQEQ LEESGGGLVKPEGSLLTCTASGVSFSSSYWIYWVRQAPGKGLEWIAICIYTGSS GSTYYASWAKGRFTVSETSTTVTLQMTSLTAADTATYFCARASAWTYGMDLWG PGTLVTVSSGGGGSGGGSGGGSGGGSVHMPGLGFLGPRQARVNVGGSGGGSGGGSG QVQLQESGPGLVKPSSETLSLTCTVSGGSIYYSWTWVRQAPAGGLEWIGRIHIS GVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAFDIW QGQTMVTVSSGGGGSGGGSGGGSGGGSGGGSEI VLTQSPGTLTSLSPGERSTLSR ASQSFTSNYLAWYQQRPQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLTISR LEPEDFAVYYCQQYGSFPWTFGQGTKEIKTTTPAPRPPTPAPTIASQPLSLRP EACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKKLLY IFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCELRVKFSRSADAPAYQQGNQLY NELNLGRREEYDVLDKRRGRDPEMGKPRRKNPQEGLYNELQDKKMAEAYSEIG MKGERRRGKGDGLYQGLSTATKDTYDALHMQALPPR
122N-HA-4131scFv-TPS5-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGSVTIKC QASQSFYNLLAWYQQKPGQPPKLLIYDASDLASGVPSRFKSGSGTDFTLTISD LECADAAAYYCQSDGSSYAFGGGTEVTVKGGGGSGGGSGGGSGGGSGGGSGQEQ LEESGGGLVKPEGSLLTCTASGVSFSSSYWIYWVRQAPGKGLEWIAICIYTGSS GSTYYASWAKGRFTVSETSTTVTLQMTSLTAADTATYFCARASAWTYGMDLWG PGTLVTVSSGGGGSGGGSGGGSGGGSVHMPGLGFLGPGSARVNVGGSGGGSGGGSG QVQLQESGPGLVKPSSETLSLTCTVSGGSIYYSWTWVRQAPAGGLEWIGRIHIS GVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAFDIW QGQTMVTVSSGGGGSGGGSGGGSGGGSGGGSEI VLTQSPGTLTSLSPGERSTLSR ASQSFTSNYLAWYQQRPQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLTISR LEPEDFAVYYCQQYGSFPWTFGQGTKEIKTTTPAPRPPTPAPTIASQPLSLRP EACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKKLLY IFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCELRVKFSRSADAPAYQQGNQLY NELNLGRREEYDVLDKRRGRDPEMGKPRRKNPQEGLYNELQDKKMAEAYSEIG MKGERRRGKGDGLYQGLSTATKDTYDALHMQALPPR
123N-HA-4131scFv-TPS6-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGSVTIKC QASQSFYNLLAWYQQKPGQPPKLLIYDASDLASGVPSRFKSGSGTDFTLTISD LECADAAAYYCQSDGSSYAFGGGTEVTVKGGGGSGGGSGGGSGGGSGGGSGQEQ LEESGGGLVKPEGSLLTCTASGVSFSSSYWIYWVRQAPGKGLEWIAICIYTGSS GSTYYASWAKGRFTVSETSTTVTLQMTSLTAADTATYFCARASAWTYGMDLWG PGTLVTVSSGGGGSGGGSGGGSGGGSGGGSPMAKGGGGSGGGSGGGSGGGSGGGSG QVQLQESGPGLVKPSSETLSLTCTVSGGSIYYSWTWVRQAPAGGLEWIGRIHIS GVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAFDIW QGQTMVTVSSGGGGSGGGSGGGSGGGSGGGSEI VLTQSPGTLTSLSPGERSTLSR ASQSFTSNYLAWYQQRPQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLTISR LEPEDFAVYYCQQYGSFPWTFGQGTKEIKTTTPAPRPPTPAPTIASQPLSLRP EACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKKLLY IFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCELRVKFSRSADAPAYQQGNQLY NELNLGRREEYDVLDKRRGRDPEMGKPRRKNPQEGLYNELQDKKMAEAYSEIG MKGERRRGKGDGLYQGLSTATKDTYDALHMQALPPR
124N-HA-4131scFv-TPS7-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGSVTIKC QASQSFYNLLAWYQQKPGQPPKLLIYDASDLASGVPSRFKSGSGTDFTLTISD LECADAAAYYCQSDGSSYAFGGGTEVTVKGGGGSGGGSGGGSGGGSGGGSGQEQ LEESGGGLVKPEGSLLTCTASGVSFSSSYWIYWVRQAPGKGLEWIAICIYTGSS GSTYYASWAKGRFTVSETSTTVTLQMTSLTAADTATYFCARASAWTYGMDLWG PGTLVTVSSGGGGSGGGSGGGSGGGSGGGSPMAKSGGGSGGGSGGGSGGGSGGGSG QVQLQESGPGLVKPSSETLSLTCTVSGGSIYYSWTWVRQAPAGGLEWIGRIHIS GVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAFDIW QGQTMVTVSSGGGGSGGGSGGGSGGGSGGGSEI VLTQSPGTLTSLSPGERSTLSR ASQSFTSNYLAWYQQRPQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLTISR LEPEDFAVYYCQQYGSFPWTFGQGTKEIKTTTPAPRPPTPAPTIASQPLSLRP EACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKKLLY IFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCELRVKFSRSADAPAYQQGNQLY NELNLGRREEYDVLDKRRGRDPEMGKPRRKNPQEGLYNELQDKKMAEAYSEIG MKGERRRGKGDGLYQGLSTATKDTYDALHMQALPPR
125N-HA-4131scFv-2XTPS4-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGSVTIKC QASQSFYNLLAWYQQKPGQPPKLLIYDASDLASGVPSRFKSGSGTDFTLTISD LECADAAAYYCQSDGSSYAFGGGTEVTVKGGGGSGGGSGGGSGGGSGGGSGQEQ RVNVGGSGGGSGGGSGGGSGGGSGGGSVHMPGLGFLGPRQARVNVGGSGGGSGGGSG QVQLQESGPGLVKPSSETLSLTCTVSGGSIYYSWTWVRQAPAGGLEWIGRIHIS GVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAFDIW QGQTMVTVSSGGGGSGGGSGGGSGGGSGGGSEI VLTQSPGTLTSLSPGERSTLSR ASQSFTSNYLAWYQQRPQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLTISR LEPEDFAVYYCQQYGSFPWTFGQGTKEIKTTTPAPRPPTPAPTIASQPLSLRP EACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKKLLY IFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCELRVKFSRSADAPAYQQGNQLY NELNLGRREEYDVLDKRRGRDPEMGKPRRKNPQEGLYNELQDKKMAEAYSEIG MKGERRRGKGDGLYQGLSTATKDTYDALHMQALPPR

TABLE 2-continued

Exemplary Protein Sequences	
SEQ ID NO: Name	Amino Acid Sequence
	PAGQGLEWIGRIHISGVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVY FCARSGGTYSAFDIWGQGTMTVSSGGGSGGGGSGGGGSGGGGSEIVLTQSPG TSLSLSPGERSTLSCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRF SGSGSGTDFTLTISRLEPEDFAVYQCQYGSSPWTFGQGTKEIKTTTPAPRPP TPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSL VITLYCKRGRKKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCELVRVKFSR SADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYN ELQDKMAEAYS EIGMKGERRRGKHDGLYQGLSTATKDTYDALHMQALPPR
126N-HA- 4131scFv- 2XTPS5-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGSVTIKC QASQSFYNLLAWYQQKPGQPPKLLIYDASDLASGVP SRFKGSGSGTDFTLTISD LECADAAYYCQSADGSSYAFGGGT EVVVKGGGSGGGGSGGGGSHMPLGFLGPGSA RVVNGSGGGGSGGGGQVQLQESGPGLVKPSSETLSLTCTVSGGSISSYSWTWVRQ GKGLEWIAICIYTGSSGSTYASWAKGRFTVSETSSTTVTLQMTSLTAADTATYF CARASAWTYGMDLWGPGLTVTVSSGGGSGGGGSGGGGSHMPLGFLGPGSARV VNGSGGGGSGGGGQVQLQESGPGLVKPSSETLSLTCTVSGGSISSYSWTWVRQ PAGQGLEWIGRIHISGVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVY FCARSGGTYSAFDIWGQGTMTVSSGGGSGGGGSGGGGSGGGGSEIVLTQSPG TSLSLSPGERSTLSCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRF SGSGSGTDFTLTISRLEPEDFAVYQCQYGSSPWTFGQGTKEIKTTTPAPRPP TPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSL VITLYCKRGRKKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCELVRVKFSR SADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYN ELQDKMAEAYS EIGMKGERRRGKHDGLYQGLSTATKDTYDALHMQALPPR
127N-HA- 4131scFv- 2XTP56-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGSVTIKC QASQSFYNLLAWYQQKPGQPPKLLIYDASDLASGVP SRFKGSGSGTDFTLTISD LECADAAYYCQSADGSSYAFGGGT EVVVKGGGSGGGGSGGGGSGMAKGGG GSGGGGSGGGGQVQLQESGPGLVKPSSETLSLTCTVSGGSISSYSWTWVRQ GKGLEWIAICIYTGSSGSTYASWAKGRFTVSETSSTTVTLQMTSLTAADTATYF CARASAWTYGMDLWGPGLTVTVSSGGGSGGGGSGGGGSGGGGSGMAKGGG GGGSGGGGSGGGGQVQLQESGPGLVKPSSETLSLTCTVSGGSISSYSWTWVRQ PAGQGLEWIGRIHISGVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVY FCARSGGTYSAFDIWGQGTMTVSSGGGSGGGGSGGGGSGGGGSEIVLTQSPG TSLSLSPGERSTLSCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRF SGSGSGTDFTLTISRLEPEDFAVYQCQYGSSPWTFGQGTKEIKTTTPAPRPP TPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSL VITLYCKRGRKKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCELVRVKFSR SADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYN ELQDKMAEAYS EIGMKGERRRGKHDGLYQGLSTATKDTYDALHMQALPPR
128N-HA- 4131scFv- 2XTP57-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGSVTIKC QASQSFYNLLAWYQQKPGQPPKLLIYDASDLASGVP SRFKGSGSGTDFTLTISD LECADAAYYCQSADGSSYAFGGGT EVVVKGGGSGGGGSGGGGSGMAKGGG GSGGGGSGGGGQVQLQESGPGLVKPSSETLSLTCTVSGGSISSYSWTWVRQ GKGLEWIAICIYTGSSGSTYASWAKGRFTVSETSSTTVTLQMTSLTAADTATYF CARASAWTYGMDLWGPGLTVTVSSGGGSGGGGSGGGGSGGGGSGMAKGGG GGGSGGGGSGGGGQVQLQESGPGLVKPSSETLSLTCTVSGGSISSYSWTWVRQ PAGQGLEWIGRIHISGVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVY FCARSGGTYSAFDIWGQGTMTVSSGGGSGGGGSGGGGSGGGGSEIVLTQSPG TSLSLSPGERSTLSCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRF SGSGSGTDFTLTISRLEPEDFAVYQCQYGSSPWTFGQGTKEIKTTTPAPRPP TPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSL VITLYCKRGRKKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCELVRVKFSR SADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYN ELQDKMAEAYS EIGMKGERRRGKHDGLYQGLSTATKDTYDALHMQALPPR
129N-HA-BTN3A1- V5-2G4STMB- 53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYAQFVSLGPSGPILAMVGEDADLPC HLFPTMSAETMELKWSSSLRQVNVYADGKEVEDRQSAPYRGRTSILRDGITA GKAALRIHNVTASDSGKYL CYFQDGDY EKALVELKVAALGSDLHVDVKYKDG GIHLECRSTGWYPQPQIQWNNKGENIPTVEAPVADGVGLYAVAASVIMRGSS GEGVCTIRSSSLGLEKTASISADPFFRSAQRWIAALAGTGGGSGGGSLVP RGSXIPNPLGLDSTLVPRGSGGGGSGGGGQVQLQESGPGLVKPSSETLSLTCT TVSGGSISSYSWTWVRQ PAGQGLEWIGRIHISGVTNHNPSLKSRSVMSIDTSRT QFSLRLTSVTAADTAVYFCARSGGTYSAFDIWGQGTMTVSSGGGSGGGGSGG GGSGGGGSEIVLTQSPGTSLSPGERSTLSCRASQSFTSNYLAWYQQRPGQAPR LLIYGASTRAIGIPDRFSGSGTDFTLTISRLEPEDFAVYQCQYGSSPWTFG QGTKEIKTTTPAPRPP TPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDI YIWAPLAGTCGVLLLSLVI TLYCKRGRKKLLYIFKQPFMRPVQTTQEEDGCSCR FPEEEEGGCELVRVKFSR SADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGRD PEMGGKPRRKNPQEGLYNELQDKMAEAYS EIGMKGERRRGKHDGLYQGLSTAT KDTYDALHMQALPPR

TABLE 2-continued

Exemplary Protein Sequences	
SEQ ID NO: Name	Amino Acid Sequence
130N-HA-RORB-V5-2G4STMB-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYADILLTQSPATLSLSPGERATLSC RASQNI GTSIQWYQQKPGQAPRLLIRSSSESISGIPSRFSGSGSGTDFTLTIS LEPEDFAVYYCQQSNTWPF TFGQGT KLEIKGGGSGGGSGGGSGGGSGGGSEVQL VESGAEVKKPGASVKVCKASGYFTFTNYIIHWVKQEPGQGLEWIGYFNPYNHGT KYNEKFKGRATLTADKSISTAYMELSSLRSEDTAVYYCARSGPYAWFDTWGQGT TVTVSSGGGSGGGSGGGSLVPRGSKPIPNPLLGLDSTLVPRGSGGGSGGGSGGGSQVQ LQESGPGLVKPSSETLSLTCTVSGGSISSYSWTWVRQPAQGQLEWIGRIHISGVT NHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAFDIWGQG TMVTVSSGGGSGGGSGGGSGGGSGGGSEIVLTQSPGTLTSLSPGERSTLSCRASQ SFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLTISRLEP EDFAVYYCQQYGSSPWTFGQGTKEIKTTTPAPRPPTPAPTIASQPLSLRPEAC RPAAGGAVHTRGLDFACDIYIWAFLAGTCGVLLLSLVI TLYCKRGRKKLLYIFK QPFMRPVQTTQEEDGCSRFEEEEGGCELRVKF SR SADAPAYQQGQNQLYNEL NLGRREYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKG ERRRGKGDGLYQGLSTATKDTYDALHMQALPPR
131N-HA-RORB-V5-3G45-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYADILLTQSPATLSLSPGERATLSC RASQNI GTSIQWYQQKPGQAPRLLIRSSSESISGIPSRFSGSGSGTDFTLTIS LEPEDFAVYYCQQSNTWPF TFGQGT KLEIKGGGSGGGSGGGSGGGSGGGSEVQL VESGAEVKKPGASVKVCKASGYFTFTNYIIHWVKQEPGQGLEWIGYFNPYNHGT KYNEKFKGRATLTADKSISTAYMELSSLRSEDTAVYYCARSGPYAWFDTWGQGT TVTVSSGGGSGGGSGGGSGGSKPIPNPLLGLDSTGGSGSGGGSGGGSGGGSQVQ LQESGPGLVKPSSETLSLTCTVSGGSISSYSWTWVRQPAQGQLEWIGRIHISGVT NHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAFDIWGQG TMVTVSSGGGSGGGSGGGSGGGSGGGSEIVLTQSPGTLTSLSPGERSTLSCRASQ SFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLTISRLEP EDFAVYYCQQYGSSPWTFGQGTKEIKTTTPAPRPPTPAPTIASQPLSLRPEAC RPAAGGAVHTRGLDFACDIYIWAFLAGTCGVLLLSLVI TLYCKRGRKKLLYIFK QPFMRPVQTTQEEDGCSRFEEEEGGCELRVKF SR SADAPAYQQGQNQLYNEL NLGRREYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKG ERRRGKGDGLYQGLSTATKDTYDALHMQALPPR
132N-HA-RORB-TPS4-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYADILLTQSPATLSLSPGERATLSC RASQNI GTSIQWYQQKPGQAPRLLIRSSSESISGIPSRFSGSGSGTDFTLTIS LEPEDFAVYYCQQSNTWPF TFGQGT KLEIKGGGSGGGSGGGSGGGSGGGSEVQL VESGAEVKKPGASVKVCKASGYFTFTNYIIHWVKQEPGQGLEWIGYFNPYNHGT KYNEKFKGRATLTADKSISTAYMELSSLRSEDTAVYYCARSGPYAWFDTWGQGT TVTVSSGGGSGGGSGGGSGGGSVHMLGELGPRQARVNVNGSGGGSGGGSGGGSQVQ LQESGPGLVKPSSETLSLTCTVSGGSISSYSWTWVRQPAQGQLEWIGRIHISGVT NHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAFDIWGQG TMVTVSSGGGSGGGSGGGSGGGSGGGSEIVLTQSPGTLTSLSPGERSTLSCRASQ SFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLTISRLEP EDFAVYYCQQYGSSPWTFGQGTKEIKTTTPAPRPPTPAPTIASQPLSLRPEAC RPAAGGAVHTRGLDFACDIYIWAFLAGTCGVLLLSLVI TLYCKRGRKKLLYIFK QPFMRPVQTTQEEDGCSRFEEEEGGCELRVKF SR SADAPAYQQGQNQLYNEL NLGRREYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKG ERRRGKGDGLYQGLSTATKDTYDALHMQALPPR
133N-HA-RORB-TPS6-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYADILLTQSPATLSLSPGERATLSC RASQNI GTSIQWYQQKPGQAPRLLIRSSSESISGIPSRFSGSGSGTDFTLTIS LEPEDFAVYYCQQSNTWPF TFGQGT KLEIKGGGSGGGSGGGSGGGSGGGSEVQL VESGAEVKKPGASVKVCKASGYFTFTNYIIHWVKQEPGQGLEWIGYFNPYNHGT KYNEKFKGRATLTADKSISTAYMELSSLRSEDTAVYYCARSGPYAWFDTWGQGT TVTVSSGGGSGGGSGGGSGGGSPMAKGGGSGGGSGGGSGGGSGGGSGGGSQVQ LQESGPGLVKPSSETLSLTCTVSGGSISSYSWTWVRQPAQGQLEWIGRIHISGVT NHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAFDIWGQG TMVTVSSGGGSGGGSGGGSGGGSGGGSEIVLTQSPGTLTSLSPGERSTLSCRASQ SFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLTISRLEP EDFAVYYCQQYGSSPWTFGQGTKEIKTTTPAPRPPTPAPTIASQPLSLRPEAC RPAAGGAVHTRGLDFACDIYIWAFLAGTCGVLLLSLVI TLYCKRGRKKLLYIFK QPFMRPVQTTQEEDGCSRFEEEEGGCELRVKF SR SADAPAYQQGQNQLYNEL NLGRREYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKG ERRRGKGDGLYQGLSTATKDTYDALHMQALPPR
134N-HA-RORB-2XTP54-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYADILLTQSPATLSLSPGERATLSC RASQNI GTSIQWYQQKPGQAPRLLIRSSSESISGIPSRFSGSGSGTDFTLTIS LEPEDFAVYYCQQSNTWPF TFGQGT KLEIKGGGSGGGSGGGSVHMLGFLGPRQAR VVNGSGGGGSEVQLVESGAEVKKPGASVKVCKASGYFTFTNYIIHWVKQEPGQ GLEWIGYFNPYNHGTKYNEKFKGRATLTADKSISTAYMELSSLRSEDTAVYYCA RSGPYAWFDTWGQGT TVTVSSGGGSGGGSGGGSGGGSVHMLGELGPRQARVNVNG SGGGSGGGGSGVQLQESGPGLVKPSSETLSLTCTVSGGSISSYSWTWVRQPAQ QGLEWIGRIHISGVTNHNPSLKSRSVMSIDTSRTQFSLRLTSVTAADTAVYFCA RSGGTYSAFDIWGQGT TMVTVSSGGGSGGGSGGGSGGGSGGGSEIVLTQSPGTL SLSPGERSTLSCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGS

TABLE 2-continued

Exemplary Protein Sequences	
SEQ ID NO: Name	Amino Acid Sequence
	GSGTDFTLTISRLEPEDFAVYYCQQYGGSPWTFGQGTKEIKTTTPAPRPPTPA PTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVIT LYCKRGRKLLYIFKQPFMRPVQTTQEEDGCSRFPPEEEGGGCELVRKFSRSAD APAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQ KDKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
135N-HA-RORB- 2XTP56-53B6	MALPVTALLPLALLLHAARPGYPYDVPDYADILLTQSPATLSLSPGERATLSC RASQNIQTSIQWYQKPGQAPRLIRSSSESISGIPSRFSGSGSGTDFTLTISS LEPEDFAVYYCQQNTWPFPTFGQGTKEIKGGGSGGGSGGGSGGSGGSGGSGG SGGGSGGGGSEVQLVESGAEVKKPGASVKVSCASGYTFTNYIIHWKQEPGQ GLEWIGYFNPNYHGTKYNEFKGRATLTADKISITAYMELSSLSRSEDVAVYYCA RSGPYAWFDTWGQGTMTVTVSSGGGSGGGSGGGSGGGSGGSGGSGGSGGSGG GSGGGSGGGGSGVQLQESGPGLVKPSSETLSLTCTVSGGSIYYSWTWVRQAPG QGLEWIGRIHISGVTNHNP SLKSRVSMIDTSRTQFSLRLTSVTAADTAVYFCA RSGGTYSAFDIWGQGTMTVTVSSGGGSGGGSGGGSGGGSGGSGGSGGSGGSGG LSPGERSTLSCRASQFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGS GSGTDFTLTISRLEPEDFAVYYCQQYGGSPWTFGQGTKEIKTTTPAPRPPTPA PTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVIT LYCKRGRKLLYIFKQPFMRPVQTTQEEDGCSRFPPEEEGGGCELVRKFSRSAD APAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQ KDKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
136V5-GS20- 4131scFv- TPS1-53B6	MALPVTALLPLALLLHAARPGKPIPNPLLGLDSTGGGSGGGSGGGSGGGSGG SDIVMTQTPASVSEPVGGSVTIKQASQSFYNLLAWYQKPGQPPKLLIYDASD LASGVPSRFKSGSGTDFTLTISDLECADAAAYCQADGSSYAFGGGTEVVVK GGGGSGGGSGGGSGGGSGGQLEESGGGLVKPEGSLTLTCTASGVSFSSSY WIYVWRQAPGKGLEWIIACIYTGSSGTYASWAKGRFTVSETSSTTVTLQMTSL TAADTATYFCARASAWTYGMDLWGPGLTVTVSSGGGSGGGSGGGSGGGSGGSD NHSPGLLAGSGGGSGGGSGGGSGGQVQLQESGPGLVKPSSETLSLTCTVSGGSI YYSWTVWRQAPGQGLEWIGRIHISGVTNHNP SLKSRVSMIDTSRTQFSLRLTS VTAADTAVYFCARSGGTYSAFDIWGQGTMTVTVSSGGGSGGGSGGGSGGGSGG EIVLTQSPGTLSPGERSTLSCRASQFTSNYLAWYQQRPGQAPRLLIYGAST RAIGIPDRFSGSGSGTDFTLTISRLEPEDFAVYYCQQYGGSPWTFGQGTKEIK TTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAG TCGVLLLSLVITLYCKRGRKLLYIFKQPFMRPVQTTQEEDGCSRFPPEEEGG CELRVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRR KNPQEGLYNELQDKMAEAYSIEIGMKGERRRGKGHDGLYQGLSTATKDTYDALH MQALPPR
137V5-GS20- 4131scFv- TPS4-53B6	MALPVTALLPLALLLHAARPGKPIPNPLLGLDSTGGGSGGGSGGGSGGGSGG SDIVMTQTPASVSEPVGGSVTIKQASQSFYNLLAWYQKPGQPPKLLIYDASD LASGVPSRFKSGSGTDFTLTISDLECADAAAYCQADGSSYAFGGGTEVVVK GGGGSGGGSGGGSGGGSGGQLEESGGGLVKPEGSLTLTCTASGVSFSSSY WIYVWRQAPGKGLEWIIACIYTGSSGTYASWAKGRFTVSETSSTTVTLQMTSL TAADTATYFCARASAWTYGMDLWGPGLTVTVSSGGGSGGGSGGGSGGGSGGSH FLGPRQARVINGSGGGSGGGSGGQVQLQESGPGLVKPSSETLSLTCTVSGGSI YYSWTVWRQAPGQGLEWIGRIHISGVTNHNP SLKSRVSMIDTSRTQFSLRLTS VTAADTAVYFCARSGGTYSAFDIWGQGTMTVTVSSGGGSGGGSGGGSGGGSGG EIVLTQSPGTLSPGERSTLSCRASQFTSNYLAWYQQRPGQAPRLLIYGAST RAIGIPDRFSGSGSGTDFTLTISRLEPEDFAVYYCQQYGGSPWTFGQGTKEIK TTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAG TCGVLLLSLVITLYCKRGRKLLYIFKQPFMRPVQTTQEEDGCSRFPPEEEGG CELRVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRR KNPQEGLYNELQDKMAEAYSIEIGMKGERRRGKGHDGLYQGLSTATKDTYDALH MQALPPR
138V5-G520- 4131scFv- TPS6-53B6	MALPVTALLPLALLLHAARPGKPIPNPLLGLDSTGGGSGGGSGGGSGGGSGG SDIVMTQTPASVSEPVGGSVTIKQASQSFYNLLAWYQKPGQPPKLLIYDASD LASGVPSRFKSGSGTDFTLTISDLECADAAAYCQADGSSYAFGGGTEVVVK GGGGSGGGSGGGSGGGSGGQLEESGGGLVKPEGSLTLTCTASGVSFSSSY WIYVWRQAPGKGLEWIIACIYTGSSGTYASWAKGRFTVSETSSTTVTLQMTSL TAADTATYFCARASAWTYGMDLWGPGLTVTVSSGGGSGGGSGGGSGGGSGGSP MAKKGSGGGSGGGSGGGSGGQVQLQESGPGLVKPSSETLSLTCTVSGGSI YYSWTVWRQAPGQGLEWIGRIHISGVTNHNP SLKSRVSMIDTSRTQFSLRLTS VTAADTAVYFCARSGGTYSAFDIWGQGTMTVTVSSGGGSGGGSGGGSGGGSGG EIVLTQSPGTLSPGERSTLSCRASQFTSNYLAWYQQRPGQAPRLLIYGAST RAIGIPDRFSGSGSGTDFTLTISRLEPEDFAVYYCQQYGGSPWTFGQGTKEIK TTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAG TCGVLLLSLVITLYCKRGRKLLYIFKQPFMRPVQTTQEEDGCSRFPPEEEGG CELRVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRR KNPQEGLYNELQDKMAEAYSIEIGMKGERRRGKGHDGLYQGLSTATKDTYDALH MQALPPR

Exemplary Protein Sequence	
SEQ ID NO: Name	Amino Acid Sequence
139V5-G520-4131scFv-2XTPS6-53B6	MALPVTALLPLALLLHAARP GKPIPNLLGLDSTGGGSGGGSGGGSGGGSDIVMTQTPASVSEPVGGSVTIKCQASQSFYNNLAWYQQKPGQP P K L L I Y D A S D L A S G V P S R F K G S G S G T D F T L T I S D L E C A D A A A Y Y C Q S A D G S S Y A F G G G T E V V V K G G G G S G G G S G G G G S P M A K K G G G S G G G S G G G S G G G S Q E Q L E E S G G L V K P E G S L T L T C T A S G V S F S S S Y I W V W R Q A P G K G L E W I A C I Y T G S S G S T Y Y A S W A K G R F T V S E T S S T T V T L Q M T S L T A A D T A T Y F C A R A S A W T Y G M D L W G P G T L V T V S S G G G S G G G S G G G S G G G S S Q V Q L Q E S G P L V K P S E T L S L T C T V S G G S I S Y S W T W V R Q P A G Q G L E W I G R I H I S G V T N H N P S L K S R V S M S I D T S R T Q F S L R L T S V T A A D T A V Y F C A R S G G T Y S A F D I W G Q G T M V T V S S G G G S G G G S G G G S S E I V L T Q S P G T L S L S P G E R S T L S C R A S Q S F T S N Y L A W Y Q R P G Q A P R L L I Y G A S T R A I G I P D R F S G S G S G T D F T L T I S R L E P E D F A V Y Y C Q Q Y G S S P W T F G Q G T K V E I K T T P A P R P P T P A P T I A S Q P L S L R P E A C R P A A G G A V H T R G L D F A C D I Y I W A P L A G T C G V L L S L V I T L Y C K R G R K K L L Y I F K Q P F M R P V Q T T Q E E D G C S C R F P E E E E G G C E L R V K F S R S A D A P A Y Q Q G Q N Q L Y N E L N L G R R E E Y D V L D K R R G R D P E M G K P R R K N P Q E G L Y N E L Q K D K M A E A Y S E I G M K G E R R R G K G H D G L Y Q G L S T A T K D T Y D A L H M Q A L P P R
140V5-GS20-4131scFv-G545-53B6	MALPVTALLPLALLLHAARP GKPIPNLLGLDSTGGGSGGGSGGGSGGGSDIVMTQTPASVSEPVGGSVTIKCQASQSFYNNLAWYQQKPGQP P K L L I Y D A S D L A S G V P S R F K G S G S G T D F T L T I S D L E C A D A A A Y Y C Q S A D G S S Y A F G G G T E V V V K G G G S G G G S G G G S G G G S Q E Q L E E S G G L V K P E G S L T L T C T A S G V S F S S S Y I W I Y W V R Q A P G K G L E W I A C I Y T G S S G S T Y Y A S W A K G R F T V S E T S S T T V T L Q M T S L T A A D T A T Y F C A R A S A W T Y G M D L W G P G T L V T V S S G G G S G G G S G G G S G G G S G G G S G G G S G G G S S Q V Q L Q E S G P L V K P S E T L S L T C T V S G G S I S Y S W T W V R Q P A G Q G L E W I G R I H I S G V T N H N P S L K S R V S M S I D T S R T Q F S L R L T S V T A A D T A V Y F C A R S G G T Y S A F D I W G Q G T M V T V S S G G G S G G G S G G G S G G G S E I V L T Q S P G T L S L S P G E R S T L S C R A S Q S F T S N Y L A W Y Q R P G Q A P R L L I Y G A S T R A I G I P D R F S G S G S G T D F T L T I S R L E P E D F A V Y Y C Q Q Y G S S P W T F G Q G T K V E I K T T P A P R P P T P A P T I A S Q P L S L R P E A C R P A A G G A V H T R G L D F A C D I Y I W A P L A G T C G V L L S L V I T L Y C K R G R K K L L Y I F K Q P F M R P V Q T T Q E E D G C S C R F P E E E E G G C E L R V K F S R S A D A P A Y Q Q G Q N Q L Y N E L N L G R R E E Y D V L D K R R G R D P E M G K P R R K N P Q E G L Y N E L Q K D K M A E A Y S E I G M K G E R R R G K G H D G L Y Q G L S T A T K D T Y D A L H M Q A L P P R
1414131scFv-V5-2XTPS1-53B6	MALPVTALLPLALLLHAARP GPYPYDVDPDYADIVMTQTPASVSEPVGGSVTIKCQASQSFYNLLAWYQQKPGQP P K L L I Y D A S D L A S G V P S R F K G S G S G T D F T L T I S D L E C A D A A A Y Y C Q S A D G S S Y A F G G G T E V V V K G G G S G G G S G S L S G R S D N H S P L G L A G S G G S G G G S Q E Q L E E S G G L V K P E G S L T L T C T A S G V S F S S S Y I W I Y W V R Q A P G K G L E W I A C I Y T G S S G T Y Y A S W A K G R F T V S E T S S T T V T L Q M T S L T A A D T A T Y F C A R A S A W T Y G M D L W G P G T L V T V S S G G G S G G G S G G S K P I P N P L L G L D S T G G S G S L S G R S D N H S P L G L A G S G G S G G G S G G G S G G G S Q V Q L Q E S G P L V K P S E T L S L T C T V S G G S I S Y S W T W V R Q P A G Q G L E W I G R I H I S G V T N H N P S L K S R V S M S I D T S R T Q F S L R L T S V T A A D T A V Y F C A R S G G T Y S A F D I W G Q G T M V T V S S G G G S G G G S G G G S G G G S S E I V L T Q S P G T L S L S P G E R S T L S C R A S Q S F T S N Y L A W Y Q R P G Q A P R L L I Y G A S T R A I G I P D R F S G S G S G T D F T L T I S R L E P E D F A V Y Y C Q Q Y G S S P W T F G Q G T K V E I K T T P A P R P P T P A P T I A S Q P L S L R P E A C R P A A G G A V H T R G L D F A C D I Y I W A P L A G T C G V L L S L V I T L Y C K R G R K K L L Y I F K Q P F M R P V Q T T Q E E D G C S C R F P E E E G G C E L R V K F S R S A D A P A Y Q Q G Q N Q L Y N E L N L G R R E E Y D V L D K R R G R D P E M G K P R R K N P Q E G L Y N E L Q K D K M A E A Y S E I G M K G E R R R G K G H D G L Y Q G L S T A T K D T Y D A L H M Q A L P P R
1424131scFv-V5-2XTPS4-53B6	MALPVTALLPLALLLHAARP GPYPYDVDPDYADIVMTQTPASVSEPVGGSVTIKCQASQSFYNLLAWYQQKPGQP P K L L I Y D A S D L A S G V P S R F K G S G S G T D F T L T I S D L E C A D A A A Y Y C Q S A D G S S Y A F G G G T E V V V K G G G S G G G S G S V H M P L G E L G P R Q A R V N G S G G G S G G G S Q E Q L E E S G G L V K P E G S L T L T C T A S G V S F S S S Y I W V W R Q A P G K G L E W I A C I Y T G S S G T Y Y A S W A K G R F T V S E T S S T T V T L Q M T S L T A A D T A T Y F C A R A S A W T Y G M D L W G P G T L V T V S S G G G S G G G S G G S K P I P N P L L G L D S T G G G S G S V H M P L G E L G P R Q A R V N G S G G G S G G G S G G G S G G G S Q V Q L Q E S G P L V K P S E T L S L T C T V S G G S I S Y S W T W V R Q P A G Q G L E W I G R I H I S G V T N H N P S L K S R V S M S I D T S R T Q F S L R L T S V T A A D T A V Y F C A R S G G T Y S A F D I W G Q G T M V T V S S G G G S G G G S G G G S G G G S S E I V L T Q S P G T L S L S P G E R S T L S C R A S Q S F T S N Y L A W Y Q R P G Q A P R L L I Y G A S T R A I G I P D R F S G S G S G T D F T L T I S R L E P E D F A V Y Y C Q Q Y G S S P W T F G Q G T K V E I K T T P A P R P P T P A P T I A S Q P L S L R P E A C R P A A G G A V H T R G L D F A C D I Y I W A P L A G T C G V L L S L V I T L Y C K R G R K K L L Y I F K Q P F M R P V Q T T Q E E D G C S C R F P E E E G G C E L R V K F S R S A D A P A Y Q Q G Q N Q L Y N E L N L G R R E E Y D V L D K R R G R D P E M G K P R R K N P Q E G L Y N E L Q K D K M A E A Y S E I G M K G E R R R G K G H D G L Y Q G L S T A T K D T Y D A L H M Q A L P P R
1434131scFv-V5-2XTPS5-53B6	MALPVTALLPLALLLHAARP GPYPYDVDPDYADIVMTQTPASVSEPVGGSVTIKCQASQSFYNLLAWYQQKPGQP P K L L I Y D A S D L A S G V P S R F K G S G S G T D F T L T I S D L E C A D A A A Y Y C Q S A D G S S Y A F G G G T E V V V K G G G S G G G S G S M P L G E L G P S A R V N G S G G G S G G G S Q E Q L E E S G G L V K P E G S L T L T C T A S G V S F S S S Y I W V W R Q A P G K G L E W I A C I Y T G S S G T Y Y A S W A K G R F T V S E T S S T T V T L Q M T S L T A A D T A T Y F C A R A S A W T Y G M D L W G P G T L V T V S S G G G S G G G S G G S K P I P N P L L G L D S T G G G S G S V H M P L G E L G P R Q A R V N G S G G G S G G G S G G G S G G G S Q V Q L Q E S G P L V K P S E T L S L T C T V S G G S I S Y S W T W V R Q P A G Q G L E W I G R I H I S G V T N H N P S L K S R V S M S I D T S R T Q F S L R L T S V T A A D T A V Y F C A R S G G T Y S A F D I W G Q G T M V T V S S G G G S G G G S G G G S G G G S G G G S S E I V L T Q S P G T L S L S P G E R S T L S C R A S Q S F T S N Y L A W Y Q R P G Q A P R L L I Y G A S T R A I G I P D R F S G S G S G T D F T L T I S R L E P E D F A V Y Y C Q Q Y G S S P W T F G Q G T K V E I K T T P A P R P P T P A P T I A S Q P L S L R P E A C R P A A G G A V H T R G L D F A C D I Y I W A P L A G T C G V L L S L V I T L Y C K R G R K K L L Y I F K Q P F M R P V Q T T Q E E D G C S C R F P E E E G G C E L R V K F S R S A D A P A Y Q Q G Q N Q L Y N E L N L G R R E E Y D V L D K R R G R D P E M G K P R R K N P Q E G L Y N E L Q K D K M A E A Y S E I G M K G E R R R G K G H D G L Y Q G L S T A T K D T Y D A L H M Q A L P P R

TABLE 2-continued

Exemplary Protein Sequences	
SEQ ID NO: Name	Amino Acid Sequence
	MPLGFLPGSARVVNGSGGGSGGGGSQVQLQESGPGLVKPSSETLSLTCTVSG GSISYYSWTWVRQPAGQGLEWIGRIHISGVTNHNPSLKS RVSMSIDTSRTQFSL RLTSVTAADTAVYFCARSGGTYSAFDIWGQGTMTVSSGGGGSGGGSGGGSGG GGGSEIVLTQSPGTLTSLSPGERSTLSCRASQSFTSNYLAWYQQRPGQAPRLLIY GASTRAIGIPDRFSGSGGTDFTLTISRLEPEDFAVYYCQYQSSPWFQGGTK VEIKTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWA PLAGTCGVLLLSLVITLYCKRGRKKLLYIFKQPFMRPVQTTQEDGCSRFPPEE EEGCELRVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPENGG KPRRKNPQEGLYNELQDKMAEAYS EIGMKGERRRGKGHDGLYQGLSTATKDTY DALHMQALPPR
144 4131scFv-V5- 2XTPS6-53B6	MALPVTALLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGSVTIKC QASQSFYNLLAWYQQKPGQPPKLLIYDASDLASGVPSRFGSGSGTDFTLTISD LECADAAAYYCQSDGSSYAFGGTEVVVKGGGGSGGGSGGGSGGSPMAKGGG GSGGGSGGGSGGQLEESGGGLVKPEGSLLTCTASGVSFSSSYWIYWRQAP KGLEWIACIYTGSSGTYASWAKGRFTVSETSTTVTLQMTSLTAADTATYF CARASAWTYGMDLWGPGLTVTVSSGGGGSGGGSGGKIPNPLLGLDSTGGGSPM AKKGGGGSGGGSGGGSGGGSGQVQLQESGPGLVKPSSETLSLTCTVSGGSIY YSWTWVRQPAGQGLEWIGRIHISGVTNHNPSLKS RVSMSIDTSRTQFSLRLTSV TAADTAVYFCARSGGTYSAFDIWGQGTMTVTVSSGGGGSGGGSGGGSGGGSE IVLTQSPGTLTSLSPGERSTLSCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTR AIGIPDRFSGSGGTDFTLTISRLEPEDFAVYYCQYQSSPWFQGGTKVEIKT TTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWA PLAGTCGVLLLSLVITLYCKRGRKKLLYIFKQPFMRPVQTTQEDGCSRFPPEEEGGC ELRVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPENGGKPRRK NPQEGLYNELQDKMAEAYS EIGMKGERRRGKGHDGLYQGLSTATKDTYDALHM QALPPR
145 4131scFv-V5- 2XTPS7-53B6	MALPVTALLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGSVTIKC QASQSFYNLLAWYQQKPGQPPKLLIYDASDLASGVPSRFGSGSGTDFTLTISD LECADAAAYYCQSDGSSYAFGGTEVVVKGGGGSGGGSGGGSGGSPMAKGGG GSGGGSGGGSGGQLEESGGGLVKPEGSLLTCTASGVSFSSSYWIYWRQAP KGLEWIACIYTGSSGTYASWAKGRFTVSETSTTVTLQMTSLTAADTATYF CARASAWTYGMDLWGPGLTVTVSSGGGGSGGGSGGKIPNPLLGLDSTGGGSPM AKGSGGGSGGGSGGGSGGGSGQVQLQESGPGLVKPSSETLSLTCTVSGGSIY YSWTWVRQPAGQGLEWIGRIHISGVTNHNPSLKS RVSMSIDTSRTQFSLRLTSV TAADTAVYFCARSGGTYSAFDIWGQGTMTVTVSSGGGGSGGGSGGGSGGGSE IVLTQSPGTLTSLSPGERSTLSCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTR AIGIPDRFSGSGGTDFTLTISRLEPEDFAVYYCQYQSSPWFQGGTKVEIKT TTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWA PLAGTCGVLLLSLVITLYCKRGRKKLLYIFKQPFMRPVQTTQEDGCSRFPPEEEGGC ELRVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPENGGKPRRK NPQEGLYNELQDKMAEAYS EIGMKGERRRGKGHDGLYQGLSTATKDTYDALHM QALPPR
146 N-HA- 4131scFv- G530-53B6	MALPVTALLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGSVTIKC QASQSFYNLLAWYQQKPGQPPKLLIYDASDLASGVPSRFGSGSGTDFTLTISD LECADAAAYYCQSDGSSYAFGGTEVVVKGGGGSGGGSGGGSGGGSGGQEQ LEESGGGLVKPEGSLLTCTASGVSFSSSYWIYWRQAPKGLEWIACIYTGSS GSTYYASWAKGRFTVSETSTTVTLQMTSLTAADTATYFCARASAWTYGMDLW PGTLVTVSSGGGGSGGGSGGGSGGGSGGGSGGGSGGQVQLQESGPGLVKPS ETLSLTCTVSGGSIYYSWTWVRQPAGQGLEWIGRIHISGVTNHNPSLKS RVSM SIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSAFDIWGQGTMTVTVSSGGGG GGGGSGGGSGGGSEIVLTQSPGTLTSLSPGERSTLSCRASQSFTSNYLAWYQ RPGQAPRLLIYGASTRAIGIPDRFSGSGGTDFTLTISRLEPEDFAVYYCQYQ SSPWFQGGTKVEIKTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRG LDFACDIYIWA PLAGTCGVLLLSLVITLYCKRGRKKLLYIFKQPFMRPVQTTQEDGCSRFPPEEEGGC ELRVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPENGGKPRRK NPQEGLYNELQDKMAEAYS EIGMKGERRRGKGHDGLYQGLSTATKDTYDALHM QALPPR
147 N-HA- 4131scFv- GS20-53B6	MALPVTALLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGSVTIKC QASQSFYNLLAWYQQKPGQPPKLLIYDASDLASGVPSRFGSGSGTDFTLTISD LECADAAAYYCQSDGSSYAFGGTEVVVKGGGGSGGGSGGGSGGGSGGQEQ LEESGGGLVKPEGSLLTCTASGVSFSSSYWIYWRQAPKGLEWIACIYTGSS GSTYYASWAKGRFTVSETSTTVTLQMTSLTAADTATYFCARASAWTYGMDLW PGTLVTVSSGGGGSGGGSGGGSGGGSGGGSGGQVQLQESGPGLVKPSSETLSLTCTV SGGSIYYSWTWVRQPAGQGLEWIGRIHISGVTNHNPSLKS RVSMSIDTSRTQFSL RLTSVTAADTAVYFCARSGGTYSAFDIWGQGTMTVTVSSGGGGSGGGSGGGSGG GGGSEIVLTQSPGTLTSLSPGERSTLSCRASQSFTSNYLAWYQQRPGQAPRLLI YGASTRAIGIPDRFSGSGGTDFTLTISRLEPEDFAVYYCQYQSSPWFQGGTK VEIKTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWA

TABLE 2-continued

Exemplary Protein Sequences	
SEQ ID NO: Name	Amino Acid Sequence
	APLAGTCGVLLSLVITLYCKRGRKLLYIFKQPFMRPVQTTQEEDGCS CRFPPE EEEGGCEL RVKFSRSADAPAYQQQNQLYNELNLGRREEYDVL DKRRGRDPEMG GKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGK GHDGLYQGLSTATKDT YDALHMQALPPR
148N-HA- 4131scFv- GS10-53B6	MALPVTALLPLALLLHAARPGYPYDVPDYADIVMTQTPASVSEPVGGSVTIKC QASQSFYNLLAWYQQKPGQPPKLLIYDASDLASGVPSRFGSGSGTDFTLTISD LECADAAYYCQSDAGSSYAFGGGTEVVVKGSGGSGGGSGGGSGGGSQEQ LEESGGGLVKPEGS LTLTCTASGVSFSSSYWIYVVRQAPGKGLEWIACIYTGSS GSTYYASWAKGRFTVSETSTTVTLQMTSLTAADTATYFCARASAWTYGMDLWG PGTLTVTVSSGGGSGGGSGGGSQVQLQESGPGLVKPS ETLTLCTVSGGSISSYSWT WVRQAPAGQGLEWIGRIHISGVTNHNPSLKSRVSMSIDTSRTQFSLRLTSVTAAD TAVYFCARSGGTYSAPDIWQGTMTVTVSSGGGSGGGSGGGSGGGSGGGSEIVLT QSPGTLSLSPGERSTLSCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGI PDRFSGSGSGTDFTLTISRLEPEDFAVYYCQQYGS SPWTFGQGTKEIKTTTPA PRPTPTPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWA PLAGTCGVLL LLSLVITLYCKRGRKLLYIFKQPFMRPVQTTQEEDGCS CRFPPEEEGGCEL RV KFSRSADAPAYQQQNQLYNELNLGRREEYDVL DKRRGRDPEMGGKPRRKNPQE GLYNELQKDKMAEAYSEIGMKGERRRGK GHDGLYQGLSTATKDTYDALHMQALP PR
149V5-GS20- 4131scFv- TPS6-35-53B6	MALPVTALLPLALLLHAARPGKPIPNPLLGLDSTGGGSGGGSGGGSGGGG SDIVMTQTPASVSEPVGGSVTIKCQASQSFYNLLAWYQQKPGQPPKLLIYDASD LASGVPSRFGSGSGTDFTLTISDLECADAAYYCQSDAGSSYAFGGGTEVVVKG GGGSGGGSGGGSGGGSGGGSGGQLEESGGGLVKPEGS LTLTCTASGVSFSSSY WIYVVRQAPGKGLEWIACIYTGSSGSTYYASWAKGRFTVSETSTTVTLQMTSL TAADTATYFCARASAWTYGMDLWGPGLTVTVSSGGGSGGGSGGGSGGGSPMAKKG GGGSGGGSGGGSGGQVQLQESGPGLVKPS ETLTLCTVSGGSISSYSWTWVRQP AGQGLEWIGRIHISGVTNHNPSLKSRVSMSIDTSRTQFSLRLTSVTAADTAVYF CARSGGTYSAPDIWQGTMTVTVSSGGGSGGGSGGGSGGGSGGGSEIVLTQSPGT LSLSPGERSTLSCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRS GSGSGTDFTLTISRLEPEDFAVYYCQQYGS SPWTFGQGTKEIKTTTPAPRPTPT PAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWA PLAGTCGVLLSLV ITLYCKRGRKLLYIFKQPFMRPVQTTQEEDGCS CRFPPEEEGGCEL RVKFSRS ADAPAYQQQNQLYNELNLGRREEYDVL DKRRGRDPEMGKPRRKNPQEGLYNE LQKDKMAEAYSEIGMKGERRRGK GHDGLYQGLSTATKDTYDALHMQALPPR
150V5-GS20- 4131scFv- TPS6-30-53B6	MALPVTALLPLALLLHAARPGKPIPNPLLGLDSTGGGSGGGSGGGSGGGG SDIVMTQTPASVSEPVGGSVTIKCQASQSFYNLLAWYQQKPGQPPKLLIYDASD LASGVPSRFGSGSGTDFTLTISDLECADAAYYCQSDAGSSYAFGGGTEVVVKG GGGSGGGSGGGSGGGSGGGSGGQLEESGGGLVKPEGS LTLTCTASGVSFSSSY WIYVVRQAPGKGLEWIACIYTGSSGSTYYASWAKGRFTVSETSTTVTLQMTSL TAADTATYFCARASAWTYGMDLWGPGLTVTVSSGGGSGGGSGGGSGGGSPMAKKG GGGSGGGSGGQVQLQESGPGLVKPS ETLTLCTVSGGSISSYSWTWVRQAPAGQL EWIGRIHISGVTNHNPSLKSRVSMSIDTSRTQFSLRLTSVTAADTAVYFCARS GTYSAFDIWQGTMTVTVSSGGGSGGGSGGGSGGGSGGGSEIVLTQSPGTLSP GERSTLSCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSG TDFTLTISRLEPEDFAVYYCQQYGS SPWTFGQGTKEIKTTTPAPRPTPTPTPT ASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWA PLAGTCGVLLSLVITLYC KGRKLLYIFKQPFMRPVQTTQEEDGCS CRFPPEEEGGCEL RVKFSRSADAPA YQQQNQLYNELNLGRREEYDVL DKRRGRDPEMGKPRRKNPQEGLYNELQKDK MAEAYS EIGMKGERRRGK GHDGLYQGLSTATKDTYDALHMQALPPR
151V5-GS20- 4131scFv- TPS6-25-53B6	MALPVTALLPLALLLHAARPGKPIPNPLLGLDSTGGGSGGGSGGGSGGGG SDIVMTQTPASVSEPVGGSVTIKCQASQSFYNLLAWYQQKPGQPPKLLIYDASD LASGVPSRFGSGSGTDFTLTISDLECADAAYYCQSDAGSSYAFGGGTEVVVKG GGGSGGGSGGGSGGGSGGGSGGQLEESGGGLVKPEGS LTLTCTASGVSFSSSY WIYVVRQAPGKGLEWIACIYTGSSGSTYYASWAKGRFTVSETSTTVTLQMTSL TAADTATYFCARASAWTYGMDLWGPGLTVTVSSGGGSGGGSGGGSPMAKKG GGGSGGQVQLQESGPGLVKPS ETLTLCTVSGGSISSYSWTWVRQAPAGQGLEWIGR IHSIGVTNHNPSLKSRVSMSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYS APDIWQGTMTVTVSSGGGSGGGSGGGSGGGSGGGSEIVLTQSPGTLSPGERST LSCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLT ISRLEPEDFAVYYCQQYGS SPWTFGQGTKEIKTTTPAPRPTPTPTPTPTPTPT SLRPEACRPAAGGAVHTRGLDFACDIYIWA PLAGTCGVLLSLVITLYCKRGRK KLLYIFKQPFMRPVQTTQEEDGCS CRFPPEEEGGCEL RVKFSRSADAPAYQQQ NQLYNELNLGRREEYDVL DKRRGRDPEMGKPRRKNPQEGLYNELQKDKMAEAY SEIGMKGERRRGK GHDGLYQGLSTATKDTYDALHMQALPPR
152V5-GS20- 4131scFv- TPS6-20-53B6	MALPVTALLPLALLLHAARPGKPIPNPLLGLDSTGGGSGGGSGGGSGGGG SDIVMTQTPASVSEPVGGSVTIKCQASQSFYNLLAWYQQKPGQPPKLLIYDASD LASGVPSRFGSGSGTDFTLTISDLECADAAYYCQSDAGSSYAFGGGTEVVVKG GGGSGGGSGGGSGGGSGGGSGGQLEESGGGLVKPEGS LTLTCTASGVSFSSSY WIYVVRQAPGKGLEWIACIYTGSSGSTYYASWAKGRFTVSETSTTVTLQMTSL

[illegible]

TABLE 2-continued

Exemplary Protein Sequences	
SEQ ID NO: Name	Amino Acid Sequence
	ITLYCKRGRKLLYIFKQPFMRPVQTTQEEDGCS CRFP EEEEGGCELRVKFSRS ADAPAYQQGQNQLYNELNLGRREEYDVL DKRRGRDPEMGGKPRRKNPQEGLYNE LQDKDMAEAYSEIGMKGERRRGK GHDGLYQGLSTATKDTYDALHMQALPPR
157V5-GS20- 4131scFv- TPS8-53B6	MALPVTALLPLALLLHAARPGKPIPNPLLGLDSTGGGSGGGSGGGSGGGG SDIVMTQTPASVSEPVGGSVTIKCQASQSFYNLLAWYQKPGQPPKLLIYDASD LASGVPSRFKSGSGTDFTLTISDLECAAAAYCQ SADGSSYAFGGGTEVVVK GGGSGSGGGSGGGSGGGSGQEQLEESGGGLVKPEGSLTLTCTASGVSFSSSY WIYVWRQAPKGLEWIACIYTGSSGSTYYASWAKGRFTVSETSTTVTLQMTSL TAADTATYFCARASAWTYGMDLWGPGLVTVSSGGGSGGGSGGSSPLGLAGS GGGSGGGSGMAKKGSGGGSGGQVQLQESGPGLVKPSSETLSLTCTVSGGSI YYSWTWVRQAPQGLEWIGRIHISGVTNHNPSLKS RVSMIDTSRTQFSLRLTS VTAADTAVYFCARSGGTYSAFDIWGQGTMTVSSGGGSGGGSGGGSGGGG EIVLTQSPGTLSSLSPGERSTLSCRASQSFTSNYLA WYQRPQAPRLLIYGAST RAIGIPDRFSGSGSGTDFTLTISRLEPEDFAVYQCQYGS SPWTFGQGTKEIK TTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAG TCGVLLLSLVITLYCKRGRKLLYIFKQPFMRPVQTTQEEDGCS CRFP EEEEGG CELRVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVL DKRRGRDPEMGGKPRR KNPQEGLYNELQDKDMAEAYSEIGMKGERRRGK GHDGLYQGLSTATKDTYDALH MQALPPR
158V5-GS20- 4131scFv- TPS9-53B6	MALPVTALLPLALLLHAARPGKPIPNPLLGLDSTGGGSGGGSGGGSGGGG SDIVMTQTPASVSEPVGGSVTIKCQASQSFYNLLAWYQKPGQPPKLLIYDASD LASGVPSRFKSGSGTDFTLTISDLECAAAAYCQ SADGSSYAFGGGTEVVVK GGGSGSGGGSGGGSGGGSGQEQLEESGGGLVKPEGSLTLTCTASGVSFSSSY WIYVWRQAPKGLEWIACIYTGSSGSTYYASWAKGRFTVSETSTTVTLQMTSL TAADTATYFCARASAWTYGMDLWGPGLVTVSSGGGSGGGSGGSSVHMLGEL GPGSGGGSGMAKKGSGGGSGGQVQLQESGPGLVKPSSETLSLTCTVSGGSI YYSWTWVRQAPQGLEWIGRIHISGVTNHNPSLKS RVSMIDTSRTQFSLRLTS VTAADTAVYFCARSGGTYSAFDIWGQGTMTVSSGGGSGGGSGGGSGGGG EIVLTQSPGTLSSLSPGERSTLSCRASQSFTSNYLA WYQRPQAPRLLIYGAST RAIGIPDRFSGSGSGTDFTLTISRLEPEDFAVYQCQYGS SPWTFGQGTKEIK TTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAG TCGVLLLSLVITLYCKRGRKLLYIFKQPFMRPVQTTQEEDGCS CRFP EEEEGG CELRVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVL DKRRGRDPEMGGKPRR KNPQEGLYNELQDKDMAEAYSEIGMKGERRRGK GHDGLYQGLSTATKDTYDALH MQALPPR
159V5-GS20- 4131scFv- TPS10-53B6	MALPVTALLPLALLLHAARPGKPIPNPLLGLDSTGGGSGGGSGGGSGGGG SDIVMTQTPASVSEPVGGSVTIKCQASQSFYNLLAWYQKPGQPPKLLIYDASD LASGVPSRFKSGSGTDFTLTISDLECAAAAYCQ SADGSSYAFGGGTEVVVK GGGSGSGGGSGGGSGGGSGQEQLEESGGGLVKPEGSLTLTCTASGVSFSSSY WIYVWRQAPKGLEWIACIYTGSSGSTYYASWAKGRFTVSETSTTVTLQMTSL TAADTATYFCARASAWTYGMDLWGPGLVTVSSGGGSGGGSGGSSPLVRGKG GGGSGGGSGMAKKGSGGGSGGQVQLQESGPGLVKPSSETLSLTCTVSGGSI YYSWTWVRQAPQGLEWIGRIHISGVTNHNPSLKS RVSMIDTSRTQFSLRLTS VTAADTAVYFCARSGGTYSAFDIWGQGTMTVSSGGGSGGGSGGGSGGGG EIVLTQSPGTLSSLSPGERSTLSCRASQSFTSNYLA WYQRPQAPRLLIYGAST RAIGIPDRFSGSGSGTDFTLTISRLEPEDFAVYQCQYGS SPWTFGQGTKEIK TTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAG TCGVLLLSLVITLYCKRGRKLLYIFKQPFMRPVQTTQEEDGCS CRFP EEEEGG CELRVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVL DKRRGRDPEMGGKPRR KNPQEGLYNELQDKDMAEAYSEIGMKGERRRGK GHDGLYQGLSTATKDTYDALH MQALPPR
160V5-GS20- 4131scFv- TPS11-53B6	MALPVTALLPLALLLHAARPGKPIPNPLLGLDSTGGGSGGGSGGGSGGGG SDIVMTQTPASVSEPVGGSVTIKCQASQSFYNLLAWYQKPGQPPKLLIYDASD LASGVPSRFKSGSGTDFTLTISDLECAAAAYCQ SADGSSYAFGGGTEVVVK GGGSGSGGGSGGGSGGGSGQEQLEESGGGLVKPEGSLTLTCTASGVSFSSSY WIYVWRQAPKGLEWIACIYTGSSGSTYYASWAKGRFTVSETSTTVTLQMTSL TAADTATYFCARASAWTYGMDLWGPGLVTVSSGGGSGGGSGGSSPLGLAGS GSGGGSRQARVVNGGGSGGGSGGQVQLQESGPGLVKPSSETLSLTCTVSGGSI YYSWTWVRQAPQGLEWIGRIHISGVTNHNPSLKS RVSMIDTSRTQFSLRLTS VTAADTAVYFCARSGGTYSAFDIWGQGTMTVSSGGGSGGGSGGGSGGGG EIVLTQSPGTLSSLSPGERSTLSCRASQSFTSNYLA WYQRPQAPRLLIYGAST RAIGIPDRFSGSGSGTDFTLTISRLEPEDFAVYQCQYGS SPWTFGQGTKEIK TTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAG TCGVLLLSLVITLYCKRGRKLLYIFKQPFMRPVQTTQEEDGCS CRFP EEEEGG CELRVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVL DKRRGRDPEMGGKPRR KNPQEGLYNELQDKDMAEAYSEIGMKGERRRGK GHDGLYQGLSTATKDTYDALH MQALPPR

Exemplary Protein Sequences

Exemplary Protein Sequences	
SEQ ID	Amino Acid Sequence
161V5-G520-4131scFv-TPS12-53B6	MALPVTALLPLALLHAARP GKPIPNPLLGLDSTGGGSGGGGSGGGSGGGGSDIVMTQTPASVSEPVGGSVTIKQASQSFYNLLAWYQKPGQPPKLLIYDASDLASGVP SRFKGSGSGTDFTLTI SDLECAADAAAYYCQ SADGSSYAFGGGTEVVVKGGGGSGGGSGGGSGGGSGGGGQEQLEESGGGLVKPEGSLTLTCTASGVFSSSYWIYVWRQAPGKGLEWIA CIYTGSSGSTYYASWAKGRFTVSETSTTVTLQM TSLTAADTATYFCARASAWTYGMDLWGPGLTVTVSSGGGSGGGSGGGSGGGSGGSGGSGRQARVVMGSGGGSGGGSGGGGQEQLEESGGGLVKPSETLSLTC TVSGGSI SYYSWTWVRQPAQGQLEWIGRIHISGVTNHNPSLKS RVSM SIDTSRTQFSLRLTSVTAADTAVFYCARSGGTYSAPDIWQGQTMVTVSSGGGSGGGSGGGSGGGSGGGSEI VLTQSPGTL SLS PGRSTLSCRASQSFSTNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLTI SRLEPEDFAVYYCQYQSGSPWTFQGQTKVEIKTTTPAPRPPTPAPT IASQPLSLRPEACRPAAGGAVHTRGLDFACDIYI WAPLAGTCGVLLSLSVITLYCKRGRKLLYIFKQPFMRPVQTTQEE DGCSCRFPEEEEGGCEL RVKFSRSADAPAYQQQGNQLYNELNLGRREEYDVL DKRRGRDPEMGGKPRRKNPQEGLYNELQDKMAEAYS EIMKGERRRGKHDGLYQGLSTATKDTYDALHMQALPPR
162V5-G520-4131scFv-TPS13-53B6	MALPVTALLPLALLHAARP GKPIPNPLLGLDSTGGGSGGGGSGGGSGGGGSDIVMTQTPASVSEPVGGSVTIKQASQSFYNLLAWYQKPGQPPKLLIYDASDLASGVP SRFKGSGSGTDFTLTI SDLECAADAAAYYCQ SADGSSYAFGGGTEVVVKGGGGSGGGSGGGSGGGSGGGGQEQLEESGGGLVKPEGSLTLTCTASGVFSSSYWIYVWRQAPGKGLEWIA CIYTGSSGSTYYASWAKGRFTVSETSTTVTLQM TSLTAADTATYFCARASAWTYGMDLWGPGLTVTVSSGGGSGGGSGGGSGGGSGGSGGSGRQARVVMGSGGGSGGGSGGGGQEQLEESGGGLVKPSETLSLTC TVSGGSI SYYSWTWVRQPAQGQLEWIGRIHISGVTNHNPSLKS RVSM SIDTSRTQFSLRLTSVTAADTAVFYCARSGGTYSAPDIWQGQTMVTVSSGGGSGGGSGGGSGGGSGGGSEI VLTQSPGT LSLSPGRSTLSCRASQSFSTNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLTI SRLEPEDFAVYYCQYQSGSPWTFQGQTKVEIKTTTPAPRPPTPAPT IASQPLSLRPEACRPAAGGAVHTRGLDFACDIYI WAPLAGTCGVLLSLSVITLYCKRGRKLLYIFKQPFMRPVQTTQEE DGCSCRFPEEEEGGCEL RVKFSRSADAPAYQQQGNQLYNELNLGRREEYDVL DKRRGRDPEMGGKPRRKNPQEGLYNELQDKMAEAYS EIMKGERRRGKHDGLYQGLSTATKDTYDALHMQALPPR
163V5-G520-4131scFv-G545-53B6	MALPVTALLPLALLHAARP GKPIPNPLLGLDSTGGGSGGGGSGGGSGGGGSDIVMTQTPASVSEPVGGSVTIKQASQSFYNLLAWYQKPGQPPKLLIYDASDLASGVP SRFKGSGSGTDFTLTI SDLECAADAAAYYCQ SADGSSYAFGGGTEVVVKGGGGSGGGSGGGSGGGSGGGGQEQLEESGGGLVKPEGSLTLTCTASGVFSSSYWIYVWRQAPGKGLEWIA CIYTGSSGSTYYASWAKGRFTVSETSTTVTLQM TSLTAADTATYFCARASAWTYGMDLWGPGLTVTVSSGGGSGGGSGGGSGGGSGGSGGSGGSGGSGGGSGGGSGGGSGGGGQEQLEESGGGLVKPSETLSLTC TVSGGSI SYYSWTWVRQPAQGQLEWIGRIHISGVTNHNPSLKS RVSM SIDTSRTQFSLRLTSVTAADTAVFYCARSGGTYSAPDIWQGQTMVTVSSGGGSGGGSGGGSGGGSGGGSEI VLTQSPGT LSLSPGRSTLSCRASQSFSTNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLTI SRLEPEDFAVYYCQYQSGSPWTFQGQTKVEIKTTTPAPRPPTPAPT IASQPLSLRPEACRPAAGGAVHTRGLDFACDIYI WAPLAGTCGVLLSLSVITLYCKRGRKLLYIFKQPFMRPVQTTQEE DGCSCRFPEEEEGGCEL RVKFSRSADAPAYQQQGNQLYNELNLGRREEYDVL DKRRGRDPEMGGKPRRKNPQEGLYNELQDKMAEAYS EIMKGERRRGKHDGLYQGLSTATKDTYDALHMQALPPR
164v5-G520-4131scFv-G535-53B6	MALPVTALLPLALLHAARP GKPIPNPLLGLDSTGGGSGGGGSGGGSGGGGSDIVMTQTPASVSEPVGGSVTIKQASQSFYNLLAWYQKPGQPPKLLIYDASDLASGVP SRFKGSGSGTDFTLTI SDLECAADAAAYYCQ SADGSSYAFGGGTEVVVKGGGGSGGGSGGGSGGGSGGGGQEQLEESGGGLVKPEGSLTLTCTASGVFSSSYWIYVWRQAPGKGLEWIA CIYTGSSGSTYYASWAKGRFTVSETSTTVTLQM TSLTAADTATYFCARASAWTYGMDLWGPGLTVTVSSGGGSGGGSGGGSGGGSGGSGGSGGSGGSGGGSGGGSGGGSGGGGQEQLEESGGGLVKPSETLSLTC TVSGGSI SYYSWTWVRQPAQGQLEWIGRIHISGVTNHNPSLKS RVSM SIDTSRTQFSLRLTSVTAADTAVFYCARSGGTYSAPDIWQGQTMVTVSSGGGSGGGSGGGSGGGSGGGSEI VLTQSPGT LSLSPGRSTLSCRASQSFSTNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLTI SRLEPEDFAVYYCQYQSGSPWTFQGQTKVEIKTTTPAPRPPTPAPT IASQPLSLRPEACRPAAGGAVHTRGLDFACDIYI WAPLAGTCGVLLSLSVITLYCKRGRKLLYIFKQPFMRPVQTTQEE DGCSCRFPEEEEGGCEL RVKFSRSADAPAYQQQGNQLYNELNLGRREEYDVL DKRRGRDPEMGGKPRRKNPQEGLYNELQDKMAEAYS EIMKGERRRGKHDGLYQGLSTATKDTYDALHMQALPPR
165V5-GS20-4131scFv-GS25-53B6	MALPVTALLPLALLHAARP GKPIPNPLLGLDSTGGGSGGGGSGGGSGGGGSDIVMTQTPASVSEPVGGSVTIKQASQSFYNLLAWYQKPGQPPKLLIYDASDLASGVP SRFKGSGSGTDFTLTI SDLECAADAAAYYCQ SADGSSYAFGGGTEVVVKGGGGSGGGSGGGSGGGSGGGGQEQLEESGGGLVKPEGSLTLTCTASGVFSSSYWIYVWRQAPGKGLEWIA CIYTGSSGSTYYASWAKGRFTVSETSTTVTLQM TSLTAADTATYFCARASAWTYGMDLWGPGLTVTVSSGGGSGGGSGGGSGGGSGGGSGGSGGSGGSGGGSGGGSGGGSGGGGQEQLEESGGGLVKPSETLSLTC TVSGGSI SYYSWTWVRQPAQGQLEWIGRIHISGVTNHNPSLKS RVSM SIDTSRTQFSLRLTSVTAADTAVFYCARSGGTYSAPDIWQGQTMVTVSSGGGSGGGSGGGSGGGSGGGSEI VLTQSPGT LSLSPGRSTLSCRASQSFSTNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLTI SRLEPEDFAVYYCQYQSGSPWTFQGQTKVEIKTTTPAPRPPTPAPT IASQPLSLRPEACRPAAGGAVHTRGLDFACDIYI WAPLAGTCGVLLSLSVITLYCKRGRKLLYIFKQPFMRPVQTTQEE DGCSCRFPEEEEGGCEL RVKFSRSADAPAYQQQGNQLYNELNLGRREEYDVL DKRRGRDPEMGGKPRRKNPQEGLYNELQDKMAEAYS EIMKGERRRGKHDGLYQGLSTATKDTYDALHMQALPPR

TABLE 2-continued

Exemplary Protein Sequences	
SEQ ID NO: Name	Amino Acid Sequence
	IHISGVTNHNPSLKSRVSMIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYSA FDIWQQTMTVTVSSGGGGSGGGSGGGSGGGSGGGSEIVLTQSPGTLTSLSPGERST LSCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTL TISRLEPEDFAVYQCQYQSSPWTFGQGTKEIKTTTPAPRPPTPAPTIASQPL SLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVIITLYCKRGRK KLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCELRVKFSRSADAPAYQQGQ NQLYNELNLGRREYDVLDKRRGRDPGEMGKPRKPNQEGLYNELQKDKMAEAY SEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
166N-HA-IgV- G545-53B6	MALPVTALLPLALLLHAARPGYPYDVPDYAQFSLGSPILAMVGEDADLPC HLFPTMSAETMELKMWSSSLRQVNVYADGKEVEDRQSAFYRGRTSILRDGITA GKAALRIHNVTASDSGKYLICYFDGDFYEKALVELKVAGGGSGGGSGGGSGGG GSGSGGGSGGGSGGGSGGGSGGGSGGGSGVQLQESGPGLVKPSSETLSLTCTVS GGSISYYSWTWVRQPAQGQLEWIGRIHISGVTNHNPSLKSRVSMIDTSRTQFS LRLTSVTAADTAVYFCARSGGTYSAFDIWQGTMTVTVSSGGGGSGGGSGGGSGGG GGGSEIVLTQSPGTLTSLSPGERSTLSCRASQSFTSNYLAWYQQRPGQAPRLLI YGASTRAIGIPDRFSGSGSGTDFTLTISRLEPEDFAVYQCQYQSSPWTFGQGT KVEIKTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIW APLAGTCGVLLLSLVIITLYCKRGRKLLYIFKQPFMRPVQTTQEEDGCSCRFPE EEEGGCELRVKFSRSADAPAYQQGQNLYNELNLGRREYDVLDKRRGRDPGEMG KPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDT YDALHMQALPPR
167N-HA-IgC- G545-53B6	MALPVTALLPLALLLHAARPGYPYDVPDYAALGSDLHVDVKYKDGGIHLECR STGWYQPPQIQWSNNKGENIPTVEAPVVADGVLGYAVAASVIMRGSSEGEVST IRSLLGLEKTASISIAADPFPRSAQRWIAALAGTGGGGSGGGSGGGSGGGSGGS GGGSGGGSGGGSGGGSGGGSGGGSGVQLQESGPGLVKPSSETLSLTCTVSGGSI SYYSWTWVRQPAQGQLEWIGRIHISGVTNHNPSLKSRVSMIDTSRTQFSLRLT SVTAADTAVYFCARSGGTYSAFDIWQGTMTVTVSSGGGGSGGGSGGGSGGGSGGG SEIVLTQSPGTLTSLSPGERSTLSCRASQSFTSNYLAWYQQRPGQAPRLLIYGAS TRAIGIPDRFSGSGSGTDFTLTISRLEPEDFAVYQCQYQSSPWTFGQGTKEIK KTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLA GTCGVLLLSLVIITLYCKRGRKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEEG GCELRVKFSRSADAPAYQQGQNLYNELNLGRREYDVLDKRRGRDPGEMGKPR RKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDAL HMQALPPR
168N-HA-BTN3A1- G545-53B6	MALPVTALLPLALLLHAARPGYPYDVPDYAQFSLGSPILAMVGEDADLPC HLFPTMSAETMELKMWSSSLRQVNVYADGKEVEDRQSAFYRGRTSILRDGITA GKAALRIHNVTASDSGKYLICYFDGDFYEKALVELKVAALGSDLHVDVKYKDG GIHLECRSTGWYQPPQIQWSNNKGENIPTVEAPVVADGVLGYAVAASVIMRGS GEGVSCITIRSLLGLEKTASISIAADPFPRSAQRWIAALAGTGGGGSGGGSGGG SGGSGSGGGSGGGSGGGSGGGSGGGSGVQLQESGPGLVKPSSETLSLTCTV TVSGGSIYYSWTWVRQPAQGQLEWIGRIHISGVTNHNPSLKSRVSMIDTSRT QFSLRLTSVTAADTAVYFCARSGGTYSAFDIWQGTMTVTVSSGGGGSGGGSGGG GGGSEIVLTQSPGTLTSLSPGERSTLSCRASQSFTSNYLAWYQQRPGQAPR LLIYGASTRAIGIPDRFSGSGSGTDFTLTISRLEPEDFAVYQCQYQSSPWTFG QGTKEIKTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDI YIWAPLAGTCGVLLLSLVIITLYCKRGRKLLYIFKQPFMRPVQTTQEEDGCSC RFPEEEEGCELRVKFSRSADAPAYQQGQNLYNELNLGRREYDVLDKRRGRDP EMGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQGLSTAT KDTYDALHMQALPPR
169N-HA-sec49k- G545-53B6	MALPVTALLPLALLLHAARPGYPYDVPDYAGFHTINATWWANITLVGPPDTPV TWYDTQGLWFCNGSRVKNPQIRHTCNDQNLTLIHVNKTYERTYMGYNRQGTKE DYKVVVIPPATVKQPEPEYVVFYMGENKTELEGPGTPVTWFNQDGKFCBG EKVHLHPEFNHTCDKQNLILLFVNFTHDGAYLGYNHQGTQRTHYEVTVLDFPDS QGMKIEHNSEETEKNDEHNHWKQGGQKQGGQKTNTKVNDRRKTAQKRPSKL KPATIEAMLTVTAGSNLTLVGPKAEGKVTFDGDLRKPCPEPNYRLRHECNNQN LTLINVTKDYEGTYGTNDKDEGKRYRVKNTTNSQSVKIOPYTRQTTPDQEHK FELQFETNGNYDSKIPGGGGSGGGSGGGSGGGSGGGSGGGSGGGSGGGSGGG GSGGGSGVQLQESGPGLVKPSSETLSLTCTVSGGSIYYSWTWVRQPAQGQLEW IGRHISGVTNHNPSLKSRVSMIDTSRTQFSLRLTSVTAADTAVYFCARSGGT YSAFDIWQGTMTVTVSSGGGGSGGGSGGGSGGGSGGGSEIVLTQSPGTLTSLSPGE RSTLSCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSGTDF TLTISRLEPEDFAVYQCQYQSSPWTFGQGTKEIKTTTPAPRPPTPAPTIAS QPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVIITLYCKR GRKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCELRVKFSRSADAPAYQ QGQNLYNELNLGRREYDVLDKRRGRDPGEMGKPRRKNPQEGLYNELQKDKMA EAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR

TABLE 2-continued

Exemplary Protein Sequences	
SEQ ID NO: Name	Amino Acid Sequence
170N-HA-UL11-GS45-53B6	MALPVTALLLPLALLLHAARPGYPYDVPDYAHDACIPVVGKIGTNTVLTNAVDFH PGDHVRWSYGGGAGYMLCVYTGSWTEYKKPDIIFKCLSNNSLLLINVTVNYTN TYRSLTSLNNWVHNQHHHKFPGNLDTCSYSLTVNENGTFFPTTTTKKPTTTTRTT TTTTTKKTTTTTTAAKKTISTTHKHSSPKSSTPNSHVEHHVGFATAAE TPLQPSQHQHVATHGGGGSGGGSGGGSGGGSGGGSGGGSGGGSGGGSGGGG SGGGSGVQLQESGPGLVKPSSETLSLTCTVSGGSISSYSWTWVRQPAQGQLEWI GRIHISGVTNHNPSLKSRSMSIDTSRTQFSRLTSVTAADTAVYFCARSGGTY SAFDIWQGTMTVTVSSGGGGSGGGSGGGSGGGSGGGSEIVLTQSPGTLSPGER STLSCRASQSFSTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSGTDF TLTISRLEPEDFAVYYCQYGGSSPWTFGQGTKEIKTTTPAPRPPTPAPTIASQ PLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVI TLYCKRG RKKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCELRVKFSRSADAPAYQQ GQNQLYNELNLGRREEYDVLDRKRRDPEMGGKPRRKNPQEGLYNELQDKDMAE AYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR

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MUC16-Specific CARs

In another aspect, provided herein is a CAR that specifically binds to MUC16, wherein the CAR comprises an extracellular ligand-binding domain comprising: a V_H region having the sequence shown in SEQ ID NO: 58, and/or a V_L region having the sequence shown in SEQ ID NO: 59. In some embodiments, the V_H and V_L are linked together by a flexible linker. In some embodiments, a flexible linker comprises the amino acid sequence shown in SEQ ID NO: 57 ((GGGGGS)₃).

In another aspect, provided is a MUC16 specific chimeric antigen receptor (CAR) comprising an extracellular ligand-binding domain, a first transmembrane domain, and an intracellular signaling domain, wherein the extracellular domain comprises a single chain Fv fragment (scFv) comprising a heavy chain variable (V_H) region and a light chain variable (V_L) region, wherein the V_H region comprises (i) a V_H complementarity determining region one (CDR1) having the amino acid sequence shown in SEQ ID NO: 60 or 63; (ii) a V_H complementarity determining region two (CDR2) having the amino acid sequence shown in SEQ ID NO: 61 or 64; and (iii) a V_H complementarity determining region three (CDR3) having the amino acid sequence shown in SEQ ID NO: 62 or 65.

In another aspect, provided is a MUC16 specific chimeric antigen receptor (CAR) comprising an extracellular ligand-binding domain, a first transmembrane domain, and an intracellular signaling domain, wherein the extracellular domain comprises a single chain Fv fragment (scFv) comprising a heavy chain variable (V_H) region and a light chain variable (V_L) region, wherein the V_L region comprises (i) a V_L complementarity determining region one (CDR1) having the amino acid sequence shown in SEQ ID NO: 66 or 69; (ii) a V_L complementarity determining region two (CDR2) having the amino acid sequence shown in SEQ ID NO: 67 or 70;

and (iii) a V_L complementarity determining region three (CDR3) having the amino acid sequence shown in SEQ ID NO: 68 or 71.

In another aspect, provided is a MUC16 specific chimeric antigen receptor (CAR) comprising an extracellular ligand-binding domain, a first transmembrane domain, and an intracellular signaling domain, wherein the extracellular domain comprises a single chain Fv fragment (scFv) comprising a heavy chain variable (V_H) region and a light chain variable (V_L) region, wherein (a) the V_H region comprises (i) a V_H complementarity determining region one (CDR1) having the amino acid sequence shown in SEQ ID NO: 60 or 63; (ii) a V_H complementarity determining region two (CDR2) having the amino acid sequence shown in SEQ ID NO: 61 or 64; and (iii) a V_H complementarity determining region three (CDR3) having the amino acid sequence shown in SEQ ID NO: 62 or 65; and (b) the V_L region comprises (i) a V_L complementarity determining region one (CDR1) having the amino acid sequence shown in SEQ ID NO: 66; (ii) a V_L complementarity determining region two (CDR2) having the amino acid sequence shown in SEQ ID NO: 67; and (iii) a V_L complementarity determining region three (CDR3) having the amino acid sequence shown in SEQ ID NO: 68.

Also provided herein are CDR portions of antigen binding domains of antibodies to MUC16 or CDR portions of extracellular ligand-binding domains of CARs to MUC16 (including Chothia, Kabat CDRs, and CDR contact regions). Determination of CDR regions is well within the skill of the art. It is understood that in some embodiments, CDRs can be a combination of the Kabat and Chothia CDR (also termed "combined CRs" or "extended CDRs"). In some embodiments, the CDRs are the Kabat CDRs. In other embodiments, the CDRs are the Chothia CDRs. In other words, in embodiments with more than one CDR, the CDRs may be any of Kabat, Chothia, combination CDRs, or combinations thereof.

TABLE 3

53B6 MUC16 CAR component sequences		
Clone	VH Sequence	SEQ ID NO:
53B6-VH	QVQLQESGPGLVKPSSETLSLTCTVSGGSISSYSWTWVRQPA QGQLEWIGRIHISGVTNHNPSLKSRSMSIDTSRTQFSRLR TSVTAADTAVYFCARSGGTYSAFDIWQGTMTVTVSS	58

TABLE 3-continued

53B6 MUC16 CAR component sequences		
Clone		SEQ ID NO:
	<u>VL sequence</u>	
53B6-VL	EIVLTQSPGTLSSLSPGERSTLSCRASQSPTSNYLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTLTISRLEPE DFAVYYCQQYGSSPWTFGQGTKVEIK	59
	<u>CDRs for 53B6 VH (Kabat definition)</u>	
CDR1	YYSWT	60
CDR2	RIHISGVTNHNPSLKS	61
CDR3	SGGTYSAFDI	62
	<u>CDRs for 53B6 VH (Chothia definition)</u>	
CDR1	SGGSISYY	63
CDR2	HISGV	64
CDR3	SGGTYSAFDI	65
	<u>CDRs for 53B6 VL (Kabat definition)</u>	
CDR1	RASQSFTSNYLA	66
CDR2	GASTRAI	67
CDR3	QQYGSSPWT	68
	<u>CDRs for 53B6 VL (Chothia definition)</u>	
CDR1	RASQSFTSNYLA	69
CDR2	GASTRAI	70
CDR3	QQYGSSPWT	71
	<u>CAR Amino Acid Sequences</u>	
CD8α signal sequence	MALPVTALLLPLALLLHAARP	1
GS linker 2	GGGGSGGGSGGGSGGGGS	72
CD8α hinge and transmembrane regions	TTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLD FACDIYIWAPLAGTCGVLLSLVIT	73
41BB cytoplasmic signaling domain	KRGRKKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCE L	33
CD3ζ cytoplasmic signaling domain	RVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGR DPEMGGKPRKPNPQEGLYNELQKDKMAEAYSEIGMKGERRR GKGHDGLYQGLSTATKDTYDALHMQALPPR	34
	<u>Mimotope sequences</u>	
Rituximab		
Mimotope	CPYSNPSLC	50
<u>QBEND-10</u>		
Epitope	ELPTQGTFSNVSTNVSPAKPTTA	74
Epitope	ELPTQGTFSNVSTNVS	179

TABLE 4

53B6 CAR sequences			
Clone Name	Name/Component	Sequence	SEQ ID NO:
53B6	CD8 α signal sequence, 53B6 scFv, CD8 α hinge and transmembrane regions, 41BB cytoplasmic signaling domain, CD3 ζ cytoplasmic signaling domain	MALPVTALLLPLALLLHAARPQVQLQESGPGLVKPS ETLSLTCTVSGGSI SYSWTWVRQAPAGGLEWIGRI HISGVTNHNPSLKS RVSMISDTSRTQFSLRLTSVTA ADTAVYFCARSGGTYS AFDIWGQGTMTVTVSSGGGGS GGGGSGGGSGGGGSEIVLTQSPGTLSSLSPGERSTL SCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAI GIPDRFSGSGSGTDFTLTISRLEPEDFAVYYCQQYG SSPWTFGQGTKVEIKTTTPAPRPPTPAPTIASQPLS LRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCG VLLLSLVITLYCKRGRKKLLYIFKQPFMRPVQTTQE EDGCSRFPPEEEEGGCELRVKFSRSADAPAYQQGQN QLYNELNLGRREEYDVLDKRRGRDPENGGKPRRKNP QEGLYNELQKDKMAEAYSEIGMKGERRRGKGDGLY QGLSTATKDTYDALHMQALPPR	75
53B6-SR2	CD8 α signal sequence, 53B6 scFv, CD20 mimotope, CD20 mimotope, CD8 α hinge and transmembrane regions, 41BB cytoplasmic signaling domain, CD3 ζ cytoplasmic signaling domain	MALPVTALLLPLALLLHAARPQVQLQESGPGLVKPS ETLSLTCTVSGGSI SYSWTWVRQAPAGGLEWIGRI HISGVTNHNPSLKS RVSMISDTSRTQFSLRLTSVTA ADTAVYFCARSGGTYS AFDIWGQGTMTVTVSSGGGGS GGGGSGGGSGGGGSEIVLTQSPGTLSSLSPGERSTL SCRASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAI GIPDRFSGSGSGTDFTLTISRLEPEDFAVYYCQQYG SSPWTFGQGTKVEIKSGGGGSCPYSNPSLCSGGGG SCPYSNPSLCSGGGGSTTPAPRPPTPAPTIASQPL SLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTC GVLLLSLVITLYCKRGRKKLLYIFKQPFMRPVQTTQ EEDGCSRFPPEEEEGGCELRVKFSRSADAPAYQQGQ NQLYNELNLGRREEYDVLDKRRGRDPENGGKPRRKN PQEGLYNELQKDKMAEAYSEIGMKGERRRGKGDGL YQGLSTATKDTYDALHMQALPPR	76
53B6-RSR	CD8 α signal sequence, CD20 mimotope, 53B6 scFv, CD20 mimotope, CD8 α hinge and transmembrane regions, 41BB cytoplasmic signaling domain, CD3 ζ cytoplasmic signaling domain	MALPVTALLLPLALLLHAARPGGGGSCPYSNPSLCS GGGSQVQLQESGPGLVKPSSETLSLTCTVSGGSI SY SWTWVRQAPAGGLEWIGRIHISGVTNHNPSLKS RV MSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYS FDIWGQGTMTVTVSSGGGSGGGGSGGGGSGGGGSEI VLTQSPGTLSSLSPGERSTLS CRASQSFTSNYLAWYQ QRPQAPRLLIYGASTRAIGIPDRFSGSGSGTDFTL TISRLEPEDFAVYYCQQYGS SPWTFGQGTKVEIKG GGSCPYSNPSLCTTPAPRPPTPAPTIASQPLSLRP EACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLL LSLVITLYCKRGRKKLLYIFKQPFMRPVQTTQEEDG CSCRFPPEEEEGGCELRVKFSRSADAPAYQQGQNQLY NELNLGRREEYDVLDKRRGRDPENGGKPRRKNPQEG LYNELQKDKMAEAYSEIGMKGERRRGKGDGLYQGL STATKDTYDALHMQALPPR	77
53B6-R2S	CD8 α signal sequence, CD20 mimotope, CD20 mimotope, 53B6 scFv, CD8 α hinge and transmembrane regions, 41BB cytoplasmic signaling domain, CD3 ζ cytoplasmic signaling domain	MALPVTALLLPLALLLHAARPGGGGSCPYSNPSLCS GGGSCPYSNPSLCSGGGSQVQLQESGPGLVKPSSETL SLTCTVSGGSI SYSWTWVRQAPAGGLEWIGRIHIS GVTNHNPSLKS RVSMISDTSRTQFSLRLTSVTAADT AVYFCARSGGTYS AFDIWGQGTMTVTVSSGGGSGGGG SGGGSGGGGSEIVLTQSPGTLSSLSPGERSTLS CR ASQSFTSNYLAWYQQRPGQAPRLLIYGASTRAIGIP DRFSGSGSGTDFTLTISRLEPEDFAVYYCQQYGS SP WTFGQGTKVEIKTTTPAPRPPTPAPTIASQPLSLRP EACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLL LSLVITLYCKRGRKKLLYIFKQPFMRPVQTTQEEDG CSCRFPPEEEEGGCELRVKFSRSADAPAYQQGQNQLY NELNLGRREEYDVLDKRRGRDPENGGKPRRKNPQEG LYNELQKDKMAEAYSEIGMKGERRRGKGDGLYQGL STATKDTYDALHMQALPPR	78
53B6-QR3	CD8 α signal sequence, CD20 mimotope, 53B6 scFv, CD20 mimotope, QBEND-10 epitope, CD20 mimotope, hinge and transmembrane regions of human CD8 α	MALPVTALLLPLALLLHAARPGGGGSCPYSNPSLCS GGGGSGGGGSGVQLQESGPGLVKPSSETLSLTCTVSG GSISYYSWTVVRQAPAGGLEWIGRIHISGVTNHNPS LKS RVSMISDTSRTQFSLRLTSVTAADTAVYFCARS GGTYS AFDIWGQGTMTVTVSSGGGSGGGGSGGGGSG GGGSEIVLTQSPGTLSSLSPGERSTLS CRASQSFTSN YLAWYQQRPGQAPRLLIYGASTRAIGIPDRFSGSGS GTDFTLTISRLEPEDFAVYYCQQYGS SPWTFGQGTK VEIKSGGGGSCPYSNPSLCSGGGGSELPTQGTFSN VSTNVS PAKPTTACPYSNPSLCTTPAPRPPTPAP TIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIW APLAGTCGVLLLSLVITLYCKRGRKKLLYIFKQPFM	79

TABLE 4-continued

53B6 CAR sequences			
Clone Name	Name/Component	Sequence	SEQ ID NO:
	molecule, 41BB	RPVQTTQEEDGCSRFPPEEEGGCELLRVKFSRSAD	
	signaling	APAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPEM	
	domain, CD3 ζ	GGKPRRKNPQEGLYNELQDKMAEAYSEIGMGKERR	
	signaling	RGKGHDGLYQGLSTATKDTYDALHMQALPPR	
	domain		
53B6- RSR (2)	CD8 α signal	MALPVTALLLPLALLLHAARPGGGGSCPYSNPSLCG	180
	sequence, CD20	GGGSQVQLQESGPGLVKPSSETLSLTCTVSGGSISYY	
	mimotope, 53B6	SWTWVRQPAQGQLEWIGRIHISGVTNHNPSLKSRVS	
	scFv, CD20	MSIDTSRTQFSLRLTSVTAADTAVYFCARSGGTYS	
	mimotope, CD8 α	FDIWGGQTMVTVSSGGGGSGGGGGSGGGGGSEI	
	hinge and	VLTQSPGTLSSLSPGERSTLSCRASQSFTSNYLAWYQ	
	transmembrane	QRPQGAPRLLIYGASTRAIGIPDRFSGSGSGTDFTL	
	regions, 41BB	TISRLEPEDFAVYQCQYGGSSPWTFGQGTKVEIKGG	
	cytoplasmic	GGSCPYSNPSLCGGGGSTTTPAPRPPTPAPTIASQP	
	signaling	LSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGT	
	domain, CD3 ζ	CGVLLLSLVITLYCKRGRKKLLYIFKQPFMRPVQTT	
	cytoplasmic	QEEDGCSRFPPEEEGGCELLRVKFSRSADAPAYQQG	
	signaling	QNQLYNELNLGRREEYDVLDKRRGRDPEMGKPRRK	
	domain	NPQEGLYNELQDKMAEAYSEIGMGERRRGKHDG	
		LYQGLSTATKDTYDALHMQALPPR	

The disclosure encompasses modifications to the CARs and polypeptides shown in Table 4, including functionally equivalent CARs having modifications which do not significantly affect their properties and variants which have enhanced or decreased activity and/or affinity. For example, the amino acid sequence may be mutated to obtain an antibody with the desired binding affinity to MUC16. Modification of polypeptides is routine practice in the art and need not be described in detail herein. Examples of modified polypeptides include polypeptides with conservative substitutions of amino acid residues, one or more deletions or additions of amino acids which do not significantly deleteriously change the functional activity, or which mature (enhance) the affinity of the polypeptide for its ligand, or use of chemical analogs.

Amino acid sequence insertions include amino- and/or carboxyl-terminal fusions ranging in length from one residue to polypeptides containing a hundred or more residues, as well as intrasequence insertions of single or multiple amino acid residues. Examples of terminal insertions include an antibody with an N-terminal methionyl residue or the antibody fused to an epitope or mimotope tag. Other insertional variants of the antibody molecule include the fusion to the N- or C-terminus of the antibody of an enzyme or a polypeptide which increases the half-life of the antibody in the blood circulation.

Substitution variants have at least one amino acid residue in the antibody molecule removed and a different residue inserted in its place. The sites of greatest interest for substitutional mutagenesis include the hypervariable regions, but FR alterations are also contemplated. Conservative substitutions are shown in Table 6 under the heading of "conservative substitutions." If such substitutions result in a change in biological activity, then more substantial changes, denominated "exemplary substitutions" in Table 6, or as further described below in reference to amino acid classes, may be introduced and the products screened.

In some embodiments, provided herein is a CAR, which specifically binds to MUC16, wherein the CAR comprises a V_H region comprising a sequence shown in SEQ ID NO: 58; and/or a V_L region comprising a sequence shown in SEQ ID NO: 59.

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In some embodiments, the provided herein are CARs comprising CDR portions of antibodies to MUC16 antibodies based on CDR contact regions. CDR contact regions are regions of an antibody that imbue specificity to the antibody for an antigen. In general, CDR contact regions include the residue positions in the CDRs and Vernier zones which are constrained in order to maintain proper loop structure for the antibody to bind a specific antigen. See, e.g., Makabe et al., J. Biol. Chem., 283:1156-1166, 2007. Determination of CDR contact regions is well within the skill of the art.

The binding affinity (K_D) of the ligand binding domain of the MUC16 specific CAR as described herein to MUC16 (such as human MUC16 (e.g., Uniprot accession number: Q8WXI7)) can be for example about 0.1 to about 1000 nM, for example between about 0.5 nM to about 500 nM, or for example between about 1nM to about 250 nM. In some embodiments, the binding affinity is about any of 1000 nM, 750 nM, 500 nM, 400 nM, 300 nM, 250 nM, 200 nM, 100 nM, 90 nM, 80 nM, 70 nM, 60 nM, 50 nM, 45 nM, 40 nM, 35 nM, 30 nM, 25 nM, 20 nM, 19 nM, 18 nM, 17 nM, 16 nM, 15 nM, 10 nM, 8 nM, 7.5 nM, 7 nM, 6.5 nM, 6 nM, 5.5 nM, 5 nM, 4 nM, 3 nM, 2 nM, 1 nM, 0.5 nM, 0.3 nM or 0.1 nM.

In some embodiments, the binding affinity is less than about any of 1000 nM, 900 nM, 800 nM, 250 nM, 200 nM, 100 nM, 50 nM, 30 nM, 20 nM, 10 nM, 7.5 nM, 7 nM, 6.5 nM, 6 nM, 5 nM.

Monoclonal Antibody-Specific Epitopes and Mimotopes

In some embodiments, the extracellular domain of any one of the MUC16 specific CARs disclosed herein may comprise one or more epitopes or mimotopes specific for (i.e., specifically recognized by) a monoclonal antibody. These epitopes or mimotopes are also referred to herein as mAb-specific epitopes or mimotopes. In these embodiments, the extracellular domain comprises the V_H and V_L polypeptides that specifically bind to MUC16 and one or more epitopes or mimotopes that bind to one or more monoclonal antibodies (mAbs). CARs comprising the mAb-specific epitopes or mimotopes can be single-chain or multi-chain.

The inclusion of epitopes or mimotopes specific for monoclonal antibodies in the extracellular domain of the CARs described herein allows sorting and depletion of engineered immune cells expressing the CARs. In some

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embodiments, this feature also promotes recovery of endogenous MUC16 expressing cells that were depleted by administration of engineered immune cells expressing the CARs.

Accordingly, in some embodiments, the present disclosure relates to a method for sorting and/or depleting the engineered immune cells endowed with the CARs comprising mAb-specific epitopes or mimotopes and a method for promoting recovery of endogenous MUC16 expressing cells, such as bone marrow progenitor cells.

Several epitope- or mimotope-monoclonal antibody couples can be used to generate CARs comprising monoclonal antibody specific epitopes or mimotopes; in particular, those already approved for medical use, such as CD20 epitope or mimotope/rituximab as a non-limiting example.

In some embodiments, the monoclonal antibody specific for the epitope or mimotope may be conjugated with a cytotoxic drug. It is also possible to promote CDC cytotoxicity by using engineered antibodies on which are grafted component(s) of the complement system. In some embodiments, activation of the CAR-T cells can be modulated by depleting the cells using an antibody which recognizes the epitope or mimotope.

The disclosure also encompasses methods for sorting the engineered immune cells endowed with the MUC16 specific CARs expressing the mAb-specific epitope(s) or mimotope(s) and therapeutic methods where the activation of the engineered immune cells endowed with these CARs is modulated by depleting the cells using an antibody that targets the external ligand binding domain of said CARs.

CARs comprising one or more epitopes or mimotopes specifically recognized by a monoclonal antibody are disclosed in WO2016/120216, which is hereby incorporated by reference in its entirety. The one or more epitopes or mimotopes can be selected from any number of epitopes and mimotopes known in the art. In some embodiments, the one or more epitopes or mimotopes can be a target of a monoclonal antibody approved for medical use, such as, for example without limitation, the CD20 epitope or mimotope recognized by rituximab. In some embodiments, the one or more epitopes or mimotopes comprises any one or more of the amino acid sequences shown in SEQ ID NOs: 50, 51 and 74.

In some embodiments, the epitope or mimotope can be located between the scFv and the hinge of a CAR. In some embodiments, two instances of the same epitope or mimotope, separated by linkers, may be used in the CAR. For example, a polypeptide comprising 2 copies of the mimotope shown in SEQ ID NO: 50, separated by linkers, as shown in GSGGGGSCPYSNPSLCSGGGGSCPYSNPSLCSGGGGGS (SEQ ID NO: 51), can be used within a CAR, located between the light chain variable region and the hinge.

In some embodiments, the extracellular binding domain of the CAR comprising the V_H and V_L polypeptides and the mAb-specific epitope(s) or mimotope(s) may have different structures depending on the position of insertion of the epitope(s) or mimotope(s) and the use of linkers. For example, the extracellular binding domain of the MUC16 specific CAR comprising mAb-specific epitopes or mimotopes may have one of the following structures (a mimotope may substitute for an epitope in the following structures):

$V_1-L_1-V_2-(L)_x$ -Epitope1-(L) $_x$;
 $V_1-L_1-V_2-(L)_x$ -Epitope1-(L) $_x$ -Epitope2-(L) $_x$;
 $V_1-L_1-V_2-(L)_x$ -Epitope1-(L) $_x$ -Epitope2-(L) $_x$ -Epitope3-(L) $_x$;
 $(L)_x$ -Epitope1-(L) $_x$ - $V_1-L_1-V_2$;
 $(L)_x$ -Epitope1-(L) $_x$ -Epitope2-(L) $_x$ - $V_1-L_1-V_2$;

Epitope1-(L) $_x$ -Epitope2-(L) $_x$ -Epitope3-(L) $_x$ - $V_1-L_1-V_2$;
 $(L)_x$ -Epitope1-(L) $_x$ - $V_1-L_1-V_2-(L)_x$ -Epitope2-(L) $_x$;
 $(L)_x$ -Epitope1-(L) $_x$ - $V_1-L_1-V_2-(L)_x$ -Epitope2-(L) $_x$ -
 Epitope3-(L) $_x$;
 $(L)_x$ -Epitope1-(L) $_x$ - $V_1-L_1-V_2-(L)_x$ -Epitope2-(L) $_x$ -
 Epitope3-(L) $_x$ -Epitope4-(L) $_x$;
 $(L)_x$ -Epitope1-(L) $_x$ -Epitope2-(L) $_x$ - $V_1-L_1-V_2-(L)_x$ -
 Epitope3-(L) $_x$;
 $(L)_x$ -Epitope1-(L) $_x$ -Epitope2-(L) $_x$ - $V_1-L_1-V_2-(L)_x$ -
 Epitope3-(L) $_x$ -Epitope4-(L) $_x$;
 $V_1-(L)_x$ -Epitope1-(L) $_x$ - V_2 ;
 $V_1-(L)_x$ -Epitope1-(L) $_x$ - $V_2-(L)_x$ -Epitope2-(L) $_x$;
 $V_1-(L)_x$ -Epitope1-(L) $_x$ - $V_2-(L)_x$ -Epitope2-(L) $_x$ -Epitope3-(L) $_x$;
 $V_1-(L)_x$ -Epitope1-(L) $_x$ - $V_2-(L)_x$ -Epitope2-(L) $_x$ -Epitope3-(L) $_x$ -
 Epitope4-(L) $_x$;
 $(L)_x$ -Epitope1-(L) $_x$ - $V_1-(L)_x$ -Epitope2-(L) $_x$ - V_2 ; or,
 $(L)_x$ -Epitope1-(L) $_x$ - $V_1-(L)_x$ -Epitope2-(L) $_x$ - $V_2-(L)_x$ -
 Epitope3-(L) $_x$;

wherein,

V_1 is V_L and V_2 is V_H or V_1 is V_H and V_2 is V_L ;
 L_1 is a linker suitable to link the V_H chain to the V_L chain;

L is a linker comprising glycine and serine residues, and each occurrence of L in the extracellular binding domain can be identical or different to other occurrence of L in the same extracellular binding domain, for example SGGGG (SEQ ID NO: 185), GGGGS (SEQ ID NO: 178) or SGGGGS (SEQ ID NO: 186) (all of which occur, e.g., in SEQ ID NO: 52), and,

x is 0 or 1 and each occurrence of x is selected independently from the others; and,

Epitope 1, Epitope 2, Epitope 3 and Epitope 4 are mAb-specific epitopes and can be identical or different.

In some embodiments, the extracellular binding domain of the CAR comprises the following sequence:

$V_1-L_1-V_2-(L)_x$ -Epitope1-(L) $_x$ -Epitope2-(L) $_x$; or,
 $(L)_x$ -Epitope1-(L) $_x$ - $V_1-L_1-V_2-(L)_x$ -Epitope2-(L) $_x$ -
 Epitope3-(L) $_x$ -Epitope4-(L) $_x$. wherein V_1 , V_2 , L_1 , L , x and Epitope 1, Epitope 2, Epitope 3 and Epitope 4 are as defined above.

In some embodiments, any one of the MUC16 specific CARs disclosed herein may comprise one or more mAb-specific epitopes or mimotopes selected from a CD52 epitope or mimotope, a CD20 epitope or mimotope, a CD3 epitope or mimotope, a CD41 epitope or mimotope, a CD25 epitope or mimotope, a CD30 epitope or mimotope, an EGFR epitope or mimotope, a TNF α epitope or mimotope, a VEGF epitope or mimotope, a complement protein C5 epitope or mimotope, a CD11a epitope or mimotope, a CD33 epitope or mimotope, an α_4 -integrin epitope or mimotope, an IgE Fc region epitope or mimotope, an RSV protein F epitope or mimotope, an IL-6 receptor epitope or mimotope, a HER2 receptor epitope or mimotope, an integrin $\alpha_4\beta_7$ epitope or mimotope, a BAFF (B-cell activator factor) epitope or mimotope, an IL-1 β epitope or mimotope, a RANKL epitope or mimotope, a CTLA4 epitope or mimotope, a CD34 epitope or mimotope, an IL-12 epitope or mimotope, and/or an IL-23 epitope or mimotope.

In some embodiments, the MUC16 specific CARs disclosed herein may comprise one or more mAb-specific epitopes or mimotopes selected from epitopes and mimotopes specifically recognized by alemtuzumab, ibritumomab tiuxetan, muromonab-CD3, tositumomab, abciximab, basiliximab, brentuximab vedotin, cetuximab, infliximab, rituximab, bevacizumab, certolizumab pegol, daclizumab, eculizumab, efalizumab, gemtuzumab, natalizumab, omalizumab, palivizumab, ranibizumab, tocilizumab,

trastuzumab, vedolizumab, adalimumab, belimumab, canakinumab, denosumab, golimumab, ipilimumab, ofatumumab, panitumumab, QBEND-10 and/or ustekinumab.

In some embodiments, the MUC16 specific CARs comprise one or more mAb-specific epitopes or mimotopes selected from the epitopes and mimotopes disclosed in Table 5:

TABLE 5

Examples of mAb-specific epitopes and mimotopes that can be used in the extracellular binding domain of the MUC16 specific CARs of the disclosure such as for example mimotopes and epitopes with their corresponding mAb.		
Rituximab		
Mimotope	SEQ ID NO: 50	CPYSNPSLC
Palivizumab		
Epitope	SEQ ID NO: 80	NSSELLSLINDMP ITNDQKKLMSNN
Cetuximab		
Mimotope 1	SEQ ID NO: 81	CQFDLSTRRLKC
Mimotope 2	SEQ ID NO: 82	CQYNLSSRALKC
Mimotope 3	SEQ ID NO: 83	CVWQRWQKSYVC
Mimotope 4	SEQ ID NO: 84	CMWDRFSRWYKC
Nivolumab		
Epitope 1	SEQ ID NO: 85	SFVLNWYRMSPS NQTDKLAAPFED R
Epitope 2	SEQ ID NO: 86	SGTYLCGAISLA PKAQIKE
QBEND-10		
Epitope	SEQ ID NO: 87	ELPTQGTFSNVS TNVSPAKPTTTA
Alemtuzumab		
Epitope	SEQ ID NO: 88	GQNDTSQTSSPS

The intracellular signaling domain of a CAR as disclosed herein is responsible for intracellular signaling following the binding of extracellular ligand-binding domain to the target resulting in the activation of the immune cell and immune response. The intracellular signaling domain has the ability to activate at least one of the normal effector functions of the immune cell in which the CAR is expressed. For example, the effector function of a T cell can be a cytolytic activity or helper activity including the secretion of cytokines.

Expression of Proteins in Cells and Related Embodiments

In some embodiments, an immune cell e.g. T cell of the disclosure comprises and/or expresses a polypeptide that comprises any one or more of the sequences listed in Tables 1-3 (SEQ ID NOs: 1-79). In some embodiments, an immune cell e.g. T cell of the disclosure comprises and/or expresses a nucleic acid e.g. a vector that encodes such a polypeptide.

In some embodiments, an immune cell e.g. T cell of the disclosure comprises and/or expresses a polypeptide that comprises the amino acid sequence of MUC16 CAR1 (53B6) (SEQ ID NO: 11). In some embodiments, an immune cell e.g. T cell of the disclosure comprises and/or expresses a polypeptide that comprises the amino acid sequence of

Sec49k (SEQ ID NO: 12). In some embodiments, an immune cell e.g. T cell of the disclosure comprises and/or expresses a polypeptide that comprises the amino acid sequence of UL11 (SEQ ID NO: 13). In some embodiments, an immune cell e.g. T cell of the disclosure comprises and/or expresses a polypeptide that comprises the amino acid sequence of BTN3A1-IgV (SEQ ID NO: 14). In some embodiments, an immune cell e.g. T cell of the disclosure comprises and/or expresses a polypeptide that comprises the amino acid sequence of BTN3A1-IgC (SEQ ID NO: 15).

In some embodiments, an immune cell e.g. T cell of the disclosure comprises and/or expresses a nucleic acid e.g. a vector that encodes a protease-activating CD45-gate CAR disclosed herein. In some embodiments, an immune cell e.g. T cell of the disclosure comprises and/or expresses a nucleic acid e.g. a vector that comprises a nucleic acid sequence that encodes the amino acid sequence of a MUC16 CAR (53B6) (any of SEQ ID NOs: 11, 56, 75, 76, 77, 78, 79, and 121-170). In some embodiments, an immune cell e.g. T cell of the disclosure comprises and/or expresses a nucleic acid e.g. a vector that comprises a nucleic acid sequence that encodes the amino acid sequence of Sec49k (SEQ ID NO: 12). In some embodiments, an immune cell e.g. T cell of the disclosure comprises and/or expresses a nucleic acid e.g. a vector that comprises a nucleic acid sequence that encodes the amino acid sequence of UL11 (SEQ ID NO: 13). In some embodiments, an immune cell e.g. T cell of the disclosure comprises and/or expresses a nucleic acid e.g. a vector that comprises a nucleic acid sequence that encodes the amino acid sequence of BTN3A1-IgV (SEQ ID NO: 14). In some embodiments, an immune cell e.g. T cell of the disclosure comprises and/or expresses a nucleic acid e.g. a vector that comprises a nucleic acid sequence that encodes the amino acid sequence of BTN3A1-IgC (SEQ ID NO: 15).

The disclosure encompasses modifications to the proteins of the disclosure embodiments shown in Table 2, including functionally equivalent proteins having modifications which do not significantly affect their properties and variants which have enhanced or decreased activity and/or affinity. Modification of polypeptides is routine practice in the art and need not be described in detail herein. Examples of modified polypeptides include polypeptides with conservative substitutions of amino acid residues, one or more deletions or additions of amino acids which do not significantly deleteriously change the functional activity, or which mature (enhance) the affinity of the polypeptide for its ligand, or use of chemical analogs.

Amino acid sequence insertions include amino- and/or carboxyl-terminal fusions ranging in length from one residue to polypeptides containing a hundred or more residues, as well as intrasequence insertions of single or multiple amino acid residues. Examples of terminal insertions include an antibody with an N-terminal methionyl residue or the antibody fused to an epitope or mimotope tag.

Substitution variants have at least one amino acid residue in the protein removed and a different residue inserted in its place. Conservative substitutions are shown in Table 6 under the heading of "conservative substitutions." If such substitutions result in a change in biological activity, then more substantial changes, denominated "exemplary substitutions" in Table 6, or as further described below in reference to amino acid classes, may be introduced and the products screened.

TABLE 6

Amino Acid Substitutions		
Original residue (naturally occurring amino acid)	Conservative substitutions	Exemplary substitutions
Ala (A)	Val	Val; Leu; Ile
Arg (R)	Lys	Lys; Gln; Asn
Asn (N)	Gln	Gln; His; Asp; Lys; Arg
Asp (D)	Glu	Glu; Asn
Cys (C)	Ser	Ser; Ala
Gln (Q)	Asn	Asn; Glu
Glu (E)	Asp	Asp; Gln
Gly (G)	Ala	Ala
His (H)	Arg	Asn; Gln; Lys; Arg
Ile (I)	Leu	Leu; Val; Met; Ala; Phe; Norleucine
Leu (L)	Ile	Norleucine; Ile; Val; Met; Ala; Phe
Lys (K)	Arg	Arg; Gln; Asn
Met (M)	Leu	Leu; Phe; Ile
Phe (F)	Tyr	Leu; Val; Ile; Ala; Tyr
Pro (P)	Ala	Ala
Ser (S)	Thr	Thr
Thr (T)	Ser	Ser
Trp (W)	Tyr	Tyr; Phe
Tyr (Y)	Phe	Trp; Phe; Thr; Ser
Val (V)	Leu	Ile; Leu; Met; Phe; Ala; Norleucine

Protease-activating CD45-gate CAR protein and protease-activating CD45-gate CAR protein derivatives may be synthesized in situ in the cell after introduction of polynucleotides encoding the proteins into the cell. Alternatively, protease-activating CD45-gate CAR protein and protease-activating CD45-gate CAR protein derivative proteins may be produced outside of cells, and then introduced into cells. Methods for introducing a polynucleotide construct into cells are known in the art. In some embodiments, stable transformation methods can be used to integrate the polynucleotide construct into the genome of the cell. In other embodiments, transient transformation methods can be used to transiently express the polynucleotide construct, and the polynucleotide construct not integrated into the genome of the cell. In other embodiments, virus-mediated methods can be used. The polynucleotides may be introduced into a cell by any suitable means such as for example, recombinant viral vectors (e.g. retroviruses e.g. lentiviruses, adenoviruses), liposomes, and the like. Transient transformation methods include, for example without limitation, microinjection, electroporation or particle bombardment. Polynucleotides may be included in vectors, such as for example plasmid vectors or viral vectors.

In some embodiments, an immune cell e.g. T cell of the disclosure can comprise at least one protease-activating CD45-gate CAR protein or protease-activating CD45-gate CAR protein derivative. In some embodiments, an immune cell e.g. T cell can comprise at least one protease-activating CD45-gate CAR protein or protease-activating CD45-gate CAR protein derivative and one or more additional CARs (gated or not gated), and each CAR may comprise different extracellular ligand-binding domains.

In some embodiments of an immune cell e.g. T cell provided herein, a CAR that the T cell expresses can comprise an extracellular ligand-binding domain (e.g., a single chain variable fragment (scFv)), a transmembrane domain, and an intracellular signaling domain. In some embodiments, the extracellular ligand-binding domain, transmembrane domain, and intracellular signaling domain

are in one polypeptide, i.e., in a single chain. Multichain CARs and polypeptides are also provided herein. In some embodiments, the multichain CARs comprise: a first polypeptide comprising a transmembrane domain and at least one extracellular ligand-binding domain, and a second polypeptide comprising a transmembrane domain and at least one intracellular signaling domain, wherein the polypeptides assemble together to form a multichain CAR. The CAR can be modified as described herein to comprise a CD45 recruiting domain that is linked to the amino terminus of the CAR's extracellular ligand-binding domain.

The extracellular ligand-binding domain specifically binds to a target of interest. In some embodiments, the target of interest can be any molecule of interest, including, for example, without limitation, BCMA, MUC16 (also known as CA125), EGFR, EGFRvIII, MUC1, Flt-3, WT-1, CD20, CD23, CD30, CD38, CD70, CD33, CD133, MHC-WT1, TSPAN10, MHC-PRAME, MHC-NY-ESO1, HER2 (ERBB2), CAIX (Carbonic anhydrase IX), LIV1, ADAM10, CHRNA2, LeY, NKG2D, CS1, CD44v6, ROR1, CD19, Claudin-18.2 (Claudin-18A2, or Claudin18 isoform 2), PSCA, DLL3 (Delta-like protein 3, Drosophila Delta homolog 3, Delta3), Mud 7 (Mucin17, Muc3, Muc3), FAP alpha (Fibroblast Activation Protein alpha), Ly6G6D (Lymphocyte antigen 6 complex locus protein G6d, c6orf23, G6D, MEGT1, NG25), PSMA, MSLN, or RNF43 (E3 ubiquitin-protein ligase RNF43, RING finger protein 43).

In some embodiments, the extracellular ligand-binding domain comprises an scFv comprising the light chain variable (V_L) region and the heavy chain variable (V_H) region of a target antigen specific monoclonal antibody joined by a flexible linker. Single chain variable region fragments are made by linking light and/or heavy chain variable regions by using a short linking peptide (Bird et al., Science 242:423-426, 1988). An example of a linking peptide is the GS linker having the amino acid sequence (GGGGS)₃ (SEQ ID NO: 57), which bridges approximately 3.5 nm between the carboxy terminus of one variable region and the amino terminus of the other variable region. Linkers of other sequences have been designed and used (Bird et al., 1988, supra). In general, linkers can be short, flexible polypeptides and preferably comprised of about 20 or fewer amino acid residues. Linkers can in turn be modified for additional functions, such as attachment of drugs or attachment to solid supports. The single chain variants can be produced either recombinantly or synthetically. For synthetic production of scFv, an automated synthesizer can be used. For recombinant production of scFv, a suitable plasmid or other vector containing a polynucleotide that encodes the scFv can be introduced into a suitable host cell, either eukaryotic, such as yeast, plant, insect or mammalian cells, or prokaryotic, such as *E. coli*. Polynucleotides encoding the scFv of interest can be made by routine manipulations such as ligation of polynucleotides. The resultant scFv can be isolated using standard protein purification techniques known in the art.

In some embodiments, a flexible linker as described herein is modified to comprise a protease cleavage site and the modified linker connects the CD45 recruiting domain of the protease-activating CD45-gate CAR described herein to the antigen binding domain of the protease-activating CD45-gate CAR. An example of such a modified linker comprises the amino acid sequence of SEQ ID NO: 9.

The intracellular signaling domain of a protease-activating CD45-gate CAR according to the disclosure is responsible for intracellular signaling following the binding of extracellular ligand-binding domain to the target resulting in the activation of the immune cell and immune response. The

intracellular signaling domain has the ability to activate at least one of the normal effector functions of the immune cell in which the protease-activating CD45-gate CAR is expressed. For example, the effector function of a T cell can be a cytolytic activity or helper activity including the secretion of cytokines.

In some embodiments, an intracellular signaling domain for use in a protease-activating CD45-gate CAR can be the cytoplasmic sequences of, for example without limitation, the T cell receptor and co-receptors that act in concert to initiate signal transduction following antigen receptor engagement, as well as any derivative or variant of these sequences and any synthetic sequence that has the same functional capability. Intracellular signaling domains comprise two distinct classes of cytoplasmic signaling sequences: those that initiate antigen-dependent primary activation, and those that act in an antigen-independent manner to provide a secondary or co-stimulatory signal. Primary cytoplasmic signaling sequences can comprise signaling motifs which are known as immunoreceptor tyrosine-based activation motifs or ITAMs. ITAMs are well defined signaling motifs found in the intracytoplasmic tail of a variety of receptors that serve as binding sites for syk/zap70 class tyrosine kinases. Examples of ITAM used in the protease-activating CD45-gate CARs of the disclosure can include as non-limiting examples those derived from TCR ζ , FcR γ , FcR β , FcR ϵ , CD3 γ , CD3 δ , CD3 ϵ , CD5, CD22, CD79a, CD79b and CD66d. In some embodiments, the intracellular signaling domain of the CAR can comprise the CD3 ζ signaling domain (e.g. comprising or consisting of the amino acid sequence of SEQ ID NO: 34).

In some embodiments the intracellular signaling domain of the protease-activating CD45-gate CAR of the disclosure comprises a domain of a co-stimulatory molecule. In some embodiments, the intracellular signaling domain of a protease-activating CD45-gate CAR of the disclosure comprises a part of a co-stimulatory molecule selected from the group consisting of a fragment of 41BB (GenBank: AAA53133, e.g. comprising or consisting of the amino acid sequence of SEQ ID NO: 33) and CD28 (NP_006130.1, e.g. comprising or consisting of the amino acid sequence of SEQ ID NO: 35). In some embodiments, the intracellular signaling domain of the CAR of the disclosure comprises an amino acid sequence which comprises at least 70%, preferably at least 80%, more preferably at least 90%, 95%, 97%, or 99% sequence identity with one or more of the amino acid sequences shown in SEQ ID NOs: 33-35.

CARs generally are expressed on the surface membrane of the cell. Thus, the protease-activating CD45-gate CAR disclosed herein can comprise a transmembrane domain. Suitable transmembrane domains for a protease-activating CD45-gate CAR disclosed herein have the ability to (a) be expressed at the surface of a cell, for example an immune cell such as, for example without limitation, lymphocyte cells (e.g. T cells) or Natural killer (NK) cells, and (b) interact with the ligand-binding domain and intracellular signaling domain for directing a cellular response of an immune cell against a predefined target cell. The transmembrane domain can be derived either from a natural or from a synthetic source. The transmembrane domain can be derived from any membrane-bound or transmembrane protein. As non-limiting examples, the transmembrane polypeptide can be a domain of the T cell receptor such as α , β , γ or δ , polypeptide constituting CD3 complex, IL-2 receptor e.g. p55 (α chain), p75 (β chain or γ chain), subunit chain of Fc receptors, in particular Fc γ receptor III or CD proteins. Alternatively, the transmembrane domain can be synthetic

and can comprise predominantly hydrophobic residues such as leucine and valine. In some embodiments said transmembrane domain is derived from the human CD8 α chain (e.g., NP_001139345.1). The protease-activating CD45-gate CAR can further comprise a stalk domain between the extracellular ligand-binding domain and said transmembrane domain. A stalk domain may comprise up to 300 amino acids, for example, from 10 to 100 amino acids or 25 to 50 amino acids. The stalk region may be derived from all or part of naturally occurring molecules, such as from all or part of the extracellular region of CD8, CD4, or CD28, or from all or part of an antibody constant region. Alternatively the stalk domain may be a synthetic sequence that corresponds to a naturally occurring stalk sequence, or may be an entirely synthetic stalk sequence. In some embodiments said stalk domain is a part of human CD8 α chain (e.g., NP_001139345.1). In another particular embodiment, the transmembrane domain comprises a part of the human CD8 α chain. In some embodiments, protease-activating CD45-gate CARs disclosed herein can comprise a CD45 recruiting domain, a protease-cleavable linker, an extracellular ligand-binding domain, CD8 α human stalk and transmembrane domains, the CD3 ζ signaling domain, and 4-1BB signaling domain. In some embodiments, a nucleic acid encoding a protease-activating CD45-gate CAR as disclosed herein can be introduced into an immune cell as a transgene via a vector e.g. a plasmid vector. In some embodiments, the vector e.g. plasmid vector can also contain, for example, a selection marker which provides for identification and/or selection of cells which received the vector.

Protease-activating CD45-gate CAR polypeptides may be synthesized in situ in the cell after introduction of polynucleotides encoding the protease-activating CD45-gate CAR polypeptides into the cell. Alternatively, protease-activating CD45-gate CAR polypeptides may be produced outside of cells, and then introduced into cells. Methods for introducing a polynucleotide construct into cells are known in the art. In some embodiments, stable transformation methods can be used to integrate the polynucleotide construct into the genome of the cell. In other embodiments, transient transformation methods can be used to transiently express the polynucleotide construct, and the polynucleotide construct not integrated into the genome of the cell. In other embodiments, virus-mediated methods can be used. The polynucleotides may be introduced into a cell by any suitable means such as for example, recombinant viral vectors (e.g. retroviruses (e.g. lentiviruses), adenoviruses), liposomes, and the like. Transient transformation methods include, for example without limitation, microinjection, electroporation or particle bombardment. Polynucleotides may be included in vectors, such as for example plasmid vectors or viral vectors.

Methods of generating engineered immune cells expressing any of the protease-activating CD45-gate CARs provided herein is described in WO/2016/166630, incorporated by reference in its entirety.

Also provided herein are immune cells e.g. T cells such as isolated T cells obtained according to any one of the methods described herein and modified e.g. engineered to comprise and express the nucleic acids, vectors and polypeptides disclosed herein. Any immune cell capable of expressing heterologous DNAs can be used for the purpose of expressing the protease-activating CD45-gate CAR protein or protease-activating CD45-gate CAR protein derivative and optionally any additional CAR of interest. In some embodiments, the immune cell is a T cell. In some embodiments, an immune cell can be derived from, for example

without limitation, a stem cell. The stem cells can be adult stem cells, non-human embryonic stem cells, more particularly non-human stem cells, cord blood stem cells, progenitor cells, bone marrow stem cells, induced pluripotent stem cells, totipotent stem cells or hematopoietic stem cells. Representative human cells are CD34+ cells. The isolated cell can also be a dendritic cell, killer dendritic cell, a mast cell, a NK-cell, a B-cell or a T cell selected from the group consisting of inflammatory T-lymphocytes, cytotoxic T-lymphocytes, regulatory T-lymphocytes or helper T-lymphocytes. In some embodiments, the cell can be derived from the group consisting of CD4+ T-lymphocytes and CD8+ T-lymphocytes.

In some embodiments, the engineered immune cell disclosed herein is an engineered B cell, mast cell, myeloid-derived phagocyte, T cell e.g. an alpha/beta and/or gamma/delta T cell, tumor infiltrating lymphocyte (TIL), NK cell, TCR-expressing cell, dendritic cell, or NK-T cell. In some embodiments, the cell is and/or is derived from an autologous T cell. In some embodiments, the cell is and/or is derived from an allogeneic T cell.

In some embodiments, the immune cells e.g. T cells such as isolated T cells are further modified e.g. genetically modified by methods described herein (e.g. known gene editing techniques that employ, for example, TALENs, CRISPR/Cas9, or megaTAL nucleases).

Prior to expansion and genetic modification, a source of cells can be obtained from a subject through a variety of non-limiting methods. Cells can be obtained from a number of non-limiting sources, including peripheral blood mononuclear cells, bone marrow, lymph node tissue, cord blood, thymus tissue, tissue from a site of infection, ascites, pleural effusion, spleen tissue, and tumors. In some embodiments, any number of T cell lines available and known to those skilled in the art, may be used. In some embodiments, cells can be derived from a healthy donor, from a subject diagnosed with cancer or from a subject diagnosed with an infection. In some embodiments, cells can be part of a mixed population of cells which present different phenotypic characteristics.

In some embodiments, an isolated cell or engineered immune cell according to the present disclosure comprises one inactivated gene selected from the group consisting of CD52, GR, PD-1, CTLA-4, LAG3, Tim3, BTLA, BY55, TIGIT, B7H5, LAIR1, SIGLEC10, 2B4, HLA, TCR α and TCR β and/or expresses a CAR, a multi-chain CAR and/or a pTa transgene. In some embodiments, an isolated cell comprises polynucleotides encoding polypeptides comprising a multi-chain CAR. In some embodiments, the isolated cell according to the present disclosure comprises two inactivated genes selected from the group consisting of: CD52 and GR, CD52 and TCR α , CDR52 and TCR β , GR and TCR α , GR and TCR β , TCR α and TCR β , PD-1 and TCR α , PD-1 and TCR β , CTLA-4 and TCR α , CTLA-4 and TCR β , LAG3 and TCR α , LAG3 and TCR β , Tim3 and TCR α , Tim3 and TCR β , BTLA and TCR α , BTLA and TCR β , BY55 and TCR α , BY55 and TCR β , TIGIT and TCR α , TIGIT and TCR β , B7H5 and TCR α , B7H5 and TCR β , LAIR1 and TCR α , LAIR1 and TCR β , SIGLEC10 and TCR α , SIGLEC10 and TCR β , 2B4 and TCR α , 2B4 and TCR β and/or expresses a CAR (e.g. the protease-activating CD45-gate CAR of the present disclosure), a multi-chain CAR and a pTa transgene.

Gene inactivation can be carried out by methods practiced by those with skill in the art. The methods include, but are

not limited to gene inactivation by use of zinc fingers, TALEN[®]s, and CRISPR/Cas-based system.

In some embodiments, the protease-activating CD45-gate CAR-containing immune cell has an inactivated CD52 gene. In some embodiments only one copy of the CD52 gene is inactivated.

In some embodiments, the protease-activating CD45-gate CAR-containing immune cell has an inactivated TCR α gene.

In some embodiments, the protease-activating CD45-gate CAR-containing immune cell has an inactivated TCR β gene.

In some embodiments, TALEN[®] is used for gene inactivation. In such embodiments, the efficiency of gene inactivation with TALEN[®] is not 100%, and resulting TCR $\alpha\beta$ -negative T-cells are enriched by depleting residual TCR $\alpha\beta$ -positive T cells before cryopreservation. However, CD52-negative cells are not purified, resulting in a cell product with varying frequencies of CD52-negative cells, typically between 60-80%. Accordingly in some embodiments, the genotype of the protease-activating CD45-gate CAR T cells of the disclosure is protease-activating CD45-gate CAR+ TCR $\alpha\beta$ -CD52+/- T-cells.

In some embodiments, TCR is rendered not functional in the cells according to the disclosure by inactivating TCR α gene and/or TCR β gene(s). In some embodiments, a method to obtain modified cells derived from an individual is provided, wherein the cells can proliferate independently of the major histocompatibility complex (MHC) signaling pathway. Modified cells, which can proliferate independently of the MHC signaling pathway, susceptible to be obtained by this method are encompassed in the scope of the present disclosure. Modified cells disclosed herein can be used in for treating subjects in need thereof against Host versus Graft (HvG) rejection and Graft versus Host Disease (GvHD); therefore in the scope of the present disclosure is a method of treating subjects in need thereof against Host versus Graft (HvG) rejection and Graft versus Host Disease (GvHD) comprising treating said subject by administering to said subject an effective amount of modified cells comprising inactivated TCR α and/or TCR β genes.

In some embodiments, the immune cells are engineered to be resistant to one or more chemotherapy drugs. The chemotherapy drug can be, for example, a purine nucleotide analogue (PNA), thus making the immune cell suitable for cancer treatment combining adoptive immunotherapy and chemotherapy. Exemplary PNAs include, for example, clofarabine, fludarabine, and cytarabine, alone or in combination. PNAs are metabolized by deoxycytidine kinase (dCK) into mono-, di-, and tri-phosphate PNA. Their tri-phosphate forms compete with ATP for DNA synthesis, act as pro-apoptotic agents, and are potent inhibitors of ribonucleotide reductase (RNR), which is involved in trinucleotide production. Provided herein are protease-activating CD45-gate CAR-T cells comprising an inactivated dCK gene. In some embodiments, the dCK knockout cells are made by transfection of T cells using polynucleotides encoding specific TAL-nuclease directed against dCK genes by, for example, electroporation of mRNA. The dCK knockout protease-activating CD45-gate CAR-T cells are resistant to PNAs, including for example clofarabine and/or fludarabine, and maintain T cell cytotoxic activity toward cells that express the ligand of the protease-activating CD45-gate CAR.

In some embodiments, isolated cells or cell lines of the disclosure can comprise a pTa or a functional variant

thereof. In some embodiments, an isolated cell or cell line can be further genetically modified by inactivating the TCR α gene.

In some embodiments, the CAR-T cell (e.g. protease-activating CD45-gate CAR-expressing T cell) comprises a polynucleotide encoding a safety switch, such as for example RQR8. See, e.g., WO2013153391A, which is hereby incorporated by reference in its entirety. In CAR-T cells comprising the polynucleotide, the safety switch polypeptide is expressed at the surface of a CAR-T cell. In some embodiments, the safety switch polypeptide comprises the amino acid sequence shown in SEQ ID NO: 48.

(SEQ ID NO: 48)

CPYSNPSLCSGGGGSELPTQGTFSNVSTNVSPAKPTTACPYSNPSLCSG

GGGSPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIW

APLAGTCGVLLLSLVITLYCNHRNRRRVCKCPRPVV

The safety switch polypeptide may also comprise a signal peptide at the amino terminus. In some embodiments, the safety switch polypeptide comprises the amino acid sequence shown in SEQ ID NO: 49.

(SEQ ID NO: 49)

MGTSLLCWMALCLLGADHADACPYSNPSLCSGGGGSELPTQGTFSNVSTN

VSPAKPTTACPYSNPSLCSGGGGSPAPRPPTPAPTIASQPLSLRPEACR

PAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCNHRNRRRVCK

KCPRPVV

When the safety switch polypeptide is expressed at the surface of a CAR-T cell, binding of rituximab to the R epitopes or mimotopes of the polypeptide causes lysis of the cell. More than one molecule of rituximab may bind per polypeptide expressed at the cell surface. Each R epitope or mimotope of the polypeptide may bind a separate molecule of rituximab. Deletion of protease-activating CD45-gate CAR-T cells may occur in vivo, for example by administering rituximab to a subject. The decision to delete the transferred cells may arise from undesirable effects being detected in the subject which are attributable to the transferred cells, such as for example, when unacceptable levels of toxicity are detected.

In some embodiments, the CAR-T cell comprises a selected epitope or mimotope within the scFv or extracellular domain (e.g. stalk domain) of the protease-activating CD45-gate CAR, the selected epitope or mimotope having a specificity to be recognized by a specific antibody. See, e.g., PCT application PCT/EP2016/051467, WO2016/120216, "mAb-DRIVEN CHIMERIC ANTIGEN RECEPTOR SYSTEMS FOR SORTING/DEPLETING ENGINEERED IMMUNE CELLS," filed on Jan. 25, 2016, which is hereby incorporated by reference in its entirety. Such an epitope or mimotope facilitates sorting and/or depleting the CAR-T cells. The epitope or mimotope can be selected from any number of epitopes and mimotopes known in the art. In some embodiments, the epitope or mimotope can be a target of a monoclonal antibody approved for medical use, such as, for example without limitation, a CD20 epitope or mimotope recognized by rituximab. In some embodiments, the epitope or mimotope comprises the amino acid sequence shown in SEQ ID NO: 50.

(SEQ ID NO: 50)

CPYSNPSLC

In some embodiments, the epitope or mimotope is located within the CAR. For example without limitation, the epitope or mimotope can be located between the scFv and the hinge of a CAR. In some embodiments, two instances of the same epitope or mimotope, separate by linkers, may be used in the CAR. For example, the polypeptide comprising the amino acid sequence shown in SEQ ID NO: 51 can be used within a CAR, located between the light chain variable region and the hinge.

(SEQ ID NO: 51)

GSGGGGSCPYSNPSLCSGGGGSCPYSNPSLCSGGGGG

In some embodiments, the epitope- or mimotope-specific antibody may be conjugated with a cytotoxic drug. It is also possible to promote CDC cytotoxicity by using engineered antibodies on which are grafted component(s) of the complement system. In some embodiments, activation of the CAR-T cells can be modulated by depleting the cells using an antibody which recognizes the epitope or mimotope.

Adverse events also may be minimized by transducing the immune cells (containing one or more CARs) with a suicide gene. It may also be desired to incorporate an inducible "on" or "accelerator" switch into the immune cells. Suitable techniques include use of inducible caspase-9 (U.S. Appl. 2011/0286980) or a thymidine kinase, before, after or at the same time, as the cells are transduced with the CAR construct of the present disclosure. Additional methods for introducing suicide genes and/or "on" switches include TALENS, zinc fingers, RNAi, siRNA, shRNA, antisense technology, and other techniques known in the art.

In accordance with the disclosure, additional on-off or other types of control switch techniques may be incorporated herein. These techniques may employ the use of dimerization domains and optional activators of such domain dimerization. These techniques include, e.g., those described by Wu et al., Science 2014 350 (6258) utilizing FKBP/Rapalog dimerization systems in certain cells, the contents of which are incorporated by reference herein in their entirety. Additional dimerization technology is described in, e.g., Fegan et al. Chem. Rev. 2010, 110, 3315-3336 as well as U.S. Pat. Nos. 5,830,462; 5,834,266; 5,869,337; and 6,165,787, the contents of which are also incorporated by reference herein in their entirety. Additional dimerization pairs may include cyclosporine-A/cyclophilin, receptor, estrogen/estrogen receptor (optionally using tamoxifen), glucocorticoids/glucocorticoid receptor, tetracycline/tetracycline receptor, vitamin D/vitamin D receptor. Further examples of dimerization technology can be found in e.g., WO 2014/127261, WO 2015/090229, US 2014/0286987, US2015/0266973, US2016/0046700, U.S. Pat. No. 8,486,693, US 2014/0171649, and US 2012/0130076, the contents of which are further incorporated by reference herein in their entirety.

Also provided herein are cell lines obtained from a modified e.g. transformed immune cell e.g. T cell according to any of the methods described herein. In some embodiments, an immune cell e.g. T cell according to the disclosure comprises a polynucleotide encoding a protease-activating CD45-gate CAR protein or protease-activating CD45-gate CAR protein derivative. In some embodiments, an immune cell e.g. T cell according to the disclosure comprises a polynucleotide encoding a protease-activating CD45-gate

CAR protein or protease-activating CD45-gate CAR protein derivative and a polynucleotide encoding an additional CAR.

The immune cells e.g. T cells of the disclosure can be activated and expanded, either prior to or after modification of the cells, using methods as generally described, for example without limitation, in U.S. Pat. Nos. 6,352,694; 6,534,055; 6,905,680; 6,692,964; 5,858,358; 6,887,466; 6,905,681; 7,144,575; 7,067,318; 7,172,869; 7,232,566; 7,175,843; 5,883,223; 6,905,874; 6,797,514; 6,867,041; and U.S. Patent Application Publication No. 20060121005. Immune cells e.g. T cells can be expanded in vitro or in vivo. Generally, the immune cells of the disclosure can be expanded, for example, by contact with an agent that stimulates a CD3 TCR complex and a co-stimulatory molecule on the surface of the immune cells to create an activation signal for the cell. For example, chemicals such as calcium ionophore A23187, phorbol 12-myristate 13-acetate (PMA), or mitogenic lectins like phytohemagglutinin (PHA) can be used to create an activation signal for the immune cell e.g. T cell.

In some embodiments, T cell populations may be stimulated in vitro by contact with, for example, an anti-CD3 antibody, or antigen-binding fragment thereof, or an anti-CD2 antibody immobilized on a surface, or by contact with a protein kinase C activator (e.g., bryostatin) in conjunction with a calcium ionophore. For co-stimulation of an accessory molecule on the surface of the T cells, a ligand that binds the accessory molecule is used. For example, a population of T cells can be contacted with an anti-CD3 antibody and an anti-CD28 antibody, under conditions appropriate for stimulating proliferation of the T cells. Conditions appropriate for T cell culture include an appropriate medium (e.g., Minimal Essential Media or RPMI Media 1640 or, X-vivo 5, (Lonza)) that may contain factors necessary for proliferation and viability, including serum (e.g., fetal bovine or human serum), interleukin-2 (IL-2), insulin, IFN- γ , IL-4, IL-7, GM-CSF, IL-10, IL-2, IL-15, a TGF β , and TNF, or any other additives for the growth of cells known to the skilled artisan. Other additives for the growth of cells include, but are not limited to, surfactant, plasmanate, and reducing agents such as N-acetyl-cysteine and 2-mercaptoethanol. Media can include RPMI 1640, AIM V, DMEM, MEM, α -MEM, F-12, X-Vivo 10, and X-Vivo 20, Optimizer, with added amino acids, sodium pyruvate, and vitamins, either serum-free or supplemented with an appropriate amount of serum (or plasma) or a defined set of hormones, and/or an amount of cytokine(s) sufficient for the growth and expansion of T cells. Antibiotics, e.g., penicillin and streptomycin, are included only in experimental cultures, not in cultures of cells that are to be infused into a subject. The target cells are maintained under conditions necessary to support growth, for example, an appropriate temperature (e.g., 37° C.) and atmosphere (e.g., air plus 5% CO₂). Immune cells e.g. T cells that have been exposed to varied stimulation times may exhibit different characteristics.

In some embodiments, the cells of the disclosure can be expanded by co-culturing with tissue or cells. The cells can also be expanded in vivo, for example in the subject's blood after administering the cell into the subject.

In another aspect, the disclosure provides compositions (such as pharmaceutical compositions) comprising any of the cells of the disclosure or any of the populations of cells of the disclosure. In some embodiments, the composition comprises a T cell comprising a polynucleotide encoding a protease-activating CD45-gate CAR protein or protease-activating CD45-gate CAR protein derivative, which may

further comprise a polynucleotide encoding a second CAR. The compositions comprise, for example, an immune cell e.g. T cell of the disclosure, e.g. an immune cell that expresses a protease-activating CD45-gate CAR and/or protease-activating CD45-gate CAR protein derivative, or comprise a population of cells that comprises an immune cell e.g. T cell of the disclosure, and one or more pharmaceutically acceptable carriers or excipients.

In some embodiments, primary cells isolated from a donor are manipulated as described herein to provide a population of cells of which a subpopulation (e.g. a proportion less than 100%, such as 10%, 20%, 30%) of the resulting cells comprise all of the desired modifications. Such a resulting population comprising a mixture of cells that comprise all of the modifications and cells that do not can be used in the methods of treatment of the disclosure and to prepare the compositions of the disclosure. Alternatively, this population of cells (the "starting population") can be manipulated by known methods e.g. cell sorting and/or expansion of cells that have the desired modifications, to provide a population of cells that is enriched for those cells comprising one or more of the desired modifications (e.g. enriched for cells that express the desired antigen binding protein, for cells that express a protease-activating CD45-gate CAR protein and/or protease-activating CD45-gate CAR protein derivative), that is, that comprises a higher percentage of such modified cells than did the starting population. The population enriched for the modified cells can then be used in the methods of treatment of the disclosure and to prepare the compositions of the disclosure, for example. In some embodiments, the enriched population of cells contains or contains at least 40%, 50%, 60%, 70%, 80%, 90%, 95%, or 99% cells that have one or more of the modifications. In other embodiments, the proportion of cells of the enriched population of cells that comprise one or more of the modifications is at least 30% higher than the proportion of cells of the starting population of cells that comprise the desired modifications.

In some embodiments, a population of immune cells comprises one or more of the engineered immune cells disclosed herein. In an embodiment, the population comprises about or at least about 1×10^4 , about or at least about 1×10^5 , about or at least about 1×10^6 , about or at least about 1×10^7 , or about or at least about 1×10^8 engineered cells, e.g. engineered immune cells, as disclosed herein, optionally wherein the population does not comprise more than about 1×10^{10} or more than about 1×10^9 or more than about 5×10^9 engineered cells, e.g. engineered immune cells, as disclosed herein. In an embodiment, a population of immune cells as disclosed herein is enriched for the engineered immune cell as disclosed herein. In various embodiments, the population of immune cells is at least 50%, e.g. 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or more than 95% engineered cells that are T cells, e.g., alpha/beta T cells and gamma/delta T cells, B cells, natural killer (NK) cells, natural killer T (NKT) cells, mast cells, and/or myeloid-derived phagocytes. In various embodiments, the population of immune cells is at least 50%, e.g. 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or more than 95% engineered T cells.

Methods of Treating

Immune cells e.g. T cells obtained by the methods described above, or cell lines derived from such immune cells or T cells, can be used as a medicament. In some embodiments, such a medicament can be used for treating a disorder such as for example a viral disease, a bacterial disease, a cancer, an inflammatory disease, an immune

disease, an autoimmune disease, an infection, or an aging-associated disease. In some embodiments, the cancer is a hematological malignancy or non-solid tumor. In some embodiments, the cancer is a solid cancer or solid tumor. In some embodiments, the cancer is a solid cancer or solid tumor. In some embodiments, the cancer is a hematological malignancy that is acute lymphoblastic leukemia (ALL), acute myeloid leukemia (AML), chronic myelogenous leukemia (CML), chronic eosinophilic leukemia (CEL), myelodysplasia syndrome (MDS), non-Hodgkin's lymphoma (NHL), or multiple myeloma (MM). In some embodiments, the cancer is a solid cancer that is biliary cancer, bladder cancer, bone or soft tissue carcinoma, brain tumor, breast cancer, cervical cancer, colon cancer, colorectal adenocarcinoma, colorectal cancer, desmoid tumor, embryonal cancer, endometrial cancer, esophageal cancer, gastric cancer, gastric adenocarcinoma, glioblastoma multiforme, gynecological tumor, head and neck squamous cell carcinoma, hepatic cancer, lung cancer, malignant melanoma, osteosarcoma, ovarian cancer, pancreatic cancer, pancreatic ductal adenocarcinoma, primary astrocytic tumor, primary thyroid cancer, prostate cancer, renal cancer, renal cell carcinoma, rhabdomyosarcoma, skin cancer, soft tissue sarcoma, testicular germ-cell tumor, urothelial cancer, uterine sarcoma, or uterine cancer. In some embodiments, the cancer can be selected from the group consisting of gastric cancer, sarcoma, osteosarcoma, rhabdomyosarcoma, tissue sarcoma, uterine sarcoma, lymphoma, leukemia, head and neck cancer, thymic cancer, epithelial cancer, salivary cancer, liver cancer, stomach cancer, thyroid cancer, lung cancer, ovarian cancer, breast cancer, prostate cancer, esophageal cancer, pancreatic cancer, glioma, leukemia, multiple myeloma, renal cell carcinoma, bladder cancer, cervical cancer, choriocarcinoma, colon cancer, oral cancer, skin cancer, and melanoma. In some embodiments, the subject is a previously treated adult subject with locally advanced or metastatic melanoma, squamous cell head and neck cancer (SCHNC), ovarian carcinoma, sarcoma, or relapsed or refractory classic Hodgkin's Lymphoma (cHL). In an embodiment, the condition is ovarian cancer and the protease-activating CD45-gate CAR comprises the amino acid sequence of any of SEQ ID NOs: 11, 56, 75, 76, 77, 78, 79, and 121-170.

In some embodiments, immune cells e.g. T cells according to the disclosure, or cell line derived from the immune cells e.g. T cells, can be used in the manufacture of a medicament for treatment of a disorder in a subject in need thereof. In some embodiments, the disorder can be, for example, a cancer, an autoimmune disorder, or an infection.

Also provided herein are methods for treating subjects. In some embodiments the method comprises administering or providing an immune cell e.g. T cell of the disclosure to a subject in need thereof. In some embodiments, the method comprises a step of administering the immune cells e.g. T cells of the disclosure to a subject in need thereof.

In another aspect, provided herein is a method of treating a condition in a patient comprising administering to the patient an engineered immune cell as disclosed herein. In an embodiment of the method, the engineered immune cell is an allogeneic engineered immune cell derived from a donor other than the patient.

In another aspect, provided herein is a method of treating a condition in a patient comprising administering to the patient a population of immune cells as disclosed herein. In an embodiment of the method, the immune cells of the population are derived from one or more allogeneic immune cells from a donor other than the patient.

In another aspect, provided herein is a method of treating a condition in a patient comprising administering to the patient a pharmaceutical composition as disclosed herein. In an embodiment of the method, the composition comprises one or more engineered allogeneic immune cells derived from a donor other than the patient.

In some embodiments, immune cells e.g. T cells of the disclosure can undergo robust in vivo cell expansion and can persist for an extended amount of time. Methods of treatment of the disclosure can be ameliorating, curative or prophylactic. The method of the disclosure may be either part of an autologous immunotherapy or part of an allogeneic immunotherapy treatment. The disclosure is particularly suitable for allogeneic immunotherapy. Immune cells e.g. T cells from donors can be transformed into non-alloreactive cells using standard protocols and reproduced as needed, thereby producing e.g. CAR-T cells which may be administered to one or several subjects. Such CAR-T cell therapy can be made available as an "off the shelf" therapeutic product.

In another aspect, the disclosure provides a method of inhibiting tumor growth or progression in a subject who has a tumor, comprising administering to the subject an effective amount of immune cells e.g. T cells as described herein. In another aspect, the disclosure provides a method of inhibiting or preventing metastasis of cancer cells in a subject, comprising administering to the subject in need thereof an effective amount of immune cells e.g. T cells as described herein. In another aspect, the disclosure provides a method of inducing tumor regression in a subject who has a tumor, comprising administering to the subject an effective amount of immune cells e.g. T cells as described herein.

In some embodiments, the immune cells e.g. T cells provided herein can be administered parenterally in a subject. In some embodiments, the subject is a human.

In some embodiments, the method can further comprise administering an effective amount of a second therapeutic agent. In some embodiments, the second therapeutic agent is, for example, crizotinib, palbociclib, an anti-CTLA4 antibody, an anti-4-1 BB antibody, a PD-1 antibody, or a PD-L1 antibody.

Also provided is the use of any of the immune cells e.g. T cells provided herein in the manufacture of a medicament for the treatment of cancer or for inhibiting tumor growth or progression in a subject in need thereof.

Also provided herein are uses and methods as described herein wherein the condition is ovarian cancer and the protease-activating CD45-gate CAR comprises the amino acid sequence of any of SEQ ID NOs: 11, 56, 75, 76, 77, 78, 79, and 121-170. Also provided herein are uses and methods as described herein wherein the condition is ovarian cancer and a CAR comprising the amino acid sequence of any of SEQ ID NOs: 11, 56, 75, 76, 77, 78, 79, and 121-170 is used in addition to or instead of the protease-activating CD45-gate CAR.

The degree to which the gate modulates the protease-activating CD45-gate CAR's activity can be assessed in various ways. One way is to measure the cytotoxicity of a T cell that expresses the protease-activating CD45-gate CAR in comparison to various controls, such as a T cell that expresses a protein that is the same as the protease-activating CD45-gate CAR but that lacks a protease cleavage site in the linker that connects the CD45 recruiting domain to the ligand binding domain (non-cleavable gate), a T cell that expresses a protein that comprises the same CAR but that has no CD45 recruiting domain (non-gated CAR), and a T cell that has not been transduced. The cytotoxicity of these

cells can be measured by incubating them with target cells that express the ligand that the CAR antigen binding domain recognizes. The assay can be adjusted by varying the ratio of the effector CAR T cell to the target ligand-expressing cell. When the effector and target cells are incubated together under the appropriate conditions, while not wishing to be bound by theory, it is expected that the CAR T cells will exhibit high cytotoxicity and both the cleavable and non-cleavable CD45-gate CAR T cells will exhibit lower cytotoxicity because the gate at least partially inactivates the CAR. It is expected that the non-transduced cells will exhibit no cytotoxicity. This form of assay serves to confirm that, under the conditions of the assay, the gate can at least partially inactivate the CAR when the CAR's ligand is present but the protease is absent. It thus mimics the in vivo condition in which the protease-activating CD45-gate CAR T cell encounters its antigen off-tumor. In such an assay, in some embodiments, the protease-activating CD45-gate CAR cytotoxicity is reduced by or by at least about 20%, 30%, 40%, 50%, 60%, 70%, 80%, or 90% relative to the non-gated CAR. The degree of reduction observed depends on the conditions of the assay.

To mimic the in vivo condition in which the protease-activating CD45-gate CAR T cell encounters antigen at the tumor site, the same assay can be repeated in the presence of a protease that can cleave the linker that connects the CD45 recruiting domain to the ligand binding domain. It is expected that the protease will not affect the non-gated CAR T cells' cytotoxicity. It is also expected that the cleavable protease-activating CD45-gate CAR will show reduced cytotoxicity in the absence of the protease and will show higher cytotoxicity in the presence of the protease as compared with in the absence of the protease, because, while not wishing to be bound by theory, in the presence of the protease, the gate that inhibited CAR activation has been removed by cleavage of the linker. It is expected that the non-cleavable gate CAR cytotoxicity will not be affected by the presence of the protease. It is expected that its cytotoxicity will be reduced both in the absence of and in the presence of the protease, because, while not wishing to be bound by theory, the gate inhibits the CAR's activation under both conditions. In such an assay, in some embodiments, in the absence of protease, the protease-activating CD45-gate CAR cytotoxicity is reduced by or by at least about 20%, 30%, 40%, 50%, 60%, 70%, 80%, or 90% relative to the non-gated CAR and, in the presence of protease, the cells' cytotoxicity is significantly higher. The degrees of reduction and increase observed depend on the conditions of the assay.

In another aspect, provided herein is a method of reducing on-target off-tumor toxicity of CAR T cells comprising administering to a patient a cell as disclosed herein, a population of cells as disclosed herein, or a composition as disclosed herein, wherein the on-target, off-tumor toxicity is lower than the on-target, off-tumor toxicity of control cells, a population of control cells or a composition of control cells, respectively, administered under the same conditions, wherein the control cells, population of control cells, or control cells of the composition comprise a CAR that lacks at least one of a CD45 recruiting domain and a linker that comprises a protease cleavage site.

In another aspect, provided herein is a method of increasing the efficiency of CAR T cells or CAR T cell therapy comprising administering to a patient a cell as disclosed herein, a population of cells as disclosed herein, or a composition as disclosed herein, wherein the efficiency is greater than the efficiency of control cells, a population of

control cells or a composition of control cells, respectively, administered under the same conditions, wherein the control cells, population of control cells, or control cells of the composition comprise a CAR that lacks at least one of a CD45 recruiting domain and a linker that comprises a protease cleavage site.

In another aspect, provided herein is a method of increasing the efficacy of CAR T cell therapy comprising administering to a patient a cell as disclosed herein, a population of cells as disclosed herein, or a composition as disclosed herein, wherein the efficacy is greater than the efficacy of control cells, a population of control cells or a composition of control cells, respectively, administered under the same conditions, wherein the control cells, population of control cells, or control cells of the composition comprise a CAR that lacks at least one of a CD45 recruiting domain and a linker that comprises a protease cleavage site.

In another aspect, provided herein is a method of increasing the efficacy of CAR T cell therapy against a solid tumor comprising administering to a patient having a solid tumor a cell as disclosed herein, a population of cells as disclosed herein, or a composition as disclosed herein, wherein the efficacy against the solid tumor is greater than the efficacy against the solid tumor of control cells, a population of control cells or a composition of control cells, respectively, administered under the same conditions, wherein the control cells, population of control cells, or control cells of the composition comprise a CAR that lacks at least one of a CD45 recruiting domain and a linker that comprises a protease cleavage site.

In another aspect, provided herein is a method of reducing the incidence of side effects in CAR T cell therapy comprising administering to a patient a cell as disclosed herein, a population of cells as disclosed herein, or a composition as disclosed herein, wherein the incidence of side effects is lower than the incidence of side effects when control cells, a population of control cells or a composition of control cells, respectively, are administered under the same conditions, wherein the control cells, population of control cells, or control cells of the composition comprise a CAR that lacks at least one of a CD45 recruiting domain and a linker that comprises a protease cleavage site.

In another aspect, provided herein is a method of effecting reversible colocalization of CD45 and a CD45-gate CAR in a CAR T cell, the reversible colocalization comprising a CD45-gate CAR recruiting domain binding to a CD45 protein on a CAR T cell surface, resulting in colocalization, which binding and colocalization can be disrupted by a protease cleaving a protease cleavage site in the CD45-gate CAR's linker, thereby reversing the colocalization, the method comprising the steps of:

- (a) providing a T cell that expresses CD45,
- (b) introducing a nucleic acid encoding a protease-activating CD45-gate CAR as disclosed herein into the T cell,
- (c) maintaining the T cell under conditions in which both CD45 and the CAR are functionally expressed at the cell surface,

resulting in a CD45-gate CAR recruiting domain binding to a CD45 protein on the CAR T cell surface and colocalization of the CAR and the CD45 protein, which colocalization can be reversed by a protease cleaving a protease cleavage site in the CD45-gate CAR's linker, wherein the cleaving produces a functional CAR no longer connected by the linker to the CD45-gate CAR recruiting domain.

In another aspect, provided herein is a method of effecting reversible reduction of CAR activity by a CD45 protein in a CAR T cell, the reversible reduction of CAR activity comprising a CD45-gate CAR recruiting domain binding to a CD45 protein on a CAR T cell surface, resulting in inactivation of the chimeric antigen receptor by the CD45, which inactivation can be disrupted by a protease cleaving a protease cleavage site in the CD45-gate CAR's linker, resulting in the disassociation of the chimeric antigen receptor and the CD45 protein and ending the inactivation of the chimeric antigen receptor by the CD45, the method comprising the steps of:

- (a) providing a T cell that expresses CD45,
- (b) introducing a nucleic acid encoding a CD45-gate CAR into the T cell,
- (c) maintaining the T cell under conditions in which both CD45 and the CAR are functionally expressed at the cell surface,

resulting in a CD45-gate CAR recruiting domain binding to a CD45 protein on the CAR T cell surface, association of the CAR and the CD45 protein, and reduction of the CAR's activity by the CD45 protein, which activity reduction can be reversed by a protease cleaving a protease cleavage site in the CD45-gate CAR's linker, thereby ending the association of the CAR and the CD45 protein, and ending the CD45 protein's reduction of CAR activity, wherein the cleaving produces a functional CAR no longer connected by the linker to the CD45-gate CAR recruiting domain.

In another aspect, provided herein is a method of treating with CAR T cell therapy a patient who has a tumor characterized by a protease-rich tumor microenvironment, comprising administering to the patient a cell as disclosed herein, a population of cells as disclosed herein, or a composition as disclosed herein, wherein CAR T cell activity is lower outside the protease-rich tumor microenvironment than in the protease-rich tumor microenvironment.

In another aspect, provided herein is a method of regulating cytotoxic activity of a CAR T cell that comprises a CAR as disclosed herein, the method comprising inhibiting the cytotoxic activity of the CAR T cell by associating the CAR with CD45 of the CAR T cell, and activating the cytotoxic activity of the CAR T cell by subjecting and/or exposing the CAR T cell to a protease that recognizes and cleaves the protease cleavage site.

In embodiments of the methods of treating a condition in a patient disclosed herein, the cell, population of cells or composition can be administered to the subject on one occasion or can be administered to the subject on two or more occasions spaced at least about 1, 2, 3, 4, 5, 6, 7, or more days apart. In some embodiments, the disorder can be a viral disease, a bacterial disease, a cancer, an inflammatory disease, an immune disease, or an aging-associated disease.

In some embodiments, the treatment can be in combination with one or more therapies against cancer selected from the group of antibodies therapy, chemotherapy, cytokines therapy, dendritic cell therapy, gene therapy, hormone therapy, laser light therapy and radiation therapy.

In some embodiments, treatment can be administered to or administrated into subjects undergoing an immunosuppressive treatment. Indeed, the disclosure may rely on cells or a population of cells which have been made resistant to at least one immunosuppressive agent due to the inactivation of a gene encoding a receptor for such immunosuppressive agent. In this aspect, the immunosuppressive treatment may help the selection and expansion of the T cells according to the disclosure within the subject.

The administration of the cells or population of cells according to the disclosure may be carried out in any convenient manner, including by aerosol inhalation, injection, ingestion, transfusion, implantation or transplantation. The compositions described herein may be administered to a subject subcutaneously, intradermally, intratumorally, intranodally, intramedullary, intramuscularly, by intravenous or intralymphatic injection, or intraperitoneally. In one embodiment, the cell compositions of the disclosure are administered by intravenous injection.

In some embodiments, the administration of the cells or population of cells can comprise administration of, for example, about 10^4 to about 10^6 cells per kg body weight including all integer values of cell numbers within those ranges. In some embodiments the administration of the cells or population of cells can comprise administration of about 10^4 to about 10^6 cells per kg body weight including all integer values of cell numbers within those ranges. The cells or population of cells can be administered in one or more doses. In some embodiments, an effective amount of cells can be administered as a single dose. In some embodiments, an effective amount of cells can be administered as more than one dose over a period time. Timing of administration is within the judgment of the managing physician and depends on the clinical condition of the subject. The cells or population of cells may be obtained from any source, such as a blood bank or a donor. While individual needs vary, determination of optimal ranges of effective amounts of a given cell type for a particular disease or conditions is within the skill of the art. An effective amount means an amount which provides a therapeutic or prophylactic benefit. The dosage administered will be dependent upon the age, health and weight of the recipient, the kind of concurrent treatment, if any, the frequency of treatment and the nature of the effect desired. In some embodiments, an effective amount of cells or composition comprising those cells are administered parenterally. In some embodiments, administration can be an intravenous administration. In some embodiments, administration can be directly done by injection within a tumor.

In some embodiments of the disclosure, cells are administered to a subject in conjunction with (e.g., before, simultaneously or following) any number of relevant treatment modalities, including but not limited to treatment with agents such as monoclonal antibody therapy, CCR2 antagonist (e.g., INC-8761), antiviral therapy, cidofovir and interleukin-2, Cytarabine (also known as ARA-C) or natalizumab treatment for MS subjects or efalizumab treatment for psoriasis subjects or other treatments for PML subjects. In some embodiments, protease-activating CD45-gate CAR-T cells are administered to a subject in conjunction with one or more of the following: an anti-PD-1 antibody (e.g., nivolumab, pembrolizumab, or PF-06801591), an anti-PD-L1 antibody (e.g., avelumab, atezolizumab, or durvalumab), an anti-OX40 antibody (e.g., PF-04518600), an anti-4-1 BB antibody (e.g., PF-05082566), an anti-MCSF antibody (e.g., PD-0360324), an anti-GITR antibody, and/or an anti-TIGIT antibody. In further embodiments, the immune cells e.g. T cells of the disclosure may be used in combination with chemotherapy, radiation, immunosuppressive agents, such as cyclosporin, azathioprine, methotrexate, mycophenolate, and FK506, antibodies, or other immunoablative agents such as CAMPATH, anti-CD3 antibodies or other antibody therapies, cytoxin, fludarabine, cyclosporin, FK506, rapamycin, mycophenolic acid, steroids, FR901228, cytokines, and/or irradiation. These drugs inhibit either the calcium dependent phosphatase calcineurin (cyclosporine and FK506) or inhibit the p70S6 kinase that is important for growth factor induced

signaling (rapamycin) (Henderson, Naya et al. Immunology. 1991 July; 73(3): 316-321; Liu, Albers et al. Biochemistry 1992 April 28; 31(16):3896-901; Bierer, Hollander et al. Curr Opin Immunol. 1993 October; 5(5):763-73).

In a further embodiment, the cell compositions of the disclosure are administered to a subject in conjunction with (e.g., before, simultaneously or following) bone marrow transplantation, T cell ablative therapy using either chemotherapy agents such as fludarabine, external-beam radiation therapy (XRT), cyclophosphamide, or antibodies such as OKT3 or CAMPATH. In some embodiments, the cell compositions of the disclosure are administered following B-cell ablative therapy such as agents that react with CD20, e.g., Rituxan. For example, in one embodiment, subjects may undergo standard treatment with high dose chemotherapy followed by peripheral blood stem cell transplantation. In certain embodiments, following the transplant, subjects receive an infusion of expanded immune cells of the disclosure. In some embodiments, expanded cells are administered before or following surgery.

Kits

The disclosure also provides kits for use in the instant methods. Kits of the disclosure include one or more containers comprising a composition of the disclosure or an immune cell e.g. a T cell of the disclosure or a population of cells comprising an immune cell e.g. a T cell of the disclosure. In various embodiments, the immune cell e.g. T cell comprises one or more polynucleotide(s) encoding a protease-activating CD45-gate CAR protein as described herein or a protease-activating CD45-gate CAR protein derivative. The kit further comprises instructions for use in accordance with any of the methods of the disclosure described herein. Generally, these instructions comprise a description of administration of the composition, immune cell e.g. T cell or population of cells for the above described therapeutic treatments. In some embodiments the engineered immune cells are formulated in a solution comprising about 5% DMSO. Further, the engineered immune cells can be provided in a frozen state.

The instructions relating to the use of the kit components generally include information as to dosage, dosing schedule, and route of administration for the intended treatment. The containers may be unit doses, bulk packages (e.g., multi-dose packages) or sub-unit doses. Instructions supplied in the kits of the disclosure are typically written instructions on a label or package insert (e.g., a paper sheet included in the kit), but machine-readable instructions (e.g., instructions carried on a magnetic or optical storage disk) are also acceptable.

The kits of this disclosure are in suitable packaging. Suitable packaging includes, but is not limited to, vials, bottles, jars, flexible packaging (e.g., sealed Mylar or plastic bags), and the like. Also contemplated are packages for use in combination with a specific device, such as an inhaler, nasal administration device (e.g., an atomizer) or an infusion device such as a minipump. A kit may have a sterile access port (for example the container may be an intravenous solution bag or a vial having a stopper pierceable by a hypodermic injection needle). The container may also have a sterile access port (for example the container may be an intravenous solution bag or a vial having a stopper pierceable by a hypodermic injection needle). At least one active agent in the composition is an immune cell e.g. T cell according to the disclosure. The container may further comprise a second pharmaceutically active agent.

Kits may optionally provide additional components such as buffers and interpretive information. Normally, the kit

comprises a container and a label or package insert(s) on or associated with the container.

EXAMPLES

The following examples are offered for illustrative purposes only, and are not intended to limit the scope of the disclosure in any way. Indeed, various modifications of the disclosure in addition to those shown and described herein will become apparent to those skilled in the art from the foregoing description and fall within the scope of the appended claims.

Example 1: Synthesis and Demonstration of CD45-Gate CAR

CD45-Gate CAR Structure

FIG. 1A shows a schematic representation of a protease-activating CD45-gate CAR. To investigate how the protease-activating CD45-gate can regulate the function of CAR T cells, we used a 2nd generation CAR ("53B6") that specifically targets MUC16-expressing cells. A generic CAR protein is comprised of an extracellular target-binding domain, a transmembrane domain, and an intracellular signaling domain that, for example, comprises an activation domain (e.g. CD3zeta cytoplasmic domain) and optionally also a costimulatory domain (e.g. 4-1BB signalling domain). In general, the extracellular domain of the CD45-gate CAR disclosed herein comprises, in addition, a CD45 recruiting domain ("recruiter"), fused via a flexible linker, which linker comprises, in certain embodiments, one or more protease-cleavable sites, to the N-terminus of a generic CAR, as illustrated in FIGS. 1A and 1B (bottom construct). As diagrammed in FIG. 1B (bottom construct), an additional N-HA tag (comprising the sequence YPYDVPDYA (SEQ ID NO: 187), a variant of SEQ ID NO: 2 without the initial G) was introduced as well for detection of the CD45-gate CAR, and the CD8 signal peptide was fused to the N-terminus of each protein to enable trafficking and surface expression. Non-gated, non-N-HA-tagged 53B6 (FIG. 1B, top construct) was used as a control in some of the experiments.

CD45-Gate CAR Expression in T Cells

To evaluate CAR T cells that functionally express the protease-activating CD45-gate CAR, two variants of the CD45-gate were prepared, one comprising a first anti-CD45 scFv as the CD45 recruiter and the other comprising a second, different anti-CD45 scFv as the CD45 recruiter. A flexible protease-cleavable linker, which contained two thrombin protease cleavage sites flanking a V5 peptide motif, which can be used to detect the linker (SEQ ID NO: 8, GGGGSGGGGSLVPRGSKPIPNNLLGLDSTLVPRGSGGGGSGGGGS), joined the CD45 recruiter to the N-terminus of the CAR. These two protease-activating CD45-gate CAR clones are designated "CD45-Gate1-TMB-53B6" (comprising amino acid sequence of SEQ ID NO: 16) and "CD45-Gate2-TMB-53B6" ("TMB" indicating the thrombin-cleavable linker) (comprising amino acid sequence of SEQ ID NO: 18), where the exemplary anti-MUC16 CAR clone 53B6 was used to demonstrate the proof of concept of such a protease-activating CD45-gate CAR. In parallel, a non-cleavable linker of the same length (SEQ ID NO: 7, GGGGSGGGGSGGGSGSKPIPNNLLGLDSTLVPRGSGGGGSGGGGS) was used to generate additional control CARs, designated "CD45-Gate1-GS-53B6" (comprising amino acid sequence of SEQ ID NO: 17) and "CD45-Gate2-GS-53B6" (comprising amino acid sequence of SEQ ID NO:

19), both of which are not expected to be cleaved by a protease. CAR T cells expressing either the non-gate 53B6 CAR or the CD45-gate 53B6 CARs were generated with lentiviral vectors as previously described (14,15). In brief (and using the methods described in Example 2 below, CAR-T Production), primary T cells from a healthy donor were transduced with lentiviruses (MOI of 5) expressing the different CAR molecules and the percentage of CAR+ cells were recorded at day 8 and day 15. As the data in FIG. 2A show, CARs with the CD45-gate1 ("CD45-Gate1-TMB-53B6" and "CD45-Gate1-GS-53B6") show CAR+ rate comparable to that of the 53B6 CAR throughout the production. The CD45-gate2 CARs ("CD45-Gate2-TMB-53B6" and "CD45-Gate2-GS-53B6") exhibit lower transduction rate initially, but are able to match up to the CAR+ percentage in the end of the production (data for FIG. 2A were obtained as described below under Example 2 with respect to FIG. 7A, Flow Cytometry Analysis). The overall growth of T cells remains similar among all the clones (see FIG. 2B). These data together indicate that, although CD45 is expressed on the surface of CAR T cells, surprisingly the CD45-gate CAR can be produced and expressed in the T cells successfully without causing severe fratricide.

CD45-Gate CAR Protease Cleavage and Ligand Binding

To evaluate whether the CD45-gate can be released by the corresponding protease, the detection of V5 tag that is imbedded in the cleavable or non-cleavable linkers was assessed. The protease cleavage was carried out by incubating the CAR T cells in PBS with 2 microgram/ml of thrombin at 370 for 30 mins. The successful cleavage of the protease linker was verified by the removal of V5 tag after cleavage. As the data in FIG. 3A show, the removal of V5 tag was only observed in the clone of CD45-Gate1-TMB-53B6 and CD45-Gate2-TMB-53B6 after cleavage, indicating that the CD45-gate can be released from CAR by the specific cleavage by the protease.

Next, whether the CD45-gate directly blocks the ligand binding was evaluated by assessing the binding of soluble MUC16 ligand to each CAR clone, before or after protease cleavage. It was observed that 53B6, CD45-Gate1-TMB-53B6, CD45-Gate1-GS-53B6 show very similar ligand-binding intensity, regardless of protease cleavage (see data in FIG. 3B). CD45-Gate2-TMB-53B6 and CD45-Gate2-GS-53B6 exhibited decreased ligand-binding intensity, likely due to inferior expression of the modified CAR molecule (see data in FIG. 3B). Nonetheless, the binding was not affected by protease cleavage and the release of CD45-gate. Overall, these data demonstrate that CD45 gate does not block binding to soluble target.

CD45-Gate CAR T Cell Cytotoxicity

To understand whether the CD45-gate can regulate the CAR function, two target cell lines, Ovar3 (MUC16High) and Cov644 (MUC16Low), were used to assess the cytotoxicity of each clone. Both target cells are incorporated with luciferase gene for facile quantification of viable cells. In brief, CAR T cells were normalized to the same CAR+ percentage and incubated with 10,000 target cells at E:T (Effector:Target) ratio of 3:1, 1:1, or 1:3 in 200 μ l of RPMI medium with 10% of FBS. After the 48-hour co-culture, 100 μ l of the supernatant containing CAR T cells was transferred to new wells with 10,000 fresh target cells, while the remaining samples were used to quantify the viable target cells by luciferase assay. After another 48-hour co-culture, the second round of cytotoxicity assessment was carried out in the same way. As FIGS. 4A-4B show, CD45-Gate1 moderately inhibited the CAR function to kill the Ovar3 (MUC16High) target cells in the first round of cytotoxicity

assay and the inhibition became more pronounced in the second round. However, the CD45-Gate2 showed very limited inhibition of the CAR activity; a reduction in cytotoxic activity was observed only at low E:T ratio (FIGS. 4A and 4B). In contrast, both CD45-Gate1 and CD45-Gate2 substantially inhibited the CAR cytotoxicity against the Cov644 (MUC16Low) target cells (FIGS. 5A-5B). These data together illustrate that the CD45-gate can effectively control CAR function and the level or degree of inhibition can be affected by and vary with the antigen density.

Regulation of CD45-Gate CAR T Cell Cytotoxicity

Next, whether the protease cleavage and the removal of CD45-gate from CAR can restore its cytotoxicity was evaluated. To test this, the cytotoxicity assay was carried out with the MUC16High Ovar3 target cells similarly as above, except that serum-free medium was used in the assay to allow thrombin activity. The cytotoxicity of each clone was evaluated against the target cells at E:T of 1:1, with or without thrombin protease (final concentration of 2 μ g/ml). In this condition, the regular 53B6 CAR T cells still can efficiently kill the target cells, albeit that the activity was weaker than their previous performance in the serum-rich medium. There is no change in 53B6 CAR activity when thrombin is introduced into the culture (FIG. 6A). Importantly, data from the clone of CD45-Gate1-TMB-53B6 and CD45-Gate2-TMB-53B6 demonstrates that the CD45-gate effectively inhibited the CAR activity and the cytotoxicity was mostly restored when the protease was present. In contrast, the CD45-gate in the clone of CD45-Gate1-GS-53B6 or CD45-Gate2-GS-53B6 could not be efficiently released and thus the CAR activity remained inhibited despite the presence of the protease.

Example 2. Expression and Activity of Anti-MUC16 CAR

CAR T Production

T cells were isolated from human peripheral blood mononuclear cells (PBMCs) obtained by density gradient centrifugation (Ficoll Paque, GE Healthcare, Pittsburgh, PA) using a Pan T cell isolation kit (Miltenyi Biotec, Auburn, CA) and cryopreserved.

To make lentivirus encoding 53B6 CAR, HEK-293T cells were plated at 0.45 million cells per mL in 2 mL of DMEM (Gibco) supplemented with 10% FBS (Hyclone or JR Scientific) per well of a 6-well plate on Day 0. On Day 1, the lentivirus was prepared by mixing together lentiviral packaging vectors 1.5 μ g psPAX2, 0.5 μ g pMD2G, and 1 μ g of the appropriate transfer CAR vector in 250 μ L Opti-MEM (Gibco) per well of the 6-well plate ("DNA mix"). 10 μ L Lipofectamine 2000 (Invitrogen) in 250 μ L Opti-MEM was incubated at room temperature for 5 minutes and then added to the DNA mix. The mixture was incubated at room temperature for 20 minutes and the total volume of 500 μ L was slowly added to the sides of the wells containing HEK-293T. Purified T cells were activated in T cell transduction media (X-Vivo-15 medium [Lonza] with 10% FBS) supplemented with 20 ng/ml of human IL-2 (Miltenyi Biotec) and human T cell TransAct reagent (Miltenyi Biotec), as recommended by the manufacturer. On Day 2, the media from each well of the 6-well plate was replaced with 2 mL per well of T cell transduction media. On Day 3, T cells were resuspended at 0.33 million cells per mL in 1 mL of T cell transduction media. The lentiviral supernatant from HEK293T cells was harvested and passed through 0.45 micron filter (EMD Millipore) to remove cell debris, and then added to 0.5 million activated T cells along with 20

ng/ml human IL-2. Alternatively, to enhance the transduction efficiency, lentiviruses can be prepared with scaled up ratio as previous protocol and concentrated with Lenti-X™ Concentrator (Takara), following the manufacturer's protocol. Activated T cells were transduced at approximately MOI of 5 to generate CAR T cells.

On Day 5, 4.5 mL of T cell expansion media (X-Vivo-15 with 5% human AB serum [Gemini Bio]) was added to each well of a Grex-24 (Wilson Wolf) plate. IL-2 (20 ng/ml) was supplemented every 2-3 days. On Day 9 and Day 13, transduction efficiency was determined by detecting the percentage of T cells that recognize recombinant MUC16 protein (in-house) using flow cytometry. On Day 14, 53B6 CAR-T cells were cryopreserved.

Flow Cytometry Analysis (FIGS. 7A-7B)

To determine the percentage of T cells that were successfully transduced with 53B6 CAR, T cells were first incubated with 3 µg/ml biotinylated recombinant MUC16 protein (in house) in FACS buffer (PBS+0.5% BSA with 2 mM EDTA) for 30 minutes at 4° C. The cells were then washed with FACS buffer, stained with streptavidin-PE labelled protein (Thermo Fisher, Cat #21627) at 1:250 ratio and analyzed using flow cytometry. As example, FIG. 7A shows that 53B6 CAR T cells were successfully generated from multiple productions using different T cell donors. The flow cytometry plot of 53B6 CAR T cells was from one of the productions. Empty vector (EV) control T cells were included in the studies.

Alternatively, another recombinant MUC16 conjugated with Alexa Fluor®647 were used to directly stain CAR T cells at approximately final concentration of 3 µg/ml to evaluate the binding of 53B6 CAR in FIGS. 2A and 3B.

Phenotypes (FIG. 7B) of the CAR T cells were also determined on Day 13 according to CD62L (Biolegend) and CD45RO (Biolegend) expression within the CAR+ T cell population as follows: stem cell memory (Tscm; CD45RO-/CD62L+), central memory (Tcm; CD45RO+/CD62L+), effector memory (Tem; CD45RO+/CD62L-), effector cells (Teff; CD45RO-/CD62L-). Lineage markers using antibodies toward CD3 (BD Biosciences), CD4 (Biolegend) and CD8 (Biolegend) were also included in the stain.

Cytotoxicity Assay (FIGS. 8A-8C and FIGS. 9A-9B)

Short Term Killing Assay

This example describes experiments used to determine the specificity and in vitro activity of 53B6 CAR.

OVCAR3, COV644 and FUV1 ovarian cell lines were purchased from ATCC or Sigma, and surface expression of MUC16 protein was determined using a quantitative analysis kit (Agilent Cat. #K007811-8), according to manufacturer's recommendation. To determine the activity of 53B6 CAR T cells, 1×10^4 luciferase-expressing ovarian target cells were co-cultured in RPMI 1640 medium (Gibco) supplemented with 10% FBS with 53B6 CAR+ T cells at defined Effector:Target (E:T) ratio in 96-well assay plates (Corning). Empty vector (EV) control T cells were also included in the study. Cell viability was measured after 72 hours using One-Glo reagent (Promega). Each condition was assayed in replicate wells. Percent target cell survival after being exposed to T cells was determined by comparing to target cells alone. Average percentage of live cells and standard deviation are shown (FIGS. 8A-8C).

Long Term Serial Killing Assay

A serial killing assay involves repeated exposure of CAR T cells to their target causing the CAR T cells to undergo proliferation and in certain cases, differentiation and exhaustion.

To determine the persistent killing of 53B6 CAR T cells, 1×10^4 luciferase-expressing OVCAR3 target cells were co-cultured in RPMI 1640 medium (Gibco) supplemented with 10% FBS with 53B6 CAR+ T cells at Effector:Target ratio of 1:1 in a serial killing assay using 96-well assay plates. Every 2-3 days thereafter, 100 µl medium from each well were transferred to freshly plated target cells (1×10^4 cells per well) and percent target cell survival of the previously plated target cells after being exposed to CAR T cells was determined by comparing to target cells alone using One-Glo reagent. Each condition was assayed in 3 to 6 replicates. Empty vector (EV) control T cells were also included in the study. Average percentage of live cells and standard deviation are shown. Data presented are the activity of 53B6 CAR T cells from two different donors (FIGS. 9A-9B).

In Vivo Activity (FIGS. 10A-10B)

To test the anti-tumor activity of 53B6 CAR T cells, two different orthotopic ovarian tumor models were used. Luciferase-labeled OVCAR3 and COV644 ovarian cells were cultured in complete growth medium (RPMI 1640 supplemented with 10% FBS). Cells were collected and diluted to 15×10^6 viable cells/ml in PBS, and cell suspension was kept on ice until inoculation. NSG mice (Jackson Laboratory), aged 7-8 weeks old, were injected intra-peritoneally (IP) with 200 µl (3×10^6 target cells per mouse) of OVCAR3 or COV644. Tumor burden was monitored using the IVIS Spectrum (Perkin Elmer), where the animals were injected intra-peritoneally with 200 µl of luciferin (15 mg/ml; Pierce) and bioluminescent intensity was measured. Mice were randomized into groups of 5 based on total body bioluminescence about two weeks post tumor implantation. 53B6 CAR T cells were then thawed and resuspended in PBS. OVCAR3-tumor bearing mice received 1×10^6 53B6 CAR+ T cells per animal, whereas COV644-tumor bearing mice received 3×10^6 53B6 CAR+ T cells per animal. Empty-vector control T cells were also included in the study. All animals infused with T cells were dosed via ip injection in a volume of 200 µl/mouse. The in vivo activity of the 53B6 CAR T cells against the tumor was monitored by bioluminescence, and the results demonstrated anti-tumor activity in both models (FIGS. 10A-10B).

Example 3. Test of CD45-Gate CAR T Cell Activity

3.1. Production of New CD45-Gate CAR T (Effector) Cells (FIGS. 12A-12D and FIGS. 13A-13D)

CAR T cells expressing a CD45-gate MUC16 CAR comprising a TPS linker selected from a panel of TPS linkers (similar to the CD45-gate CARs diagrammed in FIG. 11), were successfully generated and characterized using flow cytometry according to methods described in Example 2 and below. In addition to cells expressing CARs comprising one of the cleavable TPS linkers, cells expressing a CAR comprising a non-cleavable GS linker were prepared as a negative control expected not to exhibit CAR activity because the CD45-gate cannot be released by proteases. Cells expressing a non-gated MUC16 CAR (53B6) were included as a second, positive control expected to be active in the presence or absence of proteases.

Each CAR+ T cell population was detected with AlexaFluor 647-conjugated recombinant MUC16 protein (SEA1-4) using flow cytometry (FIG. 12A). The overall expansion of T cells remains similar among all the clones and is comparable to non-transduced control (NTD), indicating no obvious fratricide (FIG. 12B). Autoactivation and memory phenotypes of the CAR T cells were also deter-

mined using flow cytometry by analyzing the expression of CD25/41BB and CD45RO/CD62L on Day 9 and Day 14, respectively. Overall, CD45-gate CAR T cells contain a higher percentage of stem-cell memory subset (CD45RO-/CD62L+, FIG. 12C) and exhibit less tonic signaling (CD25+/41BB+, FIG. 12D) than naked CAR T cells, suggesting a potential benefit of CD45-gate CARs in controlling T cell exhaustion. At the end of the production (Day 14-16), all T cells were cryopreserved in 90% FBS/10% DMSO using rate-controlled freezing chambers and stored in liquid nitrogen.

A second set of CAR T cell populations, each expressing a CD45-gate MUC16 CAR comprising a TPS linker selected from a second panel of TPS linkers, were generated successfully and characterized according to methods described in Example 2 and this section (results shown in FIGS. 13A-13D). The success in generating CD45-gate CAR T cells was surprising because T cells express endogenous CD45, however no noticeable fratricide was observed that would have hindered the production of the CD45-gate CAR T cells.

3.2. Generation and Analysis of Reagent Cell Lines (FIGS. 14A-14E)

Target cell lines were developed in house based on three cell lines that are positive for (i.e. express active) tumor-associated proteases, including an ovarian cancer cell line OVCAR3, a breast cancer cell line MDA-MB-231, and a lung carcinoma cell line NCI-H292. The surface expression of matriptase (3 µg/mL; R&D, BAF3946) was confirmed using flow cytometry (FIG. 14A) and the secretion of urokinase (uPA) was detected in conditioned media by an ELISA kit (R&D, DUPA00) (FIG. 14D). In addition, the activity of surface expressed matriptase was assessed through the detection of a V5 tag that is attached to CAR via a matriptase cleavable linker TPS6, as previously described in Example 1 (FIG. 14B). Compared to the recombinant matriptase which effectively mediated linker cleavage and removal of V5 tag, the endogenous matriptase showed limited activity, which is, without wishing to be bound by specific mechanisms, believed to be due to the lack of activation of matriptase in the in vitro assay settings. For example, FIG. 14B shows that in this experiment in vitro matriptase is not active, consistent with the CD45-Gate CAR with TPS6 also not being active.

To study the target-mediated CAR activation, MDA-MB-231 and H292 cell lines engineered to overexpress human MUC16 SEA1-4 and OVCAR3 cell line expressing endogenous MUC16 were used as target cells. The expression of the target protein MUC16 was validated by flow cytometry (FIG. 14C). All three cell lines were then engineered to co-express a firefly luciferase and a nuclear GFP using a 2A peptide to enable sensitive measurement of tumor killing using luminescence reading or Incucyte imaging.

3.3 In Vitro Analysis of CD45-Gate CAR T Cell Activity, Protease Dependence and Sensitivity (FIGS. 15A-15F and FIGS. 16A-16F)

A Jurkat-NFAT reporter cell line was used to evaluate the activity of CD45-gate CARs. The Jurkat-NFAT reporter cell line contains a nuclear factor of activated T-cell (NFAT) promoter upstream of a luciferase gene, which allows the quantitative measurement of signaling activation upon stimulation. To generate CAR reporter cells, Jurkat-NFAT cells were transduced with lentivirus encoding CD45-gate CARs and expression was analyzed after 6 days using recombinant MUC16 protein (SEA1-4). Transduced cells were then cocultured with target cells (1×10⁴/well) at a 1:1 E:T ratio in 96-well white flat-bottom plates. Wells contain-

ing only Jurkat cells also were included as negative controls. All cocultures were performed in CTS OpTmizer T cell expansion SFM (Gibco) with or without human recombinant matriptase (0.2 µg/mL, R&D, 3946-SEB-010). After overnight incubation at 37° C., the reporter gene activation was detected by luminescence and normalized to the untreated control signal when no target cells were added.

As shown in FIGS. 15A-15F, exposure to MUC16 positive target cells led to the activation of 53B6 CAR while no activation was detected with a CD45-gate CAR with non-cleavable linker. However, the activation can be restored via the cleavable linkers upon exposure to endogenous ("−Matriptase" in FIGS. 15A-15C) or exogenous proteases ("+Matriptase" in FIGS. 15A-15C). This shows that the CD45-gate suppresses CAR activity and the suppression can be protease-sensitive when the linker comprises one or more protease sites. The data comprise clones 1241 (2×TPS6), 1466 (2×TPS6), 1467 (2×TPS6), 1468 (2×TPS6), and 1469 (2×TPS6), each of which utilizes linkers with various lengths (45, 35, 30, 25, 20-mers) and yet they enable highly similar functionality. The following constructs were used in FIGS. 15D-15F:

0975 (TMB) refers to SEQ ID NO: 16 (N-HA-4131scFv-V5-2G4STMB-53B6, comprising the linker GSTMB (45 aa) having the amino acid sequence of SEQ ID NO: 8

0976 (NC) refers to SEQ ID NO: 17 (non-cleavable)

1102 (2×TPS3) refers to SEQ ID NO: 24 (N-HA-4131-VL-RS-VH-TPS1-53B6, comprising the linker GSTPS1 (45 aa), having the amino acid sequence of SEQ ID NO: 9, and comprising the linker TPS cleavage, having the amino acid sequence of SEQ ID NO: 53 flanked on both ends by GS sequences

1399 (TPS4) refers to SEQ ID NO: 137, a construct containing the elements V5-GS20-4131scFv-TPS4-53B6, comprising the linker TPS4 45 aa (SEQ ID NO: 107) and a GS sequence

1401 (TPS6) refers to SEQ ID NO: 138, a construct containing the elements V5-GS20-4131scFv-TPS6-53B6, comprising the linker TPS6 45 aa having the amino acid sequence of SEQ ID NO: 113

1241 (2×TPS6) refers to SEQ ID NO: 144, a construct containing the elements N-HA-4131scFv-2×TPS6-53B6, comprising the linker TPS6 35aa having the amino acid sequence of SEQ ID NO: 112 and a second linker comprising the cleavage site MTSP (3) (SEQ ID NO: 95) flanked on both ends by a GS sequence

1466 (2×TPS6) refers to SEQ ID NO: 153, a construct containing the elements V5-GS20-4131scFv-2×TPS6-35-53B6, comprising 2 copies of the linker TPS6 35 aa, each copy having the amino acid sequence of SEQ ID NO: 112

1467 (2×TPS6) refers to SEQ ID NO: 154, a construct containing the elements V5-GS20-4131scFv-2×TPS6-30-53B6, comprising the linker TPS6 35 aa (SEQ ID NO: 112) and the linker TPS6 30 aa (SEQ ID NO: 111)

1468 (2×TPS6) refers to SEQ ID NO: 155, a construct containing the elements V5-GS20-4131scFv-2×TPS6-25-53B6, comprising the linker TPS6 35 aa (SEQ ID NO: 112) and the linker TPS6 25 aa (SEQ ID NO: 110)

1469 (2×TPS6) refers to SEQ ID NO: 156, a construct containing the elements V5-GS20-4131scFv-2×TPS6-20-53B6, comprising the linker TPS6 35aa (SEQ ID NO: 112) and the linker TPS6 20aa (SEQ ID NO: 109)

3.4. Cytotoxicity of CD45-Gate CAR T Cells

The short-term cytotoxicity of CD45-gate CAR T cells were characterized using Incucyte. For kinetic analysis of

tumor cell killing, target cells MDA-MB-231-MUC16, H292-MUC16 and OVCAR3 cells engineered to express nuclear GFP were plated at a concentration of 1×10^4 cells per well in 96-well black flat-bottom plates. The following day, T cells were thawed and added at an effector:target (E:T) ratio of 3:1. All cocultures were performed in CTS OpTmizer T cell expansion SFM (Serum-Free Medium; Gibco) with or without human recombinant matriptase (0.2 $\mu\text{g/mL}$, R&D, 3946-SEB-010). The number of viable target cells was monitored by fluorescent imaging every 6 hours over 60 hours using the IncuCyte Live Cell Analysis System (Essen BioScience). Live-cell numbers were quantified by IncuCyte S3 software (Essen BioScience) and normalized to the initial counts at time 0 (time zero). Average fold change in target cell count and standard deviation are shown in FIGS. 16A-16F. CD45-gate CARs with non-cleavable GS linker effectively inhibited CAR function. In the presence of endogenous proteases ("w/o [without] exogenous matriptase") (top panels of FIGS. 16A-16F), CD45-gate CARs with various TPS linkers showed a spectrum of activity, with some of the linkers (e.g., TPS4) exhibiting activity comparable to that of non-gated 53B6 CAR. In the presence of exogenous matriptase ("w/[with] exogenous matriptase") (bottom panels of FIGS. 16A-16F), all CD45-gate CARs with TPS linkers showed cytotoxic activities. FIGS. 16A-16C show results using linkers from the first set of linkers (TPS3, TPS4, TPS6) plus controls; FIGS. 16D-16F show results using linkers from the first and second sets of linkers (TPS3, TPS4, TPS6, TPS8-TPS13) plus controls.

3.5 Long-Term Killing Assay (FIGS. 17A-17B)

To determine the persistent killing of CD45-gate CAR T cells upon repeated exposure to target, luciferase-expressing OVCAR3 cells were plated in CTS OpTmizer T cell expansion SFM (Gibco) at a concentration of 1×10^4 cells per well in 96-well white flat-bottom plates. The following day, T cells were thawed and added at a E:T ratio of 3:1. All cocultures were performed with or without human recombinant matriptase (0.2 $\mu\text{g/mL}$, R&D, 3946-SEB-010). Every 2-3 days, half of the supernatant containing CAR T cells from each well was transferred to freshly plated target cells (1×10^4 /well) and target cell viability was measured by luminescence using One-Glo assay (Promega) as previously described. Data presented are the cytotoxicity of CD45-gate CAR T cells, showing that those with cleavable linkers maintained long-term cytotoxicity in the presence of endogenous (FIG. 17A) or exogenous (FIG. 17B) proteases. In Vivo Assay (FIGS. 18A-18C and FIGS. 19A-19C)

In vivo anti-tumor efficacy of CD45-gate CAR T cells was evaluated in a breast cancer xenograft model (FIG. 18A shows schematic outline of experiment). 7-8 weeks old NSG mice (Jackson Laboratory) were injected subcutaneously (SC) with 200 μL of MDA-MB-231-MUC16 cells (5×10^6 cells mixed with Matrigel at 1:1 ratio). Mice were randomized into 6 groups ($n=7-8$) when average tumor size reached 290 mm^3 on day 17. Each animal received 5 million CAR+ cells via tail vein injection. Total T cell numbers were kept constant across all groups by normalizing with non-transduced T cells. Tumors were measured twice a week and tumor volume was calculated as length (mm) $\times$ width (mm) $\times$ height (mm)/2. Statistical analysis of observed differences was performed by one-way ANOVA followed by Dunnett's test for multiple comparisons.

As shown in FIG. 18B, anti-tumor CAR activity cannot be activated in a CD45-gate CAR that comprises only a non-cleavable GS linker in the MDA-MB-231 model. In contrast, anti-tumor CAR activity can be activated in CD45-gate

CARs with cleavable linkers (e.g., TPS4). Tumor growth was reduced in mice treated with the CD45-gate CAR aCD45-TPS4-53B6.

Treatment with CD45-gate CAR T cells had no effect on the body weight of the mice (FIG. 18C).

In vivo anti-tumor efficacy of CD45-gate CAR T cells was also evaluated in a non-small cell lung cancer xenograft model (FIG. 19A shows schematic outline of experiment). NSG mice (Jackson Laboratory) were injected subcutaneously (SC) with 200 μL of NCI-H292-MUC16 cells (6×10^6 cells mixed with Matrigel at 1:1 ratio). Mice were randomized into 6 groups ($n=6-7$) when average tumor size reached 150 mm^3 on day 14. CAR T cells were then thawed and infused immediately (5×10^6 CAR+ cells per animal) via tail vein injection. Total T cell numbers were kept constant across all groups by normalizing with non-transduced T cells. Tumors were measured twice a week and tumor volume was calculated as length (mm) $\times$ width (mm) $\times$ height (mm)/2. Statistical analysis of observed differences was performed by one-way ANOVA followed by Dunnett's test for multiple comparisons. Without wishing to be limited to specific mechanisms, the results likely reflect the various levels of protease activity at the tumor site in this in vivo tumor model.

As shown in FIG. 19B, a CD45-gate CAR with a non-cleavable GS linker lacked any CAR anti-tumor activity in the H292 model. In contrast, a CD45-gate CAR with a cleavable linker (TPS4) showed anti-tumor activity.

Treatment with CD45-gate CAR T cells had no effect on the body weight of the mice (FIG. 19C), indicating that the CD45-gate CAR T cells did not lead to general toxicity in vivo.

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SEQUENCE LISTING

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 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

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His Ala Ala Arg Pro
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 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

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 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

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Gly Ser Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Phe Tyr Asn Leu
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Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile
 35 40 45

Tyr Asp Ala Ser Asp Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly
 50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys
 65 70 75 80

Ala Asp Ala Ala Ala Tyr Tyr Cys Gln Ser Ala Asp Gly Ser Ser Tyr

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85	90	95
Ala Phe Gly Gly Gly Thr Glu Val Val Val Lys Gly Gly Gly Gly Gly		
100	105	110
Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser		
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Gln Glu Gln Leu Glu Glu Ser Gly Gly Gly Leu Val Lys Pro Glu Gly		
130	135	140
Ser Leu Thr Leu Thr Cys Thr Ala Ser Gly Val Ser Phe Ser Ser Ser		
145	150	155
Tyr Trp Ile Tyr Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp		
165	170	175
Ile Ala Cys Ile Tyr Thr Gly Ser Ser Gly Ser Thr Tyr Tyr Ala Ser		
180	185	190
Trp Ala Lys Gly Arg Phe Thr Val Ser Glu Thr Ser Ser Thr Thr Val		
195	200	205
Thr Leu Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe		
210	215	220
Cys Ala Arg Ala Ser Ala Trp Thr Tyr Gly Met Asp Leu Trp Gly Pro		
225	230	235
Gly Thr Leu Val Thr Val Ser Ser		
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Gly Thr Val Thr Ile Met Cys Gln Ala Ser Gln Ser Ile Ser Asn Trp		
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Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile		
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Tyr Gln Ala Ser Lys Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly		
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Ser Gly Ser Gly Thr Glu Tyr Thr Leu Thr Ile Ser Asp Leu Glu Cys		
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Ala Asp Ala Ala Thr Tyr Tyr Cys Gln Ser Tyr Tyr Asp Ser Gly Ser		
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Asn Val Phe Phe Ala Phe Gly Gly Gly Thr Lys Val Val Val Glu Gly		
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Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly		
115	120	125
Gly Gly Gly Ser Leu Ser Leu Glu Glu Ser Gly Gly Asp Leu Val Lys		
130	135	140
Pro Gly Ala Ser Leu Thr Leu Thr Cys Thr Ala Ser Gly Phe Ser Phe		
145	150	155
Ser Ala Gly Tyr Trp Ile Cys Trp Val Arg Gln Ala Pro Gly Lys Gly		
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Leu Glu Trp Ile Ala Cys Thr Tyr Ala Gly Arg Ser Gly Ser Thr Tyr		
180	185	190

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Tyr Ala Asn Trp Val Asn Gly Arg Phe Thr Ile Pro Lys Thr Ser Ser
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Thr Thr Val Thr Leu Gln Met Thr Ser Leu Ser Gly Ala Asp Thr Ala
 210 215 220

Ser Tyr Phe Cys Ala Arg Gly Asn Ala Gly Val Ala Val Gly Ala Leu
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Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser
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 1 5 10 15

Gly Ser Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Phe Tyr Asn Leu
 20 25 30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile
 35 40 45

Tyr Asp Ala Ser Asp Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly
 50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys
 65 70 75 80

Ala Asp Ala Ala Ala Tyr Tyr Cys Gln Ser Ala Asp Gly Ser Ser Tyr
 85 90 95

Ala Phe Gly Gly Gly Thr Glu Val Val Val Lys Gly Gly Gly Gly Gly
 100 105 110

Ser Gly Gly Gly Gly Ser Leu Ser Gly Arg Ser Asp Asn His Ser Pro
 115 120 125

Leu Gly Leu Ala Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
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Gln Glu Gln Leu Glu Glu Ser Gly Gly Gly Leu Val Lys Pro Glu Gly
 145 150 155 160

Ser Leu Thr Leu Thr Cys Thr Ala Ser Gly Val Ser Phe Ser Ser Ser
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Tyr Trp Ile Tyr Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp
 180 185 190

Ile Ala Cys Ile Tyr Thr Gly Ser Ser Gly Ser Thr Tyr Tyr Ala Ser
 195 200 205

Trp Ala Lys Gly Arg Phe Thr Val Ser Glu Thr Ser Ser Thr Thr Val
 210 215 220

Thr Leu Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe
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Gly Thr Leu Val Thr Val Ser Ser
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<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

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20           25           30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile
35           40           45
Tyr Gln Ala Ser Lys Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly
50           55           60
Ser Gly Ser Gly Thr Glu Tyr Thr Leu Thr Ile Ser Asp Leu Glu Cys
65           70           75           80
Ala Asp Ala Ala Thr Tyr Tyr Cys Gln Ser Tyr Tyr Asp Ser Gly Ser
85           90           95
Asn Val Phe Phe Ala Phe Gly Gly Gly Thr Lys Val Val Val Glu Gly
100          105          110
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Leu Ser Gly Arg Ser Asp
115          120          125
Asn His Ser Pro Leu Gly Leu Ala Gly Ser Gly Gly Gly Ser Gly
130          135          140
Gly Gly Gly Ser Leu Ser Leu Glu Glu Ser Gly Gly Asp Leu Val Lys
145          150          155          160
Pro Gly Ala Ser Leu Thr Leu Thr Cys Thr Ala Ser Gly Phe Ser Phe
165          170          175
Ser Ala Gly Tyr Trp Ile Cys Trp Val Arg Gln Ala Pro Gly Lys Gly
180          185          190
Leu Glu Trp Ile Ala Cys Thr Tyr Ala Gly Arg Ser Gly Ser Thr Tyr
195          200          205
Tyr Ala Asn Trp Val Asn Gly Arg Phe Thr Ile Pro Lys Thr Ser Ser
210          215          220
Thr Thr Val Thr Leu Gln Met Thr Ser Leu Ser Gly Ala Asp Thr Ala
225          230          235          240
Ser Tyr Phe Cys Ala Arg Gly Asn Ala Gly Val Ala Val Gly Ala Leu
245          250          255
Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser
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<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

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1           5           10           15
Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu Asp Ser Thr Gly Gly Gly
20           25           30
Ser Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
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<210> SEQ ID NO 8

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<400> SEQUENCE: 8

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 1 5 10 15

Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu Asp Ser Thr Leu Val Pro
 20 25 30

Arg Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 35 40 45

<210> SEQ ID NO 9
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 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 9

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly Leu
 1 5 10 15

Ser Gly Arg Ser Asp Asn His Ser Pro Leu Gly Leu Ala Gly Ser Gly
 20 25 30

Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 35 40 45

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<400> SEQUENCE: 10

Gly Gly Gly Gly Ser Leu Ser Gly Arg Ser Asp Asn His Ser Pro Leu
 1 5 10 15

Gly Leu Ala Gly Ser Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu Asp
 20 25 30

Ser Thr Leu Ser Gly Arg Ser Asp Asn His Ser Pro Leu Gly Leu Ala
 35 40 45

Gly Ser Gly Gly Gly Gly Ser
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<400> SEQUENCE: 11

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 1 5 10 15

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 20 25 30

Ser Trp Thr Trp Val Arg Gln Pro Ala Gly Gln Gly Leu Glu Trp Ile

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65					70					75					80
Arg	Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Phe	Cys	Ala
				85					90					95	
Arg	Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly	Gln	Gly	Thr
		100						105					110		
Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser
		115					120					125			
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr	Gln
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Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser	Thr	Leu	Ser
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Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu	Ala	Trp	Tyr	Gln
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Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Ser	Thr
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Arg	Ala	Ile	Gly	Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr
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Thr	Lys	Val	Glu	Ile	Lys	Thr	Thr	Thr	Pro	Ala	Pro	Arg	Pro	Pro	Thr
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Pro	Ala	Pro	Thr	Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu	Arg	Pro	Glu	Ala
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Cys	Arg	Pro	Ala	Ala	Gly	Gly	Ala	Val	His	Thr	Arg	Gly	Leu	Asp	Phe
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Ala	Cys	Asp	Ile	Tyr	Ile	Trp	Ala	Pro	Leu	Ala	Gly	Thr	Cys	Gly	Val
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Lys	Leu	Leu	Tyr	Ile	Phe	Lys	Gln	Pro	Phe	Met	Arg	Pro	Val	Gln	Thr
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Glu	Met	Gly	Gly	Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr
				405					410					415	
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			420					425					430		
Met	Lys	Gly	Glu	Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr	Gln
		435					440					445			
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Ala Leu Pro Pro Arg
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Phe Cys Asn Gly Ser Arg Val Lys Asn Pro Gln Ile Arg His Thr Cys
35 40 45
Asn Asp Gln Asn Leu Thr Leu Ile His Val Asn Lys Thr Tyr Glu Arg
50 55 60
Thr Tyr Met Gly Tyr Asn Arg Gln Gly Thr Lys Lys Glu Asp Tyr Lys
65 70 75 80
Val Val Val Ile Pro Pro Pro Ala Thr Val Lys Pro Gln Pro Glu
85 90 95
Pro Glu Tyr Val Phe Val Tyr Met Gly Glu Asn Lys Thr Leu Glu Gly
100 105 110
Pro Pro Gly Thr Pro Val Thr Trp Phe Asn Gln Asp Gly Lys Lys Phe
115 120 125
Cys Glu Gly Glu Lys Val Leu His Pro Glu Phe Asn His Thr Cys Asp
130 135 140
Lys Gln Asn Leu Ile Leu Leu Phe Val Asn Phe Thr His Asp Gly Ala
145 150 155 160
Tyr Leu Gly Tyr Asn His Gln Gly Thr Gln Arg Thr His Tyr Glu Val
165 170 175
Thr Val Leu Asp Leu Phe Pro Asp Ser Gly Gln Met Lys Ile Glu Asn
180 185 190
His Ser Glu Glu Thr Glu Gln Lys Asn Asp Glu His His Asn Trp Gln
195 200 205
Lys Gln Gly Gly Gln Lys Gln Gly Gly Gln Lys Thr Asn Gln Thr Lys
210 215 220
Val Asn Asp Arg Arg Lys Thr Ala Gln Lys Arg Pro Ser Lys Leu Lys
225 230 235 240
Pro Ala Thr Ile Glu Ala Met Leu Val Thr Val Thr Ala Gly Ser Asn
245 250 255
Leu Thr Leu Val Gly Pro Lys Ala Glu Gly Lys Val Thr Trp Phe Asp
260 265 270
Gly Asp Leu Lys Arg Pro Cys Glu Pro Asn Tyr Arg Leu Arg His Glu
275 280 285
Cys Asn Asn Gln Asn Leu Thr Leu Ile Asn Val Thr Lys Asp Tyr Glu
290 295 300
Gly Thr Tyr Tyr Gly Thr Asn Asp Lys Asp Glu Gly Lys Arg Tyr Arg
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Val Lys Val Asn Thr Thr Asn Ser Gln Ser Val Lys Ile Gln Pro Tyr
325 330 335
Thr Arg Gln Thr Thr Pro Asp Gln Glu His Lys Phe Glu Leu Gln Phe

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Glu Thr Asn Gly Asn Tyr Asp Ser Lys Ile Pro
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 polypeptide

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Leu Asn Ala Val Asp Phe His Pro Gly Asp His Val Arg Trp Ser Tyr	
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Gly Pro Gly Gly Ala Gly Tyr Met Leu Cys Val Tyr Thr Gly Ser Trp	
35 40 45	
Thr Glu Tyr Lys Lys Pro Asp Ile Ile Phe Lys Cys Leu Ser Asn Asn	
50 55 60	
Ser Leu Leu Leu Ile Asn Val Thr Val Asn Tyr Thr Asn Thr Tyr Arg	
65 70 75 80	
Thr Leu Thr Ser Leu Asn Asn Trp Val His Asn Gln His His His Lys	
85 90 95	
Phe Pro Gly Trp Asn Leu Asp Thr Cys Tyr Ser Leu Thr Val Asn Glu	
100 105 110	
Asn Gly Thr Phe Pro Thr Thr Thr Thr Lys Lys Pro Thr Thr Thr Thr	
115 120 125	
Arg Thr Thr Thr Thr Thr Thr Thr Lys Lys Thr Thr Thr Thr Arg Thr	
130 135 140	
Thr Thr Ala Ala Lys Lys Thr Thr Ile Ser Thr Thr His His Lys His	
145 150 155 160	
Ser Ser Pro Lys Lys Ser Ser Thr Pro Asn Ser His Val Glu His His	
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Val Gly Phe Glu Ala Thr Ala Ala Glu Thr Pro Leu Gln Pro Ser Pro	
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Gln His Gln His Val Ala Thr His	
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 polypeptide

<400> SEQUENCE: 14

Gln Phe Ser Val Leu Gly Pro Ser Gly Pro Ile Leu Ala Met Val Gly	
1 5 10 15	
Glu Asp Ala Asp Leu Pro Cys His Leu Phe Pro Thr Met Ser Ala Glu	
20 25 30	
Thr Met Glu Leu Lys Trp Val Ser Ser Ser Leu Arg Gln Val Val Asn	
35 40 45	
Val Tyr Ala Asp Gly Lys Glu Val Glu Asp Arg Gln Ser Ala Pro Tyr	
50 55 60	
Arg Gly Arg Thr Ser Ile Leu Arg Asp Gly Ile Thr Ala Gly Lys Ala	

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65	70	75	80
Ala Leu Arg Ile His Asn Val Thr Ala Ser Asp Ser Gly Lys Tyr Leu			
	85	90	95
Cys Tyr Phe Gln Asp Gly Asp Phe Tyr Glu Lys Ala Leu Val Glu Leu			
	100	105	110
Lys Val Ala			
	115		

<210> SEQ ID NO 15
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 15

Ala Leu Gly Ser Asp Leu His Val Asp Val Lys Gly Tyr Lys Asp Gly			
1	5	10	15
Gly Ile His Leu Glu Cys Arg Ser Thr Gly Trp Tyr Pro Gln Pro Gln			
	20	25	30
Ile Gln Trp Ser Asn Asn Lys Gly Glu Asn Ile Pro Thr Val Glu Ala			
	35	40	45
Pro Val Val Ala Asp Gly Val Gly Leu Tyr Ala Val Ala Ala Ser Val			
	50	55	60
Ile Met Arg Gly Ser Ser Gly Glu Gly Val Ser Cys Thr Ile Arg Ser			
65	70	75	80
Ser Leu Leu Gly Leu Glu Lys Thr Ala Ser Ile Ser Ile Ala Asp Pro			
	85	90	95
Phe Phe Arg Ser Ala Gln Arg Trp Ile Ala Ala Leu Ala Gly Thr			
	100	105	110

<210> SEQ ID NO 16
 <211> LENGTH: 793
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 16

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu			
1	5	10	15
His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp			
	20	25	30
Ile Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly Gly			
	35	40	45
Ser Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Phe Tyr Asn Leu Leu			
	50	55	60
Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr			
65	70	75	80
Asp Ala Ser Asp Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly Ser			
	85	90	95
Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala			
	100	105	110
Asp Ala Ala Ala Tyr Tyr Cys Gln Ser Ala Asp Gly Ser Ser Tyr Ala			
	115	120	125
Phe Gly Gly Gly Thr Glu Val Val Val Lys Gly Gly Gly Gly Ser			

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130					135					140					
Gly 145	Gly	Gly	Gly	Ser	Gly 150	Gly	Gly	Gly	Ser	Gly 155	Gly	Gly	Gly	Ser	Gln 160
Glu	Gln	Leu	Glu	Glu 165	Ser	Gly	Gly	Gly	Leu 170	Val	Lys	Pro	Glu	Gly 175	Ser
Leu	Thr	Leu	Thr	Cys 180	Thr	Ala	Ser	Gly 185	Val	Ser	Phe	Ser	Ser	Ser	Tyr
Trp	Ile	Tyr	Trp	Val 195	Arg	Gln	Ala 200	Pro	Gly	Lys	Gly	Leu 205	Glu	Trp	Ile
Ala	Cys 210	Ile	Tyr	Thr	Gly	Ser 215	Ser	Gly	Ser	Thr	Tyr 220	Tyr	Ala	Ser	Trp
Ala 225	Lys	Gly	Arg	Phe	Thr 230	Val	Ser	Glu	Thr	Ser 235	Ser	Thr	Thr	Val	Thr 240
Leu	Gln	Met	Thr	Ser 245	Leu	Thr	Ala	Ala	Asp 250	Thr	Ala	Thr	Tyr	Phe 255	Cys
Ala	Arg	Ala	Ser 260	Ala	Trp	Thr	Tyr	Gly 265	Met	Asp	Leu	Trp	Gly 270	Pro	Gly
Thr	Leu	Val	Thr 275	Val	Ser	Ser	Gly 280	Gly	Gly	Gly	Ser	Gly 285	Gly	Gly	Gly
Ser	Leu 290	Val	Pro	Arg	Gly	Ser 295	Lys	Pro	Ile	Pro	Asn 300	Pro	Leu	Leu	Gly
Leu 305	Asp	Ser	Thr	Leu	Val 310	Pro	Arg	Gly	Ser	Gly 315	Gly	Gly	Gly	Ser	Gly 320
Gly	Gly	Gly	Ser	Gln 325	Val	Gln	Leu	Gln	Glu 330	Ser	Gly	Pro	Gly	Leu 335	Val
Lys	Pro	Ser	Glu 340	Thr	Leu	Ser	Leu	Thr 345	Cys	Thr	Val	Ser	Gly 350	Gly	Ser
Ile	Ser	Tyr 355	Tyr	Ser	Trp	Thr	Trp 360	Val	Arg	Gln	Pro	Ala 365	Gly	Gln	Gly
Leu 370	Glu	Trp	Ile	Gly	Arg	Ile 375	His	Ile	Ser	Gly	Val 380	Thr	Asn	His	Asn
Pro 385	Ser	Leu	Lys	Ser	Arg 390	Val	Ser	Met	Ser	Ile 395	Asp	Thr	Ser	Arg	Thr 400
Gln	Phe	Ser	Leu	Arg 405	Leu	Thr	Ser	Val	Thr 410	Ala	Ala	Asp	Thr	Ala 415	Val
Tyr	Phe	Cys	Ala 420	Arg	Ser	Gly	Gly	Thr 425	Tyr	Ser	Ala	Phe	Asp 430	Ile	Trp
Gly	Gln	Gly 435	Thr	Met	Val	Thr 440	Val	Ser	Ser	Gly	Gly	Gly 445	Gly	Ser	Gly
Gly 450	Gly	Gly	Ser	Gly	Gly	Gly 455	Gly	Ser	Gly	Gly	Gly 460	Gly	Ser	Glu	Ile
Val 465	Leu	Thr	Gln	Ser	Pro 470	Gly	Thr	Leu	Ser	Leu 475	Ser	Pro	Gly	Glu	Arg 480
Ser	Thr	Leu	Ser	Cys 485	Arg	Ala	Ser	Gln	Ser	Phe 490	Thr	Ser	Asn	Tyr 495	Leu
Ala	Trp	Tyr	Gln 500	Gln	Arg	Pro	Gly	Gln 505	Ala	Pro	Arg	Leu	Leu	Ile	Tyr
Gly	Ala	Ser 515	Thr	Arg	Ala	Ile	Gly 520	Ile	Pro	Asp	Arg	Phe 525	Ser	Gly	Ser
Gly	Ser	Gly 530	Thr	Asp	Phe	Thr 535	Leu	Thr	Ile	Ser	Arg 540	Leu	Glu	Pro	Glu
Asp 545	Phe	Ala	Val	Tyr	Tyr 550	Cys	Gln	Gln	Tyr	Gly 555	Ser	Ser	Pro	Trp	Thr 560

Phe	Gly	Gln	Gly	Thr 565	Lys	Val	Glu	Ile	Lys 570	Thr	Thr	Thr	Pro	Ala	Pro 575
Arg	Pro	Pro	Thr 580	Pro	Ala	Pro	Thr	Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu 590
Arg	Pro	Glu	Ala	Cys	Arg	Pro	Ala 600	Ala	Gly	Gly	Ala	Val 605	His	Thr	Arg
Gly	Leu	Asp	Phe	Ala	Cys	Asp 615	Ile	Tyr	Ile	Trp	Ala	Pro 620	Leu	Ala	Gly
Thr 625	Cys	Gly	Val	Leu	Leu	Leu	Ser	Leu	Val	Ile 635	Thr	Leu	Tyr	Cys	Lys 640
Arg	Gly	Arg	Lys	Lys 645	Leu	Leu	Tyr	Ile	Phe 650	Lys	Gln	Pro	Phe	Met 655	Arg
Pro	Val	Gln	Thr	Thr 660	Gln	Glu	Glu	Asp 665	Gly	Cys	Ser	Cys	Arg	Phe	Pro 670
Glu	Glu	Glu	Glu	Gly	Gly	Cys	Glu 680	Leu	Arg	Val	Lys	Phe 685	Ser	Arg	Ser
Ala	Asp 690	Ala	Pro	Ala	Tyr	Gln 695	Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu
Leu 705	Asn	Leu	Gly	Arg	Arg 710	Glu	Glu	Tyr	Asp 715	Val	Leu	Asp	Lys	Arg	Arg 720
Gly	Arg	Asp	Pro	Glu	Met	Gly	Gly	Lys	Pro 730	Arg	Arg	Lys	Asn	Pro 735	Gln
Glu	Gly	Leu	Tyr	Asn	Glu	Leu	Gln	Lys 745	Asp	Lys	Met	Ala	Glu	Ala	Tyr 750
Ser	Glu	Ile	Gly	Met	Lys	Gly	Glu 760	Arg	Arg	Arg	Gly	Lys 765	Gly	His	Asp
Gly	Leu	Tyr	Gln	Gly	Leu	Ser	Thr 775	Ala	Thr	Lys	Asp 780	Thr	Tyr	Asp	Ala
Leu 785	His	Met	Gln	Ala	Leu	Pro	Pro	Arg							

<400> SEQUENCE: 17

Met	Ala	Leu	Pro	Val	Thr	Ala	Leu	Leu	Leu	Pro	Leu	Ala	Leu	Leu	Leu
1				5					10					15	
His	Ala	Ala	Arg	Pro	Gly	Tyr	Pro	Tyr	Asp	Val	Pro	Asp	Tyr	Ala	Asp
			20					25					30		
Ile	Val	Met	Thr	Gln	Thr	Pro	Ala	Ser	Val	Ser	Glu	Pro	Val	Gly	Gly
		35					40					45			
Ser	Val	Thr	Ile	Lys	Cys	Gln	Ala	Ser	Gln	Ser	Phe	Tyr	Asn	Leu	Leu
	50					55					60				
Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Pro	Pro	Lys	Leu	Leu	Ile	Tyr
65					70				75					80	
Asp	Ala	Ser	Asp	Leu	Ala	Ser	Gly	Val	Pro	Ser	Arg	Phe	Lys	Gly	Ser
			85						90					95	
Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Asp	Leu	Glu	Cys	Ala
			100					105					110		
Asp	Ala	Ala	Ala	Tyr	Tyr	Cys	Gln	Ser	Ala	Asp	Gly	Ser	Ser	Tyr	Ala

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115					120					125				
Phe	Gly	Gly	Gly	Thr	Glu	Val	Val	Val	Lys	Gly	Gly	Gly	Gly	Ser
130						135					140			
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Gln
145					150					155				160
Glu	Gln	Leu	Glu	Glu	Ser	Gly	Gly	Gly	Leu	Val	Lys	Pro	Glu	Ser
					165				170					175
Leu	Thr	Leu	Thr	Cys	Thr	Ala	Ser	Gly	Val	Ser	Phe	Ser	Ser	Tyr
					180				185					190
Trp	Ile	Tyr	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp
195								200				205		
Ala	Cys	Ile	Tyr	Thr	Gly	Ser	Ser	Gly	Ser	Thr	Tyr	Tyr	Ala	Ser
210						215					220			Trp
Ala	Lys	Gly	Arg	Phe	Thr	Val	Ser	Glu	Thr	Ser	Ser	Thr	Thr	Val
225					230					235				240
Leu	Gln	Met	Thr	Ser	Leu	Thr	Ala	Ala	Asp	Thr	Ala	Thr	Tyr	Phe
					245				250					255
Ala	Arg	Ala	Ser	Ala	Trp	Thr	Tyr	Gly	Met	Asp	Leu	Trp	Gly	Pro
					260				265				270	Gly
Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly
					275				280				285	
Ser	Gly	Gly	Gly	Ser	Gly	Ser	Lys	Pro	Ile	Pro	Asn	Pro	Leu	Gly
					290				295			300		
Leu	Asp	Ser	Thr	Gly	Gly	Gly	Ser	Gly	Ser	Gly	Gly	Gly	Gly	Ser
305					310					315				320
Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu
					325				330					335
Lys	Pro	Ser	Glu	Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Gly
					340				345					350
Ile	Ser	Tyr	Tyr	Ser	Trp	Thr	Trp	Val	Arg	Gln	Pro	Ala	Gly	Gln
					355				360				365	Gly
Leu	Glu	Trp	Ile	Gly	Arg	Ile	His	Ile	Ser	Gly	Val	Thr	Asn	His
					370				375				380	Asn
Pro	Ser	Leu	Lys	Ser	Arg	Val	Ser	Met	Ser	Ile	Asp	Thr	Ser	Arg
					385				390					400
Gln	Phe	Ser	Leu	Arg	Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala
					405				410					415
Tyr	Phe	Cys	Ala	Arg	Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe	Asp	Ile
					420				425				430	Trp
Gly	Gln	Gly	Thr	Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser
					435				440				445	
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu
					450				455				460	Ile
Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu
					465				470					480
Ser	Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser	Asn	Tyr
					485				490					495
Ala	Trp	Tyr	Gln	Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile
					500				505				510	Tyr
Gly	Ala	Ser	Thr	Arg	Ala	Ile	Gly	Ile	Pro	Asp	Arg	Phe	Ser	Gly
					515				520				525	Ser
Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Arg	Leu	Glu	Pro
					530				535					540

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Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro Trp Thr
 545 550 555 560
 Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Thr Thr Thr Pro Ala Pro
 565 570 575
 Arg Pro Pro Thr Pro Ala Pro Thr Ile Ala Ser Gln Pro Leu Ser Leu
 580 585 590
 Arg Pro Glu Ala Cys Arg Pro Ala Ala Gly Gly Ala Val His Thr Arg
 595 600 605
 Gly Leu Asp Phe Ala Cys Asp Ile Tyr Ile Trp Ala Pro Leu Ala Gly
 610 615 620
 Thr Cys Gly Val Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr Cys Lys
 625 630 635 640
 Arg Gly Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met Arg
 645 650 655
 Pro Val Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe Pro
 660 665 670
 Glu Glu Glu Glu Gly Gly Cys Glu Leu Arg Val Lys Phe Ser Arg Ser
 675 680 685
 Ala Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu
 690 695 700
 Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg
 705 710 715 720
 Gly Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln
 725 730 735
 Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr
 740 745 750
 Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp
 755 760 765
 Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala
 770 775 780
 Leu His Met Gln Ala Leu Pro Pro Arg
 785 790

<210> SEQ ID NO 18

<211> LENGTH: 796

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 18

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15
 His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp
 20 25 30
 Ile Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly Gly
 35 40 45
 Thr Val Thr Ile Met Cys Gln Ala Ser Gln Ser Ile Ser Asn Trp Leu
 50 55 60
 Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr
 65 70 75 80
 Gln Ala Ser Lys Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly Ser
 85 90 95
 Gly Ser Gly Thr Glu Tyr Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala

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100							105					110				
Asp	Ala	Ala	Thr	Tyr	Tyr	Cys	Gln	Ser	Tyr	Tyr	Asp	Ser	Gly	Ser	Asn	
		115					120					125				
Val	Phe	Phe	Ala	Phe	Gly	Gly	Gly	Thr	Lys	Val	Val	Val	Glu	Gly	Gly	
	130					135					140					
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	
145					150					155					160	
Gly	Gly	Ser	Leu	Ser	Leu	Glu	Glu	Ser	Gly	Gly	Asp	Leu	Val	Lys	Pro	
			165						170					175		
Gly	Ala	Ser	Leu	Thr	Leu	Thr	Cys	Thr	Ala	Ser	Gly	Phe	Ser	Phe	Ser	
			180						185				190			
Ala	Gly	Tyr	Trp	Ile	Cys	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	
		195					200					205				
Glu	Trp	Ile	Ala	Cys	Thr	Tyr	Ala	Gly	Arg	Ser	Gly	Ser	Thr	Tyr	Tyr	
	210					215					220					
Ala	Asn	Trp	Val	Asn	Gly	Arg	Phe	Thr	Ile	Pro	Lys	Thr	Ser	Ser	Thr	
225					230					235					240	
Thr	Val	Thr	Leu	Gln	Met	Thr	Ser	Leu	Ser	Gly	Ala	Asp	Thr	Ala	Ser	
			245						250					255		
Tyr	Phe	Cys	Ala	Arg	Gly	Asn	Ala	Gly	Val	Ala	Val	Gly	Ala	Leu	Trp	
			260					265					270			
Gly	Pro	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	
		275					280					285				
Gly	Gly	Gly	Ser	Leu	Val	Pro	Arg	Gly	Ser	Lys	Pro	Ile	Pro	Asn	Pro	
	290					295					300					
Leu	Leu	Gly	Leu	Asp	Ser	Thr	Leu	Val	Pro	Arg	Gly	Ser	Gly	Gly	Gly	
305					310					315					320	
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	
			325					330						335		
Gly	Leu	Val	Lys	Pro	Ser	Glu	Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	
			340					345					350			
Gly	Gly	Ser	Ile	Ser	Tyr	Tyr	Ser	Trp	Thr	Trp	Val	Arg	Gln	Pro	Ala	
		355					360					365				
Gly	Gln	Gly	Leu	Glu	Trp	Ile	Gly	Arg	Ile	His	Ile	Ser	Gly	Val	Thr	
	370					375					380					
Asn	His	Asn	Pro	Ser	Leu	Lys	Ser	Arg	Val	Ser	Met	Ser	Ile	Asp	Thr	
385					390					395					400	
Ser	Arg	Thr	Gln	Phe	Ser	Leu	Arg	Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp	
			405						410					415		
Thr	Ala	Val	Tyr	Phe	Cys	Ala	Arg	Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe	
			420					425					430			
Asp	Ile	Trp	Gly	Gln	Gly	Thr	Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	
	435						440					445				
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	
	450					455					460					
Ser	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	
465					470					475					480	
Gly	Glu	Arg	Ser	Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser	
			485						490					495		
Asn	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu	
		500						505					510			
Leu	Ile	Tyr	Gly	Ala	Ser	Thr	Arg	Ala	Ile	Gly	Ile	Pro	Asp	Arg	Phe	
	515						520					525				

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Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu
 530 535 540
 Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser
 545 550 555 560
 Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Thr Thr Thr
 565 570 575
 Pro Ala Pro Arg Pro Pro Thr Pro Ala Pro Thr Ile Ala Ser Gln Pro
 580 585 590
 Leu Ser Leu Arg Pro Glu Ala Cys Arg Pro Ala Ala Gly Gly Ala Val
 595 600 605
 His Thr Arg Gly Leu Asp Phe Ala Cys Asp Ile Tyr Ile Trp Ala Pro
 610 615 620
 Leu Ala Gly Thr Cys Gly Val Leu Leu Leu Ser Leu Val Ile Thr Leu
 625 630 635 640
 Tyr Cys Lys Arg Gly Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro
 645 650 655
 Phe Met Arg Pro Val Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys
 660 665 670
 Arg Phe Pro Glu Glu Glu Glu Gly Gly Cys Glu Leu Arg Val Lys Phe
 675 680 685
 Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu
 690 695 700
 Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp
 705 710 715 720
 Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys
 725 730 735
 Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala
 740 745 750
 Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys
 755 760 765
 Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr
 770 775 780
 Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg
 785 790 795

<210> SEQ ID NO 19

<211> LENGTH: 796

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 19

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15
 His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp
 20 25 30
 Ile Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly Gly
 35 40 45
 Thr Val Thr Ile Met Cys Gln Ala Ser Gln Ser Ile Ser Asn Trp Leu
 50 55 60
 Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr
 65 70 75 80
 Gln Ala Ser Lys Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly Ser

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85								90				95			
Gly	Ser	Gly	Thr	Glu	Tyr	Thr	Leu	Thr	Ile	Ser	Asp	Leu	Glu	Cys	Ala
			100					105					110		
Asp	Ala	Ala	Thr	Tyr	Tyr	Cys	Gln	Ser	Tyr	Tyr	Asp	Ser	Gly	Ser	Asn
		115					120					125			
Val	Phe	Phe	Ala	Phe	Gly	Gly	Gly	Thr	Lys	Val	Val	Val	Glu	Gly	Gly
		130				135					140				
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly
145					150					155					160
Gly	Gly	Ser	Leu	Ser	Leu	Glu	Glu	Ser	Gly	Gly	Asp	Leu	Val	Lys	Pro
			165					170						175	
Gly	Ala	Ser	Leu	Thr	Leu	Thr	Cys	Thr	Ala	Ser	Gly	Phe	Ser	Phe	Ser
			180					185					190		
Ala	Gly	Tyr	Trp	Ile	Cys	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu
		195					200					205			
Glu	Trp	Ile	Ala	Cys	Thr	Tyr	Ala	Gly	Arg	Ser	Gly	Ser	Thr	Tyr	Tyr
	210					215					220				
Ala	Asn	Trp	Val	Asn	Gly	Arg	Phe	Thr	Ile	Pro	Lys	Thr	Ser	Ser	Thr
225					230					235					240
Thr	Val	Thr	Leu	Gln	Met	Thr	Ser	Leu	Ser	Gly	Ala	Asp	Thr	Ala	Ser
			245					250						255	
Tyr	Phe	Cys	Ala	Arg	Gly	Asn	Ala	Gly	Val	Ala	Val	Gly	Ala	Leu	Trp
			260				265						270		
Gly	Pro	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly
	275						280					285			
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Ser	Gly	Ser	Lys	Pro	Ile	Pro	Asn	Pro
	290					295					300				
Leu	Leu	Gly	Leu	Asp	Ser	Thr	Gly	Gly	Gly	Ser	Gly	Ser	Gly	Gly	Gly
305					310					315					320
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro
			325					330						335	
Gly	Leu	Val	Lys	Pro	Ser	Glu	Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser
			340					345					350		
Gly	Gly	Ser	Ile	Ser	Tyr	Tyr	Ser	Trp	Thr	Trp	Val	Arg	Gln	Pro	Ala
		355					360					365			
Gly	Gln	Gly	Leu	Glu	Trp	Ile	Gly	Arg	Ile	His	Ile	Ser	Gly	Val	Thr
	370					375					380				
Asn	His	Asn	Pro	Ser	Leu	Lys	Ser	Arg	Val	Ser	Met	Ser	Ile	Asp	Thr
385					390					395					400
Ser	Arg	Thr	Gln	Phe	Ser	Leu	Arg	Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp
			405					410						415	
Thr	Ala	Val	Tyr	Phe	Cys	Ala	Arg	Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe
		420						425					430		
Asp	Ile	Trp	Gly	Gln	Gly	Thr	Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly
	435						440					445			
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly
	450					455					460				
Ser	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro
465					470					475					480
Gly	Glu	Arg	Ser	Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser
			485					490						495	
Asn	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu
		500						505					510		

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Leu Ile Tyr Gly Ala Ser Thr Arg Ala Ile Gly Ile Pro Asp Arg Phe
 515 520 525
 Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu
 530 535 540
 Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser
 545 550 555 560
 Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Thr Thr Thr
 565 570 575
 Pro Ala Pro Arg Pro Pro Thr Pro Ala Pro Thr Ile Ala Ser Gln Pro
 580 585 590
 Leu Ser Leu Arg Pro Glu Ala Cys Arg Pro Ala Ala Gly Gly Ala Val
 595 600 605
 His Thr Arg Gly Leu Asp Phe Ala Cys Asp Ile Tyr Ile Trp Ala Pro
 610 615 620
 Leu Ala Gly Thr Cys Gly Val Leu Leu Leu Ser Leu Val Ile Thr Leu
 625 630 635 640
 Tyr Cys Lys Arg Gly Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro
 645 650 655
 Phe Met Arg Pro Val Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys
 660 665 670
 Arg Phe Pro Glu Glu Glu Glu Gly Gly Cys Glu Leu Arg Val Lys Phe
 675 680 685
 Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu
 690 695 700
 Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp
 705 710 715 720
 Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys
 725 730 735
 Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala
 740 745 750
 Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys
 755 760 765
 Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr
 770 775 780
 Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg
 785 790 795

<210> SEQ ID NO 20

<211> LENGTH: 793

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 20

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15
 His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp
 20 25 30
 Ile Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly Gly
 35 40 45
 Ser Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Phe Tyr Asn Leu Leu
 50 55 60
 Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr

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65	70	75	80
Asp Ala Ser Asp Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly Ser	85	90	95
Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala	100	105	110
Asp Ala Ala Ala Tyr Tyr Cys Gln Ser Ala Asp Gly Ser Ser Tyr Ala	115	120	125
Phe Gly Gly Gly Thr Glu Val Val Val Lys Gly Gly Gly Gly Gly Ser	130	135	140
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln	145	150	155
Glu Gln Leu Glu Glu Ser Gly Gly Gly Leu Val Lys Pro Glu Gly Ser	165	170	175
Leu Thr Leu Thr Cys Thr Ala Ser Gly Val Ser Phe Ser Ser Ser Tyr	180	185	190
Trp Ile Tyr Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile	195	200	205
Ala Cys Ile Tyr Thr Gly Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp	210	215	220
Ala Lys Gly Arg Phe Thr Val Ser Glu Thr Ser Ser Thr Thr Val Thr	225	230	235
Leu Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys	245	250	255
Ala Arg Ala Ser Ala Trp Thr Tyr Gly Met Asp Leu Trp Gly Pro Gly	260	265	270
Thr Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly	275	280	285
Ser Gly Gly Gly Ser Gly Leu Ser Gly Arg Ser Asp Asn His Ser Pro	290	295	300
Leu Gly Leu Ala Gly Ser Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly	305	310	315
Gly Gly Gly Ser Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val	325	330	335
Lys Pro Ser Glu Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser	340	345	350
Ile Ser Tyr Tyr Ser Trp Thr Trp Val Arg Gln Pro Ala Gly Gln Gly	355	360	365
Leu Glu Trp Ile Gly Arg Ile His Ile Ser Gly Val Thr Asn His Asn	370	375	380
Pro Ser Leu Lys Ser Arg Val Ser Met Ser Ile Asp Thr Ser Arg Thr	385	390	395
Gln Phe Ser Leu Arg Leu Thr Ser Val Thr Ala Ala Asp Thr Ala Val	405	410	415
Tyr Phe Cys Ala Arg Ser Gly Gly Thr Tyr Ser Ala Phe Asp Ile Trp	420	425	430
Gly Gln Gly Thr Met Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly	435	440	445
Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Glu Ile	450	455	460
Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly Glu Arg	465	470	475
Ser Thr Leu Ser Cys Arg Ala Ser Gln Ser Phe Thr Ser Asn Tyr Leu	485	490	495

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Ala Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Arg Leu Leu Ile Tyr
 500 505 510
 Gly Ala Ser Thr Arg Ala Ile Gly Ile Pro Asp Arg Phe Ser Gly Ser
 515 520 525
 Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu Pro Glu
 530 535 540
 Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro Trp Thr
 545 550 555 560
 Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Thr Thr Thr Pro Ala Pro
 565 570 575
 Arg Pro Pro Thr Pro Ala Pro Thr Ile Ala Ser Gln Pro Leu Ser Leu
 580 585 590
 Arg Pro Glu Ala Cys Arg Pro Ala Ala Gly Gly Ala Val His Thr Arg
 595 600 605
 Gly Leu Asp Phe Ala Cys Asp Ile Tyr Ile Trp Ala Pro Leu Ala Gly
 610 615 620
 Thr Cys Gly Val Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr Cys Lys
 625 630 635 640
 Arg Gly Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met Arg
 645 650 655
 Pro Val Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe Pro
 660 665 670
 Glu Glu Glu Glu Gly Gly Cys Glu Leu Arg Val Lys Phe Ser Arg Ser
 675 680 685
 Ala Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu
 690 695 700
 Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg
 705 710 715 720
 Gly Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln
 725 730 735
 Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr
 740 745 750
 Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp
 755 760 765
 Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala
 770 775 780
 Leu His Met Gln Ala Leu Pro Pro Arg
 785 790

<210> SEQ ID NO 21
 <211> LENGTH: 796
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 21

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15
 His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp
 20 25 30
 Ile Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly Gly
 35 40 45
 Thr Val Thr Ile Met Cys Gln Ala Ser Gln Ser Ile Ser Asn Trp Leu

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50	55	60
Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr 65 70 75 80		
Gln Ala Ser Lys Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly Ser 85 90 95		
Gly Ser Gly Thr Glu Tyr Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala 100 105 110		
Asp Ala Ala Thr Tyr Tyr Cys Gln Ser Tyr Tyr Asp Ser Gly Ser Asn 115 120 125		
Val Phe Phe Ala Phe Gly Gly Gly Thr Lys Val Val Val Glu Gly Gly 130 135 140		
Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly 145 150 155 160		
Gly Gly Ser Leu Ser Leu Glu Glu Ser Gly Gly Asp Leu Val Lys Pro 165 170 175		
Gly Ala Ser Leu Thr Leu Thr Cys Thr Ala Ser Gly Phe Ser Phe Ser 180 185 190		
Ala Gly Tyr Trp Ile Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu 195 200 205		
Glu Trp Ile Ala Cys Thr Tyr Ala Gly Arg Ser Gly Ser Thr Tyr Tyr 210 215 220		
Ala Asn Trp Val Asn Gly Arg Phe Thr Ile Pro Lys Thr Ser Ser Thr 225 230 235 240		
Thr Val Thr Leu Gln Met Thr Ser Leu Ser Gly Ala Asp Thr Ala Ser 245 250 255		
Tyr Phe Cys Ala Arg Gly Asn Ala Gly Val Ala Val Gly Ala Leu Trp 260 265 270		
Gly Pro Gly Thr Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly 275 280 285		
Gly Gly Gly Ser Gly Gly Gly Ser Gly Leu Ser Gly Arg Ser Asp Asn 290 295 300		
His Ser Pro Leu Gly Leu Ala Gly Ser Gly Gly Gly Ser Gly Gly Gly 305 310 315 320		
Gly Ser Gly Gly Gly Gly Ser Gln Val Gln Leu Gln Glu Ser Gly Pro 325 330 335		
Gly Leu Val Lys Pro Ser Glu Thr Leu Ser Leu Thr Cys Thr Val Ser 340 345 350		
Gly Gly Ser Ile Ser Tyr Tyr Ser Trp Thr Trp Val Arg Gln Pro Ala 355 360 365		
Gly Gln Gly Leu Glu Trp Ile Gly Arg Ile His Ile Ser Gly Val Thr 370 375 380		
Asn His Asn Pro Ser Leu Lys Ser Arg Val Ser Met Ser Ile Asp Thr 385 390 395 400		
Ser Arg Thr Gln Phe Ser Leu Arg Leu Thr Ser Val Thr Ala Ala Asp 405 410 415		
Thr Ala Val Tyr Phe Cys Ala Arg Ser Gly Gly Thr Tyr Ser Ala Phe 420 425 430		
Asp Ile Trp Gly Gln Gly Thr Met Val Thr Val Ser Ser Gly Gly Gly 435 440 445		
Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly 450 455 460		
Ser Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro 465 470 475 480		

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35	40	45
Ser Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Phe Tyr Asn Leu Leu		
50	55	60
Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr		
65	70	75
Asp Ala Ser Asp Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly Ser		
	85	90
Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala		
	100	105
Asp Ala Ala Ala Tyr Tyr Cys Gln Ser Ala Asp Gly Ser Ser Tyr Ala		
	115	120
Phe Gly Gly Gly Thr Glu Val Val Val Lys Gly Gly Gly Gly Ser		
	130	135
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln		
	145	150
Glu Gln Leu Glu Glu Ser Gly Gly Gly Leu Val Lys Pro Glu Gly Ser		
	165	170
Leu Thr Leu Thr Cys Thr Ala Ser Gly Val Ser Phe Ser Ser Ser Tyr		
	180	185
Trp Ile Tyr Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile		
	195	200
Ala Cys Ile Tyr Thr Gly Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp		
	210	215
Ala Lys Gly Arg Phe Thr Val Ser Glu Thr Ser Ser Thr Thr Val Thr		
	225	230
Leu Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys		
	245	250
Ala Arg Ala Ser Ala Trp Thr Tyr Gly Met Asp Leu Trp Gly Pro Gly		
	260	265
Thr Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Leu Ser Gly Arg		
	275	280
Ser Asp Asn His Ser Pro Leu Gly Leu Ala Gly Ser Lys Pro Ile Pro		
	290	295
Asn Pro Leu Leu Gly Leu Asp Ser Thr Leu Ser Gly Arg Ser Asp Asn		
	305	310
His Ser Pro Leu Gly Leu Ala Gly Ser Gly Gly Gly Gly Ser Gln Val		
	325	330
Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu Thr Leu		
	340	345
Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Tyr Tyr Ser Trp		
	355	360
Thr Trp Val Arg Gln Pro Ala Gly Gln Gly Leu Glu Trp Ile Gly Arg		
	370	375
Ile His Ile Ser Gly Val Thr Asn His Asn Pro Ser Leu Lys Ser Arg		
	385	390
Val Ser Met Ser Ile Asp Thr Ser Arg Thr Gln Phe Ser Leu Arg Leu		
	405	410
Thr Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Phe Cys Ala Arg Ser		
	420	425
Gly Gly Thr Tyr Ser Ala Phe Asp Ile Trp Gly Gln Gly Thr Met Val		
	435	440
Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly		
	450	455

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Gly Gly Ser Gly Gly Gly Gly Ser Glu Ile Val Leu Thr Gln Ser Pro
 465 470 475 480
 Gly Thr Leu Ser Leu Ser Pro Gly Glu Arg Ser Thr Leu Ser Cys Arg
 485 490 495
 Ala Ser Gln Ser Phe Thr Ser Asn Tyr Leu Ala Trp Tyr Gln Gln Arg
 500 505 510
 Pro Gly Gln Ala Pro Arg Leu Leu Ile Tyr Gly Ala Ser Thr Arg Ala
 515 520 525
 Ile Gly Ile Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe
 530 535 540
 Thr Leu Thr Ile Ser Arg Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr
 545 550 555 560
 Cys Gln Gln Tyr Gly Ser Ser Pro Trp Thr Phe Gly Gln Gly Thr Lys
 565 570 575
 Val Glu Ile Lys Thr Thr Thr Pro Ala Pro Arg Pro Pro Thr Pro Ala
 580 585 590
 Pro Thr Ile Ala Ser Gln Pro Leu Ser Leu Arg Pro Glu Ala Cys Arg
 595 600 605
 Pro Ala Ala Gly Gly Ala Val His Thr Arg Gly Leu Asp Phe Ala Cys
 610 615 620
 Asp Ile Tyr Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly Val Leu Leu
 625 630 635 640
 Leu Ser Leu Val Ile Thr Leu Tyr Cys Lys Arg Gly Arg Lys Lys Leu
 645 650 655
 Leu Tyr Ile Phe Lys Gln Pro Phe Met Arg Pro Val Gln Thr Thr Gln
 660 665 670
 Glu Glu Asp Gly Cys Ser Cys Arg Phe Pro Glu Glu Glu Gly Gly
 675 680 685
 Cys Glu Leu Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr
 690 695 700
 Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg
 705 710 715 720
 Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met
 725 730 735
 Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu
 740 745 750
 Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys
 755 760 765
 Gly Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu
 770 775 780
 Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu
 785 790 795 800
 Pro Pro Arg

<210> SEQ ID NO 23

<211> LENGTH: 806

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 23

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15

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His	Ala	Ala	Arg	Pro	Gly	Tyr	Pro	Tyr	Asp	Val	Pro	Asp	Tyr	Ala	Asp
			20					25					30		
Ile	Val	Met	Thr	Gln	Thr	Pro	Ala	Ser	Val	Ser	Glu	Pro	Val	Gly	Gly
		35				40					45				
Thr	Val	Thr	Ile	Met	Cys	Gln	Ala	Ser	Gln	Ser	Ile	Ser	Asn	Trp	Leu
	50				55						60				
Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Pro	Pro	Lys	Leu	Leu	Ile	Tyr
65					70					75				80	
Gln	Ala	Ser	Lys	Leu	Ala	Ser	Gly	Val	Pro	Ser	Arg	Phe	Lys	Gly	Ser
			85						90					95	
Gly	Ser	Gly	Thr	Glu	Tyr	Thr	Leu	Thr	Ile	Ser	Asp	Leu	Glu	Cys	Ala
			100					105					110		
Asp	Ala	Ala	Thr	Tyr	Tyr	Cys	Gln	Ser	Tyr	Tyr	Asp	Ser	Gly	Ser	Asn
		115					120					125			
Val	Phe	Phe	Ala	Phe	Gly	Gly	Gly	Thr	Lys	Val	Val	Val	Glu	Gly	Gly
	130				135					140					
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly
145				150						155				160	
Gly	Gly	Ser	Leu	Ser	Leu	Glu	Glu	Ser	Gly	Gly	Asp	Leu	Val	Lys	Pro
			165					170						175	
Gly	Ala	Ser	Leu	Thr	Leu	Thr	Cys	Thr	Ala	Ser	Gly	Phe	Ser	Phe	Ser
			180					185					190		
Ala	Gly	Tyr	Trp	Ile	Cys	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu
		195				200						205			
Glu	Trp	Ile	Ala	Cys	Thr	Tyr	Ala	Gly	Arg	Ser	Gly	Ser	Thr	Tyr	Tyr
	210				215						220				
Ala	Asn	Trp	Val	Asn	Gly	Arg	Phe	Thr	Ile	Pro	Lys	Thr	Ser	Ser	Thr
225				230						235					240
Thr	Val	Thr	Leu	Gln	Met	Thr	Ser	Leu	Ser	Gly	Ala	Asp	Thr	Ala	Ser
			245					250						255	
Tyr	Phe	Cys	Ala	Arg	Gly	Asn	Ala	Gly	Val	Ala	Val	Gly	Ala	Leu	Trp
		260					265						270		
Gly	Pro	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Leu
		275				280					285				
Ser	Gly	Arg	Ser	Asp	Asn	His	Ser	Pro	Leu	Gly	Leu	Ala	Gly	Ser	Lys
	290				295						300				
Pro	Ile	Pro	Asn	Pro	Leu	Leu	Gly	Leu	Asp	Ser	Thr	Leu	Ser	Gly	Arg
305				310						315				320	
Ser	Asp	Asn	His	Ser	Pro	Leu	Gly	Leu	Ala	Gly	Ser	Gly	Gly	Gly	Gly
		325				330							335		
Ser	Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser
		340						345					350		
Glu	Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile	Ser	Tyr
	355					360						365			
Tyr	Ser	Trp	Thr	Trp	Val	Arg	Gln	Pro	Ala	Gly	Gln	Gly	Leu	Glu	Trp
	370				375						380				
Ile	Gly	Arg	Ile	His	Ile	Ser	Gly	Val	Thr	Asn	His	Asn	Pro	Ser	Leu
385				390						395					400
Lys	Ser	Arg	Val	Ser	Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln	Phe	Ser
		405						410					415		
Leu	Arg	Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Phe	Cys
		420					425						430		

Ala	Arg	Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly	Gln	Gly	
435						440				445						
Thr	Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	
450						455				460						
Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr	
465						470				475					480	
Gln	Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser	Thr	Leu	
				485				490						495		
Ser	Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu	Ala	Trp	Tyr	
		500						505				510				
Gln	Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Ser	
		515				520						525				
Thr	Arg	Ala	Ile	Gly	Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	
530						535						540				
Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp	Phe	Ala	
545						550				555					560	
Val	Tyr	Tyr	Cys	Gln	Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe	Gly	Gln	
				565				570						575		
Gly	Thr	Lys	Val	Glu	Ile	Lys	Thr	Thr	Thr	Pro	Ala	Pro	Arg	Pro	Pro	
		580						585				590				
Thr	Pro	Ala	Pro	Thr	Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu	Arg	Pro	Glu	
		595				600						605				
Ala	Cys	Arg	Pro	Ala	Ala	Gly	Gly	Ala	Val	His	Thr	Arg	Gly	Leu	Asp	
610						615						620				
Phe	Ala	Cys	Asp	Ile	Tyr	Ile	Trp	Ala	Pro	Leu	Ala	Gly	Thr	Cys	Gly	
625				630						635					640	
Val	Leu	Leu	Leu	Ser	Leu	Val	Ile	Thr	Leu	Tyr	Cys	Lys	Arg	Gly	Arg	
				645				650						655		
Lys	Lys	Leu	Leu	Tyr	Ile	Phe	Lys	Gln	Pro	Phe	Met	Arg	Pro	Val	Gln	
		660						665				670				
Thr	Thr	Gln	Glu	Glu	Asp	Gly	Cys	Ser	Cys	Arg	Phe	Pro	Glu	Glu	Glu	
675						680						685				
Glu	Gly	Gly	Cys	Glu	Leu	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	
690						695				700						
Pro	Ala	Tyr	Gln	Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	
705						710				715					720	
Gly	Arg	Arg	Glu	Glu	Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	
		725						730				735				
Pro	Glu	Met	Gly	Gly	Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	
		740						745				750				
Tyr	Asn	Glu	Leu	Gln	Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	
		755				760						765				
Gly	Met	Lys	Gly	Glu	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr		
770						775				780						
Gln	Gly	Leu	Ser	Thr	Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His	Met	
785						790				795					800	
Gln	Ala	Leu	Pro	Pro	Arg				</							

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<210> SEQ ID NO 24
<211> LENGTH: 809
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
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polypeptide

<400> SEQUENCE: 24

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Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1           5           10           15

His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp
      20           25           30

Ile Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly Gly
      35           40           45

Ser Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Phe Tyr Asn Leu Leu
      50           55           60

Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr
      65           70           75           80

Asp Ala Ser Asp Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly Ser
      85           90           95

Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala
      100          105          110

Asp Ala Ala Ala Tyr Tyr Cys Gln Ser Ala Asp Gly Ser Ser Tyr Ala
      115          120          125

Phe Gly Gly Gly Thr Glu Val Val Val Lys Gly Gly Gly Gly Gly Ser
      130          135          140

Gly Gly Gly Gly Ser Leu Ser Gly Arg Ser Asp Asn His Ser Pro Leu
      145          150          155          160

Gly Leu Ala Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln
      165          170          175

Glu Gln Leu Glu Glu Ser Gly Gly Gly Leu Val Lys Pro Glu Gly Ser
      180          185          190

Leu Thr Leu Thr Cys Thr Ala Ser Gly Val Ser Phe Ser Ser Ser Tyr
      195          200          205

Trp Ile Tyr Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile
      210          215          220

Ala Cys Ile Tyr Thr Gly Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp
      225          230          235          240

Ala Lys Gly Arg Phe Thr Val Ser Glu Thr Ser Ser Thr Thr Val Thr
      245          250          255

Leu Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys
      260          265          270

Ala Arg Ala Ser Ala Trp Thr Tyr Gly Met Asp Leu Trp Gly Pro Gly
      275          280          285

Thr Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly
      290          295          300

Ser Gly Gly Gly Ser Gly Leu Ser Gly Arg Ser Asp Asn His Ser Pro
      305          310          315          320

Leu Gly Leu Ala Gly Ser Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly
      325          330          335

Gly Gly Gly Ser Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val
      340          345          350

Lys Pro Ser Glu Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser
      355          360          365

Ile Ser Tyr Tyr Ser Trp Thr Trp Val Arg Gln Pro Ala Gly Gln Gly
      370          375          380

Leu Glu Trp Ile Gly Arg Ile His Ile Ser Gly Val Thr Asn His Asn
      385          390          395          400

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Pro	Ser	Leu	Lys	Ser	Arg	Val	Ser	Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr	
				405					410					415		
Gln	Phe	Ser	Leu	Arg	Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	
				420					425					430		
Tyr	Phe	Cys	Ala	Arg	Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp	
				435					440					445		
Gly	Gln	Gly	Thr	Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	
				450					455					460		
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	
				465					470					475		
Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	
				485					490					495		
Ser	Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu	
				500					505					510		
Ala	Trp	Tyr	Gln	Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	
				515					520					525		
Gly	Ala	Ser	Thr	Arg	Ala	Ile	Gly	Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	
				530					535					540		
Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu	
				545					550					555		
Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr	
				565					570					575		
Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile	Lys	Thr	Thr	Thr	Pro	Ala	Pro	
				580					585					590		
Arg	Pro	Pro	Thr	Pro	Ala	Pro	Thr	Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu	
				595					600					605		
Arg	Pro	Glu	Ala	Cys	Arg	Pro	Ala	Ala	Gly	Gly	Ala	Val	His	Thr	Arg	
				610					615					620		
Gly	Leu	Asp	Phe	Ala	Cys	Asp	Ile	Tyr	Ile	Trp	Ala	Pro	Leu	Ala	Gly	
				625					630					635		
Thr	Cys	Gly	Val	Leu	Leu	Leu	Ser	Leu	Val	Ile	Thr	Leu	Tyr	Cys	Lys	
				645					650					655		
Arg	Gly	Arg	Lys	Lys	Leu	Leu	Tyr	Ile	Phe	Lys	Gln	Pro	Phe	Met	Arg	
				660					665					670		
Pro	Val	Gln	Thr	Thr	Gln	Glu	Glu	Asp	Gly	Cys	Ser	Cys	Arg	Phe	Pro	
				675					680					685		
Glu	Glu	Glu	Glu	Gly	Gly	Cys	Glu	Leu	Arg	Val	Lys	Phe	Ser	Arg	Ser	
				690					695					700		
Ala	Asp	Ala	Pro	Ala	Tyr	Gln	Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	
				705					710					715		
Leu	Asn	Leu	Gly	Arg	Arg	Glu	Glu	Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	
				725					730					735		
Gly	Arg	Asp	Pro	Glu	Met	Gly	Gly	Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	
				740					745					750		
Glu	Gly	Leu	Tyr	Asn	Glu	Leu	Gln	Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	
				755					760					765		
Ser	Glu	Ile	Gly	Met	Lys	Gly	Glu	Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	
				770					775					780		
Gly	Leu	Tyr	Gln	Gly	Leu	Ser	Thr	Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	
				785					790					795		
Leu	His	Met	Gln	Ala	Leu	Pro	Pro	Arg								
				805												

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<210> SEQ ID NO 25
<211> LENGTH: 812
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 25

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1             5             10             15

His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp
      20             25             30

Ile Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly Gly
      35             40             45

Thr Val Thr Ile Met Cys Gln Ala Ser Gln Ser Ile Ser Asn Trp Leu
 50             55             60

Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr
 65             70             75             80

Gln Ala Ser Lys Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly Ser
      85             90             95

Gly Ser Gly Thr Glu Tyr Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala
      100            105            110

Asp Ala Ala Thr Tyr Tyr Cys Gln Ser Tyr Tyr Asp Ser Gly Ser Asn
      115            120            125

Val Phe Phe Ala Phe Gly Gly Gly Thr Lys Val Val Val Glu Gly Gly
      130            135            140

Gly Gly Gly Ser Gly Gly Gly Gly Ser Leu Ser Gly Arg Ser Asp Asn
      145            150            155            160

His Ser Pro Leu Gly Leu Ala Gly Ser Gly Gly Gly Gly Ser Gly Gly
      165            170            175

Gly Gly Ser Leu Ser Leu Glu Glu Ser Gly Gly Asp Leu Val Lys Pro
      180            185            190

Gly Ala Ser Leu Thr Leu Thr Cys Thr Ala Ser Gly Phe Ser Phe Ser
      195            200            205

Ala Gly Tyr Trp Ile Cys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu
      210            215            220

Glu Trp Ile Ala Cys Thr Tyr Ala Gly Arg Ser Gly Ser Thr Tyr Tyr
      225            230            235            240

Ala Asn Trp Val Asn Gly Arg Phe Thr Ile Pro Lys Thr Ser Ser Thr
      245            250            255

Thr Val Thr Leu Gln Met Thr Ser Leu Ser Gly Ala Asp Thr Ala Ser
      260            265            270

Tyr Phe Cys Ala Arg Gly Asn Ala Gly Val Ala Val Gly Ala Leu Trp
      275            280            285

Gly Pro Gly Thr Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly
      290            295            300

Gly Gly Gly Ser Gly Gly Gly Ser Gly Leu Ser Gly Arg Ser Asp Asn
      305            310            315            320

His Ser Pro Leu Gly Leu Ala Gly Ser Gly Gly Gly Ser Gly Gly Gly
      325            330            335

Gly Ser Gly Gly Gly Gly Ser Gln Val Gln Leu Gln Glu Ser Gly Pro
      340            345            350

Gly Leu Val Lys Pro Ser Glu Thr Leu Ser Leu Thr Cys Thr Val Ser
      355            360            365

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Gly 370	Ser 375	Ile 380	Ser 385	Tyr 390	Tyr 395	Ser 400	Trp 405	Trp 410	Thr 415	Trp 420	Val 425	Arg 430	Gln 435	Pro 440	Ala 445
Gly 385	Gln 390	Gly 395	Leu 400	Glu 405	Trp 410	Ile 415	Gly 420	Arg 425	Ile 430	His 435	Ile 440	Ser 445	Gly 450	Val 455	Thr 460
Asn 405	His 410	Asn 415	Pro 420	Ser 425	Leu 430	Lys 435	Ser 440	Arg 445	Val 450	Ser 455	Met 460	Ser 465	Ile 470	Asp 475	Thr 480
Ser 420	Arg 425	Thr 430	Gln 435	Phe 440	Ser 445	Leu 450	Arg 455	Leu 460	Thr 465	Ser 470	Val 475	Thr 480	Ala 485	Ala 490	Asp 495
Thr 440	Ala 445	Val 450	Tyr 455	Phe 460	Cys 465	Ala 470	Arg 475	Ser 480	Gly 485	Gly 490	Thr 495	Tyr 500	Ser 505	Ala 510	Phe 515
Asp 460	Ile 465	Trp 470	Gly 475	Gln 480	Gly 485	Thr 490	Met 495	Val 500	Thr 505	Val 510	Ser 515	Ser 520	Gly 525	Gly 530	Gly 535
Gly 465	Ser 470	Gly 475	Gly 480	Gly 485	Gly 490	Ser 495	Gly 500	Gly 505	Gly 510	Gly 515	Ser 520	Gly 525	Gly 530	Gly 535	Gly 540
Ser 480	Glu 485	Ile 490	Val 495	Leu 500	Thr 505	Gln 510	Ser 515	Pro 520	Gly 525	Thr 530	Leu 535	Ser 540	Leu 545	Ser 550	Pro 555
Gly 500	Glu 505	Arg 510	Ser 515	Thr 520	Leu 525	Ser 530	Cys 535	Arg 540	Ala 545	Ser 550	Gln 555	Ser 560	Phe 565	Thr 570	Ser 575
Asn 515	Tyr 520	Leu 525	Ala 530	Trp 535	Tyr 540	Gln 545	Gln 550	Arg 555	Pro 560	Gly 565	Gln 570	Ala 575	Pro 580	Arg 585	Leu 590
Leu 530	Ile 535	Tyr 540	Gly 545	Ala 550	Ser 555	Thr 560	Arg 565	Ala 570	Ile 575	Gly 580	Ile 585	Pro 590	Asp 595	Arg 600	Phe 605
Ser 545	Gly 550	Ser 555	Gly 560	Ser 565	Gly 570	Thr 575	Asp 580	Phe 585	Thr 590	Leu 595	Thr 600	Ile 605	Ser 610	Arg 615	Leu 620
Glu 560	Pro 565	Glu 570	Asp 575	Phe 580	Ala 585	Val 590	Tyr 595	Tyr 600	Cys 605	Gln 610	Gln 615	Tyr 620	Gly 625	Ser 630	Ser 635
Pro 580	Trp 585	Thr 590	Phe 595	Gly 600	Gln 605	Gly 610	Thr 615	Lys 620	Val 625	Glu 630	Ile 635	Lys 640	Thr 645	Thr 650	Thr 655
Pro 595	Ala 600	Pro 605	Arg 610	Pro 615	Pro 620	Thr 625	Pro 630	Ala 635	Pro 640	Thr 645	Ile 650	Ala 655	Ser 660	Gln 665	Pro 670
Leu 610	Ser 615	Leu 620	Arg 625	Pro 630	Glu 635	Ala 640	Cys 645	Arg 650	Pro 655	Ala 660	Ala 665	Gly 670	Gly 675	Ala 680	Val 685
His 625	Thr 630	Arg 635	Gly 640	Leu 645	Asp 650	Phe 655	Ala 660	Cys 665	Asp 670	Ile 675	Tyr 680	Ile 685	Trp 690	Ala 695	Pro 700
Leu 645	Ala 650	Gly 655	Thr 660	Cys 665	Gly 670	Val 675	Leu 680	Leu 685	Leu 690	Ser 695	Leu 700	Val 705	Ile 710	Thr 715	Leu 720
Tyr 660	Cys 665	Lys 670	Arg 675	Gly 680	Arg 685	Lys 690	Lys 695	Leu 700	Leu 705	Tyr 710	Ile 715	Phe 720	Lys 725	Gln 730	Pro 735
Phe 675	Met 680	Arg 685	Pro 690	Val 695	Gln 700	Thr 705	Thr 710	Gln 715	Glu 720	Glu 725	Asp 730	Gly 735	Cys 740	Ser 745	Cys 750
Arg 690	Phe 695	Pro 700	Glu 705	Glu 710	Glu 715	Gly 720	Gly 725	Gly 730	Cys 735	Glu 740	Leu 745	Arg 750	Val 755	Lys 760	Phe 765
Ser 705	Arg 710	Ser 715	Ala 720	Asp 725	Ala 730	Pro 735	Ala 740	Tyr 745	Gln 750	Gln 755	Gly 760	Gln 765	Asn 770	Gln 775	Leu 780
Tyr 720	Asn 725	Glu 730	Leu 735	Asn 740	Leu 745	Gly 750	Arg 755	Arg 760	Glu 765	Glu 770	Tyr 775	Asp 780	Val 785	Leu 790	Asp 795
Lys 735	Arg 740	Arg 745	Gly 750	Arg 755	Asp 760	Pro 765	Glu 770	Met 775	Gly 780	Gly 785	Lys 790	Pro 795	Arg 800	Arg 805	Lys 810
Asn 755	Pro 760	Gln 765	Glu 770	Gly 775	Leu 780	Tyr 785	As								

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785	790	795	800
Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg			
	805	810	

<210> SEQ ID NO 26
 <211> LENGTH: 799
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 26

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu			
1	5	10	15
His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp			
	20	25	30
Ile Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly Gly			
	35	40	45
Ser Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Phe Tyr Asn Leu Leu			
	50	55	60
Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr			
	65	70	75
Asp Ala Ser Asp Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly Ser			
	85	90	95
Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala			
	100	105	110
Asp Ala Ala Ala Tyr Tyr Cys Gln Ser Ala Asp Gly Ser Ser Tyr Ala			
	115	120	125
Phe Gly Gly Gly Thr Glu Val Val Val Lys Gly Gly Gly Gly Gly Ser			
	130	135	140
Gly Gly Gly Gly Ser Leu Val Pro Arg Gly Ser Gly Gly Gly Gly Ser			
	145	150	155
Gly Gly Gly Gly Ser Gln Glu Gln Leu Glu Glu Ser Gly Gly Gly Leu			
	165	170	175
Val Lys Pro Glu Gly Ser Leu Thr Leu Thr Cys Thr Ala Ser Gly Val			
	180	185	190
Ser Phe Ser Ser Ser Tyr Trp Ile Tyr Trp Val Arg Gln Ala Pro Gly			
	195	200	205
Lys Gly Leu Glu Trp Ile Ala Cys Ile Tyr Thr Gly Ser Ser Gly Ser			
	210	215	220
Thr Tyr Tyr Ala Ser Trp Ala Lys Gly Arg Phe Thr Val Ser Glu Thr			
	225	230	235
Ser Ser Thr Thr Val Thr Leu Gln Met Thr Ser Leu Thr Ala Ala Asp			
	245	250	255
Thr Ala Thr Tyr Phe Cys Ala Arg Ala Ser Ala Trp Thr Tyr Gly Met			
	260	265	270
Asp Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser Gly Gly Gly			
	275	280	285
Gly Ser Gly Gly Gly Gly Ser Leu Val Pro Arg Gly Ser Lys Pro Ile			
	290	295	300
Pro Asn Pro Leu Leu Gly Leu Asp Ser Thr Leu Val Pro Arg Gly Ser			
	305	310	315
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln Val Gln Leu Gln Glu			
	325	330	335

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Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Glu	Thr	Leu	Ser	Leu	Thr	Cys
			340					345					350		
Thr	Val	Ser	Gly	Gly	Ser	Ile	Ser	Tyr	Tyr	Ser	Trp	Thr	Trp	Val	Arg
		355					360					365			
Gln	Pro	Ala	Gly	Gln	Gly	Leu	Glu	Trp	Ile	Gly	Arg	Ile	His	Ile	Ser
	370					375					380				
Gly	Val	Thr	Asn	His	Asn	Pro	Ser	Leu	Lys	Ser	Arg	Val	Ser	Met	Ser
385					390					395					400
Ile	Asp	Thr	Ser	Arg	Thr	Gln	Phe	Ser	Leu	Arg	Leu	Thr	Ser	Val	Thr
				405					410					415	
Ala	Ala	Asp	Thr	Ala	Val	Tyr	Phe	Cys	Ala	Arg	Ser	Gly	Gly	Thr	Tyr
			420					425					430		
Ser	Ala	Phe	Asp	Ile	Trp	Gly	Gln	Gly	Thr	Met	Val	Thr	Val	Ser	Ser
		435					440					445			
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	
	450				455					460					
Gly	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu	Ser
465					470					475					480
Leu	Ser	Pro	Gly	Glu	Arg	Ser	Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	Ser
				485					490					495	
Phe	Thr	Ser	Asn	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Arg	Pro	Gly	Gln	Ala
			500					505					510		
Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Ser	Thr	Arg	Ala	Ile	Gly	Ile	Pro
		515					520					525			
Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile
	530				535						540				
Ser	Arg	Leu	Glu	Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Tyr
545					550					555					560
Gly	Ser	Ser	Pro	Trp	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile	Lys
				565					570					575	
Thr	Thr	Thr	Pro	Ala	Pro	Arg	Pro	Pro	Thr	Pro	Ala	Pro	Thr	Ile	Ala
			580					585					590		
Ser	Gln	Pro	Leu	Ser	Leu	Arg	Pro	Glu	Ala	Cys	Arg	Pro	Ala	Ala	Gly
		595					600					605			
Gly	Ala	Val	His	Thr	Arg	Gly	Leu	Asp	Phe	Ala	Cys	Asp	Ile	Tyr	Ile
	610					615					620				
Trp	Ala	Pro	Leu	Ala	Gly	Thr	Cys	Gly	Val	Leu	Leu	Leu	Ser	Leu	Val
625					630					635					640
Ile	Thr	Leu	Tyr	Cys	Lys	Arg	Gly	Arg	Lys	Lys	Leu	Leu	Tyr	Ile	Phe
				645					650					655	
Lys	Gln	Pro	Phe	Met	Arg	Pro	Val	Gln	Thr	Thr	Gln	Glu	Glu	Asp	Gly
			660					665					670		
Cys	Ser	Cys	Arg	Phe	Pro	Glu	Glu	Glu	Gly	Gly	Cys	Glu	Leu	Arg	
		675					680				685				
Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln	Gln	Gly	Gln
	690					695					700				
Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu	Glu	Tyr	Asp
705					710					715					720
Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met	Gly	Gly	Lys	Pro
				725					730					735	
Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn	Glu	Leu	Gln	Lys	Asp
			740					745					750		
Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys	Gly	Glu	Arg	Arg

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755	760	765
Arg Gly Lys Gly His Asp	Gly Leu Tyr Gln Gly	Leu Ser Thr Ala Thr
770	775	780

Lys Asp Thr Tyr Asp	Ala Leu His Met Gln	Ala Leu Pro Pro Arg
785	790	795

<210> SEQ ID NO 27

<211> LENGTH: 802

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 27

Met Ala Leu Pro Val Thr	Ala Leu Leu Leu Pro	Leu Ala Leu Leu Leu
1	5	10

His Ala Ala Arg Pro Gly Tyr	Pro Tyr Asp Val Pro	Asp Tyr Ala Asp
20	25	30

Ile Val Met Thr Gln Thr	Pro Ala Ser Val Ser	Glu Pro Val Gly Gly
35	40	45

Thr Val Thr Ile Met Cys	Gln Ala Ser Gln Ser	Ile Ser Asn Trp Leu
50	55	60

Ala Trp Tyr Gln Gln Lys	Pro Gly Gln Pro Pro	Lys Leu Leu Ile Tyr
65	70	75

Gln Ala Ser Lys Leu Ala	Ser Gly Val Pro Ser	Arg Phe Lys Gly Ser
85	90	95

Gly Ser Gly Thr Glu Tyr	Thr Leu Thr Ile Ser	Asp Leu Glu Cys Ala
100	105	110

Asp Ala Ala Thr Tyr Tyr	Cys Gln Ser Tyr Tyr	Asp Ser Gly Ser Asn
115	120	125

Val Phe Phe Ala Phe Gly	Gly Gly Thr Lys Val	Val Val Glu Gly Gly
130	135	140

Gly Gly Gly Ser Gly Gly	Gly Gly Ser Leu Val	Pro Arg Gly Ser Gly
145	150	155

Gly Gly Gly Ser Gly Gly	Gly Gly Ser Leu Ser	Leu Glu Glu Ser Gly
165	170	175

Gly Asp Leu Val Lys Pro	Gly Ala Ser Leu Thr	Leu Thr Cys Thr Ala
180	185	190

Ser Gly Phe Ser Phe Ser	Ala Gly Tyr Trp Ile	Cys Trp Val Arg Gln
195	200	205

Ala Pro Gly Lys Gly Leu	Glu Trp Ile Ala Cys	Thr Tyr Ala Gly Arg
210	215	220

Ser Gly Ser Thr Tyr Tyr	Ala Asn Trp Val Asn	Gly Arg Phe Thr Ile
225	230	235

Pro Lys Thr Ser Ser Thr	Thr Val Thr Leu Gln	Met Thr Ser Leu Ser
245	250	255

Gly Ala Asp Thr Ala Ser	Tyr Phe Cys Ala Arg	Gly Asn Ala Gly Val
260	265	270

Ala Val Gly Ala Leu Trp	Gly Pro Gly Thr Leu	Val Thr Val Ser Ser
275	280	285

Gly Gly Gly Gly Ser Gly	Gly Gly Gly Ser Leu	Val Pro Arg Gly Ser
290	295	300

Lys Pro Ile Pro Asn Pro	Leu Leu Gly Leu Asp	Ser Thr Leu Val Pro
305	310	315

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Arg	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Val	Gln
			325						330				335		
Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Glu	Thr	Leu	Ser
			340				345						350		
Leu	Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile	Ser	Tyr	Tyr	Ser	Trp	Thr
			355				360				365				
Trp	Val	Arg	Gln	Pro	Ala	Gly	Gln	Gly	Leu	Glu	Trp	Ile	Gly	Arg	Ile
			370		375						380				
His	Ile	Ser	Gly	Val	Thr	Asn	His	Asn	Pro	Ser	Leu	Lys	Ser	Arg	Val
385					390				395				400		
Ser	Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln	Phe	Ser	Leu	Arg	Leu	Thr
			405						410				415		
Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Phe	Cys	Ala	Arg	Ser	Gly
			420				425						430		
Gly	Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly	Gln	Gly	Thr	Met	Val	Thr
			435				440				445				
Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly
			450		455						460				
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly
465					470				475				480		
Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser	Thr	Leu	Ser	Cys	Arg	Ala
			485						490				495		
Ser	Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Arg	Pro
			500				505						510		
Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Ser	Thr	Arg	Ala	Ile
			515				520				525				
Gly	Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr
			530		535						540				
Leu	Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys
545					550				555				560		
Gln	Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val
			565						570				575		
Glu	Ile	Lys	Thr	Thr	Thr	Pro	Ala	Pro	Arg	Pro	Pro	Thr	Pro	Ala	Pro
			580				585						590		
Thr	Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu	Arg	Pro	Glu	Ala	Cys	Arg	Pro
			595		600						605				
Ala	Ala	Gly	Gly	Ala	Val	His	Thr	Arg	Gly	Leu	Asp	Phe	Ala	Cys	Asp
610					615						620				
Ile	Tyr	Ile	Trp	Ala	Pro	Leu	Ala	Gly	Thr	Cys	Gly	Val	Leu	Leu	Leu
625					630		635						640		
Ser	Leu	Val	Ile	Thr	Leu	Tyr	Cys	Lys	Arg	Gly	Arg	Lys	Lys	Leu	Leu
			645				650						655		
Tyr	Ile	Phe	Lys	Gln	Pro	Phe	Met	Arg	Pro	Val	Gln	Thr	Thr	Gln	Glu
			660				665				670				
Glu	Asp	Gly	Cys	Ser	Cys	Arg	Phe	Pro	Glu	Glu	Glu	Glu	Gly	Gly	Cys
			675		680						685				
Glu	Leu	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln
690					695						700				
Gln	Gly	Gln	Asn	Gln	Leu										

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740					745					750					
Gln	Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys	Gly
	755					760					765				
Glu	Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr	Gln	Gly	Leu	Ser
	770					775					780				
Thr	Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His	Met	Gln	Ala	Leu	Pro
	785					790					795				800

Pro Arg

<210> SEQ ID NO 28

<211> LENGTH: 908

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 28

Met	Ala	Leu	Pro	Val	Thr	Ala	Leu	Leu	Leu	Pro	Leu	Ala	Leu	Leu	Leu
1			5						10					15	
His	Ala	Ala	Arg	Pro	Gly	Tyr	Pro	Tyr	Asp	Val	Pro	Asp	Tyr	Ala	Gly
			20					25					30		
Phe	His	Thr	Ile	Asn	Ala	Thr	Trp	Trp	Ala	Asn	Ile	Thr	Leu	Val	Gly
		35					40					45			
Pro	Pro	Asp	Thr	Pro	Val	Thr	Trp	Tyr	Asp	Thr	Gln	Gly	Leu	Trp	Phe
	50					55					60				
Cys	Asn	Gly	Ser	Arg	Val	Lys	Asn	Pro	Gln	Ile	Arg	His	Thr	Cys	Asn
65					70					75				80	
Asp	Gln	Asn	Leu	Thr	Leu	Ile	His	Val	Asn	Lys	Thr	Tyr	Glu	Arg	Thr
			85						90					95	
Tyr	Met	Gly	Tyr	Asn	Arg	Gln	Gly	Thr	Lys	Lys	Glu	Asp	Tyr	Lys	Val
		100						105					110		
Val	Val	Ile	Pro	Pro	Pro	Pro	Ala	Thr	Val	Lys	Pro	Gln	Pro	Glu	Pro
		115					120					125			
Glu	Tyr	Val	Phe	Val	Tyr	Met	Gly	Glu	Asn	Lys	Thr	Leu	Glu	Gly	Pro
	130					135					140				
Pro	Gly	Thr	Pro	Val	Thr	Trp	Phe	Asn	Gln	Asp	Gly	Lys	Lys	Phe	Cys
145				150					155					160	
Glu	Gly	Glu	Lys	Val	Leu	His	Pro	Glu	Phe	Asn	His	Thr	Cys	Asp	Lys
			165					170						175	
Gln	Asn	Leu	Ile	Leu	Leu	Phe	Val	Asn	Phe	Thr	His	Asp	Gly	Ala	Tyr
		180					185						190		
Leu	Gly	Tyr	Asn	His	Gln	Gly	Thr	Gln	Arg	Thr	His	Tyr	Glu	Val	Thr
		195				200						205			
Val	Leu	Asp	Leu	Phe	Pro	Asp	Ser	Gly	Gln	Met	Lys	Ile	Glu	Asn	His
	210					215					220				
Ser	Glu	Glu	Thr	Glu	Gln	Lys	Asn	Asp	Glu	His	His	Asn	Trp	Gln	Lys
225					230				235					240	
Gln	Gly	Gly	Gln	Lys	Gln	Gly	Gly	Gln	Lys	Thr	Asn	Gln	Thr	Lys	Val
			245					250					255		
Asn	Asp	Arg	Arg	Lys	Thr	Ala	Gln	Lys	Arg	Pro	Ser	Lys	Leu	Lys	Pro
		260					265					270			
Ala	Thr	Ile	Glu	Ala	Met	Leu	Val	Thr	Val	Thr	Ala	Gly	Ser	Asn	Leu
	275					280						285			
Thr	Leu	Val	Gly	Pro	Lys	Ala	Glu	Gly	Lys	Val	Thr	Trp	Phe	Asp	Gly

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290					295					300					
Asp 305	Leu	Lys	Arg	Pro	Cys 310	Glu	Pro	Asn	Tyr	Arg 315	Leu	Arg	His	Glu	Cys 320
Asn	Asn	Gln	Asn	Leu 325	Thr	Leu	Ile	Asn	Val 330	Thr	Lys	Asp	Tyr	Glu 335	Gly
Thr	Tyr	Tyr	Gly 340	Thr	Asn	Asp	Lys	Asp 345	Glu	Gly	Lys	Arg	Tyr 350	Arg	Val
Lys	Val	Asn 355	Thr	Thr	Asn	Ser	Gln 360	Ser	Val	Lys	Ile	Gln 365	Pro	Tyr	Thr
Arg	Gln 370	Thr	Thr	Pro	Asp	Gln 375	Glu	His	Lys	Phe	Glu 380	Leu	Gln	Phe	Glu
Thr 385	Asn	Gly	Asn	Tyr	Asp 390	Ser	Lys	Ile	Pro	Gly 395	Gly	Gly	Gly	Ser	Gly 400
Gly	Gly	Gly	Ser	Leu 405	Val	Pro	Arg	Gly	Ser 410	Lys	Pro	Ile	Pro	Asn 415	Pro
Leu	Leu	Gly	Leu 420	Asp	Ser	Thr	Leu	Val 425	Pro	Arg	Gly	Ser	Gly 430	Gly	Gly
Gly	Ser	Gly 435	Gly	Gly	Gly	Ser	Gln 440	Val	Gln	Leu	Gln	Glu 445	Ser	Gly	Pro
Gly	Leu 450	Val	Lys	Pro	Ser	Glu 455	Thr	Leu	Ser	Leu	Thr	Cys 460	Thr	Val	Ser
Gly 465	Gly	Ser	Ile	Ser	Tyr 470	Tyr	Ser	Trp	Thr	Trp 475	Val	Arg	Gln	Pro	Ala 480
Gly	Gln	Gly	Leu 485	Glu	Trp	Ile	Gly	Arg	Ile 490	His	Ile	Ser	Gly	Val 495	Thr
Asn	His	Asn	Pro 500	Ser	Leu	Lys	Ser	Arg 505	Val	Ser	Met	Ser	Ile 510	Asp	Thr
Ser	Arg	Thr 515	Gln	Phe	Ser	Leu	Arg 520	Leu	Thr	Ser	Val	Thr 525	Ala	Ala	Asp
Thr	Ala 530	Val	Tyr	Phe	Cys	Ala 535	Arg	Ser	Gly	Gly	Thr 540	Tyr	Ser	Ala	Phe
Asp 545	Ile	Trp	Gly	Gln	Gly 550	Thr	Met	Val	Thr	Val 555	Ser	Ser	Gly	Gly	Gly 560
Gly	Ser	Gly	Gly	Gly 565	Gly	Ser	Gly	Gly	Gly 570	Gly	Ser	Gly	Gly	Gly 575	Gly
Ser	Glu	Ile	Val 580	Leu	Thr	Gln	Ser	Pro 585	Gly	Thr	Leu	Ser	Leu 590	Ser	Pro
Gly	Glu	Arg 595	Ser	Thr	Leu	Ser	Cys 600	Arg	Ala	Ser	Gln	Ser 605	Phe	Thr	Ser
Asn	Tyr 610	Leu	Ala	Trp	Tyr	Gln 615	Gln	Arg	Pro	Gly	Gln 620	Ala	Pro	Arg	Leu
Leu 625	Ile	Tyr	Gly	Ala	Ser 630	Thr	Arg	Ala	Ile	Gly 635	Ile	Pro	Asp	Arg	Phe 640
Ser	Gly	Ser	Gly	Ser 645	Gly	Thr	Asp	Phe	Thr 650	Leu	Thr	Ile	Ser	Arg 655	Leu
Glu	Pro	Glu	Asp 660	Phe	Ala	Val	Tyr	Tyr	Cys 665	Gln	Gln	Tyr	Gly 670	Ser	Ser
Pro	Trp	Thr 675	Phe	Gly	Gln	Gly	Thr 680	Lys	Val	Glu	Ile	Lys 685	Thr	Thr	Thr
Pro	Ala 690	Pro	Arg	Pro	Pro	Thr 695	Pro	Ala	Pro	Thr	Ile 700	Ala	Ser	Gln	Pro
Leu 705	Ser	Leu	Arg	Pro	Glu 710	Ala	Cys	Arg	Pro	Ala 715	Ala	Gly	Gly	Ala	Val 720

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His Thr Arg Gly Leu Asp Phe Ala Cys Asp Ile Tyr Ile Trp Ala Pro
 725 730 735
 Leu Ala Gly Thr Cys Gly Val Leu Leu Leu Ser Leu Val Ile Thr Leu
 740 745 750
 Tyr Cys Lys Arg Gly Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro
 755 760 765
 Phe Met Arg Pro Val Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys
 770 775 780
 Arg Phe Pro Glu Glu Glu Glu Gly Gly Cys Glu Leu Arg Val Lys Phe
 785 790 795 800
 Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu
 805 810 815
 Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp
 820 825 830
 Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys
 835 840 845
 Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala
 850 855 860
 Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys
 865 870 875 880
 Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr
 885 890 895
 Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg
 900 905

<210> SEQ ID NO 29

<211> LENGTH: 745

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 29

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15
 His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala His
 20 25 30
 Asp Ala Cys Ile Pro Val Val Gly Lys Ile Gly Thr Asn Val Thr Leu
 35 40 45
 Asn Ala Val Asp Phe His Pro Gly Asp His Val Arg Trp Ser Tyr Gly
 50 55 60
 Pro Gly Gly Ala Gly Tyr Met Leu Cys Val Tyr Thr Gly Ser Trp Thr
 65 70 75 80
 Glu Tyr Lys Lys Pro Asp Ile Ile Phe Lys Cys Leu Ser Asn Asn Ser
 85 90 95
 Leu Leu Leu Ile Asn Val Thr Val Asn Tyr Thr Asn Thr Tyr Arg Thr
 100 105 110
 Leu Thr Ser Leu Asn Asn Trp Val His Asn Gln His His His Lys Phe
 115 120 125
 Pro Gly Trp Asn Leu Asp Thr Cys Tyr Ser Leu Thr Val Asn Glu Asn
 130 135 140
 Gly Thr Phe Pro Thr Thr Thr Lys Lys Pro Thr Thr Thr Thr Arg
 145 150 155 160
 Thr Thr Thr Thr Thr Thr Thr Lys Lys Thr Thr Thr Thr Arg Thr Thr

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165								170					175				
Thr	Ala	Ala	Lys	Lys	Thr	Thr	Ile	Ser	Thr	Thr	His	His	Lys	His	Ser		
			180						185						190		
Ser	Pro	Lys	Lys	Ser	Ser	Thr	Pro	Asn	Ser	His	Val	Glu	His	His	Val		
			195						200						205		
Gly	Phe	Glu	Ala	Thr	Ala	Ala	Glu	Thr	Pro	Leu	Gln	Pro	Ser	Pro	Gln		
			210						215						220		
His	Gln	His	Val	Ala	Thr	His	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly		
			225						230						235		
Ser	Leu	Val	Pro	Arg	Gly	Ser	Lys	Pro	Ile	Pro	Asn	Pro	Leu	Leu	Gly		
			245						250						255		
Leu	Asp	Ser	Thr	Leu	Val	Pro	Arg	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly		
			260						265						270		
Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val		
			275						280						285		
Lys	Pro	Ser	Glu	Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser		
			290						295						300		
Ile	Ser	Tyr	Tyr	Ser	Trp	Thr	Trp	Val	Arg	Gln	Pro	Ala	Gly	Gln	Gly		
			305						310						315		
Leu	Glu	Trp	Ile	Gly	Arg	Ile	His	Ile	Ser	Gly	Val	Thr	Asn	His	Asn		
			325						330						335		
Pro	Ser	Leu	Lys	Ser	Arg	Val	Ser	Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr		
			340						345						350		
Gln	Phe	Ser	Leu	Arg	Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val		
			355						360						365		
Tyr	Phe	Cys	Ala	Arg	Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp		
			370						375						380		
Gly	Gln	Gly	Thr	Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly		
			385						390						395		
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile		
			405						410						415		
Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg		
			420						425						430		
Ser	Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu		
			435						440						445		
Ala	Trp	Tyr	Gln	Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr		
			450						455						460		
Gly	Ala	Ser	Thr	Arg	Ala	Ile	Gly	Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser		
			465						470						475		
Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu		
			485						490						495		
Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr		
			500						505						510		
Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile	Lys	Thr	Thr	Thr	Pro	Ala	Pro		
			515						520						525		
Arg	Pro	Pro	Thr	Pro	Ala	Pro	Thr	Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu		
			530						535						540		
Arg	Pro	Glu	Ala	Cys	Arg	Pro	Ala	Ala	Gly	Gly	Ala	Val	His	Thr	Arg		
			545						550						555		
Gly	Leu	Asp	Phe	Ala	Cys	Asp	Ile	Tyr	Ile	Trp	Ala	Pro	Leu	Ala	Gly		
			565						570						575		
Thr	Cys	Gly	Val	Leu	Leu	Leu	Ser	Leu	Val	Ile	Thr	Leu	Tyr	Cys	Lys		
			580						585						590		

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Arg Gly Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met Arg
 595 600 605
 Pro Val Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe Pro
 610 615 620
 Glu Glu Glu Glu Gly Gly Cys Glu Leu Arg Val Lys Phe Ser Arg Ser
 625 630 635 640
 Ala Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu
 645 650 655
 Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg
 660 665 670
 Gly Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln
 675 680 685
 Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr
 690 695 700
 Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp
 705 710 715 720
 Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala
 725 730 735
 Leu His Met Gln Ala Leu Pro Pro Arg
 740 745

<210> SEQ ID NO 30
 <211> LENGTH: 660
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 30

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15
 His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Gln
 20 25 30
 Phe Ser Val Leu Gly Pro Ser Gly Pro Ile Leu Ala Met Val Gly Glu
 35 40 45
 Asp Ala Asp Leu Pro Cys His Leu Phe Pro Thr Met Ser Ala Glu Thr
 50 55 60
 Met Glu Leu Lys Trp Val Ser Ser Ser Leu Arg Gln Val Val Asn Val
 65 70 75 80
 Tyr Ala Asp Gly Lys Glu Val Glu Asp Arg Gln Ser Ala Pro Tyr Arg
 85 90 95
 Gly Arg Thr Ser Ile Leu Arg Asp Gly Ile Thr Ala Gly Lys Ala Ala
 100 105 110
 Leu Arg Ile His Asn Val Thr Ala Ser Asp Ser Gly Lys Tyr Leu Cys
 115 120 125
 Tyr Phe Gln Asp Gly Asp Phe Tyr Glu Lys Ala Leu Val Glu Leu Lys
 130 135 140
 Val Ala Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Leu Val Pro Arg
 145 150 155 160
 Gly Ser Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu Asp Ser Thr Leu
 165 170 175
 Val Pro Arg Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln
 180 185 190
 Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu Thr

195					200					205					
Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile	Ser	Tyr	Tyr	Ser
210					215					220					
Trp	Thr	Trp	Val	Arg	Gln	Pro	Ala	Gly	Gln	Gly	Leu	Glu	Trp	Ile	Gly
225					230					235					240
Arg	Ile	His	Ile	Ser	Gly	Val	Thr	Asn	His	Asn	Pro	Ser	Leu	Lys	Ser
				245					250					255	
Arg	Val	Ser	Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln	Phe	Ser	Leu	Arg
			260					265					270		
Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Phe	Cys	Ala	Arg
			275				280					285			
Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly	Gln	Gly	Thr	Met
	290					295					300				
Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	
305					310				315						320
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr	Gln	Ser
			325						330					335	
Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser	Thr	Leu	Ser	Cys
			340					345					350		
Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln
			355				360					365			
Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Ser	Thr	Arg
	370					375					380				
Ala	Ile	Gly	Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp
385					390				395						400
Phe	Thr	Leu	Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp	Phe	Ala	Val	Tyr
			405					410						415	
Tyr	Cys	Gln	Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe	Gly	Gln	Gly	Thr
			420					425					430		
Lys	Val	Glu	Ile	Lys	Thr	Thr	Thr	Pro	Ala	Pro	Arg	Pro	Pro	Thr	Pro
	435						440					445			
Ala	Pro	Thr	Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu	Arg	Pro	Glu	Ala	Cys
	450					455					460				
Arg	Pro	Ala	Ala	Gly	Gly	Ala	Val	His	Thr	Arg	Gly	Leu	Asp	Phe	Ala
465					470					475					480
Cys	Asp	Ile	Tyr	Ile	Trp	Ala	Pro	Leu	Ala	Gly	Thr	Cys	Gly	Val	Leu
				485					490					495	
Leu	Leu	Ser	Leu	Val	Ile	Thr	Leu	Tyr	Cys	Lys	Arg	Gly	Arg	Lys	Lys
			500					505					510		
Leu	Leu	Tyr	Ile	Phe	Lys	Gln	Pro	Phe	Met	Arg	Pro	Val	Gln	Thr	Thr
			515				520					525			
Gln	Glu	Glu	Asp	Gly	Cys	Ser	Cys	Arg	Phe	Pro	Glu	Glu	Glu	Glu	Gly
	530														

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Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly
625 630 635 640

Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala
645 650 655

Leu Pro Pro Arg
660

<210> SEQ ID NO 31

<211> LENGTH: 656

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 31

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
1 5 10 15

His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Ala
20 25 30

Leu Gly Ser Asp Leu His Val Asp Val Lys Gly Tyr Lys Asp Gly Gly
35 40 45

Ile His Leu Glu Cys Arg Ser Thr Gly Trp Tyr Pro Gln Pro Gln Ile
50 55 60

Gln Trp Ser Asn Asn Lys Gly Glu Asn Ile Pro Thr Val Glu Ala Pro
65 70 75 80

Val Val Ala Asp Gly Val Gly Leu Tyr Ala Val Ala Ala Ser Val Ile
85 90 95

Met Arg Gly Ser Ser Gly Glu Gly Val Ser Cys Thr Ile Arg Ser Ser
100 105 110

Leu Leu Gly Leu Glu Lys Thr Ala Ser Ile Ser Ile Ala Asp Pro Phe
115 120 125

Phe Arg Ser Ala Gln Arg Trp Ile Ala Ala Leu Ala Gly Thr Gly Gly
130 135 140

Gly Gly Ser Gly Gly Gly Gly Ser Leu Val Pro Arg Gly Ser Lys Pro
145 150 155 160

Ile Pro Asn Pro Leu Leu Gly Leu Asp Ser Thr Leu Val Pro Arg Gly
165 170 175

Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln Val Gln Leu Gln
180 185 190

Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu Thr Leu Ser Leu Thr
195 200 205

Cys Thr Val Ser Gly Gly Ser Ile Ser Tyr Tyr Ser Trp Thr Trp Val
210 215 220

Arg Gln Pro Ala Gly Gln Gly Leu Glu Trp Ile Gly Arg Ile His Ile
225 230 235 240

Ser Gly Val Thr Asn His Asn Pro Ser Leu Lys Ser Arg Val Ser Met
245 250 255

Ser Ile Asp Thr Ser Arg Thr Gln Phe Ser Leu Arg Leu Thr Ser Val
260 265 270

Thr Ala Ala Asp Thr Ala Val Tyr Phe Cys Ala Arg Ser Gly Gly Thr
275 280 285

Tyr Ser Ala Phe Asp Ile Trp Gly Gln Gly Thr Met Val Thr Val Ser
290 295 300

Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser

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305	310	315	320
Gly Gly Gly Gly Ser Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu	325	330	335
Ser Leu Ser Pro Gly Glu Arg Ser Thr Leu Ser Cys Arg Ala Ser Gln	340	345	350
Ser Phe Thr Ser Asn Tyr Leu Ala Trp Tyr Gln Gln Arg Pro Gly Gln	355	360	365
Ala Pro Arg Leu Leu Ile Tyr Gly Ala Ser Thr Arg Ala Ile Gly Ile	370	375	380
Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr	385	390	395
Ile Ser Arg Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln	405	410	415
Tyr Gly Ser Ser Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile	420	425	430
Lys Thr Thr Thr Pro Ala Pro Arg Pro Pro Thr Pro Ala Pro Thr Ile	435	440	445
Ala Ser Gln Pro Leu Ser Leu Arg Pro Glu Ala Cys Arg Pro Ala Ala	450	455	460
Gly Gly Ala Val His Thr Arg Gly Leu Asp Phe Ala Cys Asp Ile Tyr	465	470	475
Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly Val Leu Leu Leu Ser Leu	485	490	495
Val Ile Thr Leu Tyr Cys Lys Arg Gly Arg Lys Lys Leu Leu Tyr Ile	500	505	510
Phe Lys Gln Pro Phe Met Arg Pro Val Gln Thr Thr Gln Glu Glu Asp	515	520	525
Gly Cys Ser Cys Arg Phe Pro Glu Glu Glu Glu Gly Gly Cys Glu Leu	530	535	540
Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln Gly	545	550	555
Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr	565	570	575
Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly Lys	580	585	590
Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys	595	600	605
Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg	610	615	620
Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala	625	630	635
Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg	645	650	655

<210> SEQ ID NO 32

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 32

Leu Val Pro Arg Gly Ser

1

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<210> SEQ ID NO 33
<211> LENGTH: 42
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 33

Lys Arg Gly Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met
1 5 10 15
Arg Pro Val Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe
20 25 30
Pro Glu Glu Glu Glu Gly Gly Cys Glu Leu
35 40

<210> SEQ ID NO 34
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 34

Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln Gly
1 5 10 15
Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr
20 25 30
Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly Lys
35 40 45
Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys
50 55 60
Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg
65 70 75 80
Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala
85 90 95
Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg
100 105 110

<210> SEQ ID NO 35
<211> LENGTH: 41
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 35

Arg Ser Lys Arg Ser Arg Gly Gly His Ser Asp Tyr Met Asn Met Thr
1 5 10 15
Pro Arg Arg Pro Gly Pro Thr Arg Lys His Tyr Gln Pro Tyr Ala Pro
20 25 30
Pro Arg Asp Phe Ala Ala Tyr Arg Ser
35 40

<210> SEQ ID NO 36
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

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peptide

<400> SEQUENCE: 36

Gly Leu Ala Val Ser Thr Ile Ser Ser Phe Phe Pro Pro Gly Tyr Gln
 1 5 10 15

<210> SEQ ID NO 37

<211> LENGTH: 45

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 37

Thr Thr Thr Pro Ala Pro Arg Pro Pro Thr Pro Ala Pro Thr Ile Ala
 1 5 10 15

Ser Gln Pro Leu Ser Leu Arg Pro Glu Ala Cys Arg Pro Ala Ala Gly
 20 25 30

Gly Ala Val His Thr Arg Gly Leu Asp Phe Ala Cys Asp
 35 40 45

<210> SEQ ID NO 38

<211> LENGTH: 231

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 38

Glu Pro Lys Ser Pro Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala
 1 5 10 15

Pro Pro Val Ala Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys
 20 25 30

Asp Thr Leu Met Ile Ala Arg Thr Pro Glu Val Thr Cys Val Val Val
 35 40 45

Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp
 50 55 60

Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr
 65 70 75 80

Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp
 85 90 95

Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu
 100 105 110

Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg
 115 120 125

Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys
 130 135 140

Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp
 145 150 155 160

Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys
 165 170 175

Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser
 180 185 190

Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser
 195 200 205

Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser
 210 215 220

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Leu Ser Leu Ser Pro Gly Lys
225 230

<210> SEQ ID NO 39
<211> LENGTH: 24
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 39

Ile Tyr Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly Val Leu Leu Leu
1 5 10 15

Ser Leu Val Ile Thr Leu Tyr Cys
20

<210> SEQ ID NO 40
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 40

Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln Gly
1 5 10 15

Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr
20 25 30

Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly Lys
35 40 45

Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys
50 55 60

Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg
65 70 75 80

Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala
85 90 95

Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg
100 105 110

<210> SEQ ID NO 41
<211> LENGTH: 52
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 41

Phe Phe Ile Pro Leu Leu Val Val Ile Leu Phe Ala Val Asp Thr Gly
1 5 10 15

Leu Phe Ile Ser Thr Gln Gln Gln Val Thr Phe Leu Leu Lys Ile Lys
20 25 30

Arg Thr Arg Lys Gly Phe Arg Leu Leu Asn Pro His Pro Lys Pro Asn
35 40 45

Pro Lys Asn Asn
50

<210> SEQ ID NO 42
<211> LENGTH: 215

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                             polypeptide

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<400> SEQUENCE: 42

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Met Asp Thr Glu Ser Asn Arg Arg Ala Asn Leu Ala Leu Pro Gln Glu
 1             5             10             15

Pro Ser Ser Val Pro Ala Phe Glu Val Leu Glu Ile Ser Pro Gln Glu
      20             25             30

Val Ser Ser Gly Arg Leu Leu Lys Ser Ala Ser Ser Pro Pro Leu His
      35             40             45

Thr Trp Leu Thr Val Leu Lys Lys Glu Gln Glu Phe Leu Gly Val Thr
 50             55             60

Gln Ile Leu Thr Ala Met Ile Cys Leu Cys Phe Gly Thr Val Val Cys
65             70             75             80

Ser Val Leu Asp Ile Ser His Ile Glu Gly Asp Ile Phe Ser Ser Phe
      85             90             95

Lys Ala Gly Tyr Pro Phe Trp Gly Ala Ile Phe Phe Ser Ile Ser Gly
      100            105            110

Met Leu Ser Ile Ile Ser Glu Arg Arg Asn Ala Thr Tyr Leu Val Arg
      115            120            125

Gly Ser Leu Gly Ala Asn Thr Ala Ser Ser Ile Ala Gly Gly Thr Gly
      130            135            140

Ile Thr Ile Leu Ile Ile Asn Leu Lys Lys Ser Leu Ala Tyr Ile His
      145            150            155            160

Ile His Ser Cys Gln Lys Phe Phe Glu Thr Lys Cys Phe Met Ala Ser
      165            170            175

Phe Ser Thr Glu Ile Val Val Met Met Leu Phe Leu Thr Ile Leu Gly
      180            185            190

Leu Gly Ser Ala Val Ser Leu Thr Ile Cys Gly Ala Gly Glu Glu Leu
      195            200            205

Lys Gly Asn Lys Val Pro Glu
      210            215

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<210> SEQ ID NO 43
<211> LENGTH: 41
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                             polypeptide

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<400> SEQUENCE: 43

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Arg Ser Lys Arg Ser Arg Gly Gly His Ser Asp Tyr Met Asn Met Thr
 1             5             10             15

Pro Arg Arg Pro Gly Pro Thr Arg Lys His Tyr Gln Pro Tyr Ala Pro
      20             25             30

Pro Arg Asp Phe Ala Ala Tyr Arg Ser
      35             40

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<210> SEQ ID NO 44
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                             peptide

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-continued

<400> SEQUENCE: 44

Met Ile Pro Ala Val Val Leu Leu Leu Leu Leu Val Glu Gln Ala
1 5 10 15

Ala Ala

<210> SEQ ID NO 45

<211> LENGTH: 43

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 45

Leu Gly Glu Pro Gln Leu Cys Tyr Ile Leu Asp Ala Ile Leu Phe Leu
1 5 10 15

Tyr Gly Ile Val Leu Thr Leu Leu Tyr Cys Arg Leu Lys Ile Gln Val
20 25 30

Arg Lys Ala Ala Ile Thr Ser Tyr Glu Lys Ser
35 40

<210> SEQ ID NO 46

<211> LENGTH: 22

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 46

Gly Ser Gly Ala Thr Asn Phe Ser Leu Leu Lys Gln Ala Gly Asp Val
1 5 10 15

Glu Glu Asn Pro Gly Pro
20

<210> SEQ ID NO 47

<211> LENGTH: 21

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 47

Gly Ser Gly Glu Gly Arg Gly Ser Leu Leu Thr Cys Gly Asp Val Glu
1 5 10 15

Glu Asn Pro Gly Pro
20

<210> SEQ ID NO 48

<211> LENGTH: 136

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 48

Cys Pro Tyr Ser Asn Pro Ser Leu Cys Ser Gly Gly Gly Gly Ser Glu
1 5 10 15

Leu Pro Thr Gln Gly Thr Phe Ser Asn Val Ser Thr Asn Val Ser Pro
20 25 30

Ala Lys Pro Thr Thr Thr Ala Cys Pro Tyr Ser Asn Pro Ser Leu Cys

-continued

35	40	45
Ser Gly Gly Gly Gly Ser	Pro Ala Pro Arg Pro	Pro Thr Pro Ala Pro
50	55	60
Thr Ile Ala Ser Gln Pro	Leu Ser Leu Arg Pro	Glu Ala Cys Arg Pro
65	70	75 80
Ala Ala Gly Gly Ala Val His	Thr Arg Gly Leu Asp Phe	Ala Cys Asp
85	90	95
Ile Tyr Ile Trp Ala Pro	Leu Ala Gly Thr Cys Gly	Val Leu Leu Leu
100	105	110
Ser Leu Val Ile Thr Leu Tyr	Cys Asn His Arg Asn	Arg Arg Arg Val
115	120	125
Cys Lys Cys Pro Arg Pro	Val Val	
130	135	

<210> SEQ ID NO 49
 <211> LENGTH: 157
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 49

Met Gly Thr Ser Leu Leu Cys Trp Met	Ala Leu Cys Leu Leu Gly Ala
1	5 10 15
Asp His Ala Asp Ala Cys Pro Tyr Ser	Asn Pro Ser Leu Cys Ser Gly
20	25 30
Gly Gly Gly Ser Glu Leu Pro Thr Gln Gly Thr	Phe Ser Asn Val Ser
35	40 45
Thr Asn Val Ser Pro Ala Lys Pro Thr Thr Thr	Ala Cys Pro Tyr Ser
50	55 60
Asn Pro Ser Leu Cys Ser Gly Gly Gly Gly Ser	Pro Ala Pro Arg Pro
65	70 75 80
Pro Thr Pro Ala Pro Thr Ile Ala Ser Gln Pro	Leu Ser Leu Arg Pro
85	90 95
Glu Ala Cys Arg Pro Ala Ala Gly Gly Ala Val His	Thr Arg Gly Leu
100	105 110
Asp Phe Ala Cys Asp Ile Tyr Ile Trp Ala Pro	Leu Ala Gly Thr Cys
115	120 125
Gly Val Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr	Cys Asn His Arg
130	135 140
Asn Arg Arg Arg Val Cys Lys Cys Pro Arg Pro	Val Val
145	150 155

<210> SEQ ID NO 50
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 50

Cys Pro Tyr Ser Asn Pro Ser Leu Cys
1 5

<210> SEQ ID NO 51
 <211> LENGTH: 37
 <212> TYPE: PRT

-continued

<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 51

Gly Ser Gly Gly Gly Gly Ser Cys Pro Tyr Ser Asn Pro Ser Leu Cys
1 5 10 15

Ser Gly Gly Gly Gly Ser Cys Pro Tyr Ser Asn Pro Ser Leu Cys Ser
20 25 30

Gly Gly Gly Gly Ser
35

<210> SEQ ID NO 52
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 52

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly
1 5 10 15

<210> SEQ ID NO 53
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 53

Leu Ser Gly Arg Ser Asp Asn His Ser Pro Leu Gly Leu Ala Gly Ser
1 5 10 15

<210> SEQ ID NO 54
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 54

Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
1 5 10

<210> SEQ ID NO 55
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 55

Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu Asp Ser Thr
1 5 10

<210> SEQ ID NO 56
<211> LENGTH: 267
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

-continued

polypeptide

<400> SEQUENCE: 56

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15

His Ala Ala Arg Pro Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu
 20 25 30

Val Lys Pro Ser Glu Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly
 35 40 45

Ser Ile Ser Tyr Tyr Ser Trp Thr Trp Val Arg Gln Pro Ala Gly Gln
 50 55 60

Gly Leu Glu Trp Ile Gly Arg Ile His Ile Ser Gly Val Thr Asn His
 65 70 75 80

Asn Pro Ser Leu Lys Ser Arg Val Ser Met Ser Ile Asp Thr Ser Arg
 85 90 95

Thr Gln Phe Ser Leu Arg Leu Thr Ser Val Thr Ala Ala Asp Thr Ala
 100 105 110

Val Tyr Phe Cys Ala Arg Ser Gly Gly Thr Tyr Ser Ala Phe Asp Ile
 115 120 125

Trp Gly Gln Gly Thr Met Val Thr Val Ser Ser Gly Gly Gly Gly Ser
 130 135 140

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Glu
 145 150 155 160

Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly Glu
 165 170 175

Arg Ser Thr Leu Ser Cys Arg Ala Ser Gln Ser Phe Thr Ser Asn Tyr
 180 185 190

Leu Ala Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Arg Leu Leu Ile
 195 200 205

Tyr Gly Ala Ser Thr Arg Ala Ile Gly Ile Pro Asp Arg Phe Ser Gly
 210 215 220

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu Pro
 225 230 235 240

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro Trp
 245 250 255

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
 260 265

<210> SEQ ID NO 57
 <211> LENGTH: 15
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 57

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 1 5 10 15

<210> SEQ ID NO 58
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 58

-continued

Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu
 1 5 10 15
 Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Tyr Tyr
 20 25 30
 Ser Trp Thr Trp Val Arg Gln Pro Ala Gly Gln Gly Leu Glu Trp Ile
 35 40 45
 Gly Arg Ile His Ile Ser Gly Val Thr Asn His Asn Pro Ser Leu Lys
 50 55 60
 Ser Arg Val Ser Met Ser Ile Asp Thr Ser Arg Thr Gln Phe Ser Leu
 65 70 75 80
 Arg Leu Thr Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Phe Cys Ala
 85 90 95
 Arg Ser Gly Gly Thr Tyr Ser Ala Phe Asp Ile Trp Gly Gln Gly Thr
 100 105 110
 Met Val Thr Val Ser Ser
 115

<210> SEQ ID NO 59
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 59

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15
 Glu Arg Ser Thr Leu Ser Cys Arg Ala Ser Gln Ser Phe Thr Ser Asn
 20 25 30
 Tyr Leu Ala Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Arg Leu Leu
 35 40 45
 Ile Tyr Gly Ala Ser Thr Arg Ala Ile Gly Ile Pro Asp Arg Phe Ser
 50 55 60
 Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu
 65 70 75 80
 Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro
 85 90 95
 Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 60
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 60

Tyr Tyr Ser Trp Thr
 1 5

<210> SEQ ID NO 61
 <211> LENGTH: 16
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

-continued

<400> SEQUENCE: 61

Arg Ile His Ile Ser Gly Val Thr Asn His Asn Pro Ser Leu Lys Ser
1 5 10 15

<210> SEQ ID NO 62

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 62

Ser Gly Gly Thr Tyr Ser Ala Phe Asp Ile
1 5 10

<210> SEQ ID NO 63

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 63

Ser Gly Gly Ser Ile Ser Tyr Tyr
1 5

<210> SEQ ID NO 64

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 64

His Ile Ser Gly Val
1 5

<210> SEQ ID NO 65

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 65

Ser Gly Gly Thr Tyr Ser Ala Phe Asp Ile
1 5 10

<210> SEQ ID NO 66

<211> LENGTH: 12

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 66

Arg Ala Ser Gln Ser Phe Thr Ser Asn Tyr Leu Ala
1 5 10

<210> SEQ ID NO 67

<211> LENGTH: 7

-continued

<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 67

Gly Ala Ser Thr Arg Ala Ile
1 5

<210> SEQ ID NO 68
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 68

Gln Gln Tyr Gly Ser Ser Pro Trp Thr
1 5

<210> SEQ ID NO 69
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 69

Arg Ala Ser Gln Ser Phe Thr Ser Asn Tyr Leu Ala
1 5 10

<210> SEQ ID NO 70
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 70

Gly Ala Ser Thr Arg Ala Ile
1 5

<210> SEQ ID NO 71
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 71

Gln Gln Tyr Gly Ser Ser Pro Trp Thr
1 5

<210> SEQ ID NO 72
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 72

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly

-continued

1 5 10 15

Gly Gly Gly Ser
20

<210> SEQ ID NO 73
 <211> LENGTH: 66
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 73

Thr Thr Thr Pro Ala Pro Arg Pro Pro Thr Pro Ala Pro Thr Ile Ala
 1 5 10 15

Ser Gln Pro Leu Ser Leu Arg Pro Glu Ala Cys Arg Pro Ala Ala Gly
 20 25 30

Gly Ala Val His Thr Arg Gly Leu Asp Phe Ala Cys Asp Ile Tyr Ile
 35 40 45

Trp Ala Pro Leu Ala Gly Thr Cys Gly Val Leu Leu Leu Ser Leu Val
 50 55 60

Ile Thr
 65

<210> SEQ ID NO 74
 <211> LENGTH: 24
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 74

Glu Leu Pro Thr Gln Gly Thr Phe Ser Asn Val Ser Thr Asn Val Ser
 1 5 10 15

Pro Ala Lys Pro Thr Thr Thr Ala
 20

<210> SEQ ID NO 75
 <211> LENGTH: 490
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 75

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15

His Ala Ala Arg Pro Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu
 20 25 30

Val Lys Pro Ser Glu Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly
 35 40 45

Ser Ile Ser Tyr Tyr Ser Trp Thr Trp Val Arg Gln Pro Ala Gly Gln
 50 55 60

Gly Leu Glu Trp Ile Gly Arg Ile His Ile Ser Gly Val Thr Asn His
 65 70 75 80

Asn Pro Ser Leu Lys Ser Arg Val Ser Met Ser Ile Asp Thr Ser Arg
 85 90 95

Thr Gln Phe Ser Leu Arg Leu Thr Ser Val Thr Ala Ala Asp Thr Ala
 100 105 110

Val	Tyr	Phe 115	Cys	Ala	Arg	Ser	Gly 120	Gly	Thr	Tyr	Ser	Ala 125	Phe	Asp	Ile
Trp	Gly 130	Gln	Gly	Thr	Met	Val 135	Thr	Val	Ser	Ser	Gly 140	Gly	Gly	Gly	Ser
Gly 145	Gly	Gly	Gly	Ser	Gly 150	Gly	Gly	Gly	Ser	Gly 155	Gly	Gly	Gly	Ser	Glu 160
Ile	Val	Leu	Thr	Gln 165	Ser	Pro	Gly	Thr	Leu 170	Ser	Leu	Ser	Pro	Gly 175	Glu
Arg	Ser	Thr	Leu 180	Ser	Cys	Arg	Ala 185	Ser	Gln	Ser	Phe	Thr	Ser	Asn	Tyr
Leu	Ala	Trp 195	Tyr	Gln	Gln	Arg	Pro 200	Gly	Gln	Ala	Pro	Arg 205	Leu	Leu	Ile
Tyr	Gly 210	Ala	Ser	Thr	Arg	Ala 215	Ile	Gly	Ile	Pro	Asp 220	Arg	Phe	Ser	Gly
Ser 225	Gly	Ser	Gly	Thr	Asp 230	Phe	Thr	Leu	Thr	Ile 235	Ser	Arg	Leu	Glu	Pro 240
Glu	Asp	Phe	Ala	Val 245	Tyr	Tyr	Cys	Gln	Gln 250	Tyr	Gly	Ser	Ser	Pro 255	Trp
Thr	Phe	Gly	Gln 260	Gly	Thr	Lys	Val	Glu 265	Ile	Lys	Thr	Thr 270	Thr	Pro	Ala
Pro	Arg	Pro 275	Pro	Thr	Pro	Ala	Pro 280	Thr	Ile	Ala	Ser	Gln 285	Pro	Leu	Ser
Leu	Arg 290	Pro	Glu	Ala	Cys	Arg 295	Pro	Ala	Ala	Gly	Gly 300	Ala	Val	His	Thr
Arg 305	Gly	Leu	Asp	Phe	Ala 310	Cys	Asp	Ile	Tyr	Ile 315	Trp	Ala	Pro	Leu	Ala 320
Gly	Thr	Cys	Gly	Val 325	Leu	Leu	Leu	Ser	Leu 330	Val	Ile	Thr	Leu	Tyr 335	Cys
Lys	Arg	Gly	Arg 340	Lys	Lys	Leu	Leu	Tyr 345	Ile	Phe	Lys	Gln 350	Pro	Phe	Met
Arg	Pro	Val 355	Gln	Thr	Thr	Gln	Glu 360	Glu	Asp	Gly	Cys 365	Ser	Cys	Arg	Phe
Pro	Glu	Glu 370	Glu	Glu	Gly	Gly 375	Cys	Glu	Leu	Arg	Val 380	Lys	Phe	Ser	Arg
Ser 385	Ala	Asp	Ala	Pro	Ala 390	Tyr	Gln	Gln	Gly	Gln 395	Asn	Gln	Leu	Tyr	Asn 400
Glu	Leu	Asn	Leu 405	Gly	Arg	Arg	Glu	Glu	Tyr 410	Asp	Val	Leu	Asp	Lys 415	Arg
Arg	Gly	Arg 420	Pro	Glu	Met	Gly	Gly 425	Lys	Pro	Arg	Arg 430	Lys	Asn	Pro	
Gln	Glu 435	Gly	Leu	Tyr	Asn	Glu 440	Leu	Gln	Lys	Asp	Lys 445	Met	Ala	Glu	Ala
Tyr	Ser 450	Glu	Ile	Gly	Met	Lys 455	Gly	Glu	Arg	Arg 460	Arg	Gly	Lys	Gly	His
Asp 465	Gly	Leu	Tyr	Gln	Gly 470	Leu	Ser	Thr	Ala 475	Thr	Lys	Asp	Thr	Tyr	Asp 480
Ala	Leu	His	Met	Gln 485	Ala	Leu	Pro	Pro	Arg 490						

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<210> SEQ ID NO 76
<211> LENGTH: 527
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 76

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Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
1      5      10      15
His Ala Ala Arg Pro Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu
20     25     30
Val Lys Pro Ser Glu Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly
35     40     45
Ser Ile Ser Tyr Tyr Ser Trp Thr Trp Val Arg Gln Pro Ala Gly Gln
50     55     60
Gly Leu Glu Trp Ile Gly Arg Ile His Ile Ser Gly Val Thr Asn His
65     70     75     80
Asn Pro Ser Leu Lys Ser Arg Val Ser Met Ser Ile Asp Thr Ser Arg
85     90     95
Thr Gln Phe Ser Leu Arg Leu Thr Ser Val Thr Ala Ala Asp Thr Ala
100    105    110
Val Tyr Phe Cys Ala Arg Ser Gly Gly Thr Tyr Ser Ala Phe Asp Ile
115    120    125
Trp Gly Gln Gly Thr Met Val Thr Val Ser Ser Gly Gly Gly Gly Ser
130    135    140
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Glu
145    150    155    160
Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly Glu
165    170    175
Arg Ser Thr Leu Ser Cys Arg Ala Ser Gln Ser Phe Thr Ser Asn Tyr
180    185    190
Leu Ala Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Arg Leu Leu Ile
195    200    205
Tyr Gly Ala Ser Thr Arg Ala Ile Gly Ile Pro Asp Arg Phe Ser Gly
210    215    220
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu Pro
225    230    235    240
Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro Trp
245    250    255
Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Gly Ser Gly Gly Gly
260    265    270
Gly Ser Cys Pro Tyr Ser Asn Pro Ser Leu Cys Ser Gly Gly Gly Gly
275    280    285
Ser Cys Pro Tyr Ser Asn Pro Ser Leu Cys Ser Gly Gly Gly Gly Ser
290    295    300
Thr Thr Thr Pro Ala Pro Arg Pro Pro Thr Pro Ala Pro Thr Ile Ala
305    310    315    320
Ser Gln Pro Leu Ser Leu Arg Pro Glu Ala Cys Arg Pro Ala Ala Gly
325    330    335
Gly Ala Val His Thr Arg Gly Leu Asp Phe Ala Cys Asp Ile Tyr Ile
340    345    350
Trp Ala Pro Leu Ala Gly Thr Cys Gly Val Leu Leu Leu Ser Leu Val
355    360    365
Ile Thr Leu Tyr Cys Lys Arg Gly Arg Lys Lys Leu Leu Tyr Ile Phe
370    375    380
Lys Gln Pro Phe Met Arg Pro Val Gln Thr Thr Gln Glu Glu Asp Gly
385    390    395    400

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Cys Ser Cys Arg Phe Pro Glu Glu Glu Glu Gly Gly Cys Glu Leu Arg
405 410 415

Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln Gly Gln
420 425 430

Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp
435 440 445

Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly Lys Pro
450 455 460

Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp
465 470 475 480

Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg Arg
485 490 495

Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr
500 505 510

Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg
515 520 525

<210> SEQ ID NO 77

<211> LENGTH: 523

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 77

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
1 5 10 15

His Ala Ala Arg Pro Gly Gly Gly Gly Ser Cys Pro Tyr Ser Asn Pro
20 25 30

Ser Leu Cys Gly Gly Gly Gly Ser Gln Val Gln Leu Gln Glu Ser Gly
35 40 45

Pro Gly Leu Val Lys Pro Ser Glu Thr Leu Ser Leu Thr Cys Thr Val
50 55 60

Ser Gly Gly Ser Ile Ser Tyr Tyr Ser Trp Thr Trp Val Arg Gln Pro
65 70 75 80

Ala Gly Gln Gly Leu Glu Trp Ile Gly Arg Ile His Ile Ser Gly Val
85 90 95

Thr Asn His Asn Pro Ser Leu Lys Ser Arg Val Ser Met Ser Ile Asp
100 105 110

Thr Ser Arg Thr Gln Phe Ser Leu Arg Leu Thr Ser Val Thr Ala Ala
115 120 125

Asp Thr Ala Val Tyr Phe Cys Ala Arg Ser Gly Gly Thr Tyr Ser Ala
130 135 140

Phe Asp Ile Trp Gly Gln Gly Thr Met Val Thr Val Ser Ser Gly Gly
145 150 155 160

Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
165 170 175

Gly Ser Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser
180 185 190

Pro Gly Glu Arg Ser Thr Leu Ser Cys Arg Ala Ser Gln Ser Phe Thr
195 200 205

Ser Asn Tyr Leu Ala Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Arg
210 215 220

Leu Leu Ile Tyr Gly Ala Ser Thr Arg Ala Ile Gly Ile Pro Asp Arg

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225	230	235	240
Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg			
	245	250	255
Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser			
	260	265	270
Ser Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Gly Gly			
	275	280	285
Gly Gly Ser Cys Pro Tyr Ser Asn Pro Ser Leu Cys Thr Thr Thr Pro			
	290	295	300
Ala Pro Arg Pro Pro Thr Pro Ala Pro Thr Ile Ala Ser Gln Pro Leu			
305	310	315	320
Ser Leu Arg Pro Glu Ala Cys Arg Pro Ala Ala Gly Gly Ala Val His			
	325	330	335
Thr Arg Gly Leu Asp Phe Ala Cys Asp Ile Tyr Ile Trp Ala Pro Leu			
	340	345	350
Ala Gly Thr Cys Gly Val Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr			
	355	360	365
Cys Lys Arg Gly Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe			
	370	375	380
Met Arg Pro Val Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg			
385	390	395	400
Phe Pro Glu Glu Glu Glu Gly Gly Cys Glu Leu Arg Val Lys Phe Ser			
	405	410	415
Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr			
	420	425	430
Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys			
	435	440	445
Arg Arg Gly Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn			
	450	455	460
Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu			
465	470	475	480
Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly			
	485	490	495
His Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr			
	500	505	510
Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg			
	515	520	

<210> SEQ ID NO 78

<211> LENGTH: 523

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 78

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu			
1	5	10	15
His Ala Ala Arg Pro Gly Gly Gly Gly Ser Cys Pro Tyr Ser Asn Pro			
	20	25	30
Ser Leu Cys Gly Gly Gly Gly Ser Cys Pro Tyr Ser Asn Pro Ser Leu			
	35	40	45
Cys Gly Gly Gly Gly Ser Gln Val Gln Leu Gln Glu Ser Gly Pro Gly			
50	55	60	

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Leu Val Lys Pro Ser Glu Thr Leu Ser Leu Thr Cys Thr Val Ser Gly	65	70	75	80
Gly Ser Ile Ser Tyr Tyr Ser Trp Thr Trp Val Arg Gln Pro Ala Gly		85	90	95
Gln Gly Leu Glu Trp Ile Gly Arg Ile His Ile Ser Gly Val Thr Asn		100	105	110
His Asn Pro Ser Leu Lys Ser Arg Val Ser Met Ser Ile Asp Thr Ser		115	120	125
Arg Thr Gln Phe Ser Leu Arg Leu Thr Ser Val Thr Ala Ala Asp Thr		130	135	140
Ala Val Tyr Phe Cys Ala Arg Ser Gly Gly Thr Tyr Ser Ala Phe Asp		145	150	155
Ile Trp Gly Gln Gly Thr Met Val Thr Val Ser Ser Gly Gly Gly Gly		165	170	175
Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser		180	185	190
Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly		195	200	205
Glu Arg Ser Thr Leu Ser Cys Arg Ala Ser Gln Ser Phe Thr Ser Asn		210	215	220
Tyr Leu Ala Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Arg Leu Leu		225	230	235
Ile Tyr Gly Ala Ser Thr Arg Ala Ile Gly Ile Pro Asp Arg Phe Ser		245	250	255
Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu		260	265	270
Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro		275	280	285
Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Thr Thr Thr Pro		290	295	300
Ala Pro Arg Pro Pro Thr Pro Ala Pro Thr Ile Ala Ser Gln Pro Leu		305	310	315
Ser Leu Arg Pro Glu Ala Cys Arg Pro Ala Ala Gly Gly Ala Val His		325	330	335
Thr Arg Gly Leu Asp Phe Ala Cys Asp Ile Tyr Ile Trp Ala Pro Leu		340	345	350
Ala Gly Thr Cys Gly Val Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr		355	360	365
Cys Lys Arg Gly Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe		370	375	380
Met Arg Pro Val Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg		385	390	395
Phe Pro Glu Glu Glu Glu Gly Gly Cys Glu Leu Arg Val Lys Phe Ser		405	410	415
Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr		420	425	430
Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys		435	440	445
Arg Arg Gly Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn		450	455	460
Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu		465	470	475
Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly				480

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	485		490		495
His Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr					
	500		505		510
Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg					
	515		520		

<210> SEQ ID NO 79

<211> LENGTH: 571

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 79

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu					
1	5		10		15
His Ala Ala Arg Pro Gly Gly Gly Gly Ser Cys Pro Tyr Ser Asn Pro					
	20		25		30
Ser Leu Cys Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln Val					
	35		40		45
Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu Thr Leu					
	50		55		60
Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Tyr Tyr Ser Trp					
65	70		75		80
Thr Trp Val Arg Gln Pro Ala Gly Gln Gly Leu Glu Trp Ile Gly Arg					
	85		90		95
Ile His Ile Ser Gly Val Thr Asn His Asn Pro Ser Leu Lys Ser Arg					
	100		105		110
Val Ser Met Ser Ile Asp Thr Ser Arg Thr Gln Phe Ser Leu Arg Leu					
	115		120		125
Thr Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Phe Cys Ala Arg Ser					
	130		135		140
Gly Gly Thr Tyr Ser Ala Phe Asp Ile Trp Gly Gln Gly Thr Met Val					
	145		150		155
Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly					
	165		170		175
Gly Gly Ser Gly Gly Gly Gly Ser Glu Ile Val Leu Thr Gln Ser Pro					
	180		185		190
Gly Thr Leu Ser Leu Ser Pro Gly Glu Arg Ser Thr Leu Ser Cys Arg					
	195		200		205
Ala Ser Gln Ser Phe Thr Ser Asn Tyr Leu Ala Trp Tyr Gln Gln Arg					
	210		215		220
Pro Gly Gln Ala Pro Arg Leu Leu Ile Tyr Gly Ala Ser Thr Arg Ala					
	225		230		235
Ile Gly Ile Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe					
	245		250		255
Thr Leu Thr Ile Ser Arg Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr					
	260		265		270
Cys Gln Gln Tyr Gly Ser Ser Pro Trp Thr Phe Gly Gln Gly Thr Lys					
	275		280		285
Val Glu Ile Lys Gly Ser Gly Gly Gly Gly Ser Cys Pro Tyr Ser Asn					
	290		295		300
Pro Ser Leu Cys Ser Gly Gly Gly Gly Ser Glu Leu Pro Thr Gln Gly					
	305		310		315
					320

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Thr Phe Ser Asn Val Ser Thr Asn Val Ser Pro Ala Lys Pro Thr Thr
 325 330 335
 Thr Ala Cys Pro Tyr Ser Asn Pro Ser Leu Cys Thr Thr Thr Pro Ala
 340 345 350
 Pro Arg Pro Pro Thr Pro Ala Pro Thr Ile Ala Ser Gln Pro Leu Ser
 355 360 365
 Leu Arg Pro Glu Ala Cys Arg Pro Ala Ala Gly Gly Ala Val His Thr
 370 375 380
 Arg Gly Leu Asp Phe Ala Cys Asp Ile Tyr Ile Trp Ala Pro Leu Ala
 385 390 395 400
 Gly Thr Cys Gly Val Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr Cys
 405 410 415
 Lys Arg Gly Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met
 420 425 430
 Arg Pro Val Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe
 435 440 445
 Pro Glu Glu Glu Glu Gly Gly Cys Glu Leu Leu Arg Val Lys Phe Ser
 450 455 460
 Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr
 465 470 475 480
 Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys
 485 490 495
 Arg Arg Gly Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn
 500 505 510
 Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu
 515 520 525
 Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly
 530 535 540
 His Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr
 545 550 555 560
 Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg
 565 570

<210> SEQ ID NO 80
 <211> LENGTH: 24
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 80

Asn Ser Glu Leu Leu Ser Leu Ile Asn Asp Met Pro Ile Thr Asn Asp
 1 5 10 15
 Gln Lys Lys Leu Met Ser Asn Asn
 20

<210> SEQ ID NO 81
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 81

Cys Gln Phe Asp Leu Ser Thr Arg Arg Leu Lys Cys
 1 5 10

-continued

<210> SEQ ID NO 82
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 82

Cys Gln Tyr Asn Leu Ser Ser Arg Ala Leu Lys Cys
1 5 10

<210> SEQ ID NO 83
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 83

Cys Val Trp Gln Arg Trp Gln Lys Ser Tyr Val Cys
1 5 10

<210> SEQ ID NO 84
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 84

Cys Met Trp Asp Arg Phe Ser Arg Trp Tyr Lys Cys
1 5 10

<210> SEQ ID NO 85
<211> LENGTH: 25
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 85

Ser Phe Val Leu Asn Trp Tyr Arg Met Ser Pro Ser Asn Gln Thr Asp
1 5 10 15
Lys Leu Ala Ala Phe Pro Glu Asp Arg
20 25

<210> SEQ ID NO 86
<211> LENGTH: 19
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 86

Ser Gly Thr Tyr Leu Cys Gly Ala Ile Ser Leu Ala Pro Lys Ala Gln
1 5 10 15

Ile Lys Glu

<210> SEQ ID NO 87
<211> LENGTH: 24
<212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 87

Glu Leu Pro Thr Gln Gly Thr Phe Ser Asn Val Ser Thr Asn Val Ser
1 5 10 15

Pro Ala Lys Pro Thr Thr Thr Ala
20

<210> SEQ ID NO 88
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 88

Gly Gln Asn Asp Thr Ser Gln Thr Ser Ser Pro Ser
1 5 10

<210> SEQ ID NO 89
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 89

Val His Met Pro Leu Gly Phe Leu Gly Pro Arg Gln Ala Arg Val Val
1 5 10 15

Asn Gly

<210> SEQ ID NO 90
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 90

Val His Met Pro Leu Gly Phe Leu Gly Pro Gly Ser Ala Arg Val Val
1 5 10 15

Asn Gly

<210> SEQ ID NO 91
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 91

Arg Gln Ala Arg Val Val Asn Gly
1 5

<210> SEQ ID NO 92
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 92

Arg Gln Ala Arg Val Gly Ser Gly
1 5

<210> SEQ ID NO 93

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 93

Val His Met Pro Leu Gly Phe Leu Gly Pro
1 5 10

<210> SEQ ID NO 94

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 94

Val His Met Pro Leu Ser Phe Leu Gly Pro
1 5 10

<210> SEQ ID NO 95

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 95

Pro Met Ala Lys Lys
1 5

<210> SEQ ID NO 96

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 96

Pro Met Ala Lys Gly Ser
1 5

<210> SEQ ID NO 97

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 97

Leu Ser Gly Arg Ser Asp Asn His
1 5

-continued

<210> SEQ ID NO 98
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 98

Leu Ser Gly Arg Ser Asp Ser His
1 5

<210> SEQ ID NO 99
<211> LENGTH: 25
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 99

Gly Gly Gly Gly Ser Val His Met Pro Leu Gly Phe Leu Gly Pro Arg
1 5 10 15

Gln Ala Arg Val Val Asn Gly Gly Ser
20 25

<210> SEQ ID NO 100
<211> LENGTH: 25
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 100

Gly Gly Gly Gly Ser Val His Met Pro Leu Gly Phe Leu Gly Pro Gly
1 5 10 15

Ser Ala Arg Val Val Asn Gly Gly Ser
20 25

<210> SEQ ID NO 101
<211> LENGTH: 25
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 101

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Pro Met Ala Lys Lys Gly
1 5 10 15

Gly Gly Gly Ser Gly Gly Gly Gly Ser
20 25

<210> SEQ ID NO 102
<211> LENGTH: 25
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 102

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Pro Met Ala Lys Gly Ser
1 5 10 15

Gly Gly Gly Ser Gly Gly Gly Gly Ser

-continued

20

25

<210> SEQ ID NO 103
 <211> LENGTH: 8
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 103

Ser Pro Leu Gly Leu Ala Gly Ser
 1 5

<210> SEQ ID NO 104
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 104

Pro Leu Gly Val Arg Gly Lys
 1 5

<210> SEQ ID NO 105
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 105

Pro Met Ala Lys Gly
 1 5

<210> SEQ ID NO 106
 <211> LENGTH: 8
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 106

Gly Ser Ala Arg Val Val Asn Gly
 1 5

<210> SEQ ID NO 107
 <211> LENGTH: 45
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 107

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Val
 1 5 10 15

His Met Pro Leu Gly Phe Leu Gly Pro Arg Gln Ala Arg Val Val Asn
 20 25 30

Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 35 40 45

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<210> SEQ ID NO 108
<211> LENGTH: 45
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 108

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Val
1 5 10 15

His Met Pro Leu Gly Phe Leu Gly Pro Gly Ser Ala Arg Val Val Asn
20 25 30

Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
35 40 45

<210> SEQ ID NO 109
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 109

Gly Gly Gly Gly Ser Pro Met Ala Lys Lys Gly Gly Gly Gly Ser Gly
1 5 10 15

Gly Gly Gly Ser
20

<210> SEQ ID NO 110
<211> LENGTH: 25
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 110

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Pro Met Ala Lys Lys Gly
1 5 10 15

Gly Gly Gly Ser Gly Gly Gly Gly Ser
20 25

<210> SEQ ID NO 111
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 111

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Pro Met Ala Lys Lys Gly
1 5 10 15

Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
20 25 30

<210> SEQ ID NO 112
<211> LENGTH: 35
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

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<400> SEQUENCE: 112

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Pro
 1 5 10 15
 Met Ala Lys Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly
 20 25 30
 Gly Gly Ser
 35

<210> SEQ ID NO 113

<211> LENGTH: 45

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 113

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly
 1 5 10 15
 Gly Gly Gly Ser Pro Met Ala Lys Lys Gly Gly Gly Gly Ser Gly Gly
 20 25 30
 Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 35 40 45

<210> SEQ ID NO 114

<211> LENGTH: 45

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 114

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly
 1 5 10 15
 Gly Gly Gly Ser Pro Met Ala Lys Gly Ser Gly Gly Gly Ser Gly Gly
 20 25 30
 Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 35 40 45

<210> SEQ ID NO 115

<211> LENGTH: 45

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 115

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Ser Ser Pro Leu Gly
 1 5 10 15
 Leu Ala Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Pro Met
 20 25 30
 Ala Lys Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 35 40 45

<210> SEQ ID NO 116

<211> LENGTH: 45

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

-continued

<400> SEQUENCE: 116

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Ser Val His Met
 1 5 10 15

Pro Leu Gly Phe Leu Gly Pro Gly Gly Ser Gly Gly Gly Ser Pro Met
 20 25 30

Ala Lys Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 35 40 45

<210> SEQ ID NO 117

<211> LENGTH: 45

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 117

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Ser Pro Leu Gly
 1 5 10 15

Val Arg Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Pro Met
 20 25 30

Ala Lys Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 35 40 45

<210> SEQ ID NO 118

<211> LENGTH: 45

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 118

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Ser Ser Pro Leu
 1 5 10 15

Gly Leu Ala Gly Ser Gly Ser Gly Gly Gly Ser Arg Gln Ala Arg Val
 20 25 30

Val Asn Gly Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 35 40 45

<210> SEQ ID NO 119

<211> LENGTH: 45

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 119

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Ser Val His Met
 1 5 10 15

Pro Leu Gly Phe Leu Gly Pro Gly Gly Gly Ser Arg Gln Ala Arg Val
 20 25 30

Val Asn Gly Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 35 40 45

<210> SEQ ID NO 120

<211> LENGTH: 45

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

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polypeptide

<400> SEQUENCE: 120

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Ser Pro Leu Gly
 1 5 10 15
 Val Arg Gly Lys Gly Gly Ser Gly Gly Gly Ser Arg Gln Ala Arg Val
 20 25 30
 Val Asn Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 35 40 45

<210> SEQ ID NO 121

<211> LENGTH: 793

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 121

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15
 His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp
 20 25 30
 Ile Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly Gly
 35 40 45
 Ser Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Phe Tyr Asn Leu Leu
 50 55 60
 Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr
 65 70 75 80
 Asp Ala Ser Asp Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly Ser
 85 90 95
 Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala
 100 105 110
 Asp Ala Ala Ala Tyr Tyr Cys Gln Ser Ala Asp Gly Ser Ser Tyr Ala
 115 120 125
 Phe Gly Gly Gly Thr Glu Val Val Val Lys Gly Gly Gly Gly Gly Ser
 130 135 140
 Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln
 145 150 155 160
 Glu Gln Leu Glu Glu Ser Gly Gly Gly Leu Val Lys Pro Glu Gly Ser
 165 170 175
 Leu Thr Leu Thr Cys Thr Ala Ser Gly Val Ser Phe Ser Ser Ser Tyr
 180 185 190
 Trp Ile Tyr Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile
 195 200 205
 Ala Cys Ile Tyr Thr Gly Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp
 210 215 220
 Ala Lys Gly Arg Phe Thr Val Ser Glu Thr Ser Ser Thr Thr Val Thr
 225 230 235 240
 Leu Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys
 245 250 255
 Ala Arg Ala Ser Ala Trp Thr Tyr Gly Met Asp Leu Trp Gly Pro Gly
 260 265 270
 Thr Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly
 275 280 285
 Ser Gly Gly Gly Gly Ser Val His Met Pro Leu Gly Phe Leu Gly Pro

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290					295					300					
Arg 305	Gln	Ala	Arg	Val	Val 310	Asn	Gly	Gly	Ser	Gly 315	Gly	Gly	Gly	Ser	Gly 320
Gly	Gly	Gly	Ser	Gln 325	Val	Gln	Leu	Gln	Glu 330	Ser	Gly	Pro	Gly	Leu 335	Val
Lys	Pro	Ser	Glu 340	Thr	Leu	Ser	Leu	Thr 345	Cys	Thr	Val	Ser	Gly 350	Gly	Ser
Ile	Ser	Tyr 355	Tyr	Ser	Trp	Thr 360	Trp	Val	Arg	Gln	Pro	Ala 365	Gly	Gln	Gly
Leu	Glu 370	Trp	Ile	Gly	Arg	Ile 375	His	Ile	Ser	Gly	Val 380	Thr	Asn	His	Asn
Pro 385	Ser	Leu	Lys	Ser	Arg 390	Val	Ser	Met	Ser	Ile 395	Asp	Thr	Ser	Arg	Thr 400
Gln	Phe	Ser	Leu	Arg 405	Leu	Thr	Ser	Val	Thr 410	Ala	Ala	Asp	Thr	Ala 415	Val
Tyr	Phe	Cys	Ala 420	Arg	Ser	Gly	Gly	Thr 425	Tyr	Ser	Ala	Phe	Asp 430	Ile	Trp
Gly	Gln	Gly	Thr 435	Met	Val	Thr	Val 440	Ser	Ser	Gly	Gly	Gly 445	Gly	Ser	Gly
Gly	Gly	Gly	Ser 450	Gly	Gly	Gly 455	Gly	Ser	Gly	Gly	Gly 460	Gly	Ser	Glu	Ile
Val 465	Leu	Thr	Gln	Ser	Pro 470	Gly	Thr	Leu	Ser	Leu 475	Ser	Pro	Gly	Glu	Arg 480
Ser	Thr	Leu	Ser	Cys 485	Arg	Ala	Ser	Gln	Ser 490	Phe	Thr	Ser	Asn	Tyr 495	Leu
Ala	Trp	Tyr	Gln 500	Gln	Arg	Pro	Gly	Gln 505	Ala	Pro	Arg	Leu 510	Leu	Ile	Tyr
Gly	Ala	Ser	Thr 515	Arg	Ala	Ile	Gly 520	Ile	Pro	Asp	Arg	Phe 525	Ser	Gly	Ser
Gly	Ser	Gly	Thr 530	Asp	Phe	Thr 535	Leu	Thr	Ile	Ser	Arg 540	Leu	Glu	Pro	Glu
Asp 545	Phe	Ala	Val	Tyr	Tyr 550	Cys	Gln	Gln	Tyr	Gly 555	Ser	Ser	Pro	Trp	Thr 560
Phe	Gly	Gln	Gly	Thr 565	Lys	Val	Glu	Ile	Lys 570	Thr	Thr	Thr	Pro	Ala 575	Pro
Arg	Pro	Pro	Thr 580	Pro	Ala	Pro	Thr	Ile 585	Ala	Ser	Gln	Pro	Leu 590	Ser	Leu
Arg	Pro	Glu	Ala 595	Cys	Arg	Pro	Ala 600	Ala	Gly	Gly	Ala 605	Val	His	Thr	Arg
Gly	Leu	Asp	Phe 610	Ala	Cys	Asp 615	Ile	Tyr	Ile	Trp	Ala 620	Pro	Leu	Ala	Gly
Thr 625	Cys	Gly	Val	Leu	Leu 630	Leu	Ser	Leu	Val	Ile 635	Thr	Leu	Tyr	Cys	Lys 640
Arg	Gly	Arg	Lys 645	Lys	Leu	Leu	Tyr	Ile	Phe 650	Lys	Gln	Pro	Phe 655	Met	Arg
Pro	Val	Gln	Thr 660	Thr	Gln	Glu	Glu	Asp 665	Gly	Cys	Ser	Cys 670	Arg	Phe	Pro
Glu	Glu	Glu	Glu 675	Gly	Gly	Cys	Glu 680	Leu	Arg	Val	Lys	Phe 685	Ser	Arg	Ser
Ala	Asp	Ala	Pro 690	Ala	Tyr	Gln 695	Gln	Gly	Gln	Asn	Gln 700	Leu	Tyr	Asn	Glu
Leu 705	Asn	Leu	Gly	Arg	Arg 710	Glu	Glu	Tyr	Asp	Val 715	Leu	Asp	Lys	Arg	Arg 720

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Gly Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln
725 730 735

Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr
740 745 750

Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp
755 760 765

Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala
770 775 780

Leu His Met Gln Ala Leu Pro Pro Arg
785 790

<210> SEQ ID NO 122
<211> LENGTH: 793
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 122

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
1 5 10 15

His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp
20 25 30

Ile Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly Gly
35 40 45

Ser Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Phe Tyr Asn Leu Leu
50 55 60

Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr
65 70 75 80

Asp Ala Ser Asp Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly Ser
85 90 95

Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala
100 105 110

Asp Ala Ala Ala Tyr Tyr Cys Gln Ser Ala Asp Gly Ser Ser Tyr Ala
115 120 125

Phe Gly Gly Gly Thr Glu Val Val Val Lys Gly Gly Gly Gly Gly Ser
130 135 140

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln
145 150 155 160

Glu Gln Leu Glu Glu Ser Gly Gly Gly Leu Val Lys Pro Glu Gly Ser
165 170 175

Leu Thr Leu Thr Cys Thr Ala Ser Gly Val Ser Phe Ser Ser Ser Tyr
180 185 190

Trp Ile Tyr Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile
195 200 205

Ala Cys Ile Tyr Thr Gly Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp
210 215 220

Ala Lys Gly Arg Phe Thr Val Ser Glu Thr Ser Ser Thr Thr Val Thr
225 230 235 240

Leu Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys
245 250 255

Ala Arg Ala Ser Ala Trp Thr Tyr Gly Met Asp Leu Trp Gly Pro Gly
260 265 270

Thr Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly

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275					280					285					
Ser	Gly	Gly	Gly	Gly	Ser	Val	His	Met	Pro	Leu	Gly	Phe	Leu	Gly	Pro
290					295					300					
Gly	Ser	Ala	Arg	Val	Val	Asn	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly
305					310					315					320
Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val
				325					330					335	
Lys	Pro	Ser	Glu	Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser
			340					345					350		
Ile	Ser	Tyr	Tyr	Ser	Trp	Thr	Trp	Val	Arg	Gln	Pro	Ala	Gly	Gln	Gly
		355					360					365			
Leu	Glu	Trp	Ile	Gly	Arg	Ile	His	Ile	Ser	Gly	Val	Thr	Asn	His	Asn
370						375					380				
Pro	Ser	Leu	Lys	Ser	Arg	Val	Ser	Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr
385					390					395					400
Gln	Phe	Ser	Leu	Arg	Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val
				405					410					415	
Tyr	Phe	Cys	Ala	Arg	Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp
			420					425					430		
Gly	Gln	Gly	Thr	Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly
		435					440					445			
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile
450						455					460				
Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg
465					470					475					480
Ser	Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu
				485					490					495	
Ala	Trp	Tyr	Gln	Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr
			500					505					510		
Gly	Ala	Ser	Thr	Arg	Ala	Ile	Gly	Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser
		515					520					525			
Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu
530						535					540				
Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr
545					550					555					560
Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile	Lys	Thr	Thr	Thr	Pro	Ala	Pro
				565					570					575	
Arg	Pro	Pro	Thr	Pro	Ala	Pro	Thr	Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu
			580					585					590		
Arg	Pro	Glu	Ala	Cys	Arg	Pro	Ala	Ala	Gly	Gly	Ala	Val	His	Thr	Arg
		595					600					605			
Gly	Leu	Asp	Phe	Ala	Cys	Asp	Ile	Tyr	Ile	Trp	Ala	Pro	Leu	Ala	Gly
610					615					620					
Thr	Cys	Gly	Val	Leu	Leu	Leu	Ser	Leu	Val	Ile	Thr	Leu	Tyr	Cys	Lys
625				630						635					640
Arg	Gly	Arg	Lys	Lys	Leu	Leu	Tyr	Ile	Phe	Lys	Gln	Pro	Phe	Met	Arg
				645					650					655	
Pro	Val	Gln	Thr	Thr	Gln	Glu	Glu	Asp	Gly	Cys	Ser	Cys	Arg	Phe	Pro
				660				665					670		
Glu	Glu	Glu	Glu	Gly	Gly	Cys	Glu	Leu	Arg	Val	Lys	Phe	Ser	Arg	Ser
		675					680					685			
Ala	Asp	Ala	Pro	Ala	Tyr	Gln	Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu
690						695					700				

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Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg
 705 710 715 720
 Gly Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln
 725 730 735
 Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr
 740 745 750
 Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp
 755 760 765
 Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala
 770 775 780
 Leu His Met Gln Ala Leu Pro Pro Arg
 785 790

<210> SEQ ID NO 123
 <211> LENGTH: 793
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 123

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15
 His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp
 20 25 30
 Ile Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly Gly
 35 40 45
 Ser Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Phe Tyr Asn Leu Leu
 50 55 60
 Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr
 65 70 75 80
 Asp Ala Ser Asp Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly Ser
 85 90 95
 Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala
 100 105 110
 Asp Ala Ala Ala Tyr Tyr Cys Gln Ser Ala Asp Gly Ser Ser Tyr Ala
 115 120 125
 Phe Gly Gly Gly Thr Glu Val Val Val Lys Gly Gly Gly Gly Gly Ser
 130 135 140
 Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln
 145 150 155 160
 Glu Gln Leu Glu Glu Ser Gly Gly Gly Leu Val Lys Pro Glu Gly Ser
 165 170 175
 Leu Thr Leu Thr Cys Thr Ala Ser Gly Val Ser Phe Ser Ser Ser Tyr
 180 185 190
 Trp Ile Tyr Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile
 195 200 205
 Ala Cys Ile Tyr Thr Gly Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp
 210 215 220
 Ala Lys Gly Arg Phe Thr Val Ser Glu Thr Ser Ser Thr Thr Val Thr
 225 230 235 240
 Leu Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys
 245 250 255
 Ala Arg Ala Ser Ala Trp Thr Tyr Gly Met Asp Leu Trp Gly Pro Gly

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260					265					270				
Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly
	275						280					285		
Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Pro	Met	Ala	Lys
	290					295					300			Lys
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser
305					310					315				320
Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu
			325					330						335
Lys	Pro	Ser	Glu	Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Gly
			340					345					350	Ser
Ile	Ser	Tyr	Tyr	Ser	Trp	Thr	Trp	Val	Arg	Gln	Pro	Ala	Gly	Gln
	355						360					365		Gly
Leu	Glu	Trp	Ile	Gly	Arg	Ile	His	Ile	Ser	Gly	Val	Thr	Asn	His
370						375					380		Asn	
Pro	Ser	Leu	Lys	Ser	Arg	Val	Ser	Met	Ser	Ile	Asp	Thr	Ser	Arg
385					390					395				400
Gln	Phe	Ser	Leu	Arg	Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala
			405						410					415
Tyr	Phe	Cys	Ala	Arg	Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe	Asp	Ile
	420							425					430	Trp
Gly	Gln	Gly	Thr	Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser
	435						440					445		Gly
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu
450						455					460			Ile
Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu
465					470					475				480
Ser	Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser	Asn	Tyr
			485					490						495
Ala	Trp	Tyr	Gln	Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile
	500							505						510
Gly	Ala	Ser	Thr	Arg	Ala	Ile	Gly	Ile	Pro	Asp	Arg	Phe	Ser	Gly
	515						520					525		Ser
Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Arg	Leu	Glu	Pro
	530					535					540			Glu
Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Tyr	Gly	Ser	Ser	Pro	Trp
545					550					555				560
Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile	Lys	Thr	Thr	Thr	Pro	Ala
			565					570						575
Arg	Pro	Pro	Thr	Pro	Ala	Pro	Thr	Ile	Ala	Ser	Gln	Pro	Leu	Ser
			580					585						590
Arg	Pro	Glu	Ala	Cys	Arg	Pro	Ala	Ala	Gly	Gly	Ala	Val	His	Thr
	595						600					605		Arg
Gly	Leu	Asp	Phe	Ala	Cys	Asp	Ile	Tyr	Ile	Trp	Ala	Pro	Leu	Ala
	610					615					620			Gly
Thr	Cys	Gly	Val	Leu	Leu	Leu	Ser	Leu	Val	Ile	Thr	Leu	Tyr	Cys
625					630					635				640
Arg	Gly	Arg	Lys	Lys	Leu	Leu	Tyr	Ile	Phe	Lys	Gln	Pro	Phe	Met
			645						650					655
Pro	Val	Gln	Thr	Thr	Gln	Glu	Glu	Asp	Gly	Cys	Ser	Cys	Arg	Phe
			660					665					670	Pro
Glu	Glu	Glu	Glu	Gly	Gly	Cys	Glu	Leu	Arg	Val	Lys	Phe	Ser	Arg
	675						680					685		Ser

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Ala Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu
690 695 700

Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg
705 710 715 720

Gly Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln
725 730 735

Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr
740 745 750

Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp
755 760 765

Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala
770 775 780

Leu His Met Gln Ala Leu Pro Pro Arg
785 790

<210> SEQ ID NO 124
<211> LENGTH: 793
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 124

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
1 5 10 15

His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp
20 25 30

Ile Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly Gly
35 40 45

Ser Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Phe Tyr Asn Leu Leu
50 55 60

Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr
65 70 75 80

Asp Ala Ser Asp Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly Ser
85 90 95

Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala
100 105 110

Asp Ala Ala Ala Tyr Tyr Cys Gln Ser Ala Asp Gly Ser Ser Tyr Ala
115 120 125

Phe Gly Gly Gly Thr Glu Val Val Val Lys Gly Gly Gly Gly Gly Ser
130 135 140

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln
145 150 155 160

Glu Gln Leu Glu Glu Ser Gly Gly Gly Leu Val Lys Pro Glu Gly Ser
165 170 175

Leu Thr Leu Thr Cys Thr Ala Ser Gly Val Ser Phe Ser Ser Ser Tyr
180 185 190

Trp Ile Tyr Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile
195 200 205

Ala Cys Ile Tyr Thr Gly Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp
210 215 220

Ala Lys Gly Arg Phe Thr Val Ser Glu Thr Ser Ser Thr Thr Val Thr
225 230 235 240

Leu Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys

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245								250				255			
Ala	Arg	Ala	Ser	Ala	Trp	Thr	Tyr	Gly	Met	Asp	Leu	Trp	Gly	Pro	Gly
			260					265					270		
Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly
		275					280					285			
Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Pro	Met	Ala	Lys	Gly
	290					295					300				
Ser	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly
305					310					315					320
Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val
			325					330						335	
Lys	Pro	Ser	Glu	Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser
			340					345					350		
Ile	Ser	Tyr	Tyr	Ser	Trp	Thr	Trp	Val	Arg	Gln	Pro	Ala	Gly	Gln	Gly
		355					360					365			
Leu	Glu	Trp	Ile	Gly	Arg	Ile	His	Ile	Ser	Gly	Val	Thr	Asn	His	Asn
	370					375					380				
Pro	Ser	Leu	Lys	Ser	Arg	Val	Ser	Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr
385					390					395					400
Gln	Phe	Ser	Leu	Arg	Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val
			405						410					415	
Tyr	Phe	Cys	Ala	Arg	Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp
			420					425					430		
Gly	Gln	Gly	Thr	Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly
		435					440					445			
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile
	450					455					460				
Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg
465					470					475					480
Ser	Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu
			485					490						495	
Ala	Trp	Tyr	Gln	Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr
			500					505					510		
Gly	Ala	Ser	Thr	Arg	Ala	Ile	Gly	Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser
		515					520					525			
Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu
	530					535					540				
Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr
545					550					555					560
Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile	Lys	Thr	Thr	Thr	Pro	Ala	Pro
			565					570						575	
Arg	Pro	Pro	Thr	Pro	Ala	Pro	Thr	Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu
			580					585					590		
Arg	Pro	Glu	Ala	Cys	Arg	Pro	Ala	Ala	Gly	Gly	Ala	Val	His	Thr	Arg
		595					600					605			
Gly	Leu	Asp	Phe	Ala	Cys	Asp	Ile	Tyr	Ile	Trp	Ala	Pro	Leu	Ala	Gly
	610					615					620				
Thr	Cys	Gly	Val	Leu	Leu	Leu	Ser	Leu	Val	Ile	Thr	Leu	Tyr	Cys	Lys
625					630					635					640
Arg	Gly	Arg	Lys	Lys	Leu	Leu	Tyr	Ile	Phe	Lys	Gln	Pro	Phe	Met	Arg
			645					650						655	
Pro	Val	Gln	Thr	Thr	Gln	Glu	Glu	Asp	Gly	Cys	Ser	Cys	Arg	Phe	Pro
			660					665					670		

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Glu Glu Glu Glu Gly Gly Cys Glu Leu Arg Val Lys Phe Ser Arg Ser
 675 680 685
 Ala Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu
 690 695 700
 Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg
 705 710 715 720
 Gly Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln
 725 730 735
 Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr
 740 745 750
 Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp
 755 760 765
 Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala
 770 775 780
 Leu His Met Gln Ala Leu Pro Pro Arg
 785 790

<210> SEQ ID NO 125
 <211> LENGTH: 808
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 125

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15
 His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp
 20 25 30
 Ile Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly Gly
 35 40 45
 Ser Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Phe Tyr Asn Leu Leu
 50 55 60
 Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr
 65 70 75 80
 Asp Ala Ser Asp Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly Ser
 85 90 95
 Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala
 100 105 110
 Asp Ala Ala Ala Tyr Tyr Cys Gln Ser Ala Asp Gly Ser Ser Tyr Ala
 115 120 125
 Phe Gly Gly Gly Thr Glu Val Val Val Lys Gly Gly Gly Gly Gly Ser
 130 135 140
 Gly Gly Gly Gly Ser Val His Met Pro Leu Gly Phe Leu Gly Pro Arg
 145 150 155 160
 Gln Ala Arg Val Val Asn Gly Gly Ser Gly Gly Gly Gly Ser Gln Glu
 165 170 175
 Gln Leu Glu Glu Ser Gly Gly Gly Leu Val Lys Pro Glu Gly Ser Leu
 180 185 190
 Thr Leu Thr Cys Thr Ala Ser Gly Val Ser Phe Ser Ser Ser Tyr Trp
 195 200 205
 Ile Tyr Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile Ala
 210 215 220
 Cys Ile Tyr Thr Gly Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp Ala

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225					230					235					240
Lys	Gly	Arg	Phe	Thr	Val	Ser	Glu	Thr	Ser	Ser	Thr	Thr	Val	Thr	Leu
				245					250					255	
Gln	Met	Thr	Ser	Leu	Thr	Ala	Ala	Asp	Thr	Ala	Thr	Tyr	Phe	Cys	Ala
			260					265					270		
Arg	Ala	Ser	Ala	Trp	Thr	Tyr	Gly	Met	Asp	Leu	Trp	Gly	Pro	Gly	Thr
		275					280					285			
Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser
	290					295					300				
Gly	Gly	Gly	Gly	Ser	Val	His	Met	Pro	Leu	Gly	Phe	Leu	Gly	Pro	Arg
305					310					315					320
Gln	Ala	Arg	Val	Val	Asn	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly
			325					330						335	
Gly	Gly	Ser	Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys
			340					345					350		
Pro	Ser	Glu	Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile
		355					360					365			
Ser	Tyr	Tyr	Ser	Trp	Thr	Trp	Val	Arg	Gln	Pro	Ala	Gly	Gln	Gly	Leu
	370					375					380				
Glu	Trp	Ile	Gly	Arg	Ile	His	Ile	Ser	Gly	Val	Thr	Asn	His	Asn	Pro
385					390					395					400
Ser	Leu	Lys	Ser	Arg	Val	Ser	Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln
			405					410						415	
Phe	Ser	Leu	Arg	Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr
			420					425					430		
Phe	Cys	Ala	Arg	Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly
	435				440						445				
Gln	Gly	Thr	Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly
	450				455						460				
Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val
465				470					475						480
Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser
			485					490						495	
Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu	Ala
			500				505						510		
Trp	Tyr	Gln	Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly
	515					520						525			
Ala	Ser	Thr	Arg	Ala	Ile	Gly	Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly
	530				535						540				
Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp
545				550						555					560
Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe
			565					570						575	
Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile	Lys	Thr	Thr	Thr	Pro	Ala	Pro	Arg
			580					585					590		
Pro	Pro	Thr	Pro	Ala	Pro	Thr	Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu	Arg
		595				600						605			
Pro	Glu	Ala	Cys	Arg	Pro	Ala	Ala	Gly	Gly	Ala	Val	His	Thr	Arg	Gly
	610					615					620				
Leu	Asp	Phe	Ala	Cys	Asp	Ile	Tyr	Ile	Trp	Ala	Pro	Leu	Ala	Gly	Thr
625					630					635					640
Cys	Gly	Val	Leu	Leu	Leu	Ser	Leu	Val	Ile	Thr	Leu	Tyr	Cys	Lys	Arg
			645					650						655	

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Gly Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met Arg Pro
 660 665 670
 Val Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe Pro Glu
 675 680 685
 Glu Glu Glu Gly Gly Cys Glu Leu Arg Val Lys Phe Ser Arg Ser Ala
 690 695 700
 Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu
 705 710 715 720
 Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly
 725 730 735
 Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu
 740 745 750
 Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser
 755 760 765
 Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp Gly
 770 775 780
 Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu
 785 790 795 800
 His Met Gln Ala Leu Pro Pro Arg
 805

<210> SEQ ID NO 126

<211> LENGTH: 808

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 126

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15
 His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp
 20 25 30
 Ile Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly Gly
 35 40 45
 Ser Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Phe Tyr Asn Leu Leu
 50 55 60
 Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr
 65 70 75 80
 Asp Ala Ser Asp Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly Ser
 85 90 95
 Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala
 100 105 110
 Asp Ala Ala Ala Tyr Tyr Cys Gln Ser Ala Asp Gly Ser Ser Tyr Ala
 115 120 125
 Phe Gly Gly Gly Thr Glu Val Val Val Lys Gly Gly Gly Gly Gly Ser
 130 135 140
 Gly Gly Gly Gly Ser Val His Met Pro Leu Gly Phe Leu Gly Pro Gly
 145 150 155 160
 Ser Ala Arg Val Val Asn Gly Gly Ser Gly Gly Gly Gly Ser Gln Glu
 165 170 175
 Gln Leu Glu Glu Ser Gly Gly Gly Leu Val Lys Pro Glu Gly Ser Leu
 180 185 190
 Thr Leu Thr Cys Thr Ala Ser Gly Val Ser Phe Ser Ser Ser Tyr Trp

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195					200					205					
Ile	Tyr	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Ile	Ala
210						215					220				
Cys	Ile	Tyr	Thr	Gly	Ser	Ser	Gly	Ser	Thr	Tyr	Tyr	Ala	Ser	Trp	Ala
225					230					235					240
Lys	Gly	Arg	Phe	Thr	Val	Ser	Glu	Thr	Ser	Ser	Thr	Thr	Val	Thr	Leu
				245					250					255	
Gln	Met	Thr	Ser	Leu	Thr	Ala	Ala	Asp	Thr	Ala	Thr	Tyr	Phe	Cys	Ala
			260					265					270		
Arg	Ala	Ser	Ala	Trp	Thr	Tyr	Gly	Met	Asp	Leu	Trp	Gly	Pro	Gly	Thr
		275					280					285			
Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser
	290					295					300				
Gly	Gly	Gly	Gly	Ser	Val	His	Met	Pro	Leu	Gly	Phe	Leu	Gly	Pro	Gly
305					310					315					320
Ser	Ala	Arg	Val	Val	Asn	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly
				325					330					335	
Gly	Gly	Ser	Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys
			340					345					350		
Pro	Ser	Glu	Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile
		355					360					365			
Ser	Tyr	Tyr	Ser	Trp	Thr	Trp	Val	Arg	Gln	Pro	Ala	Gly	Gln	Gly	Leu
	370					375					380				
Glu	Trp	Ile	Gly	Arg	Ile	His	Ile	Ser	Gly	Val	Thr	Asn	His	Asn	Pro
385					390					395					400
Ser	Leu	Lys	Ser	Arg	Val	Ser	Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln
				405					410					415	
Phe	Ser	Leu	Arg	Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr
			420					425					430		
Phe	Cys	Ala	Arg	Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly
	435						440				445				
Gln	Gly	Thr	Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly
	450					455					460				
Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val
	465				470				475					480	
Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser
				485					490					495	
Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu	Ala
			500					505					510		
Trp	Tyr	Gln	Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly
	515					520						525			
Ala	Ser	Thr	Arg	Ala	Ile	Gly	Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly
	530					535					540				
Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp
545					550					555					560
Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe
				565					570					575	
Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile	Lys	Thr	Thr	Thr	Pro	Ala	Pro	Arg
			580					585					590		
Pro	Pro	Thr	Pro	Ala	Pro	Thr	Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu	Arg
		595					600					605			
Pro	Glu	Ala	Cys	Arg	Pro	Ala	Ala	Gly	Gly	Ala	Val	His	Thr	Arg	Gly
	610					615					620				

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Leu Asp Phe Ala Cys Asp Ile Tyr Ile Trp Ala Pro Leu Ala Gly Thr
 625 630 635 640
 Cys Gly Val Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr Cys Lys Arg
 645 650 655
 Gly Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met Arg Pro
 660 665 670
 Val Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe Pro Glu
 675 680 685
 Glu Glu Glu Gly Gly Cys Glu Leu Arg Val Lys Phe Ser Arg Ser Ala
 690 695 700
 Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu
 705 710 715 720
 Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly
 725 730 735
 Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu
 740 745 750
 Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser
 755 760 765
 Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp Gly
 770 775 780
 Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu
 785 790 795 800
 His Met Gln Ala Leu Pro Pro Arg
 805

<210> SEQ ID NO 127

<211> LENGTH: 808

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 127

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15
 His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp
 20 25 30
 Ile Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly Gly
 35 40 45
 Ser Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Phe Tyr Asn Leu Leu
 50 55 60
 Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr
 65 70 75 80
 Asp Ala Ser Asp Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly Ser
 85 90 95
 Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala
 100 105 110
 Asp Ala Ala Ala Tyr Tyr Cys Gln Ser Ala Asp Gly Ser Ser Tyr Ala
 115 120 125
 Phe Gly Gly Gly Thr Glu Val Val Val Lys Gly Gly Gly Gly Ser
 130 135 140
 Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Pro Met Ala Lys Lys Gly
 145 150 155 160
 Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gln Glu

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165								170				175			
Gln	Leu	Glu	Glu	Ser	Gly	Gly	Gly	Leu	Val	Lys	Pro	Glu	Gly	Ser	Leu
			180					185				190			
Thr	Leu	Thr	Cys	Thr	Ala	Ser	Gly	Val	Ser	Phe	Ser	Ser	Ser	Tyr	Trp
			195					200				205			
Ile	Tyr	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Ile	Ala
			210					215				220			
Cys	Ile	Tyr	Thr	Gly	Ser	Ser	Gly	Ser	Thr	Tyr	Tyr	Ala	Ser	Trp	Ala
			225					230				235			
Lys	Gly	Arg	Phe	Thr	Val	Ser	Glu	Thr	Ser	Ser	Thr	Thr	Val	Thr	Leu
			245					250				255			
Gln	Met	Thr	Ser	Leu	Thr	Ala	Ala	Asp	Thr	Ala	Thr	Tyr	Phe	Cys	Ala
			260					265				270			
Arg	Ala	Ser	Ala	Trp	Thr	Tyr	Gly	Met	Asp	Leu	Trp	Gly	Pro	Gly	Thr
			275					280				285			
Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser
			290					295				300			
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Pro	Met	Ala	Lys	Lys	Gly
			305					310				315			
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly
			325					330				335			
Gly	Gly	Ser	Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys
			340					345				350			
Pro	Ser	Glu	Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile
			355					360				365			
Ser	Tyr	Tyr	Ser	Trp	Thr	Trp	Val	Arg	Gln	Pro	Ala	Gly	Gln	Gly	Leu
			370					375				380			
Glu	Trp	Ile	Gly	Arg	Ile	His	Ile	Ser	Gly	Val	Thr	Asn	His	Asn	Pro
			385					390				395			
Ser	Leu	Lys	Ser	Arg	Val	Ser	Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln
			405					410				415			
Phe	Ser	Leu	Arg	Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr
			420					425				430			
Phe	Cys	Ala	Arg	Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly
			435					440				445			
Gln	Gly	Thr	Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly
			450					455				460			
Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val
			465					470				475			
Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser
			485					490				495			
Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu	Ala
			500					505				510			
Trp	Tyr	Gln	Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly
			515					520				525			
Ala	Ser	Thr	Arg	Ala	Ile	Gly	Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly
			530					535				540			
Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp
			545					550				555			
Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe
			565					570				575			
Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile	Lys	Thr	Thr	Thr	Pro	Ala	Pro	Arg
			580					585				590			

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Pro Pro Thr Pro Ala Pro Thr Ile Ala Ser Gln Pro Leu Ser Leu Arg
 595 600 605
 Pro Glu Ala Cys Arg Pro Ala Ala Gly Gly Ala Val His Thr Arg Gly
 610 615 620
 Leu Asp Phe Ala Cys Asp Ile Tyr Ile Trp Ala Pro Leu Ala Gly Thr
 625 630 635 640
 Cys Gly Val Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr Cys Lys Arg
 645 650 655
 Gly Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met Arg Pro
 660 665 670
 Val Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe Pro Glu
 675 680 685
 Glu Glu Glu Gly Gly Cys Glu Leu Arg Val Lys Phe Ser Arg Ser Ala
 690 695 700
 Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu
 705 710 715 720
 Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly
 725 730 735
 Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu
 740 745 750
 Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser
 755 760 765
 Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp Gly
 770 775 780
 Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu
 785 790 795 800
 His Met Gln Ala Leu Pro Pro Arg
 805

<210> SEQ ID NO 128

<211> LENGTH: 808

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 128

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15
 His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp
 20 25 30
 Ile Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly Gly
 35 40 45
 Ser Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Phe Tyr Asn Leu Leu
 50 55 60
 Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr
 65 70 75 80
 Asp Ala Ser Asp Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly Ser
 85 90 95
 Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala
 100 105 110
 Asp Ala Ala Ala Tyr Tyr Cys Gln Ser Ala Asp Gly Ser Ser Tyr Ala
 115 120 125
 Phe Gly Gly Gly Thr Glu Val Val Val Lys Gly Gly Gly Gly Gly Ser

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130					135					140					
Gly 145	Gly	Gly	Gly	Ser	Gly 150	Gly	Gly	Gly	Ser	Pro 155	Met	Ala	Lys	Gly	Ser 160
Gly	Gly	Gly	Ser	Gly 165	Gly	Gly	Gly	Ser	Gly 170	Gly	Gly	Gly	Ser	Gln 175	Glu
Gln	Leu	Glu	Glu	Ser 180	Gly	Gly	Gly	Leu 185	Val	Lys	Pro	Glu	Gly 190	Ser	Leu
Thr	Leu	Thr	Cys	Thr 195	Ala	Ser	Gly 200	Val	Ser	Phe	Ser	Ser 205	Ser	Tyr	Trp
Ile	Tyr	Trp	Val	Arg 210	Gln	Ala 215	Pro	Gly	Lys	Gly	Leu 220	Glu	Trp	Ile	Ala
Cys 225	Ile	Tyr	Thr	Gly 230	Ser	Ser	Gly	Ser	Thr	Tyr 235	Tyr	Ala	Ser	Trp	Ala 240
Lys	Gly	Arg	Phe	Thr 245	Val	Ser	Glu	Thr	Ser 250	Ser	Thr	Thr	Val	Thr 255	Leu
Gln	Met	Thr	Ser	Leu 260	Thr	Ala	Ala	Asp 265	Thr	Ala	Thr	Tyr	Phe 270	Cys	Ala
Arg	Ala	Ser	Ala	Trp 275	Thr	Tyr	Gly 280	Met	Asp	Leu	Trp	Gly 285	Pro	Gly	Thr
Leu	Val	Thr	Val	Ser 290	Ser	Gly 295	Gly	Gly	Gly	Ser	Gly 300	Gly	Gly	Gly	Ser
Gly 305	Gly	Gly	Gly	Ser 310	Gly	Gly	Gly	Gly	Ser	Pro 315	Met	Ala	Lys	Gly	Ser 320
Gly	Gly	Gly	Ser	Gly 325	Gly	Gly	Gly	Ser	Gly 330	Gly	Gly	Gly	Ser	Gly 335	Gly
Gly	Gly	Ser	Gln 340	Val	Gln	Leu	Gln	Glu 345	Ser	Gly	Pro	Gly	Leu 350	Val	Lys
Pro	Ser	Glu	Thr	Leu 355	Ser	Leu	Thr 360	Cys	Thr	Val	Ser	Gly 365	Gly	Ser	Ile
Ser	Tyr	Tyr	Ser	Trp 370	Thr	Trp 375	Val	Arg	Gln	Pro	Ala 380	Gly	Gln	Gly	Leu
Glu 385	Trp	Ile	Gly	Arg 390	Ile	His	Ile	Ser	Gly 395	Val	Thr	Asn	His	Asn	Pro 400
Ser	Leu	Lys	Ser	Arg 405	Val	Ser	Met	Ser	Ile 410	Asp	Thr	Ser	Arg	Thr 415	Gln
Phe	Ser	Leu	Arg	Leu 420	Thr	Ser	Val	Thr 425	Ala	Ala	Asp	Thr	Ala 430	Val	Tyr
Phe	Cys	Ala	Arg	Ser 435	Gly	Gly	Thr 440	Tyr	Ser	Ala	Phe	Asp 445	Ile	Trp	Gly
Gln	Gly	Thr	Met	Val 450	Thr	Val 455	Ser	Ser	Gly	Gly	Gly 460	Gly	Ser	Gly	Gly
Gly 465	Gly	Ser	Gly	Gly 470	Gly	Gly	Ser	Gly	Gly	Gly 475	Gly	Ser	Glu	Ile 480	Val
Leu	Thr	Gln	Ser	Pro 485	Gly	Thr	Leu	Ser	Leu 490	Ser	Pro	Gly	Glu 495	Arg	Ser
Thr	Leu	Ser	Cys	Arg 500	Ala	Ser	Gln	Ser	Phe 505	Thr	Ser	Asn	Tyr 510	Leu	Ala
Trp	Tyr	Gln	Gln	Arg 515	Pro	Gly	Gln 520	Ala	Pro	Arg	Leu 525	Leu	Ile	Tyr	Gly
Ala	Ser	Thr	Arg	Ala 530	Ile	Gly 535	Ile	Pro	Asp	Arg	Phe 540	Ser	Gly	Ser	Gly
Ser 545	Gly	Thr	Asp	Phe 550	Thr	Leu	Thr	Ile	Ser	Arg 555	Leu	Glu	Pro	Glu	Asp 560

Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe
				565					570						
Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile	Lys	Thr	Thr	Thr	Pro	Ala	Pro	Arg
				580					585						
Pro	Pro	Thr	Pro	Ala	Pro	Thr	Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu	Arg
				595					600						
Pro	Glu	Ala	Cys	Arg	Pro	Ala	Ala	Gly	Gly	Ala	Val	His	Thr	Arg	Gly
				610					615						
Leu	Asp	Phe	Ala	Cys	Asp	Ile	Tyr	Ile	Trp	Ala	Pro	Leu	Ala	Gly	Thr
625					630					635					
Cys	Gly	Val	Leu	Leu	Leu	Ser	Leu	Val	Ile	Thr	Leu	Tyr	Cys	Lys	Arg
				645					650						
Gly	Arg	Lys	Lys	Leu	Leu	Tyr	Ile	Phe	Lys	Gln	Pro	Phe	Met	Arg	Pro
				660					665						
Val	Gln	Thr	Thr	Gln	Glu	Glu	Asp	Gly	Cys	Ser	Cys	Arg	Phe	Pro	Glu
				675					680						
Glu	Glu	Glu	Gly	Gly	Cys	Glu	Leu	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala
				690					695						
Asp	Ala	Pro	Ala	Tyr	Gln	Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu
705					710					715					
Asn	Leu	Gly	Arg	Arg	Glu	Glu	Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly
				725					730						
Arg	Asp	Pro	Glu	Met	Gly	Gly	Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu
				740					745						
Gly	Leu	Tyr	Asn	Glu	Leu	Gln	Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser
				755					760						
Glu	Ile	Gly	Met	Lys	Gly	Glu	Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly
				770					775						
Leu	Tyr	Gln	Gly	Leu	Ser	Thr	Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu
785					790					795					
His	Met	Gln	Ala	Leu	Pro	Pro	Arg								
				805											

<400> SEQUENCE: 129

Met	Ala	Leu	Pro	Val	Thr	Ala	Leu	Leu	Leu	Pro	Leu	Ala	Leu	Leu	Leu
1				5					10					15	
His	Ala	Ala	Arg	Pro	Gly	Tyr	Pro	Tyr	Asp	Val	Pro	Asp	Tyr	Ala	Gln
			20					25					30		
Phe	Ser	Val	Leu	Gly	Pro	Ser	Gly	Pro	Ile	Leu	Ala	Met	Val	Gly	Glu
		35					40					45			
Asp	Ala	Asp	Leu	Pro	Cys	His	Leu	Phe	Pro	Thr	Met	Ser	Ala	Glu	Thr
	50					55					60				
Met	Glu	Leu	Lys	Trp	Val	Ser	Ser	Ser	Leu	Arg	Gln	Val	Val	Asn	Val
65					70					75					80
Tyr	Ala	Asp	Gly	Lys	Glu	Val	Glu	Asp	Arg	Gln	Ser	Ala	Pro	Tyr	Arg
				85					90					95	
Gly	Arg	Thr	Ser	Ile	Leu	Arg	Asp	Gly	Ile	Thr	Ala	Gly	Lys	Ala	Ala

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100										105					110				
Leu	Arg	Ile	His	Asn	Val	Thr	Ala	Ser	Asp	Ser	Gly	Lys	Tyr	Leu	Cys				
		115					120					125							
Tyr	Phe	Gln	Asp	Gly	Asp	Phe	Tyr	Glu	Lys	Ala	Leu	Val	Glu	Leu	Lys				
	130					135					140								
Val	Ala	Ala	Leu	Gly	Ser	Asp	Leu	His	Val	Asp	Val	Lys	Gly	Tyr	Lys				
145					150					155					160				
Asp	Gly	Gly	Ile	His	Leu	Glu	Cys	Arg	Ser	Thr	Gly	Trp	Tyr	Pro	Gln				
			165						170					175					
Pro	Gln	Ile	Gln	Trp	Ser	Asn	Asn	Lys	Gly	Glu	Asn	Ile	Pro	Thr	Val				
			180					185					190						
Glu	Ala	Pro	Val	Val	Ala	Asp	Gly	Val	Gly	Leu	Tyr	Ala	Val	Ala	Ala				
	195						200					205							
Ser	Val	Ile	Met	Arg	Gly	Ser	Ser	Gly	Glu	Gly	Val	Ser	Cys	Thr	Ile				
	210					215					220								
Arg	Ser	Ser	Leu	Leu	Gly	Leu	Glu	Lys	Thr	Ala	Ser	Ile	Ser	Ile	Ala				
225					230					235					240				
Asp	Pro	Phe	Phe	Arg	Ser	Ala	Gln	Arg	Trp	Ile	Ala	Ala	Leu	Ala	Gly				
				245					250					255					
Thr	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Leu	Val	Pro	Arg	Gly				
			260					265					270						
Ser	Lys	Pro	Ile	Pro	Asn	Pro	Leu	Leu	Gly	Leu	Asp	Ser	Thr	Leu	Val				
		275					280					285							
Pro	Arg	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Val				
	290					295					300								
Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Glu	Thr	Leu				
305					310					315					320				
Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile	Ser	Tyr	Tyr	Ser	Trp				
			325						330					335					
Thr	Trp	Val	Arg	Gln	Pro	Ala	Gly	Gln	Gly	Leu	Glu	Trp	Ile	Gly	Arg				
		340						345					350						
Ile	His	Ile	Ser	Gly	Val	Thr	Asn	His	Asn	Pro	Ser	Leu	Lys	Ser	Arg				
	355						360					365							
Val	Ser	Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln	Phe	Ser	Leu	Arg	Leu				
	370					375					380								
Thr	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Phe	Cys	Ala	Arg	Ser				
385				390						395					400				
Gly	Gly	Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly	Gln	Gly	Thr	Met	Val				
			405						410					415					
Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly				
		420						425					430						
Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro				
	435						440					445							
Gly	Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser	Thr	Leu	Ser	Cys	Arg				
	450					455					460								
Ala	Ser	Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Arg				
465					470					475					480				
Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Ser	Thr	Arg	Ala				
			485						490					495					
Ile	Gly	Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe				
		500						505					510						
Thr	Leu	Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr				
	515						520					525							

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Cys Gln Gln Tyr Gly Ser Ser Pro Trp Thr Phe Gly Gln Gly Thr Lys
 530 535 540
 Val Glu Ile Lys Thr Thr Thr Pro Ala Pro Arg Pro Pro Thr Pro Ala
 545 550 555 560
 Pro Thr Ile Ala Ser Gln Pro Leu Ser Leu Arg Pro Glu Ala Cys Arg
 565 570 575
 Pro Ala Ala Gly Gly Ala Val His Thr Arg Gly Leu Asp Phe Ala Cys
 580 585 590
 Asp Ile Tyr Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly Val Leu Leu
 595 600 605
 Leu Ser Leu Val Ile Thr Leu Tyr Cys Lys Arg Gly Arg Lys Lys Leu
 610 615 620
 Leu Tyr Ile Phe Lys Gln Pro Phe Met Arg Pro Val Gln Thr Thr Gln
 625 630 635 640
 Glu Glu Asp Gly Cys Ser Cys Arg Phe Pro Glu Glu Glu Gly Gly
 645 650 655
 Cys Glu Leu Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr
 660 665 670
 Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg
 675 680 685
 Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met
 690 695 700
 Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu
 705 710 715 720
 Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys
 725 730 735
 Gly Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu
 740 745 750
 Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu
 755 760 765
 Pro Pro Arg
 770

<210> SEQ ID NO 130

<211> LENGTH: 790

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 130

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15
 His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp
 20 25 30
 Ile Leu Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly Glu
 35 40 45
 Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Asn Ile Gly Thr Ser Ile
 50 55 60
 Gln Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile Arg
 65 70 75 80
 Ser Ser Ser Glu Ser Ile Ser Gly Ile Pro Ser Arg Phe Ser Gly Ser
 85 90 95
 Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Glu Pro Glu

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100							105							110						
Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Ser	Asn	Thr	Trp	Pro	Phe	Thr					
		115					120					125								
Phe	Gly	Gln	Gly	Thr	Lys	Leu	Glu	Ile	Lys	Gly	Gly	Gly	Gly	Ser	Gly					
	130					135					140									
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Val					
145					150					155				160						
Gln	Leu	Val	Glu	Ser	Gly	Ala	Glu	Val	Lys	Lys	Pro	Gly	Ala	Ser	Val					
				165					170					175						
Lys	Val	Ser	Cys	Lys	Ala	Ser	Gly	Tyr	Thr	Phe	Thr	Asn	Tyr	Ile	Ile					
			180					185					190							
His	Trp	Val	Lys	Gln	Glu	Pro	Gly	Gln	Gly	Leu	Glu	Trp	Ile	Gly	Tyr					
	195						200					205								
Phe	Asn	Pro	Tyr	Asn	His	Gly	Thr	Lys	Tyr	Asn	Glu	Lys	Phe	Lys	Gly					
	210					215					220									
Arg	Ala	Thr	Leu	Thr	Ala	Asp	Lys	Ser	Ile	Ser	Thr	Ala	Tyr	Met	Glu					
225					230					235					240					
Leu	Ser	Ser	Leu	Arg	Ser	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala	Arg					
				245					250					255						
Ser	Gly	Pro	Tyr	Ala	Trp	Phe	Asp	Thr	Trp	Gly	Gln	Gly	Thr	Thr	Val					
			260					265					270							
Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Leu	Val					
	275					280						285								
Pro	Arg	Gly	Ser	Lys	Pro	Ile	Pro	Asn	Pro	Leu	Leu	Gly	Leu	Asp	Ser					
	290					295					300									
Thr	Leu	Val	Pro	Arg	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly					
305					310					315					320					
Ser	Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser					
				325					330					335						
Glu	Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile	Ser	Tyr					
			340					345					350							
Tyr	Ser	Trp	Thr	Trp	Val	Arg	Gln	Pro	Ala	Gly	Gln	Gly	Leu	Glu	Trp					
	355						360					365								
Ile	Gly	Arg	Ile	His	Ile	Ser	Gly	Val	Thr	Asn	His	Asn	Pro	Ser	Leu					
	370					375					380									
Lys	Ser	Arg	Val	Ser	Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln	Phe	Ser					
385					390					395					400					
Leu	Arg	Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Phe	Cys					
				405					410					415						
Ala	Arg	Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly	Gln	Gly					
			420					425					430							
Thr	Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly					
	435						440					445								
Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr					
	450					455					460									
Gln	Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser	Thr	Leu					
465					470					475					480					
Ser	Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu	Ala	Trp	Tyr					
				485					490					495						
Gln	Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Ser					
			500					505					510							
Thr	Arg	Ala	Ile	Gly	Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly					
	515						520					525								

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Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu Pro Glu Asp Phe Ala
 530 535 540
 Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro Trp Thr Phe Gly Gln
 545 550 555 560
 Gly Thr Lys Val Glu Ile Lys Thr Thr Thr Pro Ala Pro Arg Pro Pro
 565 570 575
 Thr Pro Ala Pro Thr Ile Ala Ser Gln Pro Leu Ser Leu Arg Pro Glu
 580 585 590
 Ala Cys Arg Pro Ala Ala Gly Gly Ala Val His Thr Arg Gly Leu Asp
 595 600 605
 Phe Ala Cys Asp Ile Tyr Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly
 610 615 620
 Val Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr Cys Lys Arg Gly Arg
 625 630 635 640
 Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met Arg Pro Val Gln
 645 650 655
 Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe Pro Glu Glu Glu
 660 665 670
 Glu Gly Gly Cys Glu Leu Arg Val Lys Phe Ser Arg Ser Ala Asp Ala
 675 680 685
 Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu
 690 695 700
 Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp
 705 710 715 720
 Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu
 725 730 735
 Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile
 740 745 750
 Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr
 755 760 765
 Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met
 770 775 780
 Gln Ala Leu Pro Pro Arg
 785 790

<210> SEQ ID NO 131

<211> LENGTH: 790

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 131

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15
 His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp
 20 25 30
 Ile Leu Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly Glu
 35 40 45
 Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Asn Ile Gly Thr Ser Ile
 50 55 60
 Gln Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile Arg
 65 70 75 80
 Ser Ser Ser Glu Ser Ile Ser Gly Ile Pro Ser Arg Phe Ser Gly Ser

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85								90				95				
Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Ser	Leu	Glu	Pro	Glu	
			100							105				110		
Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Ser	Asn	Thr	Trp	Pro	Phe	Thr	
			115							120				125		
Phe	Gly	Gln	Gly	Thr	Lys	Leu	Glu	Ile	Lys	Gly	Gly	Gly	Gly	Ser	Gly	
			130							135				140		
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Val	
			145							150				155		
Gln	Leu	Val	Glu	Ser	Gly	Ala	Glu	Val	Lys	Lys	Pro	Gly	Ala	Ser	Val	
			165							170				175		
Lys	Val	Ser	Cys	Lys	Ala	Ser	Gly	Tyr	Thr	Phe	Thr	Asn	Tyr	Ile	Ile	
			180							185				190		
His	Trp	Val	Lys	Gln	Glu	Pro	Gly	Gln	Gly	Leu	Glu	Trp	Ile	Gly	Tyr	
			195							200				205		
Phe	Asn	Pro	Tyr	Asn	His	Gly	Thr	Lys	Tyr	Asn	Glu	Lys	Phe	Lys	Gly	
			210							215				220		
Arg	Ala	Thr	Leu	Thr	Ala	Asp	Lys	Ser	Ile	Ser	Thr	Ala	Tyr	Met	Glu	
			225							230				235		
Leu	Ser	Ser	Leu	Arg	Ser	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala	Arg	
			245							250				255		
Ser	Gly	Pro	Tyr	Ala	Trp	Phe	Asp	Thr	Trp	Gly	Gln	Gly	Thr	Thr	Val	
			260							265				270		
Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	
			275							280				285		
Gly	Ser	Gly	Ser	Lys	Pro	Ile	Pro	Asn	Pro	Leu	Leu	Gly	Leu	Asp	Ser	
			290							295				300		
Thr	Gly	Gly	Gly	Ser	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	
			305							310				315		
Ser	Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	
			325							330				335		
Glu	Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile	Ser	Tyr	
			340							345				350		
Tyr	Ser	Trp	Thr	Trp	Val	Arg	Gln	Pro	Ala	Gly	Gln	Gly	Leu	Glu	Trp	
			355							360				365		
Ile	Gly	Arg	Ile	His	Ile	Ser	Gly	Val	Thr	Asn	His	Asn	Pro	Ser	Leu	
			370							375				380		
Lys	Ser	Arg	Val	Ser	Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln	Phe	Ser	
			385							390				395		
Leu	Arg	Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Phe	Cys	
			405							410				415		
Ala	Arg	Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly	Gln	Gly	
			420							425				430		
Thr	Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	
			435							440				445		
Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr	
			450							455				460		
Gln	Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser	Thr	Leu	
			465							470				475		
Ser	Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu	Ala	Trp	Tyr	
			485							490				495		
Gln	Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Ser	
			500							505				510		

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Thr Arg Ala Ile Gly Ile Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly
 515 520 525
 Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu Pro Glu Asp Phe Ala
 530 535 540
 Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro Trp Thr Phe Gly Gln
 545 550 555 560
 Gly Thr Lys Val Glu Ile Lys Thr Thr Thr Pro Ala Pro Arg Pro Pro
 565 570 575
 Thr Pro Ala Pro Thr Ile Ala Ser Gln Pro Leu Ser Leu Arg Pro Glu
 580 585 590
 Ala Cys Arg Pro Ala Ala Gly Gly Ala Val His Thr Arg Gly Leu Asp
 595 600 605
 Phe Ala Cys Asp Ile Tyr Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly
 610 615 620
 Val Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr Cys Lys Arg Gly Arg
 625 630 635 640
 Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met Arg Pro Val Gln
 645 650 655
 Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe Pro Glu Glu Glu
 660 665 670
 Glu Gly Gly Cys Glu Leu Arg Val Lys Phe Ser Arg Ser Ala Asp Ala
 675 680 685
 Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu
 690 695 700
 Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp
 705 710 715 720
 Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu
 725 730 735
 Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile
 740 745 750
 Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr
 755 760 765
 Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met
 770 775 780
 Gln Ala Leu Pro Pro Arg
 785 790

<210> SEQ ID NO 132

<211> LENGTH: 790

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 132

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15
 His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp
 20 25 30
 Ile Leu Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly Glu
 35 40 45
 Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Asn Ile Gly Thr Ser Ile
 50 55 60
 Gln Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile Arg

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65				70					75					80			
Ser	Ser	Ser	Glu	Ser	Ile	Ser	Gly	Ile	Pro	Ser	Arg	Phe	Ser	Gly	Ser		
85				90					95								
Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Ser	Leu	Glu	Pro	Glu		
100				105					110								
Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Ser	Asn	Thr	Trp	Pro	Phe	Thr		
115				120					125								
Phe	Gly	Gln	Gly	Thr	Lys	Leu	Glu	Ile	Lys	Gly	Gly	Gly	Gly	Ser	Gly		
130				135					140								
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Val		
145				150					155					160			
Gln	Leu	Val	Glu	Ser	Gly	Ala	Glu	Val	Lys	Lys	Pro	Gly	Ala	Ser	Val		
165				170					175								
Lys	Val	Ser	Cys	Lys	Ala	Ser	Gly	Tyr	Thr	Phe	Thr	Asn	Tyr	Ile	Ile		
180				185					190								
His	Trp	Val	Lys	Gln	Glu	Pro	Gly	Gln	Gly	Leu	Glu	Trp	Ile	Gly	Tyr		
195				200					205								
Phe	Asn	Pro	Tyr	Asn	His	Gly	Thr	Lys	Tyr	Asn	Glu	Lys	Phe	Lys	Gly		
210				215					220								
Arg	Ala	Thr	Leu	Thr	Ala	Asp	Lys	Ser	Ile	Ser	Thr	Ala	Tyr	Met	Glu		
225				230					235					240			
Leu	Ser	Ser	Leu	Arg	Ser	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala	Arg		
245				250					255								
Ser	Gly	Pro	Tyr	Ala	Trp	Phe	Asp	Thr	Trp	Gly	Gln	Gly	Thr	Thr	Val		
260				265					270								
Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly		
275				280					285								
Gly	Gly	Ser	Val	His	Met	Pro	Leu	Gly	Phe	Leu	Gly	Pro	Arg	Gln	Ala		
290				295					300								
Arg	Val	Val	Asn	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly		
305				310					315					320			
Ser	Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser		
325				330					335								
Glu	Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile	Ser	Tyr		
340				345					350								
Tyr	Ser	Trp	Thr	Trp	Val	Arg	Gln	Pro	Ala	Gly	Gln	Gly	Leu	Glu	Trp		
355				360					365								
Ile	Gly	Arg	Ile	His	Ile	Ser	Gly	Val	Thr	Asn	His	Asn	Pro	Ser	Leu		
370				375					380								
Lys	Ser	Arg	Val	Ser	Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln	Phe	Ser		
385				390					395					400			
Leu	Arg	Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Phe	Cys		
405				410					415								
Ala	Arg	Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly	Gln	Gly		
420				425					430								
Thr	Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly		
435				440					445								
Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr		
450				455					460								
Gln	Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser	Thr	Leu		
465				470					475					480			
Ser	Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu	Ala	Trp	Tyr		
485				490					495								

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Gln Gln Arg Pro Gly Gln Ala Pro Arg Leu Leu Ile Tyr Gly Ala Ser
 500 505 510
 Thr Arg Ala Ile Gly Ile Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly
 515 520 525
 Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu Pro Glu Asp Phe Ala
 530 535 540
 Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro Trp Thr Phe Gly Gln
 545 550 555 560
 Gly Thr Lys Val Glu Ile Lys Thr Thr Thr Pro Ala Pro Arg Pro Pro
 565 570 575
 Thr Pro Ala Pro Thr Ile Ala Ser Gln Pro Leu Ser Leu Arg Pro Glu
 580 585 590
 Ala Cys Arg Pro Ala Ala Gly Gly Ala Val His Thr Arg Gly Leu Asp
 595 600 605
 Phe Ala Cys Asp Ile Tyr Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly
 610 615 620
 Val Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr Cys Lys Arg Gly Arg
 625 630 635 640
 Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met Arg Pro Val Gln
 645 650 655
 Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe Pro Glu Glu Glu
 660 665 670
 Glu Gly Gly Cys Glu Leu Arg Val Lys Phe Ser Arg Ser Ala Asp Ala
 675 680 685
 Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu
 690 695 700
 Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp
 705 710 715 720
 Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu
 725 730 735
 Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile
 740 745 750
 Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr
 755 760 765
 Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met
 770 775 780
 Gln Ala Leu Pro Pro Arg
 785 790

<210> SEQ ID NO 133

<211> LENGTH: 790

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 133

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15
 His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp
 20 25 30
 Ile Leu Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly Glu
 35 40 45
 Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Asn Ile Gly Thr Ser Ile

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50	55	60
Gln Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile Arg		
65	70	75 80
Ser Ser Ser Glu Ser Ile Ser Gly Ile Pro Ser Arg Phe Ser Gly Ser		
	85	90 95
Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Glu Pro Glu		
	100	105 110
Asp Phe Ala Val Tyr Tyr Cys Gln Gln Ser Asn Thr Trp Pro Phe Thr		
	115	120 125
Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Gly Gly Gly Ser Gly		
	130	135 140
Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Glu Val		
	145	150 155 160
Gln Leu Val Glu Ser Gly Ala Glu Val Lys Lys Pro Gly Ala Ser Val		
	165	170 175
Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asn Tyr Ile Ile		
	180	185 190
His Trp Val Lys Gln Glu Pro Gly Gln Gly Leu Glu Trp Ile Gly Tyr		
	195	200 205
Phe Asn Pro Tyr Asn His Gly Thr Lys Tyr Asn Glu Lys Phe Lys Gly		
	210	215 220
Arg Ala Thr Leu Thr Ala Asp Lys Ser Ile Ser Thr Ala Tyr Met Glu		
	225	230 235 240
Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys Ala Arg		
	245	250 255
Ser Gly Pro Tyr Ala Trp Phe Asp Thr Trp Gly Gln Gly Thr Thr Val		
	260	265 270
Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly		
	275	280 285
Gly Gly Ser Gly Gly Gly Gly Ser Pro Met Ala Lys Lys Gly Gly Gly		
	290	295 300
Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly		
	305	310 315 320
Ser Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser		
	325	330 335
Glu Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Tyr		
	340	345 350
Tyr Ser Trp Thr Trp Val Arg Gln Pro Ala Gly Gln Gly Leu Glu Trp		
	355	360 365
Ile Gly Arg Ile His Ile Ser Gly Val Thr Asn His Asn Pro Ser Leu		
	370	375 380
Lys Ser Arg Val Ser Met Ser Ile Asp Thr Ser Arg Thr Gln Phe Ser		
	385	390 395 400
Leu Arg Leu Thr Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Phe Cys		
	405	410 415
Ala Arg Ser Gly Gly Thr Tyr Ser Ala Phe Asp Ile Trp Gly Gln Gly		
	420	425 430
Thr Met Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly		
	435	440 445
Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Glu Ile Val Leu Thr		
	450	455 460
Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly Glu Arg Ser Thr Leu		
	465	470 475 480

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35					40					45					
Arg	Ala	Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	Asn	Ile	Gly	Thr	Ser	Ile
50						55					60				
Gln	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile	Arg
65					70					75					80
Ser	Ser	Ser	Glu	Ser	Ile	Ser	Gly	Ile	Pro	Ser	Arg	Phe	Ser	Gly	Ser
				85					90					95	
Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Ser	Leu	Glu	Pro	Glu
			100					105					110		
Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Ser	Asn	Thr	Trp	Pro	Phe	Thr
		115					120					125			
Phe	Gly	Gln	Gly	Thr	Lys	Leu	Glu	Ile	Lys	Gly	Gly	Gly	Gly	Ser	Gly
	130					135					140				
Gly	Gly	Gly	Ser	Val	His	Met	Pro	Leu	Gly	Phe	Leu	Gly	Pro	Arg	Gln
145					150					155					160
Ala	Arg	Val	Val	Asn	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Val	Gln
				165					170					175	
Leu	Val	Glu	Ser	Gly	Ala	Glu	Val	Lys	Lys	Pro	Gly	Ala	Ser	Val	Lys
		180						185					190		
Val	Ser	Cys	Lys	Ala	Ser	Gly	Tyr	Thr	Phe	Thr	Asn	Tyr	Ile	Ile	His
		195					200					205			
Trp	Val	Lys	Gln	Glu	Pro	Gly	Gln	Gly	Leu	Glu	Trp	Ile	Gly	Tyr	Phe
	210					215					220				
Asn	Pro	Tyr	Asn	His	Gly	Thr	Lys	Tyr	Asn	Glu	Lys	Phe	Lys	Gly	Arg
225					230					235					240
Ala	Thr	Leu	Thr	Ala	Asp	Lys	Ser	Ile	Ser	Thr	Ala	Tyr	Met	Glu	Leu
				245					250					255	
Ser	Ser	Leu	Arg	Ser	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala	Arg	Ser
		260						265					270		
Gly	Pro	Tyr	Ala	Trp	Phe	Asp	Thr	Trp	Gly	Gln	Gly	Thr	Thr	Val	Thr
		275					280					285			
Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly
	290				295					300					
Gly	Ser	Val	His	Met	Pro	Leu	Gly	Phe	Leu	Gly	Pro	Arg	Gln	Ala	Arg
305					310					315					320
Val	Val	Asn	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser
			325					330						335	
Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Glu
			340					345					350		
Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile	Ser	Tyr	Tyr
	355						360					365			
Ser	Trp	Thr	Trp	Val	Arg	Gln	Pro	Ala	Gly	Gln	Gly	Leu	Glu	Trp	Ile
	370					375					380				
Gly	Arg	Ile	His	Ile	Ser	Gly	Val	Thr	Asn	His	Asn	Pro	Ser	Leu	Lys
385					390					395					400
Ser	Arg	Val	Ser	Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln	Phe	Ser	Leu
				405					410					415	
Arg	Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Phe	Cys	Ala
			420					425					430		
Arg	Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly	Gln	Gly	Thr
		435					440					445			
Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser
450						455					460				

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Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Glu Ile Val Leu Thr Gln
 465 470 475 480
 Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly Glu Arg Ser Thr Leu Ser
 485 490 495
 Cys Arg Ala Ser Gln Ser Phe Thr Ser Asn Tyr Leu Ala Trp Tyr Gln
 500 505 510
 Gln Arg Pro Gly Gln Ala Pro Arg Leu Leu Ile Tyr Gly Ala Ser Thr
 515 520 525
 Arg Ala Ile Gly Ile Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr
 530 535 540
 Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu Pro Glu Asp Phe Ala Val
 545 550 555 560
 Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro Trp Thr Phe Gly Gln Gly
 565 570 575
 Thr Lys Val Glu Ile Lys Thr Thr Thr Pro Ala Pro Arg Pro Pro Thr
 580 585 590
 Pro Ala Pro Thr Ile Ala Ser Gln Pro Leu Ser Leu Arg Pro Glu Ala
 595 600 605
 Cys Arg Pro Ala Ala Gly Gly Ala Val His Thr Arg Gly Leu Asp Phe
 610 615 620
 Ala Cys Asp Ile Tyr Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly Val
 625 630 635 640
 Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr Cys Lys Arg Gly Arg Lys
 645 650 655
 Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met Arg Pro Val Gln Thr
 660 665 670
 Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe Pro Glu Glu Glu Glu
 675 680 685
 Gly Gly Cys Glu Leu Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro
 690 695 700
 Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly
 705 710 715 720
 Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro
 725 730 735
 Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr
 740 745 750
 Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly
 755 760 765
 Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln
 770 775 780
 Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln
 785 790 795 800
 Ala Leu Pro Pro Arg
 805

<210> SEQ ID NO 135

<211> LENGTH: 805

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 135

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu

-continued

1	5	10	15
His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp	20	25	30
Ile Leu Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly Glu	35	40	45
Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Asn Ile Gly Thr Ser Ile	50	55	60
Gln Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile Arg	65	70	75
Ser Ser Ser Glu Ser Ile Ser Gly Ile Pro Ser Arg Phe Ser Gly Ser	85	90	95
Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Glu Pro Glu	100	105	110
Asp Phe Ala Val Tyr Tyr Cys Gln Gln Ser Asn Thr Trp Pro Phe Thr	115	120	125
Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Gly Gly Gly Gly Ser Gly	130	135	140
Gly Gly Gly Ser Gly Gly Gly Gly Ser Pro Met Ala Lys Lys Gly Gly	145	150	155
Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Glu Val Gln	165	170	175
Leu Val Glu Ser Gly Ala Glu Val Lys Lys Pro Gly Ala Ser Val Lys	180	185	190
Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asn Tyr Ile Ile His	195	200	205
Trp Val Lys Gln Glu Pro Gly Gln Gly Leu Glu Trp Ile Gly Tyr Phe	210	215	220
Asn Pro Tyr Asn His Gly Thr Lys Tyr Asn Glu Lys Phe Lys Gly Arg	225	230	235
Ala Thr Leu Thr Ala Asp Lys Ser Ile Ser Thr Ala Tyr Met Glu Leu	245	250	255
Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys Ala Arg Ser	260	265	270
Gly Pro Tyr Ala Trp Phe Asp Thr Trp Gly Gln Gly Thr Thr Val Thr	275	280	285
Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly	290	295	300
Gly Ser Gly Gly Gly Gly Gly Ser Pro Met Ala Lys Lys Gly Gly Gly	305	310	315
Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser	325	330	335
Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu	340	345	350
Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Tyr Tyr	355	360	365
Ser Trp Thr Trp Val Arg Gln Pro Ala Gly Gln Gly Leu Glu Trp Ile	370	375	380
Gly Arg Ile His Ile Ser Gly Val Thr Asn His Asn Pro Ser Leu Lys	385	390	395
Ser Arg Val Ser Met Ser Ile Asp Thr Ser Arg Thr Gln Phe Ser Leu	405	410	415
Arg Leu Thr Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Phe Cys Ala	420	425	430

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Arg Ser Gly Gly Thr Tyr Ser Ala Phe Asp Ile Trp Gly Gln Gly Thr
    435                      440                      445

Met Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
    450                      455                      460

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Glu Ile Val Leu Thr Gln
    465                      470                      475                      480

Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly Glu Arg Ser Thr Leu Ser
    485                      490                      495

Cys Arg Ala Ser Gln Ser Phe Thr Ser Asn Tyr Leu Ala Trp Tyr Gln
    500                      505                      510

Gln Arg Pro Gly Gln Ala Pro Arg Leu Leu Ile Tyr Gly Ala Ser Thr
    515                      520                      525

Arg Ala Ile Gly Ile Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr
    530                      535                      540

Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu Pro Glu Asp Phe Ala Val
    545                      550                      555                      560

Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro Trp Thr Phe Gly Gln Gly
    565                      570                      575

Thr Lys Val Glu Ile Lys Thr Thr Thr Pro Ala Pro Arg Pro Pro Thr
    580                      585                      590

Pro Ala Pro Thr Ile Ala Ser Gln Pro Leu Ser Leu Arg Pro Glu Ala
    595                      600                      605

Cys Arg Pro Ala Ala Gly Gly Ala Val His Thr Arg Gly Leu Asp Phe
    610                      615                      620

Ala Cys Asp Ile Tyr Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly Val
    625                      630                      635                      640

Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr Cys Lys Arg Gly Arg Lys
    645                      650                      655

Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met Arg Pro Val Gln Thr
    660                      665                      670

Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe Pro Glu Glu Glu Glu
    675                      680                      685

Gly Gly Cys Glu Leu Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro
    690                      695                      700

Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly
    705                      710                      715                      720

Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro
    725                      730                      735

Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr
    740                      745                      750

Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly
    755                      760                      765

Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln
    770                      775                      780

Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln
    785                      790                      795                      800

Ala Leu Pro Pro Arg
    805

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<210> SEQ ID NO 136

<211> LENGTH: 817

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

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<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 136

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Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
1      5      10      15
His Ala Ala Arg Pro Gly Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu
20     25     30
Asp Ser Thr Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
35     40     45
Gly Ser Gly Gly Gly Gly Ser Asp Ile Val Met Thr Gln Thr Pro Ala
50     55     60
Ser Val Ser Glu Pro Val Gly Gly Ser Val Thr Ile Lys Cys Gln Ala
65     70     75     80
Ser Gln Ser Phe Tyr Asn Leu Leu Ala Trp Tyr Gln Gln Lys Pro Gly
85     90     95
Gln Pro Pro Lys Leu Leu Ile Tyr Asp Ala Ser Asp Leu Ala Ser Gly
100    105    110
Val Pro Ser Arg Phe Lys Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu
115    120    125
Thr Ile Ser Asp Leu Glu Cys Ala Asp Ala Ala Tyr Tyr Cys Gln
130    135    140
Ser Ala Asp Gly Ser Ser Tyr Ala Phe Gly Gly Gly Thr Glu Val Val
145    150    155    160
Val Lys Gly Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
165    170    175
Gly Ser Gly Gly Gly Gly Ser Gln Glu Gln Leu Glu Glu Ser Gly Gly
180    185    190
Gly Leu Val Lys Pro Glu Gly Ser Leu Thr Leu Thr Cys Thr Ala Ser
195    200    205
Gly Val Ser Phe Ser Ser Ser Tyr Trp Ile Tyr Trp Val Arg Gln Ala
210    215    220
Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys Ile Tyr Thr Gly Ser Ser
225    230    235    240
Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly Arg Phe Thr Val Ser
245    250    255
Glu Thr Ser Ser Thr Thr Val Thr Leu Gln Met Thr Ser Leu Thr Ala
260    265    270
Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ala Ser Ala Trp Thr Tyr
275    280    285
Gly Met Asp Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser Gly
290    295    300
Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Ser Gly Leu Ser
305    310    315    320
Gly Arg Ser Asp Asn His Ser Pro Leu Gly Leu Ala Gly Ser Gly Gly
325    330    335
Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln Val Gln Leu
340    345    350
Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu Thr Leu Ser Leu
355    360    365
Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Tyr Tyr Ser Trp Thr Trp
370    375    380
Val Arg Gln Pro Ala Gly Gln Gly Leu Glu Trp Ile Gly Arg Ile His
385    390    395    400

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Ile	Ser	Gly	Val	Thr	Asn	His	Asn	Pro	Ser	Leu	Lys	Ser	Arg	Val	Ser	
				405					410					415		
Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln	Phe	Ser	Leu	Arg	Leu	Thr	Ser	
				420					425					430		
Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Phe	Cys	Ala	Arg	Ser	Gly	Gly	
				435					440					445		
Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly	Gln	Gly	Thr	Met	Val	Thr	Val	
				450					455					460		
Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	
				465					470					475		
Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	
				485					490					495		
Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser	Thr	Leu	Ser	Cys	Arg	Ala	Ser	
				500					505					510		
Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Arg	Pro	Gly	
				515					520					525		
Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Ser	Thr	Arg	Ala	Ile	Gly	
				530					535					540		
Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	
				545					550					555		
Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	
				565					570					575		
Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	
				580					585					590		
Ile	Lys	Thr	Thr	Thr	Pro	Ala	Pro	Arg	Pro	Pro	Thr	Pro	Ala	Pro	Thr	
				595					600					605		
Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu	Arg	Pro	Glu	Ala	Cys	Arg	Pro	Ala	
				610					615					620		
Ala	Gly	Gly	Ala	Val	His	Thr	Arg	Gly	Leu	Asp	Phe	Ala	Cys	Asp	Ile	
				625					630					635		
Tyr	Ile	Trp	Ala	Pro	Leu	Ala	Gly	Thr	Cys	Gly	Val	Leu	Leu	Leu	Ser	
				645					650					655		
Leu	Val	Ile	Thr	Leu	Tyr	Cys	Lys	Arg	Gly	Arg	Lys	Lys	Leu	Leu	Tyr	
				660					665					670		
Ile	Phe	Lys	Gln	Pro	Phe	Met	Arg	Pro	Val	Gln	Thr	Thr	Gln	Glu	Glu	
				675					680					685		
Asp	Gly	Cys	Ser	Cys	Arg	Phe	Pro	Glu	Glu	Glu	Glu	Gly	Gly	Cys	Glu	
				690					695					700		
Leu	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln	Gln	
				705					710					715		
Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu	Glu	
				725					730					735		
Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met	Gly	Gly	
				740					745					750		
Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn	Glu	Leu	Gln	
				755					760					765		
Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys	Gly	Glu	
				770					775					780		
Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr	Gln	Gly	Leu	Ser	Thr	
				785					790					795		
Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His	Met	Gln	Ala	Leu	Pro	Pro	

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Arg

<210> SEQ ID NO 137
 <211> LENGTH: 817
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 137

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Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1             5             10             15

His Ala Ala Arg Pro Gly Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu
 20             25             30

Asp Ser Thr Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
 35             40             45

Gly Ser Gly Gly Gly Gly Ser Asp Ile Val Met Thr Gln Thr Pro Ala
 50             55             60

Ser Val Ser Glu Pro Val Gly Gly Ser Val Thr Ile Lys Cys Gln Ala
 65             70             75             80

Ser Gln Ser Phe Tyr Asn Leu Leu Ala Trp Tyr Gln Gln Lys Pro Gly
 85             90             95

Gln Pro Pro Lys Leu Leu Ile Tyr Asp Ala Ser Asp Leu Ala Ser Gly
 100            105            110

Val Pro Ser Arg Phe Lys Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu
 115            120            125

Thr Ile Ser Asp Leu Glu Cys Ala Asp Ala Ala Tyr Tyr Cys Gln
 130            135            140

Ser Ala Asp Gly Ser Ser Tyr Ala Phe Gly Gly Gly Thr Glu Val Val
 145            150            155            160

Val Lys Gly Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
 165            170            175

Gly Ser Gly Gly Gly Gly Ser Gln Glu Gln Leu Glu Glu Ser Gly Gly
 180            185            190

Gly Leu Val Lys Pro Glu Gly Ser Leu Thr Leu Thr Cys Thr Ala Ser
 195            200            205

Gly Val Ser Phe Ser Ser Ser Tyr Trp Ile Tyr Trp Val Arg Gln Ala
 210            215            220

Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys Ile Tyr Thr Gly Ser Ser
 225            230            235            240

Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly Arg Phe Thr Val Ser
 245            250            255

Glu Thr Ser Ser Thr Thr Val Thr Leu Gln Met Thr Ser Leu Thr Ala
 260            265            270

Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ala Ser Ala Trp Thr Tyr
 275            280            285

Gly Met Asp Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser Gly
 290            295            300

Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Val His
 305            310            315            320

Met Pro Leu Gly Phe Leu Gly Pro Arg Gln Ala Arg Val Val Asn Gly
 325            330            335

Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln Val Gln Leu
 340            345            350
  
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Gln		Glu	Ser 355	Gly	Pro	Gly	Leu	Val 360	Lys	Pro	Ser	Glu	Thr 365	Leu	Ser	Leu
Thr	Cys 370	Thr	Val	Ser	Gly	Gly 375	Ser	Ile	Ser	Tyr	Tyr 380	Ser	Trp	Thr	Trp	
Val	Arg	Gln	Pro	Ala	Gly 390	Gln	Gly	Leu	Glu	Trp 395	Ile	Gly	Arg	Ile	His 400	
Ile	Ser	Gly	Val	Thr 405	Asn	His	Asn	Pro	Ser 410	Leu	Lys	Ser	Arg	Val 415	Ser	
Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln	Phe 425	Ser	Leu	Arg	Leu	Thr	Ser	
Val	Thr	Ala	Ala	Asp	Thr	Ala	Val 440	Tyr	Phe	Cys	Ala	Arg 445	Ser	Gly	Gly	
Thr	Tyr 450	Ser	Ala	Phe	Asp	Ile 455	Trp	Gly	Gln	Gly	Thr 460	Met	Val	Thr	Val	
Ser	Ser	Gly	Gly	Gly 470	Gly	Ser	Gly	Gly	Gly	Gly 475	Ser	Gly	Gly	Gly	Gly 480	
Ser	Gly	Gly	Gly	Gly 485	Ser	Glu	Ile	Val	Leu 490	Thr	Gln	Ser	Pro	Gly 495	Thr	
Leu	Ser	Leu	Ser 500	Pro	Gly	Glu	Arg	Ser 505	Thr	Leu	Ser	Cys 510	Arg	Ala	Ser	
Gln	Ser	Phe 515	Thr	Ser	Asn	Tyr	Leu 520	Ala	Trp	Tyr	Gln	Gln 525	Arg	Pro	Gly	
Gln	Ala 530	Pro	Arg	Leu	Leu	Ile 535	Tyr	Gly	Ala	Ser	Thr 540	Arg	Ala	Ile	Gly	
Ile	Pro	Asp	Arg	Phe 550	Ser	Gly	Ser	Gly	Ser	Gly 555	Thr	Asp	Phe	Thr	Leu 560	
Thr	Ile	Ser	Arg	Leu 565	Glu	Pro	Glu	Asp	Phe 570	Ala	Val	Tyr	Tyr	Cys 575	Gln	
Gln	Tyr	Gly	Ser 580	Ser	Pro	Trp	Thr	Phe 585	Gly	Gln	Gly	Thr	Lys 590	Val	Glu	
Ile	Lys 595	Thr	Thr	Thr	Pro	Ala	Pro 600	Arg	Pro	Pro	Thr	Pro 605	Ala	Pro	Thr	
Ile	Ala 610	Ser	Gln	Pro	Leu	Ser 615	Leu	Arg	Pro	Glu	Ala 620	Cys	Arg	Pro	Ala	
Ala	Gly	Gly	Ala	Val	His 630	Thr	Arg	Gly	Leu	Asp 635	Phe	Ala	Cys	Asp	Ile 640	
Tyr	Ile	Trp	Ala	Pro 645	Leu	Ala	Gly	Thr	Cys 650	Gly	Val	Leu	Leu	Leu	Ser 655	
Leu	Val	Ile	Thr 660	Leu	Tyr	Cys	Lys	Arg 665	Gly	Arg	Lys	Lys 670	Leu	Leu	Tyr	
Ile	Phe 675	Lys	Gln	Pro	Phe	Met	Arg 680	Pro	Val	Gln	Thr	Thr 685	Gln	Glu	Glu	
Asp	Gly 690	Cys	Ser	Cys	Arg	Phe 695	Pro	Glu	Glu	Glu	Glu 700	Gly	Gly	Cys	Glu	
Leu	Arg	Val	Lys	Phe 710	Ser	Arg	Ser	Ala	Asp 715	Ala	Pro	Ala	Tyr	Gln	Gln 720	
Gly	Gln	Asn	Gln	Leu 725	Tyr	Asn	Glu	Leu	Asn 730	Leu	Gly	Arg	Arg	Glu	Glu 735	
Tyr	Asp	Val	Leu 740	Asp	Lys	Arg	Arg	Gly 745	Arg	Asp	Pro	Glu	Met	Gly	Gly	
Lys	Pro	Arg	Arg 755	Lys	Asn	Pro	Gln 760	Glu	Gly	Leu	Tyr	Asn 765	Glu	Leu	Gln	
Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys	Gly	Glu	

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770	775	780
Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr		
785	790	795 800
Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro		
	805	810 815
Arg		
<210> SEQ ID NO 138		
<211> LENGTH: 817		
<212> TYPE: PRT		
<213> ORGANISM: Artificial Sequence		
<220> FEATURE:		
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide		
<400> SEQUENCE: 138		
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His Ala Ala Arg Pro Gly Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu		
	20	25 30
Asp Ser Thr Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly		
	35	40 45
Gly Ser Gly Gly Gly Gly Ser Asp Ile Val Met Thr Gln Thr Pro Ala		
	50	55 60
Ser Val Ser Glu Pro Val Gly Gly Ser Val Thr Ile Lys Cys Gln Ala		
	65	70 75 80
Ser Gln Ser Phe Tyr Asn Leu Leu Ala Trp Tyr Gln Gln Lys Pro Gly		
	85	90 95
Gln Pro Pro Lys Leu Leu Ile Tyr Asp Ala Ser Asp Leu Ala Ser Gly		
	100	105 110
Val Pro Ser Arg Phe Lys Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu		
	115	120 125
Thr Ile Ser Asp Leu Glu Cys Ala Asp Ala Ala Tyr Tyr Cys Gln		
	130	135 140
Ser Ala Asp Gly Ser Ser Tyr Ala Phe Gly Gly Gly Thr Glu Val Val		
	145	150 155 160
Val Lys Gly Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly		
	165	170 175
Gly Ser Gly Gly Gly Gly Ser Gln Glu Gln Leu Glu Glu Ser Gly Gly		
	180	185 190
Gly Leu Val Lys Pro Glu Gly Ser Leu Thr Leu Thr Cys Thr Ala Ser		
	195	200 205
Gly Val Ser Phe Ser Ser Ser Tyr Trp Ile Tyr Trp Val Arg Gln Ala		
	210	215 220
Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys Ile Tyr Thr Gly Ser Ser		
	225	230 235 240
Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly Arg Phe Thr Val Ser		
	245	250 255
Glu Thr Ser Ser Thr Thr Val Thr Leu Gln Met Thr Ser Leu Thr Ala		
	260	265 270
Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ala Ser Ala Trp Thr Tyr		
	275	280 285
Gly Met Asp Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser Gly		
	290	295 300
Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly		

305				310				315				320			
Gly 305	Gly	Ser	Pro	Met 325	Ala	Lys	Lys	Gly 330	Gly	Gly	Ser	Gly 335	Gly		
Gly	Ser	Gly 340	Gly	Gly	Ser	Gly	Gly 345	Gly	Gly	Ser	Gln	Val 350	Gln	Leu	
Gln	Glu 355	Ser	Gly	Pro	Gly	Leu	Val 360	Lys	Pro	Ser	Glu 365	Thr	Leu	Ser	Leu
Thr 370	Cys	Thr	Val	Ser	Gly 375	Gly	Ser	Ile	Ser	Tyr 380	Tyr	Ser	Trp	Thr	Trp
Val 385	Arg	Gln	Pro	Ala 390	Gly	Gln	Gly	Leu	Glu 395	Trp	Ile	Gly	Arg	Ile 400	His
Ile	Ser	Gly	Val 405	Thr	Asn	His	Asn	Pro 410	Ser	Leu	Lys	Ser	Arg	Val 415	Ser
Met	Ser	Ile	Asp 420	Thr	Ser	Arg	Thr	Gln 425	Phe	Ser	Leu	Arg	Leu 430	Thr	Ser
Val	Thr 435	Ala	Ala	Asp	Thr	Ala	Val 440	Tyr	Phe	Cys	Ala	Arg 445	Ser	Gly	Gly
Thr 450	Tyr	Ser	Ala	Phe	Asp	Ile 455	Trp	Gly	Gln	Gly	Thr 460	Met	Val	Thr	Val
Ser 465	Ser	Gly	Gly	Gly	Gly 470	Ser	Gly	Gly	Gly	Gly 475	Ser	Gly	Gly	Gly 480	Gly
Ser	Gly	Gly	Gly 485	Gly	Ser	Glu	Ile	Val	Leu 490	Thr	Gln	Ser	Pro	Gly 495	Thr
Leu	Ser	Leu	Ser 500	Pro	Gly	Glu	Arg	Ser 505	Thr	Leu	Ser	Cys	Arg 510	Ala	Ser
Gln	Ser	Phe 515	Thr	Ser	Asn	Tyr	Leu 520	Ala	Trp	Tyr	Gln	Gln 525	Arg	Pro	Gly
Gln 530	Ala	Pro	Arg	Leu	Leu	Ile 535	Tyr	Gly	Ala	Ser	Thr 540	Arg	Ala	Ile	Gly
Ile 545	Pro	Asp	Arg	Phe	Ser 550	Gly	Ser	Gly	Ser	Gly 555	Thr	Asp	Phe	Thr	Leu
Thr	Ile	Ser	Arg	Leu 565	Glu	Pro	Glu	Asp	Phe 570	Ala	Val	Tyr	Tyr	Cys 575	Gln
Gln	Tyr	Gly	Ser 580	Ser	Pro	Trp	Thr	Phe 585	Gly	Gln	Gly	Thr	Lys	Val	Glu
Ile	Lys 595	Thr	Thr	Thr	Pro	Ala	Pro 600	Arg	Pro	Pro	Thr	Pro 605	Ala	Pro	Thr
Ile 610	Ala	Ser	Gln	Pro	Leu	Ser 615	Leu	Arg	Pro	Glu	Ala 620	Cys	Arg	Pro	Ala
Ala 625	Gly	Gly	Ala	Val	His 630	Thr	Arg	Gly	Leu	Asp 635	Phe	Ala	Cys	Asp	Ile 640
Tyr	Ile	Trp	Ala 645	Pro	Leu	Ala	Gly	Thr	Cys 650	Gly	Val	Leu	Leu	Leu 655	Ser
Leu	Val	Ile	Thr 660	Leu	Tyr	Cys	Lys	Arg 665	Gly	Arg	Lys	Lys	Leu	Leu 670	Tyr
Ile	Phe 675	Lys	Gln	Pro	Phe	Met	Arg 680	Pro	Val	Gln	Thr	Thr 685	Gln	Glu	Glu
Asp 690	Gly	Cys	Ser	Cys	Arg	Phe 695	Pro	Glu	Glu	Glu	Glu 700	Gly	Gly	Cys	Glu
Leu 705	Arg	Val	Lys	Phe	Ser 710	Arg	Ser	Ala	Asp 715	Ala	Pro	Ala	Tyr	Gln	Gln
Gly	Gln	Asn	Gln 725	Leu	Tyr	Asn	Glu	Leu	Asn 730	Leu	Gly	Arg	Arg	Glu 735	Glu

Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met	Gly	Gly
			740					745					750		
Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn	Glu	Leu	Gln
		755					760					765			
Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys	Gly	Glu
	770					775					780				
Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr	Gln	Gly	Leu	Ser	Thr
785					790					795					800
Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His	Met	Gln	Ala	Leu	Pro	Pro
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<210> SEQ ID NO 139
<211> LENGTH: 832
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide
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His	Ala	Ala	Arg 20	Pro	Gly	Lys	Pro	Ile 25	Pro	Asn	Pro	Leu 30	Leu	Gly	Leu
Asp	Ser	Thr 35	Gly	Gly	Gly	Gly	Ser 40	Gly	Gly	Gly	Gly	Ser 45	Gly	Gly	Gly
Gly	Ser 50	Gly	Gly	Gly	Gly	Ser 55	Asp	Ile	Val	Met	Thr 60	Gln	Thr	Pro	Ala
Ser 65	Val	Ser	Glu	Pro	Val 70	Gly	Gly	Ser	Val	Thr 75	Ile	Lys	Cys	Gln	Ala 80
Ser	Gln	Ser	Phe 85	Tyr	Asn	Leu	Leu	Ala	Trp 90	Tyr	Gln	Gln	Lys	Pro 95	Gly
Gln	Pro	Pro	Lys 100	Leu	Leu	Ile	Tyr	Asp 105	Ala	Ser	Asp	Leu	Ala	Ser	Gly
Val	Pro 115	Ser	Arg	Phe	Lys	Gly	Ser 120	Gly	Ser	Gly	Thr	Asp 125	Phe	Thr	Leu
Thr 130	Ile	Ser	Asp	Leu	Glu	Cys 135	Ala	Asp	Ala	Ala	Ala 140	Tyr	Tyr	Cys	Gln
Ser 145	Ala	Asp	Gly	Ser	Ser 150	Tyr	Ala	Phe	Gly	Gly 155	Gly	Thr	Glu	Val	Val 160
Val	Lys	Gly	Gly	Gly 165	Gly	Gly	Ser	Gly	Gly 170	Gly	Gly	Ser	Gly	Gly 175	Gly
Gly	Ser	Pro	Met 180	Ala	Lys	Lys	Gly 185	Gly	Gly	Gly	Ser	Gly 190	Gly	Gly	Gly
Ser	Gly 195	Gly	Gly	Gly	Ser	Gln	Glu 200	Gln	Leu	Glu	Glu	Ser 205	Gly	Gly	Gly
Leu 210	Val	Lys	Pro	Glu	Gly	Ser 215	Leu	Thr	Leu	Thr	Cys 220	Thr	Ala	Ser	Gly
Val 225	Ser	Phe	Ser	Ser 230	Ser	Tyr	Trp	Ile	Tyr 235	Trp	Val	Arg	Gln	Ala	Pro 240
Gly	Lys	Gly	Leu 245	Glu	Trp	Ile	Ala	Cys	Ile 250	Tyr	Thr	Gly	Ser	Ser	Gly
Ser	Thr	Tyr 260	Ala	Ser	Trp	Ala	Lys 265	Gly	Arg	Phe	Thr 270	Val	Ser	Glu	

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Thr	Ser	Ser	Thr	Thr	Val	Thr	Leu	Gln	Met	Thr	Ser	Leu	Thr	Ala	Ala	
		275					280					285				
Asp	Thr	Ala	Thr	Tyr	Phe	Cys	Ala	Arg	Ala	Ser	Ala	Trp	Thr	Tyr	Gly	
	290					295					300					
Met	Asp	Leu	Trp	Gly	Pro	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	
	305				310					315					320	
Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	
			325						330					335		
Gly	Ser	Pro	Met	Ala	Lys	Lys	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	
			340				345						350			
Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	Gln	
	355					360						365				
Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Glu	Thr	Leu	Ser	Leu	Thr	
	370					375					380					
Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile	Ser	Tyr	Tyr	Ser	Trp	Thr	Trp	Val	
	385				390					395					400	
Arg	Gln	Pro	Ala	Gly	Gln	Gly	Leu	Glu	Trp	Ile	Gly	Arg	Ile	His	Ile	
			405						410					415		
Ser	Gly	Val	Thr	Asn	His	Asn	Pro	Ser	Leu	Lys	Ser	Arg	Val	Ser	Met	
			420					425					430			
Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln	Phe	Ser	Leu	Arg	Leu	Thr	Ser	Val	
	435						440					445				
Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Phe	Cys	Ala	Arg	Ser	Gly	Gly	Thr	
	450					455					460					
Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly	Gln	Gly	Thr	Met	Val	Thr	Val	Ser	
	465				470					475					480	
Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	
			485					490						495		
Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu	
			500					505					510			
Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser	Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	
	515					520						525				
Ser	Phe	Thr	Ser	Asn	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Arg	Pro	Gly	Gln	
	530					535					540					
Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Ser	Thr	Arg	Ala	Ile	Gly	Ile	
	545				550					555					560	
Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	
			565					570					575			
Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	
			580				585						590			
Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile	
	595						600					605				
Lys	Thr	Thr	Thr	Pro	Ala	Pro	Arg	Pro	Pro	Thr	Pro	Ala	Pro	Thr	Ile	
	610					615					620					
Ala	Ser	Gln	Pro	Leu	Ser	Leu	Arg	Pro	Glu	Ala	Cys	Arg	Pro	Ala	Ala	
	625				630					635					640	
Gly	Gly	Ala	Val	His	Thr	Arg	Gly	Leu	Asp	Phe	Ala	Cys	Asp	Ile	Tyr	
				645				650						655		
Ile	Trp	Ala	Pro	Leu	Ala	Gly	Thr	Cys	Gly	Val	Leu	Leu	Leu	Ser	Leu	
			660					665					670			
Val	Ile	Thr	Leu	Tyr	Cys	Lys	Arg	Gly	Arg	Lys	Lys	Leu	Leu	Tyr	Ile	
	675					680						685				

Phe 690	Lys	Gln	Pro	Phe	Met	Arg 695	Pro	Val	Gln	Thr	Thr 700	Gln	Glu	Leu	Asp
Gly 705	Cys	Ser	Cys	Arg	Phe 710	Pro	Glu	Glu	Glu	Glu 715	Gly	Gly	Cys	Glu	Leu 720
Arg	Val	Lys	Phe	Ser 725	Arg	Ser	Ala	Asp	Ala 730	Pro	Ala	Tyr	Gln	Gln	Gly 735
Gln	Asn	Gln	Leu	Tyr 740	Asn	Glu	Leu	Asn	Leu 745	Gly	Arg	Arg	Glu 750	Glu	Tyr
Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly 760	Arg	Asp	Pro	Glu	Met 765	Gly	Gly	Lys
Pro	Arg	Arg	Lys	Asn	Pro	Gln 775	Glu	Gly	Leu	Tyr	Asn 780	Glu	Leu	Gln	Lys
Asp	Lys	Met	Ala	Glu	Ala 790	Tyr	Ser	Glu	Ile	Gly 795	Met	Lys	Gly	Glu	Arg 800
Arg	Arg	Gly	Lys	Gly 805	His	Asp	Gly	Leu	Tyr 810	Gln	Gly	Leu	Ser	Thr	Ala 815
Thr	Lys	Asp	Thr 820	Tyr	Asp	Ala	Leu	His 825	Met	Gln	Ala	Leu 830	Pro	Pro	Arg
<210> SEQ ID NO 140															
<211> LENGTH: 817															
<212> TYPE: PRT															
<213> ORGANISM: Artificial Sequence															
<220> FEATURE:															
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide															
<400> SEQUENCE: 140															
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His	Ala	Ala	Arg 20	Pro	Gly	Lys	Pro	Ile 25	Pro	Asn	Pro	Leu	Leu	Gly	Leu
Asp	Ser	Thr 35	Gly	Gly	Gly	Gly	Ser 40	Gly	Gly	Gly	Gly	Ser 45	Gly	Gly	Gly
Gly 50	Ser	Gly	Gly	Gly	Gly	Ser 55	Asp	Ile	Val	Met 60	Thr	Gln	Thr	Pro	Ala
Ser 65	Val	Ser	Glu	Pro	Val 70	Gly	Gly	Ser	Val	Thr 75	Ile	Lys	Cys	Gln	Ala 80
Ser	Gln	Ser	Phe 85	Tyr	Asn	Leu	Leu	Ala	Trp 90	Tyr	Gln	Gln	Lys	Pro	Gly 95
Gln	Pro	Pro	Lys 100	Leu	Leu	Ile	Tyr	Asp	Ala 105	Ser	Asp	Leu	Ala	Ser	Gly 110
Val 115	Pro	Ser	Arg	Phe	Lys	Gly	Ser 120	Gly	Ser	Gly	Thr	Asp 125	Phe	Thr	Leu
Thr 130	Ile	Ser	Asp	Leu	Glu	Cys 135	Ala	Asp	Ala	Ala	Ala 140	Tyr	Tyr	Cys	Gln
Ser 145	Ala	Asp	Gly	Ser	Ser	Tyr	Ala	Phe	Gly	Gly	Gly 155	Thr	Glu	Val	Val 160
Val	Lys	Gly	Gly	Gly 165	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly
Gly	Ser	Gly	Gly 180	Gly	Gly	Ser	Gln	Glu	Gln 185	Leu	Glu	Glu	Ser	Gly	Gly
Gly	Leu	Val 195	Lys	Pro	Glu	Gly	Ser 200	Leu	Thr	Leu	Thr	Cys 205	Thr	Ala	Ser
Gly 210	Val	Ser	Phe	Ser	Ser	Ser 215	Tyr	Trp	Ile	Tyr	Trp 220	Val	Arg	Gln	Ala

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Pro	Gly	Lys	Gly	Leu	Glu	Trp	Ile	Ala	Cys	Ile	Tyr	Thr	Gly	Ser	Ser	
225					230					235					240	
Gly	Ser	Thr	Tyr	Tyr	Ala	Ser	Trp	Ala	Lys	Gly	Arg	Phe	Thr	Val	Ser	
				245					250					255		
Glu	Thr	Ser	Ser	Thr	Thr	Val	Thr	Leu	Gln	Met	Thr	Ser	Leu	Thr	Ala	
			260					265					270			
Ala	Asp	Thr	Ala	Thr	Tyr	Phe	Cys	Ala	Arg	Ala	Ser	Ala	Trp	Thr	Tyr	
	275						280					285				
Gly	Met	Asp	Leu	Trp	Gly	Pro	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	
	290					295					300					
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Ser	Gly	Ser	Gly	
305					310					315					320	
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Ser	
				325					330					335		
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	
			340					345					350			
Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Glu	Thr	Leu	Ser	Leu	
	355					360						365				
Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile	Ser	Tyr	Tyr	Ser	Trp	Thr	Trp	
	370					375					380					
Val	Arg	Gln	Pro	Ala	Gly	Gln	Gly	Leu	Glu	Trp	Ile	Gly	Arg	Ile	His	
385					390					395					400	
Ile	Ser	Gly	Val	Thr	Asn	His	Asn	Pro	Ser	Leu	Lys	Ser	Arg	Val	Ser	
				405					410					415		
Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln	Phe	Ser	Leu	Arg	Leu	Thr	Ser	
			420					425					430			
Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Phe	Cys	Ala	Arg	Ser	Gly	Gly	
	435						440					445				
Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly	Gln	Gly	Thr	Met	Val	Thr	Val	
	450					455					460					
Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	
465					470					475					480	
Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	
				485					490					495		
Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser	Thr	Leu	Ser	Cys	Arg	Ala	Ser	
		500						505					510			
Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Arg	Pro	Gly	
		515					520					525				
Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Ser	Thr	Arg	Ala	Ile	Gly	
	530					535					540					
Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	
545					550					555					560	
Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	
				565					570					575		
Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	
			580					585					590			
Ile	Lys	Thr	Thr	Thr	Pro	Ala	Pro	Arg	Pro	Pro	Thr	Pro	Ala	Pro	Thr	
		595					600					605				
Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu	Arg	Pro	Glu	Ala	Cys	Arg	Pro	Ala	
	610					615					620					
Ala	Gly	Gly	Ala	Val	His	Thr	Arg	Gly	Leu	Asp	Phe	Ala	Cys	Asp	Ile	
625					630					635					640	

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Tyr Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly Val Leu Leu Leu Ser
      645                               650                655

Leu Val Ile Thr Leu Tyr Cys Lys Arg Gly Arg Lys Lys Leu Leu Tyr
      660                               665                670

Ile Phe Lys Gln Pro Phe Met Arg Pro Val Gln Thr Thr Gln Glu Glu
      675                               680                685

Asp Gly Cys Ser Cys Arg Phe Pro Glu Glu Glu Glu Gly Gly Cys Glu
      690                               695                700

Leu Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln
      705                               710                715                720

Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu
      725                               730                735

Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly
      740                               745                750

Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln
      755                               760                765

Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu
      770                               775                780

Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr
      785                               790                795                800

Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro
      805                               810                815

```

Arg

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<210> SEQ ID NO 141
<211> LENGTH: 822
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

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<400> SEQUENCE: 141

```

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1      5      10      15

His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp
      20      25      30

Ile Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly Gly
      35      40      45

Ser Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Phe Tyr Asn Leu Leu
      50      55      60

Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr
      65      70      75      80

Asp Ala Ser Asp Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly Ser
      85      90      95

Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala
      100     105     110

Asp Ala Ala Ala Tyr Tyr Cys Gln Ser Ala Asp Gly Ser Ser Tyr Ala
      115     120     125

Phe Gly Gly Gly Thr Glu Val Val Val Lys Gly Gly Gly Gly Gly Ser
      130     135     140

Gly Gly Gly Gly Ser Leu Ser Gly Arg Ser Asp Asn His Ser Pro Leu
      145     150     155     160

Gly Leu Ala Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln
      165     170     175

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Glu 180	Gln	Leu	Glu 180	Glu	Ser	Gly	Gly 185	Leu	Val	Lys	Pro 190	Glu 190	Gly	Ser
Leu 195	Thr	Leu 195	Thr	Cys	Thr	Ala	Ser 200	Gly	Val	Ser	Phe 205	Ser 205	Ser	Tyr
Trp 210	Ile	Tyr	Trp	Val	Arg	Gln 215	Ala	Pro	Gly	Lys	Gly 220	Leu	Glu	Trp
Ala 225	Cys	Ile	Tyr	Thr	Gly 230	Ser	Ser	Gly	Ser	Thr 235	Tyr	Tyr	Ala	Trp 240
Ala	Lys	Gly	Arg	Phe 245	Thr	Val	Ser	Glu	Thr 250	Ser	Ser	Thr	Thr	Val 255
Leu	Gln	Met	Thr 260	Ser	Leu	Thr	Ala	Ala 265	Asp	Thr	Ala	Thr	Tyr	Phe
Ala	Arg	Ala 275	Ser	Ala	Trp	Thr	Tyr 280	Gly	Met	Asp	Leu	Trp 285	Gly	Pro
Thr 290	Leu	Val	Thr	Val	Ser	Ser 295	Gly	Gly	Gly	Gly	Ser 300	Gly	Gly	Gly
Ser 305	Lys	Pro	Ile	Pro	Asn 310	Pro	Leu	Leu	Gly	Leu	Asp 315	Ser	Thr	Gly
Gly	Ser	Gly	Leu	Ser 325	Gly	Arg	Ser	Asp	Asn 330	His	Ser	Pro	Leu	Gly
Ala	Gly	Ser	Gly 340	Gly	Gly	Ser	Gly	Gly 345	Gly	Gly	Ser	Gly	Gly	Gly
Ser	Gln	Val	Gln	Leu	Gln	Glu	Ser 360	Gly	Pro	Gly	Leu	Val 365	Lys	Pro
Glu 370	Thr	Leu	Ser	Leu	Thr	Cys 375	Thr	Val	Ser	Gly	Gly 380	Ser	Ile	Ser
Tyr 385	Ser	Trp	Thr	Trp	Val	Arg	Gln	Pro	Ala	Gly 395	Gln	Gly	Leu	Glu
Ile	Gly	Arg	Ile	His 405	Ile	Ser	Gly	Val	Thr 410	Asn	His	Asn	Pro	Ser
Lys	Ser	Arg	Val	Ser	Met	Ser	Ile	Asp 425	Thr	Ser	Arg	Thr	Gln	Phe
Leu	Arg	Leu	Thr	Ser	Val	Thr	Ala 440	Ala	Asp	Thr	Ala	Val 445	Tyr	Phe
Ala	Arg	Ser	Gly	Gly	Thr	Tyr 455	Ser	Ala	Phe	Asp	Ile	Trp	Gly	Gln
Thr 465	Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly 475	Ser	Gly	Gly	Gly
Ser	Gly	Gly	Gly	Gly 485	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val	Leu
Gln	Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser	Thr
Ser	Cys	Arg	Ala	Ser	Gln	Ser	Phe 520	Thr	Ser	Asn	Tyr	Leu	Ala	Trp
Gln	Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala
Thr 545	Arg	Ala	Ile	Gly	Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser
Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp	Phe
Val	Tyr	Tyr	Cys	Gln	Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe	Gly
Gly	Thr	Lys	Val	Glu	Ile	Lys	Thr	Thr	Thr	Pro	Ala	Pro	Arg	Pro

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595				600				605							
Thr	Pro	Ala	Pro	Thr	Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu	Arg	Pro	Glu
610						615					620				
Ala	Cys	Arg	Pro	Ala	Ala	Gly	Gly	Ala	Val	His	Thr	Arg	Gly	Leu	Asp
625					630					635					640
Phe	Ala	Cys	Asp	Ile	Tyr	Ile	Trp	Ala	Pro	Leu	Ala	Gly	Thr	Cys	Gly
			645						650					655	
Val	Leu	Leu	Leu	Ser	Leu	Val	Ile	Thr	Leu	Tyr	Cys	Lys	Arg	Gly	Arg
			660						665				670		
Lys	Lys	Leu	Leu	Tyr	Ile	Phe	Lys	Gln	Pro	Phe	Met	Arg	Pro	Val	Gln
		675					680						685		
Thr	Thr	Gln	Glu	Glu	Asp	Gly	Cys	Ser	Cys	Arg	Phe	Pro	Glu	Glu	Glu
	690					695					700				
Glu	Gly	Gly	Cys	Glu	Leu	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala
705					710					715					720
Pro	Ala	Tyr	Gln	Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu
			725						730					735	
Gly	Arg	Arg	Glu	Glu	Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp
			740						745					750	
Pro	Glu	Met	Gly	Gly	Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu
		755					760						765		
Tyr	Asn	Glu	Leu	Gln	Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile
	770				775						780				
Gly	Met	Lys	Gly	Glu	Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr
785					790					795					800
Gln	Gly	Leu	Ser	Thr	Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His	Met
			805						810					815	
Gln	Ala	Leu	Pro	Pro	Arg										
			820												

<210> SEQ ID NO 142

<211> LENGTH: 821

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 142

Met	Ala	Leu	Pro	Val	Thr	Ala	Leu	Leu	Leu	Pro	Leu	Ala	Leu	Leu	Leu
1				5						10				15	
His	Ala	Ala	Arg	Pro	Gly	Tyr	Pro	Tyr	Asp	Val	Pro	Asp	Tyr	Ala	Asp
		20						25					30		
Ile	Val	Met	Thr	Gln	Thr	Pro	Ala	Ser	Val	Ser	Glu	Pro	Val	Gly	Gly
		35					40					45			
Ser	Val	Thr	Ile	Lys	Cys	Gln	Ala	Ser	Gln	Ser	Phe	Tyr	Asn	Leu	Leu
		50					55				60				
Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Pro	Pro	Lys	Leu	Leu	Ile	Tyr
65					70					75					80
Asp	Ala	Ser	Asp	Leu	Ala	Ser	Gly	Val	Pro	Ser	Arg	Phe	Lys	Gly	Ser
				85					90					95	
Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Asp	Leu	Glu	Cys	Ala
			100					105					110		
Asp	Ala	Ala	Ala	Tyr	Tyr	Cys	Gln	Ser	Ala	Asp	Gly	Ser	Ser	Tyr	Ala
		115					120					125			

Phe 130	Gly	Gly	Gly	Thr	Glu	Val 135	Val	Val	Lys	Gly	Gly 140	Gly	Gly	Gly	Ser
Gly 145	Gly	Gly	Gly	Ser	Val 150	His	Met	Pro	Leu	Gly 155	Phe	Leu	Gly	Pro	Arg 160
Gln	Ala	Arg	Val	Val 165	Asn	Gly	Gly	Ser	Gly 170	Gly	Gly	Gly	Ser	Gln 175	Glu
Gln	Leu	Glu	Glu	Ser	Gly	Gly	Gly	Leu 185	Val	Lys	Pro	Glu	Gly 190	Ser	Leu
Thr	Leu	Thr	Cys	Thr	Ala	Ser	Gly 200	Val	Ser	Phe	Ser	Ser 205	Ser	Tyr	Trp
Ile	Tyr	Trp	Val	Arg	Gln	Ala 215	Pro	Gly	Lys	Gly	Leu 220	Glu	Trp	Ile	Ala
Cys 225	Ile	Tyr	Thr	Gly	Ser 230	Ser	Gly	Ser	Thr	Tyr 235	Tyr	Ala	Ser	Trp	Ala 240
Lys	Gly	Arg	Phe	Thr 245	Val	Ser	Glu	Thr	Ser 250	Ser	Thr	Thr	Val 255	Thr	Leu
Gln	Met	Thr	Ser	Leu 260	Thr	Ala	Ala	Asp 265	Thr	Ala	Thr	Tyr 270	Phe	Cys	Ala
Arg	Ala	Ser 275	Ala	Trp	Thr	Tyr	Gly 280	Met	Asp	Leu	Trp	Gly 285	Pro	Gly	Thr
Leu	Val 290	Thr	Val	Ser	Ser	Gly 295	Gly	Gly	Gly	Ser	Gly 300	Gly	Gly	Gly	Ser
Lys 305	Pro	Ile	Pro	Asn 310	Pro	Leu	Leu	Gly	Leu	Asp 315	Ser	Thr	Gly	Gly	Gly 320
Gly	Ser	Val	His	Met 325	Pro	Leu	Gly	Phe	Leu 330	Gly	Pro	Arg	Gln 335	Ala	Arg
Val	Val	Asn 340	Gly	Gly	Ser	Gly	Gly	Gly 345	Gly	Ser	Gly	Gly	Gly 350	Gly	Ser
Gln	Val 355	Gln	Leu	Gln	Glu	Ser	Gly 360	Pro	Gly	Leu	Val	Lys 365	Pro	Ser	Glu
Thr 370	Leu	Ser	Leu	Thr	Cys 375	Thr	Val	Ser	Gly	Gly	Ser 380	Ile	Ser	Tyr	Tyr
Ser 385	Trp	Thr	Trp	Val	Arg 390	Gln	Pro	Ala	Gly	Gln 395	Gly	Leu	Glu	Trp	Ile 400
Gly	Arg	Ile	His	Ile 405	Ser	Gly	Val	Thr	Asn 410	His	Asn	Pro	Ser	Leu 415	Lys
Ser	Arg	Val	Ser	Met 420	Ser	Ile	Asp	Thr 425	Ser	Arg	Thr	Gln 430	Phe	Ser	Leu
Arg	Leu 435	Thr	Ser	Val	Thr	Ala	Ala 440	Asp	Thr	Ala	Val	Tyr 445	Phe	Cys	Ala
Arg	Ser 450	Gly	Gly	Thr	Tyr	Ser 455	Ala	Phe	Asp	Ile	Trp 460	Gly	Gln	Gly	Thr
Met 465	Val	Thr	Val	Ser	Ser 470	Gly	Gly	Gly	Gly	Ser 475	Gly	Gly	Gly	Gly	Ser 480
Gly	Gly	Gly	Gly	Ser 485	Gly	Gly	Gly	Gly	Ser 490	Glu	Ile	Val	Leu 495	Thr	Gln
Ser	Pro	Gly	Thr 500	Leu	Ser	Leu	Ser	Pro 505	Gly	Glu	Arg	Ser 510	Thr	Leu	Ser
Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr 520	Ser	Asn	Tyr	Leu 525	Ala	Trp	Tyr	Gln
Gln	Arg 530	Pro	Gly	Gln	Ala	Pro 535	Arg	Leu	Leu	Ile	Tyr 540	Gly	Ala	Ser	Thr
Arg	Ala	Ile	Gly	Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr

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545	550	555	560
Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu Pro Glu Asp Phe Ala Val			
	565	570	575
Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro Trp Thr Phe Gly Gln Gly			
	580	585	590
Thr Lys Val Glu Ile Lys Thr Thr Thr Pro Ala Pro Arg Pro Pro Thr			
	595	600	605
Pro Ala Pro Thr Ile Ala Ser Gln Pro Leu Ser Leu Arg Pro Glu Ala			
	610	615	620
Cys Arg Pro Ala Ala Gly Gly Ala Val His Thr Arg Gly Leu Asp Phe			
	625	630	635
Ala Cys Asp Ile Tyr Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly Val			
	645	650	655
Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr Cys Lys Arg Gly Arg Lys			
	660	665	670
Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met Arg Pro Val Gln Thr			
	675	680	685
Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe Pro Glu Glu Glu Glu			
	690	695	700
Gly Gly Cys Glu Leu Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro			
	705	710	715
Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly			
	725	730	735
Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro			
	740	745	750
Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr			
	755	760	765
Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly			
	770	775	780
Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln			
	785	790	795
Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln			
	805	810	815
Ala Leu Pro Pro Arg			
	820		

<210> SEQ ID NO 143

<211> LENGTH: 821

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 143

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu			
1	5	10	15
His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp			
	20	25	30
Ile Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly Gly			
	35	40	45
Ser Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Phe Tyr Asn Leu Leu			
	50	55	60
Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr			
	65	70	75
			80

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Asp	Ala	Ser	Asp	Leu	Ala	Ser	Gly	Val	Pro	Ser	Arg	Phe	Lys	Gly	Ser	85	90	95	
Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Asp	Leu	Glu	Cys	Ala	100	105	110	
Asp	Ala	Ala	Ala	Tyr	Tyr	Cys	Gln	Ser	Ala	Asp	Gly	Ser	Ser	Tyr	Ala	115	120	125	
Phe	Gly	Gly	Gly	Thr	Glu	Val	Val	Val	Lys	Gly	Gly	Gly	Gly	Gly	Ser	130	135	140	
Gly	Gly	Gly	Gly	Ser	Val	His	Met	Pro	Leu	Gly	Phe	Leu	Gly	Pro	Gly	145	150	155	160
Ser	Ala	Arg	Val	Val	Asn	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Glu	165	170	175	
Gln	Leu	Glu	Glu	Ser	Gly	Gly	Gly	Leu	Val	Lys	Pro	Glu	Gly	Ser	Leu	180	185	190	
Thr	Leu	Thr	Cys	Thr	Ala	Ser	Gly	Val	Ser	Phe	Ser	Ser	Ser	Tyr	Trp	195	200	205	
Ile	Tyr	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Ile	Ala	210	215	220	
Cys	Ile	Tyr	Thr	Gly	Ser	Ser	Gly	Ser	Thr	Tyr	Tyr	Ala	Ser	Trp	Ala	225	230	235	240
Lys	Gly	Arg	Phe	Thr	Val	Ser	Glu	Thr	Ser	Ser	Thr	Thr	Val	Thr	Leu	245	250	255	
Gln	Met	Thr	Ser	Leu	Thr	Ala	Ala	Asp	Thr	Ala	Thr	Tyr	Phe	Cys	Ala	260	265	270	
Arg	Ala	Ser	Ala	Trp	Thr	Tyr	Gly	Met	Asp	Leu	Trp	Gly	Pro	Gly	Thr	275	280	285	
Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	290	295	300	
Lys	Pro	Ile	Pro	Asn	Pro	Leu	Leu	Gly	Leu	Asp	Ser	Thr	Gly	Gly	Gly	305	310	315	320
Gly	Ser	Val	His	Met	Pro	Leu	Gly	Phe	Leu	Gly	Pro	Gly	Ser	Ala	Arg	325	330	335	
Val	Val	Asn	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	340	345	350	
Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Glu	355	360	365	
Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile	Ser	Tyr	Tyr	370	375	380	
Ser	Trp	Thr	Trp	Val	Arg	Gln	Pro	Ala	Gly	Gln	Gly	Leu	Glu	Trp	Ile	385	390	395	400
Gly	Arg	Ile	His	Ile	Ser	Gly	Val	Thr	Asn	His	Asn	Pro	Ser	Leu	Lys	405	410	415	
Ser	Arg	Val	Ser	Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln	Phe	Ser	Leu	420	425	430	
Arg	Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Phe	Cys	Ala	435	440	445	
Arg	Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly	Gln	Gly	Thr	450	455	460	
Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	465	470	475	480
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr	Gln	485	490	495	
Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser	Thr	Leu	Ser				

-continued

500					505					510					
Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu	Ala	Trp	Tyr	Gln
	515						520					525			
Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Ser	Thr
	530					535					540				
Arg	Ala	Ile	Gly	Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr
545						550					555				560
Asp	Phe	Thr	Leu	Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp	Phe	Ala	Val
			565						570					575	
Tyr	Tyr	Cys	Gln	Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe	Gly	Gln	Gly
			580					585					590		
Thr	Lys	Val	Glu	Ile	Lys	Thr	Thr	Thr	Pro	Ala	Pro	Arg	Pro	Pro	Thr
		595					600					605			
Pro	Ala	Pro	Thr	Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu	Arg	Pro	Glu	Ala
	610					615					620				
Cys	Arg	Pro	Ala	Ala	Gly	Gly	Ala	Val	His	Thr	Arg	Gly	Leu	Asp	Phe
625						630					635				640
Ala	Cys	Asp	Ile	Tyr	Ile	Trp	Ala	Pro	Leu	Ala	Gly	Thr	Cys	Gly	Val
			645					650						655	
Leu	Leu	Leu	Ser	Leu	Val	Ile	Thr	Leu	Tyr	Cys	Lys	Arg	Gly	Arg	Lys
			660					665					670		
Lys	Leu	Leu	Tyr	Ile	Phe	Lys	Gln	Pro	Phe	Met	Arg	Pro	Val	Gln	Thr
	675					680						685			
Thr	Gln	Glu	Glu	Asp	Gly	Cys	Ser	Cys	Arg	Phe	Pro	Glu	Glu	Glu	Glu
	690					695					700				
Gly	Gly	Cys	Glu	Leu	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro
705						710					715				720
Ala	Tyr	Gln	Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly
			725					730						735	
Arg	Arg	Glu	Glu	Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro
		740						745					750		
Glu	Met	Gly	Gly	Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr
	755					760					765				
Asn	Glu	Leu	Gln	Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly
	770					775					780				
Met	Lys	Gly	Glu	Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr	Gln
785						790					795				800
Gly	Leu	Ser	Thr	Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His	Met	Gln
			805					810						815	
Ala	Leu	Pro	Pro	Arg											
			820												

<210> SEQ ID NO 144

<211> LENGTH: 816

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 144

Met	Ala	Leu	Pro	Val	Thr	Ala	Leu	Leu	Leu	Pro	Leu	Ala	Leu	Leu	Leu
1				5					10					15	

His	Ala	Ala	Arg	Pro	Gly	Tyr	Pro	Tyr	Asp	Val	Pro	Asp	Tyr	Ala	Asp
			20					25					30		

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Ile	Val	Met	Thr	Gln	Thr	Pro	Ala	Ser	Val	Ser	Glu	Pro	Val	Gly	Gly		
		35					40					45					
Ser	Val	Thr	Ile	Lys	Cys	Gln	Ala	Ser	Gln	Ser	Phe	Tyr	Asn	Leu	Leu		
	50					55					60						
Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Pro	Pro	Lys	Leu	Leu	Ile	Tyr		
65					70					75					80		
Asp	Ala	Ser	Asp	Leu	Ala	Ser	Gly	Val	Pro	Ser	Arg	Phe	Lys	Gly	Ser		
			85						90					95			
Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Asp	Leu	Glu	Cys	Ala		
			100					105					110				
Asp	Ala	Ala	Ala	Tyr	Tyr	Cys	Gln	Ser	Ala	Asp	Gly	Ser	Ser	Tyr	Ala		
		115					120					125					
Phe	Gly	Gly	Gly	Thr	Glu	Val	Val	Val	Lys	Gly	Gly	Gly	Gly	Gly	Ser		
	130					135					140						
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Pro	Met	Ala	Lys	Lys	Gly		
145					150					155					160		
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Glu		
			165					170						175			
Gln	Leu	Glu	Glu	Ser	Gly	Gly	Gly	Leu	Val	Lys	Pro	Glu	Gly	Ser	Leu		
			180					185					190				
Thr	Leu	Thr	Cys	Thr	Ala	Ser	Gly	Val	Ser	Phe	Ser	Ser	Ser	Tyr	Trp		
		195					200					205					
Ile	Tyr	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Ile	Ala		
	210					215					220						
Cys	Ile	Tyr	Thr	Gly	Ser	Ser	Gly	Ser	Thr	Tyr	Tyr	Ala	Ser	Trp	Ala		
225					230					235					240		
Lys	Gly	Arg	Phe	Thr	Val	Ser	Glu	Thr	Ser	Ser	Thr	Thr	Val	Thr	Leu		
			245						250					255			
Gln	Met	Thr	Ser	Leu	Thr	Ala	Ala	Asp	Thr	Ala	Thr	Tyr	Phe	Cys	Ala		
			260					265					270				
Arg	Ala	Ser	Ala	Trp	Thr	Tyr	Gly	Met	Asp	Leu	Trp	Gly	Pro	Gly	Thr		
		275					280					285					
Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser		
	290					295					300						
Lys	Pro	Ile	Pro	Asn	Pro	Leu	Leu	Gly	Leu	Asp	Ser	Thr	Gly	Gly	Gly		
305					310					315					320		
Gly	Ser	Pro	Met	Ala	Lys	Lys	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly		
			325					330						335			
Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	Gln		
			340					345					350				
Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Glu	Thr	Leu	Ser	Leu	Thr		
		355					360					365					
Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile	Ser	Tyr	Tyr	Ser	Trp	Thr	Trp	Val		
	370					375					380						
Arg	Gln	Pro	Ala	Gly	Gln	Gly	Leu	Glu	Trp	Ile	Gly	Arg	Ile	His	Ile		
385					390					395					400		
Ser	Gly	Val	Thr	Asn	His	Asn	Pro	Ser	Leu	Lys	Ser	Arg	Val	Ser	Met		
			405						410					415			
Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln	Phe	Ser	Leu	Arg	Leu	Thr	Ser	Val		
			420					425					430				
Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Phe	Cys	Ala	Arg	Ser	Gly	Gly	Thr		
	435					440					445						
Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly	Gln	Gly	Thr	Met	Val	Thr	Val	Ser		

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450	455	460
Ser Gly Gly Gly Gly Ser	Gly Gly Gly Gly Ser	Gly Gly Gly Gly Ser
465	470	475 480
Gly Gly Gly Gly Ser	Glu Ile Val Leu Thr	Gln Ser Pro Gly Thr Leu
485	490	495
Ser Leu Ser Pro Gly Glu Arg	Ser Thr Leu Ser Cys Arg Ala Ser Gln	
500	505	510
Ser Phe Thr Ser Asn Tyr Leu	Ala Trp Tyr Gln Gln Arg Pro Gly Gln	
515	520	525
Ala Pro Arg Leu Leu Ile Tyr	Gly Ala Ser Thr Arg Ala Ile Gly Ile	
530	535	540
Pro Asp Arg Phe Ser Gly Ser	Gly Ser Gly Thr Asp Phe Thr Leu Thr	
545	550	555 560
Ile Ser Arg Leu Glu Pro Glu	Asp Phe Ala Val Tyr Tyr Cys Gln Gln	
565	570	575
Tyr Gly Ser Ser Pro Trp Thr	Phe Gly Gln Gly Thr Lys Val Glu Ile	
580	585	590
Lys Thr Thr Thr Pro Ala Pro	Arg Pro Pro Thr Pro Ala Pro Thr Ile	
595	600	605
Ala Ser Gln Pro Leu Ser Leu	Arg Pro Glu Ala Cys Arg Pro Ala Ala	
610	615	620
Gly Gly Ala Val His Thr Arg	Gly Leu Asp Phe Ala Cys Asp Ile Tyr	
625	630	635 640
Ile Trp Ala Pro Leu Ala Gly	Thr Cys Gly Val Leu Leu Leu Ser Leu	
645	650	655
Val Ile Thr Leu Tyr Cys Lys	Arg Gly Arg Lys Lys Leu Leu Tyr Ile	
660	665	670
Phe Lys Gln Pro Phe Met Arg	Pro Val Gln Thr Thr Gln Glu Glu Asp	
675	680	685
Gly Cys Ser Cys Arg Phe Pro	Glu Glu Glu Glu Gly Gly Cys Glu Leu	
690	695	700
Arg Val Lys Phe Ser Arg Ser	Ala Asp Ala Pro Ala Tyr Gln Gln Gly	
705	710	715 720
Gln Asn Gln Leu Tyr Asn Glu	Leu Asn Leu Gly Arg Arg Glu Glu Tyr	
725	730	735
Asp Val Leu Asp Lys Arg Arg	Gly Arg Asp Pro Glu Met Gly Gly Lys	
740	745	750
Pro Arg Arg Lys Asn Pro Gln	Glu Gly Leu Tyr Asn Glu Leu Gln Lys	
755	760	765
Asp Lys Met Ala Glu Ala Tyr	Ser Glu Ile Gly Met Lys Gly Glu Arg	
770	775	780
Arg Arg Gly Lys Gly His Asp	Gly Leu Tyr Gln Gly Leu Ser Thr Ala	
785	790	795 800
Thr Lys Asp Thr Tyr Asp Ala	Leu His Met Gln Ala Leu Pro Pro Arg	
805	810	815

<210> SEQ ID NO 145

<211> LENGTH: 816

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 145

-continued

Met	Ala	Leu	Pro	Val	Thr	Ala	Leu	Leu	Leu	Pro	Leu	Ala	Leu	Leu	Leu	1	5	10	15
His	Ala	Ala	Arg	Pro	Gly	Tyr	Pro	Tyr	Asp	Val	Pro	Asp	Tyr	Ala	Asp	20	25	30	
Ile	Val	Met	Thr	Gln	Thr	Pro	Ala	Ser	Val	Ser	Glu	Pro	Val	Gly	Gly	35	40	45	
Ser	Val	Thr	Ile	Lys	Cys	Gln	Ala	Ser	Gln	Ser	Phe	Tyr	Asn	Leu	Leu	50	55	60	
Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Pro	Pro	Lys	Leu	Leu	Ile	Tyr	65	70	75	80
Asp	Ala	Ser	Asp	Leu	Ala	Ser	Gly	Val	Pro	Ser	Arg	Phe	Lys	Gly	Ser	85	90	95	
Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Asp	Leu	Glu	Cys	Ala	100	105	110	
Asp	Ala	Ala	Ala	Tyr	Tyr	Cys	Gln	Ser	Ala	Asp	Gly	Ser	Ser	Tyr	Ala	115	120	125	
Phe	Gly	Gly	Gly	Thr	Glu	Val	Val	Val	Lys	Gly	Gly	Gly	Gly	Gly	Ser	130	135	140	
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Pro	Met	Ala	Lys	Gly	Ser	145	150	155	160
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Glu	165	170	175	
Gln	Leu	Glu	Glu	Ser	Gly	Gly	Gly	Leu	Val	Lys	Pro	Glu	Gly	Ser	Leu	180	185	190	
Thr	Leu	Thr	Cys	Thr	Ala	Ser	Gly	Val	Ser	Phe	Ser	Ser	Ser	Tyr	Trp	195	200	205	
Ile	Tyr	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Ile	Ala	210	215	220	
Cys	Ile	Tyr	Thr	Gly	Ser	Ser	Gly	Ser	Thr	Tyr	Tyr	Ala	Ser	Trp	Ala	225	230	235	240
Lys	Gly	Arg	Phe	Thr	Val	Ser	Glu	Thr	Ser	Ser	Thr	Thr	Val	Thr	Leu	245	250	255	
Gln	Met	Thr	Ser	Leu	Thr	Ala	Ala	Asp	Thr	Ala	Thr	Tyr	Phe	Cys	Ala	260	265	270	
Arg	Ala	Ser	Ala	Trp	Thr	Tyr	Gly	Met	Asp	Leu	Trp	Gly	Pro	Gly	Thr	275	280	285	
Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	290	295	300	
Lys	Pro	Ile	Pro	Asn	Pro	Leu	Leu	Gly	Leu	Asp	Ser	Thr	Gly	Gly	Gly	305	310	315	320
Gly	Ser	Pro	Met	Ala	Lys	Gly	Ser	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	325	330	335	
Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	Gln	340	345	350	
Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Glu	Thr	Leu	Ser	Leu	Thr	355	360	365	
Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile	Ser	Tyr	Tyr	Ser	Trp	Thr	Trp	Val	370	375	380	
Arg	Gln	Pro	Ala	Gly	Gln	Gly	Leu	Glu	Trp	Ile	Gly	Arg	Ile	His	Ile	385	390	395	400
Ser	Gly	Val	Thr	Asn	His	Asn	Pro	Ser	Leu	Lys	Ser	Arg	Val	Ser	Met	405	410	415	
Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln	Phe	Ser	Leu	Arg	Leu	Thr	Ser	Val				

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420				425				430							
Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Phe	Cys	Ala	Arg	Ser	Gly	Gly	Thr
	435						440						445		
Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly	Gln	Gly	Thr	Met	Val	Thr	Val	Ser
	450					455					460				
Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser
	465				470					475					480
Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu
				485					490					495	
Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser	Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln
			500						505				510		
Ser	Phe	Thr	Ser	Asn	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Arg	Pro	Gly	Gln
	515						520						525		
Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Ser	Thr	Arg	Ala	Ile	Gly	Ile
	530					535					540				
Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr
	545				550					555					560
Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln
			565						570					575	
Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile
		580							585				590		
Lys	Thr	Thr	Thr	Pro	Ala	Pro	Arg	Pro	Pro	Thr	Pro	Ala	Pro	Thr	Ile
		595					600						605		
Ala	Ser	Gln	Pro	Leu	Ser	Leu	Arg	Pro	Glu	Ala	Cys	Arg	Pro	Ala	Ala
	610					615					620				
Gly	Gly	Ala	Val	His	Thr	Arg	Gly	Leu	Asp	Phe	Ala	Cys	Asp	Ile	Tyr
	625				630					635				640	
Ile	Trp	Ala	Pro	Leu	Ala	Gly	Thr	Cys	Gly	Val	Leu	Leu	Leu	Ser	Leu
			645						650					655	
Val	Ile	Thr	Leu	Tyr	Cys	Lys	Arg	Gly	Arg	Lys	Lys	Leu	Leu	Tyr	Ile
		660							665				670		
Phe	Lys	Gln	Pro	Phe	Met	Arg	Pro	Val	Gln	Thr	Thr	Gln	Glu	Glu	Asp
	675						680						685		
Gly	Cys	Ser	Cys	Arg	Phe	Pro	Glu	Glu	Glu	Glu	Gly	Gly	Cys	Glu	Leu
	690					695					700				
Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln	Gln	Gly
	705				710					715				720	
Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu	Glu	Tyr
			725						730					735	
Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met	Gly	Gly	Lys
		740							745				750		
Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn	Glu	Leu	Gln	Lys
		755					760						765		
Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys	Gly	Glu	Arg
	770					775					780				
Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr	Gln	Gly	Leu	Ser	Thr	Ala
	785				790					795				800	
Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His	Met	Gln	Ala	Leu	Pro	Pro	Arg
			805						810					815	

<210> SEQ ID NO 146

<211> LENGTH: 778

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 146

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Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1              5              10              15

His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp
      20              25              30

Ile Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly Gly
      35              40              45

Ser Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Phe Tyr Asn Leu Leu
      50              55              60

Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr
      65              70              75              80

Asp Ala Ser Asp Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly Ser
      85              90              95

Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala
      100              105              110

Asp Ala Ala Ala Tyr Tyr Cys Gln Ser Ala Asp Gly Ser Ser Tyr Ala
      115              120              125

Phe Gly Gly Gly Thr Glu Val Val Val Lys Gly Gly Gly Gly Ser
      130              135              140

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln
      145              150              155              160

Glu Gln Leu Glu Glu Ser Gly Gly Gly Leu Val Lys Pro Glu Gly Ser
      165              170              175

Leu Thr Leu Thr Cys Thr Ala Ser Gly Val Ser Phe Ser Ser Ser Tyr
      180              185              190

Trp Ile Tyr Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile
      195              200              205

Ala Cys Ile Tyr Thr Gly Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp
      210              215              220

Ala Lys Gly Arg Phe Thr Val Ser Glu Thr Ser Ser Thr Thr Val Thr
      225              230              235              240

Leu Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys
      245              250              255

Ala Arg Ala Ser Ala Trp Thr Tyr Gly Met Asp Leu Trp Gly Pro Gly
      260              265              270

Thr Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly
      275              280              285

Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly Ser Gly Gly Gly Ser
      290              295              300

Gly Gly Gly Gly Ser Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu
      305              310              315              320

Val Lys Pro Ser Glu Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly
      325              330              335

Ser Ile Ser Tyr Tyr Ser Trp Thr Trp Val Arg Gln Pro Ala Gly Gln
      340              345              350

Gly Leu Glu Trp Ile Gly Arg Ile His Ile Ser Gly Val Thr Asn His
      355              360              365

Asn Pro Ser Leu Lys Ser Arg Val Ser Met Ser Ile Asp Thr Ser Arg
      370              375              380

Thr Gln Phe Ser Leu Arg Leu Thr Ser Val Thr Ala Ala Asp Thr Ala

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385	390	395	400
Val Tyr Phe Cys Ala Arg Ser Gly Gly Thr Tyr Ser Ala Phe Asp Ile			
	405	410	415
Trp Gly Gln Gly Thr Met Val Thr Val Ser Ser Gly Gly Gly Ser			
	420	425	430
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Glu			
	435	440	445
Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly Glu			
	450	455	460
Arg Ser Thr Leu Ser Cys Arg Ala Ser Gln Ser Phe Thr Ser Asn Tyr			
	465	470	475
Leu Ala Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Arg Leu Leu Ile			
	485	490	495
Tyr Gly Ala Ser Thr Arg Ala Ile Gly Ile Pro Asp Arg Phe Ser Gly			
	500	505	510
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu Pro			
	515	520	525
Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro Trp			
	530	535	540
Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Thr Thr Thr Pro Ala			
	545	550	555
Pro Arg Pro Pro Thr Pro Ala Pro Thr Ile Ala Ser Gln Pro Leu Ser			
	565	570	575
Leu Arg Pro Glu Ala Cys Arg Pro Ala Ala Gly Gly Ala Val His Thr			
	580	585	590
Arg Gly Leu Asp Phe Ala Cys Asp Ile Tyr Ile Trp Ala Pro Leu Ala			
	595	600	605
Gly Thr Cys Gly Val Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr Cys			
	610	615	620
Lys Arg Gly Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met			
	625	630	635
Arg Pro Val Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe			
	645	650	655
Pro Glu Glu Glu Glu Gly Gly Cys Glu Leu Arg Val Lys Phe Ser Arg			
	660	665	670
Ser Ala Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn			
	675	680	685
Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg			
	690	695	700
Arg Gly Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro			
	705	710	715
Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala			
	725	730	735
Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His			
	740	745	750
Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp			
	755	760	765
Ala Leu His Met Gln Ala Leu Pro Pro Arg			
	770	775	

<210> SEQ ID NO 147

<211> LENGTH: 768

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 147

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Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1          5          10          15

His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp
      20          25          30

Ile Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly Gly
      35          40          45

Ser Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Phe Tyr Asn Leu Leu
      50          55          60

Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr
      65          70          75          80

Asp Ala Ser Asp Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly Ser
      85          90          95

Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala
      100          105          110

Asp Ala Ala Ala Tyr Tyr Cys Gln Ser Ala Asp Gly Ser Ser Tyr Ala
      115          120          125

Phe Gly Gly Gly Thr Glu Val Val Val Lys Gly Gly Gly Gly Gly Ser
      130          135          140

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln
      145          150          155          160

Glu Gln Leu Glu Glu Ser Gly Gly Gly Leu Val Lys Pro Glu Gly Ser
      165          170          175

Leu Thr Leu Thr Cys Thr Ala Ser Gly Val Ser Phe Ser Ser Ser Tyr
      180          185          190

Trp Ile Tyr Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile
      195          200          205

Ala Cys Ile Tyr Thr Gly Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp
      210          215          220

Ala Lys Gly Arg Phe Thr Val Ser Glu Thr Ser Ser Thr Thr Val Thr
      225          230          235          240

Leu Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys
      245          250          255

Ala Arg Ala Ser Ala Trp Thr Tyr Gly Met Asp Leu Trp Gly Pro Gly
      260          265          270

Thr Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly
      275          280          285

Ser Gly Gly Gly Ser Gly Gly Gly Gly Gly Ser Gln Val Gln Leu Gln
      290          295          300

Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu Thr Leu Ser Leu Thr
      305          310          315          320

Cys Thr Val Ser Gly Gly Ser Ile Ser Tyr Tyr Ser Trp Thr Trp Val
      325          330          335

Arg Gln Pro Ala Gly Gln Gly Leu Glu Trp Ile Gly Arg Ile His Ile
      340          345          350

Ser Gly Val Thr Asn His Asn Pro Ser Leu Lys Ser Arg Val Ser Met
      355          360          365

Ser Ile Asp Thr Ser Arg Thr Gln Phe Ser Leu Arg Leu Thr Ser Val
      370          375          380

Thr Ala Ala Asp Thr Ala Val Tyr Phe Cys Ala Arg Ser Gly Gly Thr

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385	390	395	400
Tyr Ser Ala Phe Asp Ile Trp Gly Gln Gly Thr Met Val Thr Val Ser	405	410	415
Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser	420	425	430
Gly Gly Gly Gly Ser Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu	435	440	445
Ser Leu Ser Pro Gly Glu Arg Ser Thr Leu Ser Cys Arg Ala Ser Gln	450	455	460
Ser Phe Thr Ser Asn Tyr Leu Ala Trp Tyr Gln Gln Arg Pro Gly Gln	465	470	475
Ala Pro Arg Leu Leu Ile Tyr Gly Ala Ser Thr Arg Ala Ile Gly Ile	485	490	495
Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr	500	505	510
Ile Ser Arg Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln	515	520	525
Tyr Gly Ser Ser Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile	530	535	540
Lys Thr Thr Thr Pro Ala Pro Arg Pro Pro Thr Pro Ala Pro Thr Ile	545	550	555
Ala Ser Gln Pro Leu Ser Leu Arg Pro Glu Ala Cys Arg Pro Ala Ala	565	570	575
Gly Gly Ala Val His Thr Arg Gly Leu Asp Phe Ala Cys Asp Ile Tyr	580	585	590
Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly Val Leu Leu Leu Ser Leu	595	600	605
Val Ile Thr Leu Tyr Cys Lys Arg Gly Arg Lys Lys Leu Leu Tyr Ile	610	615	620
Phe Lys Gln Pro Phe Met Arg Pro Val Gln Thr Thr Gln Glu Glu Asp	625	630	635
Gly Cys Ser Cys Arg Phe Pro Glu Glu Glu Glu Gly Gly Cys Glu Leu	645	650	655
Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln Gly	660	665	670
Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr	675	680	685
Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly Lys	690	695	700
Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys	705	710	715
Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg	725	730	735
Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala	740	745	750
Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg	755	760	765

<210> SEQ ID NO 148

<211> LENGTH: 758

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

-continued

<400> SEQUENCE: 148

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15
 His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Asp
 20 25 30
 Ile Val Met Thr Gln Thr Pro Ala Ser Val Ser Glu Pro Val Gly Gly
 35 40 45
 Ser Val Thr Ile Lys Cys Gln Ala Ser Gln Ser Phe Tyr Asn Leu Leu
 50 55 60
 Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr
 65 70 75 80
 Asp Ala Ser Asp Leu Ala Ser Gly Val Pro Ser Arg Phe Lys Gly Ser
 85 90 95
 Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asp Leu Glu Cys Ala
 100 105 110
 Asp Ala Ala Ala Tyr Tyr Cys Gln Ser Ala Asp Gly Ser Ser Tyr Ala
 115 120 125
 Phe Gly Gly Gly Thr Glu Val Val Val Lys Gly Gly Gly Gly Ser
 130 135 140
 Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gln
 145 150 155 160
 Glu Gln Leu Glu Glu Ser Gly Gly Gly Leu Val Lys Pro Glu Gly Ser
 165 170 175
 Leu Thr Leu Thr Cys Thr Ala Ser Gly Val Ser Phe Ser Ser Ser Tyr
 180 185 190
 Trp Ile Tyr Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile
 195 200 205
 Ala Cys Ile Tyr Thr Gly Ser Ser Gly Ser Thr Tyr Tyr Ala Ser Trp
 210 215 220
 Ala Lys Gly Arg Phe Thr Val Ser Glu Thr Ser Ser Thr Thr Val Thr
 225 230 235 240
 Leu Gln Met Thr Ser Leu Thr Ala Ala Asp Thr Ala Thr Tyr Phe Cys
 245 250 255
 Ala Arg Ala Ser Ala Trp Thr Tyr Gly Met Asp Leu Trp Gly Pro Gly
 260 265 270
 Thr Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly
 275 280 285
 Ser Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser
 290 295 300
 Glu Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Tyr
 305 310 315 320
 Tyr Ser Trp Thr Trp Val Arg Gln Pro Ala Gly Gln Gly Leu Glu Trp
 325 330 335
 Ile Gly Arg Ile His Ile Ser Gly Val Thr Asn His Asn Pro Ser Leu
 340 345 350
 Lys Ser Arg Val Ser Met Ser Ile Asp Thr Ser Arg Thr Gln Phe Ser
 355 360 365
 Leu Arg Leu Thr Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Phe Cys
 370 375 380
 Ala Arg Ser Gly Gly Thr Tyr Ser Ala Phe Asp Ile Trp Gly Gln Gly
 385 390 395 400
 Thr Met Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly

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405					410					415					
Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr
			420					425					430		
Gln	Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser	Thr	Leu
		435					440					445			
Ser	Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu	Ala	Trp	Tyr
	450					455					460				
Gln	Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Ser
465						470					475				480
Thr	Arg	Ala	Ile	Gly	Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly
			485						490					495	
Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp	Phe	Ala
		500						505					510		
Val	Tyr	Tyr	Cys	Gln	Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe	Gly	Gln
	515					520							525		
Gly	Thr	Lys	Val	Glu	Ile	Lys	Thr	Thr	Thr	Pro	Ala	Pro	Arg	Pro	Pro
	530					535					540				
Thr	Pro	Ala	Pro	Thr	Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu	Arg	Pro	Glu
545						550					555				560
Ala	Cys	Arg	Pro	Ala	Ala	Gly	Gly	Ala	Val	His	Thr	Arg	Gly	Leu	Asp
			565						570					575	
Phe	Ala	Cys	Asp	Ile	Tyr	Ile	Trp	Ala	Pro	Leu	Ala	Gly	Thr	Cys	Gly
			580					585					590		
Val	Leu	Leu	Leu	Ser	Leu	Val	Ile	Thr	Leu	Tyr	Cys	Lys	Arg	Gly	Arg
	595					600						605			
Lys	Lys	Leu	Leu	Tyr	Ile	Phe	Lys	Gln	Pro	Phe	Met	Arg	Pro	Val	Gln
	610					615					620				
Thr	Thr	Gln	Glu	Glu	Asp	Gly	Cys	Ser	Cys	Arg	Phe	Pro	Glu	Glu	Glu
625						630					635				640
Glu	Gly	Gly	Cys	Glu	Leu	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala
			645						650					655	
Pro	Ala	Tyr	Gln	Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu
		660						665					670		
Gly	Arg	Arg	Glu	Glu	Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp
		675				680							685		
Pro	Glu	Met	Gly	Gly	Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu
	690					695					700				
Tyr	Asn	Glu	Leu	Gln	Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile
705						710					715				720
Gly	Met	Lys	Gly	Glu	Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr
			725						730					735	
Gln	Gly	Leu	Ser	Thr	Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His	Met
		740						745					750		
Gln	Ala	Leu	Pro	Pro	Arg										
		755													

<210> SEQ ID NO 149

<211> LENGTH: 807

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 149

-continued

Met	Ala	Leu	Pro	Val	Thr	Ala	Leu	Leu	Leu	Pro	Leu	Ala	Leu	Leu	Leu	1	5	10	15
His	Ala	Ala	Arg	Pro	Gly	Lys	Pro	Ile	Pro	Asn	Pro	Leu	Leu	Gly	Leu	20	25	30	
Asp	Ser	Thr	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	35	40	45	
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Asp	Ile	Val	Met	Thr	Gln	Thr	Pro	Ala	50	55	60	
Ser	Val	Ser	Glu	Pro	Val	Gly	Gly	Ser	Val	Thr	Ile	Lys	Cys	Gln	Ala	65	70	75	80
Ser	Gln	Ser	Phe	Tyr	Asn	Leu	Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	85	90	95	
Gln	Pro	Pro	Lys	Leu	Leu	Ile	Tyr	Asp	Ala	Ser	Asp	Leu	Ala	Ser	Gly	100	105	110	
Val	Pro	Ser	Arg	Phe	Lys	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	115	120	125	
Thr	Ile	Ser	Asp	Leu	Glu	Cys	Ala	Asp	Ala	Ala	Ala	Tyr	Tyr	Cys	Gln	130	135	140	
Ser	Ala	Asp	Gly	Ser	Ser	Tyr	Ala	Phe	Gly	Gly	Gly	Thr	Glu	Val	Val	145	150	155	160
Val	Lys	Gly	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	165	170	175	
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Glu	Gln	Leu	Glu	Glu	Ser	Gly	Gly	180	185	190	
Gly	Leu	Val	Lys	Pro	Glu	Gly	Ser	Leu	Thr	Leu	Thr	Cys	Thr	Ala	Ser	195	200	205	
Gly	Val	Ser	Phe	Ser	Ser	Ser	Tyr	Trp	Ile	Tyr	Trp	Val	Arg	Gln	Ala	210	215	220	
Pro	Gly	Lys	Gly	Leu	Glu	Trp	Ile	Ala	Cys	Ile	Tyr	Thr	Gly	Ser	Ser	225	230	235	240
Gly	Ser	Thr	Tyr	Tyr	Ala	Ser	Trp	Ala	Lys	Gly	Arg	Phe	Thr	Val	Ser	245	250	255	
Glu	Thr	Ser	Ser	Thr	Thr	Val	Thr	Leu	Gln	Met	Thr	Ser	Leu	Thr	Ala	260	265	270	
Ala	Asp	Thr	Ala	Thr	Tyr	Phe	Cys	Ala	Arg	Ala	Ser	Ala	Trp	Thr	Tyr	275	280	285	
Gly	Met	Asp	Leu	Trp	Gly	Pro	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	290	295	300	
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Pro	Met	305	310	315	320
Ala	Lys	Lys	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	325	330	335	
Gly	Ser	Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	340	345	350	
Ser	Glu	Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile	Ser	355	360	365	
Tyr	Tyr	Ser	Trp	Thr	Trp	Val	Arg	Gln	Pro	Ala	Gly	Gln	Gly	Leu	Glu	370	375	380	
Trp	Ile	Gly	Arg	Ile	His	Ile	Ser	Gly	Val	Thr	Asn	His	Asn	Pro	Ser	385	390	395	400
Leu	Lys	Ser	Arg	Val	Ser	Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln	Phe	405	410	415	
Ser	Leu	Arg	Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Phe				

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420				425				430							
Cys	Ala	Arg	Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly	Gln
	435						440					445			
Gly	Thr	Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly
	450					455					460				
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val	Leu
	465				470					475					480
Thr	Gln	Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser	Thr
				485						490				495	
Leu	Ser	Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu	Ala	Trp
			500						505					510	
Tyr	Gln	Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala
		515					520						525		
Ser	Thr	Arg	Ala	Ile	Gly	Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser
	530					535					540				
Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp	Phe
	545				550					555					560
Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe	Gly
				565					570						575
Gln	Gly	Thr	Lys	Val	Glu	Ile	Lys	Thr	Thr	Thr	Pro	Ala	Pro	Arg	Pro
			580						585					590	
Pro	Thr	Pro	Ala	Pro	Thr	Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu	Arg	Pro
		595					600						605		
Glu	Ala	Cys	Arg	Pro	Ala	Ala	Gly	Gly	Ala	Val	His	Thr	Arg	Gly	Leu
	610					615					620				
Asp	Phe	Ala	Cys	Asp	Ile	Tyr	Ile	Trp	Ala	Pro	Leu	Ala	Gly	Thr	Cys
	625				630					635					640
Gly	Val	Leu	Leu	Leu	Ser	Leu	Val	Ile	Thr	Leu	Tyr	Cys	Lys	Arg	Gly
			645						650					655	
Arg	Lys	Lys	Leu	Leu	Tyr	Ile	Phe	Lys	Gln	Pro	Phe	Met	Arg	Pro	Val
			660						665					670	
Gln	Thr	Thr	Gln	Glu	Glu	Asp	Gly	Cys	Ser	Cys	Arg	Phe	Pro	Glu	Glu
		675					680						685		
Glu	Glu	Gly	Gly	Cys	Glu	Leu	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp
	690					695					700				
Ala	Pro	Ala	Tyr	Gln	Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn
	705				710					715					720
Leu	Gly	Arg	Arg	Glu	Glu	Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg
			725						730					735	
Asp	Pro	Glu	Met	Gly	Gly	Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly
		740							745					750	
Leu	Tyr	Asn	Glu	Leu	Gln	Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu
		755					760						765		
Ile	Gly	Met	Lys	Gly	Glu	Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu
	770					775							780		
Tyr	Gln	Gly	Leu	Ser	Thr	Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His
	785				790					795					800
Met	Gln	Ala	Leu	Pro	Pro	Arg									
				805											

<210> SEQ ID NO 150

<211> LENGTH: 802

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 150

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Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1              5              10              15

His Ala Ala Arg Pro Gly Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu
      20              25              30

Asp Ser Thr Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
      35              40              45

Gly Ser Gly Gly Gly Gly Gly Ser Asp Ile Val Met Thr Gln Thr Pro Ala
 50              55              60

Ser Val Ser Glu Pro Val Gly Gly Ser Val Thr Ile Lys Cys Gln Ala
65              70              75              80

Ser Gln Ser Phe Tyr Asn Leu Leu Ala Trp Tyr Gln Gln Lys Pro Gly
      85              90              95

Gln Pro Pro Lys Leu Leu Ile Tyr Asp Ala Ser Asp Leu Ala Ser Gly
      100              105              110

Val Pro Ser Arg Phe Lys Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu
      115              120              125

Thr Ile Ser Asp Leu Glu Cys Ala Asp Ala Ala Tyr Tyr Cys Gln
      130              135              140

Ser Ala Asp Gly Ser Ser Tyr Ala Phe Gly Gly Gly Thr Glu Val Val
      145              150              155              160

Val Lys Gly Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
      165              170              175

Gly Ser Gly Gly Gly Gly Ser Gln Glu Gln Leu Glu Glu Ser Gly Gly
      180              185              190

Gly Leu Val Lys Pro Glu Gly Ser Leu Thr Leu Thr Cys Thr Ala Ser
      195              200              205

Gly Val Ser Phe Ser Ser Ser Tyr Trp Ile Tyr Trp Val Arg Gln Ala
      210              215              220

Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys Ile Tyr Thr Gly Ser Ser
      225              230              235              240

Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly Arg Phe Thr Val Ser
      245              250              255

Glu Thr Ser Ser Thr Thr Val Thr Leu Gln Met Thr Ser Leu Thr Ala
      260              265              270

Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ala Ser Ala Trp Thr Tyr
      275              280              285

Gly Met Asp Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser Gly
      290              295              300

Gly Gly Gly Ser Gly Gly Gly Gly Ser Pro Met Ala Lys Lys Gly Gly
      305              310              315              320

Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln Val Gln
      325              330              335

Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu Thr Leu Ser
      340              345              350

Leu Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Tyr Tyr Ser Trp Thr
      355              360              365

Trp Val Arg Gln Pro Ala Gly Gln Gly Leu Glu Trp Ile Gly Arg Ile
      370              375              380

His Ile Ser Gly Val Thr Asn His Asn Pro Ser Leu Lys Ser Arg Val

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385	390	395	400
Ser Met Ser Ile Asp Thr Ser Arg Thr Gln Phe Ser Leu Arg Leu Thr			
	405	410	415
Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Phe Cys Ala Arg Ser Gly			
	420	425	430
Gly Thr Tyr Ser Ala Phe Asp Ile Trp Gly Gln Gly Thr Met Val Thr			
	435	440	445
Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly			
	450	455	460
Gly Ser Gly Gly Gly Gly Ser Glu Ile Val Leu Thr Gln Ser Pro Gly			
	465	470	475
Thr Leu Ser Leu Ser Pro Gly Glu Arg Ser Thr Leu Ser Cys Arg Ala			
	485	490	495
Ser Gln Ser Phe Thr Ser Asn Tyr Leu Ala Trp Tyr Gln Gln Arg Pro			
	500	505	510
Gly Gln Ala Pro Arg Leu Leu Ile Tyr Gly Ala Ser Thr Arg Ala Ile			
	515	520	525
Gly Ile Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr			
	530	535	540
Leu Thr Ile Ser Arg Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys			
	545	550	555
Gln Gln Tyr Gly Ser Ser Pro Trp Thr Phe Gly Gln Gly Thr Lys Val			
	565	570	575
Glu Ile Lys Thr Thr Thr Pro Ala Pro Arg Pro Pro Thr Pro Ala Pro			
	580	585	590
Thr Ile Ala Ser Gln Pro Leu Ser Leu Arg Pro Glu Ala Cys Arg Pro			
	595	600	605
Ala Ala Gly Gly Ala Val His Thr Arg Gly Leu Asp Phe Ala Cys Asp			
	610	615	620
Ile Tyr Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly Val Leu Leu Leu			
	625	630	635
Ser Leu Val Ile Thr Leu Tyr Cys Lys Arg Gly Arg Lys Lys Leu Leu			
	645	650	655
Tyr Ile Phe Lys Gln Pro Phe Met Arg Pro Val Gln Thr Thr Gln Glu			
	660	665	670
Glu Asp Gly Cys Ser Cys Arg Phe Pro Glu Glu Glu Glu Gly Gly Cys			
	675	680	685
Glu Leu Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln			
	690	695	700
Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu			
	705	710	715
Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly			
	725	730	735
Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu			
	740	745	750
Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly			
	755	760	765
Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser			
	770	775	780
Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro			
	785	790	795
			800
Pro Arg			

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<210> SEQ ID NO 151
<211> LENGTH: 797
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide

<400> SEQUENCE: 151

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1             5             10             15

His Ala Ala Arg Pro Gly Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu
 20             25             30

Asp Ser Thr Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
 35             40             45

Gly Ser Gly Gly Gly Gly Ser Asp Ile Val Met Thr Gln Thr Pro Ala
 50             55             60

Ser Val Ser Glu Pro Val Gly Gly Ser Val Thr Ile Lys Cys Gln Ala
 65             70             75             80

Ser Gln Ser Phe Tyr Asn Leu Leu Ala Trp Tyr Gln Gln Lys Pro Gly
 85             90             95

Gln Pro Pro Lys Leu Leu Ile Tyr Asp Ala Ser Asp Leu Ala Ser Gly
100            105            110

Val Pro Ser Arg Phe Lys Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu
115            120            125

Thr Ile Ser Asp Leu Glu Cys Ala Asp Ala Ala Tyr Tyr Cys Gln
130            135            140

Ser Ala Asp Gly Ser Ser Tyr Ala Phe Gly Gly Thr Glu Val Val
145            150            155            160

Val Lys Gly Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
165            170            175

Gly Ser Gly Gly Gly Gly Ser Gln Glu Gln Leu Glu Glu Ser Gly Gly
180            185            190

Gly Leu Val Lys Pro Glu Gly Ser Leu Thr Leu Thr Cys Thr Ala Ser
195            200            205

Gly Val Ser Phe Ser Ser Ser Tyr Trp Ile Tyr Trp Val Arg Gln Ala
210            215            220

Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys Ile Tyr Thr Gly Ser Ser
225            230            235            240

Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly Arg Phe Thr Val Ser
245            250            255

Glu Thr Ser Ser Thr Thr Val Thr Leu Gln Met Thr Ser Leu Thr Ala
260            265            270

Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ala Ser Ala Trp Thr Tyr
275            280            285

Gly Met Asp Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser Gly
290            295            300

Gly Gly Gly Ser Gly Gly Gly Gly Ser Pro Met Ala Lys Lys Gly Gly
305            310            315            320

Gly Gly Ser Gly Gly Gly Gly Ser Gln Val Gln Leu Gln Glu Ser Gly
325            330            335

Pro Gly Leu Val Lys Pro Ser Glu Thr Leu Ser Leu Thr Cys Thr Val
340            345            350

Ser Gly Gly Ser Ile Ser Tyr Tyr Ser Trp Thr Trp Val Arg Gln Pro
355            360            365

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Ala Gly Gln Gly Leu Glu Trp Ile Gly Arg Ile His Ile Ser Gly Val	
370	375 380
Thr Asn His Asn Pro Ser Leu Lys Ser Arg Val Ser Met Ser Ile Asp	
385	390 395 400
Thr Ser Arg Thr Gln Phe Ser Leu Arg Leu Thr Ser Val Thr Ala Ala	
	405 410 415
Asp Thr Ala Val Tyr Phe Cys Ala Arg Ser Gly Gly Thr Tyr Ser Ala	
	420 425 430
Phe Asp Ile Trp Gly Gln Gly Thr Met Val Thr Val Ser Ser Gly Gly	
	435 440 445
Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly	
	450 455 460
Gly Ser Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser	
465	470 475 480
Pro Gly Glu Arg Ser Thr Leu Ser Cys Arg Ala Ser Gln Ser Phe Thr	
	485 490 495
Ser Asn Tyr Leu Ala Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Arg	
	500 505 510
Leu Leu Ile Tyr Gly Ala Ser Thr Arg Ala Ile Gly Ile Pro Asp Arg	
	515 520 525
Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg	
	530 535 540
Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser	
545	550 555 560
Ser Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Thr Thr	
	565 570 575
Thr Pro Ala Pro Arg Pro Pro Thr Pro Ala Pro Thr Ile Ala Ser Gln	
	580 585 590
Pro Leu Ser Leu Arg Pro Glu Ala Cys Arg Pro Ala Ala Gly Gly Ala	
	595 600 605
Val His Thr Arg Gly Leu Asp Phe Ala Cys Asp Ile Tyr Ile Trp Ala	
	610 615 620
Pro Leu Ala Gly Thr Cys Gly Val Leu Leu Leu Ser Leu Val Ile Thr	
625	630 635 640
Leu Tyr Cys Lys Arg Gly Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln	
	645 650 655
Pro Phe Met Arg Pro Val Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser	
	660 665 670
Cys Arg Phe Pro Glu Glu Glu Glu Gly Gly Cys Glu Leu Arg Val Lys	
	675 680 685
Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln	
	690 695 700
Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu	
705	710 715 720
Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg	
	725 730 735
Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met	
	740 745 750
Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly	
	755 760 765
Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp	
	770 775 780

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Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg
785 790 795

<210> SEQ ID NO 152
 <211> LENGTH: 792
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 152

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15
 His Ala Ala Arg Pro Gly Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu
 20 25 30
 Asp Ser Thr Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
 35 40 45
 Gly Ser Gly Gly Gly Gly Ser Asp Ile Val Met Thr Gln Thr Pro Ala
 50 55 60
 Ser Val Ser Glu Pro Val Gly Gly Ser Val Thr Ile Lys Cys Gln Ala
 65 70 75 80
 Ser Gln Ser Phe Tyr Asn Leu Leu Ala Trp Tyr Gln Gln Lys Pro Gly
 85 90 95
 Gln Pro Pro Lys Leu Leu Ile Tyr Asp Ala Ser Asp Leu Ala Ser Gly
 100 105 110
 Val Pro Ser Arg Phe Lys Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu
 115 120 125
 Thr Ile Ser Asp Leu Glu Cys Ala Asp Ala Ala Tyr Tyr Cys Gln
 130 135 140
 Ser Ala Asp Gly Ser Ser Tyr Ala Phe Gly Gly Gly Thr Glu Val Val
 145 150 155 160
 Val Lys Gly Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
 165 170 175
 Gly Ser Gly Gly Gly Gly Ser Gln Glu Gln Leu Glu Glu Ser Gly Gly
 180 185 190
 Gly Leu Val Lys Pro Glu Gly Ser Leu Thr Leu Thr Cys Thr Ala Ser
 195 200 205
 Gly Val Ser Phe Ser Ser Ser Tyr Trp Ile Tyr Trp Val Arg Gln Ala
 210 215 220
 Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys Ile Tyr Thr Gly Ser Ser
 225 230 235 240
 Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly Arg Phe Thr Val Ser
 245 250 255
 Glu Thr Ser Ser Thr Thr Val Thr Leu Gln Met Thr Ser Leu Thr Ala
 260 265 270
 Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ala Ser Ala Trp Thr Tyr
 275 280 285
 Gly Met Asp Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser Gly
 290 295 300
 Gly Gly Gly Ser Pro Met Ala Lys Lys Gly Gly Gly Gly Ser Gly Gly
 305 310 315 320
 Gly Gly Ser Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys
 325 330 335
 Pro Ser Glu Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile
 340 345 350

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Ser	Tyr	Tyr	Ser	Trp	Thr	Trp	Val	Arg	Gln	Pro	Ala	Gly	Gln	Gly	Leu
	355						360					365			
Glu	Trp	Ile	Gly	Arg	Ile	His	Ile	Ser	Gly	Val	Thr	Asn	His	Asn	Pro
	370					375					380				
Ser	Leu	Lys	Ser	Arg	Val	Ser	Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln
	385				390					395					400
Phe	Ser	Leu	Arg	Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr
			405						410					415	
Phe	Cys	Ala	Arg	Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly
		420						425					430		
Gln	Gly	Thr	Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly
		435					440					445			
Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val
	450					455					460				
Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser
	465				470					475					480
Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu	Ala
			485						490					495	
Trp	Tyr	Gln	Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly
		500					505						510		
Ala	Ser	Thr	Arg	Ala	Ile	Gly	Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly
		515					520					525			
Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp
	530					535					540				
Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe
	545				550					555					560
Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile	Lys	Thr	Thr	Thr	Pro	Ala	Pro	Arg
			565					570						575	
Pro	Pro	Thr	Pro	Ala	Pro	Thr	Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu	Arg
		580					585						590		
Pro	Glu	Ala	Cys	Arg	Pro	Ala	Ala	Gly	Gly	Ala	Val	His	Thr	Arg	Gly
		595				600						605			
Leu	Asp	Phe	Ala	Cys	Asp	Ile	Tyr	Ile	Trp	Ala	Pro	Leu	Ala	Gly	Thr
	610				615						620				
Cys	Gly	Val	Leu	Leu	Leu	Ser	Leu	Val	Ile	Thr	Leu	Tyr	Cys	Lys	Arg
	625				630					635				640	
Gly	Arg	Lys	Lys	Leu	Leu	Tyr	Ile	Phe	Lys	Gln	Pro	Phe	Met	Arg	Pro
			645					650					655		
Val	Gln	Thr	Thr	Gln	Glu	Glu	Asp	Gly	Cys	Ser	Cys	Arg	Phe	Pro	Glu
		660					665						670		
Glu	Glu	Glu	Gly	Gly	Cys	Glu	Leu	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala
	675					680					685				
Asp	Ala	Pro	Ala	Tyr	Gln	Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu
	690				695						700				
Asn	Leu	Gly	Arg	Arg	Glu	Glu	Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly
	705				710					715					720
Arg	Asp	Pro	Glu	Met	Gly	Gly	Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu
			725					730						735	
Gly	Leu	Tyr	Asn	Glu	Leu	Gln	Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser
		740					745					750			
Glu	Ile	Gly	Met	Lys	Gly	Glu	Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly
	755					760						765			

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Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu
770 775 780

His Met Gln Ala Leu Pro Pro Arg
785 790

<210> SEQ ID NO 153

<211> LENGTH: 822

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 153

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
1 5 10 15

His Ala Ala Arg Pro Gly Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu
20 25 30

Asp Ser Thr Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
35 40 45

Gly Ser Gly Gly Gly Gly Ser Asp Ile Val Met Thr Gln Thr Pro Ala
50 55 60

Ser Val Ser Glu Pro Val Gly Gly Ser Val Thr Ile Lys Cys Gln Ala
65 70 75 80

Ser Gln Ser Phe Tyr Asn Leu Leu Ala Trp Tyr Gln Gln Lys Pro Gly
85 90 95

Gln Pro Pro Lys Leu Leu Ile Tyr Asp Ala Ser Asp Leu Ala Ser Gly
100 105 110

Val Pro Ser Arg Phe Lys Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu
115 120 125

Thr Ile Ser Asp Leu Glu Cys Ala Asp Ala Ala Tyr Tyr Cys Gln
130 135 140

Ser Ala Asp Gly Ser Ser Tyr Ala Phe Gly Gly Gly Thr Glu Val Val
145 150 155 160

Val Lys Gly Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly
165 170 175

Gly Ser Pro Met Ala Lys Lys Gly Gly Gly Ser Gly Gly Gly Gly
180 185 190

Ser Gly Gly Gly Gly Ser Gln Glu Gln Leu Glu Glu Ser Gly Gly Gly
195 200 205

Leu Val Lys Pro Glu Gly Ser Leu Thr Leu Thr Cys Thr Ala Ser Gly
210 215 220

Val Ser Phe Ser Ser Ser Tyr Trp Ile Tyr Trp Val Arg Gln Ala Pro
225 230 235 240

Gly Lys Gly Leu Glu Trp Ile Ala Cys Ile Tyr Thr Gly Ser Ser Gly
245 250 255

Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly Arg Phe Thr Val Ser Glu
260 265 270

Thr Ser Ser Thr Thr Val Thr Leu Gln Met Thr Ser Leu Thr Ala Ala
275 280 285

Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ala Ser Ala Trp Thr Tyr Gly
290 295 300

Met Asp Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser Gly Gly
305 310 315 320

Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Pro Met Ala
325 330 335

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Lys Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly
 340 345 350
 Ser Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser
 355 360 365
 Glu Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Tyr
 370 375 380
 Tyr Ser Trp Thr Trp Val Arg Gln Pro Ala Gly Gln Gly Leu Glu Trp
 385 390 395 400
 Ile Gly Arg Ile His Ile Ser Gly Val Thr Asn His Asn Pro Ser Leu
 405 410 415
 Lys Ser Arg Val Ser Met Ser Ile Asp Thr Ser Arg Thr Gln Phe Ser
 420 425 430
 Leu Arg Leu Thr Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Phe Cys
 435 440 445
 Ala Arg Ser Gly Gly Thr Tyr Ser Ala Phe Asp Ile Trp Gly Gln Gly
 450 455 460
 Thr Met Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly
 465 470 475 480
 Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Glu Ile Val Leu Thr
 485 490 495
 Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly Glu Arg Ser Thr Leu
 500 505 510
 Ser Cys Arg Ala Ser Gln Ser Phe Thr Ser Asn Tyr Leu Ala Trp Tyr
 515 520 525
 Gln Gln Arg Pro Gly Gln Ala Pro Arg Leu Leu Ile Tyr Gly Ala Ser
 530 535 540
 Thr Arg Ala Ile Gly Ile Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly
 545 550 555 560
 Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu Pro Glu Asp Phe Ala
 565 570 575
 Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro Trp Thr Phe Gly Gln
 580 585 590
 Gly Thr Lys Val Glu Ile Lys Thr Thr Thr Pro Ala Pro Arg Pro Pro
 595 600 605
 Thr Pro Ala Pro Thr Ile Ala Ser Gln Pro Leu Ser Leu Arg Pro Glu
 610 615 620
 Ala Cys Arg Pro Ala Ala Gly Gly Ala Val His Thr Arg Gly Leu Asp
 625 630 635 640
 Phe Ala Cys Asp Ile Tyr Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly
 645 650 655
 Val Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr Cys Lys Arg Gly Arg
 660 665 670
 Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met Arg Pro Val Gln
 675 680 685
 Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe Pro Glu Glu Glu
 690 695 700
 Glu Gly Gly Cys Glu Leu Arg Val Lys Phe Ser Arg Ser Ala Asp Ala
 705 710 715 720
 Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu
 725 730 735
 Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp
 740 745 750

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Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu
   755                               760                               765

Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile
   770                               775                               780

Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr
   785                               790                               795                               800

Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met
   805                               810                               815

Gln Ala Leu Pro Pro Arg
   820

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<210> SEQ ID NO 154
<211> LENGTH: 817
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                             polypeptide

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<400> SEQUENCE: 154

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Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1           5           10           15

His Ala Ala Arg Pro Gly Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu
 20           25           30

Asp Ser Thr Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
 35           40           45

Gly Ser Gly Gly Gly Gly Ser Asp Ile Val Met Thr Gln Thr Pro Ala
 50           55           60

Ser Val Ser Glu Pro Val Gly Gly Ser Val Thr Ile Lys Cys Gln Ala
 65           70           75           80

Ser Gln Ser Phe Tyr Asn Leu Leu Ala Trp Tyr Gln Gln Lys Pro Gly
 85           90           95

Gln Pro Pro Lys Leu Leu Ile Tyr Asp Ala Ser Asp Leu Ala Ser Gly
100           105           110

Val Pro Ser Arg Phe Lys Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu
115           120           125

Thr Ile Ser Asp Leu Glu Cys Ala Asp Ala Ala Tyr Tyr Cys Gln
130           135           140

Ser Ala Asp Gly Ser Ser Tyr Ala Phe Gly Gly Thr Glu Val Val
145           150           155           160

Val Lys Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly
165           170           175

Gly Ser Pro Met Ala Lys Lys Gly Gly Gly Gly Ser Gly Gly Gly
180           185           190

Ser Gly Gly Gly Gly Ser Gln Glu Gln Leu Glu Glu Ser Gly Gly Gly
195           200           205

Leu Val Lys Pro Glu Gly Ser Leu Thr Leu Thr Cys Thr Ala Ser Gly
210           215           220

Val Ser Phe Ser Ser Ser Tyr Trp Ile Tyr Trp Val Arg Gln Ala Pro
225           230           235           240

Gly Lys Gly Leu Glu Trp Ile Ala Cys Ile Tyr Thr Gly Ser Ser Gly
245           250           255

Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly Arg Phe Thr Val Ser Glu
260           265           270

Thr Ser Ser Thr Thr Val Thr Leu Gln Met Thr Ser Leu Thr Ala Ala
275           280           285

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Asp 290	Thr	Ala	Thr	Tyr	Phe	Cys 295	Ala	Arg	Ala	Ser	Ala 300	Trp	Thr	Tyr	Gly
Met 305	Asp	Leu	Trp	Gly	Pro 310	Gly	Thr	Leu	Val	Thr 315	Val	Ser	Ser	Gly	Gly 320
Gly	Gly	Ser	Gly	Gly 325	Gly	Gly	Ser	Pro	Met 330	Ala	Lys	Lys	Gly	Gly 335	Gly
Gly	Ser	Gly	Gly 340	Gly	Gly	Ser	Gly	Gly 345	Gly	Gly	Ser	Gln	Val 350	Gln	Leu
Gln	Glu 355	Ser	Gly	Pro	Gly	Leu	Val 360	Lys	Pro	Ser	Glu	Thr 365	Leu	Ser	Leu
Thr 370	Cys	Thr	Val	Ser	Gly	Gly 375	Ser	Ile	Ser	Tyr	Tyr 380	Ser	Trp	Thr	Trp
Val 385	Arg	Gln	Pro	Ala	Gly 390	Gln	Gly	Leu	Glu	Trp 395	Ile	Gly	Arg	Ile	His 400
Ile	Ser	Gly	Val	Thr 405	Asn	His	Asn	Pro	Ser 410	Leu	Lys	Ser	Arg	Val 415	Ser
Met	Ser	Ile	Asp	Thr 420	Ser	Arg	Thr	Gln	Phe 425	Ser	Leu	Arg	Leu	Thr	Ser
Val	Thr 435	Ala	Ala	Asp	Thr	Ala	Val 440	Tyr	Phe	Cys	Ala 445	Arg	Ser	Gly	Gly
Thr 450	Tyr	Ser	Ala	Phe	Asp	Ile 455	Trp	Gly	Gln	Gly	Thr 460	Met	Val	Thr	Val
Ser 465	Ser	Gly	Gly	Gly	Gly 470	Ser	Gly	Gly	Gly	Gly 475	Ser	Gly	Gly	Gly	Gly 480
Ser	Gly	Gly	Gly	Gly 485	Ser	Glu	Ile	Val	Leu 490	Thr	Gln	Ser	Pro	Gly 495	Thr
Leu	Ser	Leu	Ser	Pro 500	Gly	Glu	Arg	Ser	Thr 505	Leu	Ser	Cys	Arg	Ala	Ser
Gln	Ser	Phe 515	Thr	Ser	Asn	Tyr	Leu 520	Ala	Trp	Tyr	Gln 525	Gln	Arg	Pro	Gly
Gln 530	Ala	Pro	Arg	Leu	Leu	Ile 535	Tyr	Gly	Ala	Ser	Thr 540	Arg	Ala	Ile	Gly
Ile 545	Pro	Asp	Arg	Phe	Ser	Gly 550	Ser	Gly	Ser	Gly 555	Thr	Asp	Phe	Thr	Leu 560
Thr	Ile	Ser	Arg	Leu 565	Glu	Pro	Glu	Asp	Phe 570	Ala	Val	Tyr	Tyr	Cys	Gln
Gln	Tyr	Gly	Ser	Ser 580	Pro	Trp	Thr	Phe	Gly 585	Gln	Gly	Thr	Lys	Val	Glu
Ile 595	Lys	Thr	Thr	Thr	Pro	Ala	Pro 600	Arg	Pro	Pro	Thr 605	Pro	Ala	Pro	Thr
Ile 610	Ala	Ser	Gln	Pro	Leu	Ser 615	Leu	Arg	Pro	Glu	Ala 620	Cys	Arg	Pro	Ala
Ala 625	Gly	Gly	Ala	Val	His 630	Thr	Arg	Gly	Leu	Asp 635	Phe	Ala	Cys	Asp	Ile 640
Tyr	Ile	Trp	Ala	Pro 645	Leu	Ala	Gly	Thr	Cys 650	Gly	Val	Leu	Leu	Leu	Ser 655
Leu	Val	Ile	Thr	Leu 660	Tyr	Cys	Lys	Arg	Gly 665	Arg	Lys	Lys	Leu	Leu	Tyr
Ile	Phe 675	Lys	Gln	Pro	Phe	Met	Arg 680	Pro	Val	Gln	Thr 685	Thr	Gln	Glu	Glu
Asp 690	Gly	Cys	Ser	Cys	Arg	Phe 695	Pro	Glu	Glu	Glu	Glu 700	Gly	Gly	Cys	Glu

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Leu Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln
705              710              715              720

Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu
              725              730              735

Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly
              740              745              750

Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln
              755              760              765

Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu
770              775              780

Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr
785              790              795              800

Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro
              805              810              815

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Arg

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<210> SEQ ID NO 155
<211> LENGTH: 812
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

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<400> SEQUENCE: 155

```

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
1          5          10          15

His Ala Ala Arg Pro Gly Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu
          20          25          30

Asp Ser Thr Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
          35          40          45

Gly Ser Gly Gly Gly Gly Ser Asp Ile Val Met Thr Gln Thr Pro Ala
          50          55          60

Ser Val Ser Glu Pro Val Gly Gly Ser Val Thr Ile Lys Cys Gln Ala
65          70          75          80

Ser Gln Ser Phe Tyr Asn Leu Leu Ala Trp Tyr Gln Gln Lys Pro Gly
          85          90          95

Gln Pro Pro Lys Leu Leu Ile Tyr Asp Ala Ser Asp Leu Ala Ser Gly
          100         105         110

Val Pro Ser Arg Phe Lys Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu
          115         120         125

Thr Ile Ser Asp Leu Glu Cys Ala Asp Ala Ala Tyr Tyr Cys Gln
          130         135         140

Ser Ala Asp Gly Ser Ser Tyr Ala Phe Gly Gly Gly Thr Glu Val Val
145         150         155         160

Val Lys Gly Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
          165         170         175

Gly Ser Pro Met Ala Lys Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly
          180         185         190

Ser Gly Gly Gly Gly Ser Gln Glu Gln Leu Glu Glu Ser Gly Gly Gly
          195         200         205

Leu Val Lys Pro Glu Gly Ser Leu Thr Leu Thr Cys Thr Ala Ser Gly
          210         215         220

Val Ser Phe Ser Ser Ser Tyr Trp Ile Tyr Trp Val Arg Gln Ala Pro
225         230         235         240

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Gly 245	Lys	Gly	Leu	Glu	Trp	Ile	Ala	Cys	Ile	Tyr	Thr	Gly	Ser	Ser	Gly
Ser	Thr	Tyr	Tyr	Ala	Ser	Trp	Ala	Lys	Gly	Arg	Phe	Thr	Val	Ser	Glu
Thr	Ser	Ser	Thr	Thr	Val	Thr	Leu	Gln	Met	Thr	Ser	Leu	Thr	Ala	Ala
Asp	Thr	Ala	Thr	Tyr	Phe	Cys	Ala	Arg	Ala	Ser	Ala	Trp	Thr	Tyr	Gly
Met	Asp	Leu	Trp	Gly	Pro	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	Gly
Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Pro	Met	Ala	Lys	Lys	Gly	Gly	Gly
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro
Gly	Leu	Val	Lys	Pro	Ser	Glu	Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser
Gly	Gly	Ser	Ile	Ser	Tyr	Tyr	Ser	Trp	Thr	Trp	Val	Arg	Gln	Pro	Ala
Gly	Gln	Gly	Leu	Glu	Trp	Ile	Gly	Arg	Ile	His	Ile	Ser	Gly	Val	Thr
Asn	His	Asn	Pro	Ser	Leu	Lys	Ser	Arg	Val	Ser	Met	Ser	Ile	Asp	Thr
Ser	Arg	Thr	Gln	Phe	Ser	Leu	Arg	Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp
Thr	Ala	Val	Tyr	Phe	Cys	Ala	Arg	Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe
Asp	Ile	Trp	Gly	Gln	Gly	Thr	Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly
Ser	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro
Gly	Glu	Arg	Ser	Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser
Asn	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu
Leu	Ile	Tyr	Gly	Ala	Ser	Thr	Arg	Ala	Ile	Gly	Ile	Pro	Asp	Arg	Phe
Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Arg	Leu
Glu	Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Tyr	Gly	Ser	Ser
Pro	Trp	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile	Lys	Thr	Thr	Thr
Pro	Ala	Pro	Arg	Pro	Pro	Thr	Pro	Ala	Pro	Thr	Ile	Ala	Ser	Gln	Pro
Leu	Ser	Leu	Arg	Pro	Glu	Ala	Cys	Arg	Pro	Ala	Ala	Gly	Gly	Ala	Val
His	Thr	Arg	Gly	Leu	Asp	Phe	Ala	Cys	Asp	Ile	Tyr	Ile	Trp	Ala	Pro
Leu	Ala	Gly	Thr	Cys	Gly	Val	Leu	Leu	Leu	Ser	Leu	Val	Ile	Thr	Leu
Tyr	Cys	Lys	Arg	Gly	Arg	Lys	Lys	Leu	Leu	Tyr	Ile	Phe	Lys	Gln	Pro

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660					665					670					
Phe	Met	Arg	Pro	Val	Gln	Thr	Thr	Gln	Glu	Glu	Asp	Gly	Cys	Ser	Cys
	675						680					685			
Arg	Phe	Pro	Glu	Glu	Glu	Gly	Gly	Cys	Glu	Leu	Arg	Val	Lys	Phe	
	690					695						700			
Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln	Gln	Gly	Gln	Asn	Gln	Leu
	705					710					715				720
Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu	Glu	Tyr	Asp	Val	Leu	Asp
			725						730					735	
Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met	Gly	Gly	Lys	Pro	Arg	Arg	Lys
			740					745					750		
Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn	Glu	Leu	Gln	Lys	Asp	Lys	Met	Ala
			755				760					765			
Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys	Gly	Glu	Arg	Arg	Arg	Gly	Lys
	770					775						780			
Gly	His	Asp	Gly	Leu	Tyr	Gln	Gly	Leu	Ser	Thr	Ala	Thr	Lys	Asp	Thr
	785					790					795				800
Tyr	Asp	Ala	Leu	His	Met	Gln	Ala	Leu	Pro	Pro	Arg				
			805						810						

<210> SEQ ID NO 156

<211> LENGTH: 807

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 156

Met	Ala	Leu	Pro	Val	Thr	Ala	Leu	Leu	Leu	Pro	Leu	Ala	Leu	Leu	Leu
1				5					10					15	
His	Ala	Ala	Arg	Pro	Gly	Lys	Pro	Ile	Pro	Asn	Pro	Leu	Leu	Gly	Leu
			20					25					30		
Asp	Ser	Thr	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly
		35					40					45			
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Asp	Ile	Val	Met	Thr	Gln	Thr	Pro	Ala
		50				55					60				
Ser	Val	Ser	Glu	Pro	Val	Gly	Gly	Ser	Val	Thr	Ile	Lys	Cys	Gln	Ala
	65				70						75				80
Ser	Gln	Ser	Phe	Tyr	Asn	Leu	Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly
			85						90					95	
Gln	Pro	Pro	Lys	Leu	Leu	Ile	Tyr	Asp	Ala	Ser	Asp	Leu	Ala	Ser	Gly
			100					105					110		
Val	Pro	Ser	Arg	Phe	Lys	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu
		115					120					125			
Thr	Ile	Ser	Asp	Leu	Glu	Cys	Ala	Asp	Ala	Ala	Ala	Tyr	Tyr	Cys	Gln
	130					135						140			
Ser	Ala	Asp	Gly	Ser	Ser	Tyr	Ala	Phe	Gly	Gly	Gly	Thr	Glu	Val	Val
	145				150						155				160
Val	Lys	Gly	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly
			165						170					175	
Gly	Ser	Pro	Met	Ala	Lys	Lys	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly
		180						185					190		
Ser	Gly	Gly	Gly	Gly	Ser	Gln	Glu	Gln	Leu	Glu	Glu	Ser	Gly	Gly	Gly
	195					200							205		

Leu 210	Val	Lys	Pro	Glu	Gly	Ser 215	Leu	Thr	Leu	Thr	Cys 220	Thr	Ala	Ser	Gly
Val 225	Ser	Phe	Ser	Ser	Ser 230	Tyr	Trp	Ile	Tyr	Trp 235	Val	Arg	Gln	Ala	Pro 240
Gly	Lys	Gly	Leu	Glu 245	Trp	Ile	Ala	Cys	Ile 250	Tyr	Thr	Gly	Ser	Ser 255	Gly
Ser	Thr	Tyr	Tyr	Ala 260	Ser	Trp	Ala	Lys	Gly 265	Arg	Phe	Thr	Val	Ser	Glu
Thr	Ser	Ser	Thr	Thr 275	Val	Thr	Leu	Gln	Met 280	Thr	Ser	Leu	Thr	Ala	Ala
Asp 290	Thr	Ala	Thr	Tyr	Phe 295	Cys	Ala	Arg	Ala	Ser 300	Ala	Trp	Thr	Tyr	Gly
Met 305	Asp	Leu	Trp	Gly 310	Pro	Gly	Thr	Leu	Val 315	Thr	Val	Ser	Ser	Gly	Gly 320
Gly	Gly	Ser	Pro	Met 325	Ala	Lys	Lys	Gly	Gly 330	Gly	Gly	Ser	Gly	Gly 335	Gly
Gly	Ser	Gln	Val	Gln 340	Leu	Gln	Glu	Ser	Gly 345	Pro	Gly	Leu	Val	Lys	Pro
Ser	Glu	Thr	Leu	Ser 355	Leu	Thr	Cys	Thr	Val 360	Ser	Gly	Gly	Ser	Ile	Ser
Tyr 370	Tyr	Ser	Trp	Thr	Trp 375	Val	Arg	Gln	Pro	Ala 380	Gly	Gln	Gly	Leu	Glu
Trp 385	Ile	Gly	Arg	Ile 390	His	Ile	Ser	Gly	Val 395	Thr	Asn	His	Asn	Pro	Ser 400
Leu	Lys	Ser	Arg	Val 405	Ser	Met	Ser	Ile	Asp 410	Thr	Ser	Arg	Thr	Gln	Phe
Ser	Leu	Arg	Leu 420	Thr	Ser	Val	Thr	Ala	Ala 425	Asp	Thr	Ala	Val	Tyr	Phe
Cys	Ala	Arg 435	Ser	Gly	Gly	Thr	Tyr 440	Ser	Ala	Phe	Asp	Ile 445	Trp	Gly	Gln
Gly 450	Thr	Met	Val	Thr	Val	Ser 455	Ser	Gly	Gly	Gly	Gly 460	Ser	Gly	Gly	Gly
Gly 465	Ser	Gly	Gly	Gly 470	Gly	Ser	Gly	Gly	Gly	Gly 475	Ser	Glu	Ile	Val	Leu 480
Thr	Gln	Ser	Pro	Gly 485	Thr	Leu	Ser	Leu	Ser 490	Pro	Gly	Glu	Arg	Ser	Thr 495
Leu	Ser	Cys 500	Arg	Ala	Ser	Gln	Ser	Phe	Thr 505	Ser	Asn	Tyr	Leu	Ala	Trp
Tyr	Gln	Gln 515	Arg	Pro	Gly	Gln	Ala 520	Pro	Arg	Leu	Leu	Ile 525	Tyr	Gly	Ala
Ser 530	Thr	Arg	Ala	Ile	Gly	Ile 535	Pro	Asp	Arg	Phe	Ser 540	Gly	Ser	Gly	Ser
Gly 545	Thr	Asp	Phe	Thr	Leu 550	Thr	Ile	Ser	Arg	Leu 555	Glu	Pro	Glu	Asp	Phe 560
Ala	Val	Tyr	Tyr	Cys 565	Gln	Gln	Tyr	Gly	Ser 570	Ser	Pro	Trp	Thr	Phe	Gly 575
Gln	Gly	Thr	Lys 580	Val	Glu	Ile	Lys	Thr	Thr 585	Thr	Pro	Ala	Pro	Arg	Pro
Pro	Thr	Pro 595	Ala	Pro	Thr	Ile	Ala 600	Ser	Gln	Pro	Leu	Ser 605	Leu	Arg	Pro
Glu 610	Ala	Cys	Arg	Pro	Ala	Ala 615	Gly	Gly	Ala	Val	His 620	Thr	Arg	Gly	Leu
Asp	Phe	Ala	Cys	Asp	Ile	Tyr	Ile	Trp	Ala	Pro	Leu	Ala	Gly	Thr	Cys

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625	630	635	640
Gly Val Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr Cys Lys Arg Gly			
	645	650	655
Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met Arg Pro Val			
	660	665	670
Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe Pro Glu Glu			
	675	680	685
Glu Glu Gly Gly Cys Glu Leu Arg Val Lys Phe Ser Arg Ser Ala Asp			
	690	695	700
Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn			
	705	710	715
Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg			
	725	730	735
Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly			
	740	745	750
Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu			
	755	760	765
Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu			
	770	775	780
Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His			
	785	790	795
Met Gln Ala Leu Pro Pro Arg			
	805		

<210> SEQ ID NO 157

<211> LENGTH: 817

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 157

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu			
1	5	10	15
His Ala Ala Arg Pro Gly Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu			
	20	25	30
Asp Ser Thr Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly			
	35	40	45
Gly Ser Gly Gly Gly Gly Ser Asp Ile Val Met Thr Gln Thr Pro Ala			
	50	55	60
Ser Val Ser Glu Pro Val Gly Gly Ser Val Thr Ile Lys Cys Gln Ala			
	65	70	75
Ser Gln Ser Phe Tyr Asn Leu Leu Ala Trp Tyr Gln Gln Lys Pro Gly			
	85	90	95
Gln Pro Pro Lys Leu Leu Ile Tyr Asp Ala Ser Asp Leu Ala Ser Gly			
	100	105	110
Val Pro Ser Arg Phe Lys Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu			
	115	120	125
Thr Ile Ser Asp Leu Glu Cys Ala Asp Ala Ala Tyr Tyr Cys Gln			
	130	135	140
Ser Ala Asp Gly Ser Ser Tyr Ala Phe Gly Gly Gly Thr Glu Val Val			
	145	150	155
Val Lys Gly Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly			
	165	170	175

Gly 180	Ser 185	Gln 190	Gly 195	Gly 200	Ser 205	Gln 210	Gly 215	Glu 220	Gln 225	Leu 230	Glu 235	Ser 240	Gln 245	Gly 250	Ser 255	Glu 260	Gln 265	Leu 270	Gly 275	Ser 280	Gln 285	Gly 290	Ser 295	Glu 300	Gln 305	Leu 310	Gly 315	Ser 320	Gln 325	Gly 330	Ser 335	Glu 340	Gln 345	Leu 350	Gly 355	Ser 360	Gln 365	Thr 370	Cys 375	Val 385	Arg 390	Gln 400	Ile 405	Ser 410	Met 415	Ser 420	Ile 425	Thr 430	Ala 435	Thr 440	Ala 445	Ser 450	Thr 455	Gly 460	Ser 465	Gly 470	Gly 475	Gly 480	Gly 485	Ser 490	Glu 495	Arg 500	Thr 505	Leu 510	Ser 515	Thr 520	Gln 525	Ala 530	Thr 535	Gly 540	Ser 545	Gln 550	Thr 555	Val 560	Ala 565	Thr 570	Cys 575	Gln 580	Val 585	Thr 590	Pro 595	Ala 600	Thr 605	Leu 610	Thr 615	Thr 620	Thr 625	Thr 630	Pro 635	Ala 640	Thr 645	Thr 650	Thr 655	Thr 660	Thr 665	Thr 670	Thr 675	Thr 680	Thr 685	Thr 690	Thr 695	Thr 700	Thr 705	Thr 710	Thr 715	Thr 720	Thr 725	Thr 730	Thr 735	Thr 740	Thr 745	Thr 750	Thr 755	Thr 760	Thr 765	Thr 770	Thr 775	Thr 780	Thr 785	Thr 790	Thr 795	Thr 800	Thr 805	Thr 810	Thr 815	Thr 820	Thr 825	Thr 830	Thr 835	Thr 840	Thr 845	Thr 850	Thr 855	Thr 860	Thr 865	Thr 870	Thr 875	Thr 880	Thr 885	Thr 890	Thr 895	Thr 900	Thr 905	Thr 910	Thr 915	Thr 920	Thr 925	Thr 930	Thr 935	Thr 940	Thr 945	Thr 950	Thr 955	Thr 960	Thr 965	Thr 970	Thr 975	Thr 980	Thr 985	Thr 990	Thr 995
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595	600	605
Ile Ala Ser Gln Pro Leu Ser Leu Arg Pro Glu Ala Cys Arg Pro Ala		
610	615	620
Ala Gly Gly Ala Val His Thr Arg Gly Leu Asp Phe Ala Cys Asp Ile		
625	630	635
Tyr Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly Val Leu Leu Leu Ser		
645	650	655
Leu Val Ile Thr Leu Tyr Cys Lys Arg Gly Arg Lys Lys Leu Leu Tyr		
660	665	670
Ile Phe Lys Gln Pro Phe Met Arg Pro Val Gln Thr Thr Gln Glu Glu		
675	680	685
Asp Gly Cys Ser Cys Arg Phe Pro Glu Glu Glu Glu Gly Gly Cys Glu		
690	695	700
Leu Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln		
705	710	715
Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu		
725	730	735
Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly		
740	745	750
Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln		
755	760	765
Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu		
770	775	780
Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr		
785	790	795
Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro		
805	810	815

Arg

<210> SEQ ID NO 158

<211> LENGTH: 817

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 158

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu		
1	5	10
His Ala Ala Arg Pro Gly Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu		
20	25	30
Asp Ser Thr Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly		
35	40	45
Gly Ser Gly Gly Gly Gly Ser Asp Ile Val Met Thr Gln Thr Pro Ala		
50	55	60
Ser Val Ser Glu Pro Val Gly Gly Ser Val Thr Ile Lys Cys Gln Ala		
65	70	75
Ser Gln Ser Phe Tyr Asn Leu Leu Ala Trp Tyr Gln Gln Lys Pro Gly		
85	90	95
Gln Pro Pro Lys Leu Leu Ile Tyr Asp Ala Ser Asp Leu Ala Ser Gly		
100	105	110
Val Pro Ser Arg Phe Lys Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu		
115	120	125
Thr Ile Ser Asp Leu Glu Cys Ala Asp Ala Ala Ala Tyr Tyr Cys Gln		

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130	135	140
Ser Ala Asp Gly Ser Ser Tyr Ala Phe Gly Gly Gly Thr Glu Val Val		
145	150	155 160
Val Lys Gly Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly		
	165	170 175
Gly Ser Gly Gly Gly Gly Ser Gln Glu Gln Leu Glu Glu Ser Gly Gly		
	180	185 190
Gly Leu Val Lys Pro Glu Gly Ser Leu Thr Leu Thr Cys Thr Ala Ser		
	195	200 205
Gly Val Ser Phe Ser Ser Ser Tyr Trp Ile Tyr Trp Val Arg Gln Ala		
	210	215 220
Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys Ile Tyr Thr Gly Ser Ser		
	225	230 235 240
Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly Arg Phe Thr Val Ser		
	245	250 255
Glu Thr Ser Ser Thr Thr Val Thr Leu Gln Met Thr Ser Leu Thr Ala		
	260	265 270
Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ala Ser Ala Trp Thr Tyr		
	275	280 285
Gly Met Asp Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser Gly		
	290	295 300
Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Ser Val His Met Pro		
	305	310 315 320
Leu Gly Phe Leu Gly Pro Gly Gly Ser Gly Gly Gly Ser Pro Met Ala		
	325	330 335
Lys Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln Val Gln Leu		
	340	345 350
Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu Thr Leu Ser Leu		
	355	360 365
Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Tyr Tyr Ser Trp Thr Trp		
	370	375 380
Val Arg Gln Pro Ala Gly Gln Gly Leu Glu Trp Ile Gly Arg Ile His		
	385	390 395 400
Ile Ser Gly Val Thr Asn His Asn Pro Ser Leu Lys Ser Arg Val Ser		
	405	410 415
Met Ser Ile Asp Thr Ser Arg Thr Gln Phe Ser Leu Arg Leu Thr Ser		
	420	425 430
Val Thr Ala Ala Asp Thr Ala Val Tyr Phe Cys Ala Arg Ser Gly Gly		
	435	440 445
Thr Tyr Ser Ala Phe Asp Ile Trp Gly Gln Gly Thr Met Val Thr Val		
	450	455 460
Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly		
	465	470 475 480
Ser Gly Gly Gly Gly Ser Glu Ile Val Leu Thr Gln Ser Pro Gly Thr		
	485	490 495
Leu Ser Leu Ser Pro Gly Glu Arg Ser Thr Leu Ser Cys Arg Ala Ser		
	500	505 510
Gln Ser Phe Thr Ser Asn Tyr Leu Ala Trp Tyr Gln Gln Arg Pro Gly		
	515	520 525
Gln Ala Pro Arg Leu Leu Ile Tyr Gly Ala Ser Thr Arg Ala Ile Gly		
	530	535 540
Ile Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu		
	545	550 555 560

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Thr Ile Ser Arg Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln
 565 570 575
 Gln Tyr Gly Ser Ser Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu
 580 585 590
 Ile Lys Thr Thr Thr Pro Ala Pro Arg Pro Pro Thr Pro Ala Pro Thr
 595 600 605
 Ile Ala Ser Gln Pro Leu Ser Leu Arg Pro Glu Ala Cys Arg Pro Ala
 610 615 620
 Ala Gly Gly Ala Val His Thr Arg Gly Leu Asp Phe Ala Cys Asp Ile
 625 630 635 640
 Tyr Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly Val Leu Leu Leu Ser
 645 650 655
 Leu Val Ile Thr Leu Tyr Cys Lys Arg Gly Arg Lys Lys Leu Leu Tyr
 660 665 670
 Ile Phe Lys Gln Pro Phe Met Arg Pro Val Gln Thr Thr Gln Glu Glu
 675 680 685
 Asp Gly Cys Ser Cys Arg Phe Pro Glu Glu Glu Glu Gly Gly Cys Glu
 690 695 700
 Leu Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln
 705 710 715 720
 Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu
 725 730 735
 Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly
 740 745 750
 Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln
 755 760 765
 Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu
 770 775 780
 Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr
 785 790 795 800
 Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro
 805 810 815

Arg

<210> SEQ ID NO 159
 <211> LENGTH: 817
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 159

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15
 His Ala Ala Arg Pro Gly Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu
 20 25 30
 Asp Ser Thr Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
 35 40 45
 Gly Ser Gly Gly Gly Gly Ser Asp Ile Val Met Thr Gln Thr Pro Ala
 50 55 60
 Ser Val Ser Glu Pro Val Gly Gly Ser Val Thr Ile Lys Cys Gln Ala
 65 70 75 80
 Ser Gln Ser Phe Tyr Asn Leu Leu Ala Trp Tyr Gln Gln Lys Pro Gly
 85 90 95

Gln	Pro		Lys	Leu	Leu	Ile	Tyr	Asp	Ala	Ser	Asp		Leu	Ala	Ser	Gly
			100					105						110		
Val	Pro	Ser	Arg	Phe	Lys	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	
		115					120					125				
Thr	Ile	Ser	Asp	Leu	Glu	Cys	Ala	Asp	Ala	Ala	Ala	Tyr	Tyr	Cys	Gln	
		130				135						140				
Ser	Ala	Asp	Gly	Ser	Ser	Tyr	Ala	Phe	Gly	Gly	Gly	Thr	Glu	Val	Val	
		145				150					155				160	
Val	Lys	Gly	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	
				165											175	
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Glu	Gln	Leu	Glu	Glu	Ser	Gly	Gly	
				180						185				190		
Gly	Leu	Val	Lys	Pro	Glu	Gly	Ser	Leu	Thr	Leu	Thr	Cys	Thr	Ala	Ser	
							200					205				
Gly	Val	Ser	Phe	Ser	Ser	Ser	Tyr	Trp	Ile	Tyr	Trp	Val	Arg	Gln	Ala	
						215						220				
Pro	Gly	Lys	Gly	Leu	Glu	Trp	Ile	Ala	Cys	Ile	Tyr	Thr	Gly	Ser	Ser	
					230					235					240	
Gly	Ser	Thr	Tyr	Tyr	Ala	Ser	Trp	Ala	Lys	Gly	Arg	Phe	Thr	Val	Ser	
					245					250				255		
Glu	Thr	Ser	Ser	Thr	Thr	Val	Thr	Leu	Gln	Met	Thr	Ser	Leu	Thr	Ala	
								265						270		
Ala	Asp	Thr	Ala	Thr	Tyr	Phe	Cys	Ala	Arg	Ala	Ser	Ala	Trp	Thr	Tyr	
						280						285				
Gly	Met	Asp	Leu	Trp	Gly	Pro	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	
						295						300				
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Ser	Pro	Leu	Gly	Val	
					310						315				320	
Arg	Gly	Lys	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Pro	Met	Ala	
					325						330			335		
Lys	Lys	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	
					340								350			
Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Glu	Thr	Leu	Ser	Leu	
							360						365			
Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile	Ser	Tyr	Tyr	Ser	Trp	Thr	Trp	
						375						380				
Val	Arg	Gln	Pro	Ala	Gly	Gln	Gly	Leu	Glu	Trp	Ile	Gly	Arg	Ile	His	
					390						395				400	
Ile	Ser	Gly	Val	Thr	Asn	His	Asn	Pro	Ser	Leu	Lys	Ser	Arg	Val	Ser	
					405					410				415		
Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln	Phe	Ser	Leu	Arg	Leu	Thr	Ser	
								425					430			
Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Phe	Cys	Ala	Arg	Ser	Gly	Gly	

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Gln Ser Phe Thr Ser Asn Tyr Leu Ala Trp Tyr Gln Gln Arg Pro Gly
 515 520 525
 Gln Ala Pro Arg Leu Leu Ile Tyr Gly Ala Ser Thr Arg Ala Ile Gly
 530 535 540
 Ile Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu
 545 550 555 560
 Thr Ile Ser Arg Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln
 565 570 575
 Gln Tyr Gly Ser Ser Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu
 580 585 590
 Ile Lys Thr Thr Thr Pro Ala Pro Arg Pro Pro Thr Pro Ala Pro Thr
 595 600 605
 Ile Ala Ser Gln Pro Leu Ser Leu Arg Pro Glu Ala Cys Arg Pro Ala
 610 615 620
 Ala Gly Gly Ala Val His Thr Arg Gly Leu Asp Phe Ala Cys Asp Ile
 625 630 635 640
 Tyr Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly Val Leu Leu Leu Ser
 645 650 655
 Leu Val Ile Thr Leu Tyr Cys Lys Arg Gly Arg Lys Lys Leu Leu Tyr
 660 665 670
 Ile Phe Lys Gln Pro Phe Met Arg Pro Val Gln Thr Thr Gln Glu Glu
 675 680 685
 Asp Gly Cys Ser Cys Arg Phe Pro Glu Glu Glu Glu Gly Gly Cys Glu
 690 695 700
 Leu Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln
 705 710 715 720
 Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu
 725 730 735
 Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly
 740 745 750
 Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln
 755 760 765
 Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu
 770 775 780
 Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr
 785 790 795 800
 Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro
 805 810 815

Arg

<210> SEQ ID NO 160

<211> LENGTH: 817

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 160

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15
 His Ala Ala Arg Pro Gly Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu
 20 25 30
 Asp Ser Thr Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
 35 40 45

Gly	Ser	Gly	Gly	Gly	Gly	Ser	Asp	Ile	Val	Met	Thr	Gln	Thr	Pro	Ala
Ser	Val	Ser	Glu	Pro	Val	Gly	Gly	Ser	Val	Thr	Ile	Lys	Cys	Gln	Ala
Ser	Gln	Ser	Phe	Tyr	Asn	Leu	Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly
Gln	Pro	Pro	Lys	Leu	Leu	Ile	Tyr	Asp	Ala	Ser	Asp	Leu	Ala	Ser	Gly
Val	Pro	Ser	Arg	Phe	Lys	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu
Thr	Ile	Ser	Asp	Leu	Glu	Cys	Ala	Asp	Ala	Ala	Ala	Tyr	Tyr	Cys	Gln
Ser	Ala	Asp	Gly	Ser	Ser	Tyr	Ala	Phe	Gly	Gly	Gly	Thr	Glu	Val	Val
Val	Lys	Gly	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Glu	Gln	Leu	Glu	Glu	Ser	Gly	Gly
Gly	Leu	Val	Lys	Pro	Glu	Gly	Ser	Leu	Thr	Leu	Thr	Cys	Thr	Ala	Ser
Gly	Val	Ser	Phe	Ser	Ser	Ser	Tyr	Trp	Ile	Tyr	Trp	Val	Arg	Gln	Ala
Pro	Gly	Lys	Gly	Leu	Glu	Trp	Ile	Ala	Cys	Ile	Tyr	Thr	Gly	Ser	Ser
Gly	Ser	Thr	Tyr	Tyr	Ala	Ser	Trp	Ala	Lys	Gly	Arg	Phe	Thr	Val	Ser
Glu	Thr	Ser	Ser	Thr	Thr	Val	Thr	Leu	Gln	Met	Thr	Ser	Leu	Thr	Ala
Ala	Asp	Thr	Ala	Thr	Tyr	Phe	Cys	Ala	Arg	Ala	Ser	Ala	Trp	Thr	Tyr
Gly	Met	Asp	Leu	Trp	Gly	Pro	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Gly
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Ser	Ser	Pro	Leu	Gly
Leu	Ala	Gly	Ser	Gly	Ser	Gly	Gly	Gly	Ser	Arg	Gln	Ala	Arg	Val	Val
Asn	Gly	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu
Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Glu	Thr	Leu	Ser	Leu
Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile	Ser	Tyr	Tyr	Ser	Trp	Thr	Trp
Val	Arg	Gln	Pro	Ala	Gly	Gln	Gly	Leu	Glu	Trp	Ile	Gly	Arg	Ile	His
Ile	Ser	Gly	Val	Thr	Asn	His	Asn	Pro	Ser	Leu	Lys	Ser	Arg	Val	Ser
Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln	Phe	Ser	Leu	Arg	Leu	Thr	Ser
Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Phe	Cys	Ala	Arg	Ser	Gly	Gly
Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly	Gln	Gly	Thr	Met	Val	Thr	Val
Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly

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465	470	475	480
Ser Gly Gly Gly Gly Ser Glu Ile Val Leu Thr Gln Ser Pro Gly Thr	485	490	495
Leu Ser Leu Ser Pro Gly Glu Arg Ser Thr Leu Ser Cys Arg Ala Ser	500	505	510
Gln Ser Phe Thr Ser Asn Tyr Leu Ala Trp Tyr Gln Gln Arg Pro Gly	515	520	525
Gln Ala Pro Arg Leu Leu Ile Tyr Gly Ala Ser Thr Arg Ala Ile Gly	530	535	540
Ile Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu	545	550	555
Thr Ile Ser Arg Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln	565	570	575
Gln Tyr Gly Ser Ser Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu	580	585	590
Ile Lys Thr Thr Thr Pro Ala Pro Arg Pro Pro Thr Pro Ala Pro Thr	595	600	605
Ile Ala Ser Gln Pro Leu Ser Leu Arg Pro Glu Ala Cys Arg Pro Ala	610	615	620
Ala Gly Gly Ala Val His Thr Arg Gly Leu Asp Phe Ala Cys Asp Ile	625	630	635
Tyr Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly Val Leu Leu Leu Ser	645	650	655
Leu Val Ile Thr Leu Tyr Cys Lys Arg Gly Arg Lys Lys Leu Leu Tyr	660	665	670
Ile Phe Lys Gln Pro Phe Met Arg Pro Val Gln Thr Thr Gln Glu Glu	675	680	685
Asp Gly Cys Ser Cys Arg Phe Pro Glu Glu Glu Glu Gly Gly Cys Glu	690	695	700
Leu Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln	705	710	715
Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu	725	730	735
Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly	740	745	750
Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln	755	760	765
Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu	770	775	780
Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr	785	790	795
Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro	805	810	815

Arg

<210> SEQ ID NO 161

<211> LENGTH: 817

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 161

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu

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1	5	10	15
His Ala Ala Arg Pro Gly Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu	20	25	30
Asp Ser Thr Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly	35	40	45
Gly Ser Gly Gly Gly Gly Ser Asp Ile Val Met Thr Gln Thr Pro Ala	50	55	60
Ser Val Ser Glu Pro Val Gly Gly Ser Val Thr Ile Lys Cys Gln Ala	65	70	75
Ser Gln Ser Phe Tyr Asn Leu Leu Ala Trp Tyr Gln Gln Lys Pro Gly	85	90	95
Gln Pro Pro Lys Leu Leu Ile Tyr Asp Ala Ser Asp Leu Ala Ser Gly	100	105	110
Val Pro Ser Arg Phe Lys Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu	115	120	125
Thr Ile Ser Asp Leu Glu Cys Ala Asp Ala Ala Ala Tyr Tyr Cys Gln	130	135	140
Ser Ala Asp Gly Ser Ser Tyr Ala Phe Gly Gly Gly Thr Glu Val Val	145	150	155
Val Lys Gly Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly	165	170	175
Gly Ser Gly Gly Gly Gly Ser Gln Glu Gln Leu Glu Glu Ser Gly Gly	180	185	190
Gly Leu Val Lys Pro Glu Gly Ser Leu Thr Leu Thr Cys Thr Ala Ser	195	200	205
Gly Val Ser Phe Ser Ser Ser Tyr Trp Ile Tyr Trp Val Arg Gln Ala	210	215	220
Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys Ile Tyr Thr Gly Ser Ser	225	230	235
Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly Arg Phe Thr Val Ser	245	250	255
Glu Thr Ser Ser Thr Thr Val Thr Leu Gln Met Thr Ser Leu Thr Ala	260	265	270
Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ala Ser Ala Trp Thr Tyr	275	280	285
Gly Met Asp Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser Gly	290	295	300
Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Ser Val His Met Pro	305	310	315
Leu Gly Phe Leu Gly Pro Gly Gly Gly Ser Arg Gln Ala Arg Val Val	325	330	335
Asn Gly Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln Val Gln Leu	340	345	350
Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu Thr Leu Ser Leu	355	360	365
Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Tyr Tyr Ser Trp Thr Trp	370	375	380
Val Arg Gln Pro Ala Gly Gln Gly Leu Glu Trp Ile Gly Arg Ile His	385	390	395
Ile Ser Gly Val Thr Asn His Asn Pro Ser Leu Lys Ser Arg Val Ser	405	410	415
Met Ser Ile Asp Thr Ser Arg Thr Gln Phe Ser Leu Arg Leu Thr Ser	420	425	430

Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Phe	Cys	Ala	Arg	Ser	Gly	Gly
		435					440					445			
Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly	Gln	Gly	Thr	Met	Val	Thr	Val
	450					455					460				
Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly
465					470					475					480
Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr
				485					490					495	
Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser	Thr	Leu	Ser	Cys	Arg	Ala	Ser
			500					505				510			
Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Arg	Pro	Gly
		515					520					525			
Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Ser	Thr	Arg	Ala	Ile	Gly
	530					535					540				
Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu
545					550					555					560
Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln
				565					570					575	
Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu
		580						585					590		
Ile	Lys	Thr	Thr	Thr	Pro	Ala	Pro	Arg	Pro	Pro	Thr	Pro	Ala	Pro	Thr
		595					600					605			
Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu	Arg	Pro	Glu	Ala	Cys	Arg	Pro	Ala
	610					615					620				
Ala	Gly	Gly	Ala	Val	His	Thr	Arg	Gly	Leu	Asp	Phe	Ala	Cys	Asp	Ile
625					630					635					640
Tyr	Ile	Trp	Ala	Pro	Leu	Ala	Gly	Thr	Cys	Gly	Val	Leu	Leu	Leu	Ser
			645						650				655		
Leu	Val	Ile	Thr	Leu	Tyr	Cys	Lys	Arg	Gly	Arg	Lys	Lys	Leu	Leu	Tyr
		660						665				670			
Ile	Phe	Lys	Gln	Pro	Phe	Met	Arg	Pro	Val	Gln	Thr	Thr	Gln	Glu	Glu
	675						680					685			
Asp	Gly	Cys	Ser	Cys	Arg	Phe	Pro	Glu	Glu	Glu	Gly	Gly	Gly	Cys	Glu
	690					695				700					
Leu	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln	Gln
705					710					715					720
Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu	Glu
			725					730						735	
Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met	Gly	Gly
		740					745					750			
Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn	Glu	Leu	Gln
	755					760						765			
Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys	Gly	Glu
	770					775					780				
Arg	Arg	Arg	Gly	Lys	Gly	His</									

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<210> SEQ ID NO 162
<211> LENGTH: 817
<212> TYPE: PRT
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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
    polypeptide

<400> SEQUENCE: 162

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1           5           10           15

His Ala Ala Arg Pro Gly Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu
      20           25           30

Asp Ser Thr Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
      35           40           45

Gly Ser Gly Gly Gly Gly Ser Asp Ile Val Met Thr Gln Thr Pro Ala
 50           55           60

Ser Val Ser Glu Pro Val Gly Gly Ser Val Thr Ile Lys Cys Gln Ala
 65           70           75           80

Ser Gln Ser Phe Tyr Asn Leu Leu Ala Trp Tyr Gln Gln Lys Pro Gly
      85           90           95

Gln Pro Pro Lys Leu Leu Ile Tyr Asp Ala Ser Asp Leu Ala Ser Gly
      100          105          110

Val Pro Ser Arg Phe Lys Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu
      115          120          125

Thr Ile Ser Asp Leu Glu Cys Ala Asp Ala Ala Tyr Tyr Cys Gln
      130          135          140

Ser Ala Asp Gly Ser Ser Tyr Ala Phe Gly Gly Gly Thr Glu Val Val
      145          150          155          160

Val Lys Gly Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
      165          170          175

Gly Ser Gly Gly Gly Gly Ser Gln Glu Gln Leu Glu Glu Ser Gly Gly
      180          185          190

Gly Leu Val Lys Pro Glu Gly Ser Leu Thr Leu Thr Cys Thr Ala Ser
      195          200          205

Gly Val Ser Phe Ser Ser Ser Tyr Trp Ile Tyr Trp Val Arg Gln Ala
      210          215          220

Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys Ile Tyr Thr Gly Ser Ser
      225          230          235          240

Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly Arg Phe Thr Val Ser
      245          250          255

Glu Thr Ser Ser Thr Thr Val Thr Leu Gln Met Thr Ser Leu Thr Ala
      260          265          270

Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ala Ser Ala Trp Thr Tyr
      275          280          285

Gly Met Asp Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser Gly
      290          295          300

Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Ser Pro Leu Gly Val
      305          310          315          320

Arg Gly Lys Gly Gly Ser Gly Gly Gly Ser Arg Gln Ala Arg Val Val
      325          330          335

Asn Gly Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln Val Gln Leu
      340          345          350

Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu Thr Leu Ser Leu
      355          360          365

Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Tyr Tyr Ser Trp Thr Trp
      370          375          380

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Val	Arg	Gln	Pro	Ala	Gly	Gln	Gly	Leu	Glu	Trp	Ile	Gly	Arg	Ile	His	385	390	395	400
Ile	Ser	Gly	Val	Thr	Asn	His	Asn	Pro	Ser	Leu	Lys	Ser	Arg	Val	Ser	405	410	415	
Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln	Phe	Ser	Leu	Arg	Leu	Thr	Ser	420	425	430	
Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Phe	Cys	Ala	Arg	Ser	Gly	Gly	435	440	445	
Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly	Gln	Gly	Thr	Met	Val	Thr	Val	450	455	460	
Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	465	470	475	480
Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	485	490	495	
Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser	Thr	Leu	Ser	Cys	Arg	Ala	Ser	500	505	510	
Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Arg	Pro	Gly	515	520	525	
Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Ser	Thr	Arg	Ala	Ile	Gly	530	535	540	
Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	545	550	555	560
Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	565	570	575	
Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	580	585	590	
Ile	Lys	Thr	Thr	Thr	Pro	Ala	Pro	Arg	Pro	Pro	Thr	Pro	Ala	Pro	Thr	595	600	605	
Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu	Arg	Pro	Glu	Ala	Cys	Arg	Pro	Ala	610	615	620	
Ala	Gly	Gly	Ala	Val	His	Thr	Arg	Gly	Leu	Asp	Phe	Ala	Cys	Asp	Ile	625	630	635	640
Tyr	Ile	Trp	Ala	Pro	Leu	Ala	Gly	Thr	Cys	Gly	Val	Leu	Leu	Leu	Ser	645	650	655	
Leu	Val	Ile	Thr	Leu	Tyr	Cys	Lys	Arg	Gly	Arg	Lys	Lys	Leu	Leu	Tyr	660	665	670	
Ile	Phe	Lys	Gln	Pro	Phe	Met	Arg	Pro	Val	Gln	Thr	Thr	Gln	Glu	Glu	675	680	685	
Asp	Gly	Cys	Ser	Cys	Arg	Phe	Pro	Glu	Glu	Glu	Glu	Gly	Gly	Cys	Glu	690	695	700	
Leu	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln	Gln	705	710	715	720
Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu	Glu	725	730	735	
Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met	Gly	Gly	740	745	750	
Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn	Glu	Leu	Gln	755	760	765	
Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys	Gly	Glu	770	775	780	
Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr	Gln	Gly	Leu	Ser	Thr	785	790	795	800
Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His	Met	Gln	Ala	Leu	Pro	Pro				

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805	810	815
Arg		
<210> SEQ ID NO 163		
<211> LENGTH: 817		
<212> TYPE: PRT		
<213> ORGANISM: Artificial Sequence		
<220> FEATURE:		
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide		
<400> SEQUENCE: 163		
Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu		
1 5 10 15		
His Ala Ala Arg Pro Gly Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu		
20 25 30		
Asp Ser Thr Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly		
35 40 45		
Gly Ser Gly Gly Gly Gly Ser Asp Ile Val Met Thr Gln Thr Pro Ala		
50 55 60		
Ser Val Ser Glu Pro Val Gly Gly Ser Val Thr Ile Lys Cys Gln Ala		
65 70 75 80		
Ser Gln Ser Phe Tyr Asn Leu Leu Ala Trp Tyr Gln Gln Lys Pro Gly		
85 90 95		
Gln Pro Pro Lys Leu Leu Ile Tyr Asp Ala Ser Asp Leu Ala Ser Gly		
100 105 110		
Val Pro Ser Arg Phe Lys Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu		
115 120 125		
Thr Ile Ser Asp Leu Glu Cys Ala Asp Ala Ala Tyr Tyr Cys Gln		
130 135 140		
Ser Ala Asp Gly Ser Ser Tyr Ala Phe Gly Gly Gly Thr Glu Val Val		
145 150 155 160		
Val Lys Gly Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly		
165 170 175		
Gly Ser Gly Gly Gly Gly Ser Gln Glu Gln Leu Glu Glu Ser Gly Gly		
180 185 190		
Gly Leu Val Lys Pro Glu Gly Ser Leu Thr Leu Thr Cys Thr Ala Ser		
195 200 205		
Gly Val Ser Phe Ser Ser Ser Tyr Trp Ile Tyr Trp Val Arg Gln Ala		
210 215 220		
Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys Ile Tyr Thr Gly Ser Ser		
225 230 235 240		
Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly Arg Phe Thr Val Ser		
245 250 255		
Glu Thr Ser Ser Thr Thr Val Thr Leu Gln Met Thr Ser Leu Thr Ala		
260 265 270		
Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ala Ser Ala Trp Thr Tyr		
275 280 285		
Gly Met Asp Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser Gly		
290 295 300		
Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Ser Gly Gly Gly		
305 310 315 320		
Ser Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly		
325 330 335		
Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gln Val Gln Leu		

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340							345					350				
Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Glu	Thr	Leu	Ser	Leu	
		355					360					365				
Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile	Ser	Tyr	Tyr	Ser	Trp	Thr	Trp	
	370					375					380					
Val	Arg	Gln	Pro	Ala	Gly	Gln	Gly	Leu	Glu	Trp	Ile	Gly	Arg	Ile	His	
	385				390					395					400	
Ile	Ser	Gly	Val	Thr	Asn	His	Asn	Pro	Ser	Leu	Lys	Ser	Arg	Val	Ser	
				405					410					415		
Met	Ser	Ile	Asp	Thr	Ser	Arg	Thr	Gln	Phe	Ser	Leu	Arg	Leu	Thr	Ser	
			420						425				430			
Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Phe	Cys	Ala	Arg	Ser	Gly	Gly	
		435					440					445				
Thr	Tyr	Ser	Ala	Phe	Asp	Ile	Trp	Gly	Gln	Gly	Thr	Met	Val	Thr	Val	
	450					455					460					
Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	
	465				470					475					480	
Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	
				485					490					495		
Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ser	Thr	Leu	Ser	Cys	Arg	Ala	Ser	
			500						505				510			
Gln	Ser	Phe	Thr	Ser	Asn	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Arg	Pro	Gly	
		515					520					525				
Gln	Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Ser	Thr	Arg	Ala	Ile	Gly	
	530					535					540					
Ile	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	
	545				550					555					560	
Thr	Ile	Ser	Arg	Leu	Glu	Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	
				565					570					575		
Gln	Tyr	Gly	Ser	Ser	Pro	Trp	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	
			580						585				590			
Ile	Lys	Thr	Thr	Thr	Pro	Ala	Pro	Arg	Pro	Pro	Thr	Pro	Ala	Pro	Thr	
		595					600					605				
Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu	Arg	Pro	Glu	Ala	Cys	Arg	Pro	Ala	
	610					615					620					
Ala	Gly	Gly	Ala	Val	His	Thr	Arg	Gly	Leu	Asp	Phe	Ala	Cys	Asp	Ile	
	625				630					635					640	
Tyr	Ile	Trp	Ala	Pro	Leu	Ala	Gly	Thr	Cys	Gly	Val	Leu	Leu	Leu	Ser	
			645						650					655		
Leu	Val	Ile	Thr	Leu	Tyr	Cys	Lys	Arg	Gly	Arg	Lys	Lys	Leu	Leu	Tyr	
			660					665					670			
Ile	Phe	Lys	Gln	Pro	Phe	Met	Arg	Pro	Val	Gln	Thr	Thr	Gln	Glu	Glu	
		675					680						685			
Asp	Gly	Cys	Ser	Cys	Arg	Phe	Pro	Glu	Glu	Glu	Glu	Gly	Gly	Cys	Glu	
	690					695					700					
Leu	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln	Gln	
	705				710					715					720	
Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu	Glu	
			725						730					735		
Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met	Gly	Gly	
			740					745					750			
Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn	Glu	Leu	Gln	
		755					760					765				

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Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu
770 775 780

Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr
785 790 795 800

Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro
805 810 815

Arg

<210> SEQ ID NO 164

<211> LENGTH: 807

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 164

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
1 5 10 15

His Ala Ala Arg Pro Gly Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu
20 25 30

Asp Ser Thr Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
35 40 45

Gly Ser Gly Gly Gly Gly Ser Asp Ile Val Met Thr Gln Thr Pro Ala
50 55 60

Ser Val Ser Glu Pro Val Gly Gly Ser Val Thr Ile Lys Cys Gln Ala
65 70 75 80

Ser Gln Ser Phe Tyr Asn Leu Leu Ala Trp Tyr Gln Gln Lys Pro Gly
85 90 95

Gln Pro Pro Lys Leu Leu Ile Tyr Asp Ala Ser Asp Leu Ala Ser Gly
100 105 110

Val Pro Ser Arg Phe Lys Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu
115 120 125

Thr Ile Ser Asp Leu Glu Cys Ala Asp Ala Ala Tyr Tyr Cys Gln
130 135 140

Ser Ala Asp Gly Ser Ser Tyr Ala Phe Gly Gly Thr Glu Val Val
145 150 155 160

Val Lys Gly Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly
165 170 175

Gly Ser Gly Gly Gly Gly Ser Gln Glu Gln Leu Glu Glu Ser Gly Gly
180 185 190

Gly Leu Val Lys Pro Glu Gly Ser Leu Thr Leu Thr Cys Thr Ala Ser
195 200 205

Gly Val Ser Phe Ser Ser Ser Tyr Trp Ile Tyr Trp Val Arg Gln Ala
210 215 220

Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys Ile Tyr Thr Gly Ser Ser
225 230 235 240

Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly Arg Phe Thr Val Ser
245 250 255

Glu Thr Ser Ser Thr Thr Val Thr Leu Gln Met Thr Ser Leu Thr Ala
260 265 270

Ala Asp Thr Ala Thr Tyr Phe Cys Ala Arg Ala Ser Ala Trp Thr Tyr
275 280 285

Gly Met Asp Leu Trp Gly Pro Gly Thr Leu Val Thr Val Ser Ser Gly
290 295 300

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Gly Gly Ser Gly Gly Gly Ser Gly Ser Gly Gly Gly Gly Ser Gly Gly
 305 310 315 320
 Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
 325 330 335
 Gly Ser Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro
 340 345 350
 Ser Glu Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile Ser
 355 360 365
 Tyr Tyr Ser Trp Thr Trp Val Arg Gln Pro Ala Gly Gln Gly Leu Glu
 370 375 380
 Trp Ile Gly Arg Ile His Ile Ser Gly Val Thr Asn His Asn Pro Ser
 385 390 395 400
 Leu Lys Ser Arg Val Ser Met Ser Ile Asp Thr Ser Arg Thr Gln Phe
 405 410 415
 Ser Leu Arg Leu Thr Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Phe
 420 425 430
 Cys Ala Arg Ser Gly Gly Thr Tyr Ser Ala Phe Asp Ile Trp Gly Gln
 435 440 445
 Gly Thr Met Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly
 450 455 460
 Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Glu Ile Val Leu
 465 470 475 480
 Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly Glu Arg Ser Thr
 485 490 495
 Leu Ser Cys Arg Ala Ser Gln Ser Phe Thr Ser Asn Tyr Leu Ala Trp
 500 505 510
 Tyr Gln Gln Arg Pro Gly Gln Ala Pro Arg Leu Leu Ile Tyr Gly Ala
 515 520 525
 Ser Thr Arg Ala Ile Gly Ile Pro Asp Arg Phe Ser Gly Ser Gly Ser
 530 535 540
 Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu Pro Glu Asp Phe
 545 550 555 560
 Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro Trp Thr Phe Gly
 565 570 575
 Gln Gly Thr Lys Val Glu Ile Lys Thr Thr Thr Pro Ala Pro Arg Pro
 580 585 590
 Pro Thr Pro Ala Pro Thr Ile Ala Ser Gln Pro Leu Ser Leu Arg Pro
 595 600 605
 Glu Ala Cys Arg Pro Ala Ala Gly Gly Ala Val His Thr Arg Gly Leu
 610 615 620
 Asp Phe Ala Cys Asp Ile Tyr Ile Trp Ala Pro Leu Ala Gly Thr Cys
 625 630 635 640
 Gly Val Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr Cys Lys Arg Gly
 645 650 655
 Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met Arg Pro Val
 660 665 670
 Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe Pro Glu Glu
 675 680 685
 Glu Glu Gly Gly Cys Glu Leu Arg Val Lys Phe Ser Arg Ser Ala Asp
 690 695 700
 Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn
 705 710 715 720

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Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg
725 730 735

Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly
740 745 750

Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu
755 760 765

Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu
770 775 780

Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His
785 790 795 800

Met Gln Ala Leu Pro Pro Arg
805

<210> SEQ ID NO 165

<211> LENGTH: 797

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 165

Met Ala Leu Pro Val Thr Ala Leu Leu Pro Leu Ala Leu Leu Leu
1 5 10 15

His Ala Ala Arg Pro Gly Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu
20 25 30

Asp Ser Thr Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
35 40 45

Gly Ser Gly Gly Gly Gly Ser Asp Ile Val Met Thr Gln Thr Pro Ala
50 55 60

Ser Val Ser Glu Pro Val Gly Gly Ser Val Thr Ile Lys Cys Gln Ala
65 70 75 80

Ser Gln Ser Phe Tyr Asn Leu Leu Ala Trp Tyr Gln Gln Lys Pro Gly
85 90 95

Gln Pro Pro Lys Leu Leu Ile Tyr Asp Ala Ser Asp Leu Ala Ser Gly
100 105 110

Val Pro Ser Arg Phe Lys Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu
115 120 125

Thr Ile Ser Asp Leu Glu Cys Ala Asp Ala Ala Tyr Tyr Cys Gln
130 135 140

Ser Ala Asp Gly Ser Ser Tyr Ala Phe Gly Gly Gly Thr Glu Val Val
145 150 155 160

Val Lys Gly Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
165 170 175

Gly Ser Gly Gly Gly Gly Ser Gln Glu Gln Leu Glu Glu Ser Gly Gly
180 185 190

Gly Leu Val Lys Pro Glu Gly Ser Leu Thr Leu Thr Cys Thr Ala Ser
195 200 205

Gly Val Ser Phe Ser Ser Ser Tyr Trp Ile Tyr Trp Val Arg Gln Ala
210 215 220

Pro Gly Lys Gly Leu Glu Trp Ile Ala Cys Ile Tyr Thr Gly Ser Ser
225 230 235 240

Gly Ser Thr Tyr Tyr Ala Ser Trp Ala Lys Gly Arg Phe Thr Val Ser
245 250 255

Glu Thr Ser Ser Thr Thr Val Thr Leu Gln Met Thr Ser Leu Thr Ala
260 265 270

Ala 275	Asp	Thr	Ala	Thr	Tyr	Phe	Cys 280	Ala	Arg	Ala	Ser	Ala 285	Trp	Thr	Tyr
Gly 290	Met	Asp	Leu	Trp	Gly	Pro 295	Gly	Thr	Leu	Val	Thr 300	Val	Ser	Ser	Gly
Gly 305	Gly	Gly	Ser	Gly	Gly	Gly 310	Gly	Ser	Gly	Gly 315	Gly	Gly	Ser	Gly	Gly 320
Gly	Gly	Ser	Gly	Gly	Gly	Gly 325	Ser	Gln	Val 330	Gln	Leu	Gln	Glu	Ser 335	Gly
Pro	Gly	Leu	Val 340	Lys	Pro	Ser	Glu	Thr 345	Leu	Ser	Leu	Thr	Cys 350	Thr	Val
Ser	Gly	Gly 355	Ser	Ile	Ser	Tyr	Tyr 360	Ser	Trp	Thr	Trp	Val 365	Arg	Gln	Pro
Ala 370	Gly	Gln	Gly	Leu	Glu	Trp 375	Ile	Gly	Arg	Ile	His 380	Ile	Ser	Gly	Val
Thr 385	Asn	His	Asn	Pro	Ser	Leu 390	Lys	Ser	Arg	Val 395	Ser	Met	Ser	Ile	Asp 400
Thr	Ser	Arg	Thr	Gln 405	Phe	Ser	Leu	Arg	Leu 410	Thr	Ser	Val	Thr	Ala 415	Ala
Asp	Thr	Ala	Val 420	Tyr	Phe	Cys	Ala	Arg 425	Ser	Gly	Gly	Thr 430	Tyr	Ser	Ala
Phe	Asp	Ile 435	Trp	Gly	Gln	Gly	Thr 440	Met	Val	Thr	Val 445	Ser	Ser	Gly	Gly
Gly 450	Gly	Ser	Gly	Gly	Gly	Gly 455	Ser	Gly	Gly	Gly	Gly 460	Ser	Gly	Gly	Gly
Gly 465	Ser	Glu	Ile	Val	Leu	Thr 470	Gln	Ser	Pro	Gly 475	Thr	Leu	Ser	Leu	Ser 480
Pro	Gly	Glu	Arg 485	Ser	Thr	Leu	Ser	Cys	Arg 490	Ala	Ser	Gln	Ser	Phe 495	Thr
Ser	Asn	Tyr	Leu 500	Ala	Trp	Tyr	Gln	Gln 505	Arg	Pro	Gly	Gln 510	Ala	Pro	Arg
Leu	Leu	Ile 515	Tyr	Gly	Ala	Ser	Thr 520	Arg	Ala	Ile	Gly	Ile 525	Pro	Asp	Arg
Phe 530	Ser	Gly	Ser	Gly	Ser	Gly 535	Thr	Asp	Phe	Thr	Leu 540	Thr	Ile	Ser	Arg
Leu 545	Glu	Pro	Glu	Asp	Phe	Ala 550	Val	Tyr	Tyr	Cys 555	Gln	Gln	Tyr	Gly	Ser 560
Ser	Pro	Trp	Thr	Phe 565	Gly	Gln	Gly	Thr	Lys 570	Val	Glu	Ile	Lys	Thr 575	Thr
Thr	Pro	Ala	Pro 580	Arg	Pro	Pro	Thr	Pro 585	Ala	Pro	Thr	Ile 590	Ala	Ser	Gln
Pro	Leu	Ser 595	Leu	Arg	Pro	Glu	Ala 600	Cys	Arg	Pro	Ala	Ala 605	Gly	Gly	Ala
Val 610	His	Thr	Arg	Gly	Leu	Asp 615	Phe	Ala	Cys	Asp	Ile 620	Tyr	Ile	Trp	Ala
Pro 625	Leu	Ala	Gly	Thr	Cys	Gly 630	Val	Leu	Leu	Leu	Ser 635	Leu	Val	Ile	Thr 640
Leu	Tyr	Cys	Lys	Arg 645	Gly	Arg	Lys	Lys	Leu 650	Leu	Tyr	Ile	Phe	Lys 655	Gln
Pro	Phe	Met	Arg 660	Pro	Val	Gln	Thr	Thr 665	Gln	Glu	Glu	Asp 670	Gly	Cys	Ser
Cys	Arg	Phe 675	Pro	Glu	Glu	Glu	Glu 680	Gly	Gly	Cys	Glu	Leu 685	Arg	Val	Lys

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Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln
 690 695 700
 Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu
 705 710 715 720
 Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg
 725 730 735
 Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met
 740 745 750
 Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly
 755 760 765
 Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp
 770 775 780
 Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg
 785 790 795

<210> SEQ ID NO 166
 <211> LENGTH: 660
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 166

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15
 His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Gln
 20 25 30
 Phe Ser Val Leu Gly Pro Ser Gly Pro Ile Leu Ala Met Val Gly Glu
 35 40 45
 Asp Ala Asp Leu Pro Cys His Leu Phe Pro Thr Met Ser Ala Glu Thr
 50 55 60
 Met Glu Leu Lys Trp Val Ser Ser Ser Leu Arg Gln Val Val Asn Val
 65 70 75 80
 Tyr Ala Asp Gly Lys Glu Val Glu Asp Arg Gln Ser Ala Pro Tyr Arg
 85 90 95
 Gly Arg Thr Ser Ile Leu Arg Asp Gly Ile Thr Ala Gly Lys Ala Ala
 100 105 110
 Leu Arg Ile His Asn Val Thr Ala Ser Asp Ser Gly Lys Tyr Leu Cys
 115 120 125
 Tyr Phe Gln Asp Gly Asp Phe Tyr Glu Lys Ala Leu Val Glu Leu Lys
 130 135 140
 Val Ala Gly Gly Gly Lys Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser
 145 150 155 160
 Gly Gly Gly Ser Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser
 165 170 175
 Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln
 180 185 190
 Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu Thr
 195 200 205
 Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Tyr Tyr Ser
 210 215 220
 Trp Thr Trp Val Arg Gln Pro Ala Gly Gln Gly Leu Glu Trp Ile Gly
 225 230 235 240
 Arg Ile His Ile Ser Gly Val Thr Asn His Asn Pro Ser Leu Lys Ser
 245 250 255

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Arg Val Ser Met Ser Ile Asp Thr Ser Arg Thr Gln Phe Ser Leu Arg
 260 265 270
 Leu Thr Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Phe Cys Ala Arg
 275 280 285
 Ser Gly Gly Thr Tyr Ser Ala Phe Asp Ile Trp Gly Gln Gly Thr Met
 290 295 300
 Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly
 305 310 315 320
 Gly Gly Gly Ser Gly Gly Gly Gly Ser Glu Ile Val Leu Thr Gln Ser
 325 330 335
 Pro Gly Thr Leu Ser Leu Ser Pro Gly Glu Arg Ser Thr Leu Ser Cys
 340 345 350
 Arg Ala Ser Gln Ser Phe Thr Ser Asn Tyr Leu Ala Trp Tyr Gln Gln
 355 360 365
 Arg Pro Gly Gln Ala Pro Arg Leu Leu Ile Tyr Gly Ala Ser Thr Arg
 370 375 380
 Ala Ile Gly Ile Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp
 385 390 395 400
 Phe Thr Leu Thr Ile Ser Arg Leu Glu Pro Glu Asp Phe Ala Val Tyr
 405 410 415
 Tyr Cys Gln Gln Tyr Gly Ser Ser Pro Trp Thr Phe Gly Gln Gly Thr
 420 425 430
 Lys Val Glu Ile Lys Thr Thr Thr Pro Ala Pro Arg Pro Pro Thr Pro
 435 440 445
 Ala Pro Thr Ile Ala Ser Gln Pro Leu Ser Leu Arg Pro Glu Ala Cys
 450 455 460
 Arg Pro Ala Ala Gly Gly Ala Val His Thr Arg Gly Leu Asp Phe Ala
 465 470 475 480
 Cys Asp Ile Tyr Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly Val Leu
 485 490 495
 Leu Leu Ser Leu Val Ile Thr Leu Tyr Cys Lys Arg Gly Arg Lys Lys
 500 505 510
 Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met Arg Pro Val Gln Thr Thr
 515 520 525
 Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe Pro Glu Glu Glu Glu Gly
 530 535 540
 Gly Cys Glu Leu Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala
 545 550 555 560
 Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg
 565 570 575
 Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu
 580 585 590
 Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn
 595 600 605
 Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met
 610 615 620
 Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly
 625 630 635 640
 Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala
 645 650 655
 Leu Pro Pro Arg
 660

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<210> SEQ ID NO 167
<211> LENGTH: 656
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide

<400> SEQUENCE: 167

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1             5             10             15

His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Ala
 20             25             30

Leu Gly Ser Asp Leu His Val Asp Val Lys Gly Tyr Lys Asp Gly Gly
 35             40             45

Ile His Leu Glu Cys Arg Ser Thr Gly Trp Tyr Pro Gln Pro Gln Ile
 50             55             60

Gln Trp Ser Asn Asn Lys Gly Glu Asn Ile Pro Thr Val Glu Ala Pro
 65             70             75             80

Val Val Ala Asp Gly Val Gly Leu Tyr Ala Val Ala Ala Ser Val Ile
 85             90             95

Met Arg Gly Ser Ser Gly Glu Gly Val Ser Cys Thr Ile Arg Ser Ser
100            105            110

Leu Leu Gly Leu Glu Lys Thr Ala Ser Ile Ser Ile Ala Asp Pro Phe
115            120            125

Phe Arg Ser Ala Gln Arg Trp Ile Ala Ala Leu Ala Gly Thr Gly Gly
130            135            140

Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser
145            150            155            160

Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly
165            170            175

Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln Val Gln Leu Gln
180            185            190

Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu Thr Leu Ser Leu Thr
195            200            205

Cys Thr Val Ser Gly Gly Ser Ile Ser Tyr Tyr Ser Trp Thr Trp Val
210            215            220

Arg Gln Pro Ala Gly Gln Gly Leu Glu Trp Ile Gly Arg Ile His Ile
225            230            235            240

Ser Gly Val Thr Asn His Asn Pro Ser Leu Lys Ser Arg Val Ser Met
245            250            255

Ser Ile Asp Thr Ser Arg Thr Gln Phe Ser Leu Arg Leu Thr Ser Val
260            265            270

Thr Ala Ala Asp Thr Ala Val Tyr Phe Cys Ala Arg Ser Gly Gly Thr
275            280            285

Tyr Ser Ala Phe Asp Ile Trp Gly Gln Gly Thr Met Val Thr Val Ser
290            295            300

Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
305            310            315            320

Gly Gly Gly Gly Ser Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu
325            330            335

Ser Leu Ser Pro Gly Glu Arg Ser Thr Leu Ser Cys Arg Ala Ser Gln
340            345            350

Ser Phe Thr Ser Asn Tyr Leu Ala Trp Tyr Gln Gln Arg Pro Gly Gln
355            360            365

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Ala Pro Arg Leu Leu Ile Tyr Gly Ala Ser Thr Arg Ala Ile Gly Ile
 370 375 380
 Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
 385 390 395 400
 Ile Ser Arg Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln
 405 410 415
 Tyr Gly Ser Ser Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile
 420 425 430
 Lys Thr Thr Thr Pro Ala Pro Arg Pro Pro Thr Pro Ala Pro Thr Ile
 435 440 445
 Ala Ser Gln Pro Leu Ser Leu Arg Pro Glu Ala Cys Arg Pro Ala Ala
 450 455 460
 Gly Gly Ala Val His Thr Arg Gly Leu Asp Phe Ala Cys Asp Ile Tyr
 465 470 475 480
 Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly Val Leu Leu Leu Ser Leu
 485 490 495
 Val Ile Thr Leu Tyr Cys Lys Arg Gly Arg Lys Lys Leu Leu Tyr Ile
 500 505 510
 Phe Lys Gln Pro Phe Met Arg Pro Val Gln Thr Thr Gln Glu Glu Asp
 515 520 525
 Gly Cys Ser Cys Arg Phe Pro Glu Glu Glu Glu Gly Gly Cys Glu Leu
 530 535 540
 Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln Gly
 545 550 555 560
 Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr
 565 570 575
 Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly Lys
 580 585 590
 Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys
 595 600 605
 Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg
 610 615 620
 Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala
 625 630 635 640
 Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg
 645 650 655

<210> SEQ ID NO 168

<211> LENGTH: 771

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 168

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu
 1 5 10 15
 His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Gln
 20 25 30
 Phe Ser Val Leu Gly Pro Ser Gly Pro Ile Leu Ala Met Val Gly Glu
 35 40 45
 Asp Ala Asp Leu Pro Cys His Leu Phe Pro Thr Met Ser Ala Glu Thr
 50 55 60
 Met Glu Leu Lys Trp Val Ser Ser Ser Leu Arg Gln Val Val Asn Val

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65	70	75	80
Tyr Ala Asp Gly Lys Glu Val Glu Asp Arg Gln Ser Ala Pro Tyr Arg			
	85	90	95
Gly Arg Thr Ser Ile Leu Arg Asp Gly Ile Thr Ala Gly Lys Ala Ala			
	100	105	110
Leu Arg Ile His Asn Val Thr Ala Ser Asp Ser Gly Lys Tyr Leu Cys			
	115	120	125
Tyr Phe Gln Asp Gly Asp Phe Tyr Glu Lys Ala Leu Val Glu Leu Lys			
	130	135	140
Val Ala Ala Leu Gly Ser Asp Leu His Val Asp Val Lys Gly Tyr Lys			
	145	150	155
Asp Gly Gly Ile His Leu Glu Cys Arg Ser Thr Gly Trp Tyr Pro Gln			
	165	170	175
Pro Gln Ile Gln Trp Ser Asn Asn Lys Gly Glu Asn Ile Pro Thr Val			
	180	185	190
Glu Ala Pro Val Val Ala Asp Gly Val Gly Leu Tyr Ala Val Ala Ala			
	195	200	205
Ser Val Ile Met Arg Gly Ser Ser Gly Glu Gly Val Ser Cys Thr Ile			
	210	215	220
Arg Ser Ser Leu Leu Gly Leu Glu Lys Thr Ala Ser Ile Ser Ile Ala			
	225	230	235
Asp Pro Phe Phe Arg Ser Ala Gln Arg Trp Ile Ala Ala Leu Ala Gly			
	245	250	255
Thr Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly			
	260	265	270
Gly Gly Ser Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly			
	275	280	285
Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gln Val			
	290	295	300
Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu Thr Leu			
	305	310	315
Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Tyr Tyr Ser Trp			
	325	330	335
Thr Trp Val Arg Gln Pro Ala Gly Gln Gly Leu Glu Trp Ile Gly Arg			
	340	345	350
Ile His Ile Ser Gly Val Thr Asn His Asn Pro Ser Leu Lys Ser Arg			
	355	360	365
Val Ser Met Ser Ile Asp Thr Ser Arg Thr Gln Phe Ser Leu Arg Leu			
	370	375	380
Thr Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Phe Cys Ala Arg Ser			
	385	390	395
Gly Gly Thr Tyr Ser Ala Phe Asp Ile Trp Gly Gln Gly Thr Met Val			
	405	410	415
Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly			
	420	425	430
Gly Gly Ser Gly Gly Gly Gly Ser Glu Ile Val Leu Thr Gln Ser Pro			
	435	440	445
Gly Thr Leu Ser Leu Ser Pro Gly Glu Arg Ser Thr Leu Ser Cys Arg			
	450	455	460
Ala Ser Gln Ser Phe Thr Ser Asn Tyr Leu Ala Trp Tyr Gln Gln Arg			
	465	470	475
Pro Gly Gln Ala Pro Arg Leu Leu Ile Tyr Gly Ala Ser Thr Arg Ala			
	485	490	495

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Ile Gly Ile Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe
      500                      505                      510

Thr Leu Thr Ile Ser Arg Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr
      515                      520                      525

Cys Gln Gln Tyr Gly Ser Ser Pro Trp Thr Phe Gly Gln Gly Thr Lys
      530                      535                      540

Val Glu Ile Lys Thr Thr Thr Pro Ala Pro Arg Pro Pro Thr Pro Ala
545                      550                      555                      560

Pro Thr Ile Ala Ser Gln Pro Leu Ser Leu Arg Pro Glu Ala Cys Arg
      565                      570                      575

Pro Ala Ala Gly Gly Ala Val His Thr Arg Gly Leu Asp Phe Ala Cys
      580                      585                      590

Asp Ile Tyr Ile Trp Ala Pro Leu Ala Gly Thr Cys Gly Val Leu Leu
      595                      600                      605

Leu Ser Leu Val Ile Thr Leu Tyr Cys Lys Arg Gly Arg Lys Lys Leu
      610                      615                      620

Leu Tyr Ile Phe Lys Gln Pro Phe Met Arg Pro Val Gln Thr Thr Gln
625                      630                      635                      640

Glu Glu Asp Gly Cys Ser Cys Arg Phe Pro Glu Glu Glu Glu Gly Gly
      645                      650                      655

Cys Glu Leu Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr
      660                      665                      670

Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg
      675                      680                      685

Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met
      690                      695                      700

Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu
705                      710                      715                      720

Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys
      725                      730                      735

Gly Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu
      740                      745                      750

Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu
      755                      760                      765

Pro Pro Arg
      770

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<210> SEQ ID NO 169

<211> LENGTH: 908

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 169

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Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
1          5          10          15

His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Gly
      20          25          30

Phe His Thr Ile Asn Ala Thr Trp Trp Ala Asn Ile Thr Leu Val Gly
      35          40          45

Pro Pro Asp Thr Pro Val Thr Trp Tyr Asp Thr Gln Gly Leu Trp Phe
      50          55          60

Cys Asn Gly Ser Arg Val Lys Asn Pro Gln Ile Arg His Thr Cys Asn

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65				70				75				80			
Asp	Gln	Asn	Leu	Thr 85	Leu	Ile	His	Val	Asn 90	Lys	Thr	Tyr	Glu	Arg 95	Thr
Tyr	Met	Gly	Tyr	Asn 100	Arg	Gln	Gly	Thr 105	Lys	Lys	Glu	Asp	Tyr	Lys 110	Val
Val	Val	Ile	Pro	Pro	Pro	Pro	Ala 120	Thr	Val	Lys	Pro	Gln	Pro	Glu	Pro
Glu	Tyr	Val	Phe	Val	Tyr	Met	Gly 135	Glu	Asn	Lys	Thr 140	Leu	Glu	Gly	Pro
Pro	Gly	Thr	Pro	Val	Thr 150	Trp	Phe	Asn	Gln	Asp 155	Gly	Lys	Lys	Phe	Cys 160
Glu	Gly	Glu	Lys	Val 165	Leu	His	Pro	Glu	Phe 170	Asn	His	Thr	Cys	Asp 175	Lys
Gln	Asn	Leu	Ile	Leu 180	Leu	Phe	Val	Asn 185	Phe	Thr	His	Asp	Gly	Ala 190	Tyr
Leu	Gly	Tyr	Asn	His 195	Gln	Gly	Thr 200	Gln	Arg	Thr	His	Tyr 205	Glu	Val	Thr
Val	Leu	Asp	Leu	Phe 210	Pro	Asp 215	Ser	Gly	Gln	Met	Lys 220	Ile	Glu	Asn	His
Ser	Glu	Glu	Thr	Glu 225	Gln 230	Lys	Asn	Asp	Glu	His 235	His	Asn	Trp	Gln	Lys 240
Gln	Gly	Gly	Gln	Lys 245	Gln	Gly	Gly	Gln	Lys 250	Thr	Asn	Gln	Thr	Lys 255	Val
Asn	Asp	Arg	Arg	Lys 260	Thr	Ala	Gln	Lys 265	Arg	Pro	Ser	Lys	Leu	Lys 270	Pro
Ala	Thr	Ile	Glu	Ala 275	Met	Leu	Val 280	Thr	Val	Thr	Ala	Gly 285	Ser	Asn	Leu
Thr	Leu	Val	Gly	Pro 290	Lys	Ala 295	Glu	Gly	Lys	Val	Thr 300	Trp	Phe	Asp	Gly
Asp	Leu	Lys	Arg	Pro 305	Cys 310	Glu	Pro	Asn	Tyr	Arg 315	Leu	Arg	His	Glu	Cys 320
Asn	Asn	Gln	Asn	Leu 325	Thr	Leu	Ile	Asn	Val 330	Thr	Lys	Asp	Tyr	Glu 335	Gly
Thr	Tyr	Tyr	Gly	Thr 340	Asn	Asp	Lys	Asp 345	Glu	Gly	Lys	Arg	Tyr	Arg 350	Val
Lys	Val	Asn	Thr	Thr 355	Asn	Ser	Gln 360	Ser	Val	Lys	Ile	Gln 365	Pro	Tyr	Thr
Arg	Gln	Thr	Thr	Pro 370	Asp	Gln 375	Glu	His	Lys	Phe	Glu 380	Leu	Gln	Phe	Glu
Thr	Asn	Gly	Asn	Tyr 385	Asp 390	Ser	Lys	Ile	Pro	Gly 395	Gly	Gly	Gly	Ser	Gly 400
Gly	Gly	Gly	Ser	Gly 405	Gly	Gly	Ser	Gly	Gly 410	Gly	Ser	Gly	Ser	Gly 415	Gly
Gly	Gly	Ser	Gly	Gly 420	Gly	Gly	Ser	Gly	Gly 425	Gly	Gly	Ser	Gly	Gly 430	Gly
Gly	Ser	Gly	Gly	Gly 435	Gly	Ser	Gln 440	Val	Gln	Leu	Gln	Glu 445	Ser	Gly	Pro
Gly	Leu	Val	Lys	Pro 450	Ser	Glu 455	Thr	Leu	Ser	Leu	Thr 460	Cys	Thr	Val	Ser
Gly	Gly	Ser	Ile	Ser 465	Tyr	Tyr 470	Ser	Trp	Thr	Trp 475	Val	Arg	Gln	Pro	Ala 480
Gly	Gln	Gly	Leu	Glu 485	Trp	Ile	Gly	Arg	Ile 490	His	Ile	Ser	Gly	Val	Thr

Asn	His		Asn	Pro	Ser	Leu	Lys	Ser	Arg	Val	Ser	Met	Ser	Ile	Asp	Thr
				500					505					510		
Ser	Arg	Thr	Gln	Phe	Ser	Leu	Arg	Leu	Thr	Ser	Val	Thr	Ala	Ala	Asp	
		515					520					525				
Thr	Ala	Val	Tyr	Phe	Cys	Ala	Arg	Ser	Gly	Gly	Thr	Tyr	Ser	Ala	Phe	
		530				535					540					
Asp	Ile	Trp	Gly	Gln	Gly	Thr	Met	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	
545					550				555						560	
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	
				565				570						575		
Ser	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu	Ser	Leu	Ser	Pro	
			580					585					590			
Gly	Glu	Arg	Ser	Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	Ser	Phe	Thr	Ser	
		595					600					605				
Asn	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Arg	Pro	Gly	Gln	Ala	Pro	Arg	Leu	
		610				615					620					
Leu	Ile	Tyr	Gly	Ala	Ser	Thr	Arg	Ala	Ile	Gly	Ile	Pro	Asp	Arg	Phe	
625					630					635					640	
Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Arg	Leu	
				645					650					655		
Glu	Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln	Tyr	Gly	Ser	Ser	
			660					665					670			
Pro	Trp	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile	Lys	Thr	Thr	Thr	
		675					680					685				
Pro	Ala	Pro	Arg	Pro	Pro	Thr	Pro	Ala	Pro	Thr	Ile	Ala	Ser	Gln	Pro	
		690				695					700					
Leu	Ser	Leu	Arg	Pro	Glu	Ala	Cys	Arg	Pro	Ala	Ala	Gly	Gly	Ala	Val	
705					710					715					720	
His	Thr	Arg	Gly	Leu	Asp	Phe	Ala	Cys	Asp	Ile	Tyr	Ile	Trp	Ala	Pro	
				725					730					735		
Leu	Ala	Gly	Thr	Cys	Gly	Val	Leu	Leu	Leu	Ser	Leu	Val	Ile	Thr	Leu	
			740					745					750			
Tyr	Cys	Lys	Arg	Gly	Arg	Lys	Lys	Leu	Leu	Tyr	Ile	Phe	Lys	Gln	Pro	
		755					760					765				
Phe	Met	Arg	Pro	Val	Gln	Thr	Thr	Gln	Glu	Glu	Asp	Gly	Cys	Ser	Cys	
		770				775					780					
Arg	Phe	Pro	Glu	Glu	Glu	Glu	Gly	Gly	Cys	Glu	Leu	Arg	Val	Lys	Phe	
785					790					795					800	
Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln	Gln	Gly	Gln	Asn	Gln	Leu	
				805					810					815		
Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu	Glu	Tyr	Asp	Val	Leu	Asp	
			820					825					830			
Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met								

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<210> SEQ ID NO 170
<211> LENGTH: 745
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide

<400> SEQUENCE: 170

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
1      5      10      15
His Ala Ala Arg Pro Gly Tyr Pro Tyr Asp Val Pro Asp Tyr Ala His
20     25     30
Asp Ala Cys Ile Pro Val Val Gly Lys Ile Gly Thr Asn Val Thr Leu
35     40     45
Asn Ala Val Asp Phe His Pro Gly Asp His Val Arg Trp Ser Tyr Gly
50     55     60
Pro Gly Gly Ala Gly Tyr Met Leu Cys Val Tyr Thr Gly Ser Trp Thr
65     70     75     80
Glu Tyr Lys Lys Pro Asp Ile Ile Phe Lys Cys Leu Ser Asn Asn Ser
85     90     95
Leu Leu Leu Ile Asn Val Thr Val Asn Tyr Thr Asn Thr Tyr Arg Thr
100    105    110
Leu Thr Ser Leu Asn Asn Trp Val His Asn Gln His His His Lys Phe
115    120    125
Pro Gly Trp Asn Leu Asp Thr Cys Tyr Ser Leu Thr Val Asn Glu Asn
130    135    140
Gly Thr Phe Pro Thr Thr Thr Lys Lys Pro Thr Thr Thr Thr Arg
145    150    155    160
Thr Thr Thr Thr Thr Thr Thr Lys Lys Thr Thr Thr Thr Arg Thr
165    170    175
Thr Ala Ala Lys Lys Thr Thr Ile Ser Thr Thr His His Lys His Ser
180    185    190
Ser Pro Lys Lys Ser Ser Thr Pro Asn Ser His Val Glu His His Val
195    200    205
Gly Phe Glu Ala Thr Ala Ala Glu Thr Pro Leu Gln Pro Ser Pro Gln
210    215    220
His Gln His Val Ala Thr His Gly Gly Gly Gly Ser Gly Gly Gly Gly
225    230    235    240
Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly Ser Gly Gly Gly Ser
245    250    255
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly
260    265    270
Gly Gly Gly Ser Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val
275    280    285
Lys Pro Ser Glu Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser
290    295    300
Ile Ser Tyr Tyr Ser Trp Thr Trp Val Arg Gln Pro Ala Gly Gln Gly
305    310    315    320
Leu Glu Trp Ile Gly Arg Ile His Ile Ser Gly Val Thr Asn His Asn
325    330    335
Pro Ser Leu Lys Ser Arg Val Ser Met Ser Ile Asp Thr Ser Arg Thr
340    345    350
Gln Phe Ser Leu Arg Leu Thr Ser Val Thr Ala Ala Asp Thr Ala Val
355    360    365

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Tyr Phe Cys Ala Arg Ser Gly Gly Thr Tyr Ser Ala Phe Asp Ile Trp
 370 375 380
 Gly Gln Gly Thr Met Val Thr Val Ser Ser Gly Gly Gly Ser Gly
 385 390 395 400
 Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Glu Ile
 405 410 415
 Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly Glu Arg
 420 425 430
 Ser Thr Leu Ser Cys Arg Ala Ser Gln Ser Phe Thr Ser Asn Tyr Leu
 435 440 445
 Ala Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Arg Leu Leu Ile Tyr
 450 455 460
 Gly Ala Ser Thr Arg Ala Ile Gly Ile Pro Asp Arg Phe Ser Gly Ser
 465 470 475 480
 Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu Pro Glu
 485 490 495
 Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Pro Trp Thr
 500 505 510
 Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Thr Thr Thr Pro Ala Pro
 515 520 525
 Arg Pro Pro Thr Pro Ala Pro Thr Ile Ala Ser Gln Pro Leu Ser Leu
 530 535 540
 Arg Pro Glu Ala Cys Arg Pro Ala Ala Gly Gly Ala Val His Thr Arg
 545 550 555 560
 Gly Leu Asp Phe Ala Cys Asp Ile Tyr Ile Trp Ala Pro Leu Ala Gly
 565 570 575
 Thr Cys Gly Val Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr Cys Lys
 580 585 590
 Arg Gly Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met Arg
 595 600 605
 Pro Val Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe Pro
 610 615 620
 Glu Glu Glu Glu Gly Gly Cys Glu Leu Arg Val Lys Phe Ser Arg Ser
 625 630 635 640
 Ala Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu
 645 650 655
 Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg
 660 665 670
 Gly Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln
 675 680 685
 Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr
 690 695 700
 Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp
 705 710 715 720
 Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala
 725 730 735
 Leu His Met Gln Ala Leu Pro Pro Arg
 740 745

<210> SEQ ID NO 171

<211> LENGTH: 245

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

-continued

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 171

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Asp Ile Leu Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
1           5           10           15
Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Asn Ile Gly Thr Ser
           20           25           30
Ile Gln Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
           35           40           45
Arg Ser Ser Ser Glu Ser Ile Ser Gly Ile Pro Ser Arg Phe Ser Gly
           50           55           60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Glu Pro
           65           70           75           80
Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Ser Asn Thr Trp Pro Phe
           85           90           95
Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Gly Gly Gly Ser
           100          105          110
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Glu
           115          120          125
Val Gln Leu Val Glu Ser Gly Ala Glu Val Lys Lys Pro Gly Ala Ser
           130          135          140
Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asn Tyr Ile
           145          150          155          160
Ile His Trp Val Lys Gln Glu Pro Gly Gln Gly Leu Glu Trp Ile Gly
           165          170          175
Tyr Phe Asn Pro Tyr Asn His Gly Thr Lys Tyr Asn Glu Lys Phe Lys
           180          185          190
Gly Arg Ala Thr Leu Thr Ala Asp Lys Ser Ile Ser Thr Ala Tyr Met
           195          200          205
Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys Ala
           210          215          220
Arg Ser Gly Pro Tyr Ala Trp Phe Asp Thr Trp Gly Gln Gly Thr Thr
           225          230          235          240
Val Thr Val Ser Ser
           245

```

<210> SEQ ID NO 172

<211> LENGTH: 25

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 172

```

Gly Ser Ser Pro Leu Gly Leu Ala Gly Ser Gly Gly Gly Ser Gly
1           5           10           15
Gly Gly Gly Ser Pro Met Ala Lys Lys
           20           25

```

<210> SEQ ID NO 173

<211> LENGTH: 25

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

-continued

<400> SEQUENCE: 173

Gly Gly Ser Val His Met Pro Leu Gly Phe Leu Gly Pro Gly Gly Ser
1 5 10 15

Gly Gly Gly Ser Pro Met Ala Lys Lys
20 25

<210> SEQ ID NO 174

<211> LENGTH: 25

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 174

Gly Gly Ser Pro Leu Gly Val Arg Gly Lys Gly Gly Gly Ser Gly
1 5 10 15

Gly Gly Gly Ser Pro Met Ala Lys Lys
20 25

<210> SEQ ID NO 175

<211> LENGTH: 25

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 175

Gly Gly Ser Ser Pro Leu Gly Leu Ala Gly Ser Gly Ser Gly Gly Gly
1 5 10 15

Ser Arg Gln Ala Arg Val Val Asn Gly
20 25

<210> SEQ ID NO 176

<211> LENGTH: 25

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 176

Gly Gly Ser Val His Met Pro Leu Gly Phe Leu Gly Pro Gly Gly Gly
1 5 10 15

Ser Arg Gln Ala Arg Val Val Asn Gly
20 25

<210> SEQ ID NO 177

<211> LENGTH: 25

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 177

Gly Gly Ser Pro Leu Gly Val Arg Gly Lys Gly Gly Ser Gly Gly Gly
1 5 10 15

Ser Arg Gln Ala Arg Val Val Asn Gly
20 25

<210> SEQ ID NO 178

<211> LENGTH: 5

-continued

<212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 178

Gly Gly Gly Gly Ser
 1 5

<210> SEQ ID NO 179
 <211> LENGTH: 16
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 179

Glu Leu Pro Thr Gln Gly Thr Phe Ser Asn Val Ser Thr Asn Val Ser
 1 5 10 15

<210> SEQ ID NO 180
 <211> LENGTH: 528
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 180

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu Leu
 1 5 10 15

His Ala Ala Arg Pro Gly Gly Gly Gly Ser Cys Pro Tyr Ser Asn Pro
 20 25 30

Ser Leu Cys Gly Gly Gly Gly Ser Gln Val Gln Leu Gln Glu Ser Gly
 35 40 45

Pro Gly Leu Val Lys Pro Ser Glu Thr Leu Ser Leu Thr Cys Thr Val
 50 55 60

Ser Gly Gly Ser Ile Ser Tyr Tyr Ser Trp Thr Trp Val Arg Gln Pro
 65 70 75 80

Ala Gly Gln Gly Leu Glu Trp Ile Gly Arg Ile His Ile Ser Gly Val
 85 90 95

Thr Asn His Asn Pro Ser Leu Lys Ser Arg Val Ser Met Ser Ile Asp
 100 105 110

Thr Ser Arg Thr Gln Phe Ser Leu Arg Leu Thr Ser Val Thr Ala Ala
 115 120 125

Asp Thr Ala Val Tyr Phe Cys Ala Arg Ser Gly Gly Thr Tyr Ser Ala
 130 135 140

Phe Asp Ile Trp Gly Gln Gly Thr Met Val Thr Val Ser Ser Gly Gly
 145 150 155 160

Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
 165 170 175

Gly Ser Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser
 180 185 190

Pro Gly Glu Arg Ser Thr Leu Ser Cys Arg Ala Ser Gln Ser Phe Thr
 195 200 205

Ser Asn Tyr Leu Ala Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Arg
 210 215 220

Leu Leu Ile Tyr Gly Ala Ser Thr Arg Ala Ile Gly Ile Pro Asp Arg

-continued

```

<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                             polypeptide

```

```

<400> SEQUENCE: 182

```

```

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly
1           5           10          15

```

```

Gly Ser Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly
20          25          30

```

```

Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
35          40          45

```

```

<210> SEQ ID NO 183
<211> LENGTH: 226
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                             polypeptide

```

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<400> SEQUENCE: 183

```

```

Gln Phe Ser Val Leu Gly Pro Ser Gly Pro Ile Leu Ala Met Val Gly
1           5           10          15

```

```

Glu Asp Ala Asp Leu Pro Cys His Leu Phe Pro Thr Met Ser Ala Glu
20          25          30

```

```

Thr Met Glu Leu Lys Trp Val Ser Ser Ser Leu Arg Gln Val Val Asn
35          40          45

```

```

Val Tyr Ala Asp Gly Lys Glu Val Glu Asp Arg Gln Ser Ala Pro Tyr
50          55          60

```

```

Arg Gly Arg Thr Ser Ile Leu Arg Asp Gly Ile Thr Ala Gly Lys Ala
65          70          75          80

```

```

Ala Leu Arg Ile His Asn Val Thr Ala Ser Asp Ser Gly Lys Tyr Leu
85          90          95

```

```

Cys Tyr Phe Gln Asp Gly Asp Phe Tyr Glu Lys Ala Leu Val Glu Leu
100         105         110

```

```

Lys Val Ala Ala Leu Gly Ser Asp Leu His Val Asp Val Lys Gly Tyr
115         120         125

```

```

Lys Asp Gly Gly Ile His Leu Glu Cys Arg Ser Thr Gly Trp Tyr Pro
130         135         140

```

```

Gln Pro Gln Ile Gln Trp Ser Asn Asn Lys Gly Glu Asn Ile Pro Thr
145         150         155         160

```

```

Val Glu Ala Pro Val Val Ala Asp Gly Val Gly Leu Tyr Ala Val Ala
165         170         175

```

```

Ala Ser Val Ile Met Arg Gly Ser Ser Gly Glu Gly Val Ser Cys Thr
180         185         190

```

```

Ile Arg Ser Ser Leu Leu Gly Leu Glu Lys Thr Ala Ser Ile Ser Ile
195         200         205

```

```

Ala Asp Pro Phe Phe Arg Ser Ala Gln Arg Trp Ile Ala Ala Leu Ala
210         215         220

```

```

Gly Thr
225

```

```

<210> SEQ ID NO 184
<211> LENGTH: 4
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

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-continued

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 184

Gly Gly Gly Ser
1

<210> SEQ ID NO 185

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 185

Ser Gly Gly Gly Gly
1 5

<210> SEQ ID NO 186

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 186

Ser Gly Gly Gly Gly Ser
1 5

<210> SEQ ID NO 187

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 187

Tyr Pro Tyr Asp Val Pro Asp Tyr Ala
1 5

What is claimed is:

1. A protease-activating CD45-gate chimeric antigen receptor (CD45-gate CAR) comprising an extracellular domain, a transmembrane domain, and an intracellular domain, wherein the extracellular domain comprises:

a CD45 recruiting domain comprising an anti-CD45 antibody, or an antigen binding fragment thereof, an antigen binding domain, and

a linker comprising one or more protease cleavage sites that is cleavable by at least one protease.

2. The protease-activating CD45-gate CAR of claim 1, wherein the intracellular domain comprises at least one signaling domain that is reversibly inactivated by CD45.

3. The protease-activating CD45-gate CAR of claim 1, wherein the linker is between the CD45 recruiting domain and the antigen binding domain.

4. The protease-activating CD45-gate CAR of claim 1, wherein the CD45 recruiting domain comprises one or more linkers.

5. The CD45-gate CAR of claim 1, wherein the linker connects the carboxy terminus of the CD45 recruiting domain to the amino terminus of the antigen binding

domain, and further wherein the intracellular domain comprises at least one signaling domain that is reversibly inactivated by CD45.

6. The protease-activating CD45-gate CAR of claim 1, wherein the CD45 recruiting domain comprises one or more of an anti-CD45 antibody antigen binding fragment.

7. The protease-activating CD45-gate CAR of claim 6, wherein the anti-CD45 antibody antigen binding fragment comprises an anti-CD45 scFv.

8. The protease-activating CD45-gate CAR of claim 1, wherein the CD45 recruiting domain comprises the amino acid sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, or SEQ ID NO: 6, without the signal sequence of SEQ ID NO:1.

9. The protease-activating CD45-gate CAR of claim 1, wherein the antigen binding domain specifically binds BCMA, MUC16, EGFR, EGFRvIII, MUC1, Flt-3, WT-1, CD20, CD23, CD30, CD38, CD70, CD33, CD133, MHC-WT1, TSPAN10, MHC-PRAME, MHC-NY-ESO1, HER2, (Carbonic anhydrase IX, LIV1, ADAM10, CHRNA2, LeY, NKG2D, CS1, CD44v6, ROR1, CD19, Claudin-18.2.

10. The protease-activating CD45 gate CAR of claim 1, wherein the linker comprising at least one protease cleavage site has a length of 15-100 amino acids.

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11. The protease-activating CD45-gate CAR of claim 1, wherein the linker comprises one or more protease cleavage sites that are cleavable by at least one protease which is present in a tumor microenvironment.

12. The protease-activating CD45-gate CAR of claim 1, wherein the linker comprises at least one protease cleavage site that is cleavable by a serine protease, a cysteine-type lysosomal protease, a metalloproteinase, a coagulation factor protease, or an aspartyl-type lysosomal protease.

13. The protease-activating CD45-gate CAR of claim 1, wherein the linker comprises at least one protease cleavage site that is cleavable by matrix metalloproteinase (MMP), matriptase (MT-SP1), trypsin, plasmin, prostate-specific antigen (PSA), urokinase plasminogen activator (uPA), urokinase plasminogen activator receptor (uPAR), legumain, a disintegrin and metalloproteinase (ADAM), a transmembrane Serine Protease (TMPRSS), Granzyme B, activated protein C, Caspase, Cathepsin, Chymase, Elastase, Guandinobenzotase, HtrA1, Human Neutrophil Elastase, Lactoferrin, Marapsin, NS3/4A, PACE4, tissue plasminogen activator (tPA), thrombin, DESC1, DPP-4, FAP, Hepsin, Matriptase-2, secretase, kallikrein-related peptidase (KLK), and tryptase.

14. The protease-activating CD45-gate CAR of claim 1, wherein the linker comprises one or more protease cleavage sites, wherein the protease cleavage site comprises the amino acid sequence of SEQ ID NO: 32, 91-98, 103-105 or SEQ ID NO: 106.

15. The protease-activating CD45-gate CAR of claim 1, wherein the linker comprises one or more amino acid sequences of SEQ ID NO: 53, 89, or 90.

16. The protease-activating CD45-gate CAR of claim 1, wherein the linker comprises one or more amino acid sequences of SEQ ID NO: 8-10, 53, 89, 90, 99-102, 107-120, 172-176 or SEQ ID NO: 177.

17. The protease-activating CD45-gate CAR of claim 1, wherein the linker comprises two or more protease cleavage sites and each cleavage site is the same as or different from any of the other cleavage sites.

18. The protease-activating CD45-gate CAR of claim 1, wherein the intracellular domain comprises the cytoplasmic signaling domain of one or more of CD3 zeta, CD28, and CD2.

19. The protease-activating CD45-gate CAR of claim 1, wherein the intracellular domain comprises at least one costimulatory domain.

20. The protease-activating CD45-gate CAR of claim 19, wherein the at least one costimulatory domain is a signaling region of CD28, OX-40, 4-1BB/CD137, CD2, CD7, CD27, CD30, CD40, inducible T cell costimulator (ICOS), lymphocyte function-associated antigen-1 (LFA-1), CD3 gamma, CD3 delta, CD3 epsilon, CD247, CD276 (B7-H3), LIGHT (TNFSF14), NKG2C, Ig alpha (CD79a), DAP-10, Fc gamma receptor, MHC class I molecule, Signaling Lymphocytic Activation Molecules (SLAM proteins), BTLA, a Toll ligand receptor, ICAM-1, CDS, GITR, BAFFR, HVEM (LIGHTR), KIRDS2, NKp80 (KLRF1), NKp44, NKp30, NKp46, CD19, CD4, CD8alpha, CD8beta, IL-2R beta,

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IL-2R gamma, IL-7R alpha, ITGA4, VLA1, ITGA6, ITGAD, ITGAE, ITGAL, IT GAM, ITGAX, ITGB1, ITGB7, NKG2D, TNFR2, TRANCE/RANKL, DNAMI (CD226), CD84, CD96 (Tactile), CEACAM1, CRT AM, Ly9 (CD229), CD 160 (BY55), PSGL1, CD100 (SEMA4D), CD69, SELPLG (CD162), LTBR, LAT, GADS, SLP-76, PAG/Cbp, CD 19a, a ligand that specifically binds with CD83, or any combination thereof.

21. The protease-activating CD45-gate CAR of claim 1, further comprising a signal sequence optionally wherein the signal sequence is a CD8 signal sequence comprising the amino acid sequence of SEQ ID NO: 1.

22. The protease-activating CD45-gate CAR of claim 1, wherein the protease-activating CD45-gate CAR comprises the amino acid sequence of any one of SEQ ID NOs: 11, 16-31, 56, 75, 76, 77, 78, 79, and 121-170 or a variant thereof, wherein the variant does not comprise the amino acid sequence of the HA tag of SEQ ID NO: 2, does not comprise the amino acid sequence of the V5 peptide motif of SEQ ID NO: 55, or does not comprise the signal sequence of SEQ ID NO: 1.

23. A nucleic acid encoding the protease-activating CD45-gate CAR of claim 1.

24. An engineered immune cell comprising the nucleic acid of claim 23.

25. A vector comprising the nucleic acid of claim 23.

26. The vector of claim 25, wherein the vector is an expression vector.

27. An engineered immune cell comprising the vector of claim 25.

28. An engineered immune cell comprising the protease-activating CD45-gate CAR of claim 1.

29. The engineered immune cell of claim 28, wherein the immune cell is a T cell, an NK cell, or a dendritic cell.

30. The engineered immune cell of claim 29, wherein the T cell is a tumor infiltrating lymphocyte, an iPSC-derived T cell, a TCR-expressing cell, or an NK-T cell.

31. A population of cells comprising at least about 1×10^4 , 1×10^5 , 1×10^6 , 1×10^7 or 1×10^8 of the engineered immune cell of claim 28.

32. A pharmaceutical composition comprising the engineered immune cell of claim 28 and a pharmaceutically acceptable carrier.

33. A method of treating cancer in a patient comprising administering to the patient the engineered immune cell of claim 28, and wherein the cancer expresses an antigen that is recognized and bound by the antigen binding domain of the CD45-gate CAR.

34. The method of claim 33, wherein the cancer is a solid tumor cancer or a liquid tumor cancer.

35. A method of treating a patient who has a tumor characterized by a protease-rich tumor microenvironment, comprising administering to the patient the engineered immune cell of claim 28, and wherein the tumor expresses an antigen that is recognized and bound by the antigen binding domain of the CD45-gate CAR.

* * * * *